



US012398407B2

(12) **United States Patent**
Soong et al.

(10) **Patent No.: US 12,398,407 B2**

(45) **Date of Patent: Aug. 26, 2025**

(54) **YEAST EXPRESSING A HETEROLOGOUS
PHOSPHOLIPASE FOR ETHANOL
PRODUCTION**

(71) Applicant: **Novozymes A/S**, Bagsvaerd (DK)

(72) Inventors: **Chee-Leong Soong**, Raleigh, NC (US);
James Ron Huffman, Wake Forest, NC
(US); **Monica Tassone**, West
Sacramento, CA (US); **Jung Yi**,
Sacramento, CA (US); **Hanlin Ouyang**,
Raleigh, NC (US)

(73) Assignee: **Novozymes A/S**, Bagsvaerd (DK)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/484,322**

(22) Filed: **Oct. 10, 2023**

(65) **Prior Publication Data**

US 2024/0110204 A1 Apr. 4, 2024

Related U.S. Application Data

(62) Division of application No. 17/283,760, filed as
application No. PCT/US2019/054996 on Oct. 7,
2019, now Pat. No. 11,807,889.

(60) Provisional application No. 62/742,805, filed on Oct.
8, 2018.

(51) **Int. Cl.**

C12P 7/06 (2006.01)

C12N 9/16 (2006.01)

C12N 9/18 (2006.01)

C12N 9/34 (2006.01)

(52) **U.S. Cl.**

CPC **C12P 7/06** (2013.01); **C12N 9/16**
(2013.01); **C12N 9/18** (2013.01); **C12N**
9/2428 (2013.01); **C12Y 301/01004** (2013.01);
C12Y 301/04003 (2013.01); **C12Y 302/01001**
(2013.01); **C12Y 302/01003** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,745,676 B2 * 8/2020 Sun C11B 3/04
2015/0368627 A1 12/2015 Sun et al.
2018/0155744 A1 6/2018 Cripwell et al.

FOREIGN PATENT DOCUMENTS

WO 1996040939 A2 12/1996
WO 2008135547 A1 11/2008
WO 2011128712 A1 10/2011
WO 2011153516 A2 12/2011
WO 2014090161 A1 6/2014
WO 2014147219 A1 9/2014
WO 2015140275 A1 9/2015
WO 2017037614 A1 3/2017
WO 2017077504 A1 5/2017
WO 2017087330 A1 5/2017
WO 2018075430 A1 4/2018
WO 2018098381 A1 5/2018

OTHER PUBLICATIONS

Accession BBO52657. Nov. 20, 2014 (Year: 2014).
Aon, 2001, Metabolic engineering, 3, 250-264.
Chica et al., Current Opinion in Biotechnology, 2005, 378-384,
16(4).
Keun, 2012, Biochemistry molecular biology, Abstract.
Liu et al, 2014, WO2014090161—Accession No. BBI75668.
Singh et al., Current Protein and Peptide Science, 2017, 1-11, 18.

* cited by examiner

Primary Examiner — Christian L Fronda

(74) *Attorney, Agent, or Firm* — Eric J. Fechter

(57) **ABSTRACT**

Described herein are recombinant fermenting organisms
having a heterologous polynucleotide encoding a phospho-
lipase. Also described are processes for producing a ferment-
ation product, such as ethanol, from starch or cellulosic-
containing material with the recombinant fermenting
organisms.

20 Claims, 4 Drawing Sheets

Specification includes a Sequence Listing.

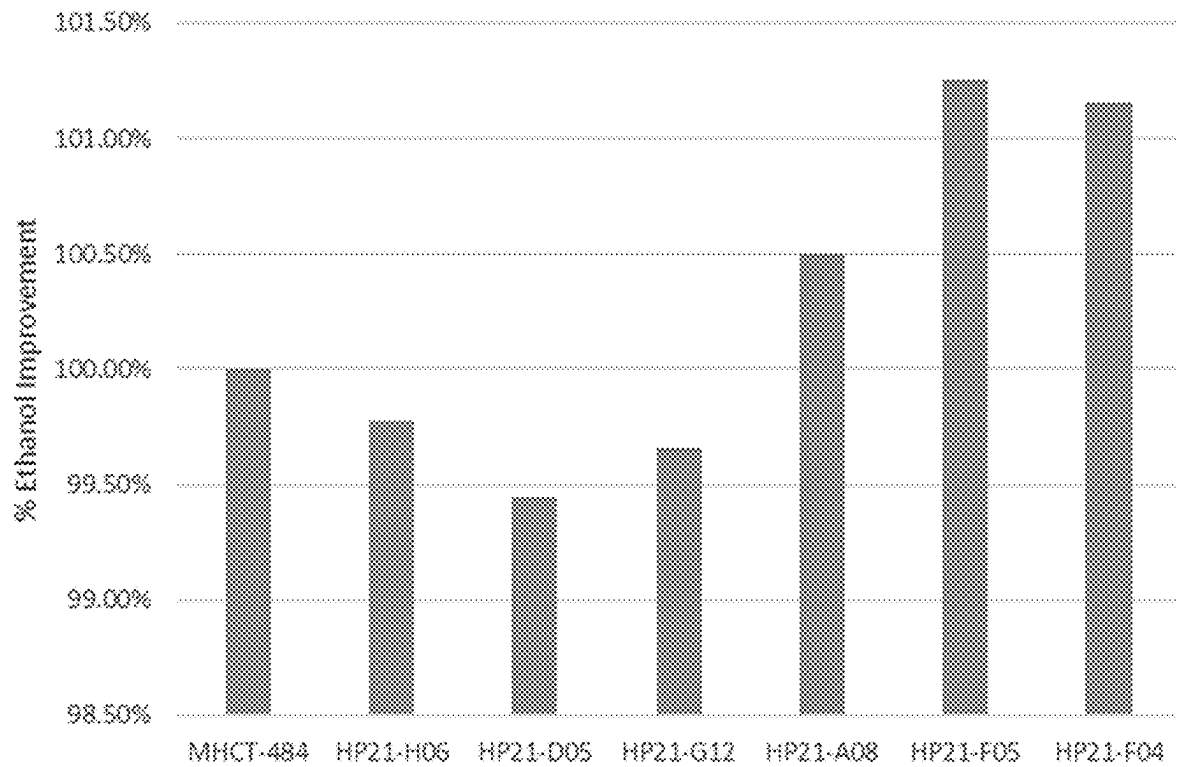


FIG. 1

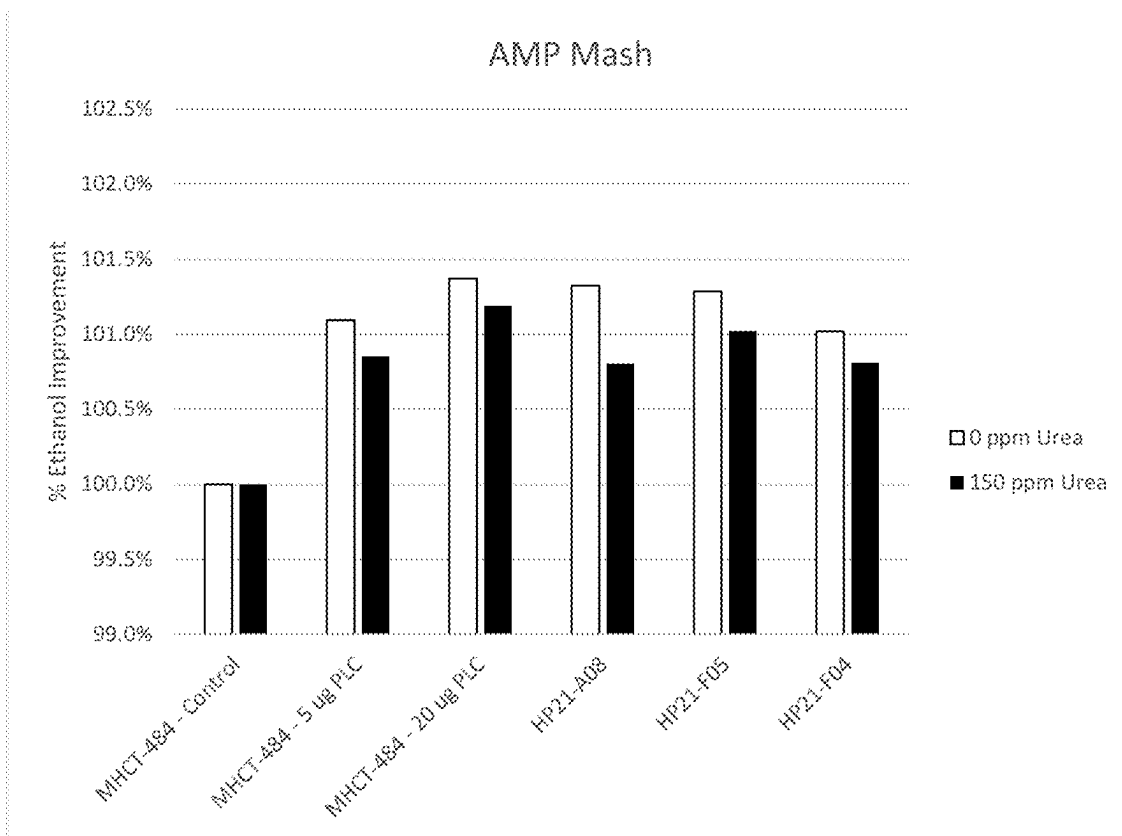


FIG. 2

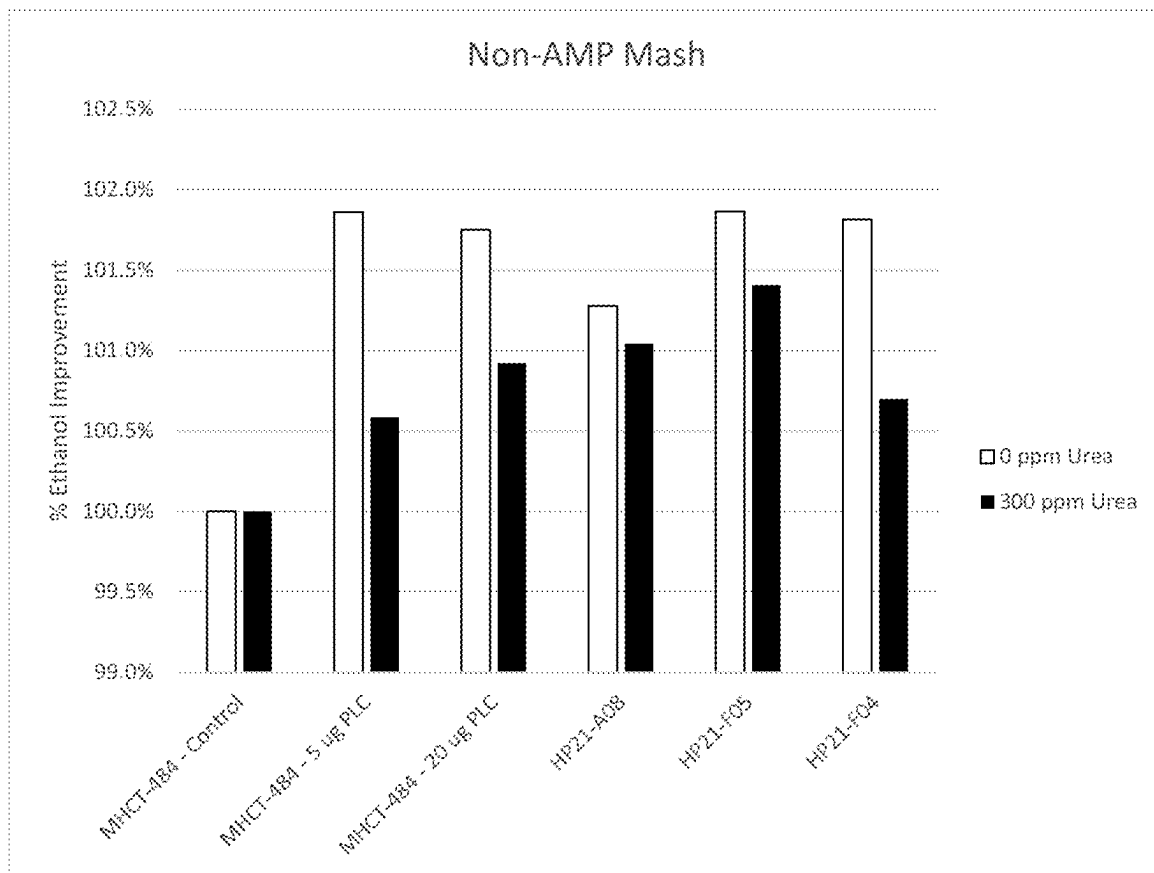


FIG. 3



FIG. 4

YEAST EXPRESSING A HETEROLOGOUS PHOSPHOLIPASE FOR ETHANOL PRODUCTION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a divisional of U.S. patent application Ser. No. 17/283,760 which is a 35 U.S.C. 371 national application of PCT/US2019/054996, filed Oct. 7, 2019, which claims priority to the benefit from U.S. Provisional Application Ser. No. 62/742,805, filed Oct. 8, 2018. The contents of these applications are fully incorporated herein by reference.

REFERENCE TO A SEQUENCE LISTING

This application contains a Sequence Listing in computer readable form, that was submitted as an xml file named SQ_ST26 (created on Oct. 3, 2023, containing 665 kb), 20 which is incorporated herein by reference.

BACKGROUND

Production of ethanol from starch and cellulosic containing materials is well-known in the art.

The most commonly industrially used commercial process for starch-containing material, often referred to as a “conventional process”, includes liquefying gelatinized starch at high temperature (about 85° C.) using typically a bacterial alpha-amylase, followed by simultaneous saccharification and fermentation (SSF) carried out anaerobically in the presence of typically a glucoamylase and a *Saccharomyces cerevisiae* yeast.

Yeasts which are used for production of ethanol for use as fuel, such as in the corn ethanol industry, require several characteristics to ensure cost effective production of the ethanol. These characteristics include ethanol tolerance, low by-product yield, rapid fermentation, and the ability to limit the amount of residual sugars remaining in the ferment. Such characteristics have a marked effect on the viability of the industrial process.

Yeast of the genus *Saccharomyces* exhibits many of the characteristics required for production of ethanol. In particular, strains of *Saccharomyces cerevisiae* are widely used for the production of ethanol in the fuel ethanol industry. Strains of *Saccharomyces cerevisiae* that are widely used in the fuel ethanol industry have the ability to produce high yields of ethanol under fermentation conditions found in, for example, the fermentation of corn mash. An example of such a strain is the yeast used in commercially available ethanol yeast product called ETHANOL RED®.

Saccharomyces cerevisiae yeast have been genetically engineered to express alpha-amylase and/or glucoamylase to improve yield and decrease the amount of exogenously added enzymes necessary during SSF (e.g., WO2018/098381, WO2017/087330, WO2017/037614, WO2011/128712, WO2011/153516, US2018/0155744). Yeast have also been engineered to express trehalase in an attempt to increase fermentation yield by breaking down residual trehalose (e.g., WO2017/077504).

WO2008/135547 concerns reducing foam in processes for production of a fermentation product by contacting the fermentation media comprising a fermenting organism with a lipolytic enzyme selected from the group consisting of phospholipase, lyso-phospholipase and lipase, and a metal salt.

WO2014/147219 concerns a phospholipase A from *Talaromyces leycettanus*.

WO2015/140275 discloses a phospholipase C from *Bacillus thuringiensis*.

5 Despite significant improvement of ethanol production processes over the past decade there is still a desire and need for providing improved processes of ethanol fermentation from starch and cellulosic containing material in an economically and commercially relevant scale.

10 For example, foam generation during ethanol fermentation is a major problem, especially in ethanol production processes where starch-containing material is liquefied with an alpha-amylase and a protease before saccharification and fermentation. Additionally, the use of nitrogen supplements (e.g., urea) is an added expense during fermentation. Therefore, there is a desire to, inter alia, reduce foam and/or reduce supplemental nitrogen requirements in ethanol fermentation.

SUMMARY

Described herein are, inter alia, methods for producing a fermentation product, such as ethanol, from starch or cellulosic-containing material, and yeast suitable for use in such processes. The Applicant has surprisingly found that yeasts expressing a phospholipase provide beneficial properties during fermentation, such as reduced foaming, improved oil extraction yield, and improved ethanol yield.

A first aspect relates to methods of producing a fermentation product from a starch-containing or cellulosic-containing material comprising: (a) saccharifying the starch-containing or cellulosic-containing material; and (b) fermenting the saccharified material of step (a) with a fermenting organism; wherein the fermenting organism comprises a heterologous polynucleotide encoding a phospholipase. In some embodiments, the phospholipase is a Phospholipase A or a Phospholipase C.

In some embodiments of the methods, fermentation and saccharification are performed simultaneously in a simultaneous saccharification and fermentation (SSF). In other embodiments, fermentation and saccharification are performed sequentially (SHF).

In some embodiments of the methods, the method comprises recovering the fermentation product from the fermentation (e.g., by distillation).

In some embodiments of the methods, the fermentation product is ethanol.

In some embodiments of the methods, fermentation is performed under reduced nitrogen conditions (e.g., less than 1000 ppm urea or ammonium hydroxide, such as less than 750 ppm, less than 500 ppm, less than 400 ppm, less than 300 ppm, less than 250 ppm, less than 200 ppm, less than 150 ppm, less than 100 ppm, less than 75 ppm, less than 50 ppm, less than 25 ppm, or less than 10 ppm).

In some embodiments of the methods, the method results in higher yield of fermentation product (e.g., ethanol) and/or reduced foam accumulation when compared to the same process using an identical cell without the heterologous polynucleotide encoding the phospholipase under the same conditions (e.g., at about or after 54 hours fermentation, such as the conditions described in Examples 3 or 4). In some embodiments, the method results in at least 0.25% (e.g., 0.5%, 0.75%, 1.0%, 1.25%, 1.5%, 1.75%, 2%, 3% or 5%) higher yield of fermentation product.

In some embodiments of the methods, the phospholipase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or

100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342. In some embodiments of the methods, the phospholipase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242. In some embodiments of the methods, the heterologous polynucleotide encodes a phospholipase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). In some embodiments of the methods, the heterologous polynucleotide encodes a phospholipase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242).

In some embodiments of the methods, saccharification of step occurs on a starch-containing material, and wherein the starch-containing material is either gelatinized or ungelatinized starch.

In some embodiments of the methods, the method comprises liquefying the starch-containing material by contacting the material with an alpha-amylase prior to saccharification.

In some embodiments of the methods, liquefying the starch-containing material and/or saccharifying the starch-containing material is conducted in presence of exogenously added protease.

In some embodiments of the methods, the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase, such as a glucoamylase having a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of a *Pycnoporus* glycoamylase (e.g., a *Pycnoporus sanguineus* glucoamylase of SEQ ID NO: 229), a *Gloeophyllum* glucoamylase (e.g. a *Gloeophyllum sepiarium* of SEQ ID NO: 8), or a glucoamylase of any one of SEQ ID NOs: 102-113 (e.g., a *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103 or 104, or a *Trichoderma reesei* glucoamylase of SEQ ID NO: 230).

In some embodiments of the methods, the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase, such as an alpha-amylase having a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231. In some embodiments of the methods, the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231. In some embodiments of the methods, the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: SEQ ID NOs: 76-101, 121-174 and 231.

In some embodiments of the methods, the fermenting organism comprises a heterologous polynucleotide encoding a trehalase, such as a trehalase having a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 175-226. In some embodiments of the methods, the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 175-226. In some embodiments of the methods, the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: SEQ ID NOs: 175-226.

In some embodiments of the methods, the fermenting organism comprises a heterologous polynucleotide encoding a protease, such as a protease having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69).

In some embodiments of the methods, saccharification of step occurs on a cellulosic-containing material, and wherein the cellulosic-containing material is pretreated (e.g. a dilute acid pretreatment).

In some embodiments of the methods, saccharification occurs on a cellulosic-containing material, and wherein the enzyme composition comprises one or more enzymes selected from a cellulase (e.g., endoglucanase, a cellobiohydrolase, or a beta-glucosidase), an AA9 polypeptide, a hemicellulase (e.g., a xylanase, an acetylxyloxyesterase, a feruloyl esterase, an arabinofuranosidase, a xylosidase, or a glucuronidase), a CIP, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin.

In some embodiments of the methods, the fermenting organism is a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell. In some embodiments, the fermenting organism is a *Saccharomyces cerevisiae* cell.

Another aspect relates to a recombinant yeast cell comprising a heterologous polynucleotide encoding a phospholipase. In some embodiments, the phospholipase is a Phospholipase A or a Phospholipase C.

In some embodiments of the yeast cell, the cell is capable of higher yield of fermentation product and/or reduced foam accumulation when compared to fermentation using the same process and an identical cell without the heterologous polynucleotide encoding the phospholipase under the same conditions (e.g., at about or after 54 hours fermentation, such as the conditions described in Examples 3 or 4). In some embodiments, the cell is capable of at least 0.25% (e.g., 0.5%, 0.75%, 1.0%, 1.25%, 1.5%, 1.75%, 2%, 3% or 5%) higher yield of fermentation product.

In some embodiments, the recombinant yeast cell is a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell. In some embodiments, the recombinant yeast cell is a *Saccharomyces cerevisiae* cell.

5

In some embodiments of the yeast cell, the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase, such as a glucoamylase having a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of a *Pycnoporus* glycoamylase (e.g., a *Pycnoporus sanguineus* glucoamylase of SEQ ID NO: 229), a *Gloeophyllum* glucoamylase (e.g., a *Gloeophyllum sepiarium* of SEQ ID NO: 8), or a glucoamylase of any one of SEQ ID NOs: 102-113 (e.g., a *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103 or 104, or a *Trichoderma reesei* glucoamylase of SEQ ID NO: 230).

In some embodiments of the yeast cell, the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase, wherein the alpha-amylase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231. In some embodiments of the methods, the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231. In some embodiments of the methods, the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: SEQ ID NOs: 76-101, 121-174 and 231.

In some embodiments of the yeast cell, the fermenting organism comprises a heterologous polynucleotide encoding a trehalase, wherein the trehalase has mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 175-226. In some embodiments of the methods, the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 175-226. In some embodiments of the methods, the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: SEQ ID NOs: 175-226.

In some embodiments of the yeast cell, the fermenting organism comprises a heterologous polynucleotide encoding a protease, such as a protease having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69).

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 shows % ethanol improvement for phospholipase-expressing yeast strains and control strain MHCT-484 described in Example 3.

6

FIG. 2 shows normalized mean ethanol improvement after 54 hours of fermentation of AMP mash at 0 and 150 ppm urea as described in Example 4.

FIG. 3 shows normalized mean ethanol improvement after 54 hours of fermentation of the non-AMP mash at 0 and 300 ppm urea as described in Example 4.

FIG. 4 shows improved defoaming capability during fermentation of a phospholipase-expressing yeast strain HP21-F04 (right) compared to control strain yMHCT48 (left) as described in Example 9.

DEFINITIONS

Unless defined otherwise or clearly indicated by context, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art.

Allelic variant: The term “allelic variant” means any of two or more alternative forms of a gene occupying the same chromosomal locus. Allelic variation arises naturally through mutation, and may result in polymorphism within populations. Gene mutations can be silent (no change in the encoded polypeptide) or may encode polypeptides having altered amino acid sequences. An allelic variant of a polypeptide is a polypeptide encoded by an allelic variant of a gene.

Alpha-amylase: The term “alpha amylase” means an 1,4-alpha-D-glucan glucanohydrolase, EC. 3.2.1.1, which catalyze hydrolysis of starch and other linear and branched 1,4-glucosidic oligo- and polysaccharides. For purposes of the present invention, alpha amylase activity can be determined using an alpha amylase assay described in the examples section below.

Auxiliary Activity 9: The term “Auxiliary Activity 9” or “AA9” means a polypeptide classified as a lytic polysaccharide monooxygenase (Quinlan et al., 2011, *Proc. Natl. Acad. Sci. USA* 208: 15079-15084; Phillips et al., 2011, *ACS Chem. Biol.* 6: 1399-1406; Lin et al., 2012, *Structure* 20: 1051-1061). AA9 polypeptides were formerly classified into the glycoside hydrolase Family 61 (GH61) according to Henrissat, 1991, *Biochem. J.* 280: 309-316, and Henrissat and Bairoch, 1996, *Biochem. J.* 316: 695-696.

AA9 polypeptides enhance the hydrolysis of a cellulosic-containing material by an enzyme having cellulolytic activity. Cellulolytic enhancing activity can be determined by measuring the increase in reducing sugars or the increase of the total of cellobiose and glucose from the hydrolysis of a cellulosic-containing material by cellulolytic enzyme under the following conditions: 1-50 mg of total protein/g of cellulose in pretreated corn stover (PCS), wherein total protein is comprised of 50-99.5% w/w cellulolytic enzyme protein and 0.5-50% w/w protein of an AA9 polypeptide for 1-7 days at a suitable temperature, such as 40 C-80° C., e.g., 50° C., 55° C., 60° C., 65° C., or 70° C., and a suitable pH, such as 4-9, e.g., 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, or 8.5, compared to a control hydrolysis with equal total protein loading without cellulolytic enhancing activity (1-50 mg of cellulolytic protein/g of cellulose in PCS).

AA9 polypeptide enhancing activity can be determined using a mixture of CELLULAST® 1.5 L (Novozymes A/S, Bagsvaerd, Denmark) and beta-glucosidase as the source of the cellulolytic activity, wherein the beta-glucosidase is present at a weight of at least 2-5% protein of the cellulase protein loading. In one embodiment, the beta-glucosidase is an *Aspergillus oryzae* beta-glucosidase (e.g., recombinantly produced in *Aspergillus oryzae* according to WO 02/095014). In another embodiment, the beta-glucosidase is

an *Aspergillus fumigatus* beta-glucosidase (e.g., recombinantly produced in *Aspergillus oryzae* as described in WO 02/095014).

AA9 polypeptide enhancing activity can also be determined by incubating an AA9 polypeptide with 0.5% phosphoric acid swollen cellulose (PASC), 100 mM sodium acetate pH 5, 1 mM MnSO₄, 0.1% gallic acid, 0.025 mg/mL of *Aspergillus fumigatus* beta-glucosidase, and 0.01% TRITON® X-100 (4-(1,1,3,3-tetramethylbutyl)phenyl-polyethylene glycol) for 24-96 hours at 40° C. followed by determination of the glucose released from the PASC.

AA9 polypeptide enhancing activity can also be determined according to WO 2013/028928 for high temperature compositions.

AA9 polypeptides enhance the hydrolysis of a cellulosic-containing material catalyzed by enzyme having cellulolytic activity by reducing the amount of cellulolytic enzyme required to reach the same degree of hydrolysis preferably at least 1.01-fold, e.g., at least 1.05-fold, at least 1.10-fold, at least 1.25-fold, at least 1.5-fold, at least 2-fold, at least 3-fold, at least 4-fold, at least 5-fold, at least 10-fold, or at least 20-fold.

Beta-glucosidase: The term “beta-glucosidase” means a beta-D-glucoside glucohydrolase (E.C. 3.2.1.21) that catalyzes the hydrolysis of terminal non-reducing beta-D-glucose residues with the release of beta-D-glucose. Beta-glucosidase activity can be determined using p-nitrophenyl-beta-D-glucopyranoside as substrate according to the procedure of Venturi et al., 2002, *J. Basic Microbiol.* 42: 55-66. One unit of beta-glucosidase is defined as 1.0 μmole of p-nitrophenolate anion produced per minute at 25° C., pH 4.8 from 1 mM p-nitrophenyl-beta-D-glucopyranoside as substrate in 50 mM sodium citrate containing 0.01% TWEEN® 20.

Beta-xylosidase: The term “beta-xylosidase” means a beta-D-xyloside xylohydrolase (E.C. 3.2.1.37) that catalyzes the exo-hydrolysis of short beta (1→4)-xylooligosaccharides to remove successive D-xylose residues from non-reducing termini. Beta-xylosidase activity can be determined using 1 mM p-nitrophenyl-beta-D-xyloside as substrate in 100 mM sodium citrate containing 0.01% TWEEN® 20 at pH 5, 40° C. One unit of beta-xylosidase is defined as 1.0 μmole of p-nitrophenolate anion produced per minute at 40° C., pH 5 from 1 mM p-nitrophenyl-beta-D-xyloside in 100 mM sodium citrate containing 0.01% TWEEN® 20.

Catalase: The term “catalase” means a hydrogen-peroxide:hydrogen-peroxide oxidoreductase (EC 1.11.1.6) that catalyzes the conversion of 2 H₂O₂ to O₂+2 H₂O. For purposes of the present invention, catalase activity is determined according to U.S. Pat. No. 5,646,025. One unit of catalase activity equals the amount of enzyme that catalyzes the oxidation of 1 μmole of hydrogen peroxide under the assay conditions.

Catalytic domain: The term “catalytic domain” means the region of an enzyme containing the catalytic machinery of the enzyme.

Cellobiohydrolase: The term “cellobiohydrolase” means a 1,4-beta-D-glucan cellobiohydrolase (E.C. 3.2.1.91 and E.C. 3.2.1.176) that catalyzes the hydrolysis of 1,4-beta-D-glucosidic linkages in cellulose, cellooigosaccharides, or any beta-1,4-linked glucose containing polymer, releasing cellobiose from the reducing end (cellobiohydrolase I) or non-reducing end (cellobiohydrolase II) of the chain (Teeri, 1997, *Trends in Biotechnology* 15: 160-167; Teen et al., 1998, *Biochem. Soc. Trans.* 26:173-178). Cellobiohydrolase activity can be determined according to the procedures

described by Lever et al., 1972, *Anal. Biochem.* 47: 273-279; van Tilbeurgh et al., 1982, *FEBS Letters* 149: 152-156; van Tilbeurgh and Claeysens, 1985, *FEBS Letters* 187: 283-288; and Tomme et al., 1988, *Eur. J. Biochem.* 170: 575-581.

Cellulolytic enzyme or cellulase: The term “cellulolytic enzyme” or “cellulase” means one or more (e.g., several) enzymes that hydrolyze a cellulosic-containing material. Such enzymes include endoglucanase(s), cellobiohydrolase(s), beta-glucosidase(s), or combinations thereof. The two basic approaches for measuring cellulolytic enzyme activity include: (1) measuring the total cellulolytic enzyme activity, and (2) measuring the individual cellulolytic enzyme activities (endoglucanases, cellobiohydrolases, and beta-glucosidases) as reviewed in Zhang et al., 2006, *Biotechnology Advances* 24: 452-481. Total cellulolytic enzyme activity can be measured using insoluble substrates, including Whatman No 1 filter paper, microcrystalline cellulose, bacterial cellulose, algal cellulose, cotton, pretreated lignocellulose, etc. The most common total cellulolytic activity assay is the filter paper assay using Whatman No 1 filter paper as the substrate. The assay was established by the International Union of Pure and Applied Chemistry (IUPAC) (Ghose, 1987, *Pure Appl. Chem.* 59: 257-68).

Cellulolytic enzyme activity can be determined by measuring the increase in production/release of sugars during hydrolysis of a cellulosic-containing material by cellulolytic enzyme(s) under the following conditions: 1-50 mg of cellulolytic enzyme protein/g of cellulose in pretreated corn stover (PCS) (or other pretreated cellulosic-containing material) for 3-7 days at a suitable temperature such as 40° C.-80° C., e.g., 50° C., 55° C., 60° C., 65° C., or 70° C., and a suitable pH such as 4-9, e.g., 5.0, 5.5, 6.0, 6.5, or 7.0, compared to a control hydrolysis without addition of cellulolytic enzyme protein. Typical conditions are 1 mL reactions, washed or unwashed PCS, 5% insoluble solids (dry weight), 50 mM sodium acetate pH 5, 1 mM MnSO₄, 50° C., 55° C., or 60° C., 72 hours, sugar analysis by AMINEX® HPX-87H column chromatography (Bio-Rad Laboratories, Inc., Hercules, CA, USA).

Coding sequence: The term “coding sequence” or “coding region” means a polynucleotide sequence, which specifies the amino acid sequence of a polypeptide. The boundaries of the coding sequence are generally determined by an open reading frame, which usually begins with the ATG start codon or alternative start codons such as GTG and TTG and ends with a stop codon such as TAA, TAG, and TGA. The coding sequence may be a sequence of genomic DNA, cDNA, a synthetic polynucleotide, and/or a recombinant polynucleotide.

Control sequence: The term “control sequence” means a nucleic acid sequence necessary for polypeptide expression. Control sequences may be native or foreign to the polynucleotide encoding the polypeptide, and native or foreign to each other. Such control sequences include, but are not limited to, a leader sequence, polyadenylation sequence, propeptide sequence, promoter sequence, signal peptide sequence, and transcription terminator sequence. The control sequences may be provided with linkers for the purpose of introducing specific restriction sites facilitating ligation of the control sequences with the coding region of the polynucleotide encoding a polypeptide.

Disruption: The term “disruption” means that a coding region and/or control sequence of a referenced gene is partially or entirely modified (such as by deletion, insertion, and/or substitution of one or more nucleotides) resulting in the absence (inactivation) or decrease in expression, and/or the absence or decrease of enzyme activity of the encoded

polypeptide. The effects of disruption can be measured using techniques known in the art such as detecting the absence or decrease of enzyme activity using from cell-free extract measurements referenced herein; or by the absence or decrease of corresponding mRNA (e.g., at least 25% decrease, at least 50% decrease, at least 60% decrease, at least 70% decrease, at least 80% decrease, or at least 90% decrease); the absence or decrease in the amount of corresponding polypeptide having enzyme activity (e.g., at least 25% decrease, at least 50% decrease, at least 60% decrease, at least 70% decrease, at least 80% decrease, or at least 90% decrease); or the absence or decrease of the specific activity of the corresponding polypeptide having enzyme activity (e.g., at least 25% decrease, at least 50% decrease, at least 60% decrease, at least 70% decrease, at least 80% decrease, or at least 90% decrease). Disruptions of a particular gene of interest can be generated by methods known in the art, e.g., by directed homologous recombination (see *Methods in Yeast Genetics* (1997 edition), Adams, Gottschling, Kaiser, and Stems, Cold Spring Harbor Press (1998)).

Endogenous gene: The term “endogenous gene” means a gene that is native to the referenced host cell. “Endogenous gene expression” means expression of an endogenous gene.

Endoglucanase: The term “endoglucanase” means a 4-(1,3;1,4)-beta-D-glucan 4-glucanohydrolase (E.C. 3.2.1.4) that catalyzes endohydrolysis of 1,4-beta-D-glycosidic linkages in cellulose, cellulose derivatives (such as carboxymethyl cellulose and hydroxyethyl cellulose), lichenin, beta-1,4 bonds in mixed beta-1,3-1,4 glucans such as cereal beta-D-glucans or xyloglucans, and other plant material containing cellulosic components. Endoglucanase activity can be determined by measuring reduction in substrate viscosity or increase in reducing ends determined by a reducing sugar assay (Zhang et al., 2006, *Biotechnology Advances* 24: 452-481). Endoglucanase activity can also be determined using carboxymethyl cellulose (CMC) as substrate according to the procedure of Ghose, 1987, *Pure and Appl. Chem.* 59: 257-268, at pH 5, 40° C.

Expression: The term “expression” includes any step involved in the production of the polypeptide including, but not limited to, transcription, post-transcriptional modification, translation, post-translational modification, and secretion. Expression can be measured—for example, to detect increased expression—by techniques known in the art, such as measuring levels of mRNA and/or translated polypeptide.

Expression vector: The term “expression vector” means a linear or circular DNA molecule that comprises a polynucleotide encoding a polypeptide and is operably linked to control sequences that provide for its expression.

Fermentable medium: The term “fermentable medium” or “fermentation medium” refers to a medium comprising one or more (e.g., two, several) sugars, such as glucose, fructose, sucrose, cellobiose, xylose, xylulose, arabinose, mannose, galactose, and/or soluble oligosaccharides, wherein the medium is capable, in part, of being converted (fermented) by a host cell into a desired product, such as ethanol. In some instances, the fermentation medium is derived from a natural source, such as sugar cane, starch, or cellulose, and may be the result of pretreating the source by enzymatic hydrolysis (saccharification). The term fermentation medium is understood herein to refer to a medium before the fermenting organism is added, such as, a medium resulting from a saccharification process, as well as a medium used in a simultaneous saccharification and fermentation process (SSF).

Glucoamylase: The term “glucoamylase” (1,4-alpha-D-glucan glucohydrolase, EC 3.2.1.3) is defined as an enzyme

that catalyzes the release of D-glucose from the non-reducing ends of starch or related oligo- and polysaccharide molecules. For purposes of the present invention, glucoamylase activity may be determined according to the procedures known in the art, such as those described in the Examples of U.S. Provisional Patent Application No. 62/703,103, filed Jul. 25, 2018.

Hemicellulolytic enzyme or hemicellulase: The term “hemicellulolytic enzyme” or “hemicellulase” means one or more (e.g., several) enzymes that hydrolyze a hemicellulosic material. See, for example, Shallom and Shoham, 2003, *Current Opinion In Microbiology* 6(3): 219-228). Hemicellulases are key components in the degradation of plant biomass. Examples of hemicellulases include, but are not limited to, an acetylmannan esterase, an acetylxyylan esterase, an arabinanase, an arabinofuranosidase, a coumaric acid esterase, a feruloyl esterase, a galactosidase, a glucuronidase, a glucuronoyl esterase, a mannanase, a mannosidase, a xylanase, and a xylosidase. The substrates for these enzymes, hemicelluloses, are a heterogeneous group of branched and linear polysaccharides that are bound via hydrogen bonds to the cellulose microfibrils in the plant cell wall, crosslinking them into a robust network. Hemicelluloses are also covalently attached to lignin, forming together with cellulose a highly complex structure. The variable structure and organization of hemicelluloses require the concerted action of many enzymes for its complete degradation. The catalytic modules of hemicellulases are either glycoside hydrolases (GHs) that hydrolyze glycosidic bonds, or carbohydrate esterases (CEs), which hydrolyze ester linkages of acetate or ferulic acid side groups. These catalytic modules, based on homology of their primary sequence, can be assigned into GH and CE families. Some families, with an overall similar fold, can be further grouped into clans, marked alphabetically (e.g., GH-A). A most informative and updated classification of these and other carbohydrate active enzymes is available in the Carbohydrate-Active Enzymes (CAZy) database. Hemicellulolytic enzyme activities can be measured according to Ghose and Bisaria, 1987, *Pure & Appl. Chem.* 59: 1739-1752, at a suitable temperature such as 40° C.-80° C., e.g., 50° C., 55° C., 60° C., 65° C., or 70° C., and a suitable pH such as 4-9, e.g., 5.0, 5.5, 6.0, 6.5, or 7.0.

Heterologous polynucleotide: The term “heterologous polynucleotide” is defined herein as a polynucleotide that is not native to the host cell; a native polynucleotide in which structural modifications have been made to the coding region; a native polynucleotide whose expression is quantitatively altered as a result of a manipulation of the DNA by recombinant DNA techniques, e.g., a different (foreign) promoter; or a native polynucleotide in a host cell having one or more extra copies of the polynucleotide to quantitatively alter expression. A “heterologous gene” is a gene comprising a heterologous polynucleotide.

High stringency conditions: The term “high stringency conditions” means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/mL sheared and denatured salmon sperm DNA, and 50% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 65° C.

Host cell: The term “host cell” means any cell type that is susceptible to transformation, transfection, transduction, and the like with a nucleic acid construct or expression vector comprising a polynucleotide described herein (e.g., a polynucleotide encoding an alpha-amylase and/or trehalase). The

term “host cell” encompasses any progeny of a parent cell that is not identical to the parent cell due to mutations that occur during replication. The term “recombinant cell” is defined herein as a non-naturally occurring host cell comprising one or more (e.g., two, several) heterologous polynucleotides.

Low stringency conditions: The term “low stringency conditions” means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/mL sheared and denatured salmon sperm DNA, and 25% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 50° C.

Mature polypeptide: The term “mature polypeptide” is defined herein as a polypeptide having biological activity that is in its final form following translation and any post-translational modifications, such as N-terminal processing, C-terminal truncation, glycosylation, phosphorylation, etc. The mature polypeptide sequence lacks a signal sequence, which may be determined using techniques known in the art (See, e.g., Zhang and Henzel, 2004, *Protein Science* 13: 2819-2824).

Medium stringency conditions: The term “medium stringency conditions” means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/mL sheared and denatured salmon sperm DNA, and 35% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 55° C.

Medium-high stringency conditions: The term “medium-high stringency conditions” means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/mL sheared and denatured salmon sperm DNA, and 35% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 60° C.

Nucleic acid construct: The term “nucleic acid construct” means a polynucleotide comprises one or more (e.g., two, several) control sequences. The polynucleotide may be single-stranded or double-stranded, and may be isolated from a naturally occurring gene, modified to contain segments of nucleic acids in a manner that would not otherwise exist in nature, or synthetic.

Operably linked: The term “operably linked” means a configuration in which a control sequence is placed at an appropriate position relative to the coding sequence of a polynucleotide such that the control sequence directs expression of the coding sequence.

Phospholipase: The term “phospholipase” means an enzyme that catalyzes the conversion of phospholipids into fatty acids and other lipophilic substances, such as phospholipase A (EC numbers 3.1.1.4, 3.1.1.5 and 3.1.1.32) or phospholipase C (EC numbers 3.1.4.3 and 3.1.4.11). For purposes of the present invention, phospholipase activity may be determined using activity assays known in the art, or according to the procedures described in the Examples herein (Example 2).

Pretreated corn stover: The term “Pretreated Corn Stover” or “PCS” means a cellulosic-containing material derived from corn stover by treatment with heat and dilute sulfuric acid, alkaline pretreatment, neutral pretreatment, or any pretreatment known in the art.

Protease: The term “protease” is defined herein as an enzyme that hydrolyses peptide bonds. It includes any

enzyme belonging to the EC 3.4 enzyme group (including each of the thirteen subclasses thereof). The EC number refers to Enzyme Nomenclature 1992 from NC-IUBMB, Academic Press, San Diego, California, including supplements 1-5 published in *Eur. J. Biochem.* 223: 1-5 (1994); *Eur. J. Biochem.* 232: 1-6 (1995); *Eur. J. Biochem.* 237: 1-5 (1996); *Eur. J. Biochem.* 250: 1-6 (1997); and *Eur. J. Biochem.* 264: 610-650 (1999); respectively. The term “subtilases” refer to a sub-group of serine protease according to Siezen et al., 1991, *Protein Engng.* 4: 719-737 and Siezen et al., 1997, *Protein Science* 6: 501-523. Serine proteases or serine peptidases is a subgroup of proteases characterised by having a serine in the active site, which forms a covalent adduct with the substrate. Further the subtilases (and the serine proteases) are characterised by having two active site amino acid residues apart from the serine, namely a histidine and an aspartic acid residue. The subtilases may be divided into 6 sub-divisions, i.e. the Subtilisin family, the Thermolysin family, the Proteinase K family, the Lantibiotic peptidase family, the Kexin family and the Pyrolysins family. The term “protease activity” means a proteolytic activity (EC 3.4). Protease activity may be determined using methods described in the art (e.g., US 2015/0125925) or using commercially available assay kits (e.g., Sigma-Aldrich).

Pullulanase: The term “pullulanase” means a starch debanching enzyme having pullulan 6-glucano-hydrolase activity (EC 3.2.1.41) that catalyzes the hydrolysis of α -1,6-glycosidic bonds in pullulan, releasing maltotriose with reducing carbohydrate ends. For purposes of the present invention, pullulanase activity can be determined according to a PHADEBAS assay or the sweet potato starch assay described in WO2016/087237.

Sequence Identity: The relatedness between two amino acid sequences or between two nucleotide sequences is described by the parameter “sequence identity”.

For purposes described herein, the degree of sequence identity between two amino acid sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, *J. Mol. Biol.* 1970, 48, 443-453) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., *Trends Genet* 2000, 16, 276-277), preferably version 3.0.0 or later. The optional parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EBLOSUM62 (EMBOSS version of BLOSUM62) substitution matrix. The output of Needle labeled “longest identity” (obtained using the -nobrief option) is used as the percent identity and is calculated as follows:

$$\frac{(\text{Identical Residues} \times 100)}{(\text{Length of the Referenced Sequence} - \text{Total Number of Gaps in Alignment})}$$

For purposes described herein, the degree of sequence identity between two deoxyribonucleotide sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, *supra*) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, *supra*), preferably version 3.0.0 or later. The optional parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EDNAFULL (EMBOSS version of NCBI NUC4.4) substitution matrix. The output of Needle labeled “longest identity” (obtained using the -nobrief option) is used as the percent identity and is calculated as follows:

$$\frac{(\text{Identical Deoxyribonucleotides} \times 100)}{(\text{Length of Referenced Sequence} - \text{Total Number of Gaps in Alignment})}$$

Signal peptide: The term “signal peptide” is defined herein as a peptide linked (fused) in frame to the amino terminus of a polypeptide having biological activity and directs the polypeptide into the cell’s secretory pathway. Signal sequences may be determined using techniques known in the art (See, e.g., Zhang and Henzel, 2004, *Protein Science* 13: 2819-2824). The polypeptides described herein may comprise any suitable signal peptide known in the art, or any signal peptide described herein (e.g., the *S. cerevisiae* MF α 1 signal peptide of SEQ ID NO: 7, the *S. cerevisiae* EXG1 signal peptide of SEQ ID NO: 227, or the *S. cerevisiae* AG2 signal peptide of SEQ ID NO: 234, or a signal peptide having at least 80%, 85%, 90%, 95%, 96%, 97%, 98% or 99% sequence identity thereof).

Trehalase: The term “trehalase” means an enzyme which degrades trehalose into its unit monosaccharides (i.e., glucose). Trehalases are classified in EC 3.2.1.28 (alpha,alpha-trehalase) and EC. 3.2.1.93 (alpha,alpha-phosphotrehalase). The EC classes are based on recommendations of the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (IUBMB). Description of EC classes can be found on the internet, e.g., on “<http://www.expasy.org/enzyme/>”. Trehalases are enzymes that catalyze the following reactions:

EC 3.2.1.28: Alpha,alpha-trehalose+H₂O $\rightleftharpoons$ 2 D-glucose;
EC 3.2.1.93: Alpha,alpha-trehalase 6-phosphate+H₂O $\rightleftharpoons$ D-glucose+D-glucose 6-phosphate.

For purposes of the present invention, trehalase activity may be determined according to the trehalase activity assay described herein in the experimental section.

Very high stringency conditions: The term “very high stringency conditions” means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5 $\times$ SSPE, 0.3% SDS, 200 micrograms/mL sheared and denatured salmon sperm DNA, and 50% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2 $\times$ SSC, 0.2% SDS at 70° C.

Very low stringency conditions: The term “very low stringency conditions” means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5 $\times$ SSPE, 0.3% SDS, 200 micrograms/mL sheared and denatured salmon sperm DNA, and 25% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2 $\times$ SSC, 0.2% SDS at 45° C.

Xylanase: The term “xylanase” means a 1,4-beta-D-xylan-xylohydrolase (E.C. 3.2.1.8) that catalyzes the endohydrolysis of 1,4-beta-D-xylosidic linkages in xylans. Xylanase activity can be determined with 0.2% AZCL-arabinoxylan as substrate in 0.01% TRITON® X-100 and 200 mM sodium phosphate pH 6 at 37° C. One unit of xylanase activity is defined as 1.0 μ mole of azurine produced per minute at 37° C., pH 6 from 0.2% AZCL-arabinoxylan as substrate in 200 mM sodium phosphate pH 6.

Xylose Isomerase: The term “Xylose Isomerase” or “XI” means an enzyme which can catalyze D-xylose into D-xylose in vivo, and convert D-glucose into D-fructose in vitro. Xylose isomerase is also known as “glucose isomerase” and is classified as E.C. 5.3.1.5. As the structure of the enzyme is very stable, the xylose isomerase is a good model for studying the relationships between protein structure and functions (Karimaki et al., Protein Eng Des Sel, 12004, 17 (12):861-869). Xylose Isomerase activity may be determined using techniques known in the art (e.g., a coupled enzyme assay using D-sorbitol dehydrogenase, as described by Verhoeven et. al., 2017, *Sci Rep* 7, 46155).

Reference to “about” a value or parameter herein includes embodiments that are directed to that value or parameter per se. For example, description referring to “about X” includes the embodiment “X”. When used in combination with measured values, “about” includes a range that encompasses at least the uncertainty associated with the method of measuring the particular value, and can include a range of plus or minus two standard deviations around the stated value.

Likewise, reference to a gene or polypeptide that is “derived from” another gene or polypeptide X, includes the gene or polypeptide X.

As used herein and in the appended claims, the singular forms “a,” “or,” and “the” include plural referents unless the context clearly dictates otherwise.

It is understood that the embodiments described herein include “consisting” and/or “consisting essentially of” embodiments. As used herein, except where the context requires otherwise due to express language or necessary implication, the word “comprise” or variations such as “comprises” or “comprising” is used in an inclusive sense, i.e. to specify the presence of the stated features but not to preclude the presence or addition of further features in various embodiments.

DETAILED DESCRIPTION

Described herein, inter alia, are methods for producing a fermentation product, such as ethanol, from starch or cellulosic containing material.

During industrial scale fermentation, yeast encounter various physiological challenges including variable concentrations of sugars, high concentrations of yeast metabolites such as ethanol, glycerol, organic acids, osmotic stress, as well as potential competition from contaminating microbes such as wild yeasts and bacteria. As a consequence, many yeasts are not suitable for use in industrial fermentation. The most widely used commercially available industrial strain of *Saccharomyces* (i.e. for industrial scale fermentation) is the *Saccharomyces cerevisiae* strain used, for example, in the product ETHANOL RED®. This strain is well suited to industrial ethanol production; however, it remains unclear how modifications to the yeast will impact performance. In particular, the functional expression of heterologous enzymes by an industrially-relevant *Saccharomyces cerevisiae* yeast is uncertain (See, for example U.S. Pat. No. 9,206,444 where the applicant was unable to functionally express numerous enzymes/enzyme classes).

The Applicant has surprisingly found that yeast expressing a phospholipase provide beneficial properties that may be useful for ethanol fermentation, such as reduced need for supplemental nitrogen.

In one aspect is a method of producing a fermentation product from a starch-containing or cellulosic-containing material comprising:

- (a) saccharifying the starch-containing or cellulosic-containing material; and
 - (b) fermenting the saccharified material of step (a) with a fermenting organism;
- wherein the fermenting organism comprises a heterologous polynucleotide encoding a phospholipase.

Steps of saccharifying and fermenting are carried out either sequentially or simultaneously (SSF). In one embodiment, steps of saccharifying and fermenting are carried out simultaneously (SSF). In another embodiment, steps of saccharifying and fermenting are carried out sequentially.

Fermenting Organism

The fermenting organism described herein may be derived from any host cell known to the skilled artisan capable of producing a fermentation product, such as ethanol. As used herein, a “derivative” of strain is derived from a referenced strain, such as through mutagenesis, recombinant DNA technology, mating, cell fusion, or cytoduction between yeast strains. Those skilled in the art will understand that the genetic alterations, including metabolic modifications exemplified herein, may be described with reference to a suitable host organism and their corresponding metabolic reactions or a suitable source organism for desired genetic material such as genes for a desired metabolic pathway. However, given the complete genome sequencing of a wide variety of organisms and the high level of skill in the area of genomics, those skilled in the art can apply the teachings and guidance provided herein to other organisms. For example, the metabolic alterations exemplified herein can readily be applied to other species by incorporating the same or analogous encoding nucleic acid from species other than the referenced species.

The host cells for preparing the recombinant cells described herein can be from any suitable host, such as a yeast strain, including, but not limited to, a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell. In particular, *Saccharomyces* host cells are contemplated, such as *Saccharomyces cerevisiae*, *bayanus* or *carlsbergensis* cells. Preferably, the yeast cell is a *Saccharomyces cerevisiae* cell. Suitable cells can, for example, be derived from commercially available strains and polyploid or aneuploid industrial strains, including but not limited to those from Superstart™, THERMOSACC®, C5 FUEL™, XyloFerm®, etc. (Lallemand); RED STAR and ETHANOL RED® (Fermentis/Lesaffre); FALI (AB Mauri); Baker’s Best Yeast, Baker’s Compressed Yeast, etc. (Fleishmann’s Yeast); BIOFERM AFT, XP, CF, and XR (North American Bioproducts Corp.); Turbo Yeast (Gert Strand AB); and FERMIOL® (DSM Specialties). Other useful yeast strains are available from biological depositories such as the American Type Culture Collection (ATCC) or the Deutsche Sammlung von Mikroorganismen und Zellkulturen GmbH (DSMZ), such as, e.g., BY4741 (e.g., ATCC 201388); Y108-1 (ATCC PTA. 10567) and NRRL YB-1952 (ARS Culture Collection). Still other *S. cerevisiae* strains suitable as host cells DBY746, [Alpha] [Eta]22, S150-2B, GPY55-15Ba, CEN.PK, USM21, TMB3500, TMB3400, VTT-A-63015, VTT-A-85068, VTT-c-79093 and their derivatives as well as *Saccharomyces* sp. 1400, 424A (LNH-ST), 259A (LNH-ST) and derivatives thereof. In one embodiment, the recombinant cell is a derivative of a strain *Saccharomyces cerevisiae* CIBTS1260 (deposited under Accession No. NRRL Y-50973 at the Agricultural Research Service Culture Collection (NRRL), Illinois 61604 U.S.A.).

The fermenting organism may be *Saccharomyces* strain, e.g., *Saccharomyces cerevisiae* strain produced using the method described and concerned in U.S. Pat. No. 8,257, 959-BB.

The strain may also be a derivative of *Saccharomyces cerevisiae* strain NMI V14/004037 (See, WO2015/143324 and WO2015/143317 each incorporated herein by reference), strain nos. V15/004035, V15/004036, and V15/004037 (See, WO 2016/153924 incorporated herein by reference), strain nos. V15/001459, V15/001460, V15/001461 (See, WO2016/138437 incorporated herein by reference), strain no. NRRL Y67342 (See, WO2017/063159

incorporated herein by reference), or any strain described in WO2017/087330 (incorporated herein by reference).

The fermenting organisms according to the invention have been generated in order to, e.g., improve fermentation yield and to improve process economy by cutting enzyme costs since part or all of the necessary enzymes needed to improve method performance are produced by the fermenting organism.

The fermenting organisms described herein may utilize expression vectors comprising the coding sequence of one or more (e.g., two, several) heterologous genes linked to one or more control sequences that direct expression in a suitable cell under conditions compatible with the control sequence(s). Such expression vectors may be used in any of the cells and methods described herein. The polynucleotides described herein may be manipulated in a variety of ways to provide for expression of a desired polypeptide. Manipulation of the polynucleotide prior to its insertion into a vector may be desirable or necessary depending on the expression vector. The techniques for modifying polynucleotides utilizing recombinant DNA methods are well known in the art.

A construct or vector (or multiple constructs or vectors) comprising the one or more (e.g., two, several) heterologous genes may be introduced into a cell so that the construct or vector is maintained as a chromosomal integrant or as a self-replicating extra-chromosomal vector as described earlier.

The various nucleotide and control sequences may be joined together to produce a recombinant expression vector that may include one or more (e.g., two, several) convenient restriction sites to allow for insertion or substitution of the polynucleotide at such sites. Alternatively, the polynucleotide(s) may be expressed by inserting the polynucleotide(s) or a nucleic acid construct comprising the sequence into an appropriate vector for expression. In creating the expression vector, the coding sequence is located in the vector so that the coding sequence is operably linked with the appropriate control sequences for expression.

The recombinant expression vector may be any vector (e.g., a plasmid or virus) that can be conveniently subjected to recombinant DNA procedures and can bring about expression of the polynucleotide. The choice of the vector will typically depend on the compatibility of the vector with the host cell into which the vector is to be introduced. The vector may be a linear or closed circular plasmid.

The vector may be an autonomously replicating vector, i.e., a vector that exists as an extrachromosomal entity, the replication of which is independent of chromosomal replication, e.g., a plasmid, an extrachromosomal element, a minichromosome, or an artificial chromosome. The vector may contain any means for assuring self-replication. Alternatively, the vector may be one that, when introduced into the host cell, is integrated into the genome and replicated together with the chromosome(s) into which it has been integrated. Furthermore, a single vector or plasmid or two or more vectors or plasmids that together contain the total DNA to be introduced into the genome of the cell, or a transposon, may be used.

The expression vector may contain any suitable promoter sequence that is recognized by a cell for expression of a gene described herein. The promoter sequence contains transcriptional control sequences that mediate the expression of the polypeptide. The promoter may be any polynucleotide that shows transcriptional activity in the cell of choice including mutant, truncated, and hybrid promoters, and may be obtained from genes encoding extracellular or intracellular polypeptides either homologous or heterologous to the cell.

Each heterologous polynucleotide described herein may be operably linked to a promoter that is foreign to the polynucleotide. For example, in one embodiment, the heterologous polynucleotide encoding the hexose transporter is operably linked to a promoter foreign to the polynucleotide. The promoters may be identical to or share a high degree of sequence identity (e.g., at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, or at least about 99%) with a selected native promoter.

Examples of suitable promoters for directing the transcription of the nucleic acid constructs in a yeast cells, include, but are not limited to, the promoters obtained from the genes for enolase, (e.g., *S. cerevisiae* enolase or *I. orientalis* enolase (ENO1)), galactokinase (e.g., *S. cerevisiae* galactokinase or *I. orientalis* galactokinase (GAL1)), alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (e.g., *S. cerevisiae* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase or *I. orientalis* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (ADH1, ADH2/GAP)), triose phosphate isomerase (e.g., *S. cerevisiae* triose phosphate isomerase or *I. orientalis* triose phosphate isomerase (TPI)), metallothionein (e.g., *S. cerevisiae* metallothionein or *I. orientalis* metallothionein (CUP1)), 3-phosphoglycerate kinase (e.g., *S. cerevisiae* 3-phosphoglycerate kinase or *I. orientalis* 3-phosphoglycerate kinase (PGK)), PDC1, xylose reductase (XR), xylitol dehydrogenase (XDH), L-(+)-lactate-cytochrome c oxidoreductase (CYB2), translation elongation factor-1 (TEF1), translation elongation factor-2 (TEF2), glyceraldehyde-3-phosphate dehydrogenase (GAPDH), and orotidine 5'-phosphate decarboxylase (URA3) genes. Other suitable promoters may be obtained from *S. cerevisiae* TDH3, HXT7, PGK1, RPL18B and CCW12 genes. Additional useful promoters for yeast host cells are described by Romanos et al., 1992, *Yeast* 8: 423-488.

The control sequence may also be a suitable transcription terminator sequence, which is recognized by a host cell to terminate transcription. The terminator sequence is operably linked to the 3'-terminus of the polynucleotide encoding the polypeptide. Any terminator that is functional in the yeast cell of choice may be used. The terminator may be identical to or share a high degree of sequence identity (e.g., at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, or at least about 99%) with the selected native terminator.

Suitable terminators for yeast host cells may be obtained from the genes for enolase (e.g., *S. cerevisiae* or *I. orientalis* enolase cytochrome C (e.g., *S. cerevisiae* or *I. orientalis* cytochrome C (CYC1)), glyceraldehyde-3-phosphate dehydrogenase (e.g., *S. cerevisiae* or *I. orientalis* glyceraldehyde-3-phosphate dehydrogenase (gpd)), PDC1, XR, XDH, transaldolase (TAL), transketolase (TKL), ribose 5-phosphate ketol-isomerase (RKI), CYB2, and the galactose family of genes (especially the GAL10 terminator). Other suitable terminators may be obtained from *S. cerevisiae* ENO2 or TEF1 genes. Additional useful terminators for yeast host cells are described by Romanos et al., 1992, *supra*.

The control sequence may also be an mRNA stabilizer region downstream of a promoter and upstream of the coding sequence of a gene which increases expression of the gene.

Examples of suitable mRNA stabilizer regions are obtained from a *Bacillus thuringiensis* cryIIIA gene (WO 94/25612) and a *Bacillus subtilis* SP82 gene (Hue et al., 1995, *Journal of Bacteriology* 177: 3465-3471).

The control sequence may also be a suitable leader sequence, when transcribed is a nontranslated region of an mRNA that is important for translation by the host cell. The leader sequence is operably linked to the 5'-terminus of the polynucleotide encoding the polypeptide. Any leader sequence that is functional in the yeast cell of choice may be used.

Suitable leaders for yeast host cells are obtained from the genes for enolase (e.g., *S. cerevisiae* or *I. orientalis* enolase (ENO-1)), 3-phosphoglycerate kinase (e.g., *S. cerevisiae* or *I. orientalis* 3-phosphoglycerate kinase), alpha-factor (e.g., *S. cerevisiae* or *I. orientalis* alpha-factor), and alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (e.g., *S. cerevisiae* or *I. orientalis* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (ADH2/GAP)).

The control sequence may also be a polyadenylation sequence; a sequence operably linked to the 3'-terminus of the polynucleotide and, when transcribed, is recognized by the host cell as a signal to add polyadenosine residues to transcribed mRNA. Any polyadenylation sequence that is functional in the host cell of choice may be used. Useful polyadenylation sequences for yeast cells are described by Guo and Sherman, 1995, *Mol. Cellular Biol.* 15: 5983-5990.

It may also be desirable to add regulatory sequences that allow the regulation of the expression of the polypeptide relative to the growth of the host cell. Examples of regulatory systems are those that cause the expression of the gene to be turned on or off in response to a chemical or physical stimulus, including the presence of a regulatory compound. Regulatory systems in prokaryotic systems include the lac, tac, and trp operator systems. In yeast, the ADH2 system or GAL1 system may be used.

The vectors may contain one or more (e.g., two, several) selectable markers that permit easy selection of transformed, transfected, transduced, or the like cells. A selectable marker is a gene the product of which provides for biocide or viral resistance, resistance to heavy metals, prototrophy to auxotrophs, and the like. Suitable markers for yeast host cells include, but are not limited to, ADE2, HIS3, LEU2, LYS2, MET3, TRP1, and URA3.

The vectors may contain one or more (e.g., two, several) elements that permit integration of the vector into the host cell's genome or autonomous replication of the vector in the cell independent of the genome.

For integration into the host cell genome, the vector may rely on the polynucleotide's sequence encoding the polypeptide or any other element of the vector for integration into the genome by homologous or non-homologous recombination. Alternatively, the vector may contain additional polynucleotides for directing integration by homologous recombination into the genome of the host cell at a precise location(s) in the chromosome(s). To increase the likelihood of integration at a precise location, the integrational elements should contain a sufficient number of nucleic acids, such as 100 to 10,000 base pairs, 400 to 10,000 base pairs, and 800 to 10,000 base pairs, which have a high degree of sequence identity to the corresponding target sequence to enhance the probability of homologous recombination. The integrational elements may be any sequence that is homologous with the target sequence in the genome of the host cell. Furthermore, the integrational elements may be non-encoding or encoding polynucleotides. On the other hand, the vector may be integrated into the genome of the host cell by non-homologous recombination. Potential integration loci include those described in the art (e.g., See US2012/0135481).

For autonomous replication, the vector may further comprise an origin of replication enabling the vector to replicate autonomously in the yeast cell. The origin of replication may be any plasmid replicator mediating autonomous replication that functions in a cell. The term "origin of replication" or "plasmid replicator" means a polynucleotide that enables a plasmid or vector to replicate in vivo. Examples of origins of replication for use in a yeast host cell are the 2 micron origin of replication, ARS1, ARS4, the combination of ARS1 and CEN3, and the combination of ARS4 and CEN6.

More than one copy of a polynucleotide described herein may be inserted into a host cell to increase production of a polypeptide. An increase in the copy number of the polynucleotide can be obtained by integrating at least one additional copy of the sequence into the yeast cell genome or by including an amplifiable selectable marker gene with the polynucleotide where cells containing amplified copies of the selectable marker gene, and thereby additional copies of the polynucleotide, can be selected for by cultivating the cells in the presence of the appropriate selectable agent.

The procedures used to ligate the elements described above to construct the recombinant expression vectors described herein are well known to one skilled in the art (see, e.g., Sambrook et al., 1989, *Molecular Cloning, A Laboratory Manual*, 2d edition, Cold Spring Harbor, New York).

Additional procedures and techniques known in the art for the preparation of recombinant cells for ethanol fermentation, are described in, e.g., WO 2016/045569, the content of which is hereby incorporated by reference.

The fermenting organism may be in the form of a composition comprising a fermenting organism (e.g., a yeast strain described herein) and a naturally occurring and/or a none naturally occurring component.

The fermenting organism described herein may be in any viable form, including crumbled, dry, including active dry and instant, compressed, cream (liquid) form etc. In one embodiment, the fermenting organism (e.g., a *Saccharomyces cerevisiae* yeast strain) is dry yeast, such as active dry yeast or instant yeast. In one embodiment, the fermenting organism (e.g., a *Saccharomyces cerevisiae* yeast strain) is crumbled yeast. In one embodiment, the fermenting organism (e.g., a *Saccharomyces cerevisiae* yeast strain) is compressed yeast. In one embodiment, the fermenting organism (e.g., a *Saccharomyces cerevisiae* yeast strain) is cream yeast.

In one embodiment is a composition comprising a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain), and one or more of the component selected from the group consisting of: surfactants, emulsifiers, gums, swelling agent, and antioxidants and other processing aids.

The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable surfactants. In one embodiment, the surfactant(s) is/are an anionic surfactant, cationic surfactant, and/or nonionic surfactant.

The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable emulsifier. In one embodiment, the emulsifier is a fatty-acid ester of sorbitan. In one embodiment, the emulsifier is selected from the group of sorbitan monostearate (SMS), citric acid esters of mono-glycerides, polyglycerolester, fatty acid esters of propylene glycol.

In one embodiment, the composition comprises a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain), and Olindronal SMS, Olindronal

SK, or Olindronal SPL including composition concerned in European Patent No. 1,724,336 (hereby incorporated by reference). These products are commercially available from Bussetti, Austria, for active dry yeast.

The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable gum. In one embodiment, the gum is selected from the group of carob, guar, tragacanth, arabic, xanthan and acacia gum, in particular for cream, compressed and dry yeast.

The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable swelling agent. In one embodiment, the swelling agent is methyl cellulose or carboxymethyl cellulose.

The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable anti-oxidant. In one embodiment, the antioxidant is butylated hydroxyanisole (BHA) and/or butylated hydroxytoluene (BHT), or ascorbic acid (vitamin C), particular for active dry yeast.

Gene Disruptions

The fermenting organisms described herein may also comprise one or more (e.g., two, several) gene disruptions, e.g., to divert sugar metabolism from undesired products to ethanol. In some aspects, the recombinant host cells produce a greater amount of ethanol compared to the cell without the one or more disruptions when cultivated under identical conditions. In some aspects, one or more of the disrupted endogenous genes is inactivated.

In certain embodiments, the fermenting organism provided herein comprises a disruption of one or more endogenous genes encoding enzymes involved in producing alternate fermentative products such as glycerol or other byproducts such as acetate or diols. For example, the cells provided herein may comprise a disruption of one or more of glycerol 3-phosphate dehydrogenase (GPD, catalyzes reaction of dihydroxyacetone phosphate to glycerol 3-phosphate), glycerol 3-phosphatase (GPP, catalyzes conversion of glycerol-3 phosphate to glycerol), glycerol kinase (catalyzes conversion of glycerol 3-phosphate to glycerol), dihydroxyacetone kinase (catalyzes conversion of dihydroxyacetone phosphate to dihydroxyacetone), glycerol dehydrogenase (catalyzes conversion of dihydroxyacetone to glycerol), and aldehyde dehydrogenase (ALD, e.g., converts acetaldehyde to acetate).

Modeling analysis can be used to design gene disruptions that additionally optimize utilization of the pathway. One exemplary computational method for identifying and designing metabolic alterations favoring biosynthesis of a desired product is the OptKnock computational framework, Burgard et al., 2003, *Biotechnol. Bioeng.* 84: 647-657.

The fermenting organisms comprising a gene disruption may be constructed using methods well known in the art, including those methods described herein. A portion of the gene can be disrupted such as the coding region or a control sequence required for expression of the coding region. Such a control sequence of the gene may be a promoter sequence or a functional part thereof, i.e., a part that is sufficient for affecting expression of the gene. For example, a promoter sequence may be inactivated resulting in no expression or a weaker promoter may be substituted for the native promoter sequence to reduce expression of the coding sequence. Other control sequences for possible modification include, but are not limited to, a leader, propeptide sequence, signal sequence, transcription terminator, and transcriptional activator.

The fermenting organisms comprising a gene disruption may be constructed by gene deletion techniques to eliminate or reduce expression of the gene. Gene deletion techniques enable the partial or complete removal of the gene thereby eliminating their expression. In such methods, deletion of the gene is accomplished by homologous recombination using a plasmid that has been constructed to contiguously contain the 5' and 3' regions flanking the gene.

The fermenting organisms comprising a gene disruption may also be constructed by introducing, substituting, and/or removing one or more (e.g., two, several) nucleotides in the gene or a control sequence thereof required for the transcription or translation thereof. For example, nucleotides may be inserted or removed for the introduction of a stop codon, the removal of the start codon, or a frame-shift of the open reading frame. Such a modification may be accomplished by site-directed mutagenesis or PCR generated mutagenesis in accordance with methods known in the art. See, for example, Botstein and Shortle, 1985, *Science* 229: 4719; Lo et al., 1985, *Proc. Natl. Acad. Sci. U.S.A.* 81: 2285; Higuchi et al., 1988, *Nucleic Acids Res* 16: 7351; Shimada, 1996, *Meth. Mol. Biol.* 57: 157; Ho et al., 1989, *Gene* 77: 61; Horton et al., 1989, *Gene* 77: 61; and Sarkar and Sommer, 1990, *BioTechniques* 8: 404.

The fermenting organisms comprising a gene disruption may also be constructed by inserting into the gene a disruptive nucleic acid construct comprising a nucleic acid fragment homologous to the gene that will create a duplication of the region of homology and incorporate construct DNA between the duplicated regions. Such a gene disruption can eliminate gene expression if the inserted construct separates the promoter of the gene from the coding region or interrupts the coding sequence such that a non-functional gene product results. A disrupting construct may be simply a selectable marker gene accompanied by 5' and 3' regions homologous to the gene. The selectable marker enables identification of transformants containing the disrupted gene.

The fermenting organisms comprising a gene disruption may also be constructed by the process of gene conversion (see, for example, Iglesias and Trautner, 1983, *Molecular General Genetics* 189: 73-76). For example, in the gene conversion method, a nucleotide sequence corresponding to the gene is mutagenized in vitro to produce a defective nucleotide sequence, which is then transformed into the recombinant strain to produce a defective gene. By homologous recombination, the defective nucleotide sequence replaces the endogenous gene. It may be desirable that the defective nucleotide sequence also comprises a marker for selection of transformants containing the defective gene.

The fermenting organisms comprising a gene disruption may be further constructed by random or specific mutagenesis using methods well known in the art, including, but not limited to, chemical mutagenesis (see, for example, Hopwood, *The Isolation of Mutants in Methods in Microbiology* (J. R. Norris and D. W. Ribbons, eds.) pp. 363-433, Academic Press, New York, 1970). Modification of the gene may be performed by subjecting the parent strain to mutagenesis and screening for mutant strains in which expression of the gene has been reduced or inactivated. The mutagenesis, which may be specific or random, may be performed, for example, by use of a suitable physical or chemical mutagenizing agent, use of a suitable oligonucleotide, or subjecting the DNA sequence to PCR generated mutagenesis. Furthermore, the mutagenesis may be performed by use of any combination of these mutagenizing methods.

Examples of a physical or chemical mutagenizing agent suitable for the present purpose include ultraviolet (UV) irradiation, hydroxylamine, N-methyl-N'-nitro-N-nitrosoguanidine (MNNG), N-methyl-N'-nitrosoguanidine (NTG) O-methyl hydroxylamine, nitrous acid, ethyl methane sulphonate (EMS), sodium bisulphite, formic acid, and nucleotide analogues. When such agents are used, the mutagenesis is typically performed by incubating the parent strain to be mutagenized in the presence of the mutagenizing agent of choice under suitable conditions, and selecting for mutants exhibiting reduced or no expression of the gene.

A nucleotide sequence homologous or complementary to a gene described herein may be used from other microbial sources to disrupt the corresponding gene in a recombinant strain of choice.

In one aspect, the modification of a gene in the recombinant cell is unmarked with a selectable marker. Removal of the selectable marker gene may be accomplished by culturing the mutants on a counter-selection medium. Where the selectable marker gene contains repeats flanking its 5' and 3' ends, the repeats will facilitate the looping out of the selectable marker gene by homologous recombination when the mutant strain is submitted to counter-selection. The selectable marker gene may also be removed by homologous recombination by introducing into the mutant strain a nucleic acid fragment comprising 5' and 3' regions of the defective gene, but lacking the selectable marker gene, followed by selecting on the counter-selection medium. By homologous recombination, the defective gene containing the selectable marker gene is replaced with the nucleic acid fragment lacking the selectable marker gene. Other methods known in the art may also be used.

Methods Using a Starch-Containing Material

In some aspects, the methods described herein produce a fermentation product from a starch-containing material. Starch-containing material is well-known in the art, containing two types of homopolysaccharides (amylose and amylopectin) and is linked by alpha-(1-4)-D-glycosidic bonds. Any suitable starch-containing starting material may be used. The starting material is generally selected based on the desired fermentation product, such as ethanol. Examples of starch-containing starting materials include cereal, tubers or grains. Specifically, the starch-containing material may be corn, wheat, barley, rye, milo, sago, cassava, tapioca, sorghum, oat, rice, peas, beans, or sweet potatoes, or mixtures thereof. Contemplated are also waxy and non-waxy types of corn and barley.

In one embodiment, the starch-containing starting material is corn. In one embodiment, the starch-containing starting material is wheat. In one embodiment, the starch-containing starting material is barley. In one embodiment, the starch-containing starting material is rye. In one embodiment, the starch-containing starting material is milo. In one embodiment, the starch-containing starting material is sago. In one embodiment, the starch-containing starting material is cassava. In one embodiment, the starch-containing starting material is tapioca. In one embodiment, the starch-containing starting material is sorghum. In one embodiment, the starch-containing starting material is rice. In one embodiment, the starch-containing starting material is peas. In one embodiment, the starch-containing starting material is beans. In one embodiment, the starch-containing starting material is sweet potatoes. In one embodiment, the starch-containing starting material is oats.

The methods using a starch-containing material may include a conventional process (e.g., including a liquefaction step described in more detail below) or a raw starch hydro-

lysis process. In some embodiments using a starch-containing material, saccharification of the starch-containing material is at a temperature above the initial gelatinization temperature. In some embodiments using a starch-containing material, saccharification of the starch-containing material is at a temperature below the initial gelatinization temperature.

Liquefaction

In aspects using a starch-containing material, the methods may further comprise a liquefaction step carried out by subjecting the starch-containing material at a temperature above the initial gelatinization temperature to an alpha-amylase and optionally a protease and/or a glucoamylase. Other enzymes such as a pullulanase and phytase may also be present and/or added in liquefaction. In some embodiments, the liquefaction step is carried out prior to steps a) and b) of the described methods.

Liquefaction step may be carried out for 0.5-5 hours, such as 1-3 hours, such as typically about 2 hours.

The term "initial gelatinization temperature" means the lowest temperature at which gelatinization of the starch-containing material commences. In general, starch heated in water begins to gelatinize between about 50° C. and 75° C.; the exact temperature of gelatinization depends on the specific starch and can readily be determined by the skilled artisan. Thus, the initial gelatinization temperature may vary according to the plant species, to the particular variety of the plant species as well as with the growth conditions. The initial gelatinization temperature of a given starch-containing material may be determined as the temperature at which birefringence is lost in 5% of the starch granules using the method described by Gorinstein and Lii, 1992, *Starch/Stärke* 44(12): 461-466.

Liquefaction is typically carried out at a temperature in the range from 70-100° C. In one embodiment, the temperature in liquefaction is between 75-95° C., such as between 75-90° C., between 80-90° C., or between 82-88° C., such as about 85° C.

A jet-cooking step may be carried out prior to liquefaction in step, for example, at a temperature between 110-145° C., 120-140° C., 125-135° C., or about 130° C. for about 1-15 minutes, for about 3-10 minutes, or about 5 minutes.

The pH during liquefaction may be between 4 and 7, such as pH 4.5-6.5, pH 5.0-6.5, pH 5.0-6.0, pH 5.2-6.2, or about 5.2, about 5.4, about 5.6, or about 5.8.

In one embodiment, the process further comprises, prior to liquefaction, the steps of:

- i) reducing the particle size of the starch-containing material, preferably by dry milling;
- ii) forming a slurry comprising the starch-containing material and water.

The starch-containing starting material, such as whole grains, may be reduced in particle size, e.g., by milling, in order to open up the structure, to increase surface area, and allowing for further processing. Generally, there are two types of processes: wet and dry milling. In dry milling whole kernels are milled and used. Wet milling gives a good separation of germ and meal (starch granules and protein). Wet milling is often applied at locations where the starch hydrolysate is used in production of, e.g., syrups. Both dry milling and wet milling are well known in the art of starch processing. In one embodiment the starch-containing material is subjected to dry milling. In one embodiment, the particle size is reduced to between 0.05 to 3.0 mm, e.g., 0.1-0.5 mm, or so that at least 30%, at least 50%, at least 70%, or at least 90% of the starch-containing material fit through a sieve with a 0.05 to 3.0 mm screen, e.g., 0.1-0.5

mm screen. In another embodiment, at least 50%, e.g., at least 70%, at least 80%, or at least 90% of the starch-containing material fit through a sieve with #6 screen.

The aqueous slurry may contain from 10-55 w/w-% dry solids (DS), e.g., 25-45 w/w-% dry solids (DS), or 30-40 w/w-% dry solids (DS) of starch-containing material.

The alpha-amylase, optionally a protease, and optionally a glucoamylase may initially be added to the aqueous slurry to initiate liquefaction (thinning). In one embodiment, only a portion of the enzymes (e.g., about 1/3) is added to the aqueous slurry, while the rest of the enzymes (e.g., about 2/3) are added during liquefaction step.

A non-exhaustive list of alpha-amylases used in liquefaction can be found below in the "Alpha-Amylases" section. Examples of suitable proteases used in liquefaction include any protease described supra in the "Proteases" section. Examples of suitable glucoamylases used in liquefaction include any glucoamylase found in the "Glucoamylases" section.

Saccharification and Fermentation of Starch-Containing Material

In aspects using a starch-containing material, a glucoamylase may be present and/or added in saccharification step a) and/or fermentation step b) or simultaneous saccharification and fermentation (SSF). The glucoamylase of the saccharification step a) and/or fermentation step b) or simultaneous saccharification and fermentation (SSF) is typically different from the glucoamylase optionally added to any liquefaction step described supra. In one embodiment, the glucoamylase is present and/or added together with a fungal alpha-amylase.

In some aspects, the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase, for example, as described in WO2017/087330, the content of which is hereby incorporated by reference.

Examples of glucoamylases can be found in the "Glucoamylases" section below.

When doing sequential saccharification and fermentation, saccharification step a) may be carried out under conditions well-known in the art. For instance, saccharification step a) may last up to from about 24 to about 72 hours. In one embodiment, pre-saccharification is done. Pre-saccharification is typically done for 40-90 minutes at a temperature between 30-65° C., typically about 60° C. Pre-saccharification is, in one embodiment, followed by saccharification during fermentation in simultaneous saccharification and fermentation (SSF). Saccharification is typically carried out at temperatures from 20-75° C., preferably from 40-70° C., typically about 60° C., and typically at a pH between 4 and 5, such as about pH 4.5.

Fermentation is carried out in a fermentation medium, as known in the art and, e.g., as described herein. The fermentation medium includes the fermentation substrate, that is, the carbohydrate source that is metabolized by the fermenting organism. With the processes described herein, the fermentation medium may comprise nutrients and growth stimulator(s) for the fermenting organism(s). Nutrient and growth stimulators are widely used in the art of fermentation and include nitrogen sources, such as ammonia; urea, vitamins and minerals, or combinations thereof.

Generally, fermenting organisms such as yeast, including *Saccharomyces cerevisiae* yeast, require an adequate source of nitrogen for propagation and fermentation. Many sources of supplemental nitrogen, if necessary, can be used and such sources of nitrogen are well known in the art. The nitrogen source may be organic, such as urea, DDGs, wet cake or

25

corn mash, or inorganic, such as ammonia or ammonium hydroxide. In one embodiment, the nitrogen source is urea.

Fermentation can be carried out under low nitrogen conditions, e.g., when using a protease-expressing yeast. In some embodiments, the fermentation step is conducted with less than 1000 ppm supplemental nitrogen (e.g., urea or ammonium hydroxide), such as less than 750 ppm, less than 500 ppm, less than 400 ppm, less than 300 ppm, less than 250 ppm, less than 200 ppm, less than 150 ppm, less than 100 ppm, less than 75 ppm, less than 50 ppm, less than 25 ppm, or less than 10 ppm, supplemental nitrogen. In some embodiments, the fermentation step is conducted with no supplemental nitrogen.

Simultaneous saccharification and fermentation ("SSF") is widely used in industrial scale fermentation product production processes, especially ethanol production processes. When doing SSF the saccharification step a) and the fermentation step b) are carried out simultaneously. There is no holding stage for the saccharification, meaning that a fermenting organism, such as yeast, and enzyme(s), may be added together. However, it is also contemplated to add the fermenting organism and enzyme(s) separately. SSF is typically carried out at a temperature from 25° C. to 40° C., such as from 28° C. to 35° C., such as from 30° C. to 34° C., or about 32° C. In one embodiment, fermentation is ongoing for 6 to 120 hours, in particular 24 to 96 hours. In one embodiment, the pH is between 4-5.

In one embodiment, a cellulolytic enzyme composition is present and/or added in saccharification, fermentation or simultaneous saccharification and fermentation (SSF). Examples of such cellulolytic enzyme compositions can be found in the "Cellulolytic Enzymes and Compositions" section below. The cellulolytic enzyme composition may be present and/or added together with a glucoamylase, such as one disclosed in the "Glucoamylases" section below.

Phospholipases

The expressed phospholipase may be any phospholipase that is suitable for the host cells and/or the methods described herein, such as a naturally occurring phospholipase (e.g., a native phospholipase from another species or an endogenous phospholipase expressed from a modified expression vector) or a variant thereof that retains phospholipase activity.

In some embodiments, the fermenting organism comprises a heterologous polynucleotide encoding a phospholipase, for example, as described in WO2018/075430, the content of which is hereby incorporated by reference. In some embodiments, the phospholipase is classified as a phospholipase A. In other embodiments, the phospholipase is classified as a phospholipase C. Any phospholipase described or referenced herein is contemplated for expression in the fermenting organism.

In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a phospholipase has an increased level of phospholipase activity compared to the host cells without the heterologous polynucleotide encoding the phospholipase, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of phospholipase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the phospholipase, when cultivated under the same conditions.

Exemplary phospholipases that can be used with the host cells and/or the methods described herein include bacterial,

26

yeast, or filamentous fungal phospholipases, e.g., derived from any of the microorganisms described or referenced herein.

Additional phospholipases that may be expressed with the fermenting organisms and used with the methods described herein are described in the examples, and include, but are not limited to phospholipases shown in Table 1 (or derivatives thereof).

TABLE 1

Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)
<i>Thermomyces lanuginosus</i>	235
<i>Talaromyces leycettanus</i>	236
<i>Penicillium emersonii</i>	237
<i>Bacillus thuringiensis</i>	238
<i>Pseudomonas</i> sp.	239
<i>Kionochaeta</i> sp.	240
<i>Mariannaea pinicola</i>	241
<i>Fictibacillus macauensis</i>	242
<i>Aspergillus wentii</i>	252
<i>Penicillium cylindrosporum</i>	253
<i>Penicillium meridianum</i>	254
<i>Penicillium bialowiezense</i>	255
<i>Penicillium sclerotiorum</i>	256
<i>Rasamsonia hyssochlamydoides</i>	257
<i>Rasamsonia eburnea</i>	258
<i>Penicillium brefeldianum</i>	259
<i>Penicillium adametzii</i>	260
<i>Rasamsonia brevistipitata</i>	261
<i>Penicillium scabrosum</i>	262
<i>Penicillium manginii</i>	263
<i>Penicillium emersonii</i>	264
<i>Rasamsonia argillacea</i>	265
<i>Penicillium parviverrucosum</i>	266
<i>Penicillium flavesces</i>	267
<i>Penicillium hispanicum</i>	268
<i>Penicillium simplicissimum</i>	269
<i>Penicillium vasconiae</i>	270
<i>Talaromyces columbinus</i>	271
<i>Talaromyces variabilis</i>	272
<i>Talaromyces rugulosus</i>	273
<i>Hamigera terricola</i>	274
<i>Penicillium piscarium</i>	275
<i>Talaromyces bacillisporus</i>	276
<i>Galactomyces candidus</i>	277
<i>Penicillium megasporum</i>	278
<i>Penicillium jensenii</i>	279
<i>Aspergillus stramentius</i>	280
<i>Bacillus pseudomycoloides</i>	281
<i>Bacillus mycoloides</i>	282
<i>Bacillus thuringiensis</i>	283
<i>Listeria innocua</i>	284
<i>Aspergillus egyptiacus</i>	285
<i>Aspergillus tamarii</i>	286
<i>Aspergillus niger</i>	287
<i>Bacillus luciferensis</i>	288
<i>Bacillus mycoloides</i>	289
<i>Bacillus mycoloides</i>	290
<i>Bacillus</i> sp.	291
<i>Bacillus drentensis</i>	292
<i>Aspergillus turcosus</i>	293
<i>Talaromyces subinflatus</i>	294
<i>Aspergillus tubingensis</i>	295
<i>Bacillus acidicer</i>	296
<i>Lysinibacillus xylanilyticus</i>	297
<i>Bacillus toyonensis</i>	298
<i>Bacillus wiedmannii</i>	299
<i>Listeria seeligeri</i>	300
<i>Penicillium swiecickii</i>	301
<i>Talaromyces boninensis</i>	302
<i>Hamigera striata</i>	303
<i>Bacillus</i> sp.	304
<i>Bacillus thuringiensis</i>	305
<i>Bacillus mycoloides</i>	306
<i>Fictibacillus macauensis</i>	307
<i>Listeria seeligeri</i>	308
<i>Penicillium donkii</i>	309

TABLE 1-continued

Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)
<i>Hamigera paravellanea</i>	310
<i>Talaromyces leycettanus</i>	311
<i>Paenibacillus</i> sp.	312
<i>Bacillus toyonensis</i>	313
<i>Bacillus thuringiensis</i>	314
<i>Bacillus thuringiensis</i>	315
<i>Talaromyces rugulosus</i>	316
<i>Penicillium</i> sp.	317
<i>Hamigera avellanea</i>	318
<i>Penicillium spikei</i>	319
<i>Paenibacillus alginolyticus</i>	320
<i>Bacillus mycoides</i>	321
<i>Bacillus bingmayongensis</i>	322
<i>Bacillus mycoides</i>	323
<i>Brevibacillus</i> sp.	324
<i>Penicillium vasconiae</i>	325
<i>Talaromyces diversus</i>	326
<i>Aspergillus wentii</i>	327
<i>Bacillus acidicer</i>	328
<i>Bacillus luti</i>	329
<i>Bacillus pseudomycoides</i>	330
<i>Bacillus mycoides</i>	331
<i>Penicillium cinnamomipureum</i>	332
<i>Talaromyces verruculosus</i>	333
<i>Talaromyces cellulolyticus</i>	334
<i>Penicillium megasporum</i>	335
<i>Bacillus toyonensis</i>	336
<i>Bacillus</i> sp.	337
<i>Bacillus manliponensis</i>	338
<i>Penicillium simplicissimum</i>	339
<i>Penicillium arenicola</i>	340
<i>Aspergillus aculeatus</i>	341
<i>Bacillus acidicer</i>	342

Additional phospholipases contemplated for use with the present invention can be found in WO2018/075430 (the content of which is incorporated herein).

Additional polynucleotides encoding suitable phospholipases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org).

The phospholipase may be a bacterial phospholipase. For example, the phospholipase may be derived from a Gram-positive bacterium such as a *Bacillus*, *Clostridium*, *Enterococcus*, *Geobacillus*, *Lactobacillus*, *Lactococcus*, *Oceanobacillus*, *Staphylococcus*, *Streptococcus*, or *Streptomyces*, or a Gram-negative bacterium such as a *Campylobacter*, *E. coli*, *Flavobacterium*, *Fusobacterium*, *Helicobacter*, *Ilyobacter*, *Neisseria*, *Pseudomonas*, *Salmonella*, or *Ureaplasma*.

In one embodiment, the phospholipase is derived from *Bacillus alkalophilus*, *Bacillus amyloliquefaciens*, *Bacillus brevis*, *Bacillus circulans*, *Bacillus clausii*, *Bacillus coagulans*, *Bacillus firmus*, *Bacillus lautus*, *Bacillus lentus*, *Bacillus licheniformis*, *Bacillus megaterium*, *Bacillus pumilus*, *Bacillus stearothermophilus*, *Bacillus subtilis*, or *Bacillus thuringiensis*.

In another embodiment, the phospholipase is derived from *Streptococcus equisimilis*, *Streptococcus pyogenes*, *Streptococcus uberis*, or *Streptococcus equi* subsp. *Zooepidemicus*.

In another embodiment, the phospholipase is derived from *Streptomyces achromogenes*, *Streptomyces avermitilis*, *Streptomyces coelicolor*, *Streptomyces griseus*, or *Streptomyces lividans*.

The phospholipase may be a fungal phospholipase. For example, the phospholipase may be derived from a yeast such as a *Candida*, *Kluyveromyces*, *Pichia*, *Saccharomyces*,

Schizosaccharomyces, *Yarrowia* or *Issatchenkia*; or derived from a filamentous fungus such as an *Acremonium*, *Agaricus*, *Alternaria*, *Aspergillus*, *Aureobasidium*, *Botryosphaeria*, *Ceriporiopsis*, *Chaetomidium*, *Chrysosporium*, *Claviceps*, *Cochliobolus*, *Coprinopsis*, *Coptotermes*, *Corynascus*, *Cryphonectria*, *Cryptococcus*, *Diplodia*, *Exidia*, *Filibasidium*, *Fusarium*, *Gibberella*, *Holomastigotoides*, *Humicola*, *Irpex*, *Lentinula*, *Leptosphaeria*, *Magnaporthe*, *Melanocarpus*, *Meripilus*, *Mucor*, *Myceliophthora*, *Neocallimastix*, *Neurospora*, *Paecilomyces*, *Penicillium*, *Phanerochaete*, *Piromyces*, *Poitrasia*, *Pseudoplectanina*, *Pseudotrichonympha*, *Rhizomucor*, *Schizophyllum*, *Scytalidium*, *Talaromyces*, *Thermoascus*, *Thielavia*, *Tolypocladium*, *Trichoderma*, *Trichophaea*, *Verticillium*, *Volvariella*, or *Xylaria*.

In another embodiment, the phospholipase is derived from *Saccharomyces carlsbergensis*, *Saccharomyces cerevisiae*, *Saccharomyces diastaticus*, *Saccharomyces douglasii*, *Saccharomyces kluyveri*, *Saccharomyces norbensis*, or *Saccharomyces oviformis*.

In another embodiment, the phospholipase is derived from *Acremonium cellulolyticus*, *Aspergillus aculeatus*, *Aspergillus awamori*, *Aspergillus foetidus*, *Aspergillus fumigatus*, *Aspergillus japonicus*, *Aspergillus nidulans*, *Aspergillus niger*, *Aspergillus oryzae*, *Chrysosporium inops*, *Chrysosporium keratinophilum*, *Chrysosporium lucknowense*, *Chrysosporium merdarium*, *Chrysosporium pannicola*, *Chrysosporium queenslandicum*, *Chrysosporium tropicum*, *Chrysosporium zonatum*, *Fusarium bactridioides*, *Fusarium cerealis*, *Fusarium crookwellense*, *Fusarium culmorum*, *Fusarium graminearum*, *Fusarium graminum*, *Fusarium heterosporum*, *Fusarium negundi*, *Fusarium oxysporum*, *Fusarium reticulatum*, *Fusarium roseum*, *Fusarium sambucinum*, *Fusarium sarcochroum*, *Fusarium sporotrichioides*, *Fusarium sulphureum*, *Fusarium torulosum*, *Fusarium trichothecioides*, *Fusarium venenatum*, *Humicola grisea*, *Humicola insolens*, *Humicola lanuginosa*, *Irpex lacteus*, *Mucor miehei*, *Myceliophthora thermophila*, *Neurospora crassa*, *Penicillium funiculosum*, *Penicillium purpurogenum*, *Phanerochaete chrysosporium*, *Thielavia achromatica*, *Thielavia albomyces*, *Thielavia albopilosa*, *Thielavia australeinsis*, *Thielavia fimeti*, *Thielavia microspora*, *Thielavia ovispora*, *Thielavia peruviana*, *Thielavia setosa*, *Thielavia spededonium*, *Thielavia subthermophila*, *Thielavia terrestris*, *Trichoderma harzianum*, *Trichoderma koningii*, *Trichoderma longibrachiatum*, *Trichoderma reesei*, or *Trichoderma viride*.

It will be understood that for the aforementioned species, the invention encompasses both the perfect and imperfect states, and other taxonomic equivalents, e.g., anamorphs, regardless of the species name by which they are known. Those skilled in the art will readily recognize the identity of appropriate equivalents.

Strains of these species are readily accessible to the public in a number of culture collections, such as the American Type Culture Collection (ATCC), Deutsche Sammlung von Mikroorganismen und Zellkulturen GmbH (DSMZ), Centraalbureau Voor Schimmelcultures (CBS), and Agricultural Research Service Patent Culture Collection, Northern Regional Research Center (NRRL).

The phospholipase coding sequences described or referenced herein, or a subsequence thereof, as well as the phospholipases described or referenced herein, or a fragment thereof, may be used to design nucleic acid probes to identify and clone DNA encoding a phospholipase from strains of different genera or species according to methods well known in the art. In particular, such probes can be used

for hybridization with the genomic DNA or cDNA of a cell of interest, following standard Southern blotting procedures, in order to identify and isolate the corresponding gene therein. Such probes can be considerably shorter than the entire sequence, but should be at least 15, e.g., at least 25, at least 35, or at least 70 nucleotides in length. Preferably, the nucleic acid probe is at least 100 nucleotides in length, e.g., at least 200 nucleotides, at least 300 nucleotides, at least 400 nucleotides, at least 500 nucleotides, at least 600 nucleotides, at least 700 nucleotides, at least 800 nucleotides, or at least 900 nucleotides in length. Both DNA and RNA probes can be used. The probes are typically labeled for detecting the corresponding gene (for example, with ^{32}P , ^3H , ^{35}S , biotin, or avidin).

A genomic DNA or cDNA library prepared from such other strains may be screened for DNA that hybridizes with the probes described above and encodes a parent. Genomic or other DNA from such other strains may be separated by agarose or polyacrylamide gel electrophoresis, or other separation techniques. DNA from the libraries or the separated DNA may be transferred to and immobilized on nitrocellulose or other suitable carrier material. In order to identify a clone or DNA that hybridizes with a coding sequence, or a subsequence thereof, the carrier material is used in a Southern blot.

In one embodiment, the nucleic acid probe is a polynucleotide, or subsequence thereof, that encodes the phospholipase of any one of SEQ ID NOs: 235-242 and 252-342 (such as the coding sequence of SEQ ID NOs: 244-251 and 343-433, respectively), or a fragment thereof.

In one embodiment, the nucleic acid probe is a polynucleotide, or subsequence thereof, that encodes the phospholipase of any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242 (such as the coding sequence of SEQ ID NO: 244, 245, 246, 247, 248, 249, 250 or 251, respectively), or a fragment thereof.

For purposes of the probes described above, hybridization indicates that the polynucleotide hybridizes to a labeled nucleic acid probe, or the full-length complementary strand thereof, or a subsequence of the foregoing; under very low to very high stringency conditions. Molecules to which the nucleic acid probe hybridizes under these conditions can be detected using, for example, X-ray film. Stringency and washing conditions are defined as described supra.

In one embodiment, the phospholipase is encoded by a polynucleotide that hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence for any one of the phospholipases described or referenced herein (e.g., the coding sequence that encodes any one of SEQ ID NOs: 235-242 and 252-342; such as the corresponding coding sequence of SEQ ID NO: 244-251 or 343-433, respectively, or the coding sequence that encodes any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242; such as the corresponding coding sequence of SEQ ID NO: 244, 245, 246, 247, 248, 249, 250 or 251, respectively). (Sambrook et al., 1989, *Molecular Cloning, A Laboratory Manual*, 2d edition, Cold Spring Harbor, New York).

The phospholipase may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, silage, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, silage, etc.) using the above-mentioned probes. Techniques for isolating microorganisms and DNA directly from natural habitats are well known in the art. The

polynucleotide encoding a phospholipase may then be derived by similarly screening a genomic or cDNA library of another microorganism or mixed DNA sample.

Once a polynucleotide encoding a phospholipase has been detected with a suitable probe as described herein, the sequence may be isolated or cloned by utilizing techniques that are known to those of ordinary skill in the art (see, e.g., Sambrook et al., 1989, *Molecular Cloning, A Laboratory Manual*, 2d edition, Cold Spring Harbor, New York). Techniques used to isolate or clone polynucleotides encoding alpha-amylases include isolation from genomic DNA, preparation from cDNA, or a combination thereof. The cloning of the polynucleotides from such genomic DNA can be effected, e.g., by using the well-known polymerase chain reaction (PCR) or antibody screening of expression libraries to detect cloned DNA fragments with shares structural features. See, e.g., Innis et al., 1990, *PCR: A Guide to Methods and Application*, Academic Press, New York. Other nucleic acid amplification procedures such as ligase chain reaction (LCR), ligated activated transcription (LAT) and nucleotide sequence-based amplification (NASBA) may be used.

In one embodiment, the phospholipase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of the phospholipases described or referenced herein (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). In another embodiment, the phospholipase has a mature polypeptide sequence that is a fragment of the any one of the phospholipases described or referenced herein (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length phospholipase (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). In other embodiments, the phospholipase may comprise the catalytic domain of any phospholipase described or referenced herein (e.g., the catalytic domain of any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242).

The phospholipase may be a variant of any one of the phospholipases described supra (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). In one embodiment, the phospholipase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the phospholipases described supra (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242).

In one embodiment, the phospholipase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the phospholipases described supra (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). In one embodiment, the phospholipase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the phospholipases described

supra (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

The amino acid changes are generally of a minor nature, that is conservative amino acid substitutions or insertions that do not significantly affect the folding and/or activity of the protein; small deletions, typically of one to about 30 amino acids; small amino-terminal or carboxyl-terminal extensions, such as an amino-terminal methionine residue; a small linker peptide of up to about 20-25 residues; or a small extension that facilitates purification by changing net charge or another function, such as a poly-histidine tract, an antigenic epitope or a binding domain.

Examples of conservative substitutions are within the group of basic amino acids (arginine, lysine and histidine), acidic amino acids (glutamic acid and aspartic acid), polar amino acids (glutamine and asparagine), hydrophobic amino acids (leucine, isoleucine and valine), aromatic amino acids (phenylalanine, tryptophan and tyrosine), and small amino acids (glycine, alanine, serine, threonine and methionine). Amino acid substitutions that do not generally alter specific activity are known in the art and are described, for example, by H. Neurath and R. L. Hill, 1979, In, *The Proteins*, Academic Press, New York. The most commonly occurring exchanges are Ala/Ser, Val/Ile, Asp/Glu, Thr/Ser, Ala/Gly, Ala/Thr, Ser/Asn, Ala/Val, Ser/Gly, Tyr/Phe, Ala/Pro, Lys/Arg, Asp/Asn, Leu/Ile, Leu/Val, Ala/Glu, and Asp/Gly.

Alternatively, the amino acid changes are of such a nature that the physico-chemical properties of the polypeptides are altered. For example, amino acid changes may improve the thermal stability of the phospholipase, alter the substrate specificity, change the pH optimum, and the like.

Essential amino acids can be identified according to procedures known in the art, such as site-directed mutagenesis or alanine-scanning mutagenesis (Cunningham and Wells, 1989, *Science* 244: 1081-1085). In the latter technique, single alanine mutations are introduced at every residue in the molecule, and the resultant mutant molecules are tested for activity to identify amino acid residues that are critical to the activity of the molecule. See also, Hilton et al., 1996, *J. Biol. Chem.* 271: 4699-4708. The active site or other biological interaction can also be determined by physical analysis of structure, as determined by such techniques as nuclear magnetic resonance, crystallography, electron diffraction, or photoaffinity labeling, in conjunction with mutation of putative contact site amino acids. See, for example, de Vos et al., 1992, *Science* 255: 306-312; Smith et al., 1992, *J. Mol. Biol.* 224: 899-904; Wlodaver et al., 1992, *FEBS Lett.* 309: 59-64. The identities of essential amino acids can also be inferred from analysis of identities with other phospholipases that are related to the referenced phospholipase.

Additional guidance on the structure-activity relationship of the polypeptides herein can be determined using multiple sequence alignment (MSA) techniques well-known in the art. Based on the teachings herein, the skilled artisan could make similar alignments with any number of phospholipases described herein or known in the art. Such alignments aid the skilled artisan to determine potentially relevant domains (e.g., binding domains or catalytic domains), as well as which amino acid residues are conserved and not conserved among the different alpha-amylase sequences. It is appreciated in the art that changing an amino acid that is conserved at a particular position between disclosed polypeptides will more likely result in a change in biological activity (Bowie

et al., 1990, *Science* 247: 1306-1310: "Residues that are directly involved in protein functions such as binding or catalysis will certainly be among the most conserved"). In contrast, substituting an amino acid that is not highly conserved among the polypeptides will not likely or significantly alter the biological activity.

Even further guidance on the structure-activity relationship for the skilled artisan can be found in published x-ray crystallography studies known in the art.

Single or multiple amino acid substitutions, deletions, and/or insertions can be made and tested using known methods of mutagenesis, recombination, and/or shuffling, followed by a relevant screening procedure, such as those disclosed by Reidhaar-Olson and Sauer, 1988, *Science* 241: 53-57; Bowie and Sauer, 1989, *Proc. Natl. Acad. Sci. USA* 86: 2152-2156; WO 95/17413; or WO 95/22625. Other methods that can be used include error-prone PCR, phage display (e.g., Lowman et al., 1991, *Biochemistry* 30: 10832-10837; U.S. Pat. No. 5,223,409; WO 92/06204), and region-directed mutagenesis (Derbyshire et al., 1986, *Gene* 46: 145; Ner et al., 1988, *DNA* 7:127).

Mutagenesis/shuffling methods can be combined with high-throughput, automated screening methods to detect activity of cloned, mutagenized polypeptides expressed by host cells (Ness et al., 1999, *Nature Biotechnology* 17: 893-896). Mutagenized DNA molecules that encode active alpha-amylases can be recovered from the host cells and rapidly sequenced using standard methods in the art. These methods allow the rapid determination of the importance of individual amino acid residues in a polypeptide.

In some embodiments, the phospholipase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the phospholipase activity of any phospholipase described or referenced herein (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242) under the same conditions.

In one embodiment, the phospholipase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any phospholipase described or referenced herein (e.g., a coding sequence for the phospholipase of any one of SEQ ID NOs: 235-242 and 252-342; such as the corresponding coding sequence of SEQ ID NO: 244-251 or 343-433, respectively; or the phospholipase of any one of SEQ ID NO: 235, 236, 237, 238, 239, 240, 241 or 242; such as the corresponding coding sequence of SEQ ID NO: 244, 245, 246, 247, 248, 249, 250 or 251, respectively). In one embodiment, the phospholipase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any phospholipase described or referenced herein (e.g., a coding sequence for the phospholipase of any one of SEQ ID NOs: 235-242 and 252-342; such as the corresponding coding sequence of SEQ ID NO: 244-251 or 343-433, respectively; or the phospholipase of any one of SEQ ID NO: 235, 236, 237, 238, 239, 240, 241 or 242; such as the corresponding coding sequence of SEQ ID NO: 244, 245, 246, 247, 248, 249, 250 or 251, respectively).

In one embodiment, the phospholipase comprises the coding sequence of any phospholipase described or refer-

enced herein (e.g., a coding sequence for the phospholipase of any one of SEQ ID NOs: 235-242 and 252-342; such as the corresponding coding sequence of SEQ ID NO: 244-251 or 343-433, respectively; or the phospholipase of any one of SEQ ID NO: 235, 236, 237, 238, 239, 240, 241 or 242; such as the corresponding coding sequence of SEQ ID NO: 244, 245, 246, 247, 248, 249, 250 or 251, respectively). In one embodiment, the phospholipase comprises a coding sequence that is a subsequence of the coding sequence from any phospholipase described or referenced herein, wherein the subsequence encodes a polypeptide having phospholipase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

The phospholipase may be a fused polypeptide or cleavable fusion polypeptide in which another polypeptide is fused at the N-terminus or the C-terminus of the alpha-amylase. A fused polypeptide may be produced by fusing a polynucleotide encoding another polypeptide to a polynucleotide encoding the phospholipase. Techniques for producing fusion polypeptides are known in the art, and include ligating the coding sequences encoding the polypeptides so that they are in frame and that expression of the fused polypeptide is under control of the same promoter(s) and terminator. Fusion proteins may also be constructed using intein technology in which fusions are created post-translationally (Cooper et al., 1993, *EMBO J.* 12: 2575-2583; Dawson et al., 1994, *Science* 266: 776-779).

Alpha-Amylases

The expressed and/or exogenous alpha-amylase may be any alpha-amylase that is suitable for the host cells and/or the methods described herein, such as a naturally occurring alpha-amylase (e.g., a native alpha-amylase from another species or an endogenous alpha-amylase expressed from a modified expression vector) or a variant thereof that retains alpha-amylase activity. Any alpha-amylase contemplated for expression by a fermenting organism described below is also contemplated for aspects of the invention involving exogenous addition of an alpha-amylase.

In some embodiments, the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase, for example, as described in WO2017/087330, the content of which is hereby incorporated by reference. Any alpha-amylase described or referenced herein is contemplated for expression in the fermenting organism.

In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding an alpha-amylase has an increased level of alpha-amylase activity compared to the host cells without the heterologous polynucleotide encoding the alpha-amylase, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of alpha-amylase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the alpha-amylase, when cultivated under the same conditions.

Exemplary alpha-amylases that can be used with the host cells and/or the methods described herein include bacterial,

yeast, or filamentous fungal alpha-amylases, e.g., derived from any of the microorganisms described or referenced herein.

The term "bacterial alpha-amylase" means any bacterial alpha-amylase classified under EC 3.2.1.1. A bacterial alpha-amylase used herein may, e.g., be derived from a strain of the genus *Bacillus*, which is sometimes also referred to as the genus *Geobacillus*. In one embodiment, the *Bacillus* alpha-amylase is derived from a strain of *Bacillus amyloliquefaciens*, *Bacillus licheniformis*, *Bacillus stearothermophilus*, or *Bacillus subtilis*, but may also be derived from other *Bacillus* sp.

Specific examples of bacterial alpha-amylases include the *Bacillus stearothermophilus* alpha-amylase (BSG) of SEQ ID NO: 3 in WO 99/19467, the *Bacillus amyloliquefaciens* alpha-amylase (BAN) of SEQ ID NO: 5 in WO 99/19467, and the *Bacillus licheniformis* alpha-amylase (BLA) of SEQ ID NO: 4 in WO 99/19467 (all sequences are hereby incorporated by reference). In one embodiment, the alpha-amylase may be an enzyme having a mature polypeptide sequence with a degree of identity of at least 60%, e.g., at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% to any of the sequences shown in SEQ ID NOs: 3, 4 or 5, in WO 99/19467.

In one embodiment, the alpha-amylase is derived from *Bacillus stearothermophilus*. The *Bacillus stearothermophilus* alpha-amylase may be a mature wild-type or a mature variant thereof. The mature *Bacillus stearothermophilus* alpha-amylases may naturally be truncated during recombinant production. For instance, the *Bacillus stearothermophilus* alpha-amylase may be a truncated at the C-terminal, so that it is from 480-495 amino acids long, such as about 491 amino acids long, e.g., so that it lacks a functional starch binding domain (compared to SEQ ID NO: 3 in WO 99/19467).

The *Bacillus* alpha-amylase may also be a variant and/or hybrid. Examples of such a variant can be found in any of WO 96/23873, WO 96/23874, WO 97/41213, WO 99/19467, WO 00/60059, and WO 02/10355 (each hereby incorporated by reference). Specific alpha-amylase variants are disclosed in U.S. Pat. Nos. 6,093,562, 6,187,576, 6,297,038, and 7,713,723 (hereby incorporated by reference) and include *Bacillus stearothermophilus* alpha-amylase (often referred to as BSG alpha-amylase) variants having a deletion of one or two amino acids at positions R179, G180, I181 and/or G182, preferably a double deletion disclosed in WO 96/23873—see, e.g., page 20, lines 1-10 (hereby incorporated by reference), such as corresponding to deletion of positions I181 and G182 compared to the amino acid sequence of *Bacillus stearothermophilus* alpha-amylase set forth in SEQ ID NO: 3 disclosed in WO 99/19467 or the deletion of amino acids R179 and G180 using SEQ ID NO: 3 in WO 99/19467 for numbering (which reference is hereby incorporated by reference). In some embodiments, the *Bacillus* alpha-amylases, such as *Bacillus stearothermophilus* alpha-amylases, have a double deletion corresponding to a deletion of positions 181 and 182 and further optionally comprise a N193F substitution (also denoted I181*+G182*+N193F) compared to the wild-type BSG alpha-amylase amino acid sequence set forth in SEQ ID NO: 3 disclosed in WO 99/19467. The bacterial alpha-amylase may also have a substitution in a position corresponding to S239 in the *Bacillus licheniformis* alpha-amylase shown in SEQ ID NO: 4 in WO 99/19467, or a S242 and/or E188P variant of the *Bacillus stearothermophilus* alpha-amylase of SEQ ID NO: 3 in WO 99/19467.

35

In one embodiment, the variant is a S242A, E or Q variant, e.g., a S242Q variant, of the *Bacillus stearothermophilus* alpha-amylase.

In one embodiment, the variant is a position E188 variant, e.g., E188P variant of the *Bacillus stearothermophilus* alpha-amylase.

The bacterial alpha-amylase may, in one embodiment, be a truncated *Bacillus* alpha-amylase. In one embodiment, the truncation is so that, e.g., the *Bacillus stearothermophilus* alpha-amylase shown in SEQ ID NO: 3 in WO 99/19467, is about 491 amino acids long, such as from 480 to 495 amino acids long, or so it lacks a functional starch bind domain.

The bacterial alpha-amylase may also be a hybrid bacterial alpha-amylase, e.g., an alpha-amylase comprising 445 C-terminal amino acid residues of the *Bacillus licheniformis* alpha-amylase (shown in SEQ ID NO: 4 of WO 99/19467) and the 37 N-terminal amino acid residues of the alpha-amylase derived from *Bacillus amyloliquefaciens* (shown in SEQ ID NO: 5 of WO 99/19467). In one embodiment, this hybrid has one or more, especially all, of the following substitutions: G48A+T49I+G107A+H156Y+A181T+N190F+I201F+A209V+Q264S (using the *Bacillus licheniformis* numbering in SEQ ID NO: 4 of WO 99/19467). In some embodiments, the variants have one or more of the following mutations (or corresponding mutations in other *Bacillus* alpha-amylases): H154Y, A181T, N190F, A209V and Q264S and/or the deletion of two residues between positions 176 and 179, e.g., deletion of E178 and G179 (using SEQ ID NO: 5 of WO 99/19467 for position numbering).

In one embodiment, the bacterial alpha-amylase is the mature part of the chimeric alpha-amylase disclosed in Richardson et al. (2002), The Journal of Biological Chemistry, Vol. 277, No 29, Issue 19 July, pp. 267501-26507, referred to as BD5088 or a variant thereof. This alpha-amylase is the same as the one shown in SEQ ID NO: 2 in WO 2007/134207. The mature enzyme sequence starts after the initial "Met" amino acid in position 1.

The alpha-amylase may be a thermostable alpha-amylase, such as a thermostable bacterial alpha-amylase, e.g., from *Bacillus stearothermophilus*. In one embodiment, the alpha-amylase used in a process described herein has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂ of at least 10 determined as described in Example 1 of WO2018/098381.

In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 15. In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of as at least 20. In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of as at least 25. In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of as at least 30. In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of as at least 40.

In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 50. In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 60. In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 10-70. In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 15-70. In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 20-70. In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 25-70.

36

In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 30-70.

In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 40-70.

In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 50-70.

In one embodiment, the thermostable alpha-amylase has a T $\frac{1}{2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 60-70.

In one embodiment, the alpha-amylase is a bacterial alpha-amylase, e.g., derived from the genus *Bacillus*, such as a strain of *Bacillus stearothermophilus*, e.g., the *Bacillus stearothermophilus* as disclosed in WO 99/019467 as SEQ ID NO: 3 with one or two amino acids deleted at positions R179, G180, I181 and/or G182, in particular with R179 and G180 deleted, or with I181 and G182 deleted, with mutations in below list of mutations.

In some embodiment, the *Bacillus stearothermophilus* alpha-amylases have double deletion I181+G182, and optional substitution N193F, further comprising one of the following substitutions or combinations of substitutions:

V59A+Q89R+G112D+E129V+K177L+R179E+K220P+N224L+Q254S;

V59A+Q89R+E129V+K177L+R179E+H208Y+K220P+N224L+Q254S;

V59A+Q89R+E129V+K177L+R179E+K220P+N224L+Q254S+D269E+D281N;

V59A+Q89R+E129V+K177L+R179E+K220P+N224L+Q254S+I270L;

V59A+Q89R+E129V+K177L+R179E+K220P+N224L+Q254S+H274K;

V59A+Q89R+E129V+K177L+R179E+K220P+N224L+Q254S+Y276F;

V59A+E129V+R157Y+K177L+R179E+K220P+N224L+S242Q+Q254S;

V59A+E129V+K177L+R179E+H208Y+K220P+N224L+S242Q+Q254S;

V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S;

V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+H274K;

V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+Y276F;

V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+D281N;

V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+M284T;

V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+G416V;

V59A+E129V+K177L+R179E+K220P+N224L+Q254S;

V59A+E129V+K177L+R179E+K220P+N224L+Q254S+M284T;

A91L+M96I+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S;

E129V+K177L+R179E;

E129V+K177L+R179E+K220P+N224L+S242Q+Q254S;

E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+Y276F+L427M;

E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+M284T;

E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+N376*+I377*;

E129V+K177L+R179E+K220P+N224L+Q254S;

E129V+K177L+R179E+K220P+N224L+Q254S+M284T;

E129V+K177L+R179E+S242Q;

37

E129V+K177L+R179V+K220P+N224L+S242Q+Q254S;
K220P+N224L+S242Q+Q254S;
M284V;
V59A+Q89R+E129V+K177L+R179E+Q254S+M284V;
and
V59A+E129V+K177L+R179E+Q254S+M284V;

In one embodiment, the alpha-amylase is selected from the group of *Bacillus stearothermophilus* alpha-amylase variants with double deletion I181*+G182*, and optionally substitution N193F, and further one of the following substitutions or combinations of substitutions:

E129V+K177L+R179E;
V59A+Q89R+E129V+K177L+R179E+H208Y+K220P+N224L+Q254S;
V59A+Q89R+E129V+K177L+R179E+Q254S+M284V;
V59A+E129V+K177L+R179E+Q254S+M284V; and
E129V+K177L+R179E+K220P+N224L+S242Q+Q254S
(using SEQ ID NO: 1 herein for numbering).

It should be understood that when referring to *Bacillus stearothermophilus* alpha-amylase and variants thereof they are normally produced in truncated form. In particular, the truncation may be so that the *Bacillus stearothermophilus* alpha-amylase shown in SEQ ID NO: 3 in WO 99/19467, or variants thereof, are truncated in the C-terminal and are typically from 480-495 amino acids long, such as about 491 amino acids long, e.g., so that it lacks a functional starch binding domain.

In one embodiment, the alpha-amylase variant may be an enzyme having a mature polypeptide sequence with a degree of identity of at least 60%, e.g., at least 70%, at least 80%, at least 90%, at least 95%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99%, but less than 100% to the sequence shown in SEQ ID NO: 3 in WO 99/19467.

In one embodiment, the bacterial alpha-amylase, e.g., *Bacillus* alpha-amylase, such as especially *Bacillus stearothermophilus* alpha-amylase, or variant thereof, is dosed to liquefaction in a concentration between 0.01-10 KNU-A/g DS, e.g., between 0.02 and 5 KNU-A/g DS, such as 0.03 and 3 KNU-A, preferably 0.04 and 2 KNU-A/g DS, such as especially 0.01 and 2 KNU-A/g DS. In one embodiment, the bacterial alpha-amylase, e.g., *Bacillus* alpha-amylase, such as especially *Bacillus stearothermophilus* alpha-amylases, or variant thereof, is dosed to liquefaction in a concentration of between 0.0001-1 mg EP (Enzyme Protein)/g DS, e.g., 0.0005-0.5 mg EP/g DS, such as 0.001-0.1 mg EP/g DS.

In one embodiment, the bacterial alpha-amylase is derived from the *Bacillus subtilis* alpha-amylase of SEQ ID NO: 76, the *Bacillus subtilis* alpha-amylase of SEQ ID NO: 82, the *Bacillus subtilis* alpha-amylase of SEQ ID NO: 83, the *Bacillus subtilis* alpha-amylase of SEQ ID NO: 84, or the *Bacillus licheniformis* alpha-amylase of SEQ ID NO: 85, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 89, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 90, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 91, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 92, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 93, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 94, the *Clostridium thermocellum* alpha-amylase of SEQ ID NO: 95, the *Thermobifida fusca* alpha-amylase of SEQ ID NO: 96, the *Thermobifida fusca* alpha-amylase of SEQ ID NO: 97, the *Anaerocellum thermophilum* of SEQ ID NO: 98, the *Anaerocellum thermophilum* of SEQ ID NO: 99, the *Anaero-*

38

cellum thermophilum of SEQ ID NO: 100, the *Streptomyces avermitilis* of SEQ ID NO: 101, or the *Streptomyces avermitilis* of SEQ ID NO: 88.

In one embodiment, the alpha-amylase is derived from *Bacillus amyloliquefaciens*, such as the *Bacillus amyloliquefaciens* alpha-amylase of SEQ ID NO: 231 (e.g., as described in WO2018/002360, or variants thereof as described in WO2017/037614).

In one embodiment, the alpha-amylase is derived from a yeast alpha-amylase, such as the *Saccharomycopsis fibuligera* alpha-amylase of SEQ ID NO: 77, the *Debaryomyces occidentalis* alpha-amylase of SEQ ID NO: 78, the *Debaryomyces occidentalis* alpha-amylase of SEQ ID NO: 79, the *Lipomyces kononenkoae* alpha-amylase of SEQ ID NO: 80, the *Lipomyces kononenkoae* alpha-amylase of SEQ ID NO: 81.

In one embodiment, the alpha-amylase is derived from a filamentous fungal alpha-amylase, such as the *Aspergillus niger* alpha-amylase of SEQ ID NO: 86, or the *Aspergillus niger* alpha-amylase of SEQ ID NO: 87.

Additional alpha-amylases that may be expressed with the fermenting organisms and used with the methods described herein are described in the examples, and include, but are not limited to alpha-amylases shown in Table 2 (or derivatives thereof).

TABLE 2

Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)
<i>Rhizomucor pusillus</i>	121
<i>Bacillus licheniformis</i>	122
<i>Aspergillus niger</i>	123
<i>Aspergillus tamarii</i>	124
<i>Acidomyces richmondensis</i>	125
<i>Aspergillus bombycis</i>	126
<i>Alternaria</i> sp	127
<i>Rhizopus microsporus</i>	128
<i>Syncephalastrum racemosum</i>	129
<i>Rhizomucor pusillus</i>	130
<i>Dichotomocladium hesseltinei</i>	131
<i>Lichtheimia ramosa</i>	132
<i>Penicillium aethiopicum</i>	133
<i>Subulispora</i> sp	134
<i>Trichoderma paraviridescens</i>	135
<i>Byssosacus striatosporus</i>	136
<i>Aspergillus brasiliensis</i>	137
<i>Penicillium subspinosum</i>	138
<i>Penicillium antarcticum</i>	139
<i>Penicillium coprophilum</i>	140
<i>Penicillium olsonii</i>	141
<i>Penicillium vasconiae</i>	142
<i>Penicillium</i> sp	143
<i>Heterocephalum aurantiacum</i>	144
<i>Neosartorya massa</i>	145
<i>Penicillium janthinellum</i>	146
<i>Aspergillus brasiliensis</i>	147
<i>Aspergillus westerdijkiae</i>	148
<i>Hamigera avellanea</i>	149
<i>Hamigera avellanea</i>	150
<i>Meripilus giganteus</i>	151
<i>Cerrena unicolor</i>	152
<i>Physalacria cryptomeriae</i>	153
<i>Lenzites betulinus</i>	154
<i>Trametes ljubarskyi</i>	155
<i>Bacillus subtilis</i>	156
<i>Bacillus subtilis</i> subsp. <i>subtilis</i>	157
<i>Schwanniomyces occidentalis</i>	158
<i>Rhizomucor pusillus</i>	159
<i>Aspergillus niger</i>	160
<i>Bacillus stearothermophilus</i>	161
<i>Bacillus halmapalus</i>	162
<i>Aspergillus oryzae</i>	163
<i>Bacillus amyloliquefaciens</i>	164

TABLE 2-continued

Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)
<i>Rhizomucor pusillus</i>	165
<i>Kionochaeta ivoriensis</i>	166
<i>Aspergillus niger</i>	167
<i>Aspergillus oryzae</i>	168
<i>Penicillium canescens</i>	169
<i>Acidomyces acidothermus</i>	170
<i>Kinochaeta ivoriensis</i>	171
<i>Aspergillus terreus</i>	172
<i>Thamnidium elegans</i>	173
<i>Meripilus giganteus</i>	174

Additional alpha-amylases contemplated for use with the present invention can be found in WO2011/153516 (the content of which is incorporated herein).

Additional polynucleotides encoding suitable alpha-amylases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org).

The alpha-amylase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding trehalases from strains of different genera or species, as described supra.

The polynucleotides encoding alpha-amylases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc) as described supra.

Techniques used to isolate or clone polynucleotides encoding alpha-amylases are described supra.

In one embodiment, the alpha-amylase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of the alpha-amylases described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In another embodiment, the alpha-amylase has a mature polypeptide sequence that is a fragment of the any one of the alpha-amylases described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length alpha-amylase (e.g. any one of SEQ ID NOs: 76-101, 121-174 and 231). In other embodiments, the alpha-amylase may comprise the catalytic domain of any alpha-amylase described or referenced herein (e.g., the catalytic domain of any one of SEQ ID NOs: 76-101, 121-174 and 231).

The alpha-amylase may be a variant of any one of the alpha-amylases described supra (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the alpha-amylase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the alpha-amylases described supra (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231).

In one embodiment, the alpha-amylase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the alpha-amylases described supra (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the alpha-amylase has an amino acid substitution, deletion, and/or insertion of

one or more (e.g., two, several) of amino acid sequence of any one of the alpha-amylases described supra (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In some embodiments, the alpha-amylase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the alpha-amylase activity of any alpha-amylase described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231) under the same conditions.

In one embodiment, the alpha-amylase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any alpha-amylase described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the alpha-amylase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any alpha-amylase described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231).

In one embodiment, the alpha-amylase comprises the coding sequence of any alpha-amylase described or referenced herein (any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the alpha-amylase comprises a coding sequence that is a subsequence of the coding sequence from any alpha-amylase described or referenced herein, wherein the subsequence encodes a polypeptide having alpha-amylase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

The alpha-amylase can also include fused polypeptides or cleavable fusion polypeptides, as described supra. Trehalases

The expressed and/or exogenous trehalase can be any trehalase that is suitable for the fermenting organisms and/or their methods of use described herein, such as a naturally occurring trehalase or a variant thereof that retains trehalase activity. Any trehalase contemplated for expression by a fermenting organism described below is also contemplated for aspects of the invention involving exogenous addition of a trehalase (e.g., added before, during or after liquefaction and/or saccharification).

In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a trehalase has an increased level of trehalase activity compared to the host cells without the heterologous polynucleotide encoding the trehalase, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of trehalase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism

without the heterologous polynucleotide encoding the trehalase, when cultivated under the same conditions.

Trehalases that may be expressed with the fermenting organisms and used with the methods described herein include, but are not limited to, trehalases shown in Table 3 (or derivatives thereof).

TABLE 3

Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)
<i>Chaetomium megalocarpum</i>	175
<i>Lecanicillium psalliotae</i>	176
<i>Doratomyces</i> sp	177
<i>Mucor moelleri</i>	178
<i>Phialophora cyclaminis</i>	179
<i>Thielavia arenaria</i>	180
<i>Thielavia antarctica</i>	181
<i>Chaetomium</i> sp	182
<i>Chaetomium nigricolor</i>	183
<i>Chaetomium jodhpurense</i>	184
<i>Chaetomium piluliferum</i>	185
<i>Myceliophthora himmulea</i>	186
<i>Chloridium virescens</i>	187
<i>Gelasinospora cratophora</i>	188
<i>Acidobacteriaceae bacterium</i>	189
<i>Acidobacterium capsulatum</i>	190
<i>Acidovorax wautersii</i>	191
<i>Xanthomonas arboricola</i>	192
<i>Kosakonia sacchari</i>	193
<i>Enterobacter</i> sp	194
<i>Saitozyma flava</i>	195
<i>Phaeotremella skinneri</i>	196
<i>Trichoderma asperellum</i>	197
<i>Corynascus sepedonium</i>	198
<i>Myceliophthora thermophila</i>	199
<i>Trichoderma reesei</i>	200
<i>Chaetomium virescens</i>	201
<i>Rhodothermus marinus</i>	202
<i>Myceliophthora sepedonium</i>	203
<i>Moelleriella libera</i>	204
<i>Acremonium dichromosporum</i>	205
<i>Fusarium sambucinum</i>	206
<i>Phoma</i> sp	207
<i>Lentinus similis</i>	208
<i>Diaporthe nobilis</i>	209
<i>Solicozozyma terricola</i>	210
<i>Dioszegia cryoxerica</i>	211
<i>Talaromyces funiculosus</i>	212
<i>Hamigera avellanea</i>	213
<i>Talaromyces ruber</i>	214
<i>Trichoderma lixii</i>	215
<i>Aspergillus cervinus</i>	216
<i>Rasamsonia brevistipitata</i>	217
<i>Acremonium curvulum</i>	218
<i>Talaromyces piceae</i>	219
<i>Penicillium</i> sp	220
<i>Talaromyces aurantiacus</i>	221
<i>Talaromyces pinophilus</i>	222
<i>Talaromyces leycettanus</i>	223
<i>Talaromyces variabilis</i>	224
<i>Aspergillus niger</i>	225
<i>Trichoderma reesei</i>	226

Additional polynucleotides encoding suitable trehalases may be derived from microorganisms of any suitable genus, including those readily available within the UniProtKB database (www.uniprot.org).

The trehalase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding trehalases from strains of different genera or species, as described supra.

The polynucleotides encoding trehalases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc) as described supra.

Techniques used to isolate or clone polynucleotides encoding trehalases are described supra.

In one embodiment, the trehalase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of the trehalases described or referenced herein (e.g., any one of SEQ ID NOs: 175-226). In another embodiment, the trehalase has a mature polypeptide sequence that is a fragment of the any one of the trehalases described or referenced herein (e.g., any one of SEQ ID NOs: 175-226). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length trehalase (e.g. any one of SEQ ID NOs: 175-226). In other embodiments, the trehalase may comprise the catalytic domain of any trehalase described or referenced herein (e.g., the catalytic domain of any one of SEQ ID NOs: 175-226).

The trehalase may be a variant of any one of the trehalases described supra (e.g., any one of SEQ ID NOs: 175-226). In one embodiment, the trehalase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the trehalases described supra (e.g., any one of SEQ ID NOs: 175-226).

In one embodiment, the trehalase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the trehalases described supra (e.g., any one of SEQ ID NOs: 175-226). In one embodiment, the trehalase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the trehalases described supra (e.g., any one of SEQ ID NOs: 175-226). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In some embodiments, the trehalase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the trehalase activity of any trehalase described or referenced herein (e.g., any one of SEQ ID NOs: 175-226) under the same conditions.

In one embodiment, the trehalase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any trehalase described or referenced herein (e.g., any one of SEQ ID NOs: 175-226). In one embodiment, the trehalase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any trehalase described or referenced herein (e.g., any one of SEQ ID NOs: 175-226).

In one embodiment, the trehalase comprises the coding sequence of any trehalase described or referenced herein (any one of SEQ ID NOs: 175-226). In one embodiment, the trehalase comprises a coding sequence that is a subsequence of the coding sequence from any trehalase described or referenced herein, wherein the subsequence encodes a polypeptide having trehalase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least

75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

The trehalase can also include fused polypeptides or cleavable fusion polypeptides, as described supra. Glucoamylases

The expressed and/or exogenous glucoamylase can be any glucoamylase that is suitable for the fermenting organisms and/or their methods of use described herein, such as a naturally occurring glucoamylase or a variant thereof that retains glucoamylase activity. Any glucoamylase contemplated for expression by a fermenting organism described below is also contemplated for aspects of the invention involving exogenous addition of a glucoamylase (e.g., added before, during or after liquefaction and/or saccharification).

In some embodiments, the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase, for example, as described in WO2017/087330, the content of which is hereby incorporated by reference. Any glucoamylase described or referenced herein is contemplated for expression in the fermenting organism.

In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a glucoamylase has an increased level of glucoamylase activity compared to the host cells without the heterologous polynucleotide encoding the glucoamylase, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of glucoamylase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the glucoamylase, when cultivated under the same conditions.

Exemplary glucoamylases that can be used with the host cells and/or the methods described herein include bacterial, yeast, or filamentous fungal glucoamylases, e.g., obtained from any of the microorganisms described or referenced herein, as described supra under the sections related to alpha-amylases.

The glucoamylase may be derived from any suitable source, e.g., derived from a microorganism or a plant. Preferred glucoamylases are of fungal or bacterial origin, selected from the group consisting of *Aspergillus* glucoamylases, in particular *Aspergillus niger* G1 or G2 glucoamylase (Boel et al. (1984), EMBO J. 3 (5), p. 1097-1102), or variants thereof, such as those disclosed in WO 92/00381, WO 00/04136 and WO 01/04273 (from Novozymes, Denmark); the *A. awamori* glucoamylase disclosed in WO 84/02921, *Aspergillus oryzae* glucoamylase (Agric. Biol. Chem. (1991), 55 (4), p. 941-949), or variants or fragments thereof. Other *Aspergillus* glucoamylase variants include variants with enhanced thermal stability: G137A and G139A (Chen et al. (1996), Prot. Eng. 9, 499-505); D257E and D293E/Q (Chen et al. (1995), Prot. Eng. 8, 575-582); N182 (Chen et al. (1994), Biochem. J. 301, 275-281); disulphide bonds, A246C (Fierobe et al. (1996), Biochemistry, 35, 8698-8704; and introduction of Pro residues in position A435 and S436 (Li et al. (1997), Protein Eng. 10, 1199-1204).

Other glucoamylases include *Athelia rolfii* (previously denoted *Corticium rolfii*) glucoamylase (see U.S. Pat. No.

4,727,026 and (Nagasaka et al. (1998) "Purification and properties of the raw-starch-degrading glucoamylases from *Corticium rolfii*, Appl Microbiol Biotechnol 50:323-330), *Talaromyces* glucoamylases, in particular derived from *Talaromyces emersonii* (WO 99/28448), *Talaromyces leycettanus* (U.S. Pat. No. Re. 32,153), *Talaromyces duponti*, *Talaromyces thermophilus* (U.S. Pat. No. 4,587,215). In one embodiment, the glucoamylase used during saccharification and/or fermentation is the *Talaromyces emersonii* glucoamylase disclosed in WO 99/28448.

Bacterial glucoamylases contemplated include glucoamylases from the genus *Clostridium*, in particular *C. thermoamylolyticum* (EP 135,138), and *C. thermohydrosulfuricum* (WO 86/01831).

Contemplated fungal glucoamylases include *Trametes cingulate*, *Pachykytospora papyracea*; and *Leucopaxillus giganteus* all disclosed in WO 2006/069289; or *Peniophora rufomarginata* disclosed in WO2007/124285; or a mixture thereof. Also hybrid glucoamylase are contemplated. Examples include the hybrid glucoamylases disclosed in WO 2005/045018.

In one embodiment, the glucoamylase is derived from a strain of the genus *Pycnoporus*, in particular a strain of *Pycnoporus* as described in WO 2011/066576 (SEQ ID NO: 2, 4 or 6 therein), including the *Pycnoporus sanguineus* glucoamylase, or from a strain of the genus *Gloeophyllum*, such as a strain of *Gloeophyllum sepiarium* or *Gloeophyllum trabeum*, in particular a strain of *Gloeophyllum* as described in WO 2011/068803 (SEQ ID NO: 2, 4, 6, 8, 10, 12, 14 or 16 therein). In one embodiment, the glucoamylase is SEQ ID NO: 2 in WO 2011/068803 (i.e. *Gloeophyllum sepiarium* glucoamylase). In one embodiment, the glucoamylase is the *Gloeophyllum sepiarium* glucoamylase of SEQ ID NO: 8. In one embodiment, the glucoamylase is the *Pycnoporus sanguineus* glucoamylase of SEQ ID NO: 229.

In one embodiment, the glucoamylase is a *Gloeophyllum trabeum* glucoamylase (disclosed as SEQ ID NO: 3 in WO2014/177546). In another embodiment, the glucoamylase is derived from a strain of the genus *Nigrospora*, in particular a strain of *Nigrospora* sp. disclosed in WO 2012/064351 (disclosed as SEQ ID NO: 2 therein).

Also contemplated are glucoamylases with a mature polypeptide sequence which exhibit a high identity to any of the above mentioned glucoamylases, i.e., at least 60%, such as at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or even 100% identity to any one of the mature polypeptide sequences mentioned above.

Glucoamylases may be added to the saccharification and/or fermentation in an amount of 0.0001-20 AGU/g DS, such as 0.001-10 AGU/g DS, 0.01-5 AGU/g DS, or 0.1-2 AGU/g DS.

Glucoamylases may be added to the saccharification and/or fermentation in an amount of 1-1,000 µg EP/g DS, such as 10-500 µg/g DS, or 25-250 µg/g DS.

Glucoamylases may be added to liquefaction in an amount of 0.1-100 µg EP/g DS, such as 0.5-50 µg EP/g DS, 1-25 µg EP/g DS, or 2-12 µg EP/g DS.

In one embodiment, the glucoamylase is added as a blend further comprising an alpha-amylase (e.g., any alpha-amylase described herein). In one embodiment, the alpha-amylase is a fungal alpha-amylase, especially an acid fungal alpha-amylase. The alpha-amylase is typically a side activity.

In one embodiment, the glucoamylase is a blend comprising *Talaromyces emersonii* glucoamylase disclosed in

WO 99/28448 as SEQ ID NO: 34 and *Trametes cingulata* glucoamylase disclosed as SEQ ID NO: 2 in WO 06/069289.

In one embodiment, the glucoamylase is a blend comprising *Talaromyces emersonii* glucoamylase disclosed in WO 99/28448, *Trametes cingulata* glucoamylase disclosed as SEQ ID NO: 2 in WO 06/069289, and an alpha-amylase.

In one embodiment, the glucoamylase is a blend comprising *Talaromyces emersonii* glucoamylase disclosed in WO 99/28448, *Trametes cingulata* glucoamylase disclosed in WO 06/069289, and *Rhizomucor pusillus* alpha-amylase with an *Aspergillus niger* glucoamylase linker and SBD disclosed as V039 in Table 5 in WO 2006/069290.

In one embodiment, the glucoamylase is a blend comprising *Gloeophyllum sepiarium* glucoamylase shown as SEQ ID NO: 2 in WO 2011/068803 and an alpha-amylase, in particular *Rhizomucor pusillus* alpha-amylase with an *Aspergillus niger* glucoamylase linker and starch-binding domain (SBD), disclosed SEQ ID NO: 3 in WO 2013/006756, in particular with the following substitutions: G128D+D143N.

In one embodiment, the alpha-amylase may be derived from a strain of the genus *Rhizomucor*, preferably a strain the *Rhizomucor pusillus*, such as the one shown in SEQ ID NO: 3 in WO 2013/006756, or the genus *Meripilus*, preferably a strain of *Meripilus giganteus*. In one embodiment, the alpha-amylase is derived from a *Rhizomucor pusillus* with an *Aspergillus niger* glucoamylase linker and starch-binding domain (SBD), disclosed as V039 in Table 5 in WO 2006/069290.

In one embodiment, the *Rhizomucor pusillus* alpha-amylase or the *Rhizomucor pusillus* alpha-amylase with an *Aspergillus niger* glucoamylase linker and starch-binding domain (SBD) has at least one of the following substitutions or combinations of substitutions: D165M; Y141W; Y141R; K136F; K192R; P224A; P224R; S123H+Y141W; G20S+Y141W; A76G+Y141W; G128D+Y141W; G128D+D143N; P219C+Y141W; N142D+D143N; Y141W+K192R; Y141W+D143N; Y141W+N383R; Y141W+P219C+ A265C; Y141W+N142D+D143N; Y141W+K192R V410A; G128D+Y141W+D143N; Y141W+D143N+P219C; Y141W+D143N+K192R; G128D+D143N+K192R; Y141W+D143N+K192R+P219C; and G128D+Y141W+D143N+K192R; or G128D+Y141W+D143N+K192R+P219C (using SEQ ID NO: 3 in WO 2013/006756 for numbering).

In one embodiment, the glucoamylase blend comprises *Gloeophyllum sepiarium* glucoamylase (e.g., SEQ ID NO: 2 in WO 2011/068803) and *Rhizomucor pusillus* alpha-amylase.

In one embodiment, the glucoamylase blend comprises *Gloeophyllum sepiarium* glucoamylase shown as SEQ ID NO: 2 in WO 2011/068803 and *Rhizomucor pusillus* with an *Aspergillus niger* glucoamylase linker and starch-binding domain (SBD), disclosed SEQ ID NO: 3 in WO 2013/006756 with the following substitutions: G128D+D143N.

Commercially available compositions comprising glucoamylase include AMG 200L; AMG 300 L; SANTM SUPER, SANTM EXTRA L, SPIRIZYME[®] PLUS, SPIRIZYME[®] FUEL, SPIRIZYME[®] B4U, SPIRIZYME[®] ULTRA, SPIRIZYME[®] EXCEL, SPIRIZYME ACHIEVE[®], and AMG[®] E (from Novozymes A/S); OPTIDEXTM 300, GC480, GC417 (from DuPont-Danisco); AMIGASETM and AMIGASETM PLUS (from DSM); G-ZYMETM G900, G-ZYMETM and G990 ZR (from DuPont-Danisco).

In one embodiment, the glucoamylase is derived from the *Debaryomyces occidentalis* glucoamylase of SEQ ID NO:

102. In one embodiment, the glucoamylase is derived from the *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103. In one embodiment, the glucoamylase is derived from the *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 104. In one embodiment, the glucoamylase is derived from the *Saccharomyces cerevisiae* glucoamylase of SEQ ID NO: 105. In one embodiment, the glucoamylase is derived from the *Aspergillus niger* glucoamylase of SEQ ID NO: 106. In one embodiment, the glucoamylase is derived from the *Aspergillus oryzae* glucoamylase of SEQ ID NO: 107. In one embodiment, the glucoamylase is derived from the *Rhizopus oryzae* glucoamylase of SEQ ID NO: 108. In one embodiment, the glucoamylase is derived from the *Clostridium thermocellum* glucoamylase of SEQ ID NO: 109. In one embodiment, the glucoamylase is derived from the *Clostridium thermocellum* glucoamylase of SEQ ID NO: 110. In one embodiment, the glucoamylase is derived from the *Arxula adeninivorans* glucoamylase of SEQ ID NO: 111. In one embodiment, the glucoamylase is derived from the *Hormoconis resinae* glucoamylase of SEQ ID NO: 112. In one embodiment, the glucoamylase is derived from the *Aureobasidium pullulans* glucoamylase of SEQ ID NO: 113.

In one embodiment, the glucoamylase is a *Trichoderma reesei* glucoamylase, such as the *Trichoderma reesei* glucoamylase of SEQ ID NO: 230.

In one embodiment, the glucoamylase has a Relative Activity heat stability at 85° C. of at least 20%, at least 30%, or at least 35% determined as described in Example 4 of WO2018/098381 (heat stability).

In one embodiment, the glucoamylase has a relative activity pH optimum at pH 5.0 of at least 90%, e.g., at least 95%, at least 97%, or 100% determined as described in Example 4 of WO2018/098381 (pH optimum).

In one embodiment, the glucoamylase has a pH stability at pH 5.0 of at least 80%, at least 85%, at least 90% determined as described in Example 4 of WO2018/098381 (pH stability).

In one embodiment, the glucoamylase used in liquefaction, such as a *Penicillium oxalicum* glucoamylase variant, has a thermostability determined as DSC Td at pH 4.0 as described in Example 15 of WO2018/098381 of at least 70° C., preferably at least 75° C., such as at least 80° C., such as at least 81° C., such as at least 82° C., such as at least 83° C., such as at least 84° C., such as at least 85° C., such as at least 86° C., such as at least 87%, such as at least 88° C., such as at least 89° C., such as at least 90° C. In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, has a thermostability determined as DSC Td at pH 4.0 as described in Example 15 of WO2018/098381 in the range between 70° C. and 95° C., such as between 80° C. and 90° C.

In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, used in liquefaction has a thermostability determined as DSC Td at pH 4.8 as described in Example 15 of WO2018/098381 of at least 70° C., preferably at least 75° C., such as at least 80° C., such as at least 81° C., such as at least 82° C., such as at least 83° C., such as at least 84° C., such as at least 85° C., such as at least 86° C., such as at least 87%, such as at least 88° C., such as at least 89° C., such as at least 90° C., such as at least 91° C. In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, has a thermostability determined as DSC Td at pH 4.8 as described in Example 15 of WO2018/098381 in the range between 70° C. and 95° C., such as between 80° C. and 90° C.

In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, used in liquefaction

has a residual activity determined as described in Example 16 of WO2018/098381, of at least 100% such as at least 105%, such as at least 110%, such as at least 115%, such as at least 120%, such as at least 125%. In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, has a thermostability determined as residual activity as described in Example 16 of WO2018/098381, in the range between 100% and 130%.

In one embodiment, the glucoamylase, e.g., of fungal origin such as a filamentous fungi, from a strain of the genus *Penicillium*, e.g., a strain of *Penicillium oxalicum*, in particular the *Penicillium oxalicum* glucoamylase disclosed as SEQ ID NO: 2 in WO 2011/127802 (which is hereby incorporated by reference).

In one embodiment, the glucoamylase has a mature polypeptide sequence of at least 80%, e.g., at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or 100% identity to the mature polypeptide shown in SEQ ID NO: 2 in WO 2011/127802.

In one embodiment, the glucoamylase is a variant of the *Penicillium oxalicum* glucoamylase disclosed as SEQ ID NO: 2 in WO 2011/127802, having a K79V substitution. The K79V glucoamylase variant has reduced sensitivity to protease degradation relative to the parent as disclosed in WO 2013/036526 (which is hereby incorporated by reference).

In one embodiment, the glucoamylase is derived from *Penicillium oxalicum*.

In one embodiment, the glucoamylase is a variant of the *Penicillium oxalicum* glucoamylase disclosed as SEQ ID NO: 2 in WO 2011/127802. In one embodiment, the *Penicillium oxalicum* glucoamylase is the one disclosed as SEQ ID NO: 2 in WO 2011/127802 having Val (V) in position 79.

Contemplated *Penicillium oxalicum* glucoamylase variants are disclosed in WO 2013/053801 which is hereby incorporated by reference.

In one embodiment, these variants have reduced sensitivity to protease degradation.

In one embodiment, these variant have improved thermostability compared to the parent.

In one embodiment, the glucoamylase has a K79V substitution (using SEQ ID NO: 2 of WO 2011/127802 for numbering), corresponding to the PE001 variant, and further comprises one of the following alterations or combinations of alterations

T65A; Q327F; E501V; Y504T; Y504*; T65A+Q327F; T65A+E501V; T65A+Y504T; T65A+Y504*; Q327F+E501V; Q327F+Y504T; Q327F+Y504*; E501V+Y504T; E501V+Y504*; T65A+Q327F+E501V; T65A+Q327F+Y504T; T65A+E501V+Y504T; Q327F+E501V+Y504T; T65A+Q327F+Y504*; T65A+E501V+Y504*; Q327F+E501V+Y504*; T65A+Q327F+E501V+Y504T; T65A+Q327F+E501V+Y504*; E501V+Y504T; T65A+K161S; T65A+Q405T; T65A+Q327W; T65A+Q327F; T65A+Q327Y; P11F+T65A+Q327F; R1K+D3W+K5Q+G7V+N8S+T10K+P11S+T65A+Q327F; P2N+P4S+P11F+T65A+Q327F; P11F+D26C+K33C+T65A+Q327F; P2N+P4S+P11F+T65A+Q327W+E501V+Y504T; R1E+D3N+P4G+G6R+G7A+N8A+T10D+P11D+T65A+Q327F; P11F+T65A+Q327W; P2N+P4S+P11F+T65A+Q327F+E501V+Y504T; P11F+T65A+Q327W+E501V+Y504T; T65A+Q327F+E501V+Y504T; T65A+S105P+Q327W; T65A+S105P+Q327F; T65A+Q327W+S364P; T65A+Q327F+S364P; T65A+S103N+Q327F; P2N+P4S+P11F+K34Y+T65A+Q327F; P2N+P4S+P11F+T65A+

Q327F+D445N+V447S; P2N+P4S+P11F+T65A+I172V+Q327F; P2N+P4S+P11F+T65A+Q327F+N502*; P2N+P4S+P11F+T65A+Q327F+N502T+P563S+K571E; P2N+P4S+P11F+R31S+K33V+T65A+Q327F+N564D+K571S; P2N+P4S+P11F+T65A+Q327F+S377T; P2N+P4S+P11F+T65A+V325T+Q327W; P2N+P4S+P11F+T65A+Q327F+D445N+V447S+E501V+Y504T; P2N+P4S+P11F+T65A+I172V+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+S377T+E501V+Y504T; P2N+P4S+P11F+D26N+K34Y+T65A+Q327F; P2N+P4S+P11F+T65A+Q327F+I375A+E501V+Y504T; P2N+P4S+P11F+T65A+K218A+K221D+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+S103N+Q327F+E501V+Y504T; P2N+P4S+T10D+T65A+Q327F+E501V+Y504T; P2N+P4S+F12Y+T65A+Q327F+E501V+Y504T; K5A+P11F+T65A+Q327F+E501V+Y504T; P2N+P4S+T10E+E18N+T65A+Q327F+E501V+Y504T; P2N+T10E+E18N+T65A+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+E501V+Y504T+T568N; P2N+P4S+P11F+T65A+Q327F+E501V+Y504T+K524T+G526A; P2N+P4S+P11F+K34Y+T65A+Q327F+D445N+V447S+E501V+Y504T; P2N+P4S+P11F+R31S+K33V+T65A+Q327F+D445N+V447S+E501V+Y504T; P2N+P4S+P11F+D26N+K34Y+T65A+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+F80*+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+K112S+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+E501V+Y504T+T516P+K524T+G526A; P2N+P4S+P11F+T65A+Q327F+E501V+N502T+Y504*; P2N+P4S+P11F+T65A+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+S103N+Q327F+E501V+Y504T; K5A+P11F+T65A+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+E501V+Y504T+T516P+K524T+G526A; P2N+P4S+P11F+T65A+V79A+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+V79G+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+V79I+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+V79L+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+V79S+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+L72V+Q327F+E501V+Y504T; S255N+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+E74N+V79K+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+G220N+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+Y245N+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+Q253N+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+D279N+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+S359N+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+D370N+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+V460S+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+V460T+P468T+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+T463N+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+S465N+E501V+Y504T; and P2N+P4S+P11F+T65A+Q327F+T477N+E501V+Y504T.

In one embodiment, the *Penicillium oxalicum* glucoamylase variant has a K79V substitution (using SEQ ID NO: 2 of WO 2011/127802 for numbering), corresponding to the PE001 variant, and further comprises one of the following substitutions or combinations of substitutions:

P11F+T65A+Q327F;
P2N+P4S+P11F+T65A+Q327F;
P11F+D26C+K33C+T65A+Q327F;
P2N+P4S+P11F+T65A+Q327W+E501V+Y504T;

P2N+P4S+P11F+T65A+Q327F+E501V+Y504T; and P11F+T65A+Q327W+E501V+Y504T.

Additional glucoamylases contemplated for use with the present invention can be found in WO2011/153516 (the content of which is incorporated herein).

Additional polynucleotides encoding suitable glucoamylases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org).

The glucoamylase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding glucoamylases from strains of different genera or species, as described supra.

The polynucleotides encoding glucoamylases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc) as described supra.

Techniques used to isolate or clone polynucleotides encoding glucoamylases are described supra.

In one embodiment, the glucoamylase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of the glucoamylases described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In another embodiment, the glucoamylase has a mature polypeptide sequence that is a fragment of the any one of the glucoamylases described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length glucoamylase (e.g. any one of SEQ ID NOs: 8, 102-113, 229 and 230). In other embodiments, the glucoamylase may comprise the catalytic domain of any glucoamylase described or referenced herein (e.g., the catalytic domain of any one of SEQ ID NOs: 8, 102-113, 229 and 230).

The glucoamylase may be a variant of any one of the glucoamylases described supra (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In one embodiment, the glucoamylase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the glucoamylases described supra (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230).

In one embodiment, the glucoamylase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the glucoamylases described supra (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In one embodiment, the glucoamylase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the glucoamylases described supra (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In some embodiments, the glucoamylase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the glucoamylase activity of any glucoamylase described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230) under the same conditions.

In one embodiment, the glucoamylase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any glucoamylase described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In one embodiment, the glucoamylase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any glucoamylase described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230).

In one embodiment, the glucoamylase comprises the coding sequence of any glucoamylase described or referenced herein (any one of SEQ ID NOs: 8, 102-113, 229 and 230). In one embodiment, the glucoamylase comprises a coding sequence that is a subsequence of the coding sequence from any glucoamylase described or referenced herein, wherein the subsequence encodes a polypeptide having glucoamylase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

The glucoamylase can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

Proteases

The expressed and/or exogenous protease can be any protease that is suitable for the fermenting organisms and/or their methods of use described herein, such as a naturally occurring protease or a variant thereof that retains protease activity. Any protease contemplated for expression by a fermenting organism described below is also contemplated for aspects of the invention involving exogenous addition of a protease.

Proteases are classified on the basis of their catalytic mechanism into the following groups: Serine proteases (S), Cysteine proteases (C), Aspartic proteases (A), Metalloproteases (M), and Unknown, or as yet unclassified, proteases (U), see Handbook of Proteolytic Enzymes, A. J. Barrett, N. D. Rawlings, J. F. Woessner (eds), Academic Press (1998), in particular the general introduction part.

Protease activity can be measured using any suitable assay, in which a substrate is employed, that includes peptide bonds relevant for the specificity of the protease in question. Assay-pH and assay-temperature are likewise to be adapted to the protease in question. Examples of assay-pH-values are pH 6, 7, 8, 9, 10, or 11. Examples of assay-temperatures are 30, 35, 37, 40, 45, 50, 55, 60, 65, 70 or 80° C.

In some aspects, the fermenting organism comprising a heterologous polynucleotide encoding a protease has an increased level of protease activity compared to the fermenting organism without the heterologous polynucleotide encoding the protease, when cultivated under the same conditions. In some aspects, the fermenting organism has an increased level of protease activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism

without the heterologous polynucleotide encoding the protease, when cultivated under the same conditions.

Exemplary proteases that may be expressed with the fermenting organisms and used with the methods described herein include, but are not limited to, proteases shown in Table 4 (or derivatives thereof).

TABLE 4

Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)	Family
<i>Aspergillus niger</i>	9	A1
<i>Trichoderma reesei</i>	10	
<i>Thermoascus aurantiacus</i>	11	M35
<i>Dichomitus squalens</i>	12	S53
<i>Nocardopsis prasina</i>	13	S1
<i>Penicillium simplicissimum</i>	14	S10
<i>Aspergillus niger</i>	15	
<i>Meriphilus giganteus</i>	16	S53
<i>Lecanicillium</i> sp. WMM742	17	S53
<i>Talaromyces proteolyticus</i>	18	S53
<i>Penicillium ranomafanaense</i>	19	A1A
<i>Aspergillus oryzae</i>	20	S53
<i>Talaromyces liani</i>	21	S10
<i>Thermoascus thermophilus</i>	22	S53
<i>Pyrococcus furiosus</i>	23	
<i>Trichoderma reesei</i>	24	
<i>Rhizomucor miehei</i>	25	
<i>Lenzites betulinus</i>	26	S53
<i>Neolentinus lepideus</i>	27	S53
<i>Thermococcus</i> sp.	28	S8
<i>Thermococcus</i> sp.	29	S8
<i>Thermomyces lanuginosus</i>	30	S53
<i>Thermococcus thioreducens</i>	31	S53
<i>Polyporus arcularius</i>	32	S53
<i>Ganoderma lucidum</i>	33	S53
<i>Ganoderma lucidum</i>	34	S53
<i>Ganoderma lucidum</i>	35	S53
<i>Trametes</i> sp. AH28-2	36	S53
<i>Cinereomyces lindbladii</i>	37	S53
<i>Trametes versicolor</i> O82DDP	38	S53
<i>Paecilomyces hepiali</i>	39	S53
<i>Isaria tenuipes</i>	40	S53
<i>Aspergillus tamarii</i>	41	S53
<i>Aspergillus brasiliensis</i>	42	S53
<i>Aspergillus iizukae</i>	43	S53
<i>Penicillium</i> sp.-72364	44	S10
<i>Aspergillus denticulatus</i>	45	S10
<i>Hamigera</i> sp. t184-6	46	S10
<i>Penicillium janthinellum</i>	47	S10
<i>Penicillium vasconiae</i>	48	S10
<i>Hamigera paravellanea</i>	49	S10
<i>Talaromyces variabilis</i>	50	S10
<i>Penicillium arenicola</i>	51	S10
<i>Nocardopsis kunsanensis</i>	52	S1
<i>Streptomyces parvulus</i>	53	S1
<i>Saccharopolyspora endophytica</i>	54	S1
<i>luteus</i> cellwall enrichments K	55	S1
<i>Saccharothrix australiensis</i>	56	S1
<i>Nocardopsis baichengensis</i>	57	S1
<i>Streptomyces</i> sp. SM15	58	S1
<i>Actinoalloteichus spitiensis</i>	59	S1
<i>Byssoschlamys verrucosa</i>	60	M35
<i>Hamigera terricola</i>	61	M35
<i>Aspergillus tamarii</i>	62	M35
<i>Aspergillus niveus</i>	63	M35
<i>Penicillium sclerotiorum</i>	64	A1
<i>Penicillium bilaiae</i>	65	A1
<i>Penicillium antarcticum</i>	66	A1
<i>Penicillium sumatrense</i>	67	A1
<i>Trichoderma lixi</i>	68	A1
<i>Trichoderma brevicompactum</i>	69	A1
<i>Penicillium cinnamopurpureum</i>	70	A1
<i>Bacillus licheniformis</i>	71	S8
<i>Bacillus subtilis</i>	72	S8
<i>Trametes cf versicolor</i>	73	S53

Additional polynucleotides encoding suitable proteases may be derived from microorganisms of any suitable genus, including those readily available within the UniProtKB database (www.uniprot.org).

In one embodiment, the protease is derived from *Aspergillus*, such as the *Aspergillus niger* protease of SEQ ID NO: 9, the *Aspergillus tamarii* protease of SEQ ID NO: 41, or the *Aspergillus denticulatus* protease of SEQ ID NO: 45. In one embodiment, the protease is derived from *Dichomitus*, such as the *Dichomitus squalens* protease of SEQ ID NO: 12. In one embodiment, the protease is derived from *Penicillium*, such as the *Penicillium simplicissimum* protease of SEQ ID NO: 14, the *Penicillium antarcticum* protease of SEQ ID NO: 66, or the *Penicillium sumatrense* protease of SEQ ID NO: 67. In one aspect, the protease is derived from *Meriphilus*, such as the *Meriphilus giganteus* protease of SEQ ID NO: 16. In one aspect, the protease is derived from *Talaromyces*, such as the *Talaromyces liani* protease of SEQ ID NO: 21. In one aspect, the protease is derived from *Thermoascus*, such as the *Thermoascus thermophilus* protease of SEQ ID NO: 22. In one aspect, the protease is derived from *Ganoderma*, such as the *Ganoderma lucidum* protease of SEQ ID NO: 33. In one aspect, the protease is derived from *Hamigera*, such as the *Hamigera terricola* protease of SEQ ID NO: 61. In one aspect, the protease is derived from *Trichoderma*, such as the *Trichoderma brevicompactum* protease of SEQ ID NO: 69.

The protease coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding proteases from strains of different genera or species, as described supra.

The polynucleotides encoding proteases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc) as described supra.

Techniques used to isolate or clone polynucleotides encoding proteases are described supra.

In one embodiment, the protease has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69). In another embodiment, the protease has a mature polypeptide sequence that is a fragment of the protease of any one of SEQ ID NOs: 9-73 (e.g., wherein the fragment has protease activity). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length protease (e.g. any one of SEQ ID NOs: 9-73). In other embodiments, the protease may comprise the catalytic domain of any protease described or referenced herein (e.g., the catalytic domain of any one of SEQ ID NOs: 9-73).

The protease may be a variant of any one of the proteases described supra (e.g., any one of SEQ ID NOs: 9-73). In one embodiment, the protease has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the proteases described supra (e.g., any one of SEQ ID NOs: 9-73).

In one embodiment, the protease has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the proteases described supra

(e.g., any one of SEQ ID NOs: 9-73). In one embodiment, the protease has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the proteases described supra (e.g., any one of SEQ ID NOs: 9-73). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In one embodiment, the protease coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any protease described or referenced herein (e.g., any one of SEQ ID NOs: 9-73). In one embodiment, the protease coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any protease described or referenced herein (e.g., any one of SEQ ID NOs: 9-73).

In one embodiment, the protease comprises the coding sequence of any protease described or referenced herein (any one of SEQ ID NOs: 9-73). In one embodiment, the protease comprises a coding sequence that is a subsequence of the coding sequence from any protease described or referenced herein, wherein the subsequence encodes a polypeptide having protease activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

The protease can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

In one embodiment, the protease used according to a process described herein is a Serine proteases. In one particular embodiment, the protease is a serine protease belonging to the family S53, e.g., an endo-protease, such as S53 protease from *Meripilus giganteus*, *Dichomitus squalens*, *Trametes versicolor*, *Polyporus arcularius*, *Lenzites betulinus*, *Ganoderma lucidum*, *Neolentinus lepideus*, or *Bacillus* sp. 19138, in a process for producing ethanol from a starch-containing material, the ethanol yield was improved, when the S53 protease was present/or added during saccharification and/or fermentation of either gelatinized or un-gelatinized starch. In one embodiment, the proteases is selected from: (a) proteases belonging to the EC 3.4.21 enzyme group; and/or (b) proteases belonging to the EC 3.4.14 enzyme group; and/or (c) Serine proteases of the peptidase family S53 that comprises two different types of peptidases: tripeptidyl aminopeptidases (exo-type) and endo-peptidases; as described in 1993, *Biochem. J.* 290:205-218 and in MEROPS protease database, release, 9.4 (31 Jan. 2011) (www.merops.ac.uk). The database is described in Rawlings, N. D., Barrett, A. J. and Bateman, A., 2010, "MEROPS: the peptidase database", *Nucl. Acids Res.* 38: D227-D233.

For determining whether a given protease is a Serine protease, and a family S53 protease, reference is made to the above Handbook and the principles indicated therein. Such determination can be carried out for all types of proteases,

be it naturally occurring or wild-type proteases; or genetically engineered or synthetic proteases.

Peptidase family S53 contains acid-acting endopeptidases and tripeptidyl-peptidases. The residues of the catalytic triad are Glu, Asp, Ser, and there is an additional acidic residue, Asp, in the oxyanion hole. The order of the residues is Glu, Asp, Asp, Ser. The Ser residue is the nucleophile equivalent to Ser in the Asp, His, Ser triad of subtilisin, and the Glu of the triad is a substitute for the general base, His, in subtilisin.

The peptidases of the S53 family tend to be most active at acidic pH (unlike the homologous subtilisins), and this can be attributed to the functional importance of carboxylic residues, notably Asp in the oxyanion hole. The amino acid sequences are not closely similar to those in family S8 (i.e. serine endopeptidase subtilisins and homologues), and this, taken together with the quite different active site residues and the resulting lower pH for maximal activity, provides for a substantial difference to that family. Protein folding of the peptidase unit for members of this family resembles that of subtilisin, having the clan type SB.

In one embodiment, the protease used according to a process described herein is a Cysteine proteases.

In one embodiment, the protease used according to a process described herein is a Aspartic proteases. Aspartic acid proteases are described in, for example, *Hand-book of Proteolytic En-zymes*, Edited by A. J. Barrett, N. D. Rawlings and J. F. Woessner, Aca-demic Press, San Diego, 1998, Chapter 270). Suitable examples of aspartic acid protease include, e.g., those disclosed in R. M. Berka et al. *Gene*, 96, 313 (1990)); (R. M. Berka et al. *Gene*, 125, 195-198 (1993)); and Gomi et al. *Biosci. Biotech. Biochem.* 57, 1095-1100 (1993), which are hereby incorporated by reference.

The protease also may be a metalloprotease, which is defined as a protease selected from the group consisting of:

- (a) proteases belonging to EC 3.4.24 (metalloendopeptidases); preferably EC 3.4.24.39 (acid metallo proteinases);
- (b) metalloproteases belonging to the M group of the above Handbook;
- (c) metalloproteases not yet assigned to clans (designation: Clan MX), or belonging to either one of clans MA, MB, MC, MD, ME, MF, MG, MH (as defined at pp. 989-991 of the above Handbook);
- (d) other families of metalloproteases (as defined at pp. 1448-1452 of the above Handbook);
- (e) metalloproteases with a HEXXH motif;
- (f) metalloproteases with an HEFTH motif;
- (g) metalloproteases belonging to either one of families M3, M26, M27, M32, M34, M35, M36, M41, M43, or M47 (as defined at pp. 1448-1452 of the above Handbook);
- (h) metalloproteases belonging to the M28E family; and
- (i) metalloproteases belonging to family M35 (as defined at pp. 1492-1495 of the above Handbook).

In other particular embodiments, metalloproteases are hydrolases in which the nucleophilic attack on a peptide bond is mediated by a water molecule, which is activated by a divalent metal cation. Examples of divalent cations are zinc, cobalt or manganese. The metal ion may be held in place by amino acid ligands. The number of ligands may be five, four, three, two, one or zero. In a particular embodiment the number is two or three, preferably three.

There are no limitations on the origin of the metalloprotease used in a process of the invention. In an embodiment the metalloprotease is classified as EC 3.4.24, preferably EC 3.4.24.39. In one embodiment, the metalloprotease is an acid-stable metalloprotease, e.g., a fungal acid-stable met-

alloprotease, such as a metalloprotease derived from a strain of the genus *Thermoascus*, preferably a strain of *Thermoascus aurantiacus*, especially *Thermoascus aurantiacus* CGMCC No. 0670 (classified as EC 3.4.24.39). In another embodiment, the metalloprotease is derived from a strain of the genus *Aspergillus*, preferably a strain of *Aspergillus oryzae*.

In one embodiment the metalloprotease has a degree of sequence identity to amino acids -178 to 177, -159 to 177, or preferably amino acids 1 to 177 (the mature polypeptide) of SEQ ID NO: 1 of WO 2010/008841 (a *Thermoascus aurantiacus* metalloprotease) of at least 80%, at least 82%, at least 85%, at least 90%, at least 95%, or at least 97%; and which have metalloprotease activity. In particular embodiments, the metalloprotease consists of an amino acid sequence with a degree of identity to SEQ ID NO: 1 as mentioned above.

The *Thermoascus aurantiacus* metalloprotease is a preferred example of a metalloprotease suitable for use in a process of the invention. Another metalloprotease is derived from *Aspergillus oryzae* and comprises the sequence of SEQ ID NO: 11 disclosed in WO 2003/048353, or amino acids -23-353; -23-374; -23-397; 1-353; 1-374; 1-397; 177-353; 177-374; or 177-397 thereof, and SEQ ID NO: 10 disclosed in WO 2003/048353.

Another metalloprotease suitable for use in a process of the invention is the *Aspergillus oryzae* metalloprotease comprising SEQ ID NO: 5 of WO 2010/008841, or a metalloprotease is an isolated polypeptide which has a degree of identity to SEQ ID NO: 5 of at least about 80%, at least 82%, at least 85%, at least 90%, at least 95%, or at least 97%; and which have metalloprotease activity. In particular embodiments, the metalloprotease consists of the amino acid sequence of SEQ ID NO: 5 of WO 2010/008841.

In a particular embodiment, a metalloprotease has an amino acid sequence that differs by forty, thirty-five, thirty, twenty-five, twenty, or by fifteen amino acids from amino acids -178 to 177, -159 to 177, or +1 to 177 of the amino acid sequences of the *Thermoascus aurantiacus* or *Aspergillus oryzae* metalloprotease.

In another embodiment, a metalloprotease has an amino acid sequence that differs by ten, or by nine, or by eight, or by seven, or by six, or by five amino acids from amino acids -178 to 177, -159 to 177, or +1 to 177 of the amino acid sequences of these metalloproteases, e.g., by four, by three, by two, or by one amino acid.

In particular embodiments, the metalloprotease a) comprises or b) consists of

- i) the amino acid sequence of amino acids -178 to 177, -159 to 177, or +1 to 177 of SEQ ID NO: 1 of WO 2010/008841;
- ii) the amino acid sequence of amino acids -23-353, -23-374, -23-397, 1-353, 1-374, 1-397, 177-353, 177-374, or 177-397 of SEQ ID NO: 3 of WO 2010/008841;
- iii) the amino acid sequence of SEQ ID NO: 5 of WO 2010/008841; or allelic variants, or fragments, of the sequences of i), ii), and iii) that have protease activity.

A fragment of amino acids -178 to 177, -159 to 177, or +1 to 177 of SEQ ID NO: 1 of WO 2010/008841 or of amino acids -23-353, -23-374, -23-397, 1-353, 1-374, 1-397, 177-353, 177-374, or 177-397 of SEQ ID NO: 3 of WO 2010/008841; is a polypeptide having one or more amino acids deleted from the amino and/or carboxyl terminus of these amino acid sequences. In one embodiment a fragment contains at least 75 amino acid residues, or at least 100 amino acid residues, or at least 125 amino acid residues, or at least 150 amino acid residues, or at least 160 amino acid

residues, or at least 165 amino acid residues, or at least 170 amino acid residues, or at least 175 amino acid residues.

To determine whether a given protease is a metalloprotease or not, reference is made to the above "Handbook of Proteolytic Enzymes" and the principles indicated therein. Such determination can be carried out for all types of proteases, be it naturally occurring or wild-type proteases; or genetically engineered or synthetic proteases.

The protease may be a variant of, e.g., a wild-type protease, having thermostability properties defined herein. In one embodiment, the thermostable protease is a variant of a metalloprotease. In one embodiment, the thermostable protease used in a process described herein is of fungal origin, such as a fungal metalloprotease, such as a fungal metalloprotease derived from a strain of the genus *Thermoascus*, preferably a strain of *Thermoascus aurantiacus*, especially *Thermoascus aurantiacus* CGMCC No. 0670 (classified as EC 3.4.24.39).

In one embodiment, the thermostable protease is a variant of the mature part of the metalloprotease shown in SEQ ID NO: 2 disclosed in WO 2003/048353 or the mature part of SEQ ID NO: 1 in WO 2010/008841 further with one of the following substitutions or combinations of substitutions:

S5*+D79L+S87P+A112P+D142L;
 D79L+S87P+A112P+T124V+D142L;
 S5*+N26R+D79L+S87P+A112P+D142L;
 N26R+T46R+D79L+S87P+A112P+D142L;
 T46R+D79L+S87P+T116V+D142L;
 D79L+P81 R+S87P+A112P+D142L;
 A27K+D79L+S87P+A112P+T124V+D142L;
 D79L+Y82F+S87P+A112P+T124V+D142L;
 D79L+Y82F+S87P+A112P+T124V+D142L;
 D79L+S87P+A112P+T124V+A126V+D142L;
 D79L+S87P+A112P+D142L;
 D79L+Y82F+S87P+A112P+D142L;
 S38T+D79L+S87P+A112P+A126V+D142L;
 D79L+Y82F+S87P+A112P+A126V+D142L;
 A27K+D79L+S87P+A112P+A126V+D142L;
 D79L+S87P+N98C+A112P+G135C+D142L;
 D79L+S87P+A112P+D142L+T141C+M161C;
 S36P+D79L+S87P+A112P+D142L;
 A37P+D79L+S87P+A112P+D142L;
 S49P+D79L+S87P+A112P+D142L;
 S50P+D79L+S87P+A112P+D142L;
 D79L+S87P+D104P+A112P+D142L;
 D79L+Y82F+S87G+A112P+D142L;
 S70V+D79L+Y82F+S87G+Y97W+A112P+D142L;
 D79L+Y82F+S87G+Y97W+D104P+A112P+D142L;
 S70V+D79L+Y82F+S87G+A112P+D142L;
 D79L+Y82F+S87G+D104P+A112P+D142L;
 D79L+Y82F+S87G+A112P+A126V+D142L;
 Y82F+S87G+S70V+D79L+D104P+A112P+D142L;
 Y82F+S87G+D79L+D104P+A112P+A126V+D142L;
 A27K+D79L+Y82F+S87G+D104P+A112P+A126V+D142L;
 A27K+Y82F+S87G+D104P+A112P+A126V+D142L;
 A27K+D79L+Y82F+D104P+A112P+A126V+D142L;
 A27K+Y82F+D104P+A112P+A126V+D142L;
 A27K+D79L+S87P+A112P+D142L; and
 D79L+S87P+D142L.

In one embodiment, the thermostable protease is a variant of the metalloprotease disclosed as the mature part of SEQ ID NO: 2 disclosed in WO 2003/048353 or the mature part of SEQ ID NO: 1 in WO 2010/008841 with one of the following substitutions or combinations of substitutions:

D79L+S87P+A112P+D142L;
 D79L+S87P+D142L; and

A27K+D79L+Y82F+S87G+D104P+A112P+A126V+D142L.

In one embodiment, the protease variant has at least 75% identity preferably at least 80%, more preferably at least 85%, more preferably at least 90%, more preferably at least 91%, more preferably at least 92%, even more preferably at least 93%, most preferably at least 94%, and even most preferably at least 95%, such as even at least 96%, at least 97%, at least 98%, at least 99%, but less than 100% identity to the mature part of the polypeptide of SEQ ID NO: 2 disclosed in WO 2003/048353 or the mature part of SEQ ID NO: 1 in WO 2010/008841.

The thermostable protease may also be derived from any bacterium as long as the protease has the thermostability properties.

In one embodiment, the thermostable protease is derived from a strain of the bacterium *Pyrococcus*, such as a strain of *Pyrococcus furiosus* (pfu protease).

In one embodiment, the protease is one shown as SEQ ID NO: 1 in U.S. Pat. No. 6,358,726-E1 (Takara Shuzo Company).

In one embodiment, the thermostable protease is a protease having a mature polypeptide sequence of at least 80% identity, such as at least 85%, such as at least 90%, such as at least 95%, such as at least 96%, such as at least 97%, such as at least 98%, such as at least 99% identity to SEQ ID NO: 1 in U.S. Pat. No. 6,358,726-B1. The *Pyrococcus furiosus* protease can be purchased from Takara Bio, Japan.

The *Pyrococcus furiosus* protease may be a thermostable protease as described in SEQ ID NO: 13 of WO2018/098381. This protease (PfuS) was found to have a thermostability of 110% (80° C./70° C.) and 103% (90° C./70° C.) at pH 4.5 determined.

In one embodiment a thermostable protease used in a process described herein has a thermostability value of more than 20% determined as Relative Activity at 80° C./70° C. determined as described in Example 2 of WO2018/098381.

In one embodiment, the protease has a thermostability of more than 30%, more than 40%, more than 50%, more than 60%, more than 70%, more than 80%, more than 90%, more than 100%, such as more than 105%, such as more than 110%, such as more than 115%, such as more than 120% determined as Relative Activity at 80° C./70° C.

In one embodiment, protease has a thermostability of between 20 and 50%, such as between 20 and 40%, such as 20 and 30% determined as Relative Activity at 80° C./70° C. In one embodiment, the protease has a thermostability between 50 and 115%, such as between 50 and 70%, such as between 50 and 60%, such as between 100 and 120%, such as between 105 and 115% determined as Relative Activity at 80° C./70° C.

In one embodiment, the protease has a thermostability value of more than 10% determined as Relative Activity at 85° C./70° C. determined as described in Example 2 of WO2018/098381.

In one embodiment, the protease has a thermostability of more than 10%, such as more than 12%, more than 14%, more than 16%, more than 18%, more than 20%, more than 30%, more than 40%, more than 50%, more than 60%, more than 70%, more than 80%, more than 90%, more than 100%, more than 110% determined as Relative Activity at 85° C./70° C.

In one embodiment, the protease has a thermostability of between 10% and 50%, such as between 10% and 30%, such as between 10% and 25% determined as Relative Activity at 85° C./70° C.

In one embodiment, the protease has more than 20%, more than 30%, more than 40%, more than 50%, more than 60%, more than 70%, more than 80%, more than 90% determined as Remaining Activity at 80° C.; and/or the protease has more than 20%, more than 30%, more than 40%, more than 50%, more than 60%, more than 70%, more than 80%, more than 90% determined as Remaining Activity at 84° C.

Determination of "Relative Activity" and "Remaining Activity" is done as described in Example 2 of WO2018/098381.

In one embodiment, the protease may have a thermostability for above 90, such as above 100 at 85° C. as determined using the Zein-BCA assay as disclosed in Example 3 of WO2018/098381.

In one embodiment, the protease has a thermostability above 60%, such as above 90%, such as above 100%, such as above 110% at 85° C. as determined using the Zein-BCA assay of WO2018/098381.

In one embodiment, protease has a thermostability between 60-120, such as between 70-120%, such as between 80-120%, such as between 90-120%, such as between 100-120%, such as 110-120% at 85° C. as determined using the Zein-BCA assay of WO2018/098381.

In one embodiment, the thermostable protease has at least 20%, such as at least 30%, such as at least 40%, such as at least 50%, such as at least 60%, such as at least 70%, such as at least 80%, such as at least 90%, such as at least 95%, such as at least 100% of the activity of the JTP196 protease variant or Protease Pfu determined by the AZCL-casein assay of WO2018/098381, and described herein.

In one embodiment, the thermostable protease has at least 20%, such as at least 30%, such as at least 40%, such as at least 50%, such as at least 60%, such as at least 70%, such as at least 80%, such as at least 90%, such as at least 95%, such as at least 100% of the protease activity of the Protease 196 variant or Protease Pfu determined by the AZCL-casein assay of WO2018/098381, and described herein.

Pullulanases
In some embodiments, a pullulanase is present and/or added in liquefaction step and/or saccharification step, or simultaneous saccharification and fermentation (SSF).

Pullulanases (E.C. 3.2.1.41, pullulan 6-glucano-hydrolase), are debranching enzymes characterized by their ability to hydrolyze the alpha-1,6-glycosidic bonds in, for example, amylopectin and pullulan.

In some embodiments, the fermenting organism comprises a heterologous polynucleotide encoding a pullulanase. Any pullulanase described or referenced herein is contemplated for expression in the fermenting organism.

The pullulanase may be any pullulanase that is suitable for the host cells and/or the methods described herein, such as a naturally occurring pullulanase or a variant thereof that retains pullulanase activity.

In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a pullulanase has an increased level of pullulanase activity compared to the host cells without the heterologous polynucleotide encoding the pullulanase, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of pullulanase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the pullulanase, when cultivated under the same conditions.

Exemplary pullulanases that can be used with the host cells and/or the methods described herein include bacterial, yeast, or filamentous fungal pullulanases, e.g., obtained from any of the microorganisms described or referenced herein, as described supra under the sections related to alpha-amylases.

Contemplated pullulanases include the pullulanases from *Bacillus amyloclavus* disclosed in U.S. Pat. No. 4,560,651 (hereby incorporated by reference), the pullulanase disclosed as SEQ ID NO: 2 in WO 01/151620 (hereby incorporated by reference), the *Bacillus deramificans* disclosed as SEQ ID NO: 4 in WO 01/151620 (hereby incorporated by reference), and the pullulanase from *Bacillus acidopullulyticus* disclosed as SEQ ID NO: 6 in WO 01/151620 (hereby incorporated by reference) and also described in FEMS Mic. Let. (1994) 115, 97-106.

Additional pullulanases contemplated include the pullulanases from *Pyrococcus woesei*, specifically from *Pyrococcus woesei* DSM No. 3773 disclosed in WO92/02614.

In one embodiment, the pullulanase is a family GH57 pullulanase. In one embodiment, the pullulanase includes an X47 domain as disclosed in U.S. 61/289,040 published as WO 2011/087836 (which are hereby incorporated by reference). More specifically the pullulanase may be derived from a strain of the genus *Thermococcus*, including *Thermococcus litoralis* and *Thermococcus hydrothermalis*, such as the *Thermococcus hydrothermalis* pullulanase truncated at site X4 right after the X47 domain (i.e., amino acids 1-782). The pullulanase may also be a hybrid of the *Thermococcus litoralis* and *Thermococcus hydrothermalis* pullulanases or a *T. hydrothermalis/T. litoralis* hybrid enzyme with truncation site X4 disclosed in U.S. 61/289,040 published as WO 2011/087836 (which is hereby incorporated by reference).

In another embodiment, the pullulanase is one comprising an X46 domain disclosed in WO 2011/076123 (Novozymes).

The pullulanase may be added in an effective amount which include the preferred amount of about 0.0001-10 mg enzyme protein per gram DS, preferably 0.0001-0.10 mg enzyme protein per gram DS, more preferably 0.0001-0.010 mg enzyme protein per gram DS. Pullulanase activity may be determined as NPUN. An Assay for determination of NPUN is described in WO2018/098381.

Suitable commercially available pullulanase products include PROMOZYME D, PROMOZYME™ D2 (Novozymes A/S, Denmark), OPTIMAX L-300 (DuPont-Danisco, USA), and AMANO 8 (Amano, Japan).

In one embodiment, the pullulanase is derived from the *Bacillus subtilis* pullulanase of SEQ ID NO: 114. In one embodiment, the pullulanase is derived from the *Bacillus licheniformis* pullulanase of SEQ ID NO: 115. In one embodiment, the pullulanase is derived from the *Oryza sativa* pullulanase of SEQ ID NO: 116. In one embodiment, the pullulanase is derived from the *Triticum aestivum* pullulanase of SEQ ID NO: 117. In one embodiment, the pullulanase is derived from the *Clostridium phytofermentans* pullulanase of SEQ ID NO: 118. In one embodiment, the pullulanase is derived from the *Streptomyces avermitilis* pullulanase of SEQ ID NO: 119. In one embodiment, the pullulanase is derived from the *Klebsiella pneumoniae* pullulanase of SEQ ID NO: 120.

Additional pullulanases contemplated for use with the present invention can be found in WO2011/153516 (the content of which is incorporated herein).

Additional polynucleotides encoding suitable pullulanases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org).

The pullulanase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding pullulanases from strains of different genera or species, as described supra.

The polynucleotides encoding pullulanases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc) as described supra.

Techniques used to isolate or clone polynucleotides encoding pullulanases are described supra.

In one embodiment, the pullulanase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of the pullulanases described or referenced herein (e.g., any one of SEQ ID NOs: 114-120).

In another embodiment, the pullulanase has a mature polypeptide sequence that is a fragment of the any one of the pullulanases described or referenced herein (e.g., any one of SEQ ID NOs: 114-120). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length pullulanase. In other embodiments, the pullulanase may comprise the catalytic domain of any pullulanase described or referenced herein (e.g., any one of SEQ ID NOs: 114-120).

The pullulanase may be a variant of any one of the pullulanases described supra (e.g., any one of SEQ ID NOs: 114-120). In one embodiment, the pullulanase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the pullulanases described supra (e.g., any one of SEQ ID NOs: 114-120).

In one embodiment, the pullulanase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the pullulanases described supra (e.g., any one of SEQ ID NOs: 114-120). In one embodiment, the pullulanase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the pullulanases described supra (e.g., any one of SEQ ID NOs: 114-120). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In some embodiments, the pullulanase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the pullulanase activity of any pullulanase described or referenced herein under the same conditions (e.g., any one of SEQ ID NOs: 114-120).

In one embodiment, the pullulanase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any pullulanase described or referenced herein (e.g., any one of SEQ ID NOs: 114-120).

In one embodiment, the pullulanase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least

92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any pullulanase described or referenced herein (e.g., any one of SEQ ID NOs: 114-120).

In one embodiment, the pullulanase comprises the coding sequence of any pullulanase described or referenced herein (e.g., any one of SEQ ID NOs: 114-120). In one embodiment, the pullulanase comprises a coding sequence that is a subsequence of the coding sequence from any pullulanase described or referenced herein, wherein the subsequence encodes a polypeptide having pullulanase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

The pullulanase can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

Methods Using a Cellulosic-Containing Material

In some aspects, the methods described herein produce a fermentation product from a cellulosic-containing material. The predominant polysaccharide in the primary cell wall of biomass is cellulose, the second most abundant is hemicellulose, and the third is pectin. The secondary cell wall, produced after the cell has stopped growing, also contains polysaccharides and is strengthened by polymeric lignin covalently cross-linked to hemicellulose. Cellulose is a homopolymer of anhydrocellobiose and thus a linear beta-(1-4)-D-glucan, while hemicelluloses include a variety of compounds, such as xylans, xyloglucans, arabinoxylans, and mannans in complex branched structures with a spectrum of substituents. Although generally polymorphous, cellulose is found in plant tissue primarily as an insoluble crystalline matrix of parallel glucan chains. Hemicelluloses usually hydrogen bond to cellulose, as well as to other hemicelluloses, which help stabilize the cell wall matrix.

Cellulose is generally found, for example, in the stems, leaves, hulls, husks, and cobs of plants or leaves, branches, and wood of trees. The cellulosic-containing material can be, but is not limited to, agricultural residue, herbaceous material (including energy crops), municipal solid waste, pulp and paper mill residue, waste paper, and wood (including forestry residue) (see, for example, Wiselogle et al., 1995, in *Handbook on Bioethanol* (Charles E. Wyman, editor), pp. 105-118, Taylor & Francis, Washington D.C.; Wyman, 1994, *Bioresource Technology* 50: 3-16; Lynd, 1990, *Applied Biochemistry and Biotechnology* 24/25: 695-719; Mosier et al., 1999, *Recent Progress in Bioconversion of Lignocellulosics*, in *Advances in Biochemical Engineering/Biotechnology*, T. Scheper, managing editor, Volume 65, pp. 23-40, Springer-Verlag, New York). It is understood herein that the cellulose may be in the form of lignocellulose, a plant cell wall material containing lignin, cellulose, and hemicellulose in a mixed matrix. In one embodiment, the cellulosic-containing material is any biomass material. In another embodiment, the cellulosic-containing material is lignocellulose, which comprises cellulose, hemicelluloses, and lignin.

In one embodiment, the cellulosic-containing material is agricultural residue, herbaceous material (including energy crops), municipal solid waste, pulp and paper mill residue, waste paper, or wood (including forestry residue).

In another embodiment, the cellulosic-containing material is *arundo*, bagasse, bamboo, corn cob, corn fiber, corn stover, *miscanthus*, rice straw, switchgrass, or wheat straw.

In another embodiment, the cellulosic-containing material is aspen, *eucalyptus*, fir, pine, poplar, spruce, or willow.

In another embodiment, the cellulosic-containing material is algal cellulose, bacterial cellulose, cotton linter, filter paper, microcrystalline cellulose (e.g., AVICEL®), or phosphoric-acid treated cellulose.

In another embodiment, the cellulosic-containing material is an aquatic biomass. As used herein the term "aquatic biomass" means biomass produced in an aquatic environment by a photosynthesis process. The aquatic biomass can be algae, emergent plants, floating-leaf plants, or submerged plants.

The cellulosic-containing material may be used as is or may be subjected to pretreatment, using conventional methods known in the art, as described herein. In a preferred embodiment, the cellulosic-containing material is pretreated.

The methods of using cellulosic-containing material can be accomplished using methods conventional in the art. Moreover, the methods of can be implemented using any conventional biomass processing apparatus configured to carry out the processes.

Cellulosic Pretreatment

In one embodiment the cellulosic-containing material is pretreated before saccharification.

In practicing the processes described herein, any pretreatment process known in the art can be used to disrupt plant cell wall components of the cellulosic-containing material (Chandra et al., 2007, *Adv. Biochem. Engin./Biotechnol.* 108: 67-93; Galbe and Zacchi, 2007, *Adv. Biochem. Engin./Biotechnol.* 108: 41-65; Hendriks and Zeeman, 2009, *Bioresource Technology* 100: 10-18; Mosier et al., 2005, *Bioresource Technology* 96: 673-686; Taherzadeh and Karimi, 2008, *Int. J. Mol. Sci.* 9: 1621-1651; Yang and Wyman, 2008, *Biofuels Bioproducts and Biorefining-Biofrp.* 2: 26-40).

The cellulosic-containing material can also be subjected to particle size reduction, sieving, pre-soaking, wetting, washing, and/or conditioning prior to pretreatment using methods known in the art.

Conventional pretreatments include, but are not limited to, steam pretreatment (with or without explosion), dilute acid pretreatment, hot water pretreatment, alkaline pretreatment, lime pretreatment, wet oxidation, wet explosion, ammonia fiber explosion, organosolv pretreatment, and biological pretreatment. Additional pretreatments include ammonia percolation, ultrasound, electroporation, microwave, supercritical CO₂, supercritical H₂O, ozone, ionic liquid, and gamma irradiation pretreatments.

In a one embodiment, the cellulosic-containing material is pretreated before saccharification (i.e., hydrolysis) and/or fermentation. Pretreatment is preferably performed prior to the hydrolysis. Alternatively, the pretreatment can be carried out simultaneously with enzyme hydrolysis to release fermentable sugars, such as glucose, xylose, and/or cellobiose. In most cases the pretreatment step itself results in some conversion of biomass to fermentable sugars (even in absence of enzymes).

In one embodiment, the cellulosic-containing material is pretreated with steam. In steam pretreatment, the cellulosic-containing material is heated to disrupt the plant cell wall components, including lignin, hemicellulose, and cellulose to make the cellulose and other fractions, e.g., hemicellulose, accessible to enzymes. The cellulosic-containing mate-

rial is passed to or through a reaction vessel where steam is injected to increase the temperature to the required temperature and pressure and is retained therein for the desired reaction time. Steam pretreatment is preferably performed at 140-250° C., e.g., 160-200° C. or 170-190° C., where the optimal temperature range depends on optional addition of a chemical catalyst. Residence time for the steam pretreatment is preferably 1-60 minutes, e.g., 1-30 minutes, 1-20 minutes, 3-12 minutes, or 4-10 minutes, where the optimal residence time depends on the temperature and optional addition of a chemical catalyst. Steam pretreatment allows for relatively high solids loadings, so that the cellulosic-containing material is generally only moist during the pretreatment. The steam pretreatment is often combined with an explosive discharge of the material after the pretreatment, which is known as steam explosion, that is, rapid flashing to atmospheric pressure and turbulent flow of the material to increase the accessible surface area by fragmentation (Duff and Murray, 1996, *Bioresource Technology* 855: 1-33; Galbe and Zacchi, 2002, *Appl. Microbiol. Biotechnol.* 59: 618-628; U.S. Patent Application No. 2002/0164730). During steam pretreatment, hemicellulose acetyl groups are cleaved and the resulting acid autocatalyzes partial hydrolysis of the hemicellulose to monosaccharides and oligosaccharides. Lignin is removed to only a limited extent.

In one embodiment, the cellulosic-containing material is subjected to a chemical pretreatment. The term "chemical treatment" refers to any chemical pretreatment that promotes the separation and/or release of cellulose, hemicellulose, and/or lignin. Such a pretreatment can convert crystalline cellulose to amorphous cellulose. Examples of suitable chemical pretreatment processes include, for example, dilute acid pretreatment, lime pretreatment, wet oxidation, ammonia fiber/freeze expansion (AFEX), ammonia percolation (APR), ionic liquid, and organosolv pretreatments.

A chemical catalyst such as H₂SO₄ or SO₂ (typically 0.3 to 5% w/w) is sometimes added prior to steam pretreatment, which decreases the time and temperature, increases the recovery, and improves enzymatic hydrolysis (Ballesteros et al., 2006, *Appl. Biochem. Biotechnol.* 129-132: 496-508; Varga et al., 2004, *Appl. Biochem. Biotechnol.* 113-116: 509-523; Sassner et al., 2006, *Enzyme Microb. Technol.* 39: 756-762). In dilute acid pretreatment, the cellulosic-containing material is mixed with dilute acid, typically H₂SO₄, and water to form a slurry, heated by steam to the desired temperature, and after a residence time flashed to atmospheric pressure. The dilute acid pretreatment can be performed with a number of reactor designs, e.g., plug-flow reactors, counter-current reactors, or continuous counter-current shrinking bed reactors (Duff and Murray, 1996, *Bioresource Technology* 855: 1-33; Schell et al., 2004, *Bioresource Technology* 91: 179-188; Lee et al., 1999, *Adv. Biochem. Eng. Biotechnol.* 65: 93-115). In a specific embodiment the dilute acid pretreatment of cellulosic-containing material is carried out using 4% w/w sulfuric acid at 180° C. for 5 minutes.

Several methods of pretreatment under alkaline conditions can also be used. These alkaline pretreatments include, but are not limited to, sodium hydroxide, lime, wet oxidation, ammonia percolation (APR), and ammonia fiber/freeze expansion (AFEX) pretreatment. Lime pretreatment is performed with calcium oxide or calcium hydroxide at temperatures of 85-150° C. and residence times from 1 hour to several days (Wyman et al., 2005, *Bioresource Technology* 96: 1959-1966; Mosier et al., 2005, *Bioresource Technology*

96: 673-686). WO 2006/110891, WO 2006/110899, WO 2006/110900, and WO 2006/110901 disclose pretreatment methods using ammonia.

Wet oxidation is a thermal pretreatment performed typically at 180-200° C. for 5-15 minutes with addition of an oxidative agent such as hydrogen peroxide or over-pressure of oxygen (Schmidt and Thomsen, 1998, *Bioresource Technology* 64: 139-151; Palonen et al., 2004, *Appl. Biochem. Biotechnol.* 117: 1-17; Varga et al., 2004, *Biotechnol. Bioeng.* 88: 567-574; Martin et al., 2006, *J. Chem. Technol. Biotechnol.* 81: 1669-1677). The pretreatment is performed preferably at 1-40% dry matter, e.g., 2-30% dry matter or 5-20% dry matter, and often the initial pH is increased by the addition of alkali such as sodium carbonate.

A modification of the wet oxidation pretreatment method, known as wet explosion (combination of wet oxidation and steam explosion) can handle dry matter up to 30%. In wet explosion, the oxidizing agent is introduced during pretreatment after a certain residence time. The pretreatment is then ended by flashing to atmospheric pressure (WO 2006/032282).

Ammonia fiber expansion (AFEX) involves treating the cellulosic-containing material with liquid or gaseous ammonia at moderate temperatures such as 90-150° C. and high pressure such as 17-20 bar for 5-10 minutes, where the dry matter content can be as high as 60% (Gollapalli et al., 2002, *Appl. Biochem. Biotechnol.* 98: 23-35; Chundawat et al., 2007, *Biotechnol. Bioeng.* 96: 219-231; Alizadeh et al., 2005, *Appl. Biochem. Biotechnol.* 121: 1133-1141; Teymouri et al., 2005, *Bioresource Technology* 96: 2014-2018). During AFEX pretreatment cellulose and hemicelluloses remain relatively intact. Lignin-carbohydrate complexes are cleaved.

Organosolv pretreatment delignifies the cellulosic-containing material by extraction using aqueous ethanol (40-60% ethanol) at 160-200° C. for 30-60 minutes (Pan et al., 2005, *Biotechnol. Bioeng.* 90: 473-481; Pan et al., 2006, *Biotechnol. Bioeng.* 94: 851-861; Kurabi et al., 2005, *Appl. Biochem. Biotechnol.* 121: 219-230). Sulphuric acid is usually added as a catalyst. In organosolv pretreatment, the majority of hemicellulose and lignin is removed.

Other examples of suitable pretreatment methods are described by Schell et al., 2003, *Appl. Biochem. Biotechnol.* 105-108: 69-85, and Mosier et al., 2005, *Bioresource Technology* 96: 673-686, and U.S. Published Application 2002/0164730.

In one embodiment, the chemical pretreatment is carried out as a dilute acid treatment, and more preferably as a continuous dilute acid treatment. The acid is typically sulfuric acid, but other acids can also be used, such as acetic acid, citric acid, nitric acid, phosphoric acid, tartaric acid, succinic acid, hydrogen chloride, or mixtures thereof. Mild acid treatment is conducted in the pH range of preferably 1-5, e.g., 1-4 or 1-2.5. In one aspect, the acid concentration is in the range from preferably 0.01 to 10 wt. % acid, e.g., 0.05 to 5 wt. % acid or 0.1 to 2 wt. % acid. The acid is contacted with the cellulosic-containing material and held at a temperature in the range of preferably 140-200° C., e.g., 165-190° C., for periods ranging from 1 to 60 minutes.

In another embodiment, pretreatment takes place in an aqueous slurry. In preferred aspects, the cellulosic-containing material is present during pretreatment in amounts preferably between 10-80 wt. %, e.g., 20-70 wt. % or 30-60 wt. %, such as around 40 wt. %. The pretreated cellulosic-containing material can be unwashed or washed using any method known in the art, e.g., washed with water.

In one embodiment, the cellulosic-containing material is subjected to mechanical or physical pretreatment. The term "mechanical pretreatment" or "physical pretreatment" refers to any pretreatment that promotes size reduction of particles. For example, such pretreatment can involve various types of grinding or milling (e.g., dry milling, wet milling, or vibratory ball milling).

The cellulosic-containing material can be pretreated both physically (mechanically) and chemically. Mechanical or physical pretreatment can be coupled with steaming/steam explosion, hydrothermolysis, dilute or mild acid treatment, high temperature, high pressure treatment, irradiation (e.g., microwave irradiation), or combinations thereof. In one aspect, high pressure means pressure in the range of preferably about 100 to about 400 psi, e.g., about 150 to about 250 psi. In another aspect, high temperature means temperature in the range of about 100 to about 300° C., e.g., about 140 to about 200° C. In a preferred aspect, mechanical or physical pretreatment is performed in a batch-process using a steam gun hydrolyzer system that uses high pressure and high temperature as defined above, e.g., a Sunds Hydrolyzer available from Sunds Defibrator AB, Sweden. The physical and chemical pretreatments can be carried out sequentially or simultaneously, as desired.

Accordingly, in one embodiment, the cellulosic-containing material is subjected to physical (mechanical) or chemical pretreatment, or any combination thereof, to promote the separation and/or release of cellulose, hemicellulose, and/or lignin.

In one embodiment, the cellulosic-containing material is subjected to a biological pretreatment. The term "biological pretreatment" refers to any biological pretreatment that promotes the separation and/or release of cellulose, hemicellulose, and/or lignin from the cellulosic-containing material. Biological pretreatment techniques can involve applying lignin-solubilizing microorganisms and/or enzymes (see, for example, Hsu, T.-A., 1996, Pretreatment of biomass, in *Handbook on Bioethanol: Production and Utilization*, Wyman, C. E., ed., Taylor & Francis, Washington, DC, 179-212; Ghosh and Singh, 1993, *Adv. Appl. Microbiol.* 39: 295-333; McMillan, J. D., 1994, Pretreating lignocellulosic biomass: a review, in *Enzymatic Conversion of Biomass for Fuels Production*, Himmel, M. E., Baker, J. O., and Overend, R. P., eds., ACS Symposium Series 566, American Chemical Society, Washington, DC, chapter 15; Gong, C. S., Cao, N. J., Du, J., and Tsao, G. T., 1999, Ethanol production from renewable resources, in *Advances in Biochemical Engineering/Biotechnology*, Scheper, T., ed., Springer-Verlag Berlin Heidelberg, Germany, 65: 207-241; Olsson and Hahn-Hagerdal, 1996, *Enz. Microb. Tech.* 18: 312-331; and Vallander and Eriksson, 1990, *Adv. Biochem. Eng./Biotechnol.* 42: 63-95).

Saccharification and Fermentation of Cellulosic-Containing Material

Saccharification (i.e., hydrolysis) and fermentation, separate or simultaneous, include, but are not limited to, separate hydrolysis and fermentation (SHF); simultaneous saccharification and fermentation (SSF); simultaneous saccharification and co-fermentation (SSCF); hybrid hydrolysis and fermentation (HHF); separate hydrolysis and co-fermentation (SHCF); hybrid hydrolysis and co-fermentation (HHCF).

SHF uses separate process steps to first enzymatically hydrolyze the cellulosic-containing material to fermentable sugars, e.g., glucose, cellobiose, and pentose monomers, and then ferment the fermentable sugars to ethanol. In SSF, the enzymatic hydrolysis of the cellulosic-containing material

and the fermentation of sugars to ethanol are combined in one step (Philippidis, G. P., 1996, Cellulose bioconversion technology, in *Handbook on Bioethanol: Production and Utilization*, Wyman, C. E., ed., Taylor & Francis, Washington, DC, 179-212). SSCF involves the co-fermentation of multiple sugars (Sheehan and Himmel, 1999, *Biotechnol. Prog.* 15: 817-827). HHF involves a separate hydrolysis step, and in addition a simultaneous saccharification and hydrolysis step, which can be carried out in the same reactor. The steps in an HHF process can be carried out at different temperatures, i.e., high temperature enzymatic saccharification followed by SSF at a lower temperature that the fermentation organism can tolerate. It is understood herein that any method known in the art comprising pretreatment, enzymatic hydrolysis (saccharification), fermentation, or a combination thereof, can be used in the practicing the processes described herein.

A conventional apparatus can include a fed-batch stirred reactor, a batch stirred reactor, a continuous flow stirred reactor with ultrafiltration, and/or a continuous plug-flow column reactor (de Castilhos Corazza et al., 2003, *Acta Scientiarum. Technology* 25: 33-38; Gusakov and Sinitsyn, 1985, *Enz. Microb. Technol.* 7: 346-352), an attrition reactor (Ryu and Lee, 1983, *Biotechnol. Bioeng.* 25: 53-65). Additional reactor types include fluidized bed, upflow blanket, immobilized, and extruder type reactors for hydrolysis and/or fermentation.

In the saccharification step (i.e., hydrolysis step), the cellulosic and/or starch-containing material, e.g., pretreated, is hydrolyzed to break down cellulose, hemicellulose, and/or starch to fermentable sugars, such as glucose, cellobiose, xylose, xylulose, arabinose, mannose, galactose, and/or soluble oligosaccharides. The hydrolysis is performed enzymatically e.g., by a cellulolytic enzyme composition. The enzymes of the compositions can be added simultaneously or sequentially.

Enzymatic hydrolysis may be carried out in a suitable aqueous environment under conditions that can be readily determined by one skilled in the art. In one aspect, hydrolysis is performed under conditions suitable for the activity of the enzymes(s), i.e., optimal for the enzyme(s). The hydrolysis can be carried out as a fed batch or continuous process where the cellulosic and/or starch-containing material is fed gradually to, for example, an enzyme containing hydrolysis solution.

The saccharification is generally performed in stirred-tank reactors or fermentors under controlled pH, temperature, and mixing conditions. Suitable process time, temperature and pH conditions can readily be determined by one skilled in the art. For example, the saccharification can last up to 200 hours, but is typically performed for preferably about 12 to about 120 hours, e.g., about 16 to about 72 hours or about 24 to about 48 hours. The temperature is in the range of preferably about 25° C. to about 70° C., e.g., about 30° C. to about 65° C., about 40° C. to about 60° C., or about 50° C. to about 55° C. The pH is in the range of preferably about 3 to about 8, e.g., about 3.5 to about 7, about 4 to about 6, or about 4.5 to about 5.5. The dry solids content is in the range of preferably about 5 to about 50 wt. %, e.g., about 10 to about 40 wt. % or about 20 to about 30 wt. %.

Saccharification may be carried out using a cellulolytic enzyme composition. Such enzyme compositions are described below in the "Cellulolytic Enzyme Composition" section below. The cellulolytic enzyme compositions can comprise any protein useful in degrading the cellulosic-containing material. In one aspect, the cellulolytic enzyme composition comprises or further comprises one or more

(e.g., several) proteins selected from the group consisting of a cellulase, an AA9 (GH61) polypeptide, a hemicellulase, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin.

In another embodiment, the cellulase is preferably one or more (e.g., several) enzymes selected from the group consisting of an endoglucanase, a cellobiohydrolase, and a beta-glucosidase.

In another embodiment, the hemicellulase is preferably one or more (e.g., several) enzymes selected from the group consisting of an acetylmannan esterase, an acetylxyylan esterase, an arabinanase, an arabinofuranosidase, a coumaric acid esterase, a feruloyl esterase, a galactosidase, a glucuronidase, a glucuronoyl esterase, a mannanase, a mannosidase, a xylanase, and a xylosidase. In another embodiment, the oxidoreductase is one or more (e.g., several) enzymes selected from the group consisting of a catalase, a laccase, and a peroxidase. The enzymes or enzyme compositions used in a processes of the present invention may be in any form suitable for use, such as, for example, a fermentation broth formulation or a cell composition, a cell lysate with or without cellular debris, a semi-purified or purified enzyme preparation, or a host cell as a source of the enzymes. The enzyme composition may be a dry powder or granulate, a non-dusting granulate, a liquid, a stabilized liquid, or a stabilized protected enzyme. Liquid enzyme preparations may, for instance, be stabilized by adding stabilizers such as a sugar, a sugar alcohol or another polyol, and/or lactic acid or another organic acid according to established processes.

In one embodiment, an effective amount of cellulolytic or hemicellulolytic enzyme composition to the cellulosic-containing material is about 0.5 to about 50 mg, e.g., about 0.5 to about 40 mg, about 0.5 to about 25 mg, about 0.75 to about 20 mg, about 0.75 to about 15 mg, about 0.5 to about 10 mg, or about 2.5 to about 10 mg per g of the cellulosic-containing material.

In one embodiment, such a compound is added at a molar ratio of the compound to glucosyl units of cellulose of about 10^{-6} to about 10, e.g., about 10^{-6} to about 7.5, about 10^{-6} to about 5, about 10^{-6} to about 2.5, about 10^{-6} to about 1, about 10^{-5} to about 1, about 10^{-5} to about 10^{-1} , about 10^{-4} to about 10^{-1} , about 10^{-3} to about 10^{-1} , or about 10^{-3} to about 10^{-2} . In another aspect, an effective amount of such a compound is about 0.1 μ M to about 1 M, e.g., about 0.5 μ M to about 0.75 M, about 0.75 μ M to about 0.5 M, about 1 μ M to about 0.25 M, about 1 μ M to about 0.1 M, about 5 μ M to about 50 mM, about 10 μ M to about 25 mM, about 50 μ M to about 25 mM, about 10 μ M to about 10 mM, about 5 μ M to about 5 mM, or about 0.1 mM to about 1 mM.

The term "liquor" means the solution phase, either aqueous, organic, or a combination thereof, arising from treatment of a lignocellulose and/or hemicellulose material in a slurry, or monosaccharides thereof, e.g., xylose, arabinose, mannose, etc., under conditions as described in WO 2012/021401, and the soluble contents thereof. A liquor for cellulolytic enhancement of an AA9 polypeptide (GH61 polypeptide) can be produced by treating a lignocellulose or hemicellulose material (or feedstock) by applying heat and/or pressure, optionally in the presence of a catalyst, e.g., acid, optionally in the presence of an organic solvent, and optionally in combination with physical disruption of the material, and then separating the solution from the residual solids. Such conditions determine the degree of cellulolytic enhancement obtainable through the combination of liquor and an AA9 polypeptide during hydrolysis of a cellulosic substrate by a cellulolytic enzyme preparation. The liquor

can be separated from the treated material using a method standard in the art, such as filtration, sedimentation, or centrifugation.

In one embodiment, an effective amount of the liquor to cellulose is about 10^{-6} to about 10 g per g of cellulose, e.g., about 10^{-6} to about 7.5 g, about 10^{-6} to about 5 g, about 10^{-6} to about 2.5 g, about 10^{-6} to about 1 g, about 10^{-5} to about 1 g, about 10^{-5} to about 10^{-1} g, about 10^{-4} to about 10^{-1} g, about 10^{-3} to about 10^{-1} g, or about 10^{-3} to about 10^{-2} g per g of cellulose.

In the fermentation step, sugars, released from the cellulosic-containing material, e.g., as a result of the pretreatment and enzymatic hydrolysis steps, are fermented to ethanol, by a fermenting organism, such as yeast described herein. Hydrolysis (saccharification) and fermentation can be separate or simultaneous.

Any suitable hydrolyzed cellulosic-containing material can be used in the fermentation step in practicing the processes described herein. Such feedstocks include, but are not limited to carbohydrates (e.g., lignocellulose, xylans, cellulose, starch, etc.). The material is generally selected based on economics, i.e., costs per equivalent sugar potential, and recalcitrance to enzymatic conversion.

Production of ethanol by a fermenting organism using cellulosic-containing material results from the metabolism of sugars (monosaccharides). The sugar composition of the hydrolyzed cellulosic-containing material and the ability of the fermenting organism to utilize the different sugars has a direct impact in process yields. Prior to Applicant's disclosure herein, strains known in the art utilize glucose efficiently but do not (or very limitedly) metabolize pentoses like xylose, a monosaccharide commonly found in hydrolyzed material.

Compositions of the fermentation media and fermentation conditions depend on the fermenting organism and can easily be determined by one skilled in the art. Typically, the fermentation takes place under conditions known to be suitable for generating the fermentation product. In some embodiments, the fermentation process is carried out under aerobic or microaerophilic (i.e., where the concentration of oxygen is less than that in air), or anaerobic conditions. In some embodiments, fermentation is conducted under anaerobic conditions (i.e., no detectable oxygen), or less than about 5, about 2.5, or about 1 mmol/L/h oxygen. In the absence of oxygen, the NADH produced in glycolysis cannot be oxidized by oxidative phosphorylation. Under anaerobic conditions, pyruvate or a derivative thereof may be utilized by the host cell as an electron and hydrogen acceptor in order to generate NAD⁺.

The fermentation process is typically run at a temperature that is optimal for the recombinant fungal cell. For example, in some embodiments, the fermentation process is performed at a temperature in the range of from about 25° C. to about 42° C. Typically the process is carried out at a temperature that is less than about 38° C., less than about 35° C., less than about 33° C., or less than about 38° C., but at least about 20° C., 22° C., or 25° C.

A fermentation stimulator can be used in a process described herein to further improve the fermentation, and in particular, the performance of the fermenting organism, such as, rate enhancement and product yield (e.g., ethanol yield). A "fermentation stimulator" refers to stimulators for growth of the fermenting organisms, in particular, yeast. Preferred fermentation stimulators for growth include vitamins and minerals. Examples of vitamins include multivitamins, biotin, pantothenate, nicotinic acid, meso-inositol, thiamine, pyridoxine, para-aminobenzoic acid, folic acid, riboflavin,

and Vitamins A, B, C, D, and E. See, for example, Alfenore et al., Improving ethanol production and viability of *Saccharomyces cerevisiae* by a vitamin feeding strategy during fed-batch process, Springer-Verlag (2002), which is hereby incorporated by reference. Examples of minerals include minerals and mineral salts that can supply nutrients comprising P, K, Mg, S, Ca, Fe, Zn, Mn, and Cu.

Cellulolytic Enzymes and Compositions

A cellulolytic enzyme or cellulolytic enzyme composition may be present and/or added during saccharification. A cellulolytic enzyme composition is an enzyme preparation containing one or more (e.g., several) enzymes that hydrolyze cellulosic-containing material. Such enzymes include endoglucanase, cellobiohydrolase, beta-glucosidase, and/or combinations thereof.

In some embodiments, the fermenting organism comprises one or more (e.g., several) heterologous polynucleotides encoding enzymes that hydrolyze cellulosic-containing material (e.g., an endoglucanase, cellobiohydrolase, beta-glucosidase or combinations thereof). Any enzyme described or referenced herein that hydrolyzes cellulosic-containing material is contemplated for expression in the fermenting organism.

The cellulolytic enzyme may be any cellulolytic enzyme that is suitable for the host cells and/or the methods described herein (e.g., an endoglucanase, cellobiohydrolase, beta-glucosidase), such as a naturally occurring cellulolytic enzyme or a variant thereof that retains cellulolytic enzyme activity.

In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a cellulolytic enzyme has an increased level of cellulolytic enzyme activity (e.g., increased endoglucanase, cellobiohydrolase, and/or beta-glucosidase) compared to the host cells without the heterologous polynucleotide encoding the cellulolytic enzyme, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of cellulolytic enzyme activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the cellulolytic enzyme, when cultivated under the same conditions.

Exemplary cellulolytic enzymes that can be used with the host cells and/or the methods described herein include bacterial, yeast, or filamentous fungal cellulolytic enzymes, e.g., obtained from any of the microorganisms described or referenced herein, as described supra under the sections related to proteases.

The cellulolytic enzyme may be of any origin. In an embodiment the cellulolytic enzyme is derived from a strain of *Trichoderma*, such as a strain of *Trichoderma reesei*; a strain of *Humicola*, such as a strain of *Humicola insolens*, and/or a strain of *Chrysosporium*, such as a strain of *Chrysosporium lucknowense*. In a preferred embodiment the cellulolytic enzyme is derived from a strain of *Trichoderma reesei*.

The cellulolytic enzyme composition may further comprise one or more of the following polypeptides, such as enzymes: AA9 polypeptide (GH61 polypeptide) having cellulolytic enhancing activity, beta-glucosidase, xylanase, beta-xylosidase, CBH I, CBH II, or a mixture of two, three, four, five or six thereof.

The further polypeptide(s) (e.g., AA9 polypeptide) and/or enzyme(s) (e.g., beta-glucosidase, xylanase, beta-xylosi-

dase, CBH I and/or CBH II may be foreign to the cellulolytic enzyme composition producing organism (e.g., *Trichoderma reesei*).

In an embodiment the cellulolytic enzyme composition comprises an AA9 polypeptide having cellulolytic enhancing activity and a beta-glucosidase.

In another embodiment the cellulolytic enzyme composition comprises an AA9 polypeptide having cellulolytic enhancing activity, a beta-glucosidase, and a CBH I.

In another embodiment the cellulolytic enzyme composition comprises an AA9 polypeptide having cellulolytic enhancing activity, a beta-glucosidase, a CBH I and a CBH II.

Other enzymes, such as endoglucanases, may also be comprised in the cellulolytic enzyme composition.

As mentioned above the cellulolytic enzyme composition may comprise a number of difference polypeptides, including enzymes.

In one embodiment, the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Thermoascus aurantiacus* AA9 (GH61A) polypeptide having cellulolytic enhancing activity (e.g., WO 2005/074656), and *Aspergillus oryzae* beta-glucosidase fusion protein (e.g., one disclosed in WO 2008/057637, in particular shown as SEQ ID NOs: 59 and 60).

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Thermoascus aurantiacus* AA9 (GH61A) polypeptide having cellulolytic enhancing activity (e.g., SEQ ID NO: 2 in WO 2005/074656), and *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499).

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO 2011/041397, and *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499).

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO 2011/041397, and *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499) or a variant disclosed in WO 2012/044915 (hereby incorporated by reference), in particular one comprising one or more such as all of the following substitutions: F100D, S283G, N456E, F512Y.

In an embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic composition, further comprising an AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one derived from a strain of *Penicillium emersonii* (e.g., SEQ ID NO: 2 in WO 2011/041397), *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 in WO 2005/047499) variant with one or more, in particular all of the following substitutions: F100D, S283G, N456E, F512Y and disclosed in WO 2012/044915; *Aspergillus fumigatus* Cel7A CBH1, e.g., the one disclosed as SEQ ID NO: 6 in WO2011/057140 and *Aspergillus fumigatus* CBH II, e.g., the one disclosed as SEQ ID NO: 18 in WO 2011/057140.

In a preferred embodiment the cellulolytic enzyme composition is a *Trichoderma reesei*, cellulolytic enzyme composition, further comprising a hemicellulase or hemicel-

lulolytic enzyme composition, such as an *Aspergillus fumigatus* xylanase and *Aspergillus fumigatus* beta-xylosidase.

In an embodiment the cellulolytic enzyme composition also comprises a xylanase (e.g., derived from a strain of the genus *Aspergillus*, in particular *Aspergillus aculeatus* or *Aspergillus fumigatus*; or a strain of the genus *Talaromyces*, in particular *Talaromyces leycettanus*) and/or a beta-xylosidase (e.g., derived from *Aspergillus*, in particular *Aspergillus fumigatus*, or a strain of *Talaromyces*, in particular *Talaromyces emersonii*).

In an embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Thermoascus aurantiacus* AA9 (GH61A) polypeptide having cellulolytic enhancing activity (e.g., WO 2005/074656), *Aspergillus oryzae* beta-glucosidase fusion protein (e.g., one disclosed in WO 2008/057637, in particular as SEQ ID NOs: 59 and 60), and *Aspergillus aculeatus* xylanase (e.g., Xyl II in WO 94/21785).

In another embodiment the cellulolytic enzyme composition comprises a *Trichoderma reesei* cellulolytic preparation, further comprising *Thermoascus aurantiacus* GH61A polypeptide having cellulolytic enhancing activity (e.g., SEQ ID NO: 2 in WO 2005/074656), *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499) and *Aspergillus aculeatus* xylanase (Xyl II disclosed in WO 94/21785).

In another embodiment the cellulolytic enzyme composition comprises a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Thermoascus aurantiacus* AA9 (GH61A) polypeptide having cellulolytic enhancing activity (e.g., SEQ ID NO: 2 in WO 2005/074656), *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499) and *Aspergillus aculeatus* xylanase (e.g., Xyl II disclosed in WO 94/21785).

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO 2011/041397, *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499) and *Aspergillus fumigatus* xylanase (e.g., Xyl III in WO 2006/078256).

In another embodiment the cellulolytic enzyme composition comprises a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO 2011/041397, *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499), *Aspergillus fumigatus* xylanase (e.g., Xyl III in WO 2006/078256), and CBH I from *Aspergillus fumigatus*, in particular Cel7A CBH1 disclosed as SEQ ID NO: 2 in WO2011/057140.

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO 2011/041397, *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499), *Aspergillus fumigatus* xylanase (e.g., Xyl III in WO 2006/078256), CBH I from *Aspergillus fumigatus*, in particular Cel7A CBH1 disclosed as SEQ ID NO: 2 in WO 2011/057140, and CBH II derived from *Aspergillus fumigatus* in particular the one disclosed as SEQ ID NO: 4 in WO 2013/028928.

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition,

further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO 2011/041397, *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499) or variant thereof with one or more, in particular all, of the following substitutions: F100D, S283G, N456E, F512Y; *Aspergillus fumigatus* xylanase (e.g., Xyl III in WO 2006/078256), CBH I from *Aspergillus fumigatus*, in particular Cel7A CBH I disclosed as SEQ ID NO: 2 in WO 2011/057140, and CBH II derived from *Aspergillus fumigatus*, in particular the one disclosed in WO 2013/028928.

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising the CBH I (GENSEQP Accession No. AZY49536 (WO2012/103293); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288); a beta-glucosidase variant (GENSEQP Accession No. AZU67153 (WO 2012/44915)), in particular with one or more, in particular all, of the following substitutions: F100D, S283G, N456E, F512Y; and AA9 (GH61 polypeptide) (GENSEQP Accession No. BAL61510 (WO 2013/028912)).

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising a CBH I (GENSEQP Accession No. AZY49536 (WO2012/103293)); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288); a GH10 xylanase (GENSEQP Accession No. BAK46118 (WO 2013/019827)); and a beta-xylosidase (GENSEQP Accession No. AZI04896 (WO 2011/057140)).

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising a CBH I (GENSEQP Accession No. AZY49536 (WO2012/103293)); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288)); and an AA9 (GH61 polypeptide; GENSEQP Accession No. BAL61510 (WO 2013/028912)).

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising a CBH I (GENSEQP Accession No. AZY49536 (WO2012/103293)); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288)), an AA9 (GH61 polypeptide; GENSEQP Accession No. BAL61510 (WO 2013/028912)), and a catalase (GENSEQP Accession No. BAC11005 (WO 2012/130120)).

In an embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising a CBH I (GENSEQP Accession No. AZY49446 (WO2012/103288); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288)), a beta-glucosidase variant (GENSEQP Accession No. AZU67153 (WO 2012/44915)), with one or more, in particular all, of the following substitutions: F100D, S283G, N456E, F512Y; an AA9 (GH61 polypeptide; GENSEQP Accession No. BAL61510 (WO 2013/028912)), a GH10 xylanase (GENSEQP Accession No. BAK46118 (WO 2013/019827)), and a beta-xylosidase (GENSEQP Accession No. AZI04896 (WO 2011/057140)).

In an embodiment the cellulolytic composition is a *Trichoderma reesei* cellulolytic enzyme preparation comprising an EG I (Swissprot Accession No. P07981), EG II (EMBL Accession No. M19373), CBH I (supra); CBH II (supra); beta-glucosidase variant (supra) with the following substitutions: F100D, S283G, N456E, F512Y; an AA9 (GH61 polypeptide; supra), GH10 xylanase (supra); and beta-xylosidase (supra).

All cellulolytic enzyme compositions disclosed in WO 2013/028928 are also contemplated and hereby incorporated by reference.

The cellulolytic enzyme composition comprises or may further comprise one or more (several) proteins selected from the group consisting of a cellulase, a AA9 (i.e., GH61) polypeptide having cellulolytic enhancing activity, a hemi-cellulase, an expansin, an esterase, a laccase, a ligninolytic enzyme, a pectinase, a peroxidase, a protease, and a swollenin.

In one embodiment the cellulolytic enzyme composition is a commercial cellulolytic enzyme composition. Examples of commercial cellulolytic enzyme compositions suitable for use in a process of the invention include: CELLIC® CTec (Novozymes A/S), CELLIC® CTec2 (Novozymes A/S), CELLIC® CTec3 (Novozymes A/S), CELLUCLAST™ (Novozymes A/S), SPEZYME™ CP (Genencor Int.), ACCELLERASE™ 1000, ACCELLERASE 1500, ACCELLERASE™ TRIO (DuPont), FILTRASE® NL (DSM); METHAPLUS® S/L 100 (DSM), ROHAMENT™ 7069 W (Röhm GmbH), or ALTERNAFUEL® CMAX3™ (Dyadic International, Inc.). The cellulolytic enzyme composition may be added in an amount effective from about 0.001 to about 5.0 wt. % of solids, e.g., about 0.025 to about 4.0 wt. % of solids or about 0.005 to about 2.0 wt. % of solids.

Additional enzymes, and compositions thereof can be found in WO2011/153516 and WO2016/045569 (the contents of which are incorporated herein).

Additional polynucleotides encoding suitable cellulolytic enzymes may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org).

The cellulolytic enzyme coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding cellulolytic enzymes from strains of different genera or species, as described supra.

The polynucleotides encoding cellulolytic enzymes may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc) as described supra.

Techniques used to isolate or clone polynucleotides encoding cellulolytic enzymes are described supra.

In one embodiment, the cellulolytic enzyme has a mature polypeptide sequence of at least 60%, e.g., at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to any cellulolytic enzyme described or referenced herein (e.g., any endoglucanase, cellobiohydrolase, or beta-glucosidase). In one aspect, the cellulolytic enzyme has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from any cellulolytic enzyme described or referenced herein. In one embodiment, the cellulolytic enzyme has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any cellulolytic enzyme described or referenced herein, allelic variant, or a fragment thereof having cellulolytic enzyme activity. In one embodiment, the cellulolytic enzyme has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) amino acids. In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In some embodiments, the cellulolytic enzyme has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least

70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the cellulolytic enzyme activity of any cellulolytic enzyme described or referenced herein (e.g., any endoglucanase, cellobiohydrolase, or beta-glucosidase) under the same conditions.

In one embodiment, the cellulolytic enzyme coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any cellulolytic enzyme described or referenced herein (e.g., any endoglucanase, cellobiohydrolase, or beta-glucosidase). In one embodiment, the cellulolytic enzyme coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any cellulolytic enzyme described or referenced herein.

In one embodiment, the polynucleotide encoding the cellulolytic enzyme comprises the coding sequence of any cellulolytic enzyme described or referenced herein (e.g., any endoglucanase, cellobiohydrolase, or beta-glucosidase). In one embodiment, the polynucleotide encoding the cellulolytic enzyme comprises a subsequence of the coding sequence from any cellulolytic enzyme described or referenced herein, wherein the subsequence encodes a polypeptide having cellulolytic enzyme activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The cellulolytic enzyme can also include fused polypeptides or cleavable fusion polypeptides, as described supra. Xylose Metabolism

In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a xylose isomerase (XI). The xylose isomerase may be any xylose isomerase that is suitable for the host cells and the methods described herein, such as a naturally occurring xylose isomerase or a variant thereof that retains xylose isomerase activity. In one embodiment, the xylose isomerase is present in the cytosol of the host cells.

In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a xylose isomerase has an increased level of xylose isomerase activity compared to the host cells without the heterologous polynucleotide encoding the xylose isomerase, when cultivated under the same conditions. In some embodiments, the fermenting organisms have an increased level of xylose isomerase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the host cells without the heterologous polynucleotide encoding the xylose isomerase, when cultivated under the same conditions.

Exemplary xylose isomerases that can be used with the recombinant host cells and methods of use described herein include, but are not limited to, XIs from the fungus *Pirromyces* sp. (WO2003/062430) or other sources (Madhavan et al., 2009, *Appl Microbiol Biotechnol.* 82(6), 1067-1078) have been expressed in *S. cerevisiae* host cells. Still other XIs suitable for expression in yeast have been described in US 2012/0184020 (an XI from *Ruminococcus flavefaciens*), WO2011/078262 (several XIs from *Reticulitermes speratus* and *Mastotermes darwiniensis*) and WO2012/009272 (con-

structs and fungal cells containing an XI from *Abiotrophia defectiva*). U.S. Pat. No. 8,586,336 describes a *S. cerevisiae* host cell expressing an XI obtained by bovine rumen fluid (shown herein as SEQ ID NO: 74).

Additional polynucleotides encoding suitable xylose isomerases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org). In one embodiment, the xylose isomerases is a bacterial, a yeast, or a filamentous fungal xylose isomerase, e.g., obtained from any of the microorganisms described or referenced herein, as described supra.

The xylose isomerase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding xylose isomerases from strains of different genera or species, as described supra.

The polynucleotides encoding xylose isomerases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc) as described supra.

Techniques used to isolate or clone polynucleotides encoding xylose isomerases are described supra.

In one embodiment, the xylose isomerase has a mature polypeptide sequence of having at least 60%, e.g., at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74). In one aspect, the xylose isomerase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74). In one embodiment, the xylose isomerase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74), allelic variant, or a fragment thereof having xylose isomerase activity. In one embodiment, the xylose isomerase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) amino acids. In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In some embodiments, the xylose isomerase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the xylose isomerase activity of any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74) under the same conditions.

In one embodiment, the xylose isomerase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74). In one embodiment, the xylose isomerase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%,

at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74).

In one embodiment, the heterologous polynucleotide encoding the xylose isomerase comprises the coding sequence of any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74). In one embodiment, the heterologous polynucleotide encoding the xylose isomerase comprises a subsequence of the coding sequence from any xylose isomerase described or referenced herein, wherein the subsequence encodes a polypeptide having xylose isomerase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The xylose isomerases can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a xylulokinase (XK). A xylulokinase, as used herein, provides enzymatic activity for converting D-xylulose to xylulose 5-phosphate. The xylulokinase may be any xylulokinase that is suitable for the host cells and the methods described herein, such as a naturally occurring xylulokinase or a variant thereof that retains xylulokinase activity. In one embodiment, the xylulokinase is present in the cytosol of the host cells.

In some embodiments, the fermenting organisms comprising a heterologous polynucleotide encoding a xylulokinase have an increased level of xylulokinase activity compared to the host cells without the heterologous polynucleotide encoding the xylulokinase, when cultivated under the same conditions. In some embodiments, the host cells have an increased level of xylose isomerase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the host cells without the heterologous polynucleotide encoding the xylulokinase, when cultivated under the same conditions.

Exemplary xylulokinases that can be used with the fermenting organisms and methods of use described herein include, but are not limited to, the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75. Additional polynucleotides encoding suitable xylulokinases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org). In one embodiment, the xylulokinases is a bacterial, a yeast, or a filamentous fungal xylulokinase, e.g., obtained from any of the microorganisms described or referenced herein, as described supra.

The xylulokinase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding xylulokinases from strains of different genera or species, as described supra.

The polynucleotides encoding xylulokinases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc) as described supra.

Techniques used to isolate or clone polynucleotides encoding xylulokinases are described supra.

In one embodiment, the xylulokinase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least

95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75). In one embodiment, the xylulokinase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75). In one embodiment, the xylulokinase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75), allelic variant, or a fragment thereof having xylulokinase activity. In one embodiment, the xylulokinase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) amino acids. In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In some embodiments, the xylulokinase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the xylulokinase activity of any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75) under the same conditions.

In one embodiment, the xylulokinase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75). In one embodiment, the xylulokinase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75).

In one embodiment, the heterologous polynucleotide encoding the xylulokinase comprises the coding sequence of any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75). In one embodiment, the heterologous polynucleotide encoding the xylulokinase comprises a subsequence of the coding sequence from any xylulokinase described or referenced herein, wherein the subsequence encodes a polypeptide having xylulokinase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The xylulokinases can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a ribulose 5 phosphate 3-epimerase (RPE1). A ribulose 5 phosphate 3-epimerase, as used herein, provides enzymatic activity for converting L-ribulose 5-phosphate to L-xylulose 5-phosphate (EC 5.1.3.22). The RPE1 may be any RPE1 that is suitable for the host cells and the methods described herein, such as a naturally occurring RPE1 or a variant

thereof that retains RPE1 activity. In one embodiment, the RPE1 is present in the cytosol of the host cells.

In one embodiment, the recombinant cell comprises a heterologous polynucleotide encoding a ribulose 5 phosphate 3-epimerase (RPE1), wherein the RPE1 is *Saccharomyces cerevisiae* RPE1, or an RPE1 having at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to a *Saccharomyces cerevisiae* RPE1.

In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a ribulose 5 phosphate isomerase (RKI1). A ribulose 5 phosphate isomerase, as used herein, provides enzymatic activity for converting ribose-5-phosphate to ribulose 5-phosphate. The RKI1 may be any RKI1 that is suitable for the host cells and the methods described herein, such as a naturally occurring RKI1 or a variant thereof that retains RKI1 activity. In one embodiment, the RKI1 is present in the cytosol of the host cells.

In one embodiment, the fermenting organism comprises a heterologous polynucleotide encoding a ribulose 5 phosphate isomerase (RKI1), wherein the RKI1 is a *Saccharomyces cerevisiae* RKI1, or an RKI1 having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to a *Saccharomyces cerevisiae* RKI1.

In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a transketolase (TKL1). The TKL1 may be any TKL1 that is suitable for the host cells and the methods described herein, such as a naturally occurring TKL1 or a variant thereof that retains TKL1 activity. In one embodiment, the TKL1 is present in the cytosol of the host cells.

In one embodiment, the fermenting organism comprises a heterologous polynucleotide encoding a transketolase (TKL1), wherein the TKL1 is a *Saccharomyces cerevisiae* TKL1, or a TKL1 having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to a *Saccharomyces cerevisiae* TKL1.

In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a transaldolase (TAL1). The TAL1 may be any TAL1 that is suitable for the host cells and the methods described herein, such as a naturally occurring TAL1 or a variant thereof that retains TAL1 activity. In one embodiment, the TAL1 is present in the cytosol of the host cells.

In one embodiment, the fermenting organism comprises a heterologous polynucleotide encoding a transketolase (TAL1), wherein the TAL1 is a *Saccharomyces cerevisiae* TAL1, or a TAL1 having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to a *Saccharomyces cerevisiae* TAL1.

55 Fermentation Products

A fermentation product can be any substance derived from the fermentation. The fermentation product can be, without limitation, an alcohol (e.g., arabinitol, n-butanol, isobutanol, ethanol, glycerol, methanol, ethylene glycol, 1,3-propanediol [propylene glycol], butanediol, glycerin, sorbitol, and xylitol); an alkane (e.g., pentane, hexane, heptane, octane, nonane, decane, undecane, and dodecane), a cycloalkane (e.g., cyclopentane, cyclohexane, cycloheptane, and cyclooctane), an alkene (e.g., pentene, hexene, heptene, and octene); an amino acid (e.g., aspartic acid, glutamic acid, glycine, lysine, serine, and threonine); a gas (e.g., methane, hydrogen (H₂), carbon dioxide (CO₂), and

carbon monoxide (CO)); isoprene; a ketone (e.g., acetone); an organic acid (e.g., acetic acid, acetonetic acid, adipic acid, ascorbic acid, citric acid, 2,5-diketo-D-gluconic acid, formic acid, fumaric acid, glucaric acid, gluconic acid, glucuronic acid, glutaric acid, 3-hydroxypropionic acid, itaconic acid, lactic acid, malic acid, malonic acid, oxalic acid, oxaloacetic acid, propionic acid, succinic acid, and xylonic acid); and polyketide.

In one aspect, the fermentation product is an alcohol. The term "alcohol" encompasses a substance that contains one or more hydroxyl moieties. The alcohol can be, but is not limited to, n-butanol, isobutanol, ethanol, methanol, arabinitol, butanediol, ethylene glycol, glycerin, glycerol, 1,3-propanediol, sorbitol, xylitol. See, for example, Gong et al., 1999, *Ethanol production from renewable resources*, in *Advances in Biochemical Engineering/Biotechnology*, Scheper, T., ed., Springer-Verlag Berlin Heidelberg, Germany, 65: 207-241; Silveira and Jonas, 2002, *Appl. Microbiol. Biotechnol.* 59: 400-408; Nigam and Singh, 1995, *Process Biochemistry* 30(2): 117-124; Ezeji et al., 2003, *World Journal of Microbiology and Biotechnology* 19(6): 595-603. In one embodiment, the fermentation product is ethanol.

In another aspect, the fermentation product is an alkane. The alkane may be an unbranched or a branched alkane. The alkane can be, but is not limited to, pentane, hexane, heptane, octane, nonane, decane, undecane, or dodecane.

In another aspect, the fermentation product is a cycloalkane. The cycloalkane can be, but is not limited to, cyclopentane, cyclohexane, cycloheptane, or cyclooctane.

In another aspect, the fermentation product is an alkene. The alkene may be an unbranched or a branched alkene. The alkene can be, but is not limited to, pentene, hexene, heptene, or octene. In another aspect, the fermentation product is an amino acid. The organic acid can be, but is not limited to, aspartic acid, glutamic acid, glycine, lysine, serine, or threonine. See, for example, Richard and Margaritis, 2004, *Biotechnology and Bioengineering* 87(4): 501-515.

In another aspect, the fermentation product is a gas. The gas can be, but is not limited to, methane, H₂, CO₂, or CO. See, for example, Kataoka et al., 1997, *Water Science and Technology* 36(6-7): 41-47; and Gunaseelan, 1997, *Biomass and Bioenergy* 13(1-2): 83-114.

In another aspect, the fermentation product is isoprene.

In another aspect, the fermentation product is a ketone. The term "ketone" encompasses a substance that contains one or more ketone moieties. The ketone can be, but is not limited to, acetone.

In another aspect, the fermentation product is an organic acid. The organic acid can be, but is not limited to, acetic acid, acetonetic acid, adipic acid, ascorbic acid, citric acid, 2,5-diketo-D-gluconic acid, formic acid, fumaric acid, glucaric acid, gluconic acid, glucuronic acid, glutaric acid, 3-hydroxypropionic acid, itaconic acid, lactic acid, malic acid, malonic acid, oxalic acid, propionic acid, succinic acid, or xylonic acid. See, for example, Chen and Lee, 1997, *Appl. Biochem. Biotechnol.* 63-65: 435-448.

In another aspect, the fermentation product is polyketide.

In some aspects, the fermenting organism (or processes thereof), provide higher yield of fermentation product (e.g., ethanol) when compared to the same process using an identical cell without the heterologous polynucleotide encoding the phospholipase under the same conditions (e.g., at about or after 54 hours fermentation, such as the conditions described in Example 3 or 4). In some embodiments, the process results in at least 0.25%, such as 0.5%, 0.75%,

1.0%, 1.25%, 1.5%, 1.75%, 2%, 3% or 5% higher yield of the fermentation product (e.g., ethanol).

Recovery

The fermentation product, e.g., ethanol, can optionally be recovered from the fermentation medium using any method known in the art including, but not limited to, chromatography, electrophoretic procedures, differential solubility, distillation, or extraction. For example, alcohol is separated from the fermented cellulosic material and purified by conventional methods of distillation. Ethanol with a purity of up to about 96 vol. % can be obtained, which can be used as, for example, fuel ethanol, drinking ethanol, i.e., potable neutral spirits, or industrial ethanol.

In some aspects of the methods, the fermentation product after being recovered is substantially pure. With respect to the methods herein, "substantially pure" intends a recovered preparation that contains no more than 15% impurity, wherein impurity intends compounds other than the fermentation product (e.g., ethanol). In one variation, a substantially pure preparation is provided wherein the preparation contains no more than 25% impurity, or no more than 20% impurity, or no more than 10% impurity, or no more than 5% impurity, or no more than 3% impurity, or no more than 1% impurity, or no more than 0.5% impurity.

Suitable assays to test for the production of ethanol and contaminants, and sugar consumption can be performed using methods known in the art. For example, ethanol product, as well as other organic compounds, can be analyzed by methods such as HPLC (High Performance Liquid Chromatography), GC-MS (Gas Chromatography Mass Spectroscopy) and LC-MS (Liquid Chromatography-Mass Spectroscopy) or other suitable analytical methods using routine procedures well known in the art. The release of ethanol in the fermentation broth can also be tested with the culture supernatant. Byproducts and residual sugar in the fermentation medium (e.g., glucose or xylose) can be quantified by HPLC using, for example, a refractive index detector for glucose and alcohols, and a UV detector for organic acids (Lin et al., *Biotechnol. Bioeng.* 90:775-779 (2005)), or using other suitable assay and detection methods well known in the art.

The invention may further be described in the following numbered paragraphs:

Paragraph [1]. A method of producing a fermentation product from a starch-containing or cellulosic-containing material comprising:

- (a) saccharifying the starch-containing or cellulosic-containing material; and
 - (b) fermenting the saccharified material of step (a) with a fermenting organism;
- wherein the fermenting organism comprises a heterologous polynucleotide encoding a phospholipase.

Paragraph [2]. The method of paragraph [1], wherein the phospholipase is a Phospholipase A or a Phospholipase C.

Paragraph [3]. The method of paragraph [1] or [2], wherein the phospholipase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242).

Paragraph [4]. The method of any one of paragraphs [1]-[3], wherein the heterologous polynucleotide encodes a phospholipase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than

four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). 5

Paragraph [5]. The method of any one of paragraphs [1]-[4], wherein the heterologous polynucleotide encodes a phospholipase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). 10

Paragraph [6]. The method of any one of paragraphs [1]-[5], wherein saccharification of step (a) occurs on a starch-containing material, and wherein the starch-containing material is either gelatinized or ungelatinized starch. 15

Paragraph [7]. The method of paragraph [6], comprising liquefying the starch-containing material by contacting the material with an alpha-amylase prior to saccharification. 20

Paragraph [8]. The method of paragraph [7], wherein liquefying the starch-containing material and/or saccharifying the starch-containing material is conducted in presence of exogenously added protease. 25

Paragraph [9]. The method of any one of paragraphs [1]-[8], wherein fermentation is performed under reduced nitrogen conditions (e.g., less than 1000 ppm urea or ammonium hydroxide, such as less than 750 ppm, less than 500 ppm, less than 400 ppm, less than 300 ppm, less than 250 ppm, less than 200 ppm, less than 150 ppm, less than 100 ppm, less than 75 ppm, less than 50 ppm, less than 25 ppm, or less than 10 ppm). 30

Paragraph [10]. The method of any one of paragraphs [1]-[9], wherein fermentation and saccharification are performed simultaneously in a simultaneous saccharification and fermentation (SSF). 35

Paragraph [11]. The method of any one of paragraphs [1]-[9], wherein fermentation and saccharification are performed sequentially (SHF). 40

Paragraph [12]. The method of any one of paragraphs paragraph [1]-[11], comprising recovering the fermentation product from the fermentation. 45

Paragraph [13]. The method of paragraph [12], wherein recovering the fermentation product from the from the fermentation comprises distillation. 50

Paragraph [14]. The method of any one of paragraphs [1]-[13], wherein the fermentation product is ethanol.

Paragraph [15]. The method of any one of paragraphs [1]-[14], wherein the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase. 55

Paragraph [16]. The method of paragraph [15], wherein the glucoamylase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of a *Pycnoporus glycoamylase* (e.g., a *Pycnoporus sanguineus* glucoamylase of SEQ ID NO: 229), a *Gloeophyllum* glucoamylase (e.g. a *Gloeophyllum sepiarium* of SEQ ID NO: 8), or a glucoamylase of any one of SEQ ID NOs: 102-113 (e.g., a *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103 or 104, or a *Trichoderma reesei* glucoamylase of SEQ ID NO: 230). 60

Paragraph [17]. The method of any one of paragraphs [1]-[16], wherein the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase. 65

Paragraph [18]. The method of paragraph [17], wherein the alpha-amylase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231.

Paragraph [19]. The method of any one of paragraphs [1]-[18], wherein the fermenting organism comprises a heterologous polynucleotide encoding a protease.

Paragraph [20]. The method of paragraph [19], wherein the protease has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69).

Paragraph [21]. The method of any one of paragraphs [1]-[20], wherein the fermenting organism comprises a heterologous polynucleotide encoding a trehalase.

Paragraph [22]. The method of paragraph [21], wherein the trehalase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 175-226.

Paragraph [23]. The method of any one of paragraphs [1]-[22], wherein saccharification of step occurs on a cellulosic-containing material, and wherein the cellulosic-containing material is pretreated.

Paragraph [24]. The method of paragraph [23], wherein the pretreatment is a dilute acid pretreatment.

Paragraph [25]. The method of any one of paragraphs [1]-[24], wherein saccharification occurs on a cellulosic-containing material, and wherein the enzyme composition comprises one or more enzymes selected from a cellulase, an AA9 polypeptide, a hemicellulase, a CIP, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin.

Paragraph [26]. The method of paragraph [25], wherein the cellulase is one or more enzymes selected from an endoglucanase, a cellobiohydrolase, and a beta-glucosidase.

Paragraph [27]. The method of paragraph [25] or [26], wherein the hemicellulase is one or more enzymes selected a xylanase, an acetylxylan esterase, a feruloyl esterase, an arabinofuranosidase, a xylosidase, and a glucuronidase.

Paragraph [28]. The method of any one of paragraphs [1]-[27], wherein the fermenting organism is a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell.

Paragraph [29]. The method of any one of paragraphs [1]-[28], wherein the fermenting organism is a *Saccharomyces cerevisiae* cell.

Paragraph [30]. The method of any one of paragraphs [1]-[29], wherein the method results in higher yield of fermentation product and/or reduced foam accumulation when compared to the same process using an identical cell without the heterologous polynucleotide encoding the phospholipase under the same conditions (e.g., at about or after 54 hours fermentation, such as the conditions described in Examples 3 or 4).

Paragraph [31]. The method of paragraph [30], wherein the method results in at least 0.25% (e.g., 0.5%, 0.75%, 1.0%, 1.25%, 1.5%, 1.75%, 2%, 3% or 5%) higher yield of fermentation product.

Paragraph [32]. A recombinant yeast cell comprising a heterologous polynucleotide encoding a phospholipase. 5

Paragraph [33]. The recombinant yeast cell of paragraph [32], wherein the phospholipase is a Phospholipase A or a Phospholipase C.

Paragraph [34]. The recombinant yeast cell of paragraph [32] or [33], wherein the phospholipase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). 10 15

Paragraph [35]. The recombinant yeast cell of any one of paragraphs [32]-[34], wherein the heterologous polynucleotide encodes a phospholipase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). 20 25

Paragraph [36]. The recombinant yeast cell of any one of paragraphs [32]-[35], wherein the heterologous polynucleotide encodes a phospholipase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). 30 35

Paragraph [37]. The recombinant yeast cell of any one of paragraphs [32]-[36], wherein the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase.

Paragraph [38]. The recombinant yeast cell of paragraph [37], wherein the glucoamylase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of a *Pycnoporus* glucoamylase (e.g., a *Pycnoporus sanguineus* glucoamylase of SEQ ID NO: 229), a *Gloeophyllum* glucoamylase (e.g. a *Gloeophyllum sepiarium* of SEQ ID NO: 8), or a glucoamylase of any one of SEQ ID NOs: 102-113 (e.g., a *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103 or 104, or a *Trichoderma reesei* glucoamylase of SEQ ID NO: 230). 40 45 50

Paragraph [39]. The recombinant yeast cell of any one of paragraphs [32]-[38], wherein the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase. 55

Paragraph [40]. The recombinant yeast cell of paragraph [39], wherein the alpha-amylase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231.

Paragraph [41]. The recombinant yeast cell of any one of paragraphs [32]-[40], wherein the fermenting organism comprises a heterologous polynucleotide encoding a protease.

Paragraph [42]. The recombinant yeast cell of paragraph [41], wherein the protease has a mature polypeptide

sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69).

Paragraph [43]. The recombinant yeast cell of any one of paragraphs [32]-[42], wherein the fermenting organism comprises a heterologous polynucleotide encoding a trehalase.

Paragraph [44]. The recombinant yeast cell of paragraph [43], wherein the trehalase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 175-226.

Paragraph [45]. The recombinant yeast of any one of paragraphs [32]-[44], wherein the cell is a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell.

Paragraph [46]. The recombinant yeast of paragraph [45], wherein the cell is a *Saccharomyces cerevisiae* cell.

Paragraph [47]. The recombinant cell of any one of paragraphs [32]-[46], wherein the cell is capable of higher yield of fermentation product and/or reduced foam accumulation when compared to fermentation using the same process and an identical cell without the heterologous polynucleotide encoding the phospholipase under the same conditions (e.g., at about or after 54 hours fermentation, such as the conditions described in Examples 3 or 4).

Paragraph [48]. The recombinant cell of paragraph [47], wherein the cell is capable of at least 0.25% (e.g., 0.5%, 0.75%, 1.0%, 1.25%, 1.5%, 1.75%, 2%, 3% or 5%) higher yield of fermentation product.

The invention described and claimed herein is not to be limited in scope by the specific aspects herein disclosed, since these aspects are intended as illustrations of several aspects of the invention. Any equivalent aspects are intended to be within the scope of this invention. Indeed, various modifications of the invention in addition to those shown and described herein will become apparent to those skilled in the art from the foregoing description. Such modifications are also intended to fall within the scope of the appended claims. In the case of conflict, the present disclosure including definitions will control. All references are specifically incorporated by reference for that which is described.

The following examples are offered to illustrate certain aspects of the present invention, but not in any way intended to limit the scope of the invention as claimed.

EXAMPLES

Materials and Methods

Chemicals used as buffers and substrates were commercial products of at least reagent grade.

ETHANOL RED® ("ER"): *Saccharomyces cerevisiae* yeast available from Fermentis/Lesaffre, USA.

YPD+clonNAT plates were composed of 10 g of yeast extract, 20 g of peptone, 20 g bacto agar, and deionized water to 960 mL, followed by autoclave treatment. 40 mL sterile 50% glucose and 1 mL clonNAT stock solution was added, followed by mixing and pouring. 65

clonNAT stock solution was composed of 2 g nourseothricin sulfate and deionized water to 20 ml.

Example 1: Construction of Yeast Strains Expressing a Heterologous Phospholipase

This example describes the construction of yeast cells containing a heterologous phospholipase under control of a *S. cerevisiae* TDH3, TEF2, PGK1, ADH1 or RPL18B promoter. Three pieces of DNA containing the promoter, gene and terminator were designed to allow for homologous recombination between the three DNA fragments and into the X-3 locus of the yeast MHCT-484 (WO2018/222990). The resulting strain has one promoter-containing fragment (left fragment), one gene-containing fragment (middle fragment) and one PRM9 terminator fragment (right fragment) integrated into the *S. cerevisiae* genome at the X-3 locus. Construction of the Promoter Containing Fragments (Left Fragments)

Synthetic linear uncloned DNA containing 60 bp homology to the X-3 site, *S. cerevisiae* promoter TDH3, TEF2, PGK1, ADH1 or RPL18B (SEQ ID NOs: 1, 2, 4, 5, and 6, respectively) and coding sequence for the *S. cerevisiae* MF1 α signal peptide (SEQ ID NO: 7) were synthesized by Thermo Fisher Scientific (Waltham, MA). To generate additional linear DNA for transformation into yeast, each of the five linear DNAs containing the left cassette above was PCR amplified.

Construction of the Terminator Containing Fragment (Right Fragment)

Synthetic linear uncloned DNA containing the *S. cerevisiae* PRM9 terminator (SEQ ID NO: 243) and 300 bp homology to the X-3 site was synthesized by Thermo Fisher Scientific.

Construction of the Gene Containing Fragment (Middle Fragment)

Synthetic linear uncloned DNA containing coding sequence for the *S. cerevisiae* MF1 α signal peptide, coding sequence for the mature polypeptide and 50 bp PRM9 terminator was synthesized by Thermo Fisher Scientific.

Integration of the Left, Middle and Right-Hand Fragments to Generate Yeast Strains with a Heterologous Phospholipase

The yeast MHCT-484 (WO2018/222990) was transformed with the left, middle and right integration fragments described above. In each transformation pool a fixed left fragment and right fragment were used as well as a fixed middle fragment containing the phospholipase gene with 100 ng of each fragment. To aid homologous recombination of the left, middle and right fragments at the genomic X-3 sites a plasmid containing Cas9 and guide RNA specific to X-3 (pMcTs442) was also used in the transformation. These four components were transformed into the *S. cerevisiae* strain MHCT-484 following a yeast electroporation protocol. Transformants were selected on YPD+cloNAT to select for transformants that contain the CRISPR/Cas9 plasmid pMcTs442. Transformants were picked using a Q-pix Colony Picking System (Molecular Devices; San Jose, CA) to inoculate one well of 96-well plate containing YPD+cloNAT media. The plates were grown for 2 days then glycerol was added to 20% final concentration and the plates were stored at -80° C. until needed. Integration of specific phospholipase construct was verified by PCR with locus specific primers and subsequent sequencing. The strains generated in this example are shown in Table 5.

TABLE 5

Strain Name	Promoter	SEQ ID NO: (mature polypeptide)	Donor Organism (catalytic domain)
5 HP21-B03	pADH1v1	236	<i>Talaromyces leycettanus</i>
HP21-C03	pADH1v1	236	<i>Talaromyces leycettanus</i>
HP21-D04	pTEF2	236	<i>Talaromyces leycettanus</i>
HP21-D05	pTDH3	236	<i>Talaromyces leycettanus</i>
HP21-E01	pPGK1	236	<i>Talaromyces leycettanus</i>
10 HP21-F01	pPGK1	236	<i>Talaromyces leycettanus</i>
HP21-G06	pRPL18B	236	<i>Talaromyces leycettanus</i>
HP21-A02	pADH1v1	237	<i>Penicillium emersonii</i>
HP21-F04	pTDH3	237	<i>Penicillium emersonii</i>
HP21-F05	pRPL18B	237	<i>Penicillium emersonii</i>
HP21-A04	pTEF2	240	<i>Kionochaeta</i> sp.
15 HP21-B01	pPGK1	240	<i>Kionochaeta</i> sp.
HP21-C05	pTDH3	240	<i>Kionochaeta</i> sp.
HP21-G02	pADH1v1	240	<i>Kionochaeta</i> sp.
HP21-B05	pTDH3	241	<i>Mariannaea pinicola</i>
HP21-B06	pRPL18B	241	<i>Mariannaea pinicola</i>
HP21-C06	pRPL18B	241	<i>Mariannaea pinicola</i>
HP21-F02	pADH1v1	241	<i>Mariannaea pinicola</i>
20 HP21-H03	pTEF2	241	<i>Mariannaea pinicola</i>
HP21-G04	pTDH3	239	<i>Pseudomonas</i> sp. 62186
HP21-G05	pRPL18B	239	<i>Pseudomonas</i> sp. 62186
HP21-H05	pRPL18B	239	<i>Pseudomonas</i> sp. 62186
HP21-A03	pADH1v1	238	<i>Bacillus thuringiensis</i>
HP21-B04	pTEF2	238	<i>Bacillus thuringiensis</i>
25 HP21-C01	pPGK1	238	<i>Bacillus thuringiensis</i>
HP21-C04	pTEF2	238	<i>Bacillus thuringiensis</i>
HP21-D01	pPGK1	238	<i>Bacillus thuringiensis</i>
HP21-D06	pRPL18B	238	<i>Bacillus thuringiensis</i>
HP21-E06	pRPL18B	238	<i>Bacillus thuringiensis</i>
HP21-F06	pRPL18B	238	<i>Bacillus thuringiensis</i>
30 HP21-H02	pADH1v1	238	<i>Bacillus thuringiensis</i>
HP21-D03	pADH1v1	235	<i>Thermomyces lanuginosus</i>
HP21-E04	pTEF2	235	<i>Thermomyces lanuginosus</i>
HP21-E05	pTDH3	235	<i>Thermomyces lanuginosus</i>
HP21-G01	pPGK1	235	<i>Thermomyces lanuginosus</i>
HP21-H06	pRPL18B	235	<i>Thermomyces lanuginosus</i>
35 HP21-A01	pPGK1	242	<i>Fictibacillus macauensis</i>
HP21-A05	pTDH3	242	<i>Fictibacillus macauensis</i>
HP21-A06	pRPL18B	242	<i>Fictibacillus macauensis</i>
HP21-F03	pTEF2	242	<i>Fictibacillus macauensis</i>
HP21-G03	pTEF2	242	<i>Fictibacillus macauensis</i>
40 HP21-H04	pTDH3	242	<i>Fictibacillus macauensis</i>

Example 2: Activity Assay and Small-Scale Fermentation of Yeast Strains Expressing Phospholipase

Preparation of Yeast Cells for Activity Assays

Yeast strains were cultivated for 24 hours in standard YPD media containing 2% glucose. After the cultivation, samples were centrifuged, and the supernatants assayed for enzyme activity as described below.

PLA (Phospholipase A) Activity Assay

PLA activity was detected by using the EnzChek® PLA1 Kit from Invitrogen (Carlsbad, CA). The EnzChek® Direct Phospholipase A Assay Kit measures phosphatidylcholine-specific phospholipase C (PC-PLC) activity. PLA1 hydrolyzes the ester linkage of phospholipids and fatty acids. The EnzChek® Phospholipase A1 substrate (PED-A1) is a dye-labeled glycerophosphoethanolamines with BODIPY® FL dye-labeled acyl chain. PLA assay measures the release of the dye at a fluorescence emission at 515 nm. Reaction conditions are described in Table 6.

Initial Preparation of Solutions for PLA1 Activity Assay:

1. Reaction Buffer: 100 mM MOPS+0.5 mM Zn pH 7
2. PLA substrate:
 - a. 40 μ l DMSO (Comp E) to one vial PLA sub (Comp A)
 - b. Protected from light
 - c. Sufficient volume for 100 rxns

87

3. 500 U/mL stock of positive control PLA (Comp D)
 - a. Dissolved Comp D vial in 100 uL of reaction buffer
4. 10 mM DOPC (Comp B)
 - a. Dissolved Comp B vial in 100 ul of ETOH
5. 10 mM DOPG (Comp C)
 - a. Dissolved Comp C vial in 100 ul of ETOH

Final Preparation of Solutions and Samples for PLA1 Activity Assay:

1. 10 U/mL positive control made by diluting 500 U/mL stock positive control in reaction buffer
 - a. Example: added 20 ul of 500 U/mL positive control to 980 mL of Reaction buffer
2. Serial dilutions of positive control to obtain 8-point standard curve with initial concentration beginning at 10 U/ml
 - a. Final positive control concentration two-fold lower (5 U/ml) when substrate added
3. When necessary, samples diluted in reaction buffer
4. Lipid Mix: 30 ul 10 mM DOPC, 30 ul 10 mM DOPG, 30 ul 1 mM PLA substrate
5. PLA substrate
 - a. 50 ul lipid mixed slowly to 5 mL reaction buffer in a smaller beaker containing a stir bar.
 - b. Stirred for ~2-5 minutes

Assay Protocol for PLA1 Activity Assay:

1. 50 ul of 8-point standard curve to columns 10, 11, 12
 - a. last row left blank for buffer
2. Added 50 ul of samples to remaining wells
3. Added 50 ul of PLA-lipid substrate to wells containing samples and controls. Mixed well without introducing bubbles.
4. Read a T0 at 505EX/515EM (450EX/515EM is ok as well)
5. Covered plate and incubate protected from light
6. Read after 5 hours of incubation

TABLE 6

PLA1 Activity Assay Condition	
Amount of yeast supernatant	50 μ L
Amount of PLA-lipid substrate	50 μ L
Substrate	PLA1 substrate from Kit
Buffer	100 mM MOPS + 0.5 mM Zn
pH	7.0 $\pm$ 0.05
Incubation temperature	22° C. (room Temperature)
Reaction time	5 hrs
Wavelength	505EX/515EM

PLC (Phospholipase C) Activity Assay

PLC activity was detected by using EnzChek® PLC Kit from Invitrogen. The EnzChek® Phospholipase A Assay Kit measures phosphatidylcholine-specific phospholipase C (PC-PLC) activity by measuring the amount of starch degraded through enzymatic hydrolysis of starch. The assay uses a proprietary substrate (glycero-phosphoethanolamine with a dye-labeled sn-2 acyl chain) to detect PLC activity. Substrate cleavage by PLC releases the dye-labeled diacylglycerol, which produces a fluorescence signal that can be measure at 516 nm emission. Reaction conditions are described in Table 7.

Initial Preparation of Solutions for PLC Activity Assay:

1. Reaction Buffer: 100 mM NaAc+0.5 mM Zn pH 5
2. 200 $\times$ stock of PLC substrate
 - a. 100 ul DMSO (Comp B) added to one vial PLC sub
 - b. Protected from light
 - c. Sufficient volume for 125 rxns

88

3. 40 U/mL stock of positive control PC-PLC (Comp E)
 - a. Dissolved Comp E vial in 200 uL of reaction buffer
- Final Secondary Preparation of Solutions and Samples for PLC Activity Assay

1. 1 U/mL positive control made by diluting 40 U/mL stock positive control 40-fold
 - a. Example: added 25 ul of 40 U/mL positive control to 0.975 mL of Reaction buffer
2. Serial dilutions of positive control to obtain 8-point standard curve with initial concentration beginning at 0.125 U/ml
 - a. Final positive control concentration two-fold lower (0.0625 U/ml) when substrate is added
3. When necessary, samples diluted in reaction buffer
4. PLC substrate: Added 40 ul of lecithin (Comp D) and 100 ul of PLC Substrate 200 $\times$ stock solution (prepared in earlier step) to 9.86 mL of reaction buffer.

Assay Protocol for PLC Activity Assay:

1. 75 ul of 8-point standard curve to columns 10, 11, 12
 - a. last row left blank for buffer
2. Added 75 ul of samples to remaining wells
3. Added 75 ul of PLC substrate to wells containing samples and controls. Mixed well without introducing bubbles.
4. Read a T0 at 509EX/516EM (490EX/520EM is ok as well)
5. Covered plate and incubate protected from light
6. Read after 5 hours

TABLE 7

PLC Activity Assay Condition	
Amount of yeast supernatant	75 μ L
Amount of PLA-lipid substrate	75 μ L
Substrate	PLC substrate from Kit
Buffer	100 mM NaAc + 0.5 mM Zn
pH	5.0 $\pm$ 0.05
Incubation temperature	22° C. (room Temperature)
Reaction time	5 hrs
Wavelength	509EX/516EM

Preparation of Yeast Culture for Microtiter Plate Fermentations

Simultaneous saccharification and fermentation (SSF) was performed via mini-scale fermentations using industrial corn mash (Avantec® Amp; Novozymes A/S). Yeast strains were cultivated overnight in YPD media with 6% glucose for 24 hours at 30° C. and 300 rpm. The corn mash was supplemented with 250 ppm of urea and dosed with 0.45 AGU/g-DS of an exogenous glucoamylase enzyme product (Spirizyme® Excel; Novozymes A/S). Approximately 0.6 mL of corn mash was dispensed per well to 96 well microtiter plates, followed by the addition of approximately 10⁸ yeast cells/g of corn mash from the overnight culture. Plates were incubated at 32° C. without shaking. Fermentation was stopped by the addition of 100 μ L of 8% H₂SO₄, followed by centrifugation at 3000 rpm for 10 min. The supernatant was analyzed for ethanol using HPLC. Fermentation reaction conditions are summarized in Table 8. SSF and activity assay results of yeast strains expressing a phospholipase are in Table 9.

TABLE 8

Microtiter plate fermentation reaction conditions	
Substrate	Avantec® Amp corn mash
Yeast pitch	10 ⁸ cells/g corn mash
Supplementary urea	250 ppm

TABLE 8-continued

Microtiter plate fermentation reaction conditions	
Exogenous glucoamylase product dose	0.15 AGU/g-DS
pH	5.0 ± 0.05
Incubation temperature	32° C.
Reaction time	48 hours

TABLE 9

Phospholipase yeast strains enzyme activity measurements and ethanol titers from SSF					
Strain ID	Promoter	SEQ ID NO: (mature polypeptide)	PLC activity units	PLA1 activity units	Mean Ethanol (g/L)
MHCT-484	none	none	99	0	107.31
HP21-D04	TEF2	236	776	23	110.67
HP21-G06	RPL18B	236	793	0	110.6105
HP21-B03	ADH1v1	236	863	43	110.2833
HP21-D05	TDH3	236	688	29	110.0453
HP21-C03	ADH1v1	236	878	47	108.885
HP21-E01	PGK1	236	710	11	108.2008
HP21-F01	PGK1	236	680	0	106.1183
HP21-F04	TDH3	237	207	0	110.67
HP21-F05	RPL18B	237	174	0	109.1528
HP21-A02	ADH1v1	237	273	0	109.0933
HP21-B01	PGK1	240	452	0	94.0695
HP21-A04	TEF2	240	451	0	83.27025
HP21-G02	ADH1v1	240	410	0	82.19925
HP21-C05	TDH3	240	426	0	81.36625
HP21-C06	RPL18B	241	236	0	112.455
HP21-B06	RPL18B	241	201	0	107.8735
HP21-B05	TDH3	241	356	0	100.436
HP21-H03	TEF2	241	275	0	99.3055
HP21-F02	ADH1v1	241	364	0	97.58
HP21-G04	TDH3	239	85	0	107.1893
HP21-G05	RPL18B	239	251	0	84.7875
HP21-H05	RPL18B	239	220	0	84.609
HP21-E06	RPL18B	238	156	0	109.599
HP21-B04	TEF2	238	152	0	108.6768
HP21-D01	PGK1	238	99	0	108.0223
HP21-C01	PGK1	238	116	0	107.5463
HP21-F06	RPL18B	238	142	0	105.077
HP21-A03	ADH1v1	238	99	0	104.5118
HP21-H02	ADH1v1	238	93	0	102.102
HP21-C04	TEF2	238	146	0	101.9235
HP21-D06	RPL18B	238	141	0	82.25875
HP21-E05	TDH3	235	846	31	110.5808
HP21-H06	RPL18B	235	685	56	109.0635
HP21-E04	TEF2	235	500	139	100.5253
HP21-G01	PGK1	235	428	123	96.06275
HP21-D03	ADH1v1	235	462	142	88.08975
HP21-F03	TEF2	242	115	0	108.171
HP21-A05	TDH3	242	126	0	108.0818
HP21-A06	RPL18B	242	135	0	106.2373
HP21-G03	TEF2	242	113	0	105.791
HP21-H04	TDH3	242	102	0	104.6903
HP21-A01	PGK1	242	91	0	104.4523

Example 3: Fermentation of Yeast Strains Expressing Phospholipase

The purpose of this experiment was to determine if expressing phospholipase in yeast would enhance ethanol yields at the end of fermentation. Yeast strains (HP21-H06, HP21-D05, HP21-G12, HP21-A08, HP21-F05 and HP21-F04) were propagated overnight, and then used to dose a fermentation of industrially liquified mash. The fermented mash was sampled at the end of fermentation and the supernatant was filtered and analyzed by HPLC to determine the final ethanol titers and residual sugars.

Yeast Propagation

Yeast strains were received as glycerol stocks and were kept frozen at -80° C. until used. The yeast strains were propagated according to the following procedure:

1. Cryo-vials of yeast were removed from the -80° C. freezer.
2. For each sample, 2x50 mL sterile Erlenmeyer flasks were obtained and labeled.
3. The flasks were placed into a sterile hood along with 6% YPD media, pipettes and tips, and cryovials.
4. Using sterile technique, ~50 mL of 6% YPD media was poured into each bottle.
5. 150 uL of the appropriate yeast sample was added to each flask using a sterile 200 uL pipette and sterile tips.
6. The flasks were capped after each addition to maintain a sterile solution.
7. The flasks were placed into a 32° C. shaking incubator and mixed at ~150 rpm overnight.

Cell Counting

The propagation slurries described above were then tested to determine the number of yeast cells present for dosing, according to the following procedure:

1. The propagated yeast samples were removed from the incubator.
2. The 50 mL samples were poured into 50 mL flip top falcon tubes.
3. The samples were centrifuged 3,500 rpm for 7 min.
4. The supernatant was discarded into the old 50 mL flasks to be autoclaved.
5. 10 mL of deionized water was added to each tube.
6. The tubes were vortex mixed to suspend the pellet.
7. The replicate samples were combined into a single 50 mL falcon tube.
8. The samples were centrifuged 3,500 rpm for 7 min.
9. The supernatant was discarded into the original 50 mL flasks to be autoclaved.
10. 10 mL of deionized water was added to each remaining tube.
11. The tubes were vortex mixed to suspend the pellet.
 - a. This slurry was saved and will be used to dose into the fermentations.
12. The yeast slurries were diluted 100x into 15 mL falcon tubes by adding 100 uL to 9.9 mL of deionized water.
13. The samples subjected to cell count using a NucleoCounter® YC-100 (Chemometec A/S) as follows:
 - a. 50 uL of the 100x slurry dilution was added to 450 uL of Lysis buffer.
 - b. The samples were vortexed and allowed to sit ~5 min for the cell lysis to occur.
 - c. A nucleo-cassette was used to sample the lysed solution, then placed into the NucleoCounter® YC-100.
 - d. The NucleoCounter® YC-100 analyzed the slurry and produced a cell count number.
 - e. The resulting number is multiplied by 1000 to get the final cell counts for use in the spreadsheet.
14. The yeasts used in this experiment, and the yeast counts that were obtained according to the procedure above are displayed in the Table 10 below.

TABLE 10

Strain ID	YCL-1	YCL-2	Average total cells
MHCT-484	1.26E+09	1.28E+09	1.27E+09
HP21-H06	1.23E+09	1.16E+09	1.19E+09
HP21-D05	1.38E+09	1.43E+09	1.40E+09

TABLE 10-continued

Strain ID	YCL-1	YCL-2	Average total cells
HP21-G12	1.14E+09	1.12E+09	1.13E+09
HP21-A08	1.38E+09	1.29E+09	1.33E+09
HP21-F05	1.28E+09	1.28E+09	1.28E+09
HP21-F04	1.41E+09	1.24E+09	1.33E+09

15. The calculation below was used to determine the number of mL of yeast slurry to add to each tube.

$$\frac{10,000,000 \text{ cells}}{\text{mL mash}} \times \frac{1 \text{ mL mash}}{1.15 \text{ g mash}} \times \text{g mash} \times \frac{\text{mL}}{\text{X cells}} \times \frac{1000 \text{ } \mu\text{L}}{\text{mL}} = \text{ } \mu\text{L yeast to add}$$

Fermentation

Fermentations of industrially liquified mash were conducted according to the following procedure:

1. Industrially liquified mash was acquired and stored frozen for future analysis. The mash used in this experiment was from a plant using Avantec® Amp and running a hydro-heater or jet cooker during liquefaction stage.
2. Two liters of the mash was thawed for approximately 2 hours prior to starting this study.
3. The mash was adjusted to 500 ppm urea and 3 ppm penicillin using stock solutions of 200 g/L urea and 1 g/L penicillin.
4. The mash was adjusted to pH 5 using 40% v/v H₂SO₄ and the dry solids content were measured on a Mettler-Toledo moisture balance.
5. 50.0+/-0.05 g of prepared mash was weighed into 250 mL media bottles.
6. The bottles were capped and stored in a refrigerator overnight.
7. The bottles were dosed with enzymes and yeast in the morning using a repeater pipette.
8. The conditions tested include Achieve® glucoamylase (0.42 AGU/gDS), trehalase (2 µg/gDS), and yeast dosed to approximately 7,000,000 cells per fermentation.
9. The bottles were capped with hole drilled caps. The holes were covered with labeling tape. A 0.24-gauge syringe needle was used to pierce a uniform vent hole to allow for CO₂ gas to escape during fermentation.
10. The bottles were placed into a 32° C. air shaker set to 150 rpm to begin fermentation.
11. Fermentation was run for 54 hours.

Sampling of Fermentations

1. After 52 hours of fermentation, 5 g of fermented mash were pipetted into a 15 mL conical falcon tube with 5 mL pipette equipped with a cut tip.
2. 50 µL of 40% H₂SO₄ were added to each of the tubes to stop fermentation and enzymatic hydrolysis.
3. The tubes were briefly mixed with a vortex mixer and centrifuged at 3,500 rpm for 7 minutes.
4. The supernatant was poured into a syringe equipped with a 0.45 µm filter and the plunger of the syringe was used to push the sample through the filter into prenumbered HPLC vials.
5. The HPLC vials were capped and the samples were submitted to HPLC for analysis of the ethanol and sugars produced during fermentation. The HPLC setup was as shown in Table 11.

TABLE 11

HPLC system	Agilent's 1100/1200 series with Chem station software Degasser Quaternary Pump Auto-Sampler Column Compartment/w Heater Refractive Index Detector (RI)
Column	Bio-Rad HPX- 87H Ion Exclusion Column 300 mm × 7.8 mm parts# 125-0140 Bio-Rad guard cartridge cation H parts# 125-0129, Holder parts# 125-0131
Method	0.005M H ₂ SO ₄ mobile phase Flow rate of 0.6 ml/min Column temperature - 65° C. RI detector temperature - 55° C.

The method quantifies analytes using calibration standards for dextrins (DP4+), maltotriose, maltose, glucose, fructose, acetic acid, lactic acid, glycerol and ethanol. A 4-point calibration including the origin was used.

Ethanol improvement for tested strains is shown in FIG. 1. Strains HP21-F05 and HP21-F04 show approximately 1.1% improvement in ethanol over control strain MHCT-484, while strain HP21-A08 showed approximately 0.5% improvement in ethanol titer of the control.

Example 4: Fermentation of Yeast Strains Expressing Phospholipase at Varying Urea Levels

The purpose of this experiment was to examine the effect of nitrogen loading (urea level) on ethanol yields of phospholipase-expressing yeasts. Yeast strains (HP21-H08, HP21-F04 and HP21-F05) were propagated overnight, and then used to dose a fermentation of industrially liquified mash. The fermented mash was sampled at the end of fermentation and the supernatant was filtered and run on HPLC to determine the final ethanol titers and residual sugars.

Yeast Propagation

Yeast strains were propagated as described above in Example 3.

Cell Counting

The propagation slurries described above were then tested to determine the number of yeast cells present for dosing according to the procedure described above in Example 3. The yeasts used in this experiment, and the corresponding yeast counts are displayed in the Table 12.

TABLE 12

Strain ID	YCL-1	YCL-2	Average total cells
MHCT-484	1.48E+09	1.44E+09	1.46E+09
HP21-A08	1.24E+09	1.24E+09	1.24E+09
HP21-F05	1.50E+09	1.48E+09	1.49E+09
HP21-F04	1.37E+09	1.40E+09	1.38E+09

Fermentation

Fermentations of industrially liquified mash were conducted according to the following procedure:

1. Industrially liquified mash was acquired and stored frozen for future analysis. The two mashses used in this experiment were from a plant using Avantec® Amp (Novozymes A/S) and a plant running another product Liquezyme® Pro (Novozymes A/S). Both mashses were from plants running a hydro-heater or jet cooker during liquefaction stage.

2. Two liters of each mash was thawed for approximately 2 hours prior to starting this study.
3. The mash was dosed with 3 ppm penicillin using a stock solution 1 g/L penicillin. Urea was added to the fermentation during enzyme dosing using stock solutions of 100 and 20 g/L.
4. The mash was adjusted to pH 5 using 40% v/v H₂SO₄ and the dry solids content were measured on a Mettler-Toledo moisture balance.
5. ~5 g of the prepared mash was added to preweighed 15 mL tubes
6. The tubes were reweighed and the weight of mash was used in calculating enzymes, yeast and urea dosing.

7. The tubes were capped and stored in a refrigerator overnight.
8. The tubes were dosed with enzymes, yeast and urea in the morning using a Biomek FX liquid handling robot.
9. The conditions tested in this experiment are displayed in Table 13 below. Yeast dose in this experiment was 7,000,000 cells per ferm.
10. The tubes were capped with holes to allow released CO₂ from fermentation.
11. Tubes were placed into a 32° C. incubator to begin fermentation.
12. Tubes were mixed with a vortex mixer twice a day for 52 hours.

TABLE 13

Treatment	#	Yeast#	Glucoamylase	Dose	Units	Trehalase	Dose	Units	PLC	Dose	Units	Urea	Dose	Units
MHCT-484-Control	1	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	0	ppm
MHCT-484-5 ug PLC	2	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	5	ug/g DS	Urea	0	ppm
MHCT-484-20 ug PLC	3	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	20	ug/g DS	Urea	0	ppm
HP21-A08	4	2	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	0	ppm
HP21-F05	5	3	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	0	ppm
HP21-F04	6	4	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	0	ppm
MHCT-484-Control	7	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	150	ppm
MHCT-484-5 ug PLC	8	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	5	ug/g DS	Urea	150	ppm
MHCT-484-20 ug PLC	9	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	20	ug/g DS	Urea	150	ppm
HP21-A08	10	2	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	150	ppm
HP21-F05	11	3	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	150	ppm
HP21-F04	12	4	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	150	ppm
MHCT-484-Control	13	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	500	ppm
MHCT-484-5 ug PLC	14	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	5	ug/g DS	Urea	500	ppm
MHCT-484-20 ug PLC	15	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	20	ug/g DS	Urea	500	ppm
HP21-A08	16	2	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	500	ppm
HP21-F05	17	3	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	500	ppm
HP21-F04	18	4	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	500	ppm
MHCT-484-Control	19	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	0	ppm
MHCT-484-5 ug PLC	20	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	5	ug/g DS	Urea	0	ppm
MHCT-484-20 ug PLC	21	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	20	ug/g DS	Urea	0	ppm
HP21-A08	22	2	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	0	ppm
HP21-F05	23	3	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	0	ppm
HP21-F04	24	4	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	0	ppm
MHCT-484-Control	25	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	300	ppm
MHCT-484-5 ug PLC	26	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	5	ug/g DS	Urea	300	ppm
MHCT-484-20 ug PLC	27	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	20	ug/g DS	Urea	300	ppm
HP21-A08	28	2	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	300	ppm
HP21-F05	29	3	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	300	ppm
HP21-F04	30	4	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	300	ppm
MHCT-484-Control	31	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	1000	ppm
MHCT-484-5 ug PLC	32	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	5	ug/g DS	Urea	1000	ppm
MHCT-484-20 ug PLC	33	1	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	20	ug/g DS	Urea	1000	ppm
HP21-A08	34	2	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	1000	ppm
HP21-F05	35	3	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	1000	ppm
HP21-F04	36	4	Achieve	0.42	AGU/gDS	Trehelase	1	ug/gDS	<i>P. emersonii</i>	0	ug/g DS	Urea	1000	ppm

Sampling of Fermentations

1. After 52 hours of fermentation, 50 μ L of 40% H_2SO_4 were added to each of the tubes to stop fermentation and enzymatic hydrolysis.
2. The tubes were briefly mixed with a vortex mixer and centrifuged at 3,500 rpm for 7 minutes.
3. The tubes were placed onto a Biomek liquid handler which was used to pipette 200 μ L of supernatant from the tubes into a 96 well 0.22 μ m filter plate.
4. The filter plate was placed on top of a round bottom 96 well polypropylene plate.
5. The filter and plate were placed into a floor centrifuge with bioseal caps on the buckets and spun at 3,500 rpm for 7 minutes or until liquid had passed through all the wells of the 96 well filter plate.
6. The plates were removed from the centrifuge and the round bottom plate with the filtered sample was heat sealed and submitted to HPLC for analysis of the ethanol and sugars produced during fermentation. The HPLC setup was as shown in Table 14.

TABLE 14

HPLC system	Agilent's 1100/1200 series with Chem station software Degasser Quaternary Pump Auto-Sampler Column Compartment/w Heater Refractive Index Detector (RI)
Column	Bio-Rad HPX- 87H Ion Exclusion Column 300 mm $\times$ 7.8 mm parts# 125-0140 Bio-Rad guard cartridge cation H parts# 125-0129, Holder parts# 125-0131
Method	0.005M H_2SO_4 mobile phase Flow rate of 0.6 ml/min Column temperature - 65° C. RI detector temperature - 55° C.

The method quantifies analytes using calibration standards for dextrans (DP4+), maltotriose, maltose, glucose, fructose, acetic acid, lactic acid, glycerol and ethanol. A 4-point calibration including the origin was used. Normalized mean ethanol values after 54 hours of fermentation of AMP mash at 0 and 150 ppm urea are shown in FIG. 2.

All tested strains expressing phospholipase showed greater increases in ethanol titer with no urea loading. Normalized mean ethanol values after 54 hours of fermentation of non-AMP mash at 0 and 150 ppm urea are shown in FIG. 3. All tested strains expressing phospholipase showed greater increases in ethanol titer with no urea loading.

Example 5: Construction of Additional Yeast Strains Expressing a Heterologous Phospholipase

This example describes the construction of yeast cells containing the remaining heterologous phospholipases of Table 1 under the control of an *S. cerevisiae* PGK1 promoter. Three pieces of DNA containing the promoter, gene and terminator were designed to allow for homologous recombination between the four DNA fragments and into the XII-2 locus of the strain Meli797 (a derivative of MBG5012 expressing both alpha-amylase and glucoamylase; WO2019/161227). The resulting strain has one promoter-containing fragment (left fragment), one gene-containing fragment (middle fragment) and one PRM9 terminator fragment (right fragment) integrated into the *S. cerevisiae* genome at the XII-2 locus.

Construction of the Promoter-Containing Fragment (Left Fragment)

Linear DNA containing 500 bp homology to the XII-2 site and the *S. cerevisiae* pPGK1 promoter was PCR amplified from HP27 plasmid DNA with primers 1229945 (5'-TCTTT TCGCG CCCTG GAAAG G-3'; SEQ ID NO: 434) and 1227122 (5'-TGTTT TATAT TTGTT GTAAA AAGTA GATAA TTA CTTC TATGA TCTG-3'; SEQ ID NO: 435). Fifty pmoles each of forward and reverse primer was used in a PCR reaction containing 5 ng of plasmid DNA as template, 0.1 mM each dATP, dGTP, dCTP, dTTP, 1 $\times$ Phusion HF Buffer (Thermo Fisher Scientific), and 2 units Phusion Hot Start DNA polymerase in a final volume of 50 μ L. The PCR was performed in a T100™ Thermal Cycler (Bio-Rad Laboratories, Inc.) programmed for one cycle at 98° C. for 30 seconds followed by 32 cycles each at 98° C. for 10 seconds, 59° C. for 20 seconds, and 72° C. for 40 seconds with a final extension at 72° C. for 10 minutes. Following thermocycling, the PCR reaction products gel isolated and cleaned up using the NucleoSpin Gel and PCR clean-up kit (Machery-Nagel).

The linear DNA containing 243 bp homology to the *S. cerevisiae* pPGK1 promoter (SEQ ID NO: 4) and the MF1 α signal peptide (SEQ ID NO: 7) was PCR amplified from DNA synthesized by GeneArt with primers 1229946 (5'-GTGAC AACAA CAGCC TGTTTC TC-3'; SEQ ID NO: 436) and 1222995 (5'-AGCTA ATGCG GAGGA TGCTG C-3'; SEQ ID NO: 437). Fifty pmoles each of forward and reverse primer was used in a PCR reaction containing 5 ng of plasmid DNA as template, 0.1 mM each dATP, dGTP, dCTP, dTTP, 1 $\times$ Phusion HF Buffer (Thermo Fisher Scientific), and 2 units Phusion Hot Start DNA polymerase in a final volume of 50 μ L. The PCR was performed in a T100™ Thermal Cycler (Bio-Rad Laboratories, Inc.) programmed for one cycle at 98° C. for 30 seconds followed by 32 cycles each at 98° C. for 10 seconds, 59° C. for 20 seconds, and 72° C. for 20 seconds with a final extension at 72° C. for 5 minutes. Following thermocycling, the PCR reaction products gel isolated and cleaned up using the NucleoSpin Gel and PCR clean-up kit (Machery-Nagel).

Construction of the Terminator-Containing Fragment (Right Fragment)

The DNA containing 250 bp of the PRM9 terminator and 500 bp of the XII-2 3'-end homology was PCR amplified from TH12 plasmid DNA (GeneArt) with primers 1221473 (5'-ACAGAAGACG GGAGA CACTA GC-3'; SEQ ID NO: 438) and 1229949 (5'-GGGGT CGCAA CTTT CCC-3'; SEQ ID NO: 439). Fifty pmoles each of forward and reverse primer was used in a PCR reaction containing 5 ng of plasmid DNA as template, 0.1 mM each dATP, dGTP, dCTP, dTTP, 1 $\times$ Phusion HF Buffer (Thermo Fisher Scientific), and 2 units Phusion Hot Start DNA polymerase in a final volume of 50 μ L. The PCR was performed in a T100™ Thermal Cycler (Bio-Rad Laboratories, Inc.) programmed for one cycle at 98° C. for 30 seconds followed by 32 cycles each at 98° C. for 10 seconds, 59° C. for 20 seconds, and 72° C. for 40 seconds with a final extension at 72° C. for 10 minutes. Following thermocycling, the PCR reaction products gel isolated and cleaned up using the NucleoSpin Gel and PCR clean-up kit (Machery-Nagel).

Construction of the Phospholipase-Containing Fragments (Middle Fragments)

Synthetic linear uncloned DNA containing the MF1 α signal peptide, phospholipase gene and 50 bp of the PRM9 terminator were obtained from Geneart or Twist Bioscience.

Integration of the Left, Middle and Right-Hand Fragments to Generate Yeast Strains with a Heterologous Phospholipase

The yeast MeJi797 was transformed with the left, middle and right integration fragments described above. In each transformation pool a fixed left fragment and right fragment with 100 ng of each fragment was used. The middle fragment consisted of a signal peptide and phospholipase gene with ~100 ng of each fragment (700 ng total). To aid homologous recombination of the left, middle and right fragments at the genomic XII-2 sites a plasmid containing MAD7 and guide RNA specific to XII-2 (pMIBa638) was also used in the transformation. These four components were transformed into the *S. cerevisiae* strain MeJi797 following a yeast electroporation protocol. Transformants were selected on YPD+cloNAT to select for transformants that contain the Mad7 plasmid pMIBa638. Transformants were picked using a Q-pix Colony Picking System (Molecular Devices) to inoculate one well of 96-well plate containing YPD+cloNAT media. The plates were grown for two days then glycerol was added to 20% final concentration and the plates were stored at -80° C. until needed. Integration of specific phospholipase construct was verified by PCR with locus specific primers and subsequent sequencing. The strains generated were used in the following examples.

Example 6: Corn Mash Fermentations of Phospholipase-Expressing Yeast Strains

The strains described in Example 5 were tested for ethanol production and residual glucose using a 96-well corn mash fermentation described below. Propagation plates were prepared by inoculating 10 µL of each strain into a 96-well seed plate containing 150 µL YP+2% glucose medium per well. Plates were incubated at 30° C. and 300 RPM overnight. The following day, 10 µL of the seed culture was transferred to 96-deep well plates containing 500 µL of East Kansas Agri-Energy Liquefact Amp corn mash supplemented with 100 ppm urea and 0.42 AGU/g Spirizyme Excel. Plates were sealed with EnzyScreen plate covers and tightly clamped to limit oxygen transfer. Corn mash plates were incubated statically at 32° C. for 56 hours. After fermentation was completed, the plates were placed at -80 C for about 10 minutes before 100 µL of 8% H₂SO₄ was added to each well of the 96-deep well corn mash plates. Plates were sealed and mixed by inversion and centrifuged at 3000 rpm for 10 minutes. Supernatants were removed and diluted to 12× in sterile deionized water prior to HPLC analysis. The average ethanol and residual glucose per integrated heterologous phospholipase gene are shown in Table 15.

TABLE 15

Donor Organism (catalytic domain)	SEQ ID NO: (coding)	SEQ (mature poly- peptide)	mean ethanol (g/L)	mean glucose (g/L)
N/A (MeJi797 control strain)	—	—	128.84	4.82
<i>Penicillium cinnamomipureum</i>	423	332	138.52	3.30
<i>Talaromyces rugulosus</i>	407	316	138.39	1.28
<i>Aspergillus turcosus</i>	384	293	136.06	1.42
<i>Aspergillus egypticus</i>	376	285	135.98	2.97
<i>Bacillus mycoides</i>	381	290	135.88	5.16
<i>Penicillium swieckii</i>	392	301	134.40	5.30
<i>Lysinibacillus xylanilyticus</i>	388	297	134.23	1.94
<i>Listeria seeligeri</i>	391	300	133.80	3.99

TABLE 15-continued

	SEQ ID NO: (coding)	SEQ (mature poly- peptide)	mean ethanol (g/L)	mean glucose (g/L)
5 Donor Organism (catalytic domain)				
<i>Penicillium spikei</i>	410	319	133.37	1.68
<i>Talaromyces leycettanus</i>	402	311	133.29	1.02
<i>Bacillus toyonensis</i>	404	313	133.10	2.35
10 <i>Bacillus mycoides</i>	373	282	132.84	5.32
<i>Bacillus mycoides</i>	397	306	132.72	5.88
<i>Rasamsonia argillacea</i>	356	265	132.72	0.96
<i>Talaromyces toyonensis</i>	427	336	132.66	4.17
<i>Rasamsonia byssoschlamyoides</i>	348	257	132.53	1.34
<i>Bacillus mycoides</i>	414	323	132.42	3.18
15 <i>Listeria innocua</i>	375	284	132.36	5.22
<i>Penicillium simplicissimum</i>	430	339	132.26	2.57
<i>Bacillus mycoides</i>	380	289	132.26	2.40
<i>Talaromyces boninensis</i>	393	302	132.05	1.08
<i>Hamigera paravellanea</i>	401	310	131.88	2.04
<i>Penicillium vasconiae</i>	416	325	131.76	1.08
<i>Rasamsonia brevistipitata</i>	352	261	131.76	2.52
20 <i>Bacillus mycoides</i>	422	331	131.68	4.36
<i>Bacillus pseudomycoides</i>	372	281	131.66	3.57
<i>Bacillus sp.</i>	428	337	131.60	2.88
<i>Bacillus acidicer</i>	419	328	131.50	4.78
<i>Bacillus manliponensis</i>	429	338	131.36	2.44
<i>Hamigera avellanea</i>	409	318	131.23	1.49
25 <i>Penicillium flavescent</i>	358	267	131.22	4.68
<i>Bacillus sp.</i>	420	329	130.98	5.96
<i>Talaromyces bacillisporus</i>	367	276	130.86	3.80
<i>Talaromyces cellulolyticus</i>	425	334	130.74	0.84
<i>Rasamsonia eburnea</i>	349	258	130.61	2.07
<i>Bacillus acidicer</i>	433	342	130.52	5.04
30 <i>Talaromyces rugulosus</i>	364	273	130.52	1.00
<i>Bacillus thuringiensis</i>	396	305	130.46	3.89
<i>Brevibacillus sp.</i>	415	324	130.44	18.72
<i>Bacillus drentensis</i>	383	292	130.41	6.75
<i>Talaromyces verruculosus</i>	424	333	129.78	3.18
<i>Penicillium piscarium</i>	366	275	129.74	4.37
35 <i>Penicillium sclerotiorum</i>	347	256	129.72	9.06
<i>Hamigera terricola</i>	365	274	129.68	10.32
<i>Penicillium arenicola</i>	431	340	129.65	1.34
<i>Aspergillus tamarii</i>	377	286	128.86	1.03
<i>Penicillium sp.</i>	408	317	128.04	1.92
<i>Bacillus sp.</i>	382	291	127.84	3.64
<i>Aspergillus tubingensis</i>	386	295	127.66	3.89
40 <i>Penicillium emersonii</i>	355	264	127.65	3.75
<i>Talaromyces subinflatus</i>	385	294	127.26	6.60
<i>Bacillus thuringiensis</i>	405	314	127.08	3.00
<i>Penicillium vasconiae</i>	361	270	126.96	12.44
<i>Talaromyces columbinus</i>	362	271	126.69	4.05
<i>Bacillus bingmayongensis</i>	413	322	126.67	6.86
45 <i>Bacillus luciferensis</i>	379	288	126.51	5.46
<i>Hamigera striata</i>	394	303	126.36	2.04
<i>Bacillus thuringiensis</i>	406	315	126.30	9.00
<i>Bacillus toyonensis</i>	389	298	126.30	11.82
<i>Bacillus mycoides</i>	412	321	126.15	13.95
<i>Bacillus acidicer</i>	387	296	125.85	5.37
50 <i>Bacillus wiedmannii</i>	390	299	125.67	7.25
<i>Talaromyces variabilis</i>	363	272	125.33	5.78
<i>Bacillus thuringiensis</i>	374	283	125.32	4.08
<i>Penicillium simplicissimum</i>	360	269	125.04	10.32
<i>Aspergillus stramentius</i>	371	280	124.92	3.52
<i>Penicillium brefeldianum</i>	350	259	124.86	17.10
55 <i>Penicillium bialowiezense</i>	346	255	124.72	3.00
<i>Penicillium scabrosum</i>	353	262	124.56	14.67
<i>Penicillium megasporum</i>	369	278	124.44	2.10
<i>Penicillium donkii</i>	400	309	123.96	0.84
<i>Penicillium jensenii</i>	370	279	123.75	5.58
<i>Galactomyces candidus</i>	368	277	123.36	14.37
60 <i>Bacillus sp.</i>	395	304	123.18	8.82
<i>Aspergillus niger</i>	378	287	123.00	8.28
<i>Bacillus pseudomycoides</i>	421	330	122.22	4.86
<i>Penicillium hispanicum</i>	359	268	118.14	19.95
<i>Penicillium manginii</i>	354	263	117.96	13.15
<i>Penicillium meridianum</i>	345	254	117.72	14.92
<i>Paenibacillus sp.</i>	403	312	110.07	38.28
65 <i>Paenibacillus alginolyticus</i>	411	320	106.52	43.96

Example 7: Enhancement of Ethanol Yield with Phospholipase-Expressing Yeast Strains

Commercial Amp corn mash was obtained from Trenton Agri at 35.9% (w/w) of dry solids content and was diluted with tape water to 32.0% (w/w). After dilution, the pH value of the mash was adjusted to 5.1 with 39% (w/v) NaOH solution. Urea and lactrol were added into the pH adjusted Amp mash to final concentration at 150 ppm and 3 ppm, respectively. The prepared corn mash was aliquoted into 250 mL flasks (100 g/flask).

To propagate yeast, 50 mL of 6% YPD and 100 μ L of yeast-glycerol stock solution were mixed in a 125-mL flask and then incubated at 32° C. for overnight. After incubation, 45 mL of propagation was transferred to a 50 mL centrifuge tube and centrifuged at 3500 rpm for 10 minutes. The liquid fraction was decanted and deionized water was used to twice wash the cells. The cells were resuspended in 10 mL of deionized water and the total and dead cell accounts were measured using a NucleoCounter® YC-100.

The exogenous α -glucoamylase (Spirizyme Achieve-TM; Novozymes A/S) was added into the flasks containing corn mash per DOE and mixed well. Then the pre-determined amount of yeast suspension was added and mixed well. The fermentation was performed at 32° C. for 54 hours. After the fermentation, 5 g of slurry was taken out and the fermentation was stopped by adding 50 μ L of 40% H₂SO₄. The liquid was separated from solid by centrifuging whole slurry at 900 g for 10 min. The ethanol concentration was measured using HPLC.

Compared with the control strain MHCT-484, both phospholipase-expressing yeast strains showed significantly increased final ethanol yield (13.30% for control strain MHCT-484, to 13.43% and to 13.42% for strains HP21-F04 and HP21-F05, respectively).

Example 8: Enhancement of Oil Extraction with Phospholipase-Expressing Yeast Strains

Commercial Amp corn mash was obtained from Trenton Agri at 35.9% (w/w) of dry solids content and was diluted water to 32.0% (w/w). After dilution, the pH value of the mash was adjusted to 5.1 with 39% (w/v) NaOH solution. Urea and lactrol were added into the pH adjusted Amp mash to final concentration at 150 ppm and 3 ppm, respectively. The prepared corn mash was aliquoted into 250-mL flasks (100 g/flask).

To propagate yeast, 50 mL of 6% YPD and 100 μ L of yeast-glycerol stock solution were mixed in a 125-mL flask and then incubated at 32° C. for overnight. After incubation, 45 mL of propagation was transferred to a 50-mL centrifuge tube and centrifuged at 3500 rpm for 10 minutes. The liquid fraction was decanted, and deionized water was used to twice wash the cells. The cells were resuspended in 10 mL of deionized water and the total and dead cell accounts were measured using a NucleoCounter® YC-100.

Exogenous α -glucoamylase (Spirizyme Achieve-TM; Novozymes A/S), was added into the flasks containing corn mash per DOE and mixed well. Then the pre-determined amount of yeast suspension was added and mixed well. The fermentation was performed at 32° C. for 54 hours. After fermentation, 10 mL of 95% hexane was added to 95 grams of whole slurry. The slurry-hexane mixture was mixed well and then centrifuged at 3000 $\times$ g for 10 minutes. After centrifuge, the top layer was transferred into a 15-mL tube using positive displacement pipettes. The oil was extracted from the slurry again with the same method. The total weight of transferred liquor was measured. The oil content and density were measured on densiometer. The final extracted oil was calculated as:

$$\text{Extracted oil (g/flask)} = \text{first extracted oil (g/flask)} + \text{second extracted oil (g/flask)}$$

Compared with the control strain MHCT-484, both lipase-expressing yeast strains HP21-F04 and HP21-F05 improved the final oil yield (0.518 g/flask and 0.966 g/flask for strains HP21-F04 and HP21-F05, respectively compared to 0.483 g/flask for control strain MHCT-484).

Example 9: Defoaming Capabilities with Phospholipase-Expressing Yeast Strains

Commercial Amp corn mash was obtained from Trenton Agri at 35.9% (w/w) of dry solids content and was diluted with tape water to 32.0% (w/w). After dilution, the pH value of the mash was adjusted to 5.1 with 39% (w/v) NaOH solution. Urea and lactrol were added into the pH adjusted Amp mash to final concentration at 150 ppm and 3 ppm, respectively. The prepared corn mash was aliquoted into 250-mL flasks (100 g/flask).

To propagate yeast, 50 mL of 6% YPD and 100 μ L of yeast-glycerol stock solution were mixed in a 125-mL flask and then incubated at 32° C. for overnight. After incubation, 45 mL of propagation was transferred to a 50-mL centrifuge tube and centrifuged at 3500 rpm for 10 minutes. The liquid fraction was decanted, and deionized water was used to twice wash the cells. The cells were resuspended in 10 mL of deionized water and the total and dead cell accounts were measured using a NucleoCounter® YC-100.

The exogenous α -glucoamylase, Spirizyme Achieve-TM (Novozymes A/S), was added into the flasks containing corn mash per DOE and mixed well. Then the pre-determined amount of yeast suspension was added and mixed well. Also, the pre-determined exogenous lipase was added and used as control. The fermentation was performed at 32° C. for 54 hours.

The defoaming capability of control MHCT-484 and lipase-expressing yeast HP21-F04 was monitored using video camera after 12-hour fermentation. Compared with control yMHCT48, lipase-expressing yeast HP21-F04 showed significant defoaming capability (FIG. 4).

SEQUENCE LISTING

```
Sequence total quantity: 439
SEQ ID NO: 1          moltype = DNA length = 621
FEATURE               Location/Qualifiers
source                1..621
                     mol_type = genomic DNA
                     organism = Saccharomyces cerevisiae

SEQUENCE: 1
```

-continued

```

caggttcgagt ttatcattat caatactgcc atttcaaaga atacgtaaat aattaatagt 60
agtgtatttc ctaactttat ttagtcaaaa aattagcctt ttaattctgc tgtaaccctg 120
acatgcccaa aatagggggc ggggttacaca gaatatataa catcgtaggt gtctgggtga 180
acagtttatt cctggcatcc actaaatata atggagcccg ctttttaagc tggcatccag 240
aaaaaaaaag aatccagca ccaaaatatt gttttcttca ccaaccatca gttcataggt 300
ccattctctt agcgcaacta cagagaacag gggcacaaac agggcaaaaa cgggcacaac 360
ctcaatggag tgatgcaacc tgccctggagt aaatgatgac acaaggcaat tgaccacgc 420
atgtatctat ctcatcttct tacaccttct attaccttct gctctctctg atttggaata 480
agctgaaaaa aaagggtgaa accagttccc tgaaattatt cccctacttg actaataagt 540
atataaagac ggtaggattt gattgtaatt ctgtaaatct atttcttaaa cttcttaaat 600
tctactttta tagttagctt t

```

```

SEQ ID NO: 2      moltype = DNA length = 644
FEATURE          Location/Qualifiers
source           1..644
                 mol_type = genomic DNA
                 organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 2
agctacctat attccacat aatatcaatc atgcggttgc tgggtgattt accaataatg 60
tttaatgtat atatatagg ggcctgtatac ttacatatag tagatgtcaa gcgtaggcgc 120
ttcccctgcc ggctgtgacg ggcgcataac caaggatatc atagaccgcc aatcagcaaa 180
ctacctccgt acattcatgt tgcacccaca catgtacaca ccagaccgc aacaaattac 240
ccataagggtt gtttgtgacg gcgtcgtaca agagaacgtg ggaacttttt aggctcacca 300
aaaaagaaag gaaaaatcac agttgctgac agaagcctca agaaaaaaa aattcttctt 360
cgactatgct ggaggcagag atgatcgagc cggtagttaa ctatatatag ctaaattggt 420
tccatcacct tcttttctgg tgcgctcct tctagtgcta tttctggctt ttcctatttt 480
ttttttttcc atttttcttt ctctctttct aatatataaa ttctcttgca ttttctattt 540
ttctctctat ctattctact tgtttattcc cttcaagggt tttttttaag gagtacttgt 600
ttttagaata tacgggtcaac gaactataat taagctagaa caaa 644

```

```

SEQ ID NO: 3      moltype = DNA length = 457
FEATURE          Location/Qualifiers
source           1..457
                 mol_type = genomic DNA
                 organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 3
ctccagaaag gcaacgcaaa attttttttc cagggaaata actttctatg acccactact 60
tctcgtagga acaatttcgg gccctctcgt gttctcttga gggtcatctt ttacatttgc 120
ttctgctgga taattttcag aggcaacaag gaaaaattag atggcaaaaa gtcgtctttc 180
aaggaaaaat cccaccatc cttcgagatc ccttgtaact tattggcaac tgaaagaatg 240
aaaaggagga aaatacaaaa tatactagaa ctgaaaaaaa aaagtataaa tagagacgat 300
atatgccaat acttcacaat gttcgaatcc attcttcatt tgcagctatt gtaaaataat 360
aaaaacatca gaacaaacaa gctcaacttg tcttttctaa gaacaaagaa taaacacaaa 420
aacaaaaagt ttttttaatt ttaatcgcta gaacaaa 457

```

```

SEQ ID NO: 4      moltype = DNA length = 700
FEATURE          Location/Qualifiers
source           1..700
                 mol_type = genomic DNA
                 organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 4
gtgagtaagg aaagagttag gaactatcgc atacctgcat ttaaagatgc cgatttgggc 60
ggaatcctt tattttggct tcacctcat actattatca gggccagaaa aaggaagtgt 120
ttccctcctt cttgaattga tgttaccctc ataaagcacg tggcctctta tcgagaaaga 180
aattaccgtt gctcgtgatt tgtttgcaaa aagaacaaaa ctgaaaaaac ccagacacgc 240
tcgacttctc gctatcctat tgattgcagc ttccaatttc gtcacacaac aaggctcctag 300
cgacgggtca caggttttgt aacaagcaat cgaagggtct ggaatggcgg gaaagggttt 360
agtaccacat gctatgatgc ccaactgtgat ctccagagca aagttcgttc gatcgtaactg 420
ttactctctc tctttcaaac agaattgtcc gaatcgtgtg acaacaacag cctgttctca 480
cacactcttt tcttcaaac aagggggttg tttagttag tagaacctcg tgaacttac 540
atttacatat atataaactt gcataaattg gcataaattg gataacata tttggtcttt 600
tctaattcgt agtttttcaa gttcttagat gctttctttt tctctttttt acagatcctc 660
aaggaaagta ttactactt tttacaacaa atataaaaca 700

```

```

SEQ ID NO: 5      moltype = DNA length = 705
FEATURE          Location/Qualifiers
source           1..705
                 mol_type = genomic DNA
                 organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 5
atccttttgt tgtttccggg tgtacaatat ggacttctc tttcttgcca accaaaccca 60
tacatcgga ttctataat accttcgttg gtctccctaa catgtagggt gcggagggga 120
gatatacaat agaacagata ccagacaaga cataatgggc taacaagac tacaccaatt 180
acactgcctc attgatgggt gtacataacg aactaatact gtacccttag acttgatagc 240
catcatcata tcgaagtctt actacccttt ttccatttgc catctattga agtaataata 300
ggcgcatgca actctttttt tttttttttt ttttctctct cccctgtgt tgtctcacca 360
tatccgcaat gacaaaaaaa tgatggaaga cactaaagga aaaaattaac gacaaagaca 420
gcaccaacag atgcgttgtt tcagagctg atgaggggta tctcgaagca cacgaaactt 480
tttcttctct tcattcacgc aactactct ctaatgagca acggtatagc gccttctctc 540

```

-continued

```

cagttacttg aatttgaat aaaaaaaagt ttgctgtctt gctatcaagt ataaatagac 600
ctgcaattat taatcttttg ttctctcgct attgttctcg ttccctttct tccttgtttc 660
ttttctcgca caatatttca agctatacca agcatacaat caact 705

```

```

SEQ ID NO: 6      moltype = DNA length = 700
FEATURE          Location/Qualifiers
source          1..700
                mol_type = genomic DNA
                organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 6
aagaggatgt ccaatatttt tttaaggaa taaggatact tcaagactag attccccctt 60
gcattcccat cagaaccgta aaccttggcg ctttccttgg gaagtattca agaagtgcct 120
tgtccggttt ctgtgggtca caaaccagcg cgcccgatat ggctttcttt tcacttatga 180
atgtaccagt acgggacaat tagaacgctc ctgtaacaat ctctttgcaa atgtgggggt 240
acattctaac catgtcacac tgctgacgaa attcaaaagta aaaaaaatg ggaccacgtc 300
ttgagaacga tagattttct ttattttaca ttgaacagtc gttgtctcag cgcgctttat 360
gttttcattc atactttcata ttataaaata acaaaagaag aatttcatat tcacgccccaa 420
gaaatcaggc tgctttccaa atgcaattga cacttcatta gccatcacac aaaactcttt 480
cttgcctggag cttcttttaa aaaagacctc agtacaccaa acacgttacc cgacctcgtt 540
attttacgac aactatgata aaattctgaa gaaaaataa aaaaattttc atactctctg 600
cttttattta aaccattgaa tgattcttt tgaacaaaac tacctgtttc accaaaggaa 660
atagaagaa aaaatcaatt agaagaaaac aaaaaacaaa 700

```

```

SEQ ID NO: 7      moltype = AA length = 19
FEATURE          Location/Qualifiers
source          1..19
                mol_type = protein
                organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 7
MRFPSIFTTV LFAASSALA 19

```

```

SEQ ID NO: 8      moltype = AA length = 556
FEATURE          Location/Qualifiers
source          1..556
                mol_type = protein
                organism = Gloeophyllum sepiarium

```

```

SEQUENCE: 8
QSVDSYVSSE GPIAKAGVLA NIGPNGSKAS GASAGVVVAS PSTSDPDYWY TWTRDSSLVF 60
KSLIDQYTTG IDGTSSSLRTL IDDFVTAEAN LQQVSNPSGT LTTGGLGEPK FNVDETAFTG 120
AWGRPQRDGP ALRSTALITY GNWLLSNGNT SYVTSNLWPI IQNDLGYVVS YWNQSTYDLW 180
EEVDSSSFPT TAVQHRALRE GAAFATAIGQ TSQVSSYTTQ ADNLLCFLQS YWNPSGGYIT 240
ANTGGGRSGK DANTLLASIH TYDPSAGCDA ATFQPCSDKA LSNLKVYVDS FRSVYSINS 300
IASNAAVATG RYPEDSYQGG NPWYLTTFAV AEQLYDALNV WESQGSLEVY STSLAFFQQF 360
SSGVTAGTYS SSSSTYSTLT SAIKSPADGF VAVNAKYTPS NGGLAEQYSK SDGSPLSAVD 420
LTWSYASALT AFEARNNTQF AGWGAAGLTV PSSCSGNSGG PTVAVTFNVN AETVWGENIY 480
LTGSVDALEN WSADNALLLS SANYPTWSIT VNLPASTAIE YKYIRKNNGA VTWESDPNNS 540
ITTPASGSTT ENDTWR 556

```

```

SEQ ID NO: 9      moltype = AA length = 374
FEATURE          Location/Qualifiers
source          1..374
                mol_type = protein
                organism = Aspergillus niger

```

```

SEQUENCE: 9
APAPTRKGFT INQIARPANK TRTINLPGMY ARSLAKFGGT VPQSVKEAAS KGSVTPPQN 60
NDEEYLPVPT VGKSTLHLDF DTGSADLWVF SDELPSSSEQT GHDLYTPSSS ATKLSGYTWD 120
ISYGDGSSAS GDVYRDVTVT GGVTTNKQAV EAASKISSEF VQNTANDGLL GLAFSSINTV 180
QPKAQTTFPD TVKSQLDSPF FAVQLKHDAP GYVDFGYIDD SKYTGSITYT DADSSQGYWG 240
FSTDGYSIGD GSSSSSGFSA IADTGTTLIL LDDEIVSAYY EQVSGAQESE EAGGYVFSCS 300
TNPPDFTTVI GDYKAVVPGK YINYAPISTG SSTCFGGIQS NSGLGLSILG DVFLKSQYVV 360
FNSEGPKLGF AAQA 374

```

```

SEQ ID NO: 10     moltype = AA length = 590
FEATURE          Location/Qualifiers
source          1..590
                mol_type = protein
                organism = Trichoderma reesei

```

```

SEQUENCE: 10
SVHLLLESLEK LPHGWKAET PSPSSQIVLQ VALTQQNIDQ LESRLAAVST PTSSTYGYKL 60
DVDEINSIFA PSDASSSAVE SWLQSHGVT S YTKQGSSIWF QTNISTANAM LSTNFHTYS 120
LTGAKKVRTL KYSIPESLIG HVDLISPTTY FGTTKAMRKL KSSGVSPAAD ALAARQEPSS 180
CKGTLVFEGE TFNVFPQDCL RTEYSVDGYT PSVKSGSRIG PGSFLNESAS FADQALFEKH 240
FNIPSQNFV VLINGGTDLP QPPSDANDGE ANLDAQITLT IAHPLPITEF ITAGSPPYFP 300
DPVEPAGTPN ENEPYLQYYE FLLSKSNAEI PQVITNSYGD EEQTVPRSYA VRVCNLIGLL 360
GLRGISVLHS SGDEGVGASC VATNSTTPQF NPIFPATCPY VTSVGGTVSF NPEVAVAGSS 420
GGFSYFYSRP WYQQEAVGTY LEKYVSAETK KYYGPYVDFS GRGFPDVAAH SVSPDYPVFQ 480
GGELTPSGGT SAASPVVAI VALLNDARLR EGKPTLGLFN PLIYHLASKG FTDITSGQSE 540
GCNGNNTQTG SPLPGAGFIA GAHWNATKGW DPTTGFGVFN LKKLLALVRF 590

```

-continued

SEQ ID NO: 11 moltype = AA length = 511
 FEATURE Location/Qualifiers
 source 1..511
 mol_type = protein
 organism = *Thermoascus aurantiacus*

SEQUENCE: 11

VPVEVAGSAQ	GLDVTLSQVG	NTRIKAVVKV	TGSEDEVTFVH	LNFFKDAAPV	QKVSLEFRNAT	60
EVQFQGQIKQR	LITEGLSDDA	LTTLAPGATI	EDEFDIASST	DLSEGGTITI	NSNGLVPITT	120
DNKVTGYIPF	TSNELSIDVD	AAEAASVTQA	VKILERRTRI	SSCSGSRQSA	LTTALRNAAS	180
LANKAADAQ	SGSASKFSEY	FKTTSSSTRQ	TVAARLRAVA	REASSSSSGA	TTYCCLDPFG	240
YCSGNVLAYT	LPSYNIIANC	PIFYTYLPPL	TSTCHAQDQA	TTVLHEFTHA	PGVYSPGTLD	300
LAYGYQAAMG	LSSSQAVMNA	DTYALYANAI	YLGCTRISSC	SGSRQSALT	ALRNAASLAN	360
AAADAAQSGS	ASKFSEYFKT	TSSSTRQTVA	ARLRAVAREA	SSSSSGATTY	YCDDPYGYCS	420
SNVLAYTLPS	YNIIANCDIF	YTYLPALTST	CHAQDQATTA	LHEFTHAPGV	YSPGTDDLAY	480
GYQAAMGLSS	SQAVMNADTY	ALYANAIYLG	C			511

SEQ ID NO: 12 moltype = AA length = 550
 FEATURE Location/Qualifiers
 source 1..550
 mol_type = protein
 organism = *Dichomitus squalens*

SEQUENCE: 12

KPTARNLKLH	ESRPSAPNGF	SLVGSADSNR	TLKLRLALAE	SNFSELERKL	YDVSTPKSAN	60
YKGHLSKAEV	QQLVAPGQDS	IDAVNAWLKE	NDITAKTISS	TGEWISFEVP	VSKANDLFDA	120
DFSVFKHDDT	GVEAIRTLSY	SIPAELQGH	DLVHPTVTFP	NPYSHLPVFQ	SPVKKTAEIQ	180
NFTAGAI PSS	CSSTITPACL	QAIYNIPTTA	ATESSNQLGV	TGFIDQYANK	KDLKTFLLKY	240
RTDISSSTTF	TLQTLDGGSN	SQTGSKAGVE	ANLDIQYTVG	VATGVPTTFI	SVGDDFQDGD	300
LEGFLDLVINA	LLDEDAPPSV	LTTSYGQDES	TISRALAVKL	CNAYAQLGAR	GVSLFASGD	360
GGVSGSQSAS	CSKFVPTPPS	GCPYMTSVGA	TQGVNVPETAA	DFSSGGFSNY	WGVDPYQSDA	420
VSTYLSALGK	TNSGKYNASG	RGFPDVSTQG	VSEFVVVDGS	VEAVDGTSCA	SPTFASIIISL	480
VNDKLVAAGK	SPLGFLNPFL	YSDGVAALND	ITSGSNPGCN	TNGPPAKKGW	DPVTGLGTPD	540
FKKLLTAVGL						550

SEQ ID NO: 13 moltype = AA length = 353
 FEATURE Location/Qualifiers
 source 1..353
 mol_type = protein
 organism = *Nocardiosis prasina*

SEQUENCE: 13

ATGALPQSPT	PEADAVMQE	ALQRDLDLTS	AEAEELLAAQ	DTAFEVDEAA	AEAAGDAYGG	60
SVFDTESLEL	TVLVTDAAAV	EAVEATGAGT	ELVSYGIDGL	DEIVQELNAA	DAVPGVVGWY	120
PDVAGDTVVL	EVLEGSGADV	SGLLADAGVD	ASAVEVTTSD	QPELYADIIG	GLAYTMGGRC	180
SVGFAATNAA	GQPGFVTAGH	CGRVGTQVTI	GNRGVFEQS	VFPGNDAAFV	RGTSNFTLTN	240
LVSRYNTGGY	ATVAGHNQAP	IGSSVCRSGS	TTGWHCGTIQ	ARGQSVSYPE	GTVTNMTRTT	300
VCAEPGDSGG	SYISGTQAQG	VTSGGSGNCR	TGGTTFYQEV	TPMVNSWQVR	LRT	353

SEQ ID NO: 14 moltype = AA length = 456
 FEATURE Location/Qualifiers
 source 1..456
 mol_type = protein
 organism = *Penicillium simplicissimum*

SEQUENCE: 14

APASTAKDSV	SSVVKNGVKY	TVFEHAATGA	KMEFVKNSGI	CETTPGVNQY	SGYLSVGSNM	60
NMWFWFPEAR	NNPQAPLAA	WFNGGPGCSS	MIGLFPQENG	CHFVNGDSTP	SLNEYSWNNY	120
ANMLYVDQPI	GVGFSYGTDD	VTSTVTAAPY	VWKLQAFYA	QFPEYESRDF	AIFTESYGGH	180
YGPEFASYIQ	EQNSAIKTS	ISGENINLVA	LGVNNGWIDS	TIQEKAYIDF	SYNNSYQQLI	240
DDSQRSTLLS	AYNSQCLPAI	QKCTKSGSNS	DCQNADSVCY	NKIEGPISSS	GDWDVYDIRE	300
PSNDPYPPST	YSTYLSNADV	VKAIGAQSSY	QECPNGPYNK	FASTGDNPRS	FLSTLSSVVK	360
SGINVLVWAG	DADWICNWL	NYEVANAVDF	SGHTEFSKAD	LAPYTVNGTE	KGMFKNVANF	420
SFLKVYAGGH	EVPPYQPDAT	LQVFEQVLQN	KPIFST			456

SEQ ID NO: 15 moltype = AA length = 502
 FEATURE Location/Qualifiers
 source 1..502
 mol_type = protein
 organism = *Aspergillus niger*

SEQUENCE: 15

LQNPERRRAVP	PPLSHRSVAS	RSVPVERRTT	DFFEYLTNKTA	RFLVNGTSIP	EVDFDVGSY	60
AGLLPNTPTG	NSSLFFWFFP	SQNPEASDEI	TIWLNGGPGC	SSLDGLLQEN	GPFLWQPGTY	120
KPVNPYPSWT	NLTNVVYIDQ	PAGTGFSPGP	STVNNEEDVA	AQFNSWFKHF	VDTFDLHGRK	180
VYITGESYAG	MYVPYIADAM	LNEEDTTYFN	LKGIQINDPS	INSDSVMMYS	PAVRHLNHYN	240
NIFQLNSTFL	SYINAKADKC	GYNAFLDKAI	TYPPSPFPPT	APEITEDCQV	WDEVVMAAYD	300
INPCFNYHYL	IDFCPYLWDV	LGFPSLASGP	NNYFNRSVDQ	KILHVPPPTY	SVCSETVIFA	360
NGDGDPSGSW	GPLPSVIERT	NNTIIGHGWL	DYLLFLNGSL	ATIQNMTWNG	KQGFQRPPE	420
PLFVPYHYGL	AELWGDDEPD	PYNLDAGAGY	LGTAHTERGL	TFSSVYLSGH	EIPQYVPGAA	480
YRQLEFLLGR	ISSLSAKGNY	TS				502

SEQ ID NO: 16 moltype = AA length = 547
 FEATURE Location/Qualifiers

-continued

```

source          1..547
                 mol_type = protein
                 organism = Meripilus giganteus

SEQUENCE: 16
TPTGRNLKLH EAREDLPAGF SLRGAASPD T LKLRIALVQ NNFAELEDKL YDVSTPSSAN 60
YGNHLSKEEV EQYIAPAPES VKAVNAWLTE NGLDAHTISP AGDWLAFEPV VSKANELFDA 120
DFSVFTHDES GLEAIRTLAY SIPAELQGH L DLVHPTVTFF NPNHLPVVR STQPIRNLTG 180
RAIPASCAST ITPACLQAIY GIPTTKATQS SNKLAVSGFI DQFANKADLK SFLAQFRKDI 240
SSSTTFSLQT LDGGENDQSP SEAGIEANLD IQYTVGLATG VPTTFISVGD DFQDGNLEGF 300
LDIINFLLGE SNPPQVLTS YGQNTTISA KLANQLCNAY AQLGARGTSI LFASGDGGVS 360
GSQSAHCSNF VPTFPSCGPF MTSVGATQGV SPETAAAFSS GGFSNVFGIP SYQASAVSGY 420
LSALGSTNSG KFNRSRGRFP DVSTQGVDFQ IVSGGQTIGV DGTSCASPTF ASVISLVNDR 480
LIAAGKSPLG FLNPFLYSSA GKAAALNDVTS GSNPGCSTNG FPAKAGWDPV TGLGTPNPAK 540
LLTAVGL 547

SEQ ID NO: 17      moltype = AA length = 541
FEATURE           Location/Qualifiers
source            1..541
                 mol_type = protein
                 organism = Lecanicillium sp.

SEQUENCE: 17
APAPHGVLVK GEITKLPK WIATGAADSD AVIKAQIGIK QNNIKGLQDK LADIADPNSP 60
NYGQWLSKEE VDKYSAPAAA DVAAVKAWLA SSGITDVTMP TNDWIEFSVP VSKMESLLGS 120
KYEFVHLET GEKVPRTKQF SVPQNLHDLI DVVTPTTVLY HNMGPFAHAS PQAADASGLT 180
SPASIKSAYN VDYGKTGMTL VGTGFLGVG ASHQDYANFA RQFSPGLTDF KDVSINGGSN 240
SGDGSALGN LDTQYCGALA APNPSEYLAH APEGSDGSSF NDAMLAPGNY LNANSNPPSA 300
VSTSYGGEED GTDPNMDRI CNEFMKAGSR GVSIFSSGD NGVGGNGESS CYNGYYPLWP 360
ASCPVVTTVG GTEPDGSGRE VVANFEQYNK NVKSPGGGFS NHFPAPSYNK NVTAYANSL 420
SAAQKQRLNP NGRGFPPDIAL VSVKYQNVN GQISQVLGTS ASSPSMAGLV GLLNDYRKQTQ 480
GKPNLGFINP LLYSDKVKPA LRPVTSGANK GCDSSGLPAK TGWDAASGLG SPDFAKRLTL 540
V 541

SEQ ID NO: 18      moltype = AA length = 633
FEATURE           Location/Qualifiers
source            1..633
                 mol_type = protein
                 organism = Talaromyces proteolyticus

SEQUENCE: 18
VPAPSKRHVV HERRDALPHS WSEPRRDVGR TQLPVRIGLT QSNIDESHDM LMDIASPSSP 60
NYRKMYTVHE VNELFAPAGE AVSAVRDWLE SAGIAAERV T QSANKQLQF DGDAAEVESL 120
LGAEYIYTH DTNGRSHMGC EKVHVPEHIS HHIDIYIPGV KSLEVREPQP AELEKRTFGF 180
RKPQPLPFA LPESLETIIN SILGGLDLCL STVITPSCIK TLYNITEGTT ATKGNELGIF 240
EDLDGYYSQT DLDLFFTLFY SQIPAGTGPT LKGDGAQAP TQTLTQAGPE SLDLDFQVSY 300
IIWPQNSILF QTDDANYEAN YTFNGFLNNF LDAIDGSYCT YSAPGIDGNT ADDPPYPDPA 360
SNGYKSLQC GVYEPTNVIS ISYGGDEAGL SVNYQKRQCN EYKKLGLQGV SVVSSGDSG 420
VAGADGCLGG GKIFNPDPFA GCPYITTVGA TYLPSGASST SDEVAVSFR PSGGGFSNIY 480
SQPSYQSDAV NTYLTQHTPP YPAYETSDNS SVGANGGIYN KAGRGYPDVA AVGDNIIVFN 540
AGAPTLIGGT SASAPIFASI LTRINEVLLA KKGTTVGFPN PTLYANPDFA HDITSGDNPG 600
CSTNGFSTAP GWDVPTGLGT PNPALLKVF LGE 633

SEQ ID NO: 19      moltype = AA length = 371
FEATURE           Location/Qualifiers
source            1..371
                 mol_type = protein
                 organism = Penicillium ranomafanaense

SEQUENCE: 19
VPTGGKKSFT VNQVAVSATK TQNFANNYAR ALAKYGAKVP THVQAAQQS GSATTTPESD 60
DEEYLTVPNV GGTTLNLDFD TGSADLVVFS SELPASEQTG HSLYKPNNGT KLSGYTWSIS 120
YGDGSSASGD VYRDVSVGG VKATGQAVEA ASTISQQFTQ DQNDGLLGL AFSSINTVKP 180
KSQTTFFDTV KSTLASPLFA VSLKHNA PGS YDFGFDKSK YTGSLTYTDV DSSQGFWGFT 240
ADSYKIGSTT GSSIKGIADT GTTLLLLDDE VVSAYYQVS GAASDSSAGG YTFDCSSTLP 300
DFTVSISGYD AVVPGSLIN YTPVSGSSKC LGGIQNSNGL GFSIFGDIPL KSQYVVFDSN 360
GPRLGFARQS S 371

SEQ ID NO: 20      moltype = AA length = 578
FEATURE           Location/Qualifiers
source            1..578
                 mol_type = protein
                 organism = Aspergillus oryzae

SEQUENCE: 20
EAFEKLSAVP KGWHYSSTPK GNTFVCLKIA LAQKDAAGFE KTVLEMSDPD HPSYGQHFTT 60
HDEMKRMLLP RDDTVDAVRQ WLENGGVTD F TQDADWINFC TTVDTANKLL NAQFKWYVSD 120
VKHIRRLRTL QYDVPESVTP HINTIQPTTR FGKISPKKAV THSKPSQLD TALAAAVVAK 180
NISHCDSIIT PTCLELYNI GDYQADANS G SKIAFASYLE EYARYADLEN FENYLAPWAK 240
GQNFVSTTFN GGLNDQNSSS DSGEANLDLQ YILGVSAPLP VTEFSTGGRG PLVPDLTQPD 300
PNSNSNEPYL EFFQNVKLKD QKDLPPQVIST SYGENEQEIP EKYARTVCNL IAQLGSRGVS 360
VLFSSGDSGV GEGCMTNDGT NRTHFPFPFP AACPVWTSVG ATFKTTPERG TYFSSGGFSD 420
YWPRPEWQDE AVSSYLETIG DTFKGLYNSS GRAFPDVAQ GMNFAVYDKG TLGEFDGTS 480
SAPAFSAVIA LLNDARLRAG KPTLGLNPNW LYKTGRQGLQ DITLGASIGC TGRARFGGAP 540

```

-continued

DGGPVVPYAS WNATQGWDPV TGLGTPDFAE LKKLALGN 578

SEQ ID NO: 21 moltype = AA length = 456
 FEATURE Location/Qualifiers
 source 1..456
 mol_type = protein
 organism = Talaromyces liani

SEQUENCE: 21
 APASTTKDNV SSVVKNQVTV TVFEHAATGA KMEFVKNSGI CETTPGVNQY SGYLSVGNNM 60
 NMWFWFFEAR NNPQTAPLAA WFNNGPGCSS MIGLPQENGP CHFVNGASTP SLNEYSWNNY 120
 ANMLYVDQPI GVGFSYGTDD VTSTVTAAPY VWKLLQAFYA QFPEYQSRDF AIFTESYGGH 180
 YGPEFAAYIQ EQNSGIAAGS VSGENINLIA LGVNNGWIDP AIQEKAYIDF SYNNSYQQLI 240
 DDSQRTNLLS DYNDQCLPAI QQCAQTGRNS DCQNADNVCI DTIEGPISSS GNWDVVDIRE 300
 PSNDPYPPST YSSYLSNSRV VKAIGAQTSY QECPNGPYNK FASTGDNPRS FLSTLSSVVQ 360
 SGIHVLVWAG DADWICNLWG NYRVANAVDF PGHAEFSAKA LAPYTVNGTE KGMFKNVDNF 420
 SFLKVVYAGH EVPPYQDATA LQVFEQILQN KSITST 456

SEQ ID NO: 22 moltype = AA length = 589
 FEATURE Location/Qualifiers
 source 1..589
 mol_type = protein
 organism = Thermoascus thermophilus

SEQUENCE: 22
 EVFERLRAVP EGWRFSAFPS DDQPIRLQIA LQQHDVEGFE RAVLDMSTPS SPNYGKHFQS 60
 HDEMKRMLLP SDDDAVDVLD WLQSAGITDI EEDADWINFR TTVGVANELL DTQFQWVSE 120
 TSSHVRRLRA LEYSIPESVT PHIHMVQPTT RFGQIGRHHT TSREKPIVSG ADIHASIAGA 180
 NNQTTGTDCN TEITPKCLQD LYKFGGYKAS ANSGSKVGF C SYLEEYARYD DLALFEEALA 240
 PYAAGQNFVS ITYNGGLNDQ HSSSDSGEAN LDLQYIVGVS APLPVTFFST GGRGELVPDL 300
 DQPNPADNSN EPYLDLQNV LKLDQKDLPO VISTSYGENE QSVPEKYARS VCNLFMQLGS 360
 RGVSVIFSSG DSGVGSACLT NDGKNQTRFM PQFPASCPWV TSVGSTQHIA PEEATYFSSG 420
 GFSDLWPMMD YQKSAVGEYL DRLGSKWAGL YNPQGRGFPD VAAQGVNFNV YDKGSLKRF 480
 GTSCSAPTFA GVIALLDNAR LRARQPPMGF LNPWLYGAGK GGLNDIVNGG STGCDGNARF 540
 GGAPNGSPV PVFASWNTQG WDPVSVGLGTP DFSRLKLAV PSRVGGRLA 589

SEQ ID NO: 23 moltype = AA length = 413
 FEATURE Location/Qualifiers
 source 1..413
 mol_type = protein
 organism = Pyrococcus furiosus

SEQUENCE: 23
 AELEGLDESA AQVMATYVWN LGYDGSIGITI GIIDTGIDAS HPDLQGVIG WVDVFNRSY 60
 PYDDHGHGTH VASIAAGTGA ASNGKYKGMA PGAKLAGIKV LGADGSGSIS TIIKGVEWAV 120
 DNKDKYGIKV INLSLGSQSD SDGTDALSQA VNAANDAGLV VVVAAGNSGP NKYTIGSPAA 180
 ASKVITVGAV DKYDVITSFS SRGPTADGRL KPEVVAPGNW IIAARASGTS MGQPINDYYT 240
 AAPGTSMATP HVAGIAALLL QAHPSWTPDK VKTALITAD IVKPDETADI AYGAGRVNAY 300
 KAINVDNYAK LVFTGYVANK GSQTHQFVIS GASFVTATLY WDNANSDDL DL YLYDPNGNQV 360
 DYSYTAAYDF EKVGYYNPTD GTWTIKVVSYS SGSANYQVDV VSDGSLSQPG SSP 413

SEQ ID NO: 24 moltype = AA length = 387
 FEATURE Location/Qualifiers
 source 1..387
 mol_type = protein
 organism = Trichoderma reesei

SEQUENCE: 24
 LPTEGQKTAS VEVQYNKNYV PHGPTALFKA KRKYGAPISD NLKSLVAARQ AKQALAKRQT 60
 GSAPNHPSDS ADSEYITSVS IGTPAQVLPL DFDTGSSDLW VFSSETPKSS ATGHAIYTPS 120
 KSSTSKKVS G ASWSISYGDG SSSSGDVYTD KVTIGGFVSN TQGVESATRV STEFVQDTVI 180
 SGLVGLAFDS GNQVRPHQPK TWFSNAASSL AEPLFTADLR HGQNGSYNFG YIDTSVAKGP 240
 VAYTPVDNSQ GFWEFTASGY SVGGGKLNRN SIDGIADTGT TLLLLDDNVV DAYYANVQSA 300
 QYDNQQEGVV FDCDEDLPSP SFGVGSSTIT IPGDLNLNLT LEEGSSSTCFG GLQSSSGIGI 360
 NIFGDVALKA ALVVFDLGNE RLGWAK 387

SEQ ID NO: 25 moltype = AA length = 408
 FEATURE Location/Qualifiers
 source 1..408
 mol_type = protein
 organism = Rhizomucor miehei

SEQUENCE: 25
 RPVSKQSESK DKLLALPLTS VSRKFSQTKF GQQQLAEKLA GLKPFSEAAA DGSVDTPGY 60
 DFDLEEYAI P VSIGTPGQDF LLLFDTGSSD TWVPHKGCTK SEGCVGSRFF DPSASSTFKA 120
 TNYNLNITYG TGGANGLYFE DSIAIGDITV TKQILAYVDN VRGPTAEQSP NADIFLDGLF 180
 GAAYPDNTAM EAEYGSTYNT VHVNLKQGL ISSPLFSVYM NTNSGTGEV FGGVNNLLG 240
 GDIAITDVM S RYGGYFWDA PVTGITVDGS AAVRFSRPOA FTIDTGTNFF IMPSSAASKI 300
 VKAALPDAT E TQGWVPCA SYQNSKSTIS IVMQKSGSSS DTIEISVPVS KMLLPVDQSN 360
 ETCMFIILPD GGNQYIVGNL FLRFFVNVYD FGNNRIGFAP LASAYENE 408

SEQ ID NO: 26 moltype = AA length = 548
 FEATURE Location/Qualifiers
 source 1..548

-continued

```

mol_type = protein
organism = Lenzites betulinus

SEQUENCE: 26
KPMGRNLKVH EAREEIPDGF SLOGSAAPDT TLKLRIALVQ SNFAELEQKL YDVSTPSSPN 60
YGAHLSKEEV EQLVAPSADS VDAVNWLKE NDLSAQTISP AGDWLAFEPV VSKANELFDA 120
DFSVFTHDQT GLEAIRMSY SIPAELQGH LVLVHPTVTFP NPYSHLPVVR SPIKASQNL 180
SRATIPASCA STITPAQLD IYGIPTTKAT QSSNKLAVSG FIDQFANSAD LATFLKKFRT 240
DISSTTTTAL QTLDGGSNSQ SGSQAGVEAN LDIQYTVGLA SGVPVTFISV GDNFQDGDLE 300
GFLDIINFLL AESAPPQVLT TSYGQNTNTI SVKLANQLCN AYAQLGARGT SILFASGDGG 360
VSGSQSSSCS KFPVTFPSGC PFMTSVGATQ GVNPEATADE SSGGFSNYFG IPSYQATAVK 420
TYLTALGTTN SGKFNTSGRA FPDVSTQGV DFEIVVDGRTE GVDGTSCASP TFAAIISLVN 480
DKLIAAGKSP LGFLNPFLYS TGASAFDIT SGSNPGCNTK GFPKAGWDP VTGLGTPNFA 540
KLLAAAGV 548

SEQ ID NO: 27      moltype = AA length = 547
FEATURE           Location/Qualifiers
source            1..547
                  mol_type = protein
                  organism = Neolentinus lepideus

SEQUENCE: 27
GPAPRNVLVH ESRDGVPEGF VKSSTASPD T LKLRIALVQ GDMASLEKAL YDVSVPSSPL 60
YQHLKSKQEV EEYVKPTQES VDAVNQWLSS EGITANTISP AGDWLQFSVP VSKANEMFDA 120
DFSVFTHTES GQQAIRLSY SIPKELVGH LVLVHPTITFP NPYSHLPVVS SPAPRNLTID 180
ASVPSSCGST ITPTCLQDLY GIPTTAATQS SNKLAVSGFI DQYANKADLK SFLTTRYKDI 240
SSSTFTTLET IDGGENPDQ SDAGVEANLD TQYTVGLATG VPTYFISVGD DYQDGDLEGF 300
LDIVNYLLSM DQPQQVLTTS YGQNTNTMSR SLANNLCNAY MQLGARGTSI LFASGDGGVS 360
GSQSGSCGSK FVPTFPSCGP YLTSVGATTG INPEVAASF SSGGFSNYGV PSYQQSVVSS 420
YISGLGSTNK GKYNSSGRGF PDVSAQGENV EIVVDGSTEG VDGTS CSSPI FASIVSLLND 480
ELIAAGKSPL GFLNPFLYS GASAFNDITS GDNPGCNTNG FSAKSGWDPV TGLGTPNYAK 540
LRTAVGF 547

SEQ ID NO: 28      moltype = AA length = 399
FEATURE           Location/Qualifiers
source            1..399
                  mol_type = protein
                  organism = Thermococcus sp.

SEQUENCE: 28
VSAEKVRVII TIDKDFNENS VFALGGNVVA RGKVFPIVIA ELSPRAVERL KNAKGVVRVE 60
YDAEVQVLKG KSPGAGKFKP SQPAQTIPWG IERIKAPDVW SITDSSSGV IEVAILDTGI 120
DYDHPDLAAN LAWGVSRLRG KVSTKPKDYK DQNGHGHVTA GTVAALNNDI GVVGVAPAVE 180
IYAVRVLDAS GRGSYSIIL GIEQALLGPD GVLDSDGDI IVGDPDDDAE EVISMSLGG 240
SDVQAFHDAI IEATNYGVVI VAASNEGAS SPSYPAAYPE VIAVGATDVN DQVPWWSNRG 300
VEVSAPGV DV LSTYPPDSYE TSGTSMATP HVSQVVALIQ AAYYNYKGSV LPVGTFFDNT 360
MSTVRGILHI TADDLGSSGW DADYGYGIVR ADLAVQAVN 399

SEQ ID NO: 29      moltype = AA length = 396
FEATURE           Location/Qualifiers
source            1..396
                  mol_type = protein
                  organism = Thermococcus sp.

SEQUENCE: 29
EKVRVITIID KDFNENSVFA LGGNVVARGK VFPIVIAELS PRAVERLKNA KGVVRVEYDA 60
EVQVLKGKSP GAGKPKSQP AQTIPTWIER IKAPDVWSIT DGSSSGVIEV AILDGTIDYD 120
HPDLAANLAW GVSVLRGKVS TKPKDYKQDN GHGTHVAGTV AALNNDIGVV GVAPAVEIYA 180
VRVLDASGRG SYSIILGIE QALLGPDGVL DSDGDGIIVG DPDDDAEVI SMSLGLSDV 240
QAFHDAIIEA YNYGVVIVAA SNEGASSPS YPAAYPEVIA VGATDVNDQV PWSNRGVEV 300
SAPGVLDVLT YPDDSYETLS GTSMATPHVS GVVALIQAY YNKYGSVLPV GTFFDNTMST 360
VRGILHITAD DLGSSGWDAD YGYGIVRADL AVQAVN 396

SEQ ID NO: 30      moltype = AA length = 572
FEATURE           Location/Qualifiers
source            1..572
                  mol_type = protein
                  organism = Thermomyces lanuginosus

SEQUENCE: 30
APFQVVERLS APPDGWIKKE KAAPSAQIQF RLGLPQNSE QLEQLALNIA TPGHELYRKH 60
LKRDEIKALV RPLASVSEKV LAWLREDEGP EDRIHDDGAW IKFTVPVSTA EKLNTTEFFV 120
FHNERTGAER IRLTEYSVPQ DIHSLVKFIQ PTHFSSSLGP QVRRVPLDV LPKLRITLED 180
CNKKITPDC LKQYKIGDYV APEDPRNRIG ISGYLEQFAR YADFEEFLES YAPDRTDANF 240
TVVSINGGRN DQNSTLDSTE ASLDIDYAVT LSYKTQAVY TTAGRGLPVP DESQDPNEV 300
SNEPYMEQLQ FLLDLPEEL PTVLTTSYGE NEQSLPGSYA DETCNMFRLL GMRGVSVIFS 360
SGDWGTGIVC KANDGSEKRI FDPVYPASCP YVTSVGGTTG VNPERAVEFS SGGFSRFPFR 420
PKYQDEAVRS YLTKLGDHWK GLYNESGRAF PDVAAQADNF VVRDQGGQWS VGGTSASAPV 480
FAAIANVNA ELLKAGKEPL GFLNPWLYGL KGRGFTDVVH GGSTGCPGTV PWTGLPAGHV 540
PYASWNATEG WDPVTGLGTP LYDELVKAAL GK 572

SEQ ID NO: 31      moltype = AA length = 397
FEATURE           Location/Qualifiers
source            1..397

```

-continued

```

mol_type = protein
organism = Thermococcus thio-reducens

SEQUENCE: 31
EKPELVVRVIV HVDRGHFNTA DVATIGGHVV YQFKLIDAVV VEVPTAVGR LKKLPGVKMV 60
EFDHKARILA GPPSWLGGGQ PSQQIPWGIS RVRAPDVWGI TDGSGGVIEV AVLDTGVDYD 120
HPDLAGNIAW CVSTLRGRVT TNPAQCKDQN GHGTHVIGTI AALNNDIGVV GVAPGVEIYS 180
IRVLDAASGS SYSDIAIGIE QALLGPDGIL DKDGDGIIVG DPDDDAEVI SMSLGGPTDD 240
QYLHDMIITA YNYGVVIVAA SGNAGASSPS YPAAYPEVIA VGASDVNDQI ASWSNRQPEV 300
SAPGVDILST YPDDTYETLS GTSMATPHVS GVVALIQAA YNKYGVLPV GTFDDMGNT 360
VRGILHVTAD DLGDAGWDIY YGYGIVRADL AVQAAIG 397

SEQ ID NO: 32      moltype = AA length = 549
FEATURE            Location/Qualifiers
source             1..549
                   mol_type = protein
                   organism = Polyporus arcularius

SEQUENCE: 32
KPMARSMKLH ESREGIPEGF SLRGAAQPEQ TIKLRLALVQ SNFAELERKL MDVSTPSSAN 60
YKGHLKAEV QQLVAPTQDS VDAVKSWLKE NDISAKTISA TGDWLSFEVP VSKANELFDA 120
DFSITHDET GTEAVRRLSY SIPAELQGH LVLHPTVTFP NPRGLPPVFT APIKAEQNL 180
TSRATIPSSC ARTITPAQLQ AIYNIPSTPA TESSNKLAVT GFIEQFANKA DLKTFLTRFR 240
TDISSTSTF LQTLDDGGSN QSSSEAGVEA NLDIYTVGV ATGVPTVFIS VGEDFQDGL 300
EGFLDVVNSL LDEDTPPFVM TTSYGQNT ISRNLANLNC NAYALGARG VSILFASGDG 360
GVAGSQSASC SKFVPTFPFG CPMTSVGAT QGFSPETAAD FSSGGFSNYF AIPDYQTSV 420
SGYKALGNT NSGKYNATGR GPDIAATQGV NFEVVGQGS GTVEGTSCSS PTLASIIISL 480
NDRLIAAGKS PLGFLNPFLY STGTSALNDI TSGSNPGCNT NGFPAKAGWD PVTGLGTPDF 540
NKLLSAVGL 549

SEQ ID NO: 33      moltype = AA length = 548
FEATURE            Location/Qualifiers
source             1..548
                   mol_type = protein
                   organism = Ganoderma lucidum

SEQUENCE: 33
KSTRNLKLH ETRQAGPSGF SHTGSADPNQ TLKRLALVQ GNTAELERKL YDVSTPSSAN 60
YKGHLKSEV QQLVAPAQGS VDAVNAWLRE NGITAKTISA AGDWLSFEVP VSKANELFDA 120
DFSVPKHDDT GVKAVRRLSY SIPAELQGH LVLHPTVTFP NPNGHMPVFQ APVKDQDAVQ 180
NFSARAVPSS CSNTITPAQL QALYNIPSDA ATQSSNKLAV TGFIEQYANQ VDLAVFLKQY 240
RADISSNTTF ALQTLDDGGS SQTNPVGVEA NLDIYTVGI ATGVPTVFIS VGDQYQDGL 300
EGFLDVINFL LDEDTPPFVV TTSYGQDEHT ISRKLQNL NAYALGARG VSILFASGDG 360
GVAGSRSSNC SKFVPTFPFG CPYMTSVGAT QGVPEAADF SSGGFSNYFG TPDYQASAVK 420
SYLSTLGSN RGKFNASGRG FPDVATQGVN FEVIDGVEV GVSPTSASP MFAAIVALLN 480
DKLIAAGKSP LGFLNPFLYS KGVESALNDIT TGSNPGCGTI GPPAKEGWDP VTGLGTPDFQ 540
KLASAAGL 548

SEQ ID NO: 34      moltype = AA length = 548
FEATURE            Location/Qualifiers
source             1..548
                   mol_type = protein
                   organism = Ganoderma lucidum

SEQUENCE: 34
KTATRNKLH ETSQAGPSGF SLTGSADPDQ TLKRLALVQ GNVAELERKL YDVSTPSSPN 60
YKGHLKSEV QQLVAPAQGS IDAINAWLKE NGISAKTISA TGDWLSFEVP VSKANELFDA 120
DFSVPKHDDT GMEVVRRLSY SIPAELQAH LVLHPTVTFP NPKGHPPVFQ APAMITNDVQ 180
NFSAGAVPSS CSSRITPAQL QALYNIPSDP ATQPSNKLAV TGYIEQYANQ DDLAVFLKEY 240
RADMSNTTF TLQTLDDGVN SQTDEAGIEA NLDVQYTVGI ATGVPTVFIS VGDQYQDGNL 300
EGFLDVVNSL LDEDTPPFVM TTSYGQDEHT MSRKLAQNL NAYALGARG VSILFASGDG 360
GVAGSRSSSC SKFVPTFPFG CPYMTSVGAT QGVPEAADF SSGGFSNYFG IPDYQASAVS 420
GYLSALGHTN KGKYNASGRG FPDVSTQGVN FEVMVDGALE GVSPTSASP TFAAVVALLN 480
DRLIAAGKSP LGFLNPFLYS KGVESALNDIT SGSNPGCRTN GPPAKEGWDP VTGLGTPDFQ 540
KLASAAGL 548

SEQ ID NO: 35      moltype = AA length = 541
FEATURE            Location/Qualifiers
source             1..541
                   mol_type = protein
                   organism = Ganoderma lucidum

SEQUENCE: 35
KPTARNLRLH ETRQAGPSGF SLTGSADPNQ TVRLRLALVQ GNTGELEERKL YDVSTPSSAN 60
YKGHLKAEV QQLVAPAQGS IDAVNAWLKE NDISAKTISA TGDWLSFEVP VSKANELFDA 120
DFSVPKHDDT GMEVVRRLSY SIPAELQGH LVLHPTVTFP NPKGNLPLFQ TPIKSKRDVP 180
ADCSNNITPA CLQALYNIPS DAATQSSNTL AVTGYIEQYA NQQLDLSFLG QFRPDISSNT 240
TFALQITDGG SNSQNGSDAG GEANLDIYTVGLATGVPTV FISVGEQYQD GDLGGLLDVI 300
NFVLAEDAPP NVITTSYGQN ITISLKLQ NLCNAYALG ARGVSILFAS GDGGVAGSQS 360
DNCTQFVPTF PSGCPYMTSV GATQGVPEA ADFSTGGFSN LFSVPDYQAA AVQSYLSALG 420
GTYQGLFNAS GRAFPDVSTQ GVNFEVTVVDG SVSGASGTS ASPTFAAIVA LLNDRLVAAG 480
KSPLGFLNPF LYSTGASALN DIATGSNPGC GTNGFSAQKG WDPVTGLGTP DFQKLAAGAAG 540
L 541

```

-continued

SEQ ID NO: 36 moltype = AA length = 547
 FEATURE Location/Qualifiers
 source 1..547
 mol_type = protein
 organism = Trametes sp.

SEQUENCE: 36

TPTGRNLKLH	EAREDIPTGY	SLRGAASPD	TLKLRLALVQ	NNFAELEDKL	YDVSTPSSAN	60
YGNHLSKEEV	EQYIAPAPES	VKAVNAWLTE	NGLDAHTISP	AGDWLAFEPV	VSKANELFDA	120
DFSVFTHDES	GLEAIRLAY	SIPAEQGH	DLVHPTVTFP	NPNALPVVR	STKPIQNLTG	180
RAIPASCAST	ITPACLQAIY	GIPTTKATQS	SNKLAVSGFI	DQFANSADLK	SFLSTFRKDI	240
SSSTTFALQT	LDGGQNNQSP	SQAGIEANLD	IQYTVGLATG	VPVTFISVGD	NFQDGDLEGF	300
LDIINFLLSE	SNPPQVLTT	YQONENTISA	KLANQLCNAY	AQLGARGTSI	LFASGDGGVA	360
GSQSSSCRNF	VPTFPGCPF	MTSVGATQGV	SPETAADFSS	GGFSNVFGIP	SYQTSVSGY	420
LSALGNTNSG	KFNRSRGFP	DVATQGVNFQ	IVSGGDTGGV	DGTSCASPTF	ASVISLINDR	480
LIAAGKSPLG	FLNPFLYSAA	GKAALNDVTS	GSNPGCNTNG	FPKAGWDPV	TGLGTPNFAK	540
LLTAVGL						547

SEQ ID NO: 37 moltype = AA length = 553
 FEATURE Location/Qualifiers
 source 1..553
 mol_type = protein
 organism = Cinereomyces lindbladii

SEQUENCE: 37

KPTARNLLVH	ESLDGVPTGF	QLVGPASPD	VLSMRIALVQ	SDPAGLEAAL	YDVSTPSSAS	60
YGNHLSKAEV	EKFVSPTES	VQAVNAWLTE	NDLTATQLSP	AGDWLGFEVP	VSKAEDLFGT	120
QFSVFTHREAT	GMQTVRTLSY	SIPSELQGH	DLVFPITNFP	DPNANLPVFR	HASKKREVT	180
LNANLTSDAV	PSSCADTITP	ACLQALYGIP	TTPATSSSTNQ	LGVSGFIDQF	ANQADLKTFL	240
QNRFTDISS	TTFLELTDG	GSNSQNRGDA	GVEANLDTQY	TVGLATDVPT	VFISVGEDNQ	300
DGSLGGFLDI	INFLDQDSP	PQVLTSYGO	NENTVSRAVA	NNLCNAYQ	GARGTSILFA	360
SGDGGVSGSQ	SASCRFVPT	FPSGCPFMTS	VGATTGINPE	TAATFSAGGF	SNYFGTSPSYQ	420
ASAVSSYLAA	LGSTNSGKFN	TSGRGYPDVS	TQGENFEIVV	SGEEGVDGT	SCASPTFASI	480
ISLVNDRLIA	AGKPPGLFLN	PFLYSTGASA	FTDITGDNP	GCNTNGPPAK	SGWDPVTGLG	540
TPNFSKLLTA	VGL					553

SEQ ID NO: 38 moltype = AA length = 559
 FEATURE Location/Qualifiers
 source 1..559
 mol_type = protein
 organism = Trametes versicolor

SEQUENCE: 38

AVASTLQLHE	ARKGIPAGFS	LHGAASPD	LNLRMALVQS	NFAGLEERLY	DVSTPSSANY	60
GKHLKAEVE	QYVAPRQSI	TAVKAWLAAN	GLSGTSSISA	GDWIAAKVPV	SKANKLLGAQ	120
FGVFNNDATG	RQIIRTLAYS	IPAEKKGHL	LVHPTITFAD	IKPLVPVSA	RRESRVLVDS	180
DLVANTIPAS	CNAAITPA	QDLYGIPSTP	ATQSSNQLGV	SGFIDQFANQ	ADLATFLTEF	240
RPDVSNSTTF	TLQTLDDGQN	PQDPSDAGVE	ANLDTQYTVG	VATNVPTTF	SVGDDTKDGI	300
FGFLDLISFL	LAAAAPQVL	TTSYGADDEG	LSANLVRNLC	QAYAQLGARG	TSILFSSGDG	360
GVSGSQAEGC	VDFVPTFPG	CPFLTSGVAT	QLTTASGLTV	ETAAGFSSGG	FSNYFPPTPY	420
QQAVVDAYIK	KTLVNGTVNE	GLFNASGRA	PDVSAVGVDY	LIVVGGGTDI	VSGTSASSPL	480
FASVIALIND	RRLAAGKEPL	GFLNPFLYSQ	AGASALNDVT	VGSNPGCASP	GFPAAQGWDP	540
VTGLGTPNFA	KLLAALAL					559

SEQ ID NO: 39 moltype = AA length = 541
 FEATURE Location/Qualifiers
 source 1..541
 mol_type = protein
 organism = Paecilomyces hepialid

SEQUENCE: 39

APAPHGPLVK	FGEIRKLPSK	WVATGAADAN	AVIKGQIGIK	QNNIQGLQAK	LADIADPNP	60
NYGQWLSKEE	VDKYSAPAAA	DVAAVKAWLA	SSGITDVTMP	TNDWIEFSVP	VSKMESLLGS	120
KYEWVHLET	GEKVPRTKEF	SVPQNLHDLI	DVVTPTTVLY	HNINPHTHSS	PQAAGAAGLT	180
SPASIKSAYN	VDYKGTGNTL	VGTTGFLGVG	ASHTDYANFG	QQFSPGLKDF	QDVSVNGGSN	240
SGDGSALEGN	LDTQYCGALA	APNPSEYLAH	APEGSDNNSF	NDAMLAFGNY	LNSARNPPSA	300
VSTSYGGEED	GVDASYLDRI	CNEFMKAGSR	GVSIFPSSGD	NGVGGNGESS	CQNGYYPLWP	360
ATCPVVTTVG	GTEFDNSGRE	VVANFEQYNK	NIKSPGGGYS	NHFAAPSYNK	AVTTSYANGL	420
AAPQKQRLNP	NGRGYPDISL	VSVKYQNVN	NQISQVLGTS	ASSPSIAGLV	GLLNDYRKQT	480
GKPNLGFINP	LLYSKVKPA	LRDVTSGSNK	GCDSVGLPAK	TGWDAASGLG	SDFGKLRTL	540
V						541

SEQ ID NO: 40 moltype = AA length = 541
 FEATURE Location/Qualifiers
 source 1..541
 mol_type = protein
 organism = Isaria tenuipes

SEQUENCE: 40

APAPHGPLVK	FGELKKLP	WVATGAAND	AVIKQIGIK	QNNIKGLQDK	LAEISDPN	60
SYGQWLSKEE	VAKYTAPADA	DVAAVKAWLS	SAGITEVTMP	TNDWLEFSVP	VSKMESLLGS	120
KYEWVHLET	GEKAPRTKEF	SVPQNLHGII	DVVTPTTVLY	HNINPNSHGN	ELASASAGLT	180
SPASIKSAYN	VDYKGTGNTL	VATTGFLGVG	ASHNDYLA	HQFSPGLKDF	QDVSVNGGSN	240
SGDGSALEGN	LDTQYCGALA	SPNPSQYLAN	SPEGSDNNSF	NDAMTAFGNY	LNSASNP	300

-continued

```

VTSYGGGEED GVDAGYLDRI CNEFMKAGSR GISVVFSSGD NGVGGNGEPS CQNGYYPLWP 360
ATCPYVTTVG GTEFDDSGRE VVANFEQYNK NVKSPGGGYS NHFPAPDYNK NVTAYANSL 420
SAAQQORLNP NGRGFPDISL VSVKYQVSLN GQTKQVLGTS ASSPSVAGLV GLLNDYRKTQ 480
GKSNLGLFLNP LLYSGKVNAAL LRDVTSGSNK GCDSVGLPAK SGWDAASGLG SFDFAKLRSL 540
I 541

```

```

SEQ ID NO: 41      moltype = AA length = 578
FEATURE           Location/Qualifiers
source            1..578
                  mol_type = protein
                  organism = Aspergillus tamarii

```

```

SEQUENCE: 41
EAFEKLSAVP KGWHYSSTPE GSTSVCLKIA LAQKDAAGFE KRVEYMSDPD HPNYGQHFTT 60
HEEMKRMLLP RDDTVDAVRQ WLENGGVTDV RQSDSWINFC TTVDTANKLL NAQFKWYVSD 120
VKHIRRLRTL QYDVPGSVAS HVNTIQPTTR FGKITPKKAV THSKPSQLDV TALAAAVVAK 180
NISHCDSIIT PTCLELNYNI GDYQADANSK SKIAFASYLE EYARYADLEN FENYLAPWAK 240
GQNFSVITYN GGLNDQNSSS DSGEANLDLQ YILGVSAPLP VTEFSTGGRG PLVPDLTQPD 300
PNANSNEPYL EFFQNVKLKD QEQLPQVIST SYGENEQEIP EKYARTVCNL IAQLGSRGVS 360
VLFSSGDSGV GEGCMTNDGT NRTHFPPQFP AACPWVTSVG ATYKTTPERA TYFSSGGFSD 420
YWARPEWQEE AVSSYLETIG DAFKGLYNAS GRAFPDVAQ GMNFAVYDKG TLGEFDGTSA 480
SAPAFSAIIA LLNDARLRAG KPTLGLFLNP LYKTGRQGLQ DITLGASTGC TGRARFGGAP 540
DGGPVVPFAS WNATQGWDPV TGLGTPDFAE LKKLALAN 578

```

```

SEQ ID NO: 42      moltype = AA length = 587
FEATURE           Location/Qualifiers
source            1..587
                  mol_type = protein
                  organism = Aspergillus brasiliensis

```

```

SEQUENCE: 42
EIFEKLSGVP NGWRYANNPQ GNEVIRLQIA LQQHDVTGFE QAVMDMSTPG HADYKGFHRT 60
HEEMKRMLLP SDTAVDSVRD WLESAGVHNI QVDADWIKFH TTVTKANALL DADFKWYVSE 120
ARHIRRLRTL QYSIPDALVS HINMIQPTTR FGQIQPNRAT MRSKPKHADE TFLTAATLAQ 180
NTSHCDSIIT PSCLKQLYNI GDYQADPKSG SKIGFASYLE EYARYADLEK FEQHLAPNAI 240
GQNFTVQQFN GGLNDQLSTK DSGEANLDLQ YILGVSAPLP VTEYSTGGRG ELVPDLSSPD 300
PNDNSNEPYL DFLQNLKLN NSDLQPQVIST SYGEDEQITP VPYARAVCNL YAQLGSRGVS 360
VIFSSGDSGV GAACTLNDGT NRTHFPPQFP ASCPWVTSVG ATSKTSPEQA VSFSSGGFSD 420
LWPRPSYQHA AVQTYLTEHL GNKFSGLFNA SGRAPPDVSA QGVNYAVYDK GILGQFDGTS 480
CSAPTFSGVI ALLNDARLRA GLPVMGFLNP FLYGAGSKLG GLNDIVTGGG VGCDDRNRFG 540
GTPNGSPVVP FASWNATGW DPVSGLGTPD FAKLKVALG ESEGDEN 587

```

```

SEQ ID NO: 43      moltype = AA length = 582
FEATURE           Location/Qualifiers
source            1..582
                  mol_type = protein
                  organism = Aspergillus iizukae

```

```

SEQUENCE: 43
EVFDTLAAVP KGWHYSHTPR ADQPISLKIA LKQHNVEGFE QAVLDMSTPG HEHYGKHFRE 60
HDEMCRMLLP SDATVDAVKD WLLAADVTDY EVDADWINLH TTVQANELL DTEFAWYVSD 120
VRAVRRLRTL RYSVPDAVAP HINMVQPTTR FGQIHPDRAT FRAGSTHFGA HILSAMSAGV 180
DVSSANVTCD DVITPLCLKE LYKVDGYRAE AEHGSKIAPA SYLEEYARYD DMVRPQEKLA 240
PYAKGENFVS ILYNGGVDDQ QSTSDSGEAN LDLQTIMGLS APLPITEYIT GGRGKLIPDL 300
SQDPDNDNSN EPYLEWQNV LKHSPEELPQ VISTSYGEDE QTIPRGYAES VCNLLAQLGS 360
RGVSVIFSSG DSGVGSACQT NDGTNTTHFP PQFPASCPWV TSVGATSKTH PEEAVYFSSG 420
GFSDLWARPA WQDDAVSTYI ESIGGKFAGL YNASGRAFPD VSAQQQNYAI FDKGRLGKMD 480
GTSCSAPAPA GIVSLLNDAR LRANRPVLGF LNPWLYGTAR EGLNDIVHGG SKGCDGRDRF 540
GGKPNPSPV PVASWNATPG WDPVSGLGTP NFATLVQVAL HD 582

```

```

SEQ ID NO: 44      moltype = AA length = 456
FEATURE           Location/Qualifiers
source            1..456
                  mol_type = protein
                  organism = Penicillium sp.

```

```

SEQUENCE: 44
APASTAKDSV SSVVKNVGKY TVFEHAATGA KMEFVKNSGI CETTPGVNQY SGYLSVGSNM 60
NMWFWFFEAR NNPQQAPLAA WFNNGPGCSS MIGLFQENGH CHFVNGDSTP SLNEYSWNNY 120
ANMLYVDQPI GVGFSGYTDV VTSTVTAAPY VWKLLQAFYA QFPEYESRDF AIFTESYGGH 180
YGPEFASIYQ DQNAAIKAGS VSGENINLVA LGVNNGWIDS TIQEKAYIDF SYNNSYKQLI 240
DDSQRSTLLS AYNDQCPLAI QKCTSSGSNS DCKNADSVCY NQIEGPISSS GWDWDVYDIRE 300
PSNDPYPPST YSTYLSNADV VKAIGAQSSY QECPNGPYNK FTSTGDNPRS FLSTLSSVVK 360
SGINVLVWAG DADWICNLWG NYEVANAVDF SGHTDFSAKD LAPYTVNGTE KGLFKNVDFN 420
SPLRVYGAGH EPPYQPDAT LQVFEQILQK KPIFST 456

```

```

SEQ ID NO: 45      moltype = AA length = 456
FEATURE           Location/Qualifiers
source            1..456
                  mol_type = protein
                  organism = Aspergillus denticulatus

```

```

SEQUENCE: 45
STASAAKDSV SSIVKNVGKY TVFEHAATGA KMEFVKNSGI CETTPGVNQY SGYLSVGDNM 60

```

-continued

```

NMWFWFFFEAR NNPQQAPLAA WFNGGPGCSS MIGLFQEHGP CHFVNGEDTP SLNEYSWNNY 120
ANMLYVDQPI  GVGFSGTDD  VTSTVTAAPY VWKLLQAFYA QFPEYESRDF AVFTESYGGH 180
YGPEFASYIQ  QQNAAIKAGT VSGENINLIA LGVNNGWIDS AIQEKAYIDF SYNNTYKQLI 240
SSSDRTRLLS  VYNSQCLPAI  QKCTSTGTTA ACRNADSVCY NNIEGPISSS GDWDVVDIRE 300
PANDPYPPAT  YSTYLADPDV  VKAIGAQTST QECPNGPYNK FASTGDNPRS FLSTLSNVVK 360
SGINVLVWAG  DADWICNWLK  NYEVANAVDY PGQSEFEAKD LAPYTVNGAE KGMFKNVDNF 420
SFLRVYGAGH  EVPPYQPETA  LQVFQQTLLK KPIFST 456

```

```

SEQ ID NO: 46      moltype = AA length = 456
FEATURE           Location/Qualifiers
source            1..456
                  mol_type = protein
                  organism = Hamigera sp.

```

```

SEQUENCE: 46
APASTAKDTL  SSIVKNGVTY  NVFEHADSGA KIEFVKNSGI CETTPGVNQY SGYLSVGDNM 60
NMWFWFFFEAR NNPQQAPLAA WFNGGPGCSS MIGLFQENGP CHFVNGENTP SLNEYSWNNY 120
ANMLYVDQPI  GVGFSGTDD  VDTSTVTAAPY VWKLLQAFYA QFPEYESRDF AIFTESYGGH 180
YGPEFAHYIQ  QQNAAIKSGS  VKGENINLIG LGVNNGWIDS AIQEKAYIDF SYNNSYKQLI 240
DFSQRTSLMR  AYKNQCLPAI  QKCYQTGTNA DCTDASSVCY NNIEGPISSS GDWDVVDIRE 300
PSNDPYPPKT  YSSYLSDPKV  VKAIGARTNY KECPNGPYNK FSTTGDNPRS FLSTLSNVVK 360
SGINVLVWAG  DADWICNWLK  GYGVANAVDY PGHAQFRAKA LAPYTVNGTE KGQFKTVDNF 420
QFLKVVYAGH  EVPPYQPETA  LQVFEQILQK KPIHST 456

```

```

SEQ ID NO: 47      moltype = AA length = 456
FEATURE           Location/Qualifiers
source            1..456
                  mol_type = protein
                  organism = Penicillium janthinellum

```

```

SEQUENCE: 47
APASTAKDTV  SSVVKDGVTY  TVFEHAATGA KMEFVKNSGI CETTPGVNQY SGYLSVGSNM 60
NMWFWFFFEAR NNPQQAPLAA WFNGGPGCSS MIGLFQENGP CHFVNGESTP SLNENSWNNY 120
ANMIYIDQPI  GVGFSGTDR  VTSTVTAAPY VWKLLQAFYA QFPEYESRDF AIFTESYGGH 180
YGPEFASYIE  QQNAAIKAGS  VTGQNVNIVA LGVNNGWIDA TIQEKAYIDF SYNNSYQQLI 240
DSSTRDSLDD  AYNQCLPAI  QQCAQSGSNS DCTNADSVCY QNIEGPISSS GDFDVVDIRE 300
PSNDPYPPKT  YSTYLSDPV  VKAIGARTNY QECPNGPYNK FASTGDNPRS FLSTLSSVVQ 360
SGINVLVWAG  DADWICNWLK  NYAVANAVDF PGNAQFSAMD LAPYTVNGVE KGQFKTVDNF 420
SFLKVVYAGH  EVPPYQPDTA  LQVFEQILQK KPISST 456

```

```

SEQ ID NO: 48      moltype = AA length = 456
FEATURE           Location/Qualifiers
source            1..456
                  mol_type = protein
                  organism = Penicillium vasconiae

```

```

SEQUENCE: 48
APASTAKDSV  SSVVKNGVKY  TVFEHAATGA KMEFVKNSGI CETTPGVNQY SGYLSVGSNM 60
NMWFWFFFEAR NNPQQAPLAA WFNGGPGCSS MIGLFQENGP CHFVNGDSTP SLNEYSWNNY 120
ANMLYVDQPI  GVGFSGTDD  VTSTVTAAPY VWKLLQAFYA QFPEYESRDF AIFTESYGGH 180
YGPEFASYIQ  EQNAAITAGS  VSGQKINLIA LGVNNGWIDS TIQEKAYIDF SYNNSYQQLI 240
DDSQRTSLLS  AYNQCLPAI  QKCTQTGSNS ACQNAANVCY NNIEGPISSS GDWDVVDIRE 300
PSNDPYPPST  YSTYLSASDV VKAIGAQSSY QECPNGPYNK FASTGDNPRS FLSTLSSVVK 360
SGINVLVWAG  DADWICNWLK  NYEVANAVDF SGHAEFSAKD LAPYTVNGAE KGMFKNVDNF 420
SFLKVVYAGH  EVPPYQPETA  LQVFEQILQK KPISST 456

```

```

SEQ ID NO: 49      moltype = AA length = 454
FEATURE           Location/Qualifiers
source            1..454
                  mol_type = protein
                  organism = Hamigera paravellanea

```

```

SEQUENCE: 49
APSLRDKRSF  VERDGVITYV  FEHAATGAKM EFVQNSGICE TTPGVNQYSG YLSVGDNMNM 60
WFWFFFEARN  PTAAPLAAWF NGGPGCSSMI GLFQENGPCP FVNGESTPSL NEYSFNNYAN 120
VLYVDQPIGT  GFSYGTDDVT  STVTAAPYVW KLLQAFYAQF PEYESRDFGI FTESYGGHYG 180
PEFASYIQEQ  NAAIKAGSVS  GDNINLVALG INNGWFDAGI QEKAYIDFSY NNSYRQIIS 240
SQRSSYLDAY  NHDCLPAIES  CASSGTSNAC KNAESVCYNG IEGPISSAAD FDVYDVQRPS 300
NDPYPPATYS  TYLQASASVR  AIGARTKYQE CPNGPYNKFE TTGDNSRSFL STLSDVVNTG 360
ITVLVWAGDA  DWICNWVGGH  AVADAVTFAR QKTFQAKPLE PYTVNGTEKG RFKTVDNFTF 420
LRVYEAGHEV  PYYQPETALQ  VFVQTMQKKA IFST 454

```

```

SEQ ID NO: 50      moltype = AA length = 453
FEATURE           Location/Qualifiers
source            1..453
                  mol_type = protein
                  organism = Talaromyces variabilis

```

```

SEQUENCE: 50
AAVPQDKRSI  VKRDGVITYN  FEHAATGAKM EFVKNSGICE TTPGVNQYSG YLSVGDNMNM 60
WFWFFESRIN  ASGAPLAAWF NGGPGCSSMI GLFQENGPCP FVNGEKKPSL NKYSFNNEYAN 120
VLYVDQPIGV  GFSYGTDDVT  STESAAPYVW KLLQAFYAQF PQYESRDFGI FTESYGGHYG 180
PEFAHYLQQQ  NEGYNKGSVD  GENINLVALG INNGWFDTLQ QEGAYIDYAY SNYKKIIDS 240
SQRSSLEDLS  KSDCLPAVKQ  CLSSGSDSDC ENASDTCGQI ESSIQQAADF DVYDVREPSN 300

```

-continued

DPYPPSTYSYD	YLADSSSVVKA	IGAKSTYKEC	PNGPYKPFSS	TGDNTRSFLS	ELSSVVQSGI	360
QVLVWAGDAD	WICNYMGVQR	VADAVEFDGS	SQFSNATLKP	YTVNGTKKGE	YKNVDNFSYL	420
RVYGAGHEVP	YYQPAVALQV	FKQTMQQQAI	KST			453

SEQ ID NO: 51	moltype = AA	length = 456
FEATURE	Location/Qualifiers	
source	1..456	
	mol_type =	protein
	organism =	Penicillium arenicola

SEQUENCE: 51						
APATHLQDKR	SIVERDGVNY	TVFEHAATGA	KLEFVTNSGI	CETTSGVNQY	SGYLSVGTNM	60
NMFWFFESR	NSPSTAPLAA	WFNGGPGCSS	MIGLFQENGP	CQFYDGASTP	SLNPYSFNEY	120
ANMIYDQPI	GVGFSYGTDD	VTSTVTAAPY	VWKLIQAFYA	SFPAYESREF	GLFTESYGGH	180
YGPEFAYYIQ	QQNAAIASGT	VTGDTIDIVA	LGINNGWIDS	ALQEKAYIEY	SYNNSYKQII	240
TSSQRTSYLS	TYTNDCLPAI	NKCTTGGSNS	ACSNAADVCY	NDIESPIMSD	ADFDVYDIRQ	300
PSNDAYPPET	YVTYLQTSV	VKAIGASSTY	QECPDAPYNK	FATTGDNDRS	FLATLSTVVQ	360
SGITVLLWAG	DADWICNVWG	NQYVADAVTW	SGQSSFAAQT	LTPYTVNGSE	VGTFKTLNDL	420
SFLRVYEAGH	EVPPYQPATA	LQAFIQTMQK	KALSSST			456

SEQ ID NO: 52	moltype = AA	length = 354
FEATURE	Location/Qualifiers	
source	1..354	
	mol_type =	protein
	organism =	Nocardiosis kunsanensis

SEQUENCE: 52						
APAPQNPTPE	AEATTMAEAL	ERDLGLNEAE	ATDLIDAEQES	ALDVAEATE	AAGEHYGSSL	60
FDTETHDLTV	LVTDSAAVPG	VEAAGAEAAV	VEHGVLEGLD	LISDLDSAGA	QEGVVGWYPE	120
VENDTVVIET	LEGADADVDA	LLSSAGVDDA	DVRVETTDEA	PEVYANIVGG	DAYTIGSSSR	180
CSVGPPASDS	YQGPGFVTAG	HCGTTGSSVS	IGNSGGVFSQ	SVFPGNDAAF	VRGTSNFSLT	240
NLVNRYNSGS	DVAVSGSTQA	PIGSQVCRSG	STTGWHCGTI	QARGQTVSYP	QGTVRDLTRT	300
SVCAEPGDSG	GSFISGSQAQ	GVTSGGSGNC	SWGTTYQYE	VNPMNLNSWN	NLST	354

SEQ ID NO: 53	moltype = AA	length = 425
FEATURE	Location/Qualifiers	
source	1..425	
	mol_type =	protein
	organism =	Streptomyces parvulus

SEQUENCE: 53						
GTAPSPAAPT	AAESLRADAA	PPALLRAMER	DLGLGREQAE	RRLGNEAEAG	AVAGRLRADL	60
GGDFAGAWVR	GAESGTLTVA	TTDAADVPAI	EARGAVAEV	RHSLADLGAA	KSRLDRAAAH	120
RDTAEAPVRY	VDVRTNTVT	QAVRPSAARA	LLAAAGVDAG	LARVETSAER	PRPLYDLRGG	180
EAYYINNSGR	CSVGFPVTKG	TQQGFATAGH	CGRAGASTSG	ANRVAQGTFO	GSVFPGRDMA	240
WVAANSQWTA	TPYVSGAGGQ	NVQVAGSTQA	PVGASVCRSG	STTGWHCGTI	QHDTSVTYP	300
EGTITGVTRT	TVCAEPGDSG	GSYISGSQAQ	GVTSGGSGNC	GSGGTTFFQP	INPLLQNYGL	360
TLKTTGGGGG	DPGEPGEPGG	TWAAGTVYRP	GDTVTYGGAT	YRCLQGHQAQ	RGWEPANVPA	420
LWQRV						425

SEQ ID NO: 54	moltype = AA	length = 350
FEATURE	Location/Qualifiers	
source	1..350	
	mol_type =	protein
	organism =	Saccharopolyspora endophytica

SEQUENCE: 54						
LTATIADPAG	PPVSPPELTA	MQRDLGLTAD	QAVARLGQEA	VAARADSALR	DALAGSYGGS	60
YFDANLGKLV	VGTTDAAKSD	EVRAAGAEPR	QVDASERQLD	GIVEALNGRG	AQVPAAVTGW	120
YADVRENADV	VTTQPGTAEQ	ATGFVRDAQV	PQESVRVWES	PAQPETYADV	VGGYAYYTAS	180
GARCSMGFAV	QGGFVTAGHC	GAPGESTTQP	TGYFAGSSFP	GNDYAFVNTG	TDDTGYPLVY	240
NYSSGYVRVS	GSAAEAPLGSS	ICRSGSTTGW	HCGTVLAKNQ	SVRYQEGTVS	GLTRTNVCAE	300
PGDSGGSFIS	GNQAQGMTSG	GWGDCRTGGE	TYYPVREAL	SAYGLTLTQ		350

SEQ ID NO: 55	moltype = AA	length = 355
FEATURE	Location/Qualifiers	
source	1..355	
	mol_type =	protein
	organism =	Luteus cellwall

SEQUENCE: 55						
ASGPLPQSPS	PTSDVATTMA	EALERDLNLT	STEAQELLTA	QEAAFEADEA	AAQAAGDAYG	60
GSVFDTETLD	LTMVMTDAAA	VQAVEATGAK	ADVVSYGIDG	LDTIIDDLNE	ADAPEGVVGW	120
YPDIDSDTVV	LEVLEGGAD	VDALLAEAGV	DASAVKVEST	TEQPELYADI	IGGLAYYMG	180
RCSVGFAATN	ASGQPGFVTA	GHCGRVGTQV	TIGNGRGVFE	RSVPFGNDAA	FVRGTSNFTL	240
TNLVSRYNSG	GYATVSGSSV	APIGSSVCRS	GSTTGWRCTG	IQARGQTVTY	PQGTIYNMTR	300
TSACAEPGDS	GGSFISGTQA	QGVTSGGSGN	CSWGGTTFYQ	EVNPMNLNSW	LRLRT	355

SEQ ID NO: 56	moltype = AA	length = 406
FEATURE	Location/Qualifiers	
source	1..406	
	mol_type =	protein
	organism =	Saccharothrix australiensis

SEQUENCE: 56						
--------------	--	--	--	--	--	--

-continued

GPPTTHQEEES	GLIAAMARDF	KITPDQARAR	LVREAKAATT	BQSLKSRLGG	HYAGAWLNEG	60
ATELVVAVD	AAQAKVVEDA	GATPKVVQRS	QIQLDELKAK	LDANKNAPKD	VPAWYVDVKT	120
NSVVVLARNT	ASAKAFARAS	GLSEADVRIE	QSTEDPRPLI	DVIGGNAYYM	SGGRCRCSVG	180
SVNGGFVTAG	HCGRVGTITT	QPSGTFAGST	FPGRDYAWVR	VSSGNTMRGL	VNRYPGTVPV	240
KGSNESSVGA	SVCRSGSTTG	WHCGTIQQKN	TSVTYPEGTI	SGVTRTNACA	EPGDSGGSWL	300
TGDQAQGVTS	GGSGNCSSGG	TTYFQPVNPI	LQAYGLQLVI	EGGPTGTGTP	TTTSSNPGGT	360
TWQPGVAYTA	GTTVTYEGVG	YECLQGHTSQ	IGWEPSAVPA	LWERVVG		406

SEQ ID NO: 57 moltype = AA length = 346
 FEATURE Location/Qualifiers
 source 1..346
 mol_type = protein
 organism = Nocardiosis baichengensis

SEQUENCE: 57

DAFPEGTEPL	AEAIEDLGV	ASGQADELLT	AEESARSLEK	EAKEAAGEAF	AGAVFDTETH	60
ELTVSVADPS	AVEAVEATGA	ETRVVEASQD	ELDAAMADLD	AASEDGVSEE	VTGWHVDLES	120
NTVVVEALEG	SEDAAEPLIA	DAGLDSAPVV	VEKADAQPET	FGAIVGGDAY	YPGNSRCSIG	180
FSVRGGFVTA	GHCSTGTSTV	SGSAGESGRV	AGSVFPGRDM	GYVRANSQWT	PSPYVNNYRG	240
GRVAVRGSNE	ASVGASVCRS	GSTTGWHCCT	IQAQNTQVNY	PQGTVRGLTR	TTACAEPGDS	300
GGSWLSGNQA	QGVTSGGSGN	CSWGGTTFQ	PVNPILSQWG	LSLT		346

SEQ ID NO: 58 moltype = AA length = 353
 FEATURE Location/Qualifiers
 source 1..353
 mol_type = protein
 organism = Streptomyces sp.

SEQUENCE: 58

NDTLTERADA	AVAEPLAGVL	DAMERDLGLS	EQEAGLQLVA	QYDASLLGET	LSADLDAYAG	60
SWLADGTDLV	VATTDRAEAA	QITEAGAKVE	IVDHTLLELE	SVKAALDEAA	ESYDITDAPV	120
WYVDITNDV	VLLTSDTAEA	KGFVEAAGVD	AGAVSIQTS	BQQAQFYDLV	GGDAYYMGGS	180
RCSVGFSVTQ	GSTPGFATAG	HCGTVGTSTT	GFNQAAQGT	EESFPQDDM	AWSVNSNWN	240
TTPTVNDGAV	TVSGSTQGA	GASICRSGST	TGWHCGTIEQ	HNTSVTYPEP	TITGVTRTSV	300
CAEPDGS	YISGSAQGV	TSGSGNCT	GGTTYHQPIN	PLLSAYGLDL	VTG	353

SEQ ID NO: 59 moltype = AA length = 353
 FEATURE Location/Qualifiers
 source 1..353
 mol_type = protein
 organism = Actinoalloteichus spitiensis

SEQUENCE: 59

DTPSPDGADA	TVASPEMLSA	MQRDLGLTEQ	EALTRVAVEA	TAVETEDELR	ASLGPAPFGGA	60
HPDGDNTLV	VGVTSAKAD	EVRAAGATPE	VVAFSADTLD	GVVSTLNETS	EVPDGVGTWY	120
VDTADNTVVV	TTALGSGEAA	ADPVAESGVN	ADAVTVVEST	EQPRTLYDII	GGDAYYFGGS	180
RCSVGFSVSV	GYVTAGHCGG	VGTATQGYNR	VSSGQVAGSV	FPQSDMGYVR	TNANWTPRPL	240
VNRYSGGATV	TVSGSNEAAV	GASICRSGST	TGWRCTVQA	KNQTVFYAQ	AVSGLTRTNA	300
CAEGDGS	WLSGSAQGV	TSGSGNCTW	GGTTYFQPLN	PILSRWGLSL	TRG	353

SEQ ID NO: 60 moltype = AA length = 338
 FEATURE Location/Qualifiers
 source 1..338
 mol_type = protein
 organism = Byssoschlamys verrucosa

SEQUENCE: 60

FPAAVDVKRA	PSSLGITLSQ	VSNTLIKAVV	QNTGRGEVSF	IHLNFFKDDA	PVKKVAVYRN	60
GSEVQFEGIG	RRYKSTGLTR	DAFTTLAPGK	TAEDVFDIAS	TCDLISGGPV	TIRSEGVVPY	120
ATANGIDIAG	YIPYSSNELT	IDVDGAIAS	VSKAIAPLNR	RTNISSCSGS	EQSTLTMAK	180
NAASLAHAAA	DAAESGSASK	FSEYFKTTAS	STRKTVAARL	RAVAQEASSS	SSGSTTYCN	240
DAYGYCTTNV	LAYTLPSHNT	IATCDLYYTN	LSALTRTCHA	QDQATTSLHE	FTHAPGVYSP	300
GTDDLAYGYA	SSTSLSSSQ	VMNADSYALY	ANAIYVGC			338

SEQ ID NO: 61 moltype = AA length = 339
 FEATURE Location/Qualifiers
 source 1..339
 mol_type = protein
 organism = Hamigera terricola

SEQUENCE: 61

SPVNVNNGRE	ELPALDVTLS	QIGNTQIKAV	VKNTGSEDVT	FMHLNFFTDS	APVKKVSVPQ	60
NNTEVEFQGI	LRRVKYTDVS	TDSTVTLAPG	ASIEDVFDIA	TTTDLASGGA	VTVKTDFGVP	120
ILASAENKVT	GYARYTSNEL	HLDVDGPSAA	TVSKAIAPLD	RRTRLSSCSG	SRSSALQAL	180
RNTVSLANAA	ANAARSGSAS	KFSEYFKTTS	SSVRSTVAAR	LSAVASEASS	TSSGSTTYYC	240
NDPYGYCSTD	VLAYTLPSYN	IIANCDIYYS	YLPALTGSCH	AQDQATTTLH	EFTHAPGVYS	300
PGTETYGYGY	NAATSLSSSQ	AVLNADSYAL	YANAIYLC			339

SEQ ID NO: 62 moltype = AA length = 334
 FEATURE Location/Qualifiers
 source 1..334
 mol_type = protein
 organism = Aspergillus tamarii

SEQUENCE: 62

-continued

```

IPVEVPASAP  GLDVTLQVQV  NTRIKAVVKN  TGSEEVTFVH  LNPFKDAAPV  QKVSLEFRNAT  60
EVQFQGIKQR  LITEGLSDEA  LTTLAPGATI  EDEFDIASTS  DLSEGGTITI  NSNGLVPITT  120
ENKVTGYIPF  ASNELSIDVD  AAEAATVSQA  VKILDRRTKV  TSCSGSRSSA  LQTALRNTVS  180
LARAAAASAAQ  SGSSSRFQEY  FKTTSSSTRS  TVAARLNAVA  NEAASTASGS  TTYYCSDVYG  240
YCSSNVLAYT  LPAYNIIANC  DLYYSYLPAL  TSTCHAQDQA  TTTLHEPHTA  PGVYSPGTTD  300
LGYGYSATA  LSASQALLNA  DTYALFANAV  NLNC  334

```

```

SEQ ID NO: 63      moltype = AA  length = 334
FEATURE            Location/Qualifiers
source             1..334
                   mol_type = protein
                   organism = Aspergillus niveus

```

```

SEQUENCE: 63
LPAKTGEQLQ  KLDVALSQVD  NTLIKAVVKN  TGSEDTFVH  LNFFRDTAPV  KKVSLFRNTT  60
EVPFHGIKQR  LRSGLSADA  LTVLAPGESI  EDEFDIAATS  DLSEGGTITI  SADGFVPIAS  120
GKNITGYVPF  SSNELSVEVD  AAQAASVASA  VKPLDRRTKV  ASCSGSRSSA  LTQALRNTVS  180
LANAAAASAAQ  SGSSSTRFQEY  FKTTSSSVRS  SVAARFRAVA  SEASSTSAGS  TTYYCTDVYG  240
YCSSNVLAYT  LPAYNIIANC  DIYYTYLPAL  TSTCHAQDQA  TTTLHEPHTA  PGVYSPGTTD  300
LGYGYDAATA  LSSSQALLNA  DSYALFANAV  NLNC  334

```

```

SEQ ID NO: 64      moltype = AA  length = 372
FEATURE            Location/Qualifiers
source             1..372
                   mol_type = protein
                   organism = Penicillium sclerotiorum

```

```

SEQUENCE: 64
IPTGGKKSSF  SVDQVAIPAT  KTKNFADTYA  RAISKFGGNV  PSHVRAAAQQ  SGAATTTPEA  60
NDEEYLTPVN  VGGTTLNLDF  DTGSADLWVF  SEQLPSSEQS  GHSVYKPNMG  TKLSGATWSI  120
SYDGGSSASG  DVYKDTVSVG  PVKATGQAVE  AASKISAQFT  RDSNNDGLLG  LAFSSINTVK  180
PKAQTTFPDT  VKSSLASPLF  AVTLKHNAPE  TYDFGFVDSS  KYTGSLAYTD  VDNSQGFWEF  240
TADSYKVGSG  GSSSIKGIAD  TGTTLLLDLD  EVVSAYYKQV  SGASDSQSAG  GYTFDCSADL  300
PDFTVTISGY  DAVVPGSLIN  YAPVSDGSST  CLGGIQSNSG  IGFSIFGDIF  LKSQYVVFDS  360
NGPRLGFAAQ  SS  372

```

```

SEQ ID NO: 65      moltype = AA  length = 371
FEATURE            Location/Qualifiers
source             1..371
                   mol_type = protein
                   organism = Penicillium bilaiae

```

```

SEQUENCE: 65
VPTGGKKSPS  INQVAIPATK  TKNFAGNYAH  AIAKYGGNVP  SHVEAAAQQS  GAATTTPESN  60
DEEYLTPVNV  GGTTLNLDFD  TGSADLWVFS  AELPSAEQSG  HALYKPSNGT  KLSGASWSIS  120
YDGGSSASGD  VYKDTVSVGS  VKATGQAVEA  ASKISAQFTK  DKNNDGLLGL  AFSSINTVKP  180
KAQTTFPDTV  KSSLASPLFA  VTLKHNAPE  YDFGFIDKSK  YTGSLAYADV  DNSQGFWEFT  240
ADSYVSGSSK  GSSIKGIADT  GTTLLLLDDE  VVSAYYKQVQ  GAQQDSSAGG  YTFDCSSKLP  300
DFTVTISGYD  AVVPGDLINF  APASEGSSTC  LGGIQSNSGI  GFSIFGDIFL  KSQYVVFDSN  360
GPRLGFAAQS  S  371

```

```

SEQ ID NO: 66      moltype = AA  length = 373
FEATURE            Location/Qualifiers
source             1..373
                   mol_type = protein
                   organism = Penicillium antarticum

```

```

SEQUENCE: 66
SPLVTPRKGF  TINQETRAVT  KSKTVNLPGV  YAQALSKYGA  TVPQHVHAAA  VSGSAVTTPE  60
ESDVEYLTPV  NVGGTTLNLD  FDTGSADLWV  FSSELTSSQQ  SGHSIYKPSS  SATKLSGSSW  120
SISYDGGSSA  SGDVIKDTVT  VGGVKATGQA  VEAASKISSA  FLQDVNNDGL  LGLAFSSINT  180
VSPRAQTTF  DTVKSQDLS  LFAVTLKHNA  PGSYDFGYID  KSKYTGSLTY  ANVDDSQGF  240
SFTASSYKIG  TTTGGSITGI  ADTGTTLLLL  PDSVVSAYYK  KVSQSQNSNY  YGGYVFPSCA  300
TLPDFTVTIN  GYNAVVPGNL  INFAQATTGS  STCYGGIQSN  SGISFSGIFD  IFLKSQYVVF  360
DSEGPRLGFA  AQA  373

```

```

SEQ ID NO: 67      moltype = AA  length = 370
FEATURE            Location/Qualifiers
source             1..370
                   mol_type = protein
                   organism = Penicillium sumatrense

```

```

SEQUENCE: 67
VPTNNVASKF  SVNQVSRPAT  KTTNFAANYG  RALSKYGAGV  PSHVEAAAAA  SGSAVTTPE  60
NDVEYLTPVS  IGGTTLNLDF  DTGSADLWVF  STELSSEQS  GHSVYNPSKS  GKKISGASWD  120
ISYDGGSSA  GDVYTDVTV  GGVTASKQAV  EAAKQISSQF  QQDNDNDGL  GLAFSSINTV  180
SPTPQKTFD  NVKSSLSQPL  FAVALKHNAP  GYVDFGFIDS  SKHTGSIAYT  SVDSSQGFWS  240
FTVDGYKVG  KSGAGFDGIA  DTGTTLLLLD  DSVVSAYYSQ  VSGAKNDNNA  GGYVFDSCAD  300
LPDFSVTIGS  YTATVPGSLI  NYGDSGDNCS  IGGIQSNSGI  GFSIFGDIFL  KSQYVVFNAN  360
GPKLGFAPQA  370

```

```

SEQ ID NO: 68      moltype = AA  length = 384
FEATURE            Location/Qualifiers
source             1..384

```


-continued

```

mol_type = protein
organism = Trichoderma lixii

SEQUENCE: 68
LPTEGQKTAS IEVTYNKNYV AHGPTALFKA KRKYGAPISD NLRAAVAACH SLTKRQTGSA 60
NTNPSDSADD EYITNSVIGT PAQVLPDLDG TGSSDLWVFS SETPKSSASG HVTYSPSKSS 120
TAKKLSGSTW SITYGDHSSS SGDVTYDVVS IGGFSVKTQA IESATKVSTQ FVQDTVISGL 180
VGLGPDVGNL VKPRAQKTWF SNAASSLAEP LFTADLRHQE TGSYNFGFID NSLAKGTIGY 240
TPADGSEGYW GFTATGYSVG GAKLGRSSIT GIADTGTLL LLLPDNVVDAY YNNVESAQYD 300
DSQEGVVFDG SEDLPSFSFG VGGQTITISG DLLNLTPIEE GSSTCFGGLQ SSADIGINIF 360
GDVALKAALV VFDLGNERLG FAQK 384

SEQ ID NO: 69      moltype = AA length = 384
FEATURE            Location/Qualifiers
source             1..384
                   mol_type = protein
                   organism = Trichoderma brevicompactum

SEQUENCE: 69
LPTEGQKTAS VEVYTNQNYA AHGPTQLYKA KRKYGAPISD NLKAIVANRK ALIKRQTGSA 60
PNHPDSADD EYITNSVIGT PAQVLPDLDG TGSSDLWVFS SETPKSSASG HTIYTPSKSS 120
TSKKLSGATW SIEYGDKSTS SGDVTYDKVT VGGFSVSTQA VESATKVSQA FVQDTANSGL 180
LGLAFDSINT VSPRQQTWF SNAANSLAQF LFTANLNHQA TGSYNFGFID TSLASGPINY 240
VPVDNSQGFQ GFTASGYSVG GGKLNRSLSL GIADTGTLL LLLPDVAVNAY YANVESAEYD 300
DEQEGVVFDG SEDLPTFSFG VSGGTITIPG DLLNLTPIDS SGQTCYGGGLQ SSSDIGINIF 360
GDVALKAALV VFDLGNERLG WAQK 384

SEQ ID NO: 70      moltype = AA length = 379
FEATURE            Location/Qualifiers
source             1..379
                   mol_type = protein
                   organism = Penicillium cinnamopurpureum

SEQUENCE: 70
IPTGVPNRKG FTVNQQVRPV TNGTKSKTLN LPAIYANALS KYGVAVPANI KAAAESGTAT 60
TTPENDDIEY LTPVDVGGTT LNLDFDFTGSA DLWVFSSELP SSESSEHSIY KPSQSGKKLD 120
GYSWKISYGD SSSASGDVYT DTVTVGGVTA DGQAVEAAKK ISQQFVQDKN NDGLLGLAFS 180
SINTVQPKAQ TTFPDTVKDQ LDSPLFAVTL KHNAPGSYDF GFIDKSKYTG SLTYADVDSK 240
DGFWAFATAD YSVSGSSSSS SRIKGIADTG TLLLLIDDEI VSAYYKQVDG AQESYSVGGY 300
TFDCSTKLDP FNIKIGDYTA VIPGDVINYA PVQGSSTCF GGIQNSGLP PSIFGDIFLK 360
SQYVVFDANG PRLGFAAAQ 379

SEQ ID NO: 71      moltype = AA length = 350
FEATURE            Location/Qualifiers
source             1..350
                   mol_type = protein
                   organism = Bacillus lichenformis

SEQUENCE: 71
AQPKNVEKD YIVGFKSGVK TASVKKDIIK ESGGKVDKQF RIINAAKAKL DKEALKEVK 60
DPDVAYVEED HVAHALAQTV PYGIPLIKAD KVQAQGFKA NVKVAVLDTG IQASHPDLNV 120
VGGASFVAGE AYNTDGNHGH THVAGTVAAL DNTTGVLGVA PSVSLYAVKV LNSSGSGSYS 180
GIVSGIEWAT TNGMDVINMS LGGASGSTAM KQAVDNAYAR GVVVVAAGN SGSSGNTNTI 240
GYPAKYDSVI AVGAVDNSN RASFSSVGAE LEVMPAGAGV YSTYPTNTYA TLNGTSMASP 300
HVAGAAALIL SKHPNLSAQ VRNRLSSTAT YLGSSFYGK GLINVEAAAQ 350

SEQ ID NO: 72      moltype = AA length = 360
FEATURE            Location/Qualifiers
source             1..360
                   mol_type = protein
                   organism = Bacillus subtilis

SEQUENCE: 72
FSNMSAQAG KSSTEKKYIV GFKQTSAMS SAKKKDVISE KGGKVQKQFK YVNAATLD 60
EKAVKELKKD PSVAYVEEDH IAHEYAQSVY YGISQIKAPA LHSQGYTGSN VKVAVIDSGI 120
DSSHDPDLNR GGASFVPSET NPYQDGSSHG THVAGTIAAL NNSIGVLGVA PSASLYAVKV 180
LDSTGSGQYS WIINGIEWAI SNNMDVINMS LGGPTGSTAL KTVVDKAVSS GIVVAAAGN 240
EGSSGSTSTV GYPAKYPSI AVGAVNSSNQ RASFSSAGSE LDVMPAGVSI QSTLPGGTYG 300
AYNGTSMATP HVAGAAALIL SKHPTWTNAQ VRDRLESTAT YLGSSFYGK GLINVQAAAQ 360

SEQ ID NO: 73      moltype = AA length = 548
FEATURE            Location/Qualifiers
source             1..548
                   mol_type = protein
                   organism = Trametes cf versicol

SEQUENCE: 73
TPTARNLKLH ESREEIPAGF SLGGAASPD TTKLRLALVQ SNFAELEDKL YDVSTPSSAN 60
YQGHLSKEEV EQLVAPSAES VNAVNAWLTE NGLTAQTISP AGDWLAFEPV VSKANELFDA 120
DFSVPETHDS GLKAVPRLAY SIPAELQGHG DLVHPTITFP NPNSHLPVVR SPVKPVQNL 180
SRAVPASCAS TITPAQLAL YGIPTTKATQ SSNKLAVSGF IDQFANSADL KTFGLGKFRD 240
ISSSTTFTLQ TLDGGSNSQS SSQAGVEANL DIQYTVGLAS AVPTIFISVG DDFQDGDLEG 300
FLDIINFLN ESAPPQVLTT SYGQNTNIS AKLANQLCNA YAQLGARGTS ILFASGDGGV 360
SSQSSSSCK FVPTFFSGCP FMTSVGATQG INPETAADF SGGFSNVFAR PSYQSTAVSS 420
YLTALGSTNS GKFNSTGRAF PDIAATQGVDF EIVVSGRTEG VDGTSASAPT LAIIISLLND 480

```

-continued

RLIAAGKSPL GFLNPFLYSA AGTAALTDIT SGSNPGCNTN GFPAKAGWDP VTGLGTPNFA 540
KLLTAVGL 548

SEQ ID NO: 74 moltype = AA length = 439
FEATURE Location/Qualifiers
source 1..439
mol_type = protein
organism = Bos taurus

SEQUENCE: 74
MAKEYFPFTG KIPFEGKDSK NVMAFHYEPE EKVVMGKKMK DWLKFAMAWW HTLGGASADQ 60
FGGQTRSYEW DKAECVPQRA KDKMDAGFEI MDKLGIEYFC FHDVDLVEEA PTIAEYEEEM 120
KAITDYAQEK MKQFPNIKLL WGTANVFGNK RYANGASTNP DFDVVARAIV QIKNSIDATI 180
KLGGTNYVWF GREGYMSLL NTDQKREKEH MATMLGMARD YARAKGFKGT FLIEPKPMPE 240
SKHQYDVTDE TVIGFLKAHG LDKDFKVNIE VNHATLAGHT FEHELACAVD AGMLGSIDAN 300
RGDAQNGWDT DQFPIDNFEL TQAMLEIIRN GGLGNGGTNF DAKIRRNSTD LEDLFIAHIS 360
GMDAMARALM NAADILENSE LPAMKKARYA SPDSGIGKDF EDGKLTFEQV YEGKKVEEP 420
KQTSKGQEKY ETIVALHCK 439

SEQ ID NO: 75 moltype = AA length = 591
FEATURE Location/Qualifiers
source 1..591
mol_type = protein
organism = Saccharomyces cerevisiae

SEQUENCE: 75
MLCSVIQRQT REVSNMTSLD SYLLGFDLST QQLKCLAINQ DLKIVHSETV EFEKDLPHYH 60
TKKGVYIHGD TIECPVAMWL GALDLVLISKY REAKFPLNKV MAVSGSCQQH GSVYSSQAE 120
SLLEQLNKKP EKDLLHYVSS VAFARQTAPN WQDHSTAKQC QEFEECIGGP EKMAQLTGSR 180
AHFRPTGPQI LKIAQLEPEA YEKTKTISLV SNFLTSLVQ HLVELEEADA CGMNLVDIRE 240
RKFMVELLHL IDSSSKDKTI RQKLMRAPMK NLIAGTICKY FIEKYGFNTN CKVSPMTGDN 300
LATICSLPLR KNDVLVSLGT STTVLLVTDK YHPSPNYHLF IHPTLPNHYM GMICYCNGSL 360
ARERIRDELN KERENNYEKT NDWTLFNQAV LDDSESSENE LGVYFPLGEI VPSVKAINKR 420
VIFNPKTGMI EREVAKPKDK RHDKNIVES QALSCRVRIS PLLSDSNASS QQRLNEDTIV 480
KFDYDESPLR DYLNKRPERT FVGGGASKND AIVKKFAQVI GATKGNFRLE TPNSCALGGC 540
YKAMWSLLYD SNKIAVPFDPK FLNDNFPWHV MESISDVNE NWIAIIPRLS P 591

SEQ ID NO: 76 moltype = AA length = 444
FEATURE Location/Qualifiers
source 1..444
mol_type = protein
organism = Bacillus subtilis

SEQUENCE: 76
ETANKSNELT APSIKSGTIL HAWNWSFNTL KHNMKDIHDA GYTAIQTSPi NQVKEGNQGD 60
KSMNWWYWLQ ISPVYENIPD YLGTQEAFKE MCAAEEYGI KVIIDAVINH TTFDYAAISN 120
EVKSIPNWTN GNTQIKNWSR RWDVTQNSLL GLYDWTQNT QVQSYLKRFL ERALNDGADG 180
PRFDAKHIE LPDDGSYGSQ FWPNTINTSA EPQYGEILQD SASRDAAYAN YMDVTASNYG 240
HSIRSALKNR NLGVSNIISHY AYDVSADKLW TWVESHDTYA NDDEESTWMS DDDIRLGWAV 300
IASRSGSTPL FFSRPEGGGR GVRFPKGSQI GDRGSALFED QSITAVNRPH NVMAQPEEL 360
SNPNGNNQIF MNQRGSHGVV LANAGSSSVS INTPTKLPDG RYDNKAGAGS FQVNDGKLTG 420
TINARSAVL YPDDIEIRCN TFFQ 444

SEQ ID NO: 77 moltype = AA length = 476
FEATURE Location/Qualifiers
source 1..476
mol_type = protein
organism = Saccharomycopsis fibuligera

SEQUENCE: 77
QPVTLFKRET NADKWSQSI YQIVTDRFAR TDGDTASACN TEDRLYCGGS FQGIKKLDY 60
IKDMGFTAIW ISPVYENIPD NTAYGYAYHG YWMKNYKIN ENFGTADDLK SLAQELHDDR 120
MLLMVDIVTN HYGSDGSGDS IDYSEYTPFN DQKYFHNLYC ISNYDDQAV QSCWEGDSSV 180
ALPDLRTEDS DVASVFNSWV KDFVGNYSID GLRIDSARKV DQGFPPDFVS ASGVYSVGEV 240
FQGDPAVTCF YQNYIPGVSN YPLYYPTTRF FKTDSSESSE LTQMISSVAS SCSDPDLLTN 300
FVENHDNERF ASMTSDQSLI SNAIAFVLLG DGIPVIYYGQ EQGLSGKSDP NNREALWLSG 360
YNKESDYKYL IAKANAARNA AVYQDSSYAT SQLSVIFSND HVIATKRGSV VSVFNGLGSS 420
GSDVVTISNT GYSSGEDLVE VLTCTVSGS SDLQVSIQGG QPQIFVPAKY ASDICS 476

SEQ ID NO: 78 moltype = AA length = 487
FEATURE Location/Qualifiers
source 1..487
mol_type = protein
organism = Debaryomyces occidentalis

SEQUENCE: 78
QPIIFDKRDV GSSADKWKDQ SIYQIVTDRF ARSDGSTTAD CLVSDRKYCG GSYKGIIDKL 60
DYIQGMGFTA IWISPVVEQI PDNTAYGYAY HGYWMKNIDE LNTNFGTAD LKQLASELHS 120
RSMMLMVDVW YNHVYAWNGD SSVYSSFTF FNQQSYFHDY CLITNYNDQT NVEDCWEGDT 180
EVSLPDLSTE DNEVIGVFQT WVSDFVQNYI IDGLRIDSAR HVDTASLTKF EDASGVYNLG 240
EYVQGDPTT CPYQSYMKGV TNYPLYPVY RFFSDTSATS SELTSMISTL QSSCDVSL 300
GNFIENHDQV RFPVSTSDTS LIKNAMAFII LGDGIPIIYY GQEQGLNGGS DPANREALWL 360
SGYNTDSEY ELISKLNQIR NQAIKKDSAY STYKSSVSS SDHYIATRK GSDANQLISIF 420
NNLGSNGSQD ITVNTGYSS GDKVIDIISC NSVSAGDFGS LSVSISGMP QVYAPSSVLS 480

-continued

 GSGICNQ 487

SEQ ID NO: 79 moltype = AA length = 487
 FEATURE Location/Qualifiers
 source 1..487
 mol_type = protein
 organism = *Debaryomyces occidentalis*

SEQUENCE: 79
 KPIFLSKRDA GSSAAAAWRS ESIYQLVTDR FARTDGTSTA TCNTGDRVYC GGTFQGIIDK 60
 LDYIQGMGFT AIWISPVVEQ IPDDTGYGYA YHGYWMKDIY AINSNFGTAD DLKNLSNELH 120
 KRNMKLMVDI VTNHYAWNGA GSSVAYSNNY PFNQQSYFHD YCLITNYDDQ TNVEDCWEGD 180
 NTVSLPDLRT EDSDVSSIFN LWVAELVSNY SIDGLRIDA KHVDESFPYS FQSAAGVYLL 240
 GEVYDGPAY TPCYQNYMSG VTNYPYYPM LRFFQGTNS VDELNAMISS LESDCKDITL 300
 LGNFIENHDQ PRLPSYTSDS ALIKNAIAFN LMSDGIPIIY YGQEQGYSGS SDPNNREALW 360
 LSGYSTSNY YKLISVNQI RNQAIYKDSK YTTYWSDVLY ASGHVIALQR GADDQRIVSV 420
 FNNLGSSGSQ TVTFSTKYSK GEKVVDVLTQ QTSYANSNST LTVSISGGAP RIYAPASLIA 480
 NSGICNF 487

SEQ ID NO: 80 moltype = AA length = 570
 FEATURE Location/Qualifiers
 source 1..570
 mol_type = protein
 organism = *Lipomyces kononenkoae*

SEQUENCE: 80
 MCGSTLSASL YVYNDDYDKI VTLYYLTSSG TTGSTLALIL PVWSNNWELW TLSAIAAGAV 60
 EITGASYVDS DTSVTYTTSL DLPLTTTSAS VPTGTANWR GRSIYQVVD RFARTDGSIT 120
 YSCDVTDVRY CGGSYRGIIN MLDYIQGMGF TAIWISPIVE NIPDDTGYGY AYHGYWMKDI 180
 FALNTNFGGA DDLIALATEL HNRGMVLMVD IVVNHFAFSG NHADVYSEY FPYSSQDYFH 240
 SPCWITDYSN QTNVEECWLG DDSVPLVDVN TQLDVTKSEY QSWVKQLIAN YSIDGLRIDT 300
 VKHVQMDFWA PFQEAAGIYT VGEVFDGDP YTCPYQENLD GVLNYPVYYP VVSAPQRVGG 360
 SSSLVDMID TLKSECIDTT LLGSFLENQD NRPFPSTSD ESLIKNAIAF TILSDGIPII 420
 YGQEQGLNG GNDPYNREAL WPTGYSTTST FYEYIASLNQ IRNHAIYIDD TYLTQNWVI 480
 YSDSTTIAMR KGFTGNQIIT VLSNLGSSGS SYTLTLSNTG YTASSVVYEI LTCTAVTVDL 540
 SGNLAVPMG GLPRVFYPES QLVGSGICSM 570

SEQ ID NO: 81 moltype = AA length = 476
 FEATURE Location/Qualifiers
 source 1..476
 mol_type = protein
 organism = *Lipomyces kononenkoae*

SEQUENCE: 81
 KTAAEWKELS IYQVITDRFA TTNLTAPDCW IRAYCGGTWK GLERKLDYIQ NMGFDAVWIS 60
 PVIHNIENV TWGFAPHGYW GDDPYRLNEH FGTAADLKSL SDSLHARGMS LMVDVVINHL 120
 ASYTLPOQVD YSLYPAPFNT SSAPFHQPCPI DFSNQSSIED CWLVTEPAPA LVDLKNEDQV 180
 ILDALINSV DLVETYDIDG IRLDTARHVP KPSLAKFQEK VGVFVTGEAL NQSVPYVAQY 240
 QGPLNSAINY PLWYALVDSF MGRTTFDYLE SVVKSEQATF SDAHALTNFL DNQDQPRFAS 300
 YLGDGNGDDV LRDENAATFL FVSGIPVIY YGFEQRFDDG FDPVNREPMW TSGYNTSTPL 360
 YNYLARLNAI RKYAASITGT QVPYSDDTVF LSGGVSHMAM QRGPLVIVLT NVGQHIIDNT 420
 GYTVTGSQFS AGDSLTLVLS CTKVKVVGAN GTFTSPSNGG KARIWIKSKY AGKPCS 476

SEQ ID NO: 82 moltype = AA length = 626
 FEATURE Location/Qualifiers
 source 1..626
 mol_type = protein
 organism = *Bacillus subtilis*

SEQUENCE: 82
 ETANKSNELT APSIKSGTIL HAWNWSFNTL KHNMKDIHDA GYTAIQTSPi NQVKEGNQGD 60
 KSMSNWWYLY QPSTYQIGNR YLGTEQEFKE MCAAEEYGI KVIIDAVINH TTSDYAAISN 120
 EVKSIPNWTG GNTQIKNWS RWDVTQNSLL GLYDWTQNT QVQSYLKRFL DRALNDGADG 180
 FRFDAKHIE LPDDGSYGSQ FWPNTINTSA EPQYGEILQD SASRDAAYAN YMDVTASNYG 240
 HSIRSAKKNR NLGVSINSHY ASDVSADKLW TWVESHDTYA NDDEESTWMS DDDIRLGWAV 300
 IASRSGSTPL FFSRPEGGN GVRFPKGSQI GDRGSALFED QAITAVNRFH NVMAGQPEEL 360
 SNPNNGNQIF MNQRGSHGV LANAGSSSVS INTATKLPDG RYDNKAGAGS FQVNDGKLTG 420
 TINARSAVAV YPDDIAKAPH VFLENYKTGV THSFNDQLTI TLRADANTTK AVYQINNPE 480
 TAFKDGQFT IGKGDPPGKT YTIMLKGTNS DGVTRTEKYS FVKRDPASAK TIGYQNPNNH 540
 SQVNAYIYKH DGSRVIELTG SWPGKPMTKN ABGIYTLTLP ADTLTTNAKV IFNNGSAQVP 600
 GQNQPGFDYV LNLGYNDSGL SGSLPH 626

SEQ ID NO: 83 moltype = AA length = 483
 FEATURE Location/Qualifiers
 source 1..483
 mol_type = protein
 organism = *Bacillus subtilis*

SEQUENCE: 83
 VNGTLMQYFE WYTPNDGQHW KRLQNDAEHL SDIGITAVWI PPAYKGLSQS DNGYGPYDLY 60
 DLGEFQQKGT VRTKYGTKE LQDAIGSLHS RNVQVYGDV LNHKAGADAT EDVTAVEVNP 120
 ANRNQETSEE YQIKAWTDFR FPGRGNTYSD FKWHWYHFDG ADWDESARKIS RIFKFRGEGK 180
 AWDWEVSSN GNIDYLYMAD VDYDNPVVA ETKKGNWYA NELSLDGFRI DAAKHIFSF 240
 LRDWVQAVRQ ATGKEMFTVA EYWQNNAGKL ENYLNKTSFN QSVFDPVPLHF NLQAASSQGG 300

-continued

GYDMRRLLDG	TVVSRHPEKA	VTFVENHDTQ	PGQSLESTVQ	TWFKPLAYAF	ILTRESGYPO	360
VFYGDMYGTK	GTSPEIPSL	KDNIEPILKA	RKEYAYGPQH	DYIDHPDIVG	WTREGDSSAA	420
KSGLAALITD	GPGGSKRMYA	GLKNAGETWY	DITGNRSDTV	KIGSDGWGEF	HVNDGSVSIY	480
VQK						483

SEQ ID NO: 84 moltype = AA length = 589
 FEATURE Location/Qualifiers
 source 1..589
 mol_type = protein
 organism = Bacillus subtilis

SEQUENCE: 84

MMEYAAIHQ	PFSTDAYSD	GRTVHIKIRT	KKGDADHIRF	IWGDPIEYND	GKWSANEQPM	60
RKIAATEMHD	YWFAEVVPPF	RRLQYAFVVT	DDHEDIFFGS	SGVCPYNEKT	LETIHYYFKF	120
PFVHEADTFQ	APIEWKSTVW	YQIFPERFAN	GREDLSPKNA	LPWGSKDPOV	NDFFGGDLQG	180
IYDKLDYLED	LGVNGIYLT	IFSAPSNHXY	DTLDYFSIDP	HFGDPEIFRT	LVSQHLQGRM	240
RIMLDVFNH	IGSASPQWQD	VVKNGDQSRV	KDWFHHSFP	VTDDNYDRFA	FTADMPKLNT	300
ANPEVQKYL	DIALYWIREF	DIDGWRLDVA	NEVDHVFVKT	FRQAVSTKPK	DVYILGEIWH	360
SAEPWLRGDE	FHAAMNYPFT	EPMIEYFADQ	TISASRMAHR	VNAHLMNGMK	QANEVVMFNLL	420
DSHDTKRLLT	RCRNDKKAR	ALLAFMFAQT	GSPCIYYGTE	IGLDGENDPL	CRKCMVWEKE	480
KQNQDMLQFM	KRLIALRKQE	NTLLTEGHLE	WNLLDDKNDF	ISFSRTLDEK	ILYFFNQGN	540
VVQHSLSREL	NIDRNNKICD	AWTEQPLHYH	DVIAVQPGEF	LILSAAAFV		589

SEQ ID NO: 85 moltype = AA length = 483
 FEATURE Location/Qualifiers
 source 1..483
 mol_type = protein
 organism = Bacillus lichenformis

SEQUENCE: 85

ANLNGTLMQY	FEWYMPNDGQ	HWKRLQNDQA	YLAHGGITAV	WIPPAYKGTG	QADVGYGAYD	60
LYDLGEPHQK	GTVRTKYGTK	GELQSAIKSL	HSRDINVYGD	VVINHKGGAD	ATEDVTAVEV	120
DPADNRNVIS	GEHLIKAWTH	FHPFGRGSTY	SDFKWHWYHF	DGTDWDESRK	LNRIYKFQK	180
AWDWEVSNN	GNIDYLMYAD	IDYDHPDVAA	EIKRWGTWYA	NELQLDGFRL	DAVKHIKFSF	240
LRDWNHVR	KTGKEMPTVA	EYQNDLGLAL	ENYLNKTNFN	HSVFDVPLHY	QPHAASTQGG	300
GVDNRKLLNG	TVVSKHPLKS	VTFVDNHDQ	PGQSLESTVQ	TWFKPLAYAF	ILTRESGYPO	360
VFYGDMYGTK	GDSQREIPAL	KKKIEPILKA	RKQYAYGAQH	DYFDHHDIVG	WTREGDSSVA	420
NSGLAALITD	GPGGAKRMYV	GRQNAGETWH	DITGNRSEPV	VINSEGWGEF	HVNGGSVSIY	480
VQR						483

SEQ ID NO: 86 moltype = AA length = 478
 FEATURE Location/Qualifiers
 source 1..478
 mol_type = protein
 organism = Aspergillus niger

SEQUENCE: 86

ATPADWRSQS	IYFLLTDRFA	RTDGTSTATC	NTADQKYCGG	TWQGIIDKLD	YIQGMGFTAI	60
WITPVTALP	QTTAYGDAYH	GYWQDIYSL	NENYGTADDL	KALSSALHER	GMVLMVDVVA	120
NHMGYDGAGS	SVDYSVFKPF	SSQDYFHPFC	FIQNYEDQTO	VEDCWLGDNT	VSLPDLDTTK	180
DVVKNWYDW	VGSLVSNYSI	DGLRIDTVKH	VQKDFWPGYN	KAAGVYCIGE	VLDGDPAYTC	240
PYQNVMDGVL	NYPIYYPLLN	AFKSTSGSMD	DLNMMINTVK	SDCPDSTLLG	TFVENHNDNR	300
FASYTNDIAL	AKNVAAPFII	NDGIPIIYAG	QEQHYAGGND	PANREATWLS	GYPTDSELYK	360
LIASRRAIRN	YAIKDTGFV	TYKNWPIYKD	DTTIPMRKGT	DGSQIVTILS	NKGASGDSYT	420
LSLSGAGYTA	GQQLTEVIGC	TTVTVGSDGN	VPVPMAGGLP	RVLYPTEKLA	GSKICSSS	478

SEQ ID NO: 87 moltype = AA length = 477
 FEATURE Location/Qualifiers
 source 1..477
 mol_type = protein
 organism = Aspergillus niger

SEQUENCE: 87

ATPADWRSQS	IYFLLTDRFA	RTDGTSTATC	NTADQKYCGG	TWQGIIDKLD	YIQGMGFTAI	60
WITPVTALP	QTTAYGDAYH	GYWQDIYSL	NENYGTADDL	KALSSALHER	GMVLMVDVVA	120
NHMGYDGAGS	SVDYSVFKPF	SSQDYFHPFC	FIQNYEDQTO	VEDCWLGDNT	VSLPDLDTTK	180
DVVKNWYDW	VGSLVSNYSI	DGLRIDTVKH	VQKDFWPGYN	KAAGVYCIGE	VLDGDPAYTC	240
PYQNVMDGVL	NYPIYYPLLN	AFKSTSGSMD	DLNMMINTVK	SDCPDSTLLG	TFVENHNDNR	300
FASYTNDIAL	AKNVAAPFII	NDGIPIIYAG	QEQHYAGGND	PANREATWLS	GYPTDSELYK	360
LIASRRAIRN	YAIKDTGFV	TYKNWPIYKD	DTTIPMRKGT	DGSQIVTILS	NKGASGDSYT	420
LSLSGAGYTA	GQQLTEVIGC	TTVTVGSDGN	VPVPMAGGLP	RVLYPTEKLA	GSKICYG	477

SEQ ID NO: 88 moltype = AA length = 431
 FEATURE Location/Qualifiers
 source 1..431
 mol_type = protein
 organism = Streptomyces avermitilis

SEQUENCE: 88

SPPGTDVTA	VLFEWKFDV	ARECTNTLGP	AGYGYVQVSP	PAEHIQGSQW	WTSYQPVSYK	60
IAGRLGDATA	FQNMINTCHT	AGVKVVVDTV	VNHMSAGSGT	GTGGSAYTKY	NYPGLYSSYD	120
MDDCTATITD	YTNRANVQNC	ELVGLADLDT	GEEYVRKTIA	GYMNTLLGYG	ADGFRVDAVK	180
HIPAADLANI	KSRLTNPSVY	WKQEVYASG	EAVQPTETYG	NGDVQEFRYA	YDLKRVFNNE	240
NLAYLKNYGE	GWGYLNSSVA	GVFVDNHDTE	RNGSTLNYKD	GANYTLNANV	MLAYPYGAPD	300

-continued

INSGYEWSDA	DAGPPGGGTV	NACWQDGWKC	QHAWPEIKAM	VAFRNATRGE	SVTNWWDNGG	360
DAIAFGRGAK	GYVAINHESG	SLTRTYQTSL	TAGTYCNVQN	NTGVTVDSSG	RFTATLGANT	420
ALALYSGKST	C					431

SEQ ID NO: 89 moltype = AA length = 643
 FEATURE Location/Qualifiers
 source 1..643
 mol_type = protein
 organism = Clostridium phytofermentans

SEQUENCE: 89

MYTLKSKLRD	LYRHPVGVDV	INKILLQAGL	SKGLIENPVI	GALPLSFLNR	IAGKKLGNF	60
FDALLALLNQ	SNDRLDPYSS	KDKKITPTWW	KEAVFYQIYP	RSPMDGNGDG	VGDLPGIISK	120
LDYLKELGVD	ALWLSPIYDS	PGDDNGYDIR	DYQKIDSQFG	TMEDFDLLLT	ELHARNMLV	180
MDLVVNHTSD	EHHWFKEALK	SSESTYRDY	FLRKEPNNWT	SFFSGSAWNH	YPEEDLWGLH	240
LFSKKQMDLN	WENPKLRQDI	YQMIRWWLEK	GVDGFRLDVI	NYISKETGLP	DGDSFIGNLM	300
GFTGIEHYFY	GPKLHNHLQE	IQKEAPTYPQ	AFSVGETPGI	GKMKGKLLTD	DSRGELNMMF	360
SFDHLETSGH	ARFDQYEDL	NYKSYIMDW	MENFADTSWM	SLFYDNHNDP	RMLSKVDHHT	420
THRQELAKML	AMIQMTLKG	PFLYQQQELG	MINKDFHEIS	NFRDVESINK	YKELCEKMPK	480
EEAFLQILAG	WSRHARTPMQ	WSAKPGCGFS	NAVPIWSDG	DELVCNAEIQ	MQDSESVLSF	540
YRDLIALRRK	TPALIYGDI	FTHKKRKIL	IYTRYLEGET	YLIICNLSND	EQKLPGNVVP	600
SESLEGLESL	SASADERKGL	VLCNYPKVM	KSLRAYEGRV	YRI		643

SEQ ID NO: 90 moltype = AA length = 1016
 FEATURE Location/Qualifiers
 source 1..1016
 mol_type = protein
 organism = Clostridium phytofermentans

SEQUENCE: 90

ATDTITIHYH	RDDGDYKWN	LWLWAEKDG	AAYYFDGEDA	PGPYVSVSLD	KSADRIGFIV	60
RTDSWEKDV	EDRFIDTSLG	DEIWISSGES	TFSYEAPEGY	EKEVSIESFQ	LKLNLYRYDE	120
EYTDISFRLT	FEDGTTDFLT	KEHMRINGI	LKAEKEVKYG	KKITLDVLKN	GLEEDYQGV	180
FSTAKIDEES	KLEMYWMQGT	GTISPADFI	KRSKEIESAL	ITSMKEITVK	LSVPCRVDI	240
KQDQKLSPK	LAVSKVEATS	TRDSEYKTIK	EGYADTFIIT	MEEPLDMSKK	YALSKTDYGS	300
RNLTLDSGLY	TSEEFEAAYT	YEGNDLGATY	SKEKTVFKVW	SPSAESISVL	FYPHGEAKDG	360
EKPEITYPMK	QTGAGVWQAE	IEGDLKKNYY	VYQVTVDGKT	KLVVDPYAKA	AGVNGERGMV	420
IDLSETDPDG	FREHSSPEFK	NPVDAVIYEI	HVRDLSMNEN	SGIENKGFKL	GFTETGTTNS	480
AGLSTGLDHM	KELGVTHVHL	LPSFDYKTI	ESKLGKFN	WGYPQNYNL	PEGSYTTDPY	540
QGEVRVREYK	EMVQALHENG	LHVMDVVYN	HTYTAGDSNF	TSLVPGYYR	TDINGNFTNG	600
SGCGNETASE	RAMVRKFIVD	SVVYWATEYK	VDGFRFDLMG	LHDIETMMNV	REALDKIDPS	660
ILLYGEGWTG	GSTPLPDSKQ	AIKNNAVELN	ERIACFSDDI	RDAIKGSVFD	ASDTGFINS	720
KRNVSNRDES	IKFGIVASVS	HPQVNLGVP	YSSRFWANEP	SQTINYASAH	DNLTLWDKLL	780
ETNMASKKEE	LVQMNKLSAA	IVLTSQGIPI	FQAGEEMART	KKGNDNSYQS	PDSINMLNWD	840
NKTEYKDLFE	YKGLIALRK	TYDAFRMQTA	EETQQKLEFV	DSOSSVIAYR	IHDAVKDGRE	900
IALIPNGTLE	EKEVLSANA	WDVLVNQDTA	GTDVITETITG	GTIKVPKAST	LVLLENKDAV	960
IKGDKDAVKG	DEIQELPTNM	QEVAKESGN	AWLWVGIAV	CVLAGGVLFW	ILKRKR	1016

SEQ ID NO: 91 moltype = AA length = 554
 FEATURE Location/Qualifiers
 source 1..554
 mol_type = protein
 organism = Clostridium phytofermentans

SEQUENCE: 91

MKNTNTLHPW	WESAAAYQIY	PRSFMDNSGD	GVGDLQGIIS	RLPYLSELGF	DLIWCPIYP	60
SPNDNDNGYDI	SDYQNIQKEY	GTMEDFEELL	HKAHERGIRV	IMDLVVNHTS	SSHFWFIESR	120
SSKDNPKRDW	YIWKDGKDNV	EPNNWESIFG	GSTWEYDEKS	GQYFLHVFGK	TMPDINWENT	180
QVKKAIFDMI	CWWLDKGIDG	FRVDAISHIK	KPDFNDMPNP	KNERYVSSFD	KHMNQSGILD	240
LLNELKENAF	SKYDIFTVAE	ANGVRIEIEI	EWVSSEKGF	NSLPQFDHLN	LWNVGSEEGK	300
ISIKKKNAL	TKWQKAAPMD	GNVALVMENH	DLVRSISRFG	SEDKYWKESA	KCLALMYMQ	360
KGVPFIYQGG	EIGMLNADYE	SHLDFRDDPT	LFAYQDRIMN	GMSPAESLQV	LKKSSRDNSR	420
TPMQWDASPH	AGFTTGTPWM	KVNQNYHWLN	AEVQKEDEDS	ILNFPYKLIK	IKKETGLIY	480
GDYKLLMEES	ESIYAYTREY	EEKNYLVVCN	LSEBELSELQI	DLDTKGEIL	ISNYEDRNSK	540
EMLLKPYECR	LYSL					554

SEQ ID NO: 92 moltype = AA length = 538
 FEATURE Location/Qualifiers
 source 1..538
 mol_type = protein
 organism = Clostridium phytofermentans

SEQUENCE: 92

MVKKWWHSSV	VYQIYPRSFN	DSNGDGIGDL	KGIEKLDYL	KNLGIDVIWL	SPVFKSPNDD	60
NGYDISDYED	IMDEFGLTED	MELLLKEANN	RGIKILMDLV	ANHTSDEHKW	FIESRKSND	120
AYRDYIWRD	PVDGHEPNDL	GSTFSGSAWE	WDEATGQYYL	HLFSKKQPD	NWENPIVREE	180
VWKSMMFWID	KGIGGFMDV	IELLGKIPDE	KIISNGPMLH	EYIREMNRNS	FGDKDLLTVG	240
ECWGATPEIA	KMYSNPDSGE	LSMVFQFEHI	GLDQIPGKDK	WDLQPLNLID	LKNVPHKQWT	300
CFHDDGWNSL	FWNNHDPRI	VSRWGNKVV	KIESAKMLAT	LLHGLKGTYP	IYQGEELGMA	360
NIKFKIDINQ	KDIETLNMKY	DRLNKGKHE	DIMESIYAKG	RDNARTPMQW	SDEIDGGFTT	420
GTPWIEVNP	FTEINAKEQV	SNPNSIYNNY	KKLIEIRKNN	EVIVYGFEM	LLPEDKNIFA	480
YVRTLKDSKI	VVVCNFYENE	VEYNIPKEYE	EKKEVLISNY	GLSLTGRLRP	FEAIMYRV	538

-continued

SEQ ID NO: 93 moltype = AA length = 555
 FEATURE Location/Qualifiers
 source 1..555
 mol_type = protein
 organism = Clostridium phytofermentans

SEQUENCE: 93

CKKADVNQNP	SELNQDESQK	EKEENDDEGT	PEVSQDETGA	VIPYDYVQNL	NIIDDNYRNF	60
YEIFVYSFYD	SNGDGIGDIN	GVISKLDYIN	DGNDATDSDL	GFNGIWLMP	MPSTTYHKYD	120
VTDDYNIDPQ	YGTLEDKFN	VSECHKRGIH	LIIDFVNHT	SAKHPWFLEA	VSYLESLKEG	180
EEPDLEKCPY	VGYYHFTKDY	NGSKTYKAG	TSNWYYEGVF	WDQMPDLALE	NENVRKEIED	240
IAKYWLDLGV	DGFRDLAAKE	YFSGEKERNI	EVLKWFSDYV	KSVKEDADIV	AEVWDEEGTI	300
AAAYESGIPS	LFNFPLSQHN	GLITNTARKL	GTSSGKNFAK	TLLRLDEKYK	EGNPKYIDAP	360
FISNHDTTRI	SAQCVNDEDQ	MKMSAGMLLT	MNGSPYVYVG	EEIGMNSKGT	KDENKRLPMQ	420
WSATDTTGIT	TPPANADSVE	QKFPVPDEQM	KDPLSLYNY	KRAVRIRNEN	PEIARGDMSV	480
IEELCTKDIS	AIKKVYQGSE	IVILYNINTE	SANILLKDG	LTELNIRGYL	SVDGNVATMS	540
DGVVSMPKYS	IVILK					555

SEQ ID NO: 94 moltype = AA length = 583
 FEATURE Location/Qualifiers
 source 1..583
 mol_type = protein
 organism = Clostridium phytofermentans

SEQUENCE: 94

MKFEAIYHRT	SDNYCYPLNE	EDLIINIKTG	HDIERVFIYY	GDPFEGGILG	GNWTWNGVEE	60
ELIYKKNLTH	HIWQTTTVKP	KFKRCKYYFK	LVANDTSYYY	FEDGFYTEAE	MNHQDKNLVY	120
FTFPWMNSID	INKTPDWVND	TVWYQIPPER	FNNGDKENDP	KNVKAWGFHT	VSNDEFYGGD	180
LQGIINRLDY	LADIGISGIY	LTPIFEANTS	HKYDTKDYMK	IDPHFGDEKV	FKNLVDTAHE	240
KGIRIMLDGV	FNHCGNQFAP	WLDVLKNGPD	SKYFNWFMIN	KWPFNKEDHN	TNDGSFYSA	300
FTSRMPKLNT	NNPEVIKYL	DVVEYVWKNF	DIDGIRLDVA	NEISHRPFCKD	LRLTKELKP	360
DFYILGELWH	DAITWLHGDE	FDGVMNYPLA	TSLADYVWYP	EKTNYDFECA	INHNFMTYMQ	420
QTNVLFNL	DSHDTNRLID	KVKDIDIFYQ	QLAVLFTMPG	SPCIYYGTEI	AMEGSYDPDC	480
RRCMPWEDID	AGLFKDRIEI	IKALIHRLKT	NNAFKSRHYH	FIEDKNNNRV	IHYIKTDEDH	540
KQVEVILNCS	KDSVVRQKRG	NELFSLNED	TILKPKGVFI	QQI		583

SEQ ID NO: 95 moltype = AA length = 575
 FEATURE Location/Qualifiers
 source 1..575
 mol_type = protein
 organism = Clostridium thermocellum

SEQUENCE: 95

MKLEAIYHKP	YSEFAPFPAP	DTLVIRLRTA	KNDVNTCILI	YHEKYDTSQR	GKVKMDKVAS	60
DGMFDYEEVE	LNVGIKRIKY	MFYLEDNYSI	KWYSSDGFDD	YMPQWGHFTY	SYICKDDIFH	120
EVEWFRNSTI	QIIFPDRFAK	FPPDTENSGK	RTIHGNGIKG	IDRFDHLVK	LGVDVVYLN	180
IPKSESYHRY	DVVYDYEIDP	MFGSKEELRE	LMDLCHKNGI	KVIFDGVFNH	SGDKFFAFRD	240
VVEKGEKSKY	ANWYFINSFP	VQGYPRPNYE	CPSFYGGMPK	LNTGNPETAK	YFLDVVKYWT	300
VEFGVDGWRL	DAAEVDKRF	WRKLRDMLKD	LNKDVVLIGE	IFDEASSWLW	GDQFDSVIN	360
PLKAMINDLF	AYRSIDVETF	RNRISGYIMK	FNKKVLSSLV	NIISTHDTPR	FLTLCNGDEK	420
RFEMAVVFQF	TFPGVPLIYY	GDEIGMEGEG	DPDCRRPMIW	DEAKWNKKT	ELYKFLIGLR	480
KRFDALRTGE	YGELPVTGCN	GILAYRRGRG	ENGIIIVAMNT	LDRKENVVVE	TGDSFDTVKA	540
FESLKDEERL	NVDKKRINIC	LNPFEWRIYK	ACGEL			575

SEQ ID NO: 96 moltype = AA length = 655
 FEATURE Location/Qualifiers
 source 1..655
 mol_type = protein
 organism = Thermobifida fusca

SEQUENCE: 96

MIGRFPILDV	SPVVDIGTAK	AVVGETFPVR	ATVFREGHEA	LGAGVVLYTP	EGQRQPLVPL	60
REIAPGTDTRY	EAEVTVTSEG	LWHFAIEAWS	DPYATWCHDA	RIKIPAGQDV	ELMLEEGARL	120
LERAAARRVPR	RPALAEIAAAA	MRDGSRSAHE	RLDLALSDLV	RDELAERPLR	ELVTRSQRFP	180
VMVSRRRALF	GSWEFFFRPS	EGAVLDTEDG	EPRSGTFATA	ARRLPAIADM	GFDVVYIPPI	240
HPVGYSFRKG	RNNSTVAQPG	DPGSVWAIGS	HEGGHDAIHP	DLGTIDDFDA	FVARARELGL	300
ETAMDALQA	SPDHPWVKEH	PEWFTVRADG	SIAYAENPPK	KYQDIYPINF	DKDPEGIFTE	360
VRRIVRYWMS	HGVRIFRVDN	PHTKPVAFWE	RLADIAATD	PDVIFLSEAF	TRPAMMHTLA	420
KIGFHQSYTY	FTWRNTKQEL	EEYLTELTEG	AAAYMRPNFF	VNTPDILHAY	LQHGGRPAPF	480
VRAILAATLS	PTWGMYSGYE	LCENRALKPG	SEELYDSEKY	QYKPRDWEAA	EAAGITITPL	540
IRKLSLRRS	HLPAQELRNL	RPHYADQPEI	ICYSKRLAGA	NHGADDTILV	VANLDPHHTR	600
EATVWLDMPA	LGAFPGDHIT	VTQLSGHSY	HWVEANYVRL	DPHVQTAHIF	TVAPA	655

SEQ ID NO: 97 moltype = AA length = 572
 FEATURE Location/Qualifiers
 source 1..572
 mol_type = protein
 organism = Thermobifida fusca

SEQUENCE: 97

APSGNRDVIIV	HLFQWRWXS	ADECRITLGP	HGFGAVQVSP	PQEHVVLPAE	DYPWWQDYQP	60
VSYKLDQTRR	GSRADFIDMV	NTCREAGVKI	YVDVINHMT	GTGSAGAGPG	SAGSSYSKYD	120
YPGIYQSQDF	NDICRRDITNW	NDKWEVQHCE	LVGLADLKTS	SPYVQDRIAA	YLNELIDLGV	180
AGFRIDAANK	IPEGDLQAIL	SRLKNVHPAW	GGGKPYIFQE	VIADSTISTG	SYTHLGSVTE	240

-continued

FQYHRDISHA	FANGNIAHLT	GLGSGLTSPD	KAVVFFVNHD	TQRYEPILTH	TDGARYDLAQ	300
KFMLAHPYGT	PKVMSSYTWS	GDDKAGPPMH	SDGTTTRPTDC	SADRWLCEHR	AVAGMVGPHN	360
AVAGQGIGSA	VTGNGRLAF	ARGSAGYAAF	NATNTAWTRT	FTTSLPDGVY	CDVANGTFVD	420
GVCDGPSYQV	SGGKFTATVP	ANGAVALHVE	APGSCGPDGC	GTPPGGGDDC	TTVTARPHAT	480
VTTWYGQEVA	VVGSIPELGS	WQPAQGVRLR	TDSGTYPVWS	GAVDLPAGVG	FEYKYVKLNP	540
DGTVEWEQGG	NRIATVDDSG	GGCSQNFYDS	WR			572

SEQ ID NO: 98 moltype = AA length = 825
 FEATURE Location/Qualifiers
 source 1..825
 mol_type = protein
 organism = Anaerocellum thermophilum

SEQUENCE: 98

MLVRAYIDDF	NEIVVVLSQL	VHSVKKEDFK	VFLNEEEIDI	EKIDKIIPHS	DNPAAEATRG	60
YEICEQKGGI	RFVLKEGHFD	YHRKPYKKPV	FVIGEMNDWQ	ISPEWEMTYS	KLRGRYELIK	120
DLKEIKIGQK	PKFAEGASQK	LWYPPGFGND	IVITEYFDRE	TAFTNMIRII	PSNRLLPNLK	180
YKVVYKSEHI	WARPREILTR	PEFFYPGELG	IKYEPYGTYP	KLWAPTAYKV	KVKVPDESEN	240
FRFEKEMARS	ENGTWNIYLT	GDLKNHYLYL	EVWHYNYDED	BGFIVYEVDP	PYSKASSSNS	300
QKSFIFDPAD	TLIEGWQQDE	FVKTIEKQDQ	AIYEMHVRD	FTIDKNSGVD	EKFRGKFLGL	360
CQKSFYKEKF	STGLLHLKEL	GITHIHLPLI	SDFGSVDDKN	PDKKYNWGYD	PVLYQCPEYW	420
YSTKSGGIEA	LKELKTMIKT	LHQNGIGVVM	DVVFNHTYHT	KGGKFSIFDK	IVPGYFYRID	480
DYGDYSNATG	CGNEIATEKP	MVRKFILDTI	IYWTEDFHID	GFRFDLMGLI	DTLTMRMIK	540
EVKRKNPYAL	IYEGEGWVMD	SMCLLEERAT	IESTAHHGYS	IGLFNDRIRD	SIRGDLDDGFK	600
TGYMHGNLSD	IERLKQGIRA	AIDDFAKEPD	ECVNVYSCHD	NLTFLDKAQK	TMVGEDIFWI	660
DRVCRLANAI	ILTSQGIPLF	HGGVEFNRSK	GGHFNNTYNAG	DNINKIDWSL	KEKPYDTFKF	720
YCDLIKLRKE	HVAFMRMRSS	EIRKYLKFLP	APDGIVAFLI	SYPYDAWKKI	IVAYNPFKEK	780
KVITLPEGVW	KIKANDGIIF	SEENELEAIG	SFEISPVSLF	IAYQK		825

SEQ ID NO: 99 moltype = AA length = 1104
 FEATURE Location/Qualifiers
 source 1..1104
 mol_type = protein
 organism = Anaerocellum thermophilum

SEQUENCE: 99

DEKTTLIHYY	YRYNEDYQGW	NLWIWPVEPV	GAEGKAYEFT	SKDDFGVKAV	VELPGKVTKV	60
GIIVRKGWNE	AKDVAVDRFI	SGISGSKEVW	LIEGEEQIYT	SQPQKTPKMT	AFIDGLNTIV	120
VKLAKKADIL	SNNRTQGFVK	TAFYEEVPIK	KVEPVLPKIN	KNFKPEEAGY	ELIDGGTKVK	180
FILKPGAGDF	KFTDTSGKLD	VYVSGTMNDW	GGTASSEGGY	KPLPAWKMTW	NAEKGYVELV	240
KELGKDGVI	GAKFKFTSWD	GTSAKWYPDG	MGNDKVI EEL	YTGNEKITKV	DTFKITTEDE	300
LEPQVPYVVS	KDSFKPTVAQ	ARNILDNPKY	YKGNLDLCT	YTKAYSARFL	WAPTAIGVIL	360
RLYDDYKTTK	YKEYEMQSQF	NGTWYLVKING	DLKGKYYQYE	VWHASNSITD	DTIRKYVVPD	420
PYSRATSANS	ERTLIFDPKD	TNPVGWEKDT	FVTLKNQEDA	IYETHVRDF	TIDASSGVRP	480
EPRGKYLFT	QTGAKGPNV	KTGIDHLKEL	GITHVHLPLT	YDFGSIDETN	PDKGYNWGYD	540
PVLYQNVEGS	YATNPNTIVR	IKEYKQMVMA	LHKAGIGIIQ	DVVFNHTFQI	GDAKFSIFDK	600
IVPGYFYRKD	KDGNYSNAG	CGNEIATEKP	MVRKFIIDTL	TYLTKEYHID	GFRFDLMAAI	660
DRVTMAKAQE	EVVKINPSAV	IYEGEWLAGS	TPLDSSLRME	IGSFNQAGLH	IGLFNDRIRE	720
AIRGNLDNES	KGFMQGNYSF	RLEDLKRGIQ	GGLGDFAADP	DECINYVSAH	DNLTLDWKLQ	780
KSVPNEDPYI	KDKMGRLANA	IVLTAQGVPP	LHGGVEFNRT	KYMNHNSYNA	GDKINKYNWN	840
LKVWYNTFK	YYQGLIALRK	AHPAFRMTTA	EDIQKYLTFI	QTPKGTGFR	LTYPKDTWND	900
IIVVYNSTKK	VQEVTLPEGN	WVVVANGDEV	GTTPIKNLTN	FVAGKALVAP	ISMVAVYKSN	960
EPFQGFTKVT	GKDPVSLESS	STVTVPKVVY	NGNIEVTFKV	KVPHGTDDDV	IYLAGSFGKA	1020
GLSDWNPGDK	DGAIELVRLQ	DGTYTIVTKL	NAGETFEYKY	TRGSWTTVEK	GANKKEIENR	1080
KLTVKDEGGG	KMIVSDTVLN	WADK				1104

SEQ ID NO: 100 moltype = AA length = 611
 FEATURE Location/Qualifiers
 source 1..611
 mol_type = protein
 organism = Anaerocellum thermophilum

SEQUENCE: 100

MRKPHIEAI	IGNTKVLGQL	DSNGILQRFY	WPAVDYYQQL	KLFLAAVFLD	GLVFFEDENF	60
KIKSGFVDDF	VYFFEYKIAD	KTIFQLDFVD	FETDSLVRWL	ETGFEDFYVF	LEPMINSSSL	120
FNAAKVDKEN	EIVYAYFKGT	YIGLAFENKI	KSFTVKNGID	DANDNQLEGW	NEATNPQIAV	180
KLKNTGKVVV	PLAFGNSKDE	IYQKLSYLKQ	KGYDEVYRQN	KAFWEKKFSK	VKLICTQDPK	240
DMQLQKRSAY	VFYVLQNSKT	GGILAASEVD	EKFFHCGGYG	FVWGRDAAFI	VSAMDELGLS	300
REVEKFFGFK	FSCQKEKEGW	DQRYITDGS	APSWGIIQDE	TASVVGWGLE	HCEKQNSLHL	360
IDLHKEQLKK	ALLFLIAAVD	SEKGVIFRSF	DLWEEREGIH	LYSNASIYAA	LKKAKYFPPE	420
LESEIEKKLK	AIKNQMTRF	YSPKLSRYVR	STDVRIPHEE	FLKLPEENRY	MQKDERYEIT	480
YYFKQDEVV	DISMLGIYYP	FEMVDSSDKA	FKATILAIER	ECQNSIVGGY	KRYSDDRYIG	540
GNPWILTTLW	LAIYKKTGQ	IDRAEKLFEW	AKAHSPLNGL	FPEQVDRITG	KPAWVPLAW	600
SHAMVVLVLY	E					611

SEQ ID NO: 101 moltype = AA length = 529
 FEATURE Location/Qualifiers
 source 1..529
 mol_type = protein
 organism = Streptomyces avermitilis

SEQUENCE: 101

MTSFRPAPAW	LADAVFYQIY	PQSFADSDGD	GIGDFNGIVQ	RLDHLVWLGV	TAVWLNPCFV	60
------------	------------	------------	------------	------------	------------	----

-continued

```

SPFRDAGYDV SDYLVNAPRY GSADDLAEV DEAGRGRGIRV LLDLVAGHTS DEHPWFTASA 120
NDPDDHRYIW APEGRPDGFV TSPGTRPGAY LPNFFDTQPA LNFGYGRKNP AEPWRQPVDA 180
AGPRANREAL RTIMDHWLGL GLAGFRVDMA ASLVKDDPGR TETARIWTEL RHWLDTAHPD 240
AVLLSEWGEF EVSVPAGFHT DFPLQFGGAT DGLPLRSLWS NGDGTVNEAW DPLDCFFDAS 300
KGKSPRPFEV AWRKASDAVG ATGFVSLPTA NHDFSRLNCG PRTAEQLPAA FAFQLTWPTL 360
PAIYYGDEIG MRYVGGLPDK EGSVLGPRYN RAGSRTPMQW DDGPGAGFST APADRLLYPL 420
DPSPRDPTVA AQRADDGSLH HVRRLVALR ASTPALGSGG SVEVLHTGYP FVYVRGGRYL 480
VVVNPQRNEV RCPYDATREA RALEASGVRV GNGTIEAEGF SYGVFDLGR 529

```

```

SEQ ID NO: 102      moltype = AA length = 431
FEATURE            Location/Qualifiers
source             1..431
                  mol_type = protein
                  organism = Streptomyces avermitilis

```

```

SEQUENCE: 102
SPPGTKDVT A VLFWEKFDV ARECTNTLGP AGYGVVQVSP PAEHIQGSQW WTSYQPVSYK 60
IAGRLGDATA PQNMINTCHT AGVKVVVDTV VNHMSAGSGT GTGGSAYTKY NYPGLYSSYD 120
MDDCTATITD YTNRANVQNC ELVGLADLDT GEEYVRKTIA GYMNTLLGYG ADGFRVDAVK 180
HIPADLANI KSRLTNPSVY WKQEVYASG EAVQPTETYG NGDVQEPFRA YDLKRVFNNE 240
NLAYLKNYGE GWGYLNSVA GVFVDNHDTE RINGSTLNYK GANYTLANVF MLAYPYGAPD 300
INSGYEWSDA DAGPPGGGTV NACWQDGWKC QHAWPEIKAM VAFRNATRGE SVTNWWDNGG 360
DAIAFGRGAK GYVAINHESG SLRTRYQTSL TAGTYCNVQN NTGVTVDSSG RFTATLGANT 420
ALALYSGKST C 431

```

```

SEQ ID NO: 103      moltype = AA length = 503
FEATURE            Location/Qualifiers
source             1..503
                  mol_type = protein
                  organism = Saccharomycopsis fibuligera

```

```

SEQUENCE: 103
LPLQEGPLNK RAYSPFEAYS NYKVDRTDLE TFLDKQKDV S LYLLQNIAY PEGQFNDGVP 60
GTVIASPSTS NPDIYYQWTR DSAITFLTIV SELEDNNFNT TLAKAVEYYI NTSYNLQRTS 120
NPSGSFDDEN HKGLGEPKFN TDGSAYTGAW GRPQNDGPAL RAYAISRYLN DVNSLNKGKL 180
VLTDSGDINF SSTEIDYKNI IKPDLEYVIG YWDSTGFDLW EENQGRHFFT SLVQQKALAY 240
AVDIAKSPDD GDFANTLSST ASTLESYLSG SDGGFVNTDV NHIVENPDLL QQNSRQGLDS 300
ATYIGPLLTH DIGESSSTPF DVDNEYVLQS YLLEEDNKD RYSVNSAYSA GAAIGRYPED 360
VYNGDGSSSE NPWFLATAYA AQVPYKLVYD AKSASNDITI NKINYDFENK YIVDLSTINS 420
GYQSSDSVTI KSGSDEFNTV ADNLVTFGDS FLQVILDHIN DDGSLNEQLN RNTGYSTSAY 480
SLTWSSGALL BAIRLRNKVK ALA 503

```

```

SEQ ID NO: 104      moltype = AA length = 497
FEATURE            Location/Qualifiers
source             1..497
                  mol_type = protein
                  organism = Saccharomycopsis fibuligera

```

```

SEQUENCE: 104
VPVELDKRNT GHFQAYSGYT VARSNFTQWI HEQPAVSWY LLQNIIDYPEG QFKSAKPGVV 60
VASPSTSEPD YFYQWTRDTA ITFLSLIAEV EDHSFSNTTL AKVVEYYISN TYTLQRVSNP 120
SGNFDSPNHD GLGEPKFNVD DTAYTASWGR PQNDGPALRA YAIRSYLNAV AKHNNKGLLL 180
AGQNGIPYSS ASDIYWKI K PDLQHVSTHW STSGFDLWEE NQGTHTFTAL VQLKALSYGI 240
PLSKTYNDPG FTSWLEKQKD ALNSYINSSG FVNSGKKHIV ESPQLSSRGG LDSATYIAAL 300
ITHDIGDDDT YTPFNVDNSY VLNSLYLLV DNKNRYKING NYKAGAAVGR YPEDVYNGVG 360
TSEGNPWQLA TAYAGQTFYT LAYNSLKNNK NLVIEKLYND LYNSFIADLS KIDSSYASKD 420
SLTLTYGSDN YKNVIKSLQ FGD SFLKVL DHIDDNGQLT EEINRYTGFO AGAVSLTWSS 480
GSLLSANRAR NKLIELL 497

```

```

SEQ ID NO: 105      moltype = AA length = 747
FEATURE            Location/Qualifiers
source             1..747
                  mol_type = protein
                  organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 105
FPTALVPRGS SSSNITSSGP SSTPFSSATE SFSTGTTVTP SSSKYPGSKT ETSVSSTTET 60
TIVPTTTTTS VITPSTTTIT TTVCSTGTNS AGETTSGCSP KTITTTVPCS TSPSETASES 120
TTTSPPTPVT TVVSTTVVTT EYASTSTKQG GEITTTFTVK NIPTTYLTIT APTSSVTTVT 180
NFTPTTITTT VCSTGTNSAG ETTSGCSPKT VTTTVPCSTG TGEYTTAATA PVTAVTTT 240
VTESSTGTN SAGKTTTSYT TKSVPPTYVF DFGKGILDQS CGGVFSNNGS SQVQLRDVVL 300
MNGTVVYDSN GAWDSSPLEE WLQRQKKVSI ERIFENIGPS AVYPSILPGV VIASPSQTHP 360
DYFYQWIRDS ALTINSIVSH SADPAIETLL QYLNVSFHLQ RTNNTLGAGI GYTNDTVALG 420
DPKWNVDNTA FTEPWGRPQN DGPALRSIAI LKIIDIYKQS GTDLGAKYPP QSTADIFDDI 480
VWRDLRFID HWNSSGFDLW EEVNGMHFFT LLVQLSAVDR SLSYFNASER SSPFVEELRQ 540
TRRDISKFLV DPANGFINGK YNYIVETPMI ADTLRSGLDI STLLAANTVH DAPSASHLPF 600
DIDDPVAVLT LHHMLMHMS IYPINDSSKN ATGIALGRYP EDVYDGYGVG EGNPWVWLATC 660
AASTTLYQLI YRHISEQHDV VVPMNNDCSN AFWSSELVFSN LTTLGNDDEG LILEFNTPAF 720
NQTIQKIFQL ADSFLVKLKA TWEQTGN 747

```

```

SEQ ID NO: 106      moltype = AA length = 621
FEATURE            Location/Qualifiers
source             1..621

```


-continued

```

mol_type = protein
organism = Aspergillus niger

SEQUENCE: 106
NVISKRAATWD SWLSNEATVA RTAILNNIGA DGAWVSGADS GIVVASPSTD NPDYFYTWTR 60
DSGLVLKTLV DLFRNGDTSL LSTIENYISA QAIVQGISNP SGDLSSGAGL GEPKFNVDTE 120
AYTGSWGRQP RDGPALRATA MIGFGQWLLD NGYTSTATDI VWPLVRNDLS YVAQYWNQTG 180
YDLWEVNGSS FFTIAQVHRA LVEGSAFATA VGSSCSWCDS QAPEILCYLQ SFWTGSFILA 240
NFDSSRSRSD ANTLLLSGIH TFDPEAACDD STFQPCSPRA LANHKEVVD SFRSIYTLNDG 300
LSDSEAVAVG RYPEDTYNG NPWFCLTLAA AEQLYDALYQ WDKQGSLEVT DVSLDFFKAL 360
YSDATGTYS SSSTYSIVD AVKTFADGFV SIVETHAASN GSMSEQYDKS DGEQLSARDL 420
TWSYAALLTA NNRRNVVPSA SWGETSASSV PGTCAATSAI GTYSSVTVTS WPSIVATGGT 480
TTTATPTGSG SVTSTSKTTA TASKTSTSTS STSCTTPTAV AVTFDLTATT TYGENIYLVG 540
SISQLGDWET SDGIALSADK YTSSDPLWYV TVTLPAGESF EYKFIRIESD DSVEWESDPN 600
REYTVPQACG TSTATVTDTW R 621

```

```

SEQ ID NO: 107      moltype = AA length = 593
FEATURE            Location/Qualifiers
source              1..593
                    mol_type = protein
                    organism = Aspergillus oryzae

```

```

SEQUENCE: 107
VQPVLRQATG LDTWLSTEAN FSRQAILNNI GADGQSAQGA SPGVVIASPS KSDPDYFYTW 60
TRDSGLVMKT LVDLFRGGDA DLLPIIEEFI SSQARIQGIS NPSGALSSGG LGPEKFNVDTE 120
TAFTGAWGRP QRDGPALRAT AMISFGEWLV ENSHTSIATD LVWVVRNDL SYVAQYWSQS 180
GFDLWEEVQG TSFTVAVSH RALVEGSSFA KTVGSSCPYC DSQAPQVRCY LQSFWTGSI 240
QANFGGGRSG KDINTVLGSI HTFDPQATCD DATFQPCSPAR ALANHKVVD SFRSIYAINS 300
GRAENQAVAV GRYPEDSYN GNPWFLTTLA AAEQLYDALY QWDKIGSLAI TDVSLPFFKA 360
LYSSAATGTY ASSTTVYKDI VSAVKAYADG YVQIVQTYAA STGSMAEQYT KTDGSQTSAR 420
DLTWSYAALL TANNRRNAV VAPWGETAAT SIPSACSTTS ASGTYSVVVI TSWPTISGYP 480
GAPDSPCQVP TTVSVTFAVK ATTVYGESIK IVGSISQLGS WNPSSATALN ADSYTTDNPL 540
WTGTINLPAG QSFYEFKIRV QNGAVTWESD PNRKYTVPST CGVKSQVQSD VWR 593

```

```

SEQ ID NO: 108      moltype = AA length = 579
FEATURE            Location/Qualifiers
source              1..579
                    mol_type = protein
                    organism = Rhizopus oryzae

```

```

SEQUENCE: 108
ASIPSSASVQ LDSYNYDGST FSGKIYVKNI AYSKKVTVIY ADGSDNWNMN GNTIAASYS 60
PISGNYEYV TFSASINGIK EFYIKYEVSG KTYIDNNNSA NYQVSTSKPT TTTATATTTT 120
APSTSTTTPP SRSEPATFPT GNSTISSWIK KQEGISRPFAM LRNINPPGSA TGFIAASLST 180
AGPDYYYAWT RDAALTSNVI VYEYNTLSG NKTILNLVD YVTFSVKTS TSTVCNCLGE 240
PKFNPDSAGY TGAWGRPNQD GPAERATTFI LFADSYLTQT KDASYVTGTL KPAIFKDLDY 300
VNVVWSNGCF DLWEEVNGVH FYTLMVMRKG LLLGADFAKR NGDSTRASTY SSTASTIANK 360
ISSFWVSSNN WIQVSQSVTG GVSKKGLDVS TLLAANLGSV DDGFPTPGSE KILATAVAVE 420
DSFASLYPIN KNLPSYLGNS IGRYPEDTYN GNGNSQGNWS FLAVTGYAEL YYRAIKEWIG 480
NGGVTVSSIS LPFFKKFDSS ATSGKKYTVG TSDFNNAQN IALAADRFLS TVQLHAHNNG 540
SLAEEDRTT GLSTGARDLT WSHASLITAS YAKAGAPAA 579

```

```

SEQ ID NO: 109      moltype = AA length = 644
FEATURE            Location/Qualifiers
source              1..644
                    mol_type = protein
                    organism = Clostridium thermocellum

```

```

SEQUENCE: 109
MANTYFNDAI IGNSGMLVCL TRNGELTRLF WPNIDYPQHF EKMATGIFYT GQKNSTSWFY 60
EDNWHHTQY VEDTNILKTI CEDGGRGLRV EQTDFVLKDR DVMVRRYVIE NIGPNEVDLG 120
FVQYSSTVST TPELRSTLFD FNVDALIHVR HNYYSISSD SEVVQFQLGN NAFDCARYTE 180
LYGYSIGMM KDGAMSPNIG KIEPGGKKT FNLICASHTL KGVKELVRWC RKMNVDEEYE 240
KTRKYWLDFL KNARLIVTGD KNIDNLYKRS ILVFKLMSDE RTGGLLASAE IDEGFTRCGR 300
YAYCWGRDAA FITGALDTAG LTEAVDKFYQ WAVMTQDDDG SWQORYHMDG NLAPSWGLOI 360
DETGTLIWGM LKHVEVTKNI DFLKSMWESI KKGVEFLTRF IDSDTGLPAP SYDLWEERVG 420
EHTYSSAAVY AGIKAGABAA RILGASEELI EKWEKAASDM KASIEKNFWR DEAGRPIRSV 480
RTKLNPNWGS HSPYTTVIKV NEKGYFRDVT LEDWTIDVSL LGVSIPIGVF DVHDERVKKT 540
VEAIERALTS HPVGGIKRYE NDNYIGGNPW VLATLWVALY YIEIKEYEKA KDYLWATKS 600
CTALGLLPEQ VSKDNGEPCW VIPLTWSHAM YVLVLAGLKE AGVL 644

```

```

SEQ ID NO: 110      moltype = AA length = 644
FEATURE            Location/Qualifiers
source              1..644
                    mol_type = protein
                    organism = Clostridium thermocellum

```

```

SEQUENCE: 110
MQKSYNNNAI TGNSSMLACF SERAELLRLF WPDIDYIQL DKMFLGLFEK NKTGSTVWLN 60
DIRCEHHQY LPDSNIIKMN VTNFFDGYKV VLYDFVHPM DVLVRRFEIE NLRGESREL 120
LMSFSAATSS DSEVACSLFD FMNEALVHYK PDSYIAVTS IPVYQFQIGN NANDAAVNTY 180
LYGKDDIGMM KDAAISWDLG VFQPHAVKTT NVYLCAADTL KSKCALVRRV KTVGGLTAFR 240
ETGRYWKDYL EKTTPKLSGN TLLDDLYKRS LLVFRMLYSK KSGGLMAAPE VDEYFTKCGK 300
YAYCWGRDAA FITGALDIG LCESVDHFFYK WAVNVQDEDG SWQORYHMNG NLGPCWGLQV 360

```

-continued

DETGTTIWM	LNHYNYTKNT	DFLKSVWDSV	KAAADFLVRF	IDSETGLPRP	SFDLWEERYG	420
EHAYSSASVC	AGLKSASEMA	RILGKPSQY	IQWETTADSI	KKAIVKYFWK	EDYRRFIRSI	480
RVKLNFGGQE	PSSDTMLIKV	NPKGYVRDVT	KEDWIVDVSL	VGLGIPFEIF	ELNDPMLRDT	540
VSLIEQVLTA	QGVGKIKRYE	NDTYIGGNPW	ILTTLWIALY	HAKSGNYKKA	KEYLIWAASG	600
KTELGLLPEQ	INRDTGKPEW	IIPLTWSHAM	YVHVSESLIN	AGVL		644

SEQ ID NO: 111 moltype = AA length = 608
 FEATURE Location/Qualifiers
 source 1..608
 mol_type = protein
 organism = *Arxula adeninivorans*

SEQUENCE: 111

DSCHTFTLAN	SPPDDKAVAL	SSSYSCGGYL	SASAFVKNLS	YDKLVTLYWT	NADNKSTPLN	60
AGSLDYVKAA	SDDQSWELWS	LNVTTPVDGV	DALLNITYVA	ASIGKTNSQQ	LNQVVEATGD	120
PIPTPQIPTI	YKPYASPSDF	SDDITNLWKP	SNDSTGTIAK	SFLFNINIP	GAAPGTVIAA	180
QSYSEPDYAY	TWVRDASLVM	DVVRNLYSSA	KSEKRLQLYE	KILPQYAKAG	AQEQNPTAI	240
SGMGEPKPYL	NNTAFTGSWG	RPQNDGPATR	AITLIEFANA	YLANGGSQDT	VREQLYDSK	300
YPQVAPIKID	LQFVASNWSS	PSFDLWEEEEE	SAHFYTRLVQ	RKALLLGADF	ANDMDHEL	360
DKLKTQASKL	SDDLPEFWKS	ARQLILYEYV	PVLRGKYSYK	DISVVLGVMH	GYANDNVFSY	420
TNDQILATAY	QVSTSFLDVY	KVANTTSDES	GKPLGIPVGR	YPEDVYDGVG	TSQGNPWYLT	480
TMAMAEFLYR	SVQEFEDAGS	IIISDTSLPF	WKYPASSVDH	KAGAKYNKND	QSFKTSLSKL	540
TGWGDAFMRR	AKYHTPSSGH	MSEEFNRTTG	EPRGAKDLTW	SYASLLSAAF	AREELRNQKN	600
YLTNVADL						608

SEQ ID NO: 112 moltype = AA length = 595
 FEATURE Location/Qualifiers
 source 1..595
 mol_type = protein
 organism = *Hormoconis resinae*

SEQUENCE: 112

APTELKARDL	SSFIASERAI	ALQGALNNIG	PDGSAPVPGAG	AGFVVASPSK	ANPDYFYTWS	60
RDSALTLKMI	IDEFILGNTT	LQTIIEQYIH	AQAVLQTVSN	PSGTFLPDGV	GLGEPKFMVD	120
GTRFNGPWGR	PQRDGPALRA	IAMMTYSNLW	IKNGQFAEAK	TKIWPPIAND	LSYVGQYWNQ	180
SGFDLWEETY	ASSFTFIQNG	HRALVEGAQL	AHDLGVTCTG	CDQAPEVLCF	LQSFNWKYI	240
VSNINVNNGR	TGLDGNLSIL	AISTFDIDAY	CDSPTLQPC	SQSLANFKVL	TDTFRNLYTI	300
NAGIPEGQVQ	AVGRYAEDVY	MGGNPWYLIT	TAAAEFLYDA	VAQWKARHVL	TVDETSLAFF	360
KDIYPEVTVR	EYKSGNANSP	FAQIMDAVTA	YADSYVAIAE	KYIPSNGLSL	EQFNRTGTPT	420
LSAIDLWTSY	AAFITMSQRR	AGQYPPSSWS	RNALPPPTTC	SASSTPGIYT	PATAAGAPNV	480
TSSCQVSTIF	NINATTYIGE	NLYVIGNSSD	LGAWNIADAY	PLSASAYTQD	RPLWSAAIPL	540
NAGEVISYQY	VRQEDCDQPY	IYETVNRTL	VPACGGAATV	TDDAWMGFVG	SSGNC	595

SEQ ID NO: 113 moltype = AA length = 601
 FEATURE Location/Qualifiers
 source 1..601
 mol_type = protein
 organism = *Aureobasidium pullulans*

SEQUENCE: 113

LSPSPESIQER	ATGSLSSWLS	SENTVALQGV	LNNIGASGSK	ASGASAGVVV	ASPSKSNPDY	60
FYTWRDSDAL	VFKALVDQLI	AGNKSLEPLI	QQYISAQAKL	QTVNNPSGGL	CSGGLAEPKF	120
EVDLTPFTGA	WGRPQRDGA	LRATAMIAYS	RYLIANGNTT	TVNNIWPPIV	QNDLSYVTQY	180
WNQTTFDLWE	EINSSSFPTT	AVQYRALVEG	NNLATQLGKS	CPNCVQSAPL	VLCFLQSYWT	240
GSYALNSGTG	GRSGKDNANS	LTSIHIFDPA	ASCDSTTFQP	CSDKALANHK	VVTDSPRSY	300
SINQIAQGS	GVAVGRYPED	SYNGNPNWYL	NTFAAAEQLY	DAVYQWKKIG	SISITSISLP	360
FPKDYSSAA	VTYSSSTVT	FTSIVNAVQT	YADSYMSIAQ	KYTPSNGALS	EQYNRADGTP	420
LSAVDLTWSY	AAFLTAYNAR	ANVLPAWSGA	SSAKLPNSCS	SGSATGPCAA	ATNTNWNPNP	480
SPSTGTPTTT	TGGSCPTPTS	IAVTFNEQKT	TSYGENIYIV	GSIPALGNWN	TANAVALSAS	540
KYTSSNPLWT	VTINFATGTS	FNKYIKKAQ	DGSVTWESDP	NRSYTVTGNC	AGTATENDSW	600
R						601

SEQ ID NO: 114 moltype = AA length = 718
 FEATURE Location/Qualifiers
 source 1..718
 mol_type = protein
 organism = *Bacillus subtilis*

SEQUENCE: 114

MVSIRRSFEA	YVDDMNIITV	LIPAEQKEIM	TPPFRLETEI	TDFPLAVREE	YSLEAKYKYV	60
CVSDHPVTFG	KIHCVRASSG	HKTDLQIGAV	IRTAAFDDEF	YYDGLGAVY	TADHTVFKVW	120
APAATSAAVK	LSHPNKGRT	FQMTREKGV	YAVVTGDLH	GYEYLFICIN	NSEWMETVDQ	180
YAKAVTVNGE	KGVVLRPDQM	KWTAPLKPFS	HPVDVAVIET	HLRDFSIHEN	SGMINKGKYL	240
ALTETDTQTA	NGSSSGLAYV	KELGVTHVEL	LPVNDFAVGD	EEKPLDAYNW	GYNPLHFFAP	300
EGSYASNPHD	QTRKRTBLKQ	MINTLHQHGL	RVILDVFNH	VYKRENSPFE	KTPVGYFFRH	360
DECGMPSNGT	GVGNDIASER	RMARKFIADC	VVYWLEEYNV	DGFRFDLLGI	LDIDTVLYMK	420
EKATKAKPGI	LLFGEGWDLA	TPLPHEQKAA	LANAPRMPGI	GFFNDMPFDA	VKGNTFHLKA	480
TGFALNGES	AQVVMHGIAG	SSGWKALAPI	VPEPSQSINY	VESHNDHTFW	DKMSFALPQE	540
NDSRKRSRQR	LAAAIILLAQ	GVPFIHSGQE	FFRTKQGVEN	SYQSSDSINQ	LDWDRRETFFK	600
EDVHYIRRLI	SLRKAHPAFR	LRSAAADIQRH	LECLTLKEHL	IAYRLYDLDE	VDEWKDIIIV	660
HHASPDSEVW	RLPNIDIPYRL	LCDPSPGFQED	PTEIKKTVAV	NGIGTVILYL	ASDLKSFA	718

SEQ ID NO: 115 moltype = AA length = 710

-continued

FEATURE
source Location/Qualifiers
1..710
mol_type = protein
organism = *Bacillus licheniformis*

SEQUENCE: 115

MPGISRPFEA	YLDEMRTITV	LVPKSRASSC	SPPFLEDDQ	GERIELSVKA	QVELEEQFKY	60
VLESSCTVPF	GRVHKVCCCE	SVWTDLQIGS	VTRSAAFDKA	FFYDGRLGAF	YSKGSTLFKV	120
WAPTASAAAI	KLEDPDSLQT	NTFQMMRRKK	GVFEVTVEGD	LNGWSYLYEL	YVNGKPLLTV	180
DPYAKAVTAN	GEKGVVLDPE	EVKVEKHRAP	RLHSPCDAVI	YEVHIRDFSI	HEDSGMRHKG	240
KYVAPTEDEGT	ETSGGFSTGI	AYLKELGVTH	IEVLPFHDFD	GVDELSPDQS	YNWGYNPLHF	300
NAPEGSYSLD	PQNPKCRITE	LKTMIQSLHK	HGFSVIMDAV	YNHVKRETS	PFEKTVPGYF	360
FRHNEYGFPS	DGTGVGNIDIA	SERLMVRKYI	LDSVRYWLEE	YDVGDIRFDL	MGILDIETVR	420
QISTLAENVK	PGVPLFGEGW	DLNTPLDSCQ	KATLQNAAGV	PAVGFFNDRF	RNAVKGSTFE	480
LSDRGYALGD	TGKKAALQHG	IAGSPGFLQP	AQSINYVECH	DNHTFWDKMA	LCFEEDADTK	540
RLRQLRAVSI	VLLSQGVFFL	HAGQEFCKTK	NGDSNSYRSG	DDINKLDWEK	RAELCEDVEY	600
VRQLIRLRRS	HPAFRLQKEE	EVKEHLSFMD	GTGEVTAYKL	KNIAAIDPWN	EIIVVHCPFA	660
KKETLKLDPQ	KQYLLHCDPF	TFNNGKVQAE	KRLRLNGIGT	YVLYEPKGIF		710

SEQ ID NO: 116 moltype = AA length = 918

FEATURE
source Location/Qualifiers
1..918
mol_type = protein
organism = *Oryza sativa*

SEQUENCE: 116

MAVGEECAAA	VASQGFVSDA	RAYWVTRSLI	AWNVDQDTS	LFLYASRDAT	MHVSDGAIHG	60
YDSKIELEPE	HASLPDNVAE	KPPFIRSIRT	FRVPSSVDVT	SLVKCQLAVA	SYDAHGRHQD	120
VTGLQLPGVL	DDMFAYTGPL	GAVFSDKDVD	LYLWAPTAQD	VRVCFYDGP	GPLLQTVQLK	180
ELNGVMSVTV	PRYRENQYLL	YEVKVYHPST	SQVEKCLADD	PYARGLSANG	TRTWLVLDINS	240
ETLKPASWDE	LSDEKPNLES	FSDISIYELH	IRDFSADHST	VDCNSRGGFR	AFTFQDSAGI	300
RHLRLKLSAAG	LTHVHLLPSF	HFASVDDNKS	NWKFVDEAQL	AKLPPGSDEQ	QAAIVSIQQE	360
DPYNWGYDVP	LWGVPKGSYA	SNPDGPSRII	EYRQMVQALN	RIGLRVVMVDV	VYNHLDSSGP	420
FGVSSVLDKI	VPGYLLRRNV	NGQIENSAAM	NNTASEHFMV	DRLIVDDLLN	WAINYKVDGF	480
RPDLMGHIMK	STMTPTVMIC	TISTI I K I K D	VFADTLIRAK	SAIRSLTRDV	HGVDGSKIYL	540
YGEWDFGEV	AQNKRGINAS	QINMSGTGIG	SFNDRIIRDSV	NGGNPFGNPL	QQGFSTGLFL	600
EPNGYYQONE	ADTRRELATY	ADHIQIGLAG	NLKDYVLRTH	TGEAKKGS DI	YTFDGSPPVGY	660
TSSPVETINY	VSADHNETLF	DIVSIKTPIG	LSIDEKCRIN	HLASSMIALS	QGIPFFHAGD	720
EILRSKSLDR	DSYNSGDWEN	KLDFTYETNN	WGVGLPPRDK	NEENWHLIK P	RLENPSFRPL	780
KNHILSVFDN	FVDILKIRYS	SPLFRLSTAS	DIEQVRVRPHN	TGPSMVPGVI	VMSIKDAQNE	840
KCKMAQLDKN	FSYVVTIFNV	CPHEVSIEIH	DLASLGLELH	PIQVNSSDAL	VRQSAYEASK	900
GRFTVPRRTT	AVFVQPRC					918

SEQ ID NO: 117 moltype = AA length = 963

FEATURE
source Location/Qualifiers
1..963
mol_type = protein
organism = *Triticum aestivum*

SEQUENCE: 117

MPMPMRTMLL	RHLSPAPALP	NPRSSASSP	QRRPARARPP	PLHSARATAL	RARRTPMAAG	60
ETGASVSUSA	AEAEAEATQA	FMPDARAYWV	TSDLIAWNVS	EQEASVLYLY	ASRTAAMGLS	120
PSNGGIQGYD	SKVELQPESA	GLPETVTQKF	PFISSYRAPR	VPSSVDVASL	VKQQLVIASF	180
GADGKHVDVT	GLQLPGVLDD	IFAYTGPLGA	VFREDSVSLH	LWAPTAQDVS	VCFDFGPAGP	240
VLETVQLKES	NGVMSVTGPR	EWENRYLYYE	VDVYHPTKAQ	VLKCLAGDPY	ARGLSANGAR	300
TWLVDINNET	LKPASWDELA	DEKPKLDSFS	DITIYELHIR	DFSADHGTVD	SDSCGGFRAP	360
ATQASAGMOH	LRKLS DAGLT	HVHLLPSFHF	AGVDDIKSNW	KFVDECKLAT	FPPGSDMQQE	420
AVVAIQEEDP	YNWGYNPVLV	GVPKGSYASD	PDGGSRIIEY	RQMVQALNRI	GLRVVMDVVY	480
NHLDSSGPCG	ISSVLDKIVP	GYVVRDRTNG	QIENSAAMMN	TASEHFMVDR	LIVDDLLNWA	540
VNYKIDGFRF	DLMGHIMKHT	MMRAKAAALQS	LTRDAHGVGD	SKIYLYGEGW	DFAEVARNQR	600
GINGSQLNMS	GTGIGSFNDR	IRDAVNGGNP	FGNPLQQGFN	TGLFLEPNGF	YQGNEDATRR	660
SLATYADQIQ	IGLAGNLRDY	VLITHTGETK	KGSEIHTFDG	LPVGYTSSPI	EIINYVSAHD	720
NETLFDVIVS	KTPMNLVSDE	RCRINHLASS	MMALSQGIPF	PHAGDEILRS	KSIDRDSYNS	780
GDWFNKLDFE	YETNNWGVGL	PPSEKNEDNW	PLMKPRLENP	SFKPAKGHIL	AALDSFVDIL	840
KIRYSSPLFR	LSTASDIKQR	VHFHNTGPSS	VPGVIVMGIE	DARDEKPEMA	QLDANFSYVV	900
TVFNVCPEHV	SMDIPALASM	RLELHPVQVN	SSDALVGKSV	YEATGRFTV	PRRTVSFVFE	960
PRC						963

SEQ ID NO: 118 moltype = AA length = 647

FEATURE
source Location/Qualifiers
1..647
mol_type = protein
organism = *Clostridium phytofermentans*

SEQUENCE: 118

MDEFWNSIDG	EKQYYDGNND	LGCTYTNRST	KLKVWAPTAS	MVVVNLYQNG	NAGKPYITEI	60
MKKEESGIWS	VCLLDLEGV	YYTYLVTVDG	QTKAEAVDPA	RTTGLNGKRA	MILDLEKTNP	120
TGFLLEDTKPK	FDLFDLAVIY	ELHIRDLSE	SDSGIKEKKG	LLGLTELNTR	NSDGLTTGLS	180
HILDGLGVTHI	HLLPCFDYAS	VDEENSSIFN	WGYDPENYNV	VEGYSSTNYP	DGAVRVKEFK	240
TLVQSLHENG	LRVIMDVVYN	HTMKTEESNF	NKIVPDYYR	KVGDKFSDAS	ACGNETASER	300
LMVRKFIVDS	IIYWAKEYHI	DGFRFDLMGI	HDIEITMNEVR	KVLDDQIDPSI	ILYGEGWVG	360
DSPLPAQORA	MKANMSMLPW	IAAFSDDFRD	GLKGSVFLAE	EKGATGSDS	KKESVKFGVV	420
ASTLHPQIDY	KKVNYSDSPW	ALEPAQCINY	VSADHNYTLW	DKIACSKCKD	TYEIRVKKNK	480

-continued

LCAAIIVFTSQ	GIPFLQAGEE	MLRNKPSSEI	AGEFVENSYN	SSDSVNCIKW	SNKANVIDVV	540
SYEGLIRFR	KEHKALRMQS	AKKISKRLTF	LPEEREDVIS	YLIQGDLDVK	TLCVIYNSSE	600
EKVTTIRLPES	DWTVYIDGNN	SGVEPLYEVK	GTTVEVEPIS	CMVLVKD		647

SEQ ID NO: 119 moltype = AA length = 1764
 FEATURE Location/Qualifiers
 source 1..1764
 mol_type = protein
 organism = Streptomyces avermitilis

SEQUENCE: 119

ATPPAPPSDA	KLAAPARHD	ATREQFYFVM	PDRFANGDTS	NDKGGLTGSR	LSTGYDPTDK	60
GFYQGGDLKG	LTRKLDYIKG	LGTTSIWLAP	IFKNQPVQGT	GKDASAGYHG	YWITDFTKVD	120
PHFGTNKDLE	TLISKAHAKG	MKVFFDVITN	HTADVVDYEE	KSYGYLSKGA	FPYLTKDGRP	180
FDDAGYTDGP	RKFPVAVDGS	FPRTPAFAAR	KKNAKVPSWL	NDPTMYHNRG	DSTFAGESST	240
HGDFSGLDL	WTERPEVVRG	MEKIYEKVV	DFGIDGFRID	TVKHVNTFEW	TQWATALDAY	300
AKKRGKDDFF	MFGFVYSADT	SVTSPYVTQG	RLDSTLDFFP	QDAARSYASQ	GGSAKKLASV	360
FGDDYKYTTD	KANAYEQVTF	LGNHDMGRIG	YFLNQDNPKA	TDAELLRKDR	LANELMFLSR	420
GNPVVYGDDE	QGFTSGGGDK	DARQTMFASK	VADYLDDEI	GTDGRHASDA	YDTSAPLYKE	480
IAALSKLRKD	NPALADGIQT	ERYAADGAGV	YAFSRTDART	GTEYVVAVNN	ADKASATFA	540
TGSADTAFKG	IHGTDVLKLS	DADKKITVTV	PAGAAVVLKA	AGRPGTPAAK	PSLTLKAPDA	600
GATGTVELSA	DVDGGRILNRV	VFAAQVGNK	WRTLGSADHA	PYRVTTQIGK	DVPAGTALRY	660
KAVVIDAAGR	TASATAASTT	GTTPAAETPT	ASSRDYAIHV	YKRPDGDYTD	WRLYAWGDLA	720
DGESTTWPAG	HDFVGRDAYG	AFAYVKLKPG	ASTVNFVID	KDGDKDVSD	RTIDVTKAGE	780
VWVEQGKETV	RTERPDYPAQ	DKTKAVIHYH	RADGDLTGWG	LHVVTGAATP	TDWSKPLEPV	840
RTDAYGAVEF	VPLTDGATSL	SYIIHKGDEK	DLSDRLDL	TADGHEVWLL	NGQENHLLPQ	900
PAGSAAALDL	TTSKAVWIDR	NTVAVNGSDA	AASTQLSSR	DGSIKVDKGS	LTSDDERWLR	960
LSKTSLSLTAQ	KAAPFHLKSY	TANVSDPRDR	DRVREALAGQ	VVASQRAANG	AVLAATGVQL	1020
AGVLDLIDYA	KADLGPTEFR	GGHPTLAVWA	PTAQSVSLEL	DGAHVRMKRN	NATGVVSVTG	1080
PASWKGKPYR	VYVVKWAPT	RKVVTNKVTD	PYSVALTTDS	ERSLVVDLDD	RSLAPSGWSS	1140
LKKPKAVPLR	DAIEQELHIR	DFSVADRTVP	AKDRGTYLAF	TDKNSDGRSR	LRQLAESGTS	1200
YVHLPLAFDI	ATIAEKKSQ	QATDCDLASY	AADSEKQDEC	LTAVAAKDAY	NWGYDPYHYT	1260
VPEGSYATDA	NGTRRTVEFR	RMVKSQNDQ	LRVVMDDVYN	HTAAAGQAGT	SVLDRIVPGY	1320
YQRLADGGSV	ATSTCCANTA	TENAMMGKLV	VDSLVTWAKE	YKVDGFRFDL	MGHQPKANIL	1380
AVRKALDALT	SAWDGVDGKK	IILYGEWGNF	GEVADDARFV	QATQKNMAGT	GIATFSDRAR	1440
DAVRGGGPPD	ADPGVQGFPS	GLYTDPNSSD	ANGTPAEQKA	RLLHYQDLIK	VGLSGNLAKY	1500
RFTDSSGKEV	TGSEVDYNGT	GAGYADAPGD	ALAYADAHDN	ESLYDALTYK	LPKGTAPADR	1560
ARMQVLAMAT	AALAQGPSLS	QAGSDLLRSK	SLDRNSYDSG	DWFNAIHWNC	QDNGNGRGL	1620
PMAADNKSXW	PYATPLLTST	KVGCQIEGT	SAGYQDLLRI	RTTEPDPSLS	TAGQVQSKLT	1680
FPLSGKDETP	GVITMKLGDL	VVVFNATPDQ	QEQTVAAALG	KDYALHPVQA	AGADPIVKSA	1740
SYTAKSGMFA	VPGRTVAFIS	QVAR				1764

SEQ ID NO: 120 moltype = AA length = 1079
 FEATURE Location/Qualifiers
 source 1..1079
 mol_type = protein
 organism = Klebsiella pneumoniae

SEQUENCE: 120

SSSSPSGSPG	SPGNPGNPGT	PGTDPDQDVV	VRLPDVAVPG	EAAQASANQA	VIHLVDIAGI	60
TSSTPADYAT	KNLYLWNNTG	CDALSAFVAD	WNDVSTTPTG	SDKYGPYVVI	PLTKESGCIN	120
VIVRDGTNKL	IDSRLRVSGF	DFTDRTVSVI	AGNSAVYDSR	ADAFRAAFGV	ALADAHWVDK	180
TTLWPGGEN	KPIVRLYYS	SSKVAADSNG	EFTDKYVKLT	PTTVSQQVSM	RFPHLASYPA	240
FKLPDDVNDV	ELLQGETVFI	SAESDGLISS	ATQVQTAGVL	DDTYAAAEEA	LSYGAQLTDS	300
GVTFRVWAPT	AQQVELVYVS	ADKKVVASHP	MTRDSASGAW	SWQGGSDLKG	AFYRYAMTVY	360
HPQSRKVEQY	EVTDPYAHSL	STNSEYSQVV	DLNDSALKPE	GWDGLTMPHA	QKTKADLAKM	420
TIHESHIRDL	SAWDQTPAE	LRGKYLALTA	QESNMVQHLK	QLSASGVTHI	ELLPVDFLAT	480
VNEFSDKVAD	IQQPFSRLCE	INSAVKSSEF	AGYCDSGSTV	EEVLTQLKQN	DSKDNPPQVQA	540
LNTLVAQTD	YNWGYDPFHY	TVPEGSYATD	PEGTARIKEF	RTMIQAIKQD	LGMNVIMDVV	600
YNHTNAAGPT	DRTSVLDKIV	PWYYQRLNET	TGSVESATCC	SDSAPEHRMF	AKLIADSLAV	660
WTTDYKIDGF	RFDLMGYHPK	AQILSAWERI	KALNPDIYFF	GEGWDSNQSD	RFEIASQINL	720
KGTGIGTFSD	RLRDAVRGGG	PFDSGDALRQ	NQGVGSGAGV	LPNELTSMTD	DQARHLADLT	780
RLGMAGNLAD	FVLIDKDGAV	KKGSEIDYNG	APGGYAADPT	EVVNYVSKHD	NQTLWDMISY	840
KAAQEADLDT	RVRMQAVSLA	TVMLGQGIAP	DQQGSELLRS	KSFTDRSDYS	GDWFRNRVDS	900
LQDNVNYVGM	PRSSDDGSNY	DIARVKDAV	ATPGETELKQ	MTAFYQBELTA	LRKSSPLFTL	960
GDGATVMQRV	DFRNTGADQQ	TGLLVMTIDD	GMQAGASLDS	RVDGIVVAIN	AAPESRTLQD	1020
FAGTSLQLSA	IQQAAGDRSL	ASGVQVAADG	SVTLPAWSVA	VLELPQGESQ	GAGLPVSSK	1079

SEQ ID NO: 121 moltype = AA length = 546
 FEATURE Location/Qualifiers
 source 1..546
 mol_type = protein
 organism = Rhizomucor pusillus

SEQUENCE: 121

ATSDDWKGA	IYQLLTDTRFG	RADDSTSNCS	NLSNYCGGTG	EGITKHLDDYI	SGMGFDIAWI	60
SPIPKNSDGG	YHGWWATDFY	QLNSNFGDES	QLKALIAAH	ERDMYVMLDV	VANHAGPTSN	120
GYSGYTFDDA	SLYHPKCTID	YNNQTSIEQC	WVADELPDID	TENSDNVAIL	NDIVSGWVGN	180
YSGDGRIDT	VKHIRKDFWT	GYAEAAGVFA	TGEVFNGDPA	YVGYPQKYL	SLINYPMYA	240
LNDVFSKSK	GFSRISEMLG	SNRRAFEDTS	VLTTFVDNHD	NPRFLNSQSD	KALFKNALTY	300
VLLGEGIPV	YSGSEQGFSG	GADPANREVL	WTNNTDSSD	LYQFIKTVNS	VRMKSNAKAV	360
MDIYVGDNAY	AFKHGDALVV	LNNYSGSGSTN	QVSFVSQKFP	DSGASLMDIV	SNITTTVSSD	420

-continued

GTVTFNLKDG LPAIFTSAGA TSPGGSSGSV EVTFDVIYATT VYQNIYITG DVSELGNWTP 480
 ANGVALSSAN YPTWSATIAL PADTTIQYKY VNIDGSTVIW EDAISNREIT TPASGTYTEK 540
 DTWDES 546

SEQ ID NO: 122 moltype = AA length = 481
 FEATURE Location/Qualifiers
 source 1..481
 mol_type = protein
 organism = Bacillus licheniformis

SEQUENCE: 122
 VNGTLMQYFE WYTPNDGQHW KRLQNDAEHL SDIGITAVWI PPAYKGTSAQ DVGYGAYDLY 60
 DLGEFPHQKGT VRTKYGTKGE LQSAIKSLHS RDINVYGDV INHKGADAT EDVTAVEVDP 120
 ADRNRVISGE HLIKAWTHFH FPGRGSTYSD FKWYWHFDG TDWDESRLN RIYKFQKKTW 180
 DWEVSNEFGN YDYLMYADID YDHPDVVAEI KRWGTWYANE LQLDGFRLDA VKHIKFSFLR 240
 DWVNVHREKT GKEMFTVAEY WSNLDGALEN YLNKTNFNHS VFDVPLHYQF HAASTQGGGY 300
 DMRKLLNGTV VSKHPLKSVT FVDNHDTPQG QSLESTVQTW FKPLAYAFIL TRESGYPQVF 360
 YGDMYGTGKD SQREIPALKH KIEPILKARK QYAYGAQHDY FDHHDIVGWT REGDSSSVANS 420
 GLAALITDGP GGAKRMVVR QNAGETWHDI TGNRSEPVVI NSEGWGEFHV NGGSVSIYVQ 480
 R 481

SEQ ID NO: 123 moltype = AA length = 484
 FEATURE Location/Qualifiers
 source 1..484
 mol_type = protein
 organism = Aspergillus niger

SEQUENCE: 123
 LSAAEWRTQS IYFLLTDRFG RTDNSTTATC DTGDDIYCGG SWQGIINHL D YIQGMGFTAI 60
 WISPIEQLP QDTADGEAYH GYWQQKIYDV NSNFGTADDL KSLSDALHAR GMYLMVDVVP 120
 NHMGYAGNGN DVDYSVDFPF DSSSYFHPYC LITDWDNLTM VQDCWEGDTI VSLPDLNTE 180
 TAVRTIWDW VADLVSNYSV DGLRIDSLE VEPDFPPGYQ EAAGVYCVGE VDNGNPALDC 240
 PYQKVLGVL NYPIYWQLLY AFESSSGSIS NLYNMIKSA SDCSDPTLLG NFIENHDNPR 300
 FASYTSDYSQ AKNVLSYIFL SDGIPIVYAG EEQHYSGGKV PYNREATWLS GYDTSALYLT 360
 WIATTNAIRK LAISADSAYI TYANDAFYTD SNTIAMRKGT SGSQVITVLS NKGSSGSSYT 420
 LTLSSGSGYS GTKLIEAYTC TSVTVDSGGD IPVPMASGLP RVLLPASVVD SSSLCGSGR 480
 LYVE 484

SEQ ID NO: 124 moltype = AA length = 586
 FEATURE Location/Qualifiers
 source 1..586
 mol_type = protein
 organism = Aspergillus tamarii

SEQUENCE: 124
 ATPADWRSQS IYFLLTDRFA RTDGSTTAAC NTEDRKYCGG TWQGIIDKLD YIQGMGFTAI 60
 WITPVTGQLP QHTAYGDAYH GYWQQDIYSL NENYGTADDL KALSSALHER GMYLMVDVVA 120
 NHMGYDGAGA SVDYSVFKPF NSQEFHSPFC LIQNYEDQTQ VENCWLGDNT VSLPDLDTTK 180
 DEVKNEWYDW VGTILVSNYSI DGLRVDTVKH VQKDFWPGYN KAAGVYCIGE VLDGDPAYTC 240
 PYQDVMDGVL NYPIIYPLLN AFKSTSGSMN DLYNMINTVK SDCPDSTLLG TFVENHDNPR 300
 FASYTNDIAL AKNVAAFIIL NDGIPIIYAG QEQHYAGGND PANREATWLS GYATDSELYK 360
 LIASANAIRS HAISKDTGFV TYKNWPIYKD DTIAMRKGT DGSQVVTLS NKGASGDSYT 420
 LSLGDTGYKA GQQLTEVIGC TTVTVGSDGK VPVPMAGGLP RVLYPTEKLA DSKICSSSGA 480
 TSPGGSSGSV EVTFDVIYATT VYQNIYITG DVSELGNWTP ANGVALSSAN YPTWSATIAL 540
 PADTTIQYKY VNIDGSTVIW EDAISNREIT TPASGTYTEK DTWDES 586

SEQ ID NO: 125 moltype = AA length = 603
 FEATURE Location/Qualifiers
 source 1..603
 mol_type = protein
 organism = Acidomyces richmondensis

SEQUENCE: 125
 LTPAEWRAQS IYQVLTDRFA LTNGSTTAPC NLNEYCGGTW QGIINKLDYI QGMGFTAIWI 60
 SPVVENIPAS DNTADGESYH GYWAQRIYEV NPNFGSAADL EALSEAIHAR GMYLMVDIVT 120
 NHMGYDGCCT CVDYSVDFPF DNQSYFHPFC LINYNNTASI QVCWEGDNIV SLPDLRTEDS 180
 DVLGMWETWI TQLVANYSID GLRVDSMQQV DQAFWQPFMS AAGDLYAVGE VFNGDPTYTC 240
 PYQQYLPGLV NYPAYWITQ AFESTSGSIG NLVNGINEMK NDCLDTLLG SFLENHDNPR 300
 FPSYTSYDSL DKNAIAFAML QDGIPIVYEG QEQHYSGGSV PNNREDIWS GYSTTSSELYT 360
 FKIINAIRT QALTKDSSYL TYKAYPVYSD SQTIAMRKGE TYPIISVFTN SGASGSAYS 420
 TLSSSDTGFS ENQSITELLT CVSTTDSGG NLVNVNISSGL PRVYPTSAI SGSTVCAEST 480
 STLTSSPIS TSTSDCTAS SVAVTFDET TTYGETIKL SGSISQLGDW NTQDAILLSA 540
 ADYKSTDNVW FVTINLPAGI VFQYKYINVD SDGDVTWEAD PNHTYTVSAT CATAATIHDT 600
 WQN 603

SEQ ID NO: 126 moltype = AA length = 588
 FEATURE Location/Qualifiers
 source 1..588
 mol_type = protein
 organism = Aspergillus bombycis

SEQUENCE: 126
 ATPADWRSQS IYFLLTDRFA RTDGSTTATC NTEDRKYCGG TWQGIIDKLD YIQGMGFTAI 60
 WITPVTGQLP QDTAYGEAYH GYWQQDIYAL NENHGTADDL KALSSALHER GMYLMVDVVA 120

-continued

NHMGYDGAGA	SVDYSVFNPF	SSQDYFHSFC	LIENYDDQTQ	SENCWLGDNS	VSLPDLDTTK	180
DEVKNEWYEW	VGNLVSNSYI	DGLRVDTVKH	VQKDFWPGYN	EAAGVYCIGE	VLNGDPAYTC	240
PYQDVMDGVL	NYPIYYPLLN	AFKSTSGSMN	DLYNMINTVK	SDCPDSTLLG	TFVENHDNPR	300
FASYTNDIAL	AKNVAAFIIL	NDGIPIIYAG	QEQHYAGGND	PANREATWLS	GYSTDSEIYK	360
LIASANAIRN	HAVSTDTEGFV	TYKNWPIYKD	DTTIAMRKGT	DGSQIVTILS	NKGASGDAYT	420
LSLGNTGYTA	GQQLTEVIGC	TTLTVGSDGN	VPVPMAGGLP	RVLYPTEKLG	DSKICSSSGR	480
GATSPGSGSS	SVEVTFDVYA	TTYVQNIYI	TGDVSELGNW	TPANGVALSS	ANYPTWSATI	540
ALPADTTIQY	KYVNIIDGSTV	IWEDAISNRE	ITTPASGTYT	EKDTWDES		588

SEQ ID NO: 127 moltype = AA length = 570
 FEATURE Location/Qualifiers
 source 1..570
 mol_type = protein
 organism = Alternaria sp.

SEQUENCE: 127

ADTSAWKSRS	IYFVLTDRIA	RSSSDTGGS	CSNLGNYCGG	TFKGLSKLD	YIKNLGFDAL	60
WITPVVANS	GGYHGYAOD	LYAVNSNYGT	AADLKSIVNT	AHSGKIYVMV	DVVANHMGQG	120
AISGNRPEPL	NQDSSYHSAC	DINYSSQTSI	EQCRIANLPD	LNTQSSQIRS	LLNTWISWLV	180
NEYSFDGVR	DTVKHVEKDF	WPGFASAAGV	YSIGEVWDGN	PTYLAEYARL	MPGLLNATY	240
YPMNNFYQK	GSSQALVDM	NTVRDTPDP	SALGTFLDNH	DNNRWLNQKN	DVTLLKNALA	300
FVILSRGIPI	VYVGTEQGYA	GGADPANRED	LWRSSFTNTA	DLYQAIIKLN	AARTSAGGLA	360
GNDHTHLYVS	SNAYAWSRAN	GNLVVLTNTA	SGGSNAQHCF	NTQKANGRWT	NVYNGGATVT	420
ADGSGNICVN	VANGEPVLL	VSTATPTSAT	PTSNPSPTTL	LTTSTACPTS	VSVSPTHRVT	480
TVFGDTIKIT	GNTAQLGNWN	PSNGVALSAA	SYTSSNPIWT	LTLPLPAGSA	IQYKFKVKSS	540
GGTVTWESDP	NRSYSVPGCQ	ASASVSSQWQ				570

SEQ ID NO: 128 moltype = AA length = 551
 FEATURE Location/Qualifiers
 source 1..551
 mol_type = protein
 organism = Rhizopus microsporus

SEQUENCE: 128

SPITIRSQNS	NDWSSRVIIQ	LLTDRFAKT	DDQSPCSDLG	NYCGGSFQGI	INHLDYIAGM	60
GPDIAWISPI	PQNAQGGYHG	YWATNFSAIN	SNFGSSNDLK	KLVAQAAHAKN	MYVMLDVVAN	120
HVGTPSTPMN	YNGYTFNQGS	YHSYCDINY	NDQTSVEQCW	LSGLPDLNTE	NDYVVNTLYS	180
TVSNWISYEG	FDGIRIDTVK	HVRKDFWDGY	VKAAGVFATG	EVLHGSVSIV	APYQSHVPSL	240
INYPLYPIY	DVFTKAATMT	RLKSGYNDIQ	SGGFSNLNLL	LNFDNHDNP	RLLSKADQSL	300
VKNALYSML	IQGIPIVYVG	TEQSYKGGND	PNNREPLWTS	GYSSSSEMYQ	FIKQVIQIRK	360
GSNATVTMDI	DQADNVVYFQ	RGNYLAVVNN	YQGGSTNSVT	VKSGSFADGT	VLKDFVSGAT	420
ATVKNKSITF	QLQNGNPAVF	SPQGATSPGG	SGGSVEVTFD	VYATTVYQON	IYITGDVSEL	480
GNWTPANGVA	LSSANYPTWS	ATIALPADTT	IQYKYNIDG	STVIWEDAIS	NREITTPASG	540
TYTEKDTWDE	S					551

SEQ ID NO: 129 moltype = AA length = 565
 FEATURE Location/Qualifiers
 source 1..565
 mol_type = protein
 organism = Syncephalastrum racemosum

SEQUENCE: 129

VPLSIIDKRA	GVTTLISKRA	ADDWKSRSIY	QIVTDRFARS	DGSTSGCGDL	SNYCGGDYKG	60
IQNLQLDYIAG	MGPDAIWISP	IPENTDGGYH	GYWAKNFEL	NTNFGSADDL	KALVTAAGHK	120
GMVVMLDVVA	NHAGFTSGGD	YSGFTFDSAS	NYHAQCDIDY	ENQTSIEQCW	VADNLDPINT	180
EDDTIVSKLH	SIVSDWVTY	DFDGIKIDTV	KHIRKDFWSG	YEEAAGVFAT	GEVFDGDAAY	240
VGPYQDLSS	LINYPYIYAI	RDVFTAGSGF	SRISDMLSSI	NSNFKDPSVL	TFVDNQDNA	300
RFLSVKSDTS	LYKNALAFIT	LTEGIPVYVY	GTEQGFKGGN	DPNNREVLWT	SNYDTSDDL	360
KFIKIVNDQV	RQKSNKTVKL	NVDVGTNTYA	FTHGKNLIVV	NNYGGSGSTAS	VTVKVGDZIA	420
DGTKLVDAVS	NITATVSGGS	ITFSLNNGLP	ALFVPSSGAT	SPGGSSGSVE	VTFDVYATT	480
YQNIYITGD	VSELGNWTPA	NGVALSSANY	PTWSATIALP	ADTTIQYKYV	NIDGSTVIWE	540
DAISNREITT	PASGTYTEKD	TWDES				565

SEQ ID NO: 130 moltype = AA length = 583
 FEATURE Location/Qualifiers
 source 1..583
 mol_type = protein
 organism = Rhizomucor pusillus

SEQUENCE: 130

ATSDDWKGKA	IYQLLTDRFG	RADDSTSNCS	NLSNYCGGTY	EGITKHLDYI	SGMGFDIAWI	60
SPIPKNSDGG	YHGFWATDFY	QLNSNFGDES	QLKALIQAAH	ERDMYVMLDV	VANHAGPTSN	120
GYSGYTFDDA	SLYHPKCTID	YNNQTSIEQC	WVADELDPID	TENSNDNVAIL	NDIVSGWVGN	180
YSFDGIRIDT	VKHIRKDFWT	GYAAGVFA	TGEVFNPDPA	YVGPYQKYL	SLINYPMYA	240
LNDVFSKSK	GFSRISSEMLG	SNRNAFEDTS	VLTTFVDNHD	NPRFLNSQSD	KALFKNALTY	300
VLLGEGPIV	YVGSEQGFSG	GADPANREVL	WTTNYDTSSD	LYQFIKTVNS	VRMKSINKAVY	360
MDIYVGDNAY	AFKHGDALVV	LNNGYSGSTN	QVSFVSQSGF	DSGASLMDIV	SNITTTVSSD	420
GTVTFNKDKG	LPAIFTSATG	GTTTATPTG	SGSVTSTSKT	TATASKTSTS	TSSTSTCTPT	480
AVAVTFDLTA	TTTYGENIYL	VGSISQLGDW	ETSDGIALSA	DKYTSSDPLW	YVTVTLPAGE	540
SFEYKFIRIE	SDDSVWESD	PNREYTPQA	CGTSTATVTD	TWR		583

SEQ ID NO: 131 moltype = AA length = 553
 FEATURE Location/Qualifiers

-continued

```

source          1..553
                mol_type = protein
                organism = Dichotomocladium hesseltinei

SEQUENCE: 131
QPVNITKRAS AADWRSRAIY QVLTDRFART DGSTSGCSNL SNYCGGTFKG ITNKLDYIAN 60
LGFDAIWISP IPTNSPGGYH GYWATDFYGI NSNFGSSNDL KELVNAAHAK GMYVMLDVVA 120
NHAGPTSNGD YSGYTFGSSG LYHNRCISINY NDQRSIEQCW VADDLPDINT ENNDNVNKP 180
NIVSTWVKTY GFDAIRIDTV KHVRFDFWPG YTSAGVFAT GEVFDGNPSY VADYQNYMES 240
LINYPYYAL NDVFASGYSF SRLSNQRVAN YHAFKDVSVL PIFIDNHDNP RFLNKKNDIA 300
QFKNALTYVL LGEIGIPVYVY GSEQAYAGGA DPANREALWS SGFSTNSDMY QFIAKLNRVR 360
QKSNKSVYMD LDVQNNVYAF MHGKSLVVLN NFGNGASRQV TVNVGAQVAS NTRLTDVVSG 420
TSVTVSGSSV TFTINNGLPA VFTVSGATSP GSSSGSVEVT FVDVATTYVG QNIYITGDVS 480
ELGNWTPANG VALSSANYPT WSATIALPAD TTIQYKYVNI DGSTVIWEDA ISNREITTPA 540
SGTYTEKDTW DES 553

SEQ ID NO: 132      moltype = AA length = 552
FEATURE            Location/Qualifiers
source             1..552
                  mol_type = protein
                  organism = Lichtheimia ramosa

SEQUENCE: 132
RPFVKRATAD DWRDRAIYQI LTDRFARSDG STDNCSDLN YCGGNYQGII QQLDYIEGMG 60
FDAIWISPIP ANADGGYHGY WATDFESLND HFGSQDDLKA LVDAAAHERGM YVMLDVVANH 120
AGPTNNGDYS GYTFGSSDLY HPQCSIDYSN QNSIEQCWVA DNLPDIDTEN DSNVNKLHDI 180
VSTWVSTYGF DGIRIDTVKH VRKDFWPGYA SAAGVFATGE VYDGDQTYVG AYQDYLDL 240
NYPLYALND VFASGQGFQR LSNQRNAIMS AFKDVSVLTS FVDNHDVARF LSTKNDQALF 300
KNALAFVLLG ETIPVYVYGS EQGFAGGADP ANREALWSSG YDTSSDLYQF IATINNNVRQ 360
KSGKKVYMDL DVQDNVYAFM HGDALVVLNN YGSGASNQVS VNVGAQVAES TSFTDAISGT 420
SITVSSGSVT FTLDNGNPAI FVPAGATSPG GSSSGSVEVTF DVIYATTYVGQ NIYITGDVSE 480
LGNWTPANGV ALSSANYPTW SATIALPADT TTIQYKYVNI DGSTVIWEDA SNREITTPAS 540
GTYTEKDTWD ES 552

SEQ ID NO: 133      moltype = AA length = 587
FEATURE            Location/Qualifiers
source             1..587
                  mol_type = protein
                  organism = Penicillium aethiopicum

SEQUENCE: 133
ARTADWPKRS IYQMTDRFA RTDGTSTSPC NTKAGLYCGG TWRGTIDHLD YIQGMGFDAV 60
MISPIIENIE GRVDYGEAYH GYWPLNLDNL NSHFGTHQDL LDLSDALHSR GMYFMMDTVI 120
NNMAYITNGS DPATDIDYSV FTPFNNADYF HPYCTMNWSV PAIAQRCNTG DDTVALPDLF 180
TEHEDVQQLL IKWANKAIKT YSIDGLRIDA AKHVPNDFLR KFSGDVDIFM TGEVLEGSVS 240
IMEDYQSNVI NSLPNYPIYF EILSAFTNGN TSQLAIAVEN MRVAIPDVNA MASFSENHDK 300
PRIASYNDDM SIAKNVLVFT MLFDGIPMIY QGQEQHLKGD GVPHNREAIW LSKYDTEAEL 360
YKLIAKLNRI RNHAGYLGSD YFEDATHPIY QGSSELAFTK QVQGRQVVMV LSNQPSSTGR 420
YALDLAVSYN AGTELMVDLN CNNYTVDNQG VLRVDMDKGE PRVFFPRKYM EGSGLCGYSG 480
ATSPGGSSGS VEVTFDVYAT TVYQGNIYIT GDVSELGNWT PANGVALSSA NYPTWSATIA 540
LPADTTIQYK YVNI DGSTVI WEDAISNREI TTPASGTYTE KDTWDES 587

SEQ ID NO: 134      moltype = AA length = 597
FEATURE            Location/Qualifiers
source             1..597
                  mol_type = protein
                  organism = Subulispora sp.

SEQUENCE: 134
LTPAEWGSQS IYQVLTDRFA LTDGSTTASC DLNTYCGGTW LGIQNHLDYI QGMGFTAIWI 60
SPIVTNIAGD SVDGDSYHGY WAQDITTVNS AFGTEQDLIN LSAALHERGM YLMVDVVNNH 120
MGYLGCGTCV DYSEYTPFNE ESYHYPCPT DYSNLTSIQV CWEGDNIVSL PDLRTEDSDV 180
RSMWYDWITP LVAKYSIDGL RMDSAEHVEK SFWPGWVSAS GVIYVGEVDE GDPTIFPDWL 240
NYIDGTLNYP AYYWITQAFQ STSGSISNLV TGVNQLKASM KTSTFGSFLE NHDQPRFP 300
TSDTDLAKNA IAFAMLADGV PIVYVQEQG YSGGGVPNDR EPLWTSGYST TSAGYTFIKT 360
INAVRHLAVT QDTAYVAYQA YPIYSDSRVI AMKKSSVLAV FSNIGSSGSG YSITLPAGAF 420
AASQALTDVA SCQTYTADAS GGLTFTFGQA PSVFYATASL AGSGLCGTTG TGGSTGTTTA 480
SETGGSSPTS TACASVPVTF NEKVTTVVGE TIKISGSVAA LGDWATGSAV ALSAASYTSS 540
NPQWDVTISF APGTVIEWKY INVASSGAVT WEADPNHTYT VPASCATAAV VSDTWQT 597

SEQ ID NO: 135      moltype = AA length = 601
FEATURE            Location/Qualifiers
source             1..601
                  mol_type = protein
                  organism = Trichoderma paraviridescens

SEQUENCE: 135
LTAAQWRSQS IYQVLTDRFS QTNGATNSAC NAGNQVYCGG TWQGIQKLD YIKSMGFTAI 60
WISPVVENLA GNSADGEAYH GYWAQDIYQV NTNFGSAADL RALSEALHNA GMYLMVDIVT 120
NHMGYLGCGT CVQYNTFNP  NSQSYHHPFC LINFNSSNMT QIQNCWEGDN TVSLPDLATE 180
NANVLMSWQT WITQLVANYT IDGLRMDSCF ELNRYGFEPF QSSANVYIVG EVDNGDPAIV 240
CPYQKNYGLN TLNYPAYYWI TQAFQSTSGS ISNLVNLMT MKSECSDTTL LGSFMENHDN 300
PRFPLSTSDI SLAKNAIAFT MLADGIPYIY EGQEQHLNGG GVPNNREAIW LSGYSTSAVL 360
YTHIKALNQI RSQAQKNSA YVTTQAAVY SDSSTIVTRK GSTGSQIVGV FSNKGANGNS 420

```

-continued

YTLTLPASDT	GFTSNEQVVE	ILSCTAYTTD	SSGNLAVAMA	GGLPRVFYAR	SSLSGSGICP	480
NLGSGGGTPT	STPPTSCTAI	PVTFDEKVTT	TFGQTIKIAG	DISALGNWNT	ANAVTLTSAAN	540
YTSSNPLWAI	TLNLAPGGVQ	EYKYINVAQN	GGVTWEADPN	HTYTVPSACT	AQPTVANTWQ	600
G						601

SEQ ID NO: 136 moltype = AA length = 598
 FEATURE Location/Qualifiers
 source 1..598
 mol_type = protein
 organism = *Byssosascus striatosporus*

SEQUENCE: 136

LSADDWRAQS	IYQLLTDRFA	LTNGSTTAPC	DTBEQIYCGG	SWQGIIDKLD	YIQGMGFTAI	60
WISPVVENLS	GDSADGEAYH	GYWAQNVYEV	NPNFGATSDL	VALSQVVHDK	GMVLMLDVVT	120
NHMGYLCGCT	CVDYSVFNPF	NEESYYHPPC	LIDYDNTTSI	EVCWEGDNIV	SLPDLRTEDS	180
DVLSTWESWV	TELSVNTYVD	GIRLDSTEEL	DQAFLPFPES	AAGVYIVGEV	DNGDPAVVCP	240
YQEVYSGVLN	YPAYYWTQA	FESTSGSIGN	LVNGINTMKS	DCSDTSLGGS	FLENHDQPRF	300
ASLTSDISLA	KNAIAFSMLQ	DGIPIVYAGE	EQHYSGGAVP	NDREALWLSG	YPTSSTLYTW	360
ITSLNQIRSH	AIATNSSYLT	YNAYPVYSD	STIVMRKGFA	DNQIVSVYTN	QGADATAYTL	420
DLPSTDTGFT	ASQSLVEIGG	CTTTATDDSG	NLAVAMASGL	PRIYYPAAGL	SGSGVCGQLG	480
SGGGTPTSTP	PTSCTAIPVT	FDEKVTTFG	QTIKIAGDIS	ALGNWNTANA	VTLSAANYTS	540
SNPLWAITLN	LAPGVVEYK	YINVAQNGGV	TWEADPNHTY	TVPSACTAQP	TVANTWQG	598

SEQ ID NO: 137 moltype = AA length = 615
 FEATURE Location/Qualifiers
 source 1..615
 mol_type = protein
 organism = *Aspergillus brasiliensis*

SEQUENCE: 137

LSAAEWRTQS	IYFLLTDRFG	RTDNSTTATC	NTGDQIYCGG	SWQGIINHLD	YIQGMGFTAI	60
WISPIEQLP	QDTSDEAYH	GYWQQKIYDV	NSNFGTADDL	KSLSDALHAR	GMVLMIDVVP	120
NHMGVAGSGN	DVDYSVFDPF	DSSSYFHPYC	LITDWDNLTM	VQDCWEGDTI	VSLPDLNTE	180
TVVRTIWDYD	VADLVSNYSV	DGLRIDSVL	VEPDFFPGYQ	EAAGVYCVGE	VDNGNPALDC	240
PYQDYLDGVL	NYPIYWQLLY	AFESSSGSIS	DLYNMIKSA	SDCSDPTLLG	NFIENHDNPR	300
FAYYTSYDSQ	AKNVLSYIFL	SDGIPIVYAG	EEQHYSGGDV	PYNREATWLS	GYDTSAELEYT	360
WIATTNAIRK	LAIAADSSYI	TYANDPIYTD	SNTIAMRKGT	SGSQVITVLS	NKGSSGSSYT	420
LTLSGSGYTS	GTKLIEAYTC	TSVTVDNSGD	IPVPMASGLP	RVLLPASVVD	DSSLGSGSGS	480
STSTTTSTAT	ATTSKSTTT	SSSSSSSCT	ASATAIPITF	EELVTTYGE	EIYLSGSISQ	540
LGDWDTSDAV	KLSADDTSS	NPEWSVTVT	PVGTTFEYFK	IKVESGGSVT	WESDPNREYT	600
VPECGSGETV	VDTWR					615

SEQ ID NO: 138 moltype = AA length = 604
 FEATURE Location/Qualifiers
 source 1..604
 mol_type = protein
 organism = *Penicillium subspinulosum*

SEQUENCE: 138

ALSAEWRTQS	IYFLLTDRFG	RTDNSTTATC	DTGDQIYCGG	SWQGVINHLD	YIQGMGFTAI	60
WISPIEQLS	GDTSDGEAYH	GYWQQKIYNV	NSNFGTADDL	VALSDALHAR	DMVLMIDVVP	120
NHMGVDGDDG	DVDYSVFDPF	DSSSYFHPYC	LITDYDDIEM	VQDCWEGDTI	VSLPDLNTE	180
TVVQDIWYAW	VADLVANYSV	DGLRIDSVL	VQPAFFPAYQ	SAAGVYCVGE	VDNGDPTLDC	240
PYQDYLDGIL	NYPIYWQLLY	AFESSSGSIS	DLYDMINSVA	SDCSDPTLLG	NFIENHDNPR	300
FAYYTSYDSQ	AKNVASFIPL	SDGIPIVYAG	QEQHYSGGDV	PYDREATWLS	GYSTTAELYT	360
WIATTNSIRK	LAIISLDDYI	TYVNDPPYTD	ENTIAMRKGT	SGLQVITVLS	NLGADGSAYT	420
LTLSGSGYDS	GTDLIEVYTC	TSVTVDSSGD	IAPVPMESGLP	RVFLPESSIK	DSDLCSGTTT	480
TTTSTAATAT	ATSTSTCTAA	TEVSIIFEEL	VTTYGEEIY	LSGSISELGS	WDTSDALELS	540
AANYTSSNPE	WYLEVTLFVG	TSFEYKFIMI	ESDGTVVWES	DPNRSYTVPS	ACSGAVETVV	600
DTWR						604

SEQ ID NO: 139 moltype = AA length = 606
 FEATURE Location/Qualifiers
 source 1..606
 mol_type = protein
 organism = *Penicillium antarcticum*

SEQUENCE: 139

LTPAEWRSQS	IYFMLTDRFG	RSDNSTTAAC	NVSDRTYCGG	TWQGIINHLD	YIQGMGFTAI	60
WITPVEQLP	QDTSDGEAYH	GYWQQNIYEV	DSNLGTAADL	LALSEALHAR	GMVLMVDVVA	120
NHMGVAGAGS	SVEYSVFHPF	SSSSYFHSYC	LISNYDDQSN	VEDCWLGDIT	VSLPDVDTTQ	180
TAVQTLWYDW	IGDLVSNYSI	DGLRIDTVKH	VQKSFWPQYN	DAAGVYCVGE	IFDGDPAITC	240
DYQNYMDGVL	NYPIYWQLLY	AFQSSSGSIS	DLYDMINSVK	SDCADPTLLG	NFIENHDNPR	300
FASYTSDYSQ	AKNVISFLFL	SDGIPIIYSG	QEQHYSGGAD	PANREATWLS	GYSTTAELYK	360
YIATTNRIRK	AAVSADSSYI	TTKNVPFYQD	SHTLAMKKGS	SASPVITVLS	NYGSSGSSYT	420
LSLSGSGYSS	GTNLMEMYTC	TSVTVDSSGN	IAPVPMASGLP	RVLMMLASSAS	SICASSTTTS	480
TATVATQTTT	LTTTGTSTCTQ	ATVLPVLFKE	LVTTTYGQNV	YISGSISQLG	SWDTSSAIAL	540
SASSYNSSNP	LQWVAITFVG	TSFSQYKFLE	KTTGSTTIQW	ESDPNRSYTV	PTGCVGTAT	600
AIATWR						606

SEQ ID NO: 140 moltype = AA length = 609
 FEATURE Location/Qualifiers
 source 1..609

-continued

```

mol_type = protein
organism = Penicillium coprophilum

SEQUENCE: 140
LTPAEWRSQS IYFLLTDRFG RTDNSVTANC NVNDRVYCGG TWQGIINQLD YIQGMGFTAI 60
WITPVTQQLS QNTDGTSTYH GYWQDDIYNV NPNHGTSDDL LALSALHAR GMYLMVDVVA 120
NHMGYAGAGN NVDYSVFTPF NSASYFHSYC LISNYNDQSN VENCWLGDTT VSLPDLDTTQ 180
SSVQTLWNWN ISDLVSKYSI DGLRVDTVKH VQKSFWPAFN RAAGVYSVGE VFDGSPSYTC 240
DYQKYMDGVL NYPMYYPLLR AFQSTSGSIS DLYNMIGTSL STCADSTLLG NFIENHDNPR 300
FPSYTSYDYSQ AKNVLSFLFL SDGIPIVYSG QBQHYSGGSD PANREALWLS KYSTTAELYK 360
YIATTNKIRK AAVAADSSYI TSKNVAFYQD SHTLAMKKGS GSSPVITVLS NAGSSGSSYT 420
LYLSGSGYSS GTQLMELYTC TSVTVDSNNK IAVPMASGLP RVFVLASSVS NSGLCGSSTP 480
TTTATTATTA TQTTTATTTA GGCTQATALP VLFKELVTS YGQDIYISGS ISQLGTWDT 540
KAVALSADSY TSSNPLWQAT ITLPVGTTFQ YKFIKKANGA ITWESDPNRS YTVPTGCSGS 600
TATVTASWK 609

SEQ ID NO: 141      moltype = AA length = 602
FEATURE            Location/Qualifiers
source             1..602
                   mol_type = protein
                   organism = Penicillium olsonii

SEQUENCE: 141
LTPAEWRSQS IYFLLTDRFG RDDNSTTATC NTGDRTYCGG TWQGIINQLD YIQGMGFTAI 60
WITPVTQQLS ANTGYGTAYH GYWQDDIYEV NPNHGSSADL KALSALHAR GMYLMVDVVA 120
NHMGYAGAGN NVDYSVFTPF SSSSYFHSYC LISNYNDQSN VENCWLGDTT VSLPDLDTTQ 180
TAVQTIWNWN ITDLVSNYSI DGLRIDTVKH VQKSFWPGFN DAAGVYSVGE IFDGNPSYTC 240
DYQNYLDGVL NYPIIYPLLY AFQSTSGSIS DLYNMINTVA SDCADSTLLG NFIENHDNPR 300
FPSYTGDSYQ AKNVISYFLF SDGIPIIYSG QBQHYSGASD PANREALWLS GYSTTAELYK 360
WIATTNKIRK LAVSADSSYI TSKNSPFYQD SHTLGMKKGS VITILSNNGA SGSSYTLSSL 420
GSGYSSGSKL MELYTCTSI TSSSGNIPVP MVSGLPRALI PASSIGSNGL CGSTTSPTTT 480
AATQTTTATT TGTCTQATAL PVLFKELVTT SYGQNVYISG SISQLGNWDA SSAIALSASS 540
YTSSNPLWQV TITLPVGTKE EYKFIKSSG SATATWESDP NRSYTVPTGC AGTTATVTAT 600
WR 602

SEQ ID NO: 142      moltype = AA length = 621
FEATURE            Location/Qualifiers
source             1..621
                   mol_type = protein
                   organism = Penicillium vasconiae

SEQUENCE: 142
LTAAEWRTQS IYFLLTDRFG RTDNSTTATC SVSDRIYCGG SWQGIINHLD YIQGMGFTAI 60
WITPVTQQLS QDTGDGEAYH GYWQDEIYNV NTNNGTAADL LALSALHSR GMYLMVDVVA 120
NHMGYDAGN TVDYSVFTPF DSSSYFHSYC EISDYSNQTN VEDCWLGDTT VSLPDLDTTL 180
SSVQTIWNWN VAALVSNYSI DGLRIDTVKH VQKSFWPGYN SAAGVYCVGE VFDGDPAYTC 240
PYQSYLDGVL NYPIIYQLLY AFESTSGSIS SLYNMINSVA SDCSDPTLLG NFIENHDNPR 300
FASYSYDYSQ AKNVISPIFF SDGIPIVYAG QBQHYSGGSD PANREATWLS GYDTTATLYK 360
YITSTNKIRS LAISKDTAYI TSKNNAFYTD SNTIAMKKGS GGSQVITVLS NRGSSGSSYT 420
LTLGSGGYSS GTQLMEMYTC TAVTVDSNGN IAVPMASGLP RIYMLASSAC SICSSSCSTT 480
TTTSTTSTST TASTLKTST TSTSTSTST TSTCTQATAL VLFKEIVTTS YGQNIYISGS 540
ISQLGSWDTN NAVALSADQY TSSNNLWYVV VTIPVGTSTF YKFIETSTGS STITWESDPN 600
RSYTVPTGCA GSTATVTATW R 621

SEQ ID NO: 143      moltype = AA length = 615
FEATURE            Location/Qualifiers
source             1..615
                   mol_type = protein
                   organism = Penicillium sp.

SEQUENCE: 143
LTAAEWRSQS IYFLLTDRFG RTDNSTTATC NVSDRIYCGG SWQGIINHLD YIQGMGFTAI 60
WITPVTQQLS QDTGDGEAYH GYWQDEIYNV NTNNGTAADL LALSALHSR GMYLMVDVVA 120
NHMGYDAGN TVDYSVFTPF DSSSYFHSYC EITDYSNQTN VEDCWLGDTT VSLPDLNTTL 180
SSVQTIWYDW VAALVSNYSI DGLRIDTVKH VQKSFWPENY SAAGVYCVGE VFDGDPAYTC 240
PYQNYLDGVL NYPIIYQLLY AFESTSGSIS DLYNMINSVA SDCSDPTLLG NFIENHDNPR 300
FASYSYDYSQ AKNVLSPIFF SDGIPIVYAG QBQHYSGGSD PANREATWLS GYDTSAELEYT 360
WITSTNKIRS LAVSKDTAYI TSKNDAFYTD SNTIAMKKGS GGSQVITVLS NRGSSGSSYT 420
LTLGSGGYSS GTKLMEYTC TAVTVDSNGN IAVPMASGLP RVYMLASSAC SICSSACSTT 480
TTSSTTSTAT TSTTLKTST TTTSTCTQA TALPVLKKEI VTTSYGQNIY ISGSISELGD 540
WDTSTNAVALS ADQYTSSNNL WYVVVTIPVG TSFEYKFIEE TSGSSSITWE SDPNRSYTVP 600
TGCAGSTATV TATWR 615

SEQ ID NO: 144      moltype = AA length = 606
FEATURE            Location/Qualifiers
source             1..606
                   mol_type = protein
                   organism = Heterocephalum aurantiacum

SEQUENCE: 144
LTAAEWRSQS IYFLLTDRFA RTDGTSTAAC NLSQRAYCGG SWQGIINHLD YIQGMGFTAI 60
WITPVTQKIE ASTSDGTAYH GYWQDDIYNI NSHYGTADDL RALSSALHSR GMYLMIDVVA 120
NHMGYPGAGT SVDYSIPTPF GSSSYFHSYC QITDYDNQSN VENCWLGDNV VSLPDLNTQN 180
SNVRNLWYDW VEELVANYSV DGLRVDTVKH VEKDFWPSYN AAAGVYCVGE VFHGDPAITC 240

```

-continued

PYQNYMDGVL	NYPIYYQLLY	AFQSSSGSIT	DLNMINNSVA	SDCKDPTTLG	NFIENHDNPR	300
FPSYTSDSMQ	AKSVIAFLFL	SDGIPIIYAG	QEQHYSGGAD	PNNREAIWLS	GYSTSSTLYQ	360
FISSTNSIRK	LAISKDSSYL	TSRNNPFYTD	SNTIAMRKGS	SGSQVITVLS	NKGSSGNSYT	420
LTLTNHGYSS	GAQLTELYTC	SSIQVASSGG	LAVPMASGLP	RVLVPSSWIQ	SGSLCGGGST	480
TTTTTATTTT	TTTTSTSSCA	AATSLAVVFN	ELVTTYGEN	IFIAGSISQL	GSWDTGKSVA	540
LSASQYTSSN	PLWTATVSLP	VGTSFQYKFI	KKEPDGQVWV	ESDPNRSYTV	PAGCAGTTQT	600
VNTSWR						606

SEQ ID NO: 145 moltype = AA length = 602
 FEATURE Location/Qualifiers
 source 1..602
 mol_type = protein
 organism = Neosartorya massa

SEQUENCE: 145

LTPAEWRSQS	IYFLLTDRFG	REDNSTTAAC	DVTQRLYCGG	SWQGIINHL	YIQGMGFTAI	60
WITPVTQEQY	EDTGDGTSYH	GYWQQNIYEV	NYNYGTAQDL	KNLADALHAR	GMVLMVDVVA	120
NHMGYDGAGN	TVDYSVFPTF	DSSSYFHPYC	LISDYSNQTN	VEDCWLGD	TVSLPDLDDTT	180
TTVRTIYDW	VKGLVANYSI	DGLRIDTVKH	VEKDFWPGYN	DAAGVYCVGE	VFGSDPTYTC	240
PYQNYLDGVL	NYPIYYQLLY	AFESTSGSIS	NLYDMINSVA	SDCADPTLLG	NFIENHDNPR	300
FASYTSDYSQ	AKNVISFIF	SDGIPIVYAG	QEQHYSGGAD	PANREAVWLS	GYSTSATLYS	360
WIASTNRIRK	LAISKDAAYI	TSKNNPFYTD	SNTLAMRKGS	IAGAQTITVL	SNKGSSGSSY	420
TLSLSGTGYS	AGASLVEMYT	CTTLTVDSGG	NLPVPMASGL	PRVLVPSSWV	SGSGLCGSGS	480
TTTTTTTATA	TTTACTSATA	LPIVFEEVVT	TTYGENVYLT	GSISQLGNWD	TSSAIALSAS	540
KYTSSNPEWY	VTVTLPVGTS	FQYKFFKKES	DGSIVWESDP	NRSYTVPTGC	AGTTVTVSDT	600
WR						602

SEQ ID NO: 146 moltype = AA length = 588
 FEATURE Location/Qualifiers
 source 1..588
 mol_type = protein
 organism = Penicillium janthinellum

SEQUENCE: 146

ATPAQWRSQS	IYFMLTDRFA	RTDGGTTAPC	DTSQRAYCGG	TWQGIIDKLD	YIQGMGFTAI	60
WITPVTGQLD	GDTGDGTAYH	GYWQQDIYSL	NSNYGTASDL	KALASALHAR	GMVLMVDVVA	120
NHMGYNGAGN	TVDYSVDFAF	NSNQYFHSYC	EVTNYSNQTN	VEDCWLGD	TVSLPDLNTEL	180
SSVQSIWYNW	VGSLVSNYSI	DGLRVDTVKH	VQKDFWPGYN	KAAGVYCVGE	VFDGDASYTC	240
PYQEVMDGVL	NYPMYPLLR	AFQSTSGSMS	DLNMINMINTVK	STCSDSTLLG	TFVENHDNPR	300
FASYTNDMSL	AKNVAAFTIM	ADGIPIIYAG	QEQHYSGGSD	PANREAVWLS	GYNTDSALYK	360
LIAKVNAIRS	YAIQSASVY	TYKNYPIYQD	ASTLAMRKGS	SGTQTITVLS	NRGASGSQYT	420
LSLGNNGYST	GTTLTEIITC	AKITVDSSGN	VPVPMASGEP	RILYPSSSIK	GSAICASSGR	480
GATSPGGSSG	SVEVTFDVYA	TTVYGQNIYI	TGDIVSELGNW	TPANGVALSS	ANYPTWSATI	540
ALPADTTTIQY	KYVNIDGSTV	IWEDAISNRE	ITTPASGTYT	EKDTWDES		588

SEQ ID NO: 147 moltype = AA length = 589
 FEATURE Location/Qualifiers
 source 1..589
 mol_type = protein
 organism = Aspergillus brasiliensis

SEQUENCE: 147

ATPAEWRSSQ	IYFLLTDRFA	RTDNSTTASC	DLSARQYCGG	SWQGIINQLD	YIQGMGFTAI	60
WVTPVTAQIP	QDTGYGQAYH	GYWQQDAYAL	NSHYGTADDL	KALATALHSR	GMVLMVDVVA	120
NHMGHNGTGS	SVEYSVYNPF	NAKKYFHNLC	WISNYDNQTN	VEDCWLGDNT	VALPDLDTTR	180
TDVKNMWDYD	VKSLVSNYSV	DGLRVDTVKN	VQKNFWPGYN	NASGVYCIGE	VFDGDASYTC	240
PYQDDLGGVL	NYPMYPLLR	AFKSTTGSIS	DLNMINMINTVK	STCKDSTLLG	TFIENHDNPR	300
FANYTSDMSL	AKNVATFTIL	ADGIPIIYAG	QEQHYSGGND	PYNREATWLS	GKTTSELYT	360
HIAASNKIRT	HAIKQDSGYL	TYKNYPIYQD	TSTLAMRKGY	NGTQTITVLS	NLGASGSSYT	420
LSLPGTGYTA	GQKITEIYTC	TNLTVNSNGS	VPVPMKSGLP	RILYPTDKLV	NGSSFCSSSG	480
RGATSPGGSS	GSVEVTFDVY	ATTVYGQNIY	ITGDVSELGN	WTPANGVALS	SANYPTWSAT	540
IALPADTTTIQY	KYVNIDGSTV	VIWEDAISNR	EITTPASGTY	TEKDTWDES		589

SEQ ID NO: 148 moltype = AA length = 588
 FEATURE Location/Qualifiers
 source 1..588
 mol_type = protein
 organism = Aspergillus westerdijkiae

SEQUENCE: 148

ATPAQWRSQS	IYFLLTDRFA	RDDGTTATC	NTEDRKYCGG	TWQGIIDQLD	YIQGMGFTAI	60
WITPVTQAQIT	EDTKYGDAYH	GYWQQDIYSL	NENYGTADDL	KALADALHER	DMVLMVDVVA	120
NHMGYAGAGD	SVDYSVFNPF	NSQDYFHSFC	LIQDYNDQTQ	SEDCWLGDNS	VSLPDLDDTTK	180
SEVQDIWYDW	VGGLVSNYSI	DGLRIDTVKH	VQKEFWPGYN	DAAGVYCIGE	ILDGDASYTC	240
PYQEVLDGVL	NYPIYYPLLN	AFKSTSGSIS	DLNMINMINTVK	SDCPDSTLLG	TFIENHDNPR	300
FASYTDDIAL	AKNVAAFTIL	ADGIPIVYAG	QEQHYAGGED	PANREATWLG	KYNTDSELYK	360
LIAASNAINR	HAISTDKEYV	NYKNYPIYKD	DSTIAMRKGF	DGAQITITVLS	NQSSSGSSYT	420
LSLGDGTGFS	GDKLTEIYTC	TAVTVDSGK	VPVPMDDGAP	RALFPTEKLS	GSSLCSGSGR	480
GATSPGGSSG	SVEVTFDVYA	TTVYGQNIYI	TGDIVSELGNW	TPANGVALSS	ANYPTWSATI	540
ALPADTTTIQY	KYVNIDGSTV	IWEDAISNRE	ITTPASGTYT	EKDTWDES		588

SEQ ID NO: 149 moltype = AA length = 476
 FEATURE Location/Qualifiers

-continued

```

source          1..476
                mol_type = protein
                organism = Hamigera avellanea

SEQUENCE: 149
ATPADWRSRS IYFILTDRFA RTDGSSTTAE DTSARAYCGG TWRGIINKLD YIQNMGFTAI 60
WITPVTAQLP GSTGHGSAYH GYWQQDIYSL EPNYGTADDL RALASALHER NMYLMVDVVA 120
NHMGWAGSGD SVDYSVFNPF DSADYFHPYC LISNYEDQTE VENCWLGD TN VALVDLDTTR 180
SDVQNIWYEW VDSL VGNYSI DGLRIDTVRH VQKDFWPGFN DAAGVYSVGE VFSGDTAYTC 240
PYQEVLDGVL NYPIYYPLLR AFQSTSGSIN DLYNMINTVK SDCADSTLMG TFLENHDNPR 300
FASYTSDVAL AKNAIAFTIL SDGIPIIYAG QEQHYSGGND PANREAVWLS GYSTDSELYS 360
FVAVTNQIRN YAISQDQGYV TWKNVPYIQD TSTLAMRKGT DGSQVITVLS NLGASGSSYT 420
LTLGGSGYSS GQQLTEIFSC ATVTVDSSGN IPVPMGSGQP KVFYPTAGLG GSGICQ 476

SEQ ID NO: 150      moltype = AA length = 586
FEATURE            Location/Qualifiers
source             1..586
                  mol_type = protein
                  organism = Hamigera avellanea

SEQUENCE: 150
ATPADWRSRS IYFILTDRFA RTDGSSTTAE DTSARAYCGG TWRGIINKLD YIQNMGFTAI 60
WITPVTAQLP GSTGHGSAYH GYWQQDIYSL EPNYGTADDL RALASALHER NMYLMVDVVA 120
NHMGWAGSGD SVDYSVFNPF DSADYFHPYC LISNYEDQTE VENCWLGD TN VALVDLDTTR 180
SDVQNIWYEW VDSL VGNYSI DGLRIDTVRH VQKDFWPGFN DAAGVYSVGE VFSGDTAYTC 240
PYQEVLDGVL NYPIYYPLLR AFQSTSGSIN DLYNMINTVK SDCADSTLMG TFLENHDNPR 300
FASYTSDVAL AKNAIAFTIL SDGIPIIYAG QEQHYSGGND PANREAVWLS GYSTDSELYS 360
FVAVTNQIRN YAISQDQGYV TWKNVPYIQD TSTLAMRKGT DGSQVITVLS NLGASGSSYT 420
LTLGGSGYSS GQQLTEIFSC ATVTVDSSGN IPVPMGSGQP KVFYPTAGLG GSGICQSSGA 480
TSPGSSSGSV EVTFDVYATT VYQNIYITG DVSELGNWTP ANGVALSSAN YPTWSATIAL 540
PADTTIQYKY VNIDGSTVIW EDAISNREIT TPASGTYTEK DTWDES 586

SEQ ID NO: 151      moltype = AA length = 576
FEATURE            Location/Qualifiers
source             1..576
                  mol_type = protein
                  organism = Meripilus giganteus

SEQUENCE: 151
RPTVFDAGAD AHS L HARAPS GSKDVIIQMF EWNWDSVAE CTNFIGPAGY GFVQVSPPQE 60
TIQGAQWWD YQPVSYTLTG KRGDRSQFAN MITTCHAAGV GVIVDTIWNH MAGVDSGTGT 120
AGSSPETHYNY PGIQNQDFH HCGLEPGDDI VNYDNAVEVQ TCELVNLADL ATDTEYVRGR 180
LAQYGNLLS LGADGLRLDA SKHIPVGDIA NILSRLSRV YITQEVIFGA GEPITPNQYT 240
GNGDVQEFY TSALKDAFLS SGISNLQDFE NRGWVPGSGA NVFVNNHDE RNASLNNNS 300
PSNTVTAT I FSLAHPYGT TILSSYDGFT NTDAGAPNNN VGTCSTSGGA NGWLCQHRWT 360
AAGMVGFERN NVGSAALNNW QAPQSQQIAF GRGALGFVAI NNADSAWSTT FTSLPDGSY 420
CDVISGKASG SSCTGSSTFTV SGGKLTATVP ARSAIAVHTG QKSGGGATPT SAPSTPTPTSG 480
TVSMTFAEQ TTTFGENIFL VGSISQLGNW NPASAIALSS AAYPTWSVSV NIPAGTTFQY 540
KFIRKETDGS VVWESDPNRQ ATAPASGTTT LTSSWR 576

SEQ ID NO: 152      moltype = AA length = 575
FEATURE            Location/Qualifiers
source             1..575
                  mol_type = protein
                  organism = Cerrena unicolor

SEQUENCE: 152
RPLNASNALD ARAPSGAKSV IIQMFETWD SVAEECTNFI GPAGYGFVQV SPPQETIQGD 60
QWWTYQPV S YILTSKRGR DQFAAMIDTC HDAGVKVIVD TIWNHMA GVE SGTGVAGSSF 120
THYNYPGIYQ NQDFPHHCGL SGDDIVNYDN AQEVQTCELV NLADLATETD YVRGLAEYG 180
NDLLSLGADG LRLDAAKHIA VGDLANIIGR LNSTPYITQE VIFGSGEPIT PNQYTGNQDV 240
QKFRYTSALK DAFNLGDISS LQDFENRGWV AGSGANVFVT NHDTERNGNS LNNNSPNNAY 300
TLAMIFSLAH PYGTSPILSS YSGFTD TDAG APNGGAGTCS GSGGSNGWLC QHRWTAVAGM 360
VGFRNTVGSA ALNNWVSPQS SQIAFGRGAL GFVAINN GDS TWSTFTTSL PDGTYCDVIT 420
GTSSSGSCTG SSFTVSGTGF TANVAARDAV AIHTGATGTG SGSGNTSTGS GGTSDTVSV 480
SFAETATTTF GENIFLVGSI SQLGAWDPAS AIALSSASYP TWTVTVTLPA GTTFEYKFIR 540
KETDGSVVWE SDPNRQATTP SSSDAATTTL STSWR 575

SEQ ID NO: 153      moltype = AA length = 562
FEATURE            Location/Qualifiers
source             1..562
                  mol_type = protein
                  organism = Physalacria cryptomeriae

SEQUENCE: 153
APFLELQVRA PSSTKQV I I Q MF EWTWDSVA AECTDFIGPA GYGYVQVSPP AEHITGDQW 60
TDYQPVSYIL TSKRGNRSQF ANMVTTCHTA GVLVIADALF NHMAGIESGT GTAGSSFTHY 120
DYPGIYQTDQ FHHCGLTSGD DISDYSSQAQ VQTCELVNLA DLATDTEYVR AKLASYANDL 180
ISLGVNGLRL DAAKHIAITD IKNILSRLSS TVYITQEVIF GSGEPVTPSM YTQNGDVQEF 240
RYTSTLQSAF SGGDISQLQN LDSKGWIAGT SANVFVANHD TERGGSSLNY KSSSNTYVTA 300
TIFSLAHPYG TPTILSSYEF SDTDAGSPNG GAGTCSTTGG ANGWLCQHRW VAFSGMVGFH 360
NNVGTASLTN WVSPQSNQIA FERSGKGFVA INNADSAWTA TFTTSLAAGS YCDVITGTSN 420
GSACSGTSYT VSGGSFSATV AARSVAIHT GATGSGSGGG GGSTGSVAIT FQETATTTLG 480
ENIFLVGSI QLRWAPASA IALSSASYPT VSVTVSIPAG TTFEYKFIRK ESDGSVVWES 540

```

-continued

 DPNRSATASS SASTQTILTS WR 562

SEQ ID NO: 154 moltype = AA length = 578
 FEATURE Location/Qualifiers
 source 1..578
 mol_type = protein
 organism = Lenzites betulinus

SEQUENCE: 154
 RPASTVFHGA ETRSLDARAP SGSKDVIIQM FEWTWDSVAA ECTNFIGPAG YGFVQGSPPQ 60
 EHIQGAQWWT DYQPVSYTLT SKRGDRTSFA NMIQTCHTAG VGVIVDTLFN HMAGVDSGTG 120
 VAGSSFTHYN YPGIYQNDQF HHCGLPEPGDD IVNYDNAVEV QTCELDNLAD LATETEVVRG 180
 RLAQYGNLLE SLGADGMRLD AAKHIAVGD I ANILSRLNRT VYITQEVIFG AGEPIITPNQY 240
 TGNQDVQEFR YTSALQDAFL NSGIANLQVL ENRGWVPGSG ANVFVTNHD T ERNGASLNNN 300
 SPSNTYVTAM IFSLAHPFGT PSILSSYSGF TDTDAGAPNG GVGTCSGSGG TNGWLCQHRW 360
 TAVAGMVGFR NQVGSAAALN WQSPQSQQIA FGRGALGFVA INNADSAWSA TFTTSLPDGS 420
 YCDVISGQTS GSTCTGSSFT VSGGSLSATV PARSIAVHT GQKGTGSGSG TGTGGGGSTG 480
 SGNVAVNFAE TATTTFGENI FVVGSIQSLG TWNTANAIAL SSPSYPTWTV SISIPAGTTF 540
 QYKFIRKETD GSVVWESDPN RQATAPASGS TTLSTSWR 578

SEQ ID NO: 155 moltype = AA length = 571
 FEATURE Location/Qualifiers
 source 1..571
 mol_type = protein
 organism = Trametes ljubarskyi

SEQUENCE: 155
 RPATFDAADA RSVQPRAPSG SKDVIIQMF EWTWDSVAAEC TNFIGPAGYG FVQGNPPQEH 60
 IQGDQWWTDY QPVSYILTSK RGDRTAFANM ISTCHAAGVG VIVDTIFNMH SGVDSGTGVA 120
 GSSFTHYNYP GIYQNDQFHH CGLEPGDDIV NYDNAVEVQT CELENLADLA TDTEYVRGRL 180
 AQYANDLLSL GADGLRLDAA KHIPTGDIAN ILSRLNRSVY ITQEVIIYGDG EPITPNQYTG 240
 NGDVQEFRYT TALKNAFLGG GISSLQSFND LGWVPGTGAN VFVTNHDTER NGNSLNNNSP 300
 SNTYVTAMIF SLAHPYGTPT ILSYSYSGFTN TDAGAPNGGT GTCGSGGGAN GWLCQHRWTA 360
 VAGMVGFRNN VGSAAALTNWQ SPQSQQIAFG RGALGFVAIN NADSAWSTTF TTSPLPDGSYC 420
 DVVSGTSSNG GCTGSSSFVS GGSLLTATVPA RSAIAIHTGE TGSGSNSGGG SGGSGVTVIN 480
 FAETATTTFG ENIFVVGSIQ QLGSWNPNANA IALSSASYPT WTVSVSVFAG TTFEYKFIRK 540
 ETDGSVVWES DPNRSDTAPA SGTQTITTSW R 571

SEQ ID NO: 156 moltype = AA length = 451
 FEATURE Location/Qualifiers
 source 1..451
 mol_type = protein
 organism = Bacillus subtilis

SEQUENCE: 156
 GPAAASAETA NKSNELTAPS IKSGETILHAW NWSFNTLKHN MKDIHDAGYT AIQTSPINQV 60
 KEGNQGDKSM SNWYWLQPT SYQIGNRYLG TEQEFKEMCA AAEYGIKVI VDAVINHTTF 120
 DYAAISNEVK SIPNWTGNT QIKNWSDRWD VTQNSLLGLY DWNTQNTQVQ SYLKRFLERA 180
 LNDGADGFRF DAAKHIELPD DGSYGSQFWP NITNTSAEFQ YGEILQDSAS RDAAYANYMD 240
 VTASNYGHSI RSALKNRNLG VSNISHYAYD VSADKLVTWV ESHDTYANDD EESTWMSDDD 300
 IRLGWAVIAS RSGSTPLFFS RPEGGGNGVR FPGKSQIGDR GSALFEDQSI TAVNRFHNVN 360
 AGQPEELSNP NGNNQIFMNQ RSGHGVVLAN AGSSSVSINT PTKLPDGRYD NKAGAGSFQV 420
 NDGKLTGTIN ARSVAVLYPD DIEIRCNTFF Q 451

SEQ ID NO: 157 moltype = AA length = 633
 FEATURE Location/Qualifiers
 source 1..633
 mol_type = protein
 organism = Bacillus subtilis

SEQUENCE: 157
 GPAAASAETA NKSNELTAPS IKSGETILHAW NWSFNTLKHN MKDIHDAGYT AIQTSPINQV 60
 KEGNQGDKSM SNWYWLQPT SYQIGNRYLG TEQEFKEMCA AAEYGIKVI VDAVINHTTS 120
 DYAAISNEVK SIPNWTGNT QIKNWSDRWD VTQNSLLGLY DWNTQNTQVQ SYLKRFLDRA 180
 LNDGADGFRF DAAKHIELPD DGSYGSQFWP NITNTSAEFQ YGEILQDSAS RDAAYANYMD 240
 VTASNYGHSI RSALKNRNLG VSNISHYASD VSADKLVTWV ESHDTYANDD EESTWMSDDD 300
 IRLGWAVIAS RSGSTPLFFS RPEGGGNGVR FPGKSQIGDR GSALFEDQAI TAVNRFHNVN 360
 AGQPEELSNP NGNNQIFMNQ RSGHGVVLAN AGSSSVSINT ATKLPDGRYD NKAGAGSFQV 420
 NDGKLTGTIN ARSVAVLYPD DIAKAPHVFL ENYKTGVTHS FNDQLTITLR ADANTTKAVY 480
 QINNGPETAF KDGQDFTIGK GDPFGKTYTI MLKGTNSDGV TRTEKYSFVK RDPASAKTIG 540
 YQNPNHWSQV NAYIYKHDGS RVIELTGSWP GKPMTKNADG IYTLTLPADT DTTNAKVFVN 600
 NGSAQVPGQN QPGFDYVLNG LYNDGLSGS LPH 633

SEQ ID NO: 158 moltype = AA length = 487
 FEATURE Location/Qualifiers
 source 1..487
 mol_type = protein
 organism = Schwanniomyces occidentalis

SEQUENCE: 158
 KPIFLSKRDA GSSAAAWRS ESIYQLVTDR FARTDGSTA TCNTGDRVYC GGTFQGIIDK 60
 LDYIQGMGFT AIWISPVVEQ IPDDTGYGYA YHGYWMKDIY AINSNFGTAD DLKNLSNELH 120
 KRNKMLMVDI VTNNHYAWNGA GSSVAYSNNY PFNQQSYPHD YCLITNYDDQ TNVEDCWEGD 180
 NTVSLPDLRT EDSDVSSIFN LWVAELVSNY SIDGLRDSA KHVDESFPYS FQSAAGVYLL 240

-continued

```

GEVYDGDPAI  TCPYQNYMSG  VTNYPLYYPM  LRFFQGTNS  VDELNAMISS  LESDCKDITL  300
LGNFIENHDQ  PRLPSYSDS  ALIKNAIAPN  LMSDGIPIIY  YGQEQGYSGS  SDPNNREALW  360
LSGYSTSNQY  YKLISVNIQI  RNQAIYKDSK  YTTYWSDVLY  ASGHVIALQR  GADDQRIQSV  420
FNNLGGSSGSQ  TVTFSTKYSG  GEKVVDVLTG  QTSYANSNST  LTVSISGGAP  RIYAPASLIA  480
NSGICNF      487

```

```

SEQ ID NO: 159      moltype = AA  length = 583
FEATURE             Location/Qualifiers
source              1..583
                    mol_type = protein
                    organism = Rhizomucor pusillus

```

```

SEQUENCE: 159
ATSDDWKGA  IYQLLTDRFG  RADDSTSNCS  NLSNYCGGTY  EGITKHLDIY  SGMGFDAIWI  60
SPIPKNSDGG  YHGYWATDFY  QLNNSNFGDES  QLKALIQAAH  ERDMYVMLDV  VANHAGPTSN  120
GYSYGTFGDA  SLYHPKCTID  YNDQTSIEQC  WVADELDPID  TENSNDVAIL  NDIVSGWVGN  180
YSFDGIRIDT  VKHIRKDFWT  GYAEAGVFA  TGEVFNQDPA  YVGPYQKYL  SLINYPMYA  240
LNDVVFVSKS  GFSRISEMLG  SNRNAFEDTS  VLTTFVDNHD  NPRFLNSQSD  KALFKNALT  300
VLLGEGPIIV  YYGSEQGFSG  GADPANREVL  WTTNYDTSSD  LYQFIKTVNS  VRMKSNAKAV  360
MDIYVGDNAY  AFKKGDALVV  LNNYSGGSTN  QVSFSVSGKF  DSGASLMDIV  SNITTTVSSD  420
GTVTFNLDKG  LPAIFTSATG  GTTTTATPTG  SGSVTSTSKT  TATASKTSTS  TSSTSTCTPT  480
AVAVTFDLTA  TTTYGENIYL  VGSISQLGDW  ETSDGIALSA  DKYTSSDPLW  YVTVTLPAGE  540
SFEYKFIRIE  SDDSVWESD  PNREYTVPQA  CGTSTATVTD  TWR          583

```

```

SEQ ID NO: 160      moltype = AA  length = 616
FEATURE             Location/Qualifiers
source              1..616
                    mol_type = protein
                    organism = Aspergillus niger

```

```

SEQUENCE: 160
AEWRTQSIYF  LLTDRFGRD  NSTTATCDTG  DQIYCGGSQ  GIINHLDYIQ  GMGFTAIWIS  60
PITEQLPQDT  ADGEAYHGYW  QQKIYDVNSN  FGTADDLKSL  SDALHARGMY  LMVDVVPNHM  120
GYAGNGNDVD  YSVDFPFDSS  SYFHPYCLIT  DWDNLTMVQD  CWEGDTIVSL  PDLNNTTETAV  180
RTIWDWVAD  LVSNSVDGL  RIDSVLEVEP  DFFPGYQEA  GVCVGEVDN  GNPALDCPYQ  240
KVLGVLNYP  IYQQLLYAFE  SSSGSIINLY  NMIKSVASDC  SDPTLLGNFI  ENHDNPRFAS  300
YTSYDYSQAK  VLSYIFLSDG  IPIVYAGEEQ  HYSGGKVPYN  REATWLSGYD  TSAELYTWIA  360
TTNAIRKLAI  SADSAIYTY  NDAFYTDSNT  IAMRKGTSGS  QVITVLSNKG  SSGSSYTLTL  420
SGSGYTSGTK  LIEATCTSV  TVDSSGDIPV  PMASGLPRVL  LPASVVDSSS  LCGSGGRTTT  480
TTTAATSTSK  ATSSSSSSA  AATSSSCTA  TSTTLPTIFE  ELVTTTYGEE  VYLSGSIQSL  540
GEWDTSDAVK  LSADDTSSN  PEWSVTVSLP  VGTTFEYKFI  KVDEGGSVTV  ESDPNREYTV  600
PECGNGSGET  VVDTWR          616

```

```

SEQ ID NO: 161      moltype = AA  length = 512
FEATURE             Location/Qualifiers
source              1..512
                    mol_type = protein
                    organism = Bacillus stearothermophilus

```

```

SEQUENCE: 161
APFNGTMMQY  FEWYLPDDGT  LWTKVANEAN  NLSSLGITAL  WLPPAYKGT  RSDVGYGVYD  60
LYDLGEGFNQ  GTVTRKYGTK  AQYLQAIQAA  HAAGMQVYAD  VVFDHKGAD  GTEWDAVEV  120
NPSDRNQEIS  GTYQIQAWTK  FDFPGRGNTY  SSFKRWYHF  DGVDWDESRK  LSRIYKFRGK  180
AWDWEDVTEF  GNYDYLMYAD  LDMDHPEVVT  ELKNWGWYV  NTTNIDGFR  DAVKHIFSF  240
FPDWLSYVRS  QTGKPLFTVG  EYWSYDINKL  HNYITKTDGT  MSLEFADPLH  KFYTASKSGG  300
AFDMRTLMTN  TLMKDQPTLA  VTFVDNHDTE  PGQALQSWVD  PWFKPLAYAF  ILTRQEGYPC  360
VPYGDYYGIP  QYNIPSLKSK  IDPLLIARRD  YAYGTQHDYL  DHSDIIGWTR  EGGTEKPGSG  420
LAALITDGGP  GSKWYVVGKQ  HAGKVFDLT  GNRSDTVTIN  SDGWGEFKVN  GGSVSVWVPR  480
KTTVSTIARP  ITTRPWTGEF  VRWTEPRLVA  WP          512

```

```

SEQ ID NO: 162      moltype = AA  length = 483
FEATURE             Location/Qualifiers
source              1..483
                    mol_type = protein
                    organism = Bacillus halmapalus

```

```

SEQUENCE: 162
HHNGTNGTMM  QYFEWHLFND  GNHWNRLRDD  ASNLNRNGIT  AIWIPPAWK  TSQNDVGYGA  60
YDLYDLGEPN  QKGTVRTKYG  TRSQLESIAH  ALKNNGVQVY  GDVVMNHKGG  ADATENVLAV  120
EVNPNNRNQE  ISGYDTIEAW  TKFDFPGRGN  TYSDFKRWY  HFDGVDWDQS  RQFQNRIFYK  180
RGKAWDEVD  SENGNYDYL  YADVMDHPE  VVNELRWGE  WYTNTLNL  FRIDAVKH  240
YSFTRDWLTH  VRNATGKEMP  AVAEFWKNDL  GALENYLNKT  NWNHVSFVDP  LHYNLYNASN  300
SGGNYDMAKL  LNGTVVQKHP  MHAVTTFVDNH  DSQPGESLES  FVQEWFKPLA  YALILTREQG  360
YPSVFGDY  GIPTHSVFAM  KAKIDPILEA  RQNFAYGTQH  DYFDHNNIIG  WTREGNTHP  420
NSGLATIMSD  GPGGKWMYV  GQNKAGQVWH  DITGNKPGTV  TINADGWANF  SVNGGSVSIW  480
VKR          483

```

```

SEQ ID NO: 163      moltype = AA  length = 478
FEATURE             Location/Qualifiers
source              1..478
                    mol_type = protein
                    organism = Aspergillus oryzae

```

```

SEQUENCE: 163

```

-continued

ATPADWRSQS	IYFLLTDRFA	RTDGSTTATC	NTADQKYCGG	TWQGIIDKLD	YIQGMGFTAI	60
WITPVTQQLP	QTTAYGDAYH	GYWQQDIYSL	NENYGTADDL	KALSSALHER	GYMLMVDVVA	120
NHMGYDAGS	SVDYSVFKPF	SSQDYFHPFC	FIQNYEDQTQ	VEDCWLGDNT	VSLPDLDTTK	180
DVVKNNEWYDW	VGSLVSNYSI	DGLRIDTVKH	VQKDFWPGYN	KAAGVYCIGE	VLDGDPAYTC	240
PYQNVMDGVL	NYPIYYPLLN	AFKSTSGSMD	DLNMMINTVK	SDCPDSTLLG	TFVENHDNPR	300
FASYTNDIAL	AKNVAAFIIL	NDGIPIIYAG	QEQHYAGGND	PANREATWLS	GYPTDSELYK	360
LIASANAIRN	YAISKDTGEV	TYKNWPIYKD	DTTIAMRKGT	DGSQIVTILS	NKGASGDSYT	420
LSLSGAGYTA	GQQLTEVIGC	TTVTVGSDGN	VPVPMAGGLP	RVLYPTEKLA	GSKICSSS	478

SEQ ID NO: 164 moltype = AA length = 483
 FEATURE Location/Qualifiers
 source 1..483
 mol_type = protein
 organism = Bacillus amyloliquefaciens

SEQUENCE: 164

VNGTLMQYFE	WYTPNDGQHW	KRLQNDAEHL	SDIGITAVWI	PPAYKGLSQS	DNGYGPYDLY	60
DLGEFQQKGT	VRTKYGTKE	LQDAIGSLHS	RNVQVYGDVV	LNHKAGADAT	EDVTAVEVNP	120
ANRNQETSEE	YQIKAWTDFR	FPGRGNTYS	FKWHWYHFDG	ADWDESRIK	RIFKFRGEGK	180
AWDWEVSSSN	SNLYDLMYAV	VDYDHPDVVA	ETKKWGIWYA	NELSLDGFR	DAAKHIKFSF	240
LRDWVQAVRQ	ATGKEMFTVA	EYWQNNAGKL	ENYLNKTSFN	QSVFVPLPHF	NLQAASSQGG	300
GVDMRRLLDG	TVVSRHPEKA	VTFVENHDTQ	PGQSLESTVQ	TWFKPLAYAF	ILTRESGYPO	360
VPYGDYMGTK	GTSPEKIPSL	KDNIEPILKA	RKEYAYGPQH	DYIDHPDVIG	WTREGDSSAA	420
KSGLAALITD	GPGGSKRMYA	GLKNAGETWY	DITGNRSDTV	KIGSDGWGEF	HVNDGSVSIY	480
VQK						483

SEQ ID NO: 165 moltype = AA length = 583
 FEATURE Location/Qualifiers
 source 1..583
 mol_type = protein
 organism = Rhizomucor pusillus

SEQUENCE: 165

ATSDDWKGA	IYQLLTDRFG	RADDSTSNCS	NLSNYCGGTY	EGITKHLDI	SGMGFDIAWI	60
SPIPKNSDGG	YHGYWATDFY	QLNSNFGDES	QLKALIAAAH	ERDMYVMLDV	VANHAGPTSN	120
GYSGYTFDDA	SLYHPKCTID	YNNQTSIEQC	WVADELDPID	TENSDNVAIL	NDIVSGWVGN	180
YSFDGIRIDT	VKHIRKDFWT	GYAEAAGVFA	TGEVFNQDPA	YVGPYQKYL	SLINYPMYA	240
LNDVFSKSK	GFSRISEMLG	SNRNAPEDTS	VLTTFVDNHD	NPRFLNSQSD	KALFKNALTY	300
VLLGEGIPIV	YYGSEQGFSG	GADPANREVL	WTTNYDTSSD	LYQFIKTVNS	VRMKSNAKAV	360
MDIYVGDNAY	AFKHGDALVV	LNNYSGSGSTN	QVSFVSQKGF	DSGASLMDIV	SNITTTVSSD	420
GTVTFNLDKG	LPAIFTSATG	GTTTTATPTG	SGSVTSTSKT	TATASKTSTS	TSSTSTCTPT	480
AVAVTFDLTA	TTYGENIYL	VGSISQLGDW	ETSDGIALSA	DKYTSSDPLW	YVTVTLPAGE	540
SPEYKFIRIE	SDDSVESWESD	PNREYTVPPQA	CGTSTATVTD	TWR		583

SEQ ID NO: 166 moltype = AA length = 623
 FEATURE Location/Qualifiers
 source 1..623
 mol_type = protein
 organism = Kionochaeta ivoriensis

SEQUENCE: 166

LSPAGWRQQS	IYQVMTDRFA	RTDGSTIASC	DTSQQAAYCGG	TWQGLINKLD	YIQGMGFTAV	60
WISPMVHOMA	GATSDGESYH	GYWAQDINTV	NSAFGTAADL	KALSAALHAR	GYMLMLDVVT	120
NHFAYDGCCT	CVDSYIFNPF	NSESYFHPFC	LIDYSNTTSI	QVCWEGDNTV	SLPDLRTEDS	180
DVRSIWNWSI	SSVISTYGV	GLRVDQAQV	ETSFWAGFEA	AAGVYVMGEV	FNGDPTYVTP	240
FQDYMDGVLN	YPAYYWITQA	FESTSGSISN	LANGMNTMKS	LAKNTSLLGS	FLENHDNPRF	300
PSLTSDMSLA	QNAIAFTMLM	DGIPPIYQGG	EQHFSGGGVP	SDREAVVLSG	YPNDTTLAYW	360
ITKLNAVRWS	AIKADSSYLA	YMAYPVYTD	HTIAMRKGDT	GYQVISVYTN	VGASGSSYSV	420
TLTSADTGFT	ASQSVVDLVG	CKTYTADSTG	SLSLSLTGGI	PIILYPAASL	TGNTICTSTG	480
GTTTTATPTG	SGSVTSTSKT	TATASKTSTS	TSSTSTCTPT	AVAVTFDLTA	TTYGENIYL	540
VGSISQLGDW	ETSDGIALSA	DKYTSSDPLW	YVTVTLPAGE	SPEYKFIRIE	SDDSVESWESD	600
PNREYTVPPQA	CGTSTATVTD	TWR				623

SEQ ID NO: 167 moltype = AA length = 588
 FEATURE Location/Qualifiers
 source 1..588
 mol_type = protein
 organism = Aspergillus niger

SEQUENCE: 167

LSAAEWRTQS	IYFLLTDRFG	RTDNSTTATC	DTGDQIYCGG	SWQGIINHLD	YIQGMGFTAI	60
WISPIEQQLP	QDTADGEAYH	GYWQQKIYDV	NSNFGTADDL	KSLSDALHAR	GYMLMVDVVP	120
NHMGYAGNGN	DVDYSVDFPF	DSSSYFHPYC	LITDWDNLTM	VQDCWEGDTI	VSLPDLNTE	180
TAVRTIWIYDW	VADLVSNYSV	DGLRIDSVL	VEPDFFFPYQ	EAAGVYCVGE	VDNGNPALDC	240
PYQKVLGVL	NYPIYQQLLY	AFESSSGSIS	NLYNMIKSA	SDCSDPTLLG	NFIENHDNPR	300
FASYTSDYSQ	AKNVLSYIFL	SDGIPIVYAG	EEQHYSGGKV	PYNREATWLS	GYDTSAELYT	360
WIATTNAIRK	LAISADSAYI	TYANDAFYTD	SENTIAMRKGT	SGSQVITVLS	NKGSSGSSYT	420
LTLSGSGYTS	KSLTLEAYTC	TSVTVDSSGD	IPVPMASGLP	RVLLPASVVD	SSSLCGSGSR	480
GATSPGGSSG	SVEVTFDVYA	TTVYGQNIYI	TGDSVSELGNW	TPANGVALSS	ANYPTWSATI	540
ALPADTTIQY	KYVNIDGSTV	IWEDAISNRE	ITTPASGTYT	EKDTWDES		588

SEQ ID NO: 168 moltype = AA length = 588
 FEATURE Location/Qualifiers

-continued

```

source          1..588
                mol_type = protein
                organism = Aspergillus oryzae

SEQUENCE: 168
ATPADWRSQS IYFLLTDRFA RTDGSSTTATC NTADQKYCGG TWQGIIDKLD YIQGMGFTAI 60
WITPVTALQLP QTTAYGDAYH GYWQQDIYSL NENYGTADDL KALSSALHER GMYLMVDVVA 120
NHMGYDGPQS SVDYSVFPF NSASYFHPFC FIQNWDNQTQ VEDCWLGDNT VSLPDLDTTK 180
DVVKNEWYDW VGSLSVNSYSI DGLRIDTVKH VQKDFWPGYN KAAGVYCIGE VLDGDPAYTC 240
PYQEVLDGVL NYPIYYPLLN AFKSTSGSMD DLYNMINTVK SDCPDSTLLG TFVENHDNPR 300
FASYTNDIAL AKNVAIFIIL NDGIPIIYAG QEQHYAGGND PANREATWLS GYPTDSELYK 360
LIASANAIRN YAIKSDTGFV TYKNWPIYKD DTTIAMRKGT DGSQIVTILS NKGASGDSYT 420
LSLSGAGYTA GQQLTEVIGC TTVTVDSGGD VPVPMAGGLP RVLYPTEKLA GSKICSSSGR 480
GATSPGSSSG SVEVTFDVYA TTVYQNIYI TGDVSELGNW TPANGVALSS ANYPTWSATI 540
ALPADTTIQY KYVNIDGSTV IWEDAISNRE ITTPASGTYT EKDTWDES 588

SEQ ID NO: 169      moltype = AA length = 583
FEATURE            Location/Qualifiers
source             1..583
                  mol_type = protein
                  organism = Penicillium canescens

SEQUENCE: 169
LTPAEWRSQS IYFMLTDRFG RSDNSTTAAC NVSDRTYCGG TWQGIINHLD YIQGMGFTAI 60
WITPVTQLP QDTGDGEAYH GYWQNIYEI DSKLGTAADL LALSALHAR GMYLMVDVVA 120
NHMGYAGSGN SVDYSVFPNF SSSSYFHSYC LISNYDDQSN VENCWLGDIT VSLPDLDTTQ 180
TAVQTIWYDW IADLVSNYSI DGLRIDTVKH VQKDFWPGYN DAAGVYCVGE IFDGDPAITC 240
DYQNYMDGVL NYPIYYQLLY AFQSSSGSIS DLYNMINSVK SDCADSTLLG NFIENHDNPR 300
FASYTSDYSQ AKNVISFLFL SDGIPIIYSG QEQHYSGGAD PANREATWLS GYSTTAELYK 360
YIATTNKIRK LAVSDSSYLT TKNVVPFYQDS HTLAMKKGSS ASPVITVLSN YGSSGSSYTL 420
SLSGSGSYSSG TKLMEMYTCT SVTVDSGGNI AVPMASGLPR VLMLASSANS LCGSSGATSP 480
GGSSSGSVEVT PDVYATTVYG QNIYITGDVS ELGNWTPANG VALSSANYPT WSATIALPAD 540
TTIQYKYVNI DGSTVIWEDA ISNREITTPA SGTYTEKDTW DES 588

SEQ ID NO: 170      moltype = AA length = 583
FEATURE            Location/Qualifiers
source             1..583
                  mol_type = protein
                  organism = Acidomyces acidothermus

SEQUENCE: 170
LTPAQWRGQS IYQVLTDRFG RTDDSTTAAC DVNDYCGGSW QGIINHLDYI QDMGFSAIWI 60
SPVVENLVGD TQDGSAYHGY WAQNIYALNP NFGTVSDLVA LSAALHQRGM YLMVDVVTNH 120
MGYDGCDCV DYSVFTPFNS QSYFHPFCLI DYNNSTSIKV CWEGDNIVSL PDMRTSDSV 180
ATEWNTWISE LVSNYSIDGL RIDSAQQVDN AFFPPFQAAA GGIHVLGEVF NGDPNVYCPY 240
QDFMSGVLNY PAYYITQAF QSTSGSISNL VNGINQMKST CTDITLLGSF LENHDNPRFP 300
SYTSDLSLDK NAITFTILQD GIPIIYEGQE QHYSGGTVPN NREAIWLSGY DKSAPLYTWI 360
ASVNIQRNQA IFKDSNYLTY MAWPIYSDAS TIAMRKGFDP LQIISVYSNK GASAASTIS 420
LESSTTGFTA NEALVEVMSC TTYTTDGSNG LAVTISGGLP AVFYPKAQLA GSGICGATSP 480
GGSSSGSVEVT PDVYATTVYG QNIYITGDVS ELGNWTPANG VALSSANYPT WSATIALPAD 540
TTIQYKYVNI DGSTVIWEDA ISNREITTPA SGTYTEKDTW DES 588

SEQ ID NO: 171      moltype = AA length = 586
FEATURE            Location/Qualifiers
source             1..586
                  mol_type = protein
                  organism = Kinochaeta ivoriensis

SEQUENCE: 171
LSPAGWRQQS IYQVMTDRFA RTDGSSTIASC DTSQQAYCGG TWQGLINKLD YIQGMGFTAV 60
WISPMVHQMA GATSDGESYH GYWAQDINTV NSAFGTAADL KALSAALHAR GMYLMVDVVT 120
NHFAVDGCGT CVDYSIFNPF NSESYFHPFC LIDYSNTTSI QVCWEGDNTV SLPDLRTEDS 180
DVRSIWNSWI SSVISTYGV DGLRVDQAQV ETSFWAGFEA AAGVYMGVEV FNGDPTYVTP 240
PDYMDGVLN YPAYWITQA FESTSGSISN LANGMNTMKS LAKNTSLGGS FLENHDNPRF 300
PSLTSDMSLA QNAIAFTMLM DGIPIIYQGG EQHFSGGGVP SDREAVWLSG YPNDTTLAYW 360
ITKLNAVRSW AIAKDSSYLA YMAYPVYTDI HTIAMRKGDT GYQVISVYTN VGASGSSYSV 420
TLTSADTGFT ASQSVVDLVG CKTYTADSTG SLSLSLTGGI PIILYPAASL TGNITICTSGA 480
TSPGSSSGSV EVTFDVYATT VYGQNIYITG DVSELGNWTP ANGVALSSAN YPTWSATIAL 540
PADTTIQYKY VNIDGSTVIW EDAISNREIT TPASGTYTEK DTWDES 586

SEQ ID NO: 172      moltype = AA length = 585
FEATURE            Location/Qualifiers
source             1..585
                  mol_type = protein
                  organism = Aspergillus terreus

SEQUENCE: 172
LTPAEWRSQS IYFLLTDRFG RTDNSTTAAC DTSDRVYCGG SWQGIINQLD YIQGMGFTAI 60
WITPVTGQFY ENTGDGTSYH GYWQDIYDL NYNYGTAQDL KNLANALHER GMYLMVDVVA 120
NHMGYDGAGN TVDYSVFPNF SSSSYFHPYC LISNYDNQTN VEDCWLGDIT VSLPDLDTTS 180
TAVRNIWYDW VADLVANYSI DGLRVDTVKH VEKDFWPGYN SAAGVYCVGE VYSGDPAYTC 240
PYQNYMDGVL NYPIYYQLLY AFESSSGSIS DLYNMISSVA SSCKDPTLLG NFIENHDNPR 300
FASYTSDYSQ AKNVITIFFL SDGIPIVYAG QEQHYSGGSD PANREATWLS GYSTSATLYT 360
WIATTNQIRS LAISKDAGYV QAKNNPFYSD SNTIAMRKGT TAGAQVITVL SNKGASGSSY 420

```

-continued

TLSSSGTGYG	AGVTLVETYT	CTTVTVDSG	NLPVPMSTGL	PRVFPVSSW	NGSALCNGAT	480
SPGGSSGSVE	VTFDVYATT	YGQNIYITGD	VSELGNWTPA	NGVALSSANY	PTWSATIALP	540
ADTTIQYKYV	NIDGSTVIWE	DAISNREITT	PASGTYTEKD	TWDES		585

SEQ ID NO: 173 moltype = AA length = 565
 FEATURE Location/Qualifiers
 source 1..565
 mol_type = protein
 organism = *Thamnidium elegans*

SEQUENCE: 173

VPLSVLDKRN	GVTTLKRAA	AADWKSRSY	QLVTRDFGRS	DGSTSACGDL	SNYCGGDYKG	60
IQNLQDYIAG	MGFDIAWIS	IPENTDGGYH	GYWAKDFEKL	NTNFGSADDL	KALVTAAGHK	120
GMVVMLDVVA	NHAGPASGGD	YSGFTFSSAS	NYHPQCTIDY	DNQTSVEQCW	VADDLPDINT	180
EDDTIVSKLH	SIVSDWVTY	DFDGIKIDTV	KHIRKDFWSG	YEEAAGVFAT	GEVFDGDAAY	240
VGPHYQDLSS	LINYPYIYAI	RDVFSAGSGF	SRISDMLSTI	KSNFKDPSVL	TTFVDNQDNA	300
RFLSVKSDMS	LYKALALFTI	LTEGIPVVY	GTEQGFGKGD	DPKNREVLWT	SNYDTSDDL	360
KFIKIVNNDV	RQKSDKTVT	DVDVGTNTYA	FTHGKNLIVV	NNYGGSGSTES	VTVKVGDSVA	420
DGTKLVDAVS	NITATVSGGS	ITFSLKDGFL	ALFVPSSGAT	SPGGSSGSVE	VTFDVYATT	480
YGQNIYITGD	VSELGNWTPA	NGVALSSANY	PTWSATIALP	ADTTIQYKYV	NIDGSTVIWE	540
DAISNREITT	PASGTYTEKD	TWDES				565

SEQ ID NO: 174 moltype = AA length = 574
 FEATURE Location/Qualifiers
 source 1..574
 mol_type = protein
 organism = *Meripilus giganteus*

SEQUENCE: 174

RPTVFDAGAD	AHSLHARAPS	GSKDVIIQMF	EWNWDSVAE	CTNFIGPAGY	GFVQVSPPE	60
TIQGAQWWD	YQPVSYTLTG	KRGDRSQFAN	MITTCHAAAGV	GVIIVDTIWNH	MAGVDSGTGT	120
AGSSPETHYNY	PGIYQNDQFH	HCGLEPGDDI	VNYDNAVEVQ	TCELVNLAADL	ATDTEYVRGR	180
LQYQGNLDS	LGADGLRLDA	SKHIPVGDIA	NILSRLSRV	YITQEVIFGA	GEPITPNQYT	240
GNGDVQEPFY	TSALKDAFLS	SGISNLQDPE	NRGWVPGSGA	NVFFVNHDE	RNGASLNNNS	300
PSNTYVTATI	FSLAHPYGT	TILSSYDGF	NTDAGAPNNN	VGTCSTSGGA	NGWLCQHRWT	360
AIAGMVGFRN	NVGSAAALNNW	QAPQSQQIAF	GRGALGFVAI	NNADSASTT	FTTSLPDGSY	420
CDVISGKASG	SSCTGSSFTV	SGGKLTATVP	ARSAIAVHTG	QKSGGGGATS	PGGSSGSVEV	480
TDFVYATTYV	GQNIYITGDV	SELGNWTPAN	GVALSSANYP	TWSATIALPA	DTTIQYKYVN	540
IDGSTVIWED	AINREITTP	ASGTYTEKDT	WDES			574

SEQ ID NO: 175 moltype = AA length = 671
 FEATURE Location/Qualifiers
 source 1..671
 mol_type = protein
 organism = *Chaetomium megalocarpum*

SEQUENCE: 175

LYINGSVTAP	CDSPIYCQGE	ILKAIELARP	FSDSKTFVDM	PTIKPVDEVI	AAFSRLSQPL	60
SNNSSELTAFL	AENFAPAGGE	LEEVPISLE	TDPSFLNKL	DVDIKEFVGK	VIDIWPDLTR	120
RYAGPGNCSQ	CANSFIPVNR	TFVVAGGRFR	EPYYWDSYWI	LEGLLRTGGA	FTQISKNMIE	180
NFLDFVETIG	FVPNGARIYY	LNRSQPPLLT	LMVKTYVDYT	NDTSILERAL	PLLVKEYEFW	240
TNNRTVSITA	TDGKEYTLNR	YSVNNNQPRP	ESYREDYVTA	NNASYAQS	IIYPVKTPLN	300
ETEKAEALYSN	LASGAESGWD	YTARWLKTPD	DAARDVYFPL	RSLNVRISVS	VDLNSILYQN	360
EVIIESEYLEQ	AGNSSEAKRF	ADAAEQRSEA	MYALMWNATH	WSYFDYNLTD	NSQRFVFPAD	420
ADTAPSDQTA	APPGQQLFD	IAQLYPPFWT	AAPASLKNP	LAVQYAYARV	ARMLDTKAGA	480
IAATNLRTGQ	QWDQPNVWPP	LQHILMKGLL	NTPATFGTED	PAYAHTQDLA	LRLAQRYLDS	540
TFCTWYATGG	STSATPQLQG	AAPGATGTMF	EKYADDATNV	AGGGGEYEVV	EGFGWTNGVL	600
IWAADVFGDR	LVRPDCGNIT	AAHTHSSEKR	TVAGASRPDA	GASGVRERRR	AVELDPWDAA	660
WTKMFRSKL	R					671

SEQ ID NO: 176 moltype = AA length = 662
 FEATURE Location/Qualifiers
 source 1..662
 mol_type = protein
 organism = *Lecanicillium psalliotae*

SEQUENCE: 176

LYQNGSVIAP	CDSPIYCHGD	ILKEVELARP	FTDSKTFVDM	PAKKPLAEIQ	AAFDKLEKPL	60
SNNTALNEFL	STYFEDAGGE	LKAVSKDKLK	TDKAFVDKLN	DTVIKEFVNK	VIDIWPDLTR	120
EYAGSATNCT	NCPNSFIPVN	RTFVVAGGRF	REPYYWDSY	IVEGLLRTGG	AFVDITKNTI	180
ENFLDLIEQF	GFIPNGARLY	YLNRSQPPL	SQMVKNIYSY	TNDTDILKRA	LPILVKEHEF	240
FMNRSVEIT	VENKTYTLNR	YAVSNTQPRP	ESFREDYNTV	NNNSYYAASG	IIYPVKTPLN	300
ESEQATLYAN	LASGAESGLD	YTTAKWSTNP	RDSMEDYFYP	LRSLNIMNIV	PVDLNSILY	360
NEKAISEFYN	ITGNSSEADS	WSKKAERQE	AIQAVFWNET	LYSYFDYNRT	SSSQHIYIPS	420
DDDTQSFEA	TAPAGMQEVF	VTQFYFPFWM	GAAPDYIRIN	PHAVKTAYS	IAKYLEKPG	480
GIPSSNLRSG	QWDQPSVWP	PLMHVLMKGL	VNTPATFGKE	DPAYKDVHKL	ALTGQRYLD	540
STFTWYATG	GSTSETQLS	GLSDSDVGIM	FEKYDDTSIN	HAGGGGEYEV	VEGFGWSNGV	600
LMWVADTFGN	ELKRPDCGNIT	TAANVHPGKR	SVSAVELSSR	DAQRVKKFGR	RAEGRMKVPG	660
VL						662

SEQ ID NO: 177 moltype = AA length = 637
 FEATURE Location/Qualifiers
 source 1..637

-continued

```

mol_type = protein
organism = Doratomyces sp.

SEQUENCE: 177
LYTNGSIIAP CDSPIYCHGD ILHQVQLAHP FPDSTFVDM PAIKSVDEIQ AAFDKLDKPL 60
KNDTALQNFL AENFAEAGHE LAEVDPSELS TDPFRFLDKVS DTVIHEFTQK VIDIWPDLTR 120
SYSPSGAGSD CPDPCNSFLP VNRTFVVAGG RFREPYWDS YWIVEGLLRT GGDFLGVSRN 180
IIENFLDFVD DFGFVPNGAR RYYLNRSQPP LLSLMVKSYV BQTNDTEILD RAVPLLIKEY 240
EFWTKNRTVE VPFNGETITL NQYNVDNTQP RPESYREDYI TATNASYISA SGEVYEEVEK 300
LNETQRATVY RNLATGAESG WDYTSRWMAR PRDAVEDVYF PLRSLNIIIEI VPDVLSNILEY 360
ANELAIAEFV RLAGADDCEE EAAAWALAE KRSADMHRIM WNDTLHSYFD YNLTSSSQNV 420
YVPADNDTAP FERPGGTPEG SQVFFSAAQF YPFWTGAAPS SLRDDPAAVQ LAYSRVASYL 480
DLRAGGIPAT NLRGQQWDQ PSVWPLMHM LTSGLLNTSP ASASDDDDPS YAPTRDLALS 540
LAQRYLDSF CTWYATGGST SETPQLDGFT DEDKGAMFEK YDASATNVAG GGGEYEVVEG 600
FGWTNGVLIW MVDTFGDELS RPDGCDIEAA NAHPARK 637

SEQ ID NO: 178      moltype = AA length = 601
FEATURE            Location/Qualifiers
source              1..601
                    mol_type = protein
                    organism = Mucor moelleri

SEQUENCE: 178
IQNHTSFSCD SPIYCEGDIL HTVQLAKIFS DSKTFVDMPT SKSESQVIEA FKAIGGRNAT 60
IAQVQQFLNE NFLTAGTEVK RLNTNITPEL NWIDNITDPD YRGWISKLNQ AWSNLTFTFD 120
TSVLCQDCAT STLPVSRFFV VPGGRFREFY YWDSFFVIKG LLLSDQVELA KNMILNFFDF 180
IDTYGFIPIG ARIYYLNRSQ PPFLTQMVFA FWEKTSDEKF MTNALPFLDK EYNFWMTNNS 240
ISVPDPKNPS KKYKMNHYVT LNTSPRPESY VEDYNTVNNG TDYSQAVKLQ LYADIAAGAE 300
TGWDYSSRWY RQKHPAPNQT EGYEMLRSIN TANIVPIDLN SLLWNTETML AEWHDRFGEK 360
SKSKKKSAYY QTQAKKRLDA MEKLMWNPTD YTYDYNLTS SSQNLETPA NLFPIWLGAL 420
PNKVMKNKTE LARVFDETEN ALRKYPGILT TSYYNNTMOW DWPNGWPPLS YVAMEGMNKV 480
EESLNGKKGS QDGKTYTTS RGTSLARLSF TLAERYAASA YCGWYKTGGS IPGILNKIDG 540
VADDGHMFEK PDVNTIGISG SQGEYVSQTG FGWTNGIALW IFDTYANLTA PDCNQITILN 600
I 601

SEQ ID NO: 179      moltype = AA length = 650
FEATURE            Location/Qualifiers
source              1..650
                    mol_type = protein
                    organism = Phialophora cyclaminis

SEQUENCE: 179
LYINGSVIAP CDSPIYCHGE LLKAIELARP FSDSKTFVDM PTRKPVDEVV AAFEKLSKPV 60
VNDTKLGEFL QANFLPAGGD LSDFPFGSLE TDPKFLDNIN DTVIREFTEA VIDIWPDLTR 120
RVVGASNCTG CANSFIPVNR TFVVGAGRFR EPYWDSEYI IEGLLRTGGN FVQISKNIIE 180
NFLDFVETIG FVPNGARIYY LNRSQPPLLT AMVKAYLEHT NDTKILDRV PLLIKEYNFW 240
ITNRSVTLTG PDGNEYTLQR YSVNNNQPRP ESYREDYNTA NNKSYYSTSG IIPENVSLN 300
DSEKKELIAN LATGAESGWD YGTRWLSRPA DALRDVYFPL RYVKTRDLVP VDLNAILYQN 360
EASISSFLVI QGNDTGAHF AKLAAARQKA MTAVLWNATL FSYFDYNMTS SSHFSFIPVD 420
DDATSIETAT APRGQQIFH VAQLYPPFWTG AAPASLRENP LAVQRAFSPV SQMLEEKAGS 480
IPATNFYTGQ QWDEPNVWPP LQHIIMEGLR NTPATFGEDD PIYQGVQDLA LRVAQRYLDS 540
TFCTWYATGG STSLTPKLAG LTDNAKGIMF EKYGDNSTNV AGGGGEYEVV EGFGWTFNGVL 600
IWAVDTFGNK LKRDPDCGDIQ PAHVEARGLO RRAVELDKWD AQWVKRFGAK 650

SEQ ID NO: 180      moltype = AA length = 654
FEATURE            Location/Qualifiers
source              1..654
                    mol_type = protein
                    organism = Thielavia arenaria

SEQUENCE: 180
LYINGSVIAP CDSPLYCHGD ILKAIELARP FQDSKTFVDM PTIRPVEEVI AAFNRLSQPL 60
SNNSELNAPL QANFAPAGGE LEEVPESELS TNPVFLDKLN DTVIKEFVAK VIDIWPDLTR 120
RYAGPGNCSE CADSFIPVNR TFVIAGGRFR EPYWDSEYI IEGLLRTGGA FTEISKNMIE 180
NFLDFVETIG HIPNGARIYY LNRSQPPLLA GMVKNYVDYT NDTSILERAL PLLIKEYEFW 240
TNNRTVQVTA SDGKTYTLHR YNVNNNQPRP ESYREDWITA NNASYAASG IIPVKTPLN 300
ESEKAALYSN LASGAESGWD YGTRFLMRPE DAARDIYFPL RHLNVRDMVT VDLNAILYQN 360
EVIISEYLEQ AGNNTAEARF ASAARQRSEA MYALMWNETL WSYFDYNLTS NSQYTFIPAD 420
VNATAAEKAN APEGQKVIHF IAQLYPPFWTG AAPDQLKNP LAVQKAYSRV AEMLDIKPGA 480
IPATNFITGQ QWDQPNVWPP LMHVLMAGLL NTPPTFGEDD PAYQAVQALA LRLGQRYLDS 540
TFCTWYATGG ETSQTPRLQG VSPDATGTMF EKYADNAINV AGGGGEYEVV EGFGWTFNGVL 600
IWAADVDFANG LKRDPDCGNIT AAHTHNGAKR AVELHPRDAA WTKKFGKRAL KKRA 654

SEQ ID NO: 181      moltype = AA length = 661
FEATURE            Location/Qualifiers
source              1..661
                    mol_type = protein
                    organism = Thielavia antarctica

SEQUENCE: 181
LYVNGSVTAP CDSPLYCQGE ILKAIELARP FADSKTFVDM PTLRPLDDVI AAFRKLKSQL 60
SNNSELNAPL AANFAPAGGE LEEVPKSELK TKPAFLDKVE DVVIKEFVGK VIDIWPDLTR 120
RYAGPGNCSE CANSFIPVNR TFVVGAGRFR EPYWDSEYI IEGLLRTGGD FTKISRNIE 180
NFLDFVDITG FVPNGARIYY LNRSQPPVLT LMVKTYVDYT NDTSILERAL PLLIKEYEFW 240

```

-continued

```

TTNRSVSVEV DGKTHTLNRY FVNNNQPRPE SYREDWITAN NASYYAASGI IYPVKTPLNG 300
TQQAELYANL ASGAESGWDY TSRWLKTPSD AARDVYFPLR SLNVINTIPV DLNSILYQNE 360
VIISEYLEQA GNKSGAKFS DAARQRSEAM YALMWNATNW SYFDYNLTSS GQNTHPADA 420
DATAAETTSS PAGTQLLFSV SQLYPFWTGA APQQLKSNPL AVTQAYARVA AMLDAKAGAI 480
PATNLLTGQG WDEPNVWPPL QHVLMMQGLLN TPATFGADDP AYQATQALAL RLAQRYLDST 540
FCTWYATGGS TSATPRLAGV SAGAEGSMFE KYGDDSTNVA GGGGEYEVVE GFGWTNGVLI 600
WAADVFAKGL QRPECGDIEA AQTHGGAARR GLGMERRAIE LDPWDARWTK AFGKRALRRR 660
A

```

```

SEQ ID NO: 182      moltype = AA length = 667
FEATURE            Location/Qualifiers
source             1..667
                   mol_type = protein
                   organism = Chaetomium sp.

```

```

SEQUENCE: 182
LYINGSVTAP CDSPLYCQGE ILKAIELARP FSDSKTFVDM PTIKPVDDVI AAFSRLSQPL 60
SNNSELNAPL AENFAPAGGE LEEVPESELE TDPAFLDKLE DTTIKEFVTK VIDIWPDLTR 120
RYAGPGNCSK CANSFIPVNR TFVVAGGRFR EPYYWDSYWI LEGLLRTGGA FTEISKNIIE 180
NFLDFVDTIG FIPNGARIYY LNRSQPPLLT LMVKTYVDYT NDTSILERAL PLLIKEHEFW 240
TTNRSVAITA ADGKKYTLQR YVNNNQPRP ESFREDYITA NNASYYAASG IIPVKTPLN 300
ETEKAELYSN LASGAEAGWD YTAARWLKTPN DAARDVYFPL RSLNVIGMVP VDLNSILYQN 360
EVIIAEYLQ AGNSSEARF ATAARKRSEA MYALMWNSTH WSYFDYNLTS NSQRIFVPTD 420
ADADPVDQTN APPGHQVLFD IAQLYPFWTG AAPASLKNNP LAVQLAYARV AHMLDTKAGA 480
IPGTNFRGTQ QWDQPNVWPP LQHVLMMQGLL NTPPTFGESD PAYQEVQRLA LRLAQRYLDS 540
TFCTWYATGG STSDMPQLQG VNPATGTMTF EKYADNATNV AGGGGEYEVV EGFWTNGVL 600
IWAADVFGEG LTRPDCGNIT AAHTHSGAKR GLDGGEGGAGL GLWGRRAVEL DPWDARWTKM 660
FGRRKRE

```

```

SEQ ID NO: 183      moltype = AA length = 671
FEATURE            Location/Qualifiers
source             1..671
                   mol_type = protein
                   organism = Chaetomium nigricolor

```

```

SEQUENCE: 183
LYINGSVTAP CDSPLYCQGE ILRAIELARP FSDSKTFVDM PTIKPLEEVI AAFNQLTQPL 60
SNNSELNFTL AENFAPAGGE LEEVPKDELN TDPGFLDKLN DTTIREFVAK VIDIWPDLTR 120
RYAGGNCSE CANSFIPVNR TFVVAGGRFR EPYYWDSYWI LEGLLRTGGA FIEISKNIIE 180
NFLDFVETIG FIPNGARIYY LNRSQPPLLT LMVKTYVDYT NDTSILERAL PLLIKEHEFW 240
VNNRSVEITA ANGQTYTLNR YHVNNNQPRP ESYREDYITA NNGSYYAASG IIPVTRTPLN 300
ETEKAELYAN LASGAESGWD YTAARWLKTPN DAANDVYFPL RSLNVRGLVP VDLNSILYQN 360
EVIIAEYLQ AGNLSLAQRF AEAARQRSEA MYALMWNATY WSYFDYNLTS NSQRIFVPLD 420
ADSRITETVG APPGHQVLFD IAQLYPFWTG AAPANLKNNP LAVQQAYSRV ASMLDAKAGA 480
IPATNFRGTQ QWDQPNVWPP LQHILMMQGLL NTPPTFGSDS PAYQHVRDLA LRLAQRYLDS 540
TFCTWYATGG STSDMPQLQG VSPDATGTMTF EKYADNATNV AGGGGEYEVV EGFWTNGVL 600
IWAADVFGDA LKRPDCGDIE AAHTHQAKKR DVEGLERRAV ELDPWDAAWT KMFGRSKLRK 660
RGAGGQKRWV S

```

```

SEQ ID NO: 184      moltype = AA length = 666
FEATURE            Location/Qualifiers
source             1..666
                   mol_type = protein
                   organism = Chaetomium jodhpurense

```

```

SEQUENCE: 184
LYFNGSVIAP CDSPLYCQGE ILKAIELARP FSDSKTFVDM PTIKPVDEVI AAFNRLSQPL 60
TNNSELNAPL AENFAPAGGE LEEVPKDELN TDPKFLDKLE DATIKEFVAK VIDIWPDLTR 120
RYAGASNCSE CANSFIPVNR TFVIAGGRFR EPYYWDSYWI LEGLLRTGGA YTQISRNMLE 180
NFLDFVETIG FIPNGARIYY LNRSQPPLLA MMIKNYVDYT NDTSILDRAL PLLIKEHEFW 240
INNRSVSITA ADGKQYTLHR YNVNNNQPRP ESYREDYITA NNASYYAASG IIPVKTPLN 300
ESEKAELYAN LATGAESGWD YTAARWLKTPN DAAKDVFYFPL RSLNVRGMVS VDLNSILYQN 360
EVIIAEYLER AGNISEAERF AAMAQRSEA MYALMWNSTH WSYFDYNLTS NSQRIFVPLD 420
DDSSTAEQAN SPPGHQVLFD IAQLYPFWTG AAPESLKNP LAVQLAYS RV ARMLDTKAGA 480
IPATNFRGTQ QWDQPNVWPP LQHVLMMQGLL NTPPTFGESD PAYQNVRLA LRLAQRYLDS 540
TFCTWYATGG ETSQTPRLQG VSPDATGTMTF EKYADNATNV AGGGGEYEVV EGFWTNGVL 600
IWAADVFDAG LKRPDCGNIT AAHTHSAKRG LERRAVELDP WDAAWTKMFG RSKLRKREEG 660
RKRWLS

```

```

SEQ ID NO: 185      moltype = AA length = 670
FEATURE            Location/Qualifiers
source             1..670
                   mol_type = protein
                   organism = Chaetomium piluliferum

```

```

SEQUENCE: 185
LYINGSVIAP CDSPLYCHGE ILKAIELARP FSDSKTFVDM PTIKPLDEVI AAFSQLSQPL 60
SNNSELNAPL AENFAPAGGE LEEVSKDELQ TDPTFLDKLD DTTIKEFVSK VIDIWPELTR 120
RYVGSDDCSG CANSFIPINR TFVVAGGRFR EPYYWDSYWI LEGLLRTGGA FVDISRNIIE 180
NFLDFVETIG FVPNGARIYY LNRSQPPLLT LMVKTYVDYT NDTSILERAL PLLVKEHEFW 240
TTNRSVSITA NGKEYTLNRY SVNNNQPRPE SYREDYITAS NESYYAESGI IIPVTRPLNE 300
TEKAELYSNL ASGAESGWDY TSRWLKTPND AANDVYFPLR SLNVGLGVVP DLNSILYQNE 360
VILAEYFEQA GNSSEAERFA AAARQRSEAM YDLMWNATHW SYFDYNLTSS SQRIFVPLDD 420

```

-continued

GASTQEQSTS	PPGYQVLFVDV	AQLYPFWTGA	APAALKSNPL	AVQHAYSRAV	DMLNTKAGAI	480
PATNRTTGGQ	WDQPNVWPPL	QHIIMQGLLN	TPPTFGESDP	AYENTQSLAL	RLAQRYLDST	540
FCTWYATGGS	TSDMPPLQGV	SAGATGTMFE	KYADDATNVA	GGGGEYEVVE	GFGWTNGVLI	600
WAADVFGDAL	KRPDCGNITA	ASTHEGATKR	DLRGLGRRAV	ELDPWDAAWT	KMFGRAKLK	660
REQGRETWVN						670

SEQ ID NO: 186 moltype = AA length = 674
 FEATURE Location/Qualifiers
 source 1..674
 mol_type = protein
 organism = Myceliophthora hinnulea

SEQUENCE: 186

LYINGSVTAP	CDSPIYCHGE	LLKGVELAHP	FVDSKTFVDM	PTLKPVDEVL	AAFSKLRQPL	60
SNNSELNNFL	AEYFAPAGHE	LEEVPDSELQ	TDPKFLDKLE	DRTIKEFVSK	VIDIWPDLTR	120
RYAGPGDCSD	CANSFIPVNR	TFVVAGGRFR	EPYYWDSYWI	LEGLLRTGGA	FTQISKNIIE	180
NFLDFIDTIG	PIPGARIYY	LNRSQPPLLT	RMVKSYYDYT	NDTSILERAL	PLLIKEHDFD	240
TNNRSVSUTA	SNGKTYTLHR	YHVENNQPRP	ESYREDYITA	NNGSYAASG	IIYPVKTPLN	300
ETEKAVLYSN	LASGAESGWD	YTAARLVRPD	DAARDVYFPL	RSLNVREIVP	VDLNSILYEN	360
EVIIAGYLEK	AGNSSEAKRF	ASAAKQRSEA	MYNLMWNATH	WSYFDYNLTS	NAQNIPIPAD	420
EDTAPFDRTA	APPGKQVLFH	IAQLYPFWTG	AAPAHKLSNP	LAVQKAYARV	SRMLDSKKGA	480
IAATNRYTGG	QWDQPNVWPP	LQHVLMOGLL	NTPATFGESD	PAYQGVQKLA	LRLAQRYLDS	540
TPCTWYATGG	STSDFPQLQG	VSPDATGIMF	EKYADSAINV	AGSGGEYEVV	EGFGWTNGVL	600
IWAADVFGNK	LKRDCGNIT	AAATHSEAKR	SLGDGGLARR	AVELDPWDAA	WTKMFGRSKL	660
RRREAEDVRK	RWSS					674

SEQ ID NO: 187 moltype = AA length = 659
 FEATURE Location/Qualifiers
 source 1..659
 mol_type = protein
 organism = Chloridium virescens

SEQUENCE: 187

LYINGSVIAP	CDSPLYCQGD	ILKAIQLAQF	FSDSKTFVDM	PTTQPVQVI	AAFNQLPQPV	60
SNNGLQNLFL	STYFAPAGGE	LTEVPKDLQ	TNPMFLNKLN	DTVIREFVTA	VIDIWPDLTR	120
TYTGASNCTG	CDSFIPVNR	TFVVAGGRFR	EPYYWDSFWI	LQGLLRTGGA	FTQISKNIIE	180
NFLDLVDQFG	FVPNGARVYY	LNRSQPPVLS	EMVRTYVAYT	NDTSILARAI	PTLIKEHNFW	240
MQNRVTNITG	ADGNTYTLNQ	YHVENTQPRP	ESYTEDYITA	NNHSYYATSG	IIYPETKPLN	300
DSEIANLYAN	LASGAESGWD	YGSRYLARPN	DAAQDVYFPL	RSLNVLEIVP	LDLNSLLYQS	360
EQNIALFLQA	TGNSSEAEQW	TSLAAQRQTA	IHALMWNETH	WSYFDYNLTS	NGQNIYVPAD	420
KDATPADTAS	APPGYQVLF	VAQFYPFWTG	AATDELRRNP	LAVRQAFTRV	DAYLTAKAGG	480
IPATNLMTGQ	QWDQPNVWPP	LMHVLMOGLL	DTPATFGAAD	PSYAALQGLA	LRLAQRYLDM	540
TPCTWYATGG	STSQTPKLQG	LGPDAVGTMF	EKYADNATNI	AGSGGEYTVV	EGFGWTNGVL	600
IWAADTFGAQ	LTRPNCGNIT	AAHVTGPKRS	VGLGRRAVEL	HESDARWVKM	FGSRAWRNA	659

SEQ ID NO: 188 moltype = AA length = 662
 FEATURE Location/Qualifiers
 source 1..662
 mol_type = protein
 organism = Gelasinospora cratophora

SEQUENCE: 188

LYVNGSVTAP	CDSPIYCYGE	LLHQVELARP	FSDSKTFVDM	PTIKPVDEVL	EAFSKLTLP	60
SNNSELHEFL	NTYGPAGGE	LEAVPTDQLH	VSPFTLDNVS	DDVVKQPVNS	VINIWPDLTR	120
KYVGAGEICT	GCANSFIPVN	RTFVVAGGRF	REPYYWDSFW	ILEGLLRTGG	AFTEISKNTI	180
ENFLDLVEQI	GFVPNGARLY	YLNRSQPPLL	TQMVRIYVEH	TNDTSILERA	VPILKKEWEW	240
WITNRTVEVT	ADGKTYSLQR	YHVDNNQPRP	ESYSEDYITA	NNHSYYATSG	IIYPETTPLN	300
DTQKAQLYAN	LASGAESGWD	YSTRWLKNPN	DAARDVYFPL	RSLNVLEIVP	VDLNSILYQN	360
EVTIGKFLAQ	QGNKDEAEW	AKKAEQRSEA	MYKLMWNSTL	WSYFDYNLTS	SSQNIYVPAD	420
PQVFPFEKPS	GTPGYQVLF	SINQMFPFWT	GAAPDQLKAN	PLAVKLTFDR	VKNYLDNKAG	480
GIPATNFVTG	QQWDEPNVWP	PLMHVLMGDL	LNTPATFGED	DPAYQETQNL	ALRLAQRYVD	540
STFCTWWATG	GSTSETPKLQ	GLGSDAKGIM	FEKYADNSTN	VAGSGGEYEV	VEGFGWTNGV	600
LIWAADKFGD	KLKRPNCGDL	TPANVGKRAV	ELDAFDPAKFT	KKFARKGKLE	KLKAKFKRRA	660
AI						662

SEQ ID NO: 189 moltype = AA length = 543
 FEATURE Location/Qualifiers
 source 1..543
 mol_type = protein
 organism = Acidobacteriaceae bacterium

SEQUENCE: 189

QTTTTAGLH	DTLAYIKRTW	HTLERSNKTL	LKSADDVKVG	QAGTLTLVVS	QDVKPQAVNA	60
SLRRELPPAD	KMRIVVRQLP	EHPEAVEPAG	LLYLPYPYV	PGGRFNEMYG	WDSYFILLGL	120
VHDDDELALAK	NMTDNFIYEI	EHYGMILNAN	RTYYLTRSQP	PFLTQMILEV	YRRTGDGKWL	180
ASTLPAIEKY	YAYWMREPHL	TPETGLSRYW	GGADTPAPEV	VHGEKDAAGH	NQYDRVREYY	240
RTHNYTAYDV	SQYYDKATDR	LKPLFYIADR	AMRESGFDP	SRYPFSADI	IHYDPVCLNS	300
LLYRMESDTA	TILKQLNRTS	EARVWEKRAT	QRAELVNRLM	WNEEKGLYFD	YDFITRRQSN	360
YHFVTTFYPL	WAGIASRQQA	DRVKNLSIF	ERAGGLQTS	YISGSQWDAP	FGWAPLQIMT	420
VEGLRRYGFN	EDADRISRKF	INMVVRDFEE	HGTIKEKYDV	VIGKSDLAAG	LKPGYTSNEA	480
GFGWTNAAVV	LFIEELAGER	PLAASLDRES	MPMLRQRHLS	PQPSVWPPFS	PQAPQYRRRD	540
PYR						543

-continued

SEQ ID NO: 190 moltype = AA length = 532
 FEATURE Location/Qualifiers
 source 1..532
 mol_type = protein
 organism = Acidobacterium capsulatum

SEQUENCE: 190

GARSSLSPHA	VAGPQPIDAY	IHTAWSTLTR	SMSDCKSVAD	PKLKSTPVLV	LPRDLAVPAN	60
VAAMQKQCHV	RVLRLPIVIT	HFDQIRESQI	ATPGLLYLPH	PYVVPGGRFN	EMYGWDSFFI	120
LKGLDDHHI	ALARGIVENF	FFEIAHYGGI	LNANRTYYFT	RSQPPFLSSM	IRAIYAAEVA	180
EGHTQAAHAW	LVEAYPYAVR	DHALWMSPIH	QAGNTGLARY	FDTGQGPVPE	MADDSTYYQD	240
VIRWLLAHPG	LHTGYLMHGS	PHLDAAARER	LAQLSCDPTL	SKVCARAHVH	GYWLTRSIFYK	300
GDRAMRESGF	DTTFRFGPFS	GSTQHPAPVG	LNALLYKYER	DLAWMAAQLG	KPGEAAKWNS	360
EAETRAEMN	HYLWNAQKKM	YFDYNFETHR	QSSYAFITTF	YPLWAGAADK	AQQQGVIALS	420
PLFEHPGGLA	ISNHDSGVQW	DLPYGWAPTE	WMAVQGLLRA	DDQHDARRIA	SEFNRTVTRTT	480
YQHDHAIYEK	YDVVNRSNDF	RVTAGYTQNV	VGFGWTNAVY	LEFQSLLAHP	GQ	532

SEQ ID NO: 191 moltype = AA length = 568
 FEATURE Location/Qualifiers
 source 1..568
 mol_type = protein
 organism = Acidovorax wautersii

SEQUENCE: 191

AAAVSSGLRI	SERHPAYSTN	PARVQVPGAP	GAPPSDHCTP	ADRYQELFVA	VQSQQIFEDS	60
KTFVDCGPIG	EPEDILAAAYR	AEHAQPDFDL	ARFVAQHFTA	PQVAANDYVG	APGMALAEHI	120
DALWPVLTRK	PEDHPVRGSA	LPLAHPYVVP	GGRFAELYVW	DSYPTMLGLA	ATGRSELVQC	180
MTDNFARLID	AFGFVPNGTR	TYILSRSQPP	LFAAMAEALGA	LVGGPPVSHY	LPQLLQEHAW	240
WMDGLHVLHP	GEARRRVVAL	PGGEILNRYW	DDRDTPREES	WREDIETASA	VDRPSADVYR	300
DLRAAAESGW	DFSTRWLRA	DAANPASLHL	SQICTDILLP	VDLNAFLYRM	EVSIKASQS	360
AGDRATATHF	HDLAHRREA	VNRLMWNEAE	GAYFDYDWR	GELRGCLTAA	TVVPLYAGMA	420
TEAQAARAVR	AVRTHLLAAG	GLATTVCSSD	QQWDRPNGWA	PLQWMAVRGL	ERYGHKDLAL	480
EVQRWIETV	RSVYQREGKL	VEKYAVGNGD	GAPLCGGGGG	EYPLQDGFGW	TNGVVQCWLD	540
PRYDTYAAQ	TVYYGPTDDG	TADSLPEA				568

SEQ ID NO: 192 moltype = AA length = 529
 FEATURE Location/Qualifiers
 source 1..529
 mol_type = protein
 organism = Xanthomonas arboricola

SEQUENCE: 192

APMDTPVVNA	PAATPPTPDL	AYPELFQAVQ	SGELFDDQKH	FVDFLPLRDP	ALINADYLAQ	60
HDHPGFDLRK	FVDANFEESP	PVQTDAIRQD	TALREHIDEL	WPKLVRSQTH	VPPYSSLLAL	120
PHPYVPPGGR	FREYVYWDY	FTMLGLVKSG	ETTLRSQMLD	NFAYLIDTYG	HIPNGNRSYY	180
LSRSQPPFFS	GEAVYQRYLP	QLRKEYAYWM	QGSDELQPGQ	AARHVVRLAD		240
GSLNRYWDE	RDTPRPEAWL	HDTRTAAEAG	DRPAAEVYRD	LRAGAESGWD	YTSRWLADGQ	300
NLRTIRTTAI	VPIDLNSLLY	HLERTLAQAC	AQPGAECQD	YAALALRRKQ	AIDAHLWNA	360
GYADYDWT	RKLSQVTA	ALYPLFTGLA	TDAHAKRTAS	TVRARLLRPG	GLATTAVKTG	420
QQWDEPNWGA	PLQWVAVDGL	RRYGEDALAR	TIGERFLTQV	QALFAREHKL	VEKYGLEADA	480
AGGGGGEYAL	QDGFGWNTGV	TLMLNLNLYP	TATKPAPAKR	ARKPEAAAR		529

SEQ ID NO: 193 moltype = AA length = 540
 FEATURE Location/Qualifiers
 source 1..540
 mol_type = protein
 organism = Kosakonia sacchari

SEQUENCE: 193

AEAQNTQPQP	DILLGPLFSD	VQTAKLFPPDQ	KTFADAVPKG	DPLMILADYR	MQRMQTSFDL	60
RHFVDVNF	PKEGEKYVPP	EGQNLREHID	GLWPVLTRTT	DSAGKWSLL	PLPKPYVVP	120
GRFREYVYWD	SYFTMLGLAE	SGHWDKIEDM	VTNFAHEIDT	WGHIPNGNRS	YYLSRSQPPF	180
FSLMVELLAT	HDGDEALKTW	LPQMEKEYQY	WMEGADTLQP	GQANKRVVKL	SDGSVLNRYW	240
DDRDTPRPES	WLDVTTAKN	NPNRPATEIY	RDLRSAAASG	WDFSSRWMD	PNQLGTIRTT	300
SIVPVDLNL	MFKMEKMLAR	GYQAAGDSAK	ASQYDALANA	RQKGIENLW	NEKEGWYADY	360
DLKTKKVRNQ	LTAALYPLF	VNAAKDRAD	KVASAAKERL	LKPGGIATTT	VNSGQQWDAP	420
NGWAPLQWVA	TAGLQNYDQ	KLAMEVSWRF	LTNVQHTYDR	EKKLVEKYDV	STTGTGGGGG	480
EYPLQDGFGW	TNGVTLKMLD	QICPKEKPCD	SVPQTQPAQQ	PAKVEPTSQ	PSKQQQAVAQ	540

SEQ ID NO: 194 moltype = AA length = 532
 FEATURE Location/Qualifiers
 source 1..532
 mol_type = protein
 organism = Enterobacter sp.

SEQUENCE: 194

DEQPAFQKNS	PDILLGPLFN	DVQSACLFPD	QKTFADAVPK	SDPLMILADY	RMQHTQSSFD	60
LRHFVEMNFT	LPAEGEKYVP	PAGQSLREHI	DDLWPVLTRT	TDKASNKWS	LLPLPKPYV	120
PGGRFREYVY	WDSYFTMLGL	AESGHWDKIS	DMVDNFAWEI	DTFGHIPNGN	RSYYLSRSQP	180
PFFSMMVELL	ATHSDALKK	YRPQMEKEYA	YWMGVDSLQ	PGQANQRVVK	LDDGSVLNRY	240
WDDRDTPRPE	SWLDDVNTAK	NNPNRPATEI	YRDLRSAAAS	GWDFSSRWMD	DPQKLTIRTT	300
TSIVPVDLNA	LMFKMEKLLA	RASQEAAGSA	AASKYEALAT	ARQKAIENHL	WNDKEGWYAD	360
YDLKSKKVRN	QLTAAALFPL	YVKAASQERA	DKVAAATSAR	LLKPGGTTTT	TINSQQQWDA	420
PNGWAPLQWV	ATEGLQNYGQ	NKVAMDVTWR	FLKNVQHTYD	REKKLVEKYD	VSTTGTGGGG	480

-continued

GEYPLQDGGF WSNQVTLKML DLVCPKEKPC DSVPENQPAA NDEAAPVKAS AQ 532

SEQ ID NO: 195 moltype = AA length = 720
 FEATURE Location/Qualifiers
 source 1..720
 mol_type = protein
 organism = Saitozyma flava

SEQUENCE: 195
 QNPASSSSFS PTPVTMVPSP PTAALNATVP GQGVYPPLQA WCNNGGNDTF CPGVLLQDVQ 60
 LSGIFPDSKT FVDKPTRGTL NETLRTFASL GNNLTVGQIE GFVNNSFKGE GLELSQVALE 120
 GFNPNPAPLD TISDPIYQGW MSVVNSYWT LIRETNRSAL CNGDCESSLI PLNNTIVVPG 180
 GRYREIYYWD SYWILIGLLE SELTAYATDL ISNFMDFIQT YGFIPNGGRK YYLNRSQPPV 240
 FTQMLNTYVQ RTGNTSILAR GLPLAHQELV WWENNRRVISV TSPYSNITRR VAHFAVNNTA 300
 PRPEGYVEDY ETAFGASPAL NETARGQLYS DLAAGAESGW DYSSRWCKQP VINLTDNFPA 360
 LRTINTAKII PVDLNSLLAG DHTLLANLYE LYGNSSNANT SVTSNSSLV PYHRQLAKNY 420
 SDAVLDLHWD AQAQFWYDFN LTNARADII HPGGTFAWQ NITPPAVASN ETYALEVVSQ 480
 ARFLLGKYTS IPSVSTLIET GLNWDFFNSW PPHVYSSIKA FETLGRAFPN ASVLSNISIP 540
 FSDVQKNQLG LDESAIPAQP ASTIGNSTSQ IAEAQKQWP QALSIEYANR YMQAAFCSWY 600
 STGGSIDGLL TQLPLSELNM TGTFFAGTTG VMFEKENVTD LDAAGGGGEY KVQIGPGWTN 660
 GVALWLGANF GQYLPQPTCP LIPIIEVQNG MNSSVYQNGT MTNSTKTQSY IFQGHRIPRK 720

SEQ ID NO: 196 moltype = AA length = 723
 FEATURE Location/Qualifiers
 source 1..723
 mol_type = protein
 organism = Phaeotremella skinneri

SEQUENCE: 196
 QTPTSSSSSS FSATPVSTSV PSPTVPLNSS VIGQGLYPPA QAWCNGGMND TFCPGVLLQD 60
 VELSGIFADS KTFPDKPTVG TLNSTLQAPA ALPANVTVGE IETFVNQYFK GEGLELEQVA 120
 IEGFVANPAI LNNITDPVFK GWVSTVNGYW QLLIRQTNES SLCNETSCAS SLIPLNHTIV 180
 VPGGRYREIY YWDSFWYLQ LKSELYTYS WDLQNFMDL IETYGFLPNG GRKYLYNRSQ 240
 PPVFIQMLDA YIKVTGNSSI LARALPIATT ELAWWSSNRT IPVTSPYGTI THLVAHYAVT 300
 NSAPRPEGYV EDFTTAMGAS PALNDSARAE LYSELASGAE TGWDYSSRW RQPLNLNTDN 360
 NPALRTLNVK AIIIPVDLNSL LAGDHALLAN LYDLYSNSSS SNSTSNSSI SNSSQAAYH 420
 RQAAQNLTAA ILDLHWSPSK SFFYDFNTSS NSQSDIYTPA SLFPLWQNT PPALVGNETA 480
 ALQLVSGVRY LLGYAGIPS VATLLATGLN WDFPNWPPH VYTSIKAFQT LYRVNPNATV 540
 LSNLTLSFAQ VQAGQLGSE TGFQIQPAST VGNTSLETAE AKGKPWPVAL GIEYANRYMQ 600
 SAFCSWYSTG GEISGVLTLQ PLSDLNSTGT FTAGQTGVMF EKFNVTDPDA AGGGGEYTVQ 660
 TGFGWNTGVV LWVAGEFGSL LPAPTCLIP IIEISGSSNS SIYANGVTN STALYRGYKI 720
 PRK 723

SEQ ID NO: 197 moltype = AA length = 663
 FEATURE Location/Qualifiers
 source 1..663
 mol_type = protein
 organism = Trichoderma asperellum

SEQUENCE: 197
 LYTNQSVIAP CDSPIYCHGD ILQDIQLAHP FSDSKTFVDM PAKRPLREIQ SAFNNLSKPL 60
 TNDSSLQVFL ETYFADAGGE LVEVPRSSLT TNPAFLNKIN DTVIREFVTK VIDIWPDLTR 120
 RYSGAVGNCS TCPNSFIPVN RTFVVAGGRF REPYYWDSYW IIEGLLRTGG AFVGIAKNTI 180
 ENFLDFIDRF GFIPNGARLY YLNRSQPPLL SQMVRIYIEH TNDTSILKRA LPLLVKHEHF 240
 WNRRTISVN IANKTYTLQ YAVQNTQPRP ESFLEDYVTA NNRSYYATSG IIPYENKHLN 300
 STEMADLYAN LATGAESGND YTSRWLANPS DAINDVYFPL RSLNNKEIVP VDLNSILYGN 360
 ELTISQFYNR TGDATAARAW AERAANRSAA IQAVFWNETL PSYFDYNLTS SSQYVVPAD 420
 KDTISLDRQT APSGKQVFFH VGQFYPFWTG AAPDYVKNNP YAVTRAYDRV TGYLDAQPGG 480
 IPASNVQTGQ QWDQPNVWPP HMHILMQGLN NVPATFTAQD PSYQDIQNL LRLGQRYLDF 540
 TFCTWLATGG STSETPKLHG LSDQDVGIMF EKYDDNSTNA AGGGGEYEVV EGFGWNTGVL 600
 LWTADTFGNK LKRQCIGNIT AGHPAPASSS SKRSVAVQLDA WDARRVKKFG KRTGRVVKRP 660
 FLF 663

SEQ ID NO: 198 moltype = AA length = 674
 FEATURE Location/Qualifiers
 source 1..674
 mol_type = protein
 organism = Corynascus sepedonium

SEQUENCE: 198
 LYINGSVTAP CDSPIYQGE LLKAVELARP FVDSKTFVDM PTIKPVDEV L AAFSKLSLPL 60
 SNNSELNAFL YENFAQAGHE LEEVPDSELE TDAKFLDKLE DRTIKEFVGK VIDIWPDLTR 120
 RYAGPSNCTE CANSFIPVNR TFVVAGGRFR EGYWDSYWI VEGLLRTGGA FTHISKNIE 180
 NFLDFVDTTG FIPNGARIYY LNRSQPPLLT LMKVSYVDYT NDTSILDRL PLLIKEHEFF 240
 MNRTVTSITG SNGKEYTLN YHVENNQPRP ESFREDYITA NNGSYAASSG IIPVKTPLN 300
 ETEKAALYSN LATGAESGWD YTSRWLGVP S DAARDVYFPL RSLNVRDIVP VDLNSILYQN 360
 EVIIAEYLEK AGNSSAAKRF ATAAEQRSEA MYSLMWNATH WSYFDYNLTD NTQHFVVPAD 420
 EDTAPQDRIE APPGQVFFH IAQLYPFWTG AAPASLKANP LAVQQAARV ARMLDIKKGA 480
 IPATNYRTGQ QWDQPNVWPP LQHILMKGLL NTPATFGKSD PAYQSVQNLA LRLAQRYLDS 540
 TFCTWYATGG STSDPQLEG VTPGATGVMF EKYADNATNV AGTGGGEYEVV EGFGWNTGVL 600
 IWAADVFGNK LKRPDGCGNIT AAHTSSAKR GLEENKLPRR AVELDPWDAA WTKMFGRSKL 660
 RRREADVVRK RWMS 674

-continued

SEQ ID NO: 199 moltype = AA length = 674
 FEATURE Location/Qualifiers
 source 1..674
 mol_type = protein
 organism = Myceliophthora thermophila

SEQUENCE: 199

LYINGSVTAP	CDSPIYCHGE	LLKGVELAHP	FVDSKTFVDM	PTLKPVDEV	AAFSKLRQPL	60
SNNSELNNFL	AEYFAPAGHE	LEEVPKGELQ	IDPKFLNKL	DRTIKEFVSK	VIDIWPDLTR	120
RYAGPGDCSG	CANSFIPVNR	TFVVAGGRFR	EPYYWDSYWI	LEGLLRTGGA	FTQISKNIIE	180
NFLDFIDTIG	PIPNGARIYY	LNRSQPPLLT	RMVKSVDYD	NDTSILERAL	PLLIKEHDF	240
TNNRSVSVTA	SNGKTYTLHR	YHVENNQPRP	ESYREDYITA	NNGSYYAASG	IIYPVKTPLN	300
ETEKAVLYSN	LASGAESGWD	YTARWLRVPD	DAARDVYFPL	RSLNVREMP	VDLNSILYEN	360
EVIIAEYLEK	AGNSSEAKRF	ASAAKQRSEA	MYNLMWNATH	WSYFDYNLTS	NAQNIFFVPAD	420
EDTASFDRYA	APPQQVLFH	VAQLYPFWTG	AAPAHLKSNP	LAVQKAYARV	SRRLDTKKGA	480
IAATNYRTGQ	QWDQPNVWPP	LQHVLMQGLL	NTPATFGESD	PAYQGVQKLA	LRLAQRYLDS	540
TPCTWYATGG	STSDFPQLQG	VSPDATGIMF	EKYADSATNV	AGGGGEYEVV	EGFGWTNGVL	600
IWAADVFGNK	LKRPDGCGNIT	AAATHSEAKR	SLGDGGLARR	AVELDPWDAA	WTKMFGRSKL	660
RRREAEDVRK	RWSS					674

SEQ ID NO: 200 moltype = AA length = 661
 FEATURE Location/Qualifiers
 source 1..661
 mol_type = protein
 organism = reesei GH37

SEQUENCE: 200

LYINGSVIAP	CDSPIYCHGD	ILREIELAHP	FSDSKTFVDM	PAKRPLSEIQ	TAFANLPKPL	60
RNDSSLQTF	ASYFADAGGE	LIQVPRANLT	TNPTFLSKIN	DTVIEQFVTO	VIDIWPDLTR	120
RYAGDAAVKN	CSSCPNSFIP	VNRTFVVAGG	RFREPYWDS	YWIVEGLLRT	GGAFFVGIARN	180
TIDNFLDFIE	RFGFVPNGAR	LYYLNRSQPP	LLSRMVKVYI	DHTNDTAILR	RALPLLVKEH	240
EFWTRNRTVD	VRVNNKTYVL	NOYAVQNTQP	RPEFREDFO	TANNRSYYAA	SGIIPATKP	300
LNESQIEELY	ANLASGAESG	NDYTARWLAD	PSDAMRDVYF	PLRSLNNKDI	VPVDLNSILY	360
GNELAIAPFY	NQTGNTTAAR	EWSSLAANRS	ASIQAVFWNE	TLFSYFDYNL	TSSSQNIYVP	420
LDKDAVALDR	QTAPPQGVLF	FHVQGFYFV	TGAPEYLRN	NPPFAVTRIFD	RVKSYLDTRP	480
GGIPASNVTM	GQWDQPNVW	PPHMHILMES	LNSVPATFSE	ADPAYQDVNR	LSLRLGQRYL	540
DFTFCTWRAT	GGSTSETPKL	OGLTDQDVGI	MFEKYNDNST	NAAGGGGEYQ	VVEGFGWTNG	600
VLLWTADTFG	SQLKRPQCGN	IMAGHPAPSK	RSAVQLDMWD	ASRVKKFGRR	AEGRMGTLHA	660
W						661

SEQ ID NO: 201 moltype = AA length = 670
 FEATURE Location/Qualifiers
 source 1..670
 mol_type = protein
 organism = Chaetomium virescens

SEQUENCE: 201

LYINGSVTAP	CDSPLYCQGE	ILKAIELARP	FSDSKTFVDM	PTIKPLEEVI	AAFGRRLKQPL	60
SNNSELTAFL	AENFAPAGGE	LEEVPKSELH	TDPVFLNKL	DAVKEFVGK	VIDIWPDLTR	120
RYAGPGNCSN	CENSFIPVNR	TFVVAGGRFR	EPYYWDSYWI	VEGLLRTGGA	FVGITKNILE	180
NFLDFIETIG	FVPNGARIYY	LNRSQPPLLT	KMIKIYVDHT	KDTSILQRAL	PLLIKEHEWW	240
TNNRSVTVTG	PNGKTYTLNR	YHVNNNQPRP	ESFREDYITA	NNGSYYATSG	IIYPVKSPLN	300
ETEKDETYAN	LATGAESGWD	YTARWLRTPN	DAAKDVYFPL	RSLNVNRMIP	VDLNSILYQN	360
EVIIGEYLEQ	AGNKSEARW	FQAAANQRSEA	MYALMWNATH	WSYFDYNLTS	NSQYIFIAND	420
EDATTAEQAN	SPPGQVLFH	ISQLYPFWTG	AAPDQLKKNP	LAVQQAAYRI	ERMLNEKAGA	480
IPSTNFRGTQ	QWDEPNVWPP	LQHILMQGLL	NTPATFGTAD	PAYAAVQNLA	LRLAQRYLDS	540
TPCTWYATGG	STSTPQLQG	VSPGATGIMF	EKYADNATNV	AGSGGEYEVV	EGFGWTNGVL	600
IWAETFQGNK	LKRPDGCDIQ	AAHTHTDKKK	RWSVDGEVRA	RERMAVELDP	WDAKWTKMFG	660
QAKGRVGRRS						670

SEQ ID NO: 202 moltype = AA length = 508
 FEATURE Location/Qualifiers
 source 1..508
 mol_type = protein
 organism = Rhodothermus marinus

SEQUENCE: 202

QDRVACQVPL	PSVERIEAVR	AYIRQSWDVL	TRSHRDLLAA	VQDPKIEHEP	GTPWPLYIAA	60
TEDSVAVVHR	LQQLPDSVL	QQIVLRVLPE	DPVAHLDEIH	PHGLLYLPEP	YVVPGGRFNE	120
MYGWDSYFTV	VGLLRDGRVD	LAKAMTDNHL	YQVRHYGKVL	NANRTYYLTR	SQPPFLSAMV	180
LAVYAHTQDR	DWLAAPVLI	ERYYAYWTP	PHLAGETGLS	RYIDLGEPGA	PEVVAGERDA	240
QGRTHYDRVR	EYYRMHEVTA	YDESLYVAE	ADSLTPLFYK	GDRSMRESGF	DPSNRFGPFS	300
VDIIHYAPVG	LNALLYRMET	DLARIHEILG	DTAAAAAWRA	RAEARREVRD	RYLWDSERGL	360
YFDYNFRGTGR	RSYVVFATTF	YPLVWGMASP	EQAARVAANL	YLLEAPGGLL	TSTHISGSQW	420
DAPYGWAPLY	LIAVEGLRRY	GYDEAADRLT	AKFVSMIVED	FERTGVILEK	YDVVQRRSDV	480
ALRYGYTSNE	IGFGWTNAVF	AELLAQMD				508

SEQ ID NO: 203 moltype = AA length = 674
 FEATURE Location/Qualifiers
 source 1..674
 mol_type = protein
 organism = Myceliophthora sepedonium

SEQUENCE: 203

-continued

LYINGSVTAP	CDSPIYCQGE	LLKAVELARP	FVDSKTFVDM	PTIKPVDEV	AAFSKLSLPL	60
SNNSLNAFL	YENFAQAGHE	LEEVDPSELE	TDKFLDKLE	DRTIKEFVGK	VIDIWPDLTR	120
RYAGPSNCTE	CANSFIPVNR	TFVVAGGRFR	EPYYWDSYWI	VEGLLRRTGGA	FTHISKNIIE	180
NFLDFVDTIG	PIPNGARIYY	LNRSQPPLLT	LMVKSVDYDT	NDTSILDRLAL	PLLIKEHEFF	240
MNNRTVSTIG	SNGKEYTLNR	YHVENNQPRP	ESFREDYITA	NNGSYYASSG	IIYPVKTPLN	300
ETEKAAALYSN	LATGAESGWD	YTSRWLGVP	DAARDVYFPL	RSLNVRDIVP	VDLNSILYQN	360
EVIIAEYLEK	AGNSSAAKRF	ATAAEQRSEA	MYSLMWNATH	WSYFDYNLTD	NTQHIFFPAD	420
EDTAPQDRIE	APPGQQVFFH	IAQLYPFWTG	AAPASLKAMP	LAVQQAYARV	ARMLDIKKGA	480
IPATNYRTGQ	QWDQPNVWPP	LQHILMKGLL	NTPATFGKSD	PAYQSVQNL	LRLAQRYLDS	540
TFCTWYATGG	STSDFPQLEG	VTPGATGVMF	EKYADNATNV	AGGGGEYEVV	EGFGWTNGVL	600
IWAADVFGNK	LKRPCGNIT	AAHTHSSAKR	GLEENKLPRR	AVELDPWDAA	WTKMFGRSKL	660
RRREAEDVRK	RWMS					674

SEQ ID NO: 204 moltype = AA length = 657
 FEATURE Location/Qualifiers
 source 1..657
 mol_type = protein
 organism = Moelleriella libera

SEQUENCE: 204

LHTNGSLIAP	CDSPIYCYGD	ILKQVELARP	FADSKTFVDM	PGVKPLAEIQ	AAFDKLEKPL	60
RNNTALQDFL	KTYFADAGQE	LQEVPKSELK	TDPQFLKTLN	DTVIREFVTK	VIDIWPDLTR	120
SYKGSNTNCT	DCPNSFIPIN	RTFVIAGGRF	REPYWDSYWI	ILEGLLRTRG	SFTQIAKNTL	180
ENFLDFVEQF	GFVPNGARIY	FLNRSQPPML	SQMVKLYIDH	TNDTAILQRA	LPLLVEKEHAW	240
WMNNRTVDVQ	VGGKTYKLNR	YAVSNTTEPR	ESYREDFETA	SNSSYYAQSG	IIYPETKKLN	300
DSQRAVLYAN	MATGGENGWD	FSSRWIANPS	DSVRDVYFPL	RTNNAQNVVP	VCLNSILYGN	360
EMTIGGFNPS	TGNTTAGQEW	AAKAKARSEA	MHATMWNETH	PSYFDYNLTS	SAQDVYTLAD	420
DDTSIYDNGT	LTGAPPQYQV	AFNGAQFYYP	WQGAAPTLYK	ENPQAVKTAY	ARVAQYLKVR	480
KGGIPATNLK	AREQWDQPNV	WPPLMHILMQ	GLLNTPTPTFG	SSDPSYKSVR	SMALTLAQRY	540
LDSTFCTWYA	TGGSTSETPK	RPGLPEKDKG	IMFEKYADNS	IDIAGSGGEY	EVVEGFGWTN	600
GVLIIWAADF	ANELKRPCDG	NGSSSSSTSS	AAKRGLSAVE	LHPADASRIK	RFSSSKG	657

SEQ ID NO: 205 moltype = AA length = 660
 FEATURE Location/Qualifiers
 source 1..660
 mol_type = protein
 organism = Acremonium dichromosporum

SEQUENCE: 205

IYVDGNITAP	CDSPVYCYGE	MLHQIQLAKP	FDDSKTFVDM	PALKPLSQIQ	AAFDRLDKPL	60
SNNSALNNFL	DEFPADAGGE	LSEVDKADLE	TDPVFLDKID	DEVVKEFTNK	VIDIWPDLTR	120
RYSGAAASNC	TNCPSFIPIL	NRTFFVAGGR	FREPYWDTY	WIEGLLRTRG	GSFTDISRDI	180
IENFLDFVDQ	FGFVPNGARI	YYLNRSQPPV	LSRMVQAYIE	HTNDTDILDR	ALPLLMKEHE	240
PFSENRTIDI	EGPNGTTYTL	NRDYVRNMQP	RPESYSEDEY	TATNTSYSH	DSGIIYPETE	300
PLNDTERANL	YSALASGAES	GWDYSSRWIA	RPRDAAEDVY	FPLRLNTNN	IVPVDLNSIM	360
YANEMAIAGF	LNQGTGNASAA	AEWEELAYNR	SEAIHALMWN	ETYMSYFDYN	LTSAAQHIYV	420
PADDVSTLE	SSTAPEGHMV	LFSVSQFYYP	WTGAAPSYIK	NNPFAIAQIY	KRVETLLDTR	480
KGGIPATNFR	TGQWDQPSV	WPPLMHILMA	GLQNTPATFG	EDDPAYQHVH	EIALRIGORY	540
LDSTFCTWRA	TGGATSETPQ	LEGFTDRDVG	IMFEKYADNS	TNIAGGGGEY	EVVEGFGWTN	600
GVLIIWTVDEF	GNEIKRPCDG	DLEAADTTER	RKRSALQLAP	RDAQRTKKFG	KRAVERQPWF	660

SEQ ID NO: 206 moltype = AA length = 667
 FEATURE Location/Qualifiers
 source 1..667
 mol_type = protein
 organism = Fusarium sambucinum

SEQUENCE: 206

LYVNGTVVAP	CDSPIYCHGD	ILEQVELARP	FSDSKTFVDM	PAIRPLSEIQ	KAFDELDKPL	60
RNNSALADFL	SENPADAGNE	LEEVPEDELK	TDPKFLDNIN	DTVIREPTEK	VIDIWPDLTR	120
RYDQDSKNC	DCPNSFIPVN	RSFVVAGGRF	REPYWDSYWI	IIEGLLRRTG	SFVNIKNTI	180
ENFLDFIEEY	GFVPNGARIY	YLNRSQPPLL	SQMVKIYIDH	TNDTDILERA	LPLLVEKEHEF	240
FMKNRSVPVY	INDETYMLNT	YVNSNTQPRP	ESYREDYITA	TNKSYYSTSG	EVYSGGEELS	300
FKQKETLYGN	LASGAESGLD	YTSKWIARPE	NAIRDNYFPL	RYLNTRNIIP	VDLNSILYGN	360
EIAIADFYEQ	TGNSSASEQW	REVAANRSYA	MHAFMWNETH	WSYFDYNLTS	KAQQIYFPAD	420
NNTVSVDTED	APKQQQVFFS	PTQFYFPWLG	AAPDYLKNNP	YAVLSAYKRV	ATYLDKREGG	480
IPASNIETGQ	QWDQPNVWPP	MMHILMAGLE	KVPATFGIMD	PSFIEIRKLA	LRLGQRYLDS	540
TFCTWYATGG	STSETPKLES	VSDKEDGIMF	EKYADNATNT	AGGGGEYEVV	EGFGWTNGVL	600
IWAVEEFGNR	LTRPKCKNLE	TAHSSDKRDT	SAVMLHARDA	KHVKKFGRRK	RAEKAACKKR	660
SSRLFHF						667

SEQ ID NO: 207 moltype = AA length = 660
 FEATURE Location/Qualifiers
 source 1..660
 mol_type = protein
 organism = Phoma sp.

SEQUENCE: 207

LYQNGSIIAP	CDSPIYCYGD	LLREIELAQP	FSDSKTFVDL	PTIRPLDEV	RAFNNLTKPI	60
QNNTELNNFL	TTYFGEAGSE	LDALPKDQLE	TQPDFLDNVN	SSVIVNFTSQ	VIDIWPDLTR	120
RYVGAGNCTG	CVSSFIPVNR	TFVVAGGRFR	EPYYWDSFWI	IEGLLRRTKGS	FTQIAENIIE	180
NFMFLVEELG	FVPINGARRY	ENRSQPPLLT	QMVRYVEYT	QNYTLLEAL	PVLEQYEFW	240
VNNRSVTLE	GGKNYTLHHY	NVSNTPQRP	SYREDYITAN	NLTYYNENGE	QFNASHPLND	300

-continued

TQKALLYAEL	ASGAETGWDY	SSRWLANPSD	AVNDDFFPLR	SLNVINTIPV	DLNSILYYNE	360
ITIAEFHRE	GNYCAARQWA	ELARNRSEAM	TALLWNAEYY	SYFDYNLTSS	AQNIYTLADN	420
TSTPLSLAGA	PAGYQVGFQL	SQLYPFWTGA	APDSIKGDPT	AIRRAFARVE	EALDTEAGAV	480
SATNLFTGQQ	WDEPNVWPPL	QYIAIQGLLN	TPLEVSEDD	BQTAEDYVWT	QNLALRLAQR	540
YDLSLFTWR	STGGATEEEP	QLPGATGNGT	IFEKYSDEAI	NARGGGGEYT	VVEGFGWSNG	600
VLIWAVDNFG	QKLTPDCGN	ITAAAPPSTA	SKRKRSAVEI	HQRDAAWIKG	TKENKMGFKK	660

SEQ ID NO: 208 moltype = AA length = 737
 FEATURE Location/Qualifiers
 source 1..737
 mol_type = protein
 organism = Lentinus similis

SEQUENCE: 208

LPQAVTPSST	SVSSQTVSTA	VPSPTASLTS	TLPSQIPLPP	KQDWCPSEIF	CAGELLQTVN	60
VAQLYPDPKT	FVDKPTARKS	QQVVSNFQNI	GGNSSNVTVG	AIEDFVNSNF	KGEGLELEPI	120
AFANFNPTPA	PLNNVSDPLV	KANSQIVHGY	WTQLTRSTND	SALCPEGTES	GSCSSLIPL	180
NHTFVPPGGR	FREQYWDYS	WIVQGLLVSE	LYDIVNDTLQ	NFMDELEHIG	FIPNGGRIYY	240
LNRSQPPLFI	HMLTSYVQAS	GDTSLKRAL	PLAEKELAWW	SANRSVQVKS	PYTNATHNVY	300
RHYVTNTAPR	PESYTYDYIT	ANDPTLQTP	TEQQRADLYA	ELATGAESGW	DYSSRWLKEP	360
LAGGSNNTSP	ALRSLNIRSL	VPVDLNSILY	KAHLNLAALY	NNESHSAAS	NHTQAAIALR	420
EAILDLHWA	NKRAFDFNI	TSNARNDFS	AATFYPLWSG	IIPDEILSDG	SGATAFGAFA	480
AVNMLNRYN	GTFPVTFLET	GLQWDAPNAW	PPHQFIIIEA	LRNLPSNITN	TPLPSAPSGN	540
STYALIPAGQ	VGIAEGDLP	QILSPGQDAT	KTGPAADINT	LNGTVVNGGN	ATSGEGWAAV	600
LQRELANRYI	ASALCSWHAT	GGEVFNLLPR	LSDSQLQITQ	SQNNNTGNMFE	KFSINDIDSA	660
GRGGYTVQA	GFGWTNGVVL	YLTHVFGDKL	VAPSCPNLVA	LSSNTATTSG	AVAQMTLPSS	720
LAVTVGAVVL	GFVGLVL					737

SEQ ID NO: 209 moltype = AA length = 683
 FEATURE Location/Qualifiers
 source 1..683
 mol_type = protein
 organism = Diaporthe nobilis

SEQUENCE: 209

QTLGGIYDYG	DNIAPCCSAL	YCYGDILDSI	QRAKPFADSK	TFVDMPTRVP	LEEVRAYDQ	60
LTKPLQNNTE	LLDFLSNFPG	PAGQEVVPVD	PGSLGVNASF	LGGIANAVNR	EFTEAVIDLW	120
PNLTRWVNES	AVCAECNDSL	LSIKRPFIVA	GGRFREPYW	DSYWLHGLL	RSNGFNTRIA	180
RNQIENFLDF	VEDYGFVPNG	ARVYYLNRSQ	PPLLAQMVRI	YIEQTGDATI	LDRAIPLLR	240
EHDFFMSNRT	VHVSVGSRNY	TLNRYNVANT	EPRPESYED	YTQVNNASY	ANDGRVPFTR	300
NTTQAEKDLQ	YKNLASGAS	GWDFSTRFMR	DPTIAANDTY	FPLASYNIN	IIPVELNSIL	360
YWNEVTIAAF	YKQQQQVFN	ATAEADADAW	DARAASRSEA	MYSVMWNETL	GGYDFDNLTS	420
GSQDVFWARD	ADSLPTEQAG	TEPGQQVVFN	IGQLSPFWTG	AAPRSLAGDP	AAVRRAFSRV	480
DEYLGSRKGG	IAPTNFVSGQ	QWDQPSVWPP	HMHILMEALL	RTAEAGGESE	GGSEDWAWAQ	540
DLALRLGQRY	DSAYCTWRA	TGGGTPSSPP	LANPPQDLGG	QMFEKYSQDS	LNEAGSGGEY	600
VVVVGFGWSN	GVLIWVADTF	RDRLQTPACG	DLTTRGEAGK	MREKRMSENG	RHGDGSNAIS	660
AVKLDYFDAA	WTSENVGGLH	GVR				683

SEQ ID NO: 210 moltype = AA length = 763
 FEATURE Location/Qualifiers
 source 1..763
 mol_type = protein
 organism = Solicocozyma terricola

SEQUENCE: 210

QGNSSMSASN	ATSSSTGTGT	SSSAAAVMTA	VPMPAPTPLSS	PVEMLPPLPPV	QPWCNGGENA	60
TYCPGSLMQL	VQLSGIYNDS	KTFPDKPTQY	NASVTYQAFD	ALPVNATVGD	VETFEVKEYFK	120
GEGQELETQV	LQNTQNPFTF	LNVIDDELYK	GFVSTVNGYW	SLLVRQTNES	ALCTNGACES	180
SLIPLNRSFI	VPGGRYREIY	YWDSFWILEG	LLKSELYLYA	YNLLENFMDL	IEKFGFLPNG	240
GRSYLNRSQ	PPVVFQMLNA	YIQVTGNVSI	LTRALPIAET	ELQWRTNRT	ISVTSPTYGT	300
NYSVARYFVT	NSAPRPEGYV	EDITTAFGGN	PALNESARSA	LYAELASGAE	TGWDYSSRWC	360
KQPLNLNTDN	DPALRTLNV	QQIPVDLNSL	LCGDHVLNAN	LYEFYMNSTI	GGSGGSSGG	420
GNSTIGMGAW	ANSTSGSNMT	SNSTSSSNMT	GLVQSHRMIA	KEFEAAINDL	MWDKQKLWWY	480
DFNMTANARA	DVYHFGGLFP	LWQNTTPSDI	VGNDTAAFGV	FAGVRYMVGM	YPGPPAPASL	540
IQTGLNWDAP	NVWPPHYVTG	IKALETVLRI	NPNSSVVPNI	TLTDFTRIPT	GQLGLNQSQL	600
PPQPASALGN	GTADLDQAIA	QMKGKWPPT	ALAIIEFANRY	LQSAPCSWYS	TGGSIPGLLQ	660
QLSPQELNLT	GSLTTGSQAM	GNIFEKENLT	NVDAAGGGGE	YTVQIGFGWS	NGVILHTAGE	720
YGQYLQVQSC	PLIAIHETAN	ATSTNATKSN	NTMVFSGYRL	PHD		763

SEQ ID NO: 211 moltype = AA length = 734
 FEATURE Location/Qualifiers
 source 1..734
 mol_type = protein
 organism = Dioszegia cryoxerica

SEQUENCE: 211

QSSSSSSFSF	TPVTTAVPSA	TAALNQTVSG	QGVYPPLQPW	CNAGENATYC	PGVILQDVQL	60
SQLFPDSKTF	VDKPTNGTQN	ATQQAQQQLG	NNITLGQLAQ	FVNTSFRGEG	LELSQVPIG	120
FVANPAAVNK	VSNPLYRGWV	STVNSYWSLL	IRETNQSAVC	TTQCESSLIP	LNITYIVVPGG	180
RYREIYYWDT	FWILEGLLKS	ELYTYAWDVL	QNFMDFVDVY	GFIPNGGRKY	YLNRSQPPVF	240
IQMLDAYVKA	TGNVTILERA	LPLASEEMRW	WINNRTTQVT	SPFTGITRRV	YFNVNVSAP	300
RPEGVYEDYE	AAGFAQPLPT	EAQRGALYAE	LATGAESGWD	YSSRWCKQP	INVTDLNPL	360
RTLNGRSIVP	VDLNSLQAGN	HALLARLYEV	YINASTTSNT	TAAQTIRANA	TQELALHKT	420

-continued

ANDYSQAVLD	LHWDPARAWF	YDFNLTSNSR	ESLYTPAGTF	ALWQNITPPG	LEGNDTAALR	480
IASGARYLLG	RYGGIQGVSS	LLVTGLNWDF	PNSWPPHTYT	SIKAFQTLGR	LVGNASIVGN	540
ATIPFSQVAT	NQLGLNETQL	PPQDPALQGN	ASLTVPSARN	VSWPLALEIE	YASRYMQGAF	600
CSWYSTGGSI	SGLLTQLPVS	QLNATGTYQA	GQTGMFEKF	NATDIDAAGG	GGEYTVQIGF	660
GWTNGVVLWL	ADNFGQYLPQ	PTCPLVVLSL	TNLNSTTAGN	STNGTSPAAP	GNATSLVAVS	720
ELQGIWEGQR	VSRD					734

SEQ ID NO: 212 moltype = AA length = 1020
 FEATURE Location/Qualifiers
 source 1..1020
 mol_type = protein
 organism = Talaromyces funiculosus

SEQUENCE: 212

LPFNERVDQV	LSRYEVTSKL	DSRSTKPSKH	GHTYQTQFLG	VTWDQRNWRL	QSTVLDQGHY	60
ESRGSIANGY	IGLNVAGAGP	LFELDSPVDG	DVINGWPLFS	RRQTFAGLAG	FYDLQPRNTG	120
TNFPWLSQYG	DDSAISGVPH	WGMVLDLGLD	GEYLDATVDN	STISDYTTY	DYKAGVLSWD	180
YKWTPKNANG	SFGISYKIFA	NKLDVNQAVV	QLSITPSTNG	SASVVNVIDG	YAAVRTDFVS	240
SGNESDVVYT	AVKPNGVTNV	TAWIYTALDG	DDAFDISSAA	LVNDKPYVHQ	NDSSIAQSVN	300
VTFTAGTTIT	INKFVGAAST	DAFPDPQSTA	REAAALSARRR	GFDDLFRSHI	SEWAQVMPDD	360
SVDDFTLANG	TLPNDTFIIE	SAVMAVVNPY	YLLQNTVGPV	ALRRVNNAPV	NDWSIPVGGL	420
TSDSYAGQIF	WDADVWMQPG	LVAAPFESAK	RITNYRAAKY	SQALENAKTA	YTSSQNQTWF	480
SPDAIYSWT	SGRVGNCTAT	GPCWDYEHYH	NGDIGISLVN	EWVSGDNET	FKNKHPIYN	540
SIATLYGDL	KKNGSYYTLT	NMTDPDEYAN	NVDAGGYTMT	LISQTLNAN	AFRKQFGMNE	600
NTTWEMADN	ILLIRENDVT	LEYTTMNSV	AVKQADVLS	TFPLDYTKNY	TTSALNDL	660
YYALKQSPDG	PGMTYAFISI	VANDVSPSGC	SAYTYAQYSY	DPYIRGPFFQ	FSEQLLDDYT	720
INGGTHPAPP	FLTGHGGANQ	VVLYGYLGLR	LLPDDMLHID	PNLPPQIPSI	KYRTFYWRGW	780
PIQAASNYTH	TTIQRATVA	PLSTADPTYA	NKSIHVSVGH	NTVNSTTYSL	SANGSALVVP	840
NRQIGSINTV	AGNVVQCKSV	LSTDAYQKQG	YPISAVDGAA	STKWQPEFAA	NISSLTVDLT	900
GSNVSSVSGF	YFDWAQAPPT	NITVLLHNSS	SAALASSGDK	PGSSAVTLNI	TISNPYNAST	960
YNANIIALPS	SNSTNYTFPA	PVPKPRYATL	FVQGNQALDE	TDTKSGNGTG	ATVAEWAILS	1020

SEQ ID NO: 213 moltype = AA length = 1063
 FEATURE Location/Qualifiers
 source 1..1063
 mol_type = protein
 organism = Hamigera avellanea

SEQUENCE: 213

ASPKSRINQC	LKKHAGQGSH	DDETSSNVYQ	TRFPGVTWDE	DNWSLTTSVL	DQGHYQSRGS	60
VANGYLGINV	ASVGPFFELD	IPLNGDVG	WPLYSRRTQF	ATISGFFDSQ	PETNGTNFGW	120
LNQYGGESVI	SGVPHVSGLI	LDLGDGTYLD	STVDNATLSG	FTSSYDFKAG	VLSWSYQWTP	180
EGKHGSYDIT	YRLFNTKLYV	NQAVVDMEIV	PTVAGKASVV	NVIEGSSAVR	TDFVESGEDD	240
GAIFSARVFW	GISNVTAYFY	ANLTVSDNVD	LSSRTLVSNK	PYVSTNESSI	AQSVVDQFIP	300
GKSVRITKVF	GAASTDAFAN	PQETAKRAAS	TAQTNGYLKS	LNSHIAEWAS	VMPDDSVEDF	360
SLPETGKLPA	DEHIIESAII	SVTNTYYLLQ	NTVGKNAIKA	SSDAALNMSD	ISVGGLTSDS	420
YAGLIWFAD	IWMQPGVLAS	HPEAAEVFTN	YRVAKYPQAV	KNIEAFASS	KNQTNFSPSA	480
AAPWTSGRY	GNCGTGTGCF	DYQYHLNGDI	GLSMINQWV	SGDTQTFRK	HFPIYDSAT	540
FFSNLVERNG	STWTLTNMTD	PDEYANHIDA	GGYTMLPIAE	TLLYANSFRK	QFGVEPNETW	600
NEIAENVLVL	RTNGVTLEFT	SMNGSAPVKQ	ADVVLVTYPL	DYNNNYSPE	SLNDLDYYAN	660
KQSEDGPAMT	WAIFSUVANE	ASPSGCSAYT	YAQYAYYPYA	RAPFFQLSEQ	MIDDASINGG	720
THPAYPFLTG	HGGANQVNL	GYLGLRLLPD	NVIHVDPNLP	PQIPHLYKRT	FYWRGWPMAS	780
ASNYTHTTIQ	RAVINVPALST	ADQKFANVSI	PVHVGPETNA	TVYRLPVNGT	LTIPNRQIAS	840
KNTVAGNLIQ	CRPVESQKDF	QPGQFPISVV	DGASSTRWQP	KHADNVSAVT	VTFADVEVGS	900
LVSIFYFDWA	QAPPVDAAVI	FHNSSLENPA	SAFSPASNSS	SSEYSVITTL	KNVEQSDPYD	960
PESDKLDIIA	IPTGNTTKVT	LPSAVPAARY	ATLFTIGNQA	LGPEIDIAAKN	GTGATVAEWA	1020
IVGQTSSATS	KRSIQTRKLQ	VRSGAALSGL	GFAQRRRQSA	EMY		1063

SEQ ID NO: 214 moltype = AA length = 1020
 FEATURE Location/Qualifiers
 source 1..1020
 mol_type = protein
 organism = Talaromyces ruber

SEQUENCE: 214

LPFNERVDHV	LRSHDLTSRL	HSRSAPKPSNH	GGTYQTQFTG	VTWDQRNWRL	QSNVLDQGHY	60
ESRGSIANGY	IGLNVAGAGP	FFELDTAVDG	DVINGWPLFS	RRQTFAGLAG	FYDLQPTTNG	120
SNFPWLDQYG	DDSVISGVPH	WGLVLDLGLN	GEYLDATVDN	STISDYTTY	DYKAGVLSWH	180
YKWTPKNANG	SFEIKYKIFA	NKLDVNQAVV	QLSITPSANG	SASVANVIDG	YSAVRTEFVA	240
SGNESDAIFT	AVKPVGVSNV	TAWIYAALDG	DDAFDFSSAT	LVNDKPYVHQ	NDSSIAQSVN	300
VTFTAGTTIT	INKFVGAAST	DAFPDPQSTA	REAAALKARRR	GFDDLFRSHI	SEWAQVMPDD	360
SVDDFTLANG	TLPNDPFIIE	SAVMAVVNPY	YLLQNTVGPV	ALRRVNNAPV	NDWSIPVGGL	420
TSDSYAGQVF	WDADVWMQPG	LVAAPFESAK	RITNYRTAIY	SQALENAKTA	YTSSQNQTSTF	480
SSDAIYSWT	SGRVGNCTAT	GPCWDYEHYH	NGDIGISLVN	QWVSGDNDT	FKNTHPIYN	540
SIATLYGDL	KKNGSYYTLT	NMTDPDEYAN	NVDAGGYTMT	LISQTLNAN	AFRKQFGMDE	600
NTTWDMADN	VLLIRENDVT	LEYTTMNSV	AVKQADVLS	TYPLDYTKNY	TTSALNDL	660
YYALKQSPDG	PGMTYAFISI	VANDVSPSGC	SAYTYAQYSY	DPYIRGPFFQ	FSEQLLDDYT	720
ANGGTHPAPP	FLTGHGGANQ	VVLYGYLGLR	LLPDNVLHID	PNLPPQIPSV	KYRTFYWRGW	780
PIQAASNYTH	TTIQRATVA	PLSTADQTYA	NRSISVQVGG	NTVNSTTYSL	PANGSAIVVP	840
NRQIGSINTV	AGNIAQCKSV	LSTDAYQPGQ	YPISAVDGAA	STKWQPEFAA	NISSLTVDLT	900
SSNASTVSGF	YFDWAQAPPT	NITVLLHNSS	NAPLTSSNSN	GGNSTVSLNV	TISNPYDASA	960
YNANVITLSS	SNTTNYTFPA	PVSKPRYATL	FVQGNQALDE	TDLKAGNGTG	ATVAEWAVALS	1020

-continued

SEQ ID NO: 215 moltype = AA length = 1055
 FEATURE Location/Qualifiers
 source 1..1055
 mol_type = protein
 organism = Trichoderma lixii

SEQUENCE: 215

ATSNRRVSEC	LGRNGSGSSTG	VHFSKNVYKT	DFAGVTWDED	NWLLSTTELK	QGAFESRSGSI	60
ANGYLGINVA	SVGPPFFELDT	EENGDVISGW	PLFSRRQSFA	TIAGFWDSQP	VMNGTNFPWI	120
SQYGSDTAIS	GIPHWSGLIL	DLGNTYLLDA	TVDNRTISNF	RSTYDYKAGV	LSWSYKWTPTK	180
GNKGSFDISY	RIFANKLYVN	QAVVDLQVTA	SKNVEASIVN	VIDGFAAVRT	DFVESGEDGN	240
AIFSAVRPVG	VANVTAYVYA	DITGSGGVDL	SSRKIVHNKP	YVHANASSIA	QAVPVKFSAG	300
RAVRVTKFVG	GAASDAFKNP	KQIAKSAAT	ALKNGYSKSL	NSHVTEWAAV	MPRESSVDSFA	360
DPKTGKLPND	NYIIDSALIA	VVNTYYLLQN	TVGKNGSKAA	NGAPVNVDSI	SVGGLTSDSY	420
AGQVFDWADL	WMQPGLLAH	PEAAERIINY	RLARYGQAKE	NVKTSYAGSQ	NETFFSASAA	480
VFPWTSGRYG	NCTATGPCWD	YEHNLNGDIG	LALVNQWVVN	GDTKDFEKNL	FPVYDSIAQL	540
YGNLLKPNGT	AWTLTNMTDP	DEYANHVDAG	GYTMPFIAET	LQNANTFRKQ	FGIEQNKTWN	600
DMASNALVLR	ENGVTLEFTT	MNGSAVVKQA	DVIMVTFPLS	YTTNNTTEDA	LNDLDYYANK	660
QSPDGPAMTY	APFSIVANEI	SPSGCSAYTY	AQYAPKPYVR	APFYQLSEQI	IDSSINGGT	720
HPAYPFLTGH	GGANQVVLFG	YLGRLVLPDD	FIHIEPNLPP	QIPYLRYRTF	YWRGWPIAW	780
SNYTHTTISR	ASGVAALDGA	DQRFAGKTIT	IHSGPESPK	AYHLPVKGSV	VVPNKQIGSQ	840
QTYAGNLVQC	HAASSPSDYV	PGQFPIAAVD	GATSTKWQPA	SADKLSSITV	SLDQEDVGS	900
VSGFHFDDWAQ	APPVNATIIF	HNEAINDPAT	VLKSQKHNSN	YKVVSSLTNI	KQSNPYIKTT	960
DLDIATPIG	NTTNVTLSQP	VAASRYASLV	IVGNQGLDHA	DVVGKNGTGA	TVAEWAIIGH	1020
SKGHTGAPGH	HGKRKLNLRA	AAAMSDPDSF	ARRRQ			1055

SEQ ID NO: 216 moltype = AA length = 1056
 FEATURE Location/Qualifiers
 source 1..1056
 mol_type = protein
 organism = Aspergillus cervinus

SEQUENCE: 216

STFSHKNDRI	LKGLKRHGDH	YSRKSNTNST	DVYQTKFDGV	TWDDDNWLLT	TTALDQGDFF	60
SRGSIANGYL	GINVASVGPF	FELDTAENG	VISGWPLFSR	RQTFATIAGF	FDSQPTNGS	120
NFPWLYQYGG	DSVISGVPHW	SGLVLDLGND	TYLDSTVDNQ	TIENFTSMYD	YKSGVLSWSY	180
TWVPAGNKGS	FDIYVRLFAH	KLVNVQAVVD	MEITPSLSFN	ATVVNILDGY	SAVRTDFVES	240
GNDNGAIFTA	VRPWGIGNVT	AYVYANLTGT	PNVDLSSRTI	VANKPYVHTN	ASSIAQAVNV	300
SCSPNETVRI	TKFVGAASSD	AFENPQETAR	QAVSAAIGAG	YLRSLGTHIA	EWASIFPDDS	360
VDRFTDPATG	KLPANKHIIIN	SAIIAVTNTY	YLLQNTVGKN	AIQAVSGAPV	NIDSISVGGL	420
TDSDYAGQVF	WDADVWQPG	LAASHPEAAQ	RITNFRAAKY	PQALANIETA	FTSSKNQTSF	480
SPSAAAFPWT	SGRFGNCTAT	GPCWDYEHYH	NGDIGLSLMY	QWIASGDTQT	FRETHFPIYD	540
STATMYSNIV	ERNGSYWTLK	NMTDPDEYAN	QIDAGGFTMP	LISQTLNLAN	AFRQQPGLDV	600
NQTWSEIANN	VLVLNDNGVT	LEYTTMNGST	VVKQADVVDL	TYPLVYSNNY	TSQNSLDDL	660
YYANQQSPDG	PAMTWAIFSI	VANDVSPSGC	SAFTYHQYSY	DPYARAPFFQ	LSEQLIDDAS	720
TNGGTHPAFP	FLTGHGGANQ	VVIFGYLGLR	LLPDEAIHID	PNLPQPIPHV	AYRTFYWRGW	780
PISAQNSNTH	TTISRAMNAS	PLDTADSRFA	NVSIPIYVGT	ESNATVFLP	PTGGLTILNR	840
QNGFNNTIPG	NVAQCRPVYS	PDDYAPGQFP	IAAVDGATST	RWQPSSANTS	SVTVTLPTDQ	900
INSPVSGFYF	NWVQLPPVNA	TVIFHDDLLE	NPAATISSSG	NSSSYRVVMT	LTNIQQSSPY	960
NAQIAALDEI	TIPTGNTTIV	QLTNHAQTSR	YATLLISGNQ	GLGDTQDGVG	ATVAEWVILG	1020
QGGGSSSSNS	NGKRKLGARS	AAALSNGWTE	RRRLI			1056

SEQ ID NO: 217 moltype = AA length = 1053
 FEATURE Location/Qualifiers
 source 1..1053
 mol_type = protein
 organism = Rasamsonia brevistipitata

SEQUENCE: 217

LRSEARVAQV	VRAYSSSTGV	EAGHGNAQY	ETRFPQVTD	QRHWRLKSTV	LDQGHFQSRG	60
SIANGYVGIN	VASAGPFFEL	DTPVDGDVIN	GWPLFSRRQT	FATIAGFYDE	QPRTNGSNFD	120
WLYQDGGESV	ISGVPHWSGL	ILDLDGTYL	DATVDDSTIS	DYSTVYDYKA	GILSWSYKWT	180
PKSSKGSFKI	SYRLFAHKLN	INQAVVRMEI	TPSADTDATV	VNVLDGYSAV	RTDFVGSQKD	240
GDAVYSASVP	WEVQNVTAAY	YAVLDGSDGV	DLSTLSLVNG	KPYVHTNESS	IAQSVNVRFR	300
AGKTVTVTKF	VGAASTDAPP	DPQQTAREAA	LAAKEEGYDA	LLRSHVAEWA	AVMPEESVDD	360
FTYHNGTLPG	DDFVESAIME	AVVNPYYLLQ	NTVGENALRE	VSHAPVNEWS	ISVGGLTSDS	420
YAGLIFWDAD	LWMHPGLAVA	FPQAATRITN	YRVAKYEQAR	QNAKTSFTGS	KNQTFWSDSA	480
AVYPWTSGRY	GNCGTGPCW	DYEHYHLNGDI	GLSLINEWVA	SGDKTKTQES	YFPIYDSIAT	540
LYADLLQQNG	SHWTLTNMGT	PDEYANAVDA	GGYTMLPIAQ	TLLYANSFRQ	QFGAQPNSTW	600
TEMASNILFL	RENDITLEYT	TMNNSVQVKQ	ADVVLVTYPL	EYTTNYNAGN	ALTDLDYYAL	660
KQSPDGPAMT	YAFISIVANE	VSPSGCSVYT	YAQYSYDPYV	RPPFFQLSEQ	LVDDYTLNGG	720
THPAYPFLTG	HGGANQVVF	GYLGLRLLPD	NVIHIDPNLP	PQIPQVKYRT	FYWRGWPIQA	780
YSNYTHTTIG	RAADVLDALD	ADQRFANTTI	PVQVGSQNSA	TVYQLPIDGV	LTVANRQVAS	840
TNTVAGNLAQ	QCPVDSSDSY	VPGQYPLAAV	DGAASTKWQP	SFAANVSSVT	VSLPESESGT	900
LVSIFYFDWA	QAPPVNATVV	FHNNTVENPT	RYMSSPTFVT	HIDNITLSSP	YNAEANPAAT	960
ILLPSSNTTN	VTLAHPMPVP	RYATLFTITG	QALSESEVQA	QNGTGATVAE	WAILASNPET	1020
GSSKRNLQLR	GVGRSALAGL	PRRANKERR	TDV			1053

SEQ ID NO: 218 moltype = AA length = 1034
 FEATURE Location/Qualifiers
 source 1..1034

-continued

```

mol_type = protein
organism = Acremonium curvulum

SEQUENCE: 218
LEERVSTQLN RHGVRSSYNR RADNGDQNR SNTSHPIYQ TSFEGVTWDS RNWRLQGTVL 60
DQGHYQSRGS IANGYFGINV ASAGPFFELD TPVDGDIVNG WPLFSRRQTF AGLAGFWATQ 120
PTTNGTNPFW LYQYGDESPI SGIPHWGGLV LDLGDDVYLD ATVDNKTIKN YRTTFDYKAG 180
VLTWDYTWSF KKRSGSGFDI TYSLFANKLD INQAAVRLSI RPSRDTKAKV VNVLEGYAAV 240
RTDFVDSGKD GDAIYSAVRP VGVNNVTAYV YAVLDADSAV DLSSAKVIDD APYLYTNKST 300
IAQSVNVFEK ANQNVITIKF VGIASDADF KPRDVAKQAA FAGKRRGYDD ALRSHVSEWA 360
QVMPDDSDVD FSSDNGTLPD DGFIIESSIM AVVNPFFLLQ NTVGANALRR VNNAAVNDYS 420
ISVGGTSDS YAGLVFWDAD IWMQPGGLAA FPEAAQRITN YRVALYPQAK RNIKTAFQSS 480
KNKTRFSDDA ALYPWTSGRW GNCTASGPCF DYEYHLNGDI GIAFVNEWIT SGDEKAFEEK 540
YFPIYDSIAT AFANLLQKNG TQWTLTNMTD PDEYANHVDA GGFTMPLISQ TLTIANLFRK 600
KFGKEENDTW ADMAENVLIL RENDVTLEYT AMNNSVEVKQ ADVVLNTFPPL DYTRDYAPSA 660
ALNDLDYYAL QCPDGPAMT YAFISIVANE VSPSGCSAYT YAQYSYSPYL RGPFHQLSEQ 720
LTDDPTTNGG THPAPYPLFQ HGGANQVVLV GYLGLRIVPD DKIHVDPNLP PQIPQVKYRT 780
FYWHGWPIAA KSNYTHTTIS RATTIKELDT ANKKYANASI QVVVGSGKSA KTYKLPANGS 840
SIVVANRKGK TVNTLEGNMI QCQPAQSFDI FVPGQFPISI NDGAASTKWQ PEFANNISAV 900
TVTVPASKSK KISGFYFDWA QAPPTNATV LHD EKMDNPT MVLLPVSKDD DKKGSVARVN 960
VTISEPWSAK DKSNFVIGLQ GGNTTNFTFS EPVAAKRYAT LFIQGNQALD KVDIKYKNGT 1020
GATVAEWGIL SDDA 1034

SEQ ID NO: 219      moltype = AA length = 1015
FEATURE            Location/Qualifiers
source              1..1015
                    mol_type = protein
                    organism = Talaromyces piceae

SEQUENCE: 219
ARVDQVLQAR DTHHPSLHSS SSYQTRFDGV TWDQRNWRLQ SRVLDQGHYQ SRGSVANGYL 60
GINVASAGFP FELDTVPDGD VINGWPLFSR RQTFAGLAGF YDRQPTTNST NYGWLNLQYGD 120
ESVISGIPHW SGLVLDLGDG HYLDAITVDS TISDYTTTYD FKAGVLSWDY NWAPRQHGGG 180
NSSPHISYQL FAHKLDINQA VVKLSITPSA SGNASVVNVI DGYSAVRTEF VKSGTDGDAV 240
YSAVSPVGVG NVTAWVYTVL DGDEAFNLSS AQLVTGKPYV YQNDSSIAQS VNVFPRAGET 300
VTITKFGVGA STDAFADPRQ TARDAAALLAK KKGFDLLRS HVSEWAVQVP DGSVDDFTRE 360
DGTLPDDEFI IESAVTAVVN PFYLLQNTVG KNALQVNSA PVNDWSIPVG GLSSDSYAGL 420
IFWDADVMMQ PGLVAAPFES AQRITNYRAA MYRQARANIQ SAFASSQNKT VFSPPGAIYP 480
WTSGRYGNCT GTGPCFDYXY HLNQDIAISL VNQWVVSQDT ETFKNEHFPI YDSIATMYAD 540
VLKKNQSFYT LTNMTDPDEY ANNKDAGGFT MPLIAKTLN ANDPRKQFNM EENSTWNEKA 600
ASVQLIREND VTLEYTTMNN SVAVKQADV LMTFPPLDYA NYSSSSALND LDYYALKQSP 660
DGGPMTYAIF SIVANQVSPS GCSAYTYAQY SYYPYARAPF FQLSEQLLDD YTANGGTHPA 720
YPFLTGHGGA NQVVLYGYLG LRLVPDETLY IDPNLPPQIP QVKYRTFYWR GWPIQAASNY 780
THHTIRRAAT VAPLDTADEK YTDEAITLHV GQQSDGNKYQ LPADGTPVTY ANRRVSSVNT 840
VKGNMIQCPQ QPSTASIQPG QPPIAAVDGA ASTKWQPEFA ANASSLTVEI PSSGKKRKTV 900
SGFFFDWAEA PPTNATVVLH NNPVSEPTLS SFEAGKGVV LRAFDIEISS PYNASTYADE 960
IIPQGNSTN YTFSHVPAP RFATLFPVQGN QALDKVDVEN KNGTGPMVAE WVILE 1015

SEQ ID NO: 220      moltype = AA length = 1022
FEATURE            Location/Qualifiers
source              1..1022
                    mol_type = protein
                    organism = Penicillium sp.

SEQUENCE: 220
LPLEERVDRV LRSYSVSGSL EARHSKYTYQ TQFDGVTWDQ QNWRLESTVL DQGHYSSRGS 60
IANGYIGLNV AGAGPFFELD APVDGDIVNG WPLFSRRQTF AGLSGFYDVQ PTTNGSNYPW 120
LDQYGYDSVI SGIPHWGGLV LDLGNGDYLD ATVDNSTISD YTTTFDYKAG VLSWDYWNTP 180
KNTSPGISYK IFSSKLDINQ AVVQLSITPS ANGASVANV IDGYAAVRTE FVTSGNDSDA 240
LFTAVKPTGI NNVTAWIYAV LDGDDAFDFS SATLVNNKPY INQNDSSIAQ AVDVEFTADS 300
TVTITKFGVA ASTDAFADPQ KTAKDAALAA RIKGFEDLLR SHVSEWAVQM PDDSVDDFSL 360
ADGTLPPDIF IIESSVMMAV NPYLLQNTV GENALRRVND APVNIWSIPV GGLTSDSYAG 420
QIFWDADLWM QPGLVAAPPE SAKRISNYRV AKYPEALANT NTSFAGSQNH TTFSSDAIY 480
SWTSGRYGNC TATGPCWDYE YHLNGDIGIS LVNQWVTSQD TETFQNDLFP IYNSVATLYA 540
DLLKLNGSYT TLNMTDPDEY YANNVDAGGY TMTLISKTLS NANAFRKHFG LDQNSTWTEM 600
AENVLVIREN DVTLEYTTMN NSVAVKQADV VLSTFPPLDYT MNYTTSDAVN LDYYALKQA 660
SDGPGMTYAI FSIVADKYSK SGCSAYTYAQ YSFDPYIRSP FFQFSEQLDD DYTNGGTHP 720
AFFFLTGHGG ANQVVLYGYL GLRLLPDNVL HIDPNLPPQI PSVKYRTFYW HGWPIQANSN 780
YHTTIRRAAT TAPLSTADQ TYSNTSITVQ VGTSGTTSY LRVDGTPPLT KNRRIGSVQT 840
ISGNIAQCQP QSPDSYEPG QYPLSVVDGA SSTKWQPESA ANISSVTVTL GSSYSPGNSS 900
NATASSILGF YFNWAQAPPD NITVVLHNAS IPYLNALSS LSNTTTTSL NITVSNPYDS 960
SADEKIIIVL SSNTTNYTFP SPVPVPRPAT LFIQGNQGLD QTDLEYGNGT GATVAEWAIL 1020
SA 1022

SEQ ID NO: 221      moltype = AA length = 1020
FEATURE            Location/Qualifiers
source              1..1020
                    mol_type = protein
                    organism = Talaromyces aurantiacus

SEQUENCE: 221
LPFQDRVDQV LRSYSAKTL D IRSAKSSSKH GNTYETQFPG VTWDQRNWRL QSTVLDQGHY 60
ESRSGIANGY IGLNVAGAGP FFELDTVPDG DIVINGWPLFS RQTFAGLAG FYDLQPTTNG 120

```

-continued

```

SNFPWLDQYG DDSAISGVPH WGLVLVDLGD GQYLDATVDN STVSDYKTTY DFKAGVLSWD 180
YKWTPKSSNV SFGISYKVA NKLVDNQAUV QLSITPSANG SASVVNVIDG YSAVRTDFVS 240
SGNESDAIYT AVKPLGVSNV TAWIYATLDG DDAFDFFSAT IVNNKPYVHQ NDSSIAQSVN 300
VTFTAGTTVT INKPVGAAST DAFDPDRSTA KEAALAGRRL GYDSSFRAHI SEWAQVMPDD 360
SVDDFTLANG TLPNDTFIIE SAVMAVVNPY YLLQNTVGPV ALRRVNNAPV NDWSISVGG 420
TSDSYAGQVF WDADVWMQPG LVAAPFESAK RITNYRAAIY SQALANAKTA YTSSQNQTSF 480
SSDAIISWT SGRYGNCTAT GPCWDYEHYH NGDIGISLVN QWVASGDNET FKNTHFPIYN 540
SIATLYGDLK KNGSYTYLT NMTDPDEYAN NVDAGGYTMT LISQTLNAN AFRKQFGMDE 600
NTTWTEADN VLLIRENDIT LEYTTMNSV AVKQADVLS TYPLDYTKNY TTSALNDL 660
YYALKQSPDG PGMTYAFISI VANDVSPSGC SAYTYAQSY DPYIRGPFQ FSEQLDDYT 720
ANGGTHPAPF FLTGHGGANQ VVLYGYLGLR LVPDDKLHID PNLPPQIPSV KYRTFYWRGW 780
PIQAASNYTH TTIQRATRVA PLSTADMTYA NKSISVQVQG NTANSTTYSL PVNGSALVIS 840
NRQIGSINTV QCNIAQCKSV QSMNGYQPGQ YPISAVDGAA STKWQPEFAA NVSSLTVDLT 900
SSNASSVSGF YFDWAQAPPV NVTVVLHNST SASLTSSAAQ NGSSTVSLNI TISPNYNASS 960
YDANVIELSS SNTTNYRFPF PVKPKRYATL FVQGSQALDE TDMKAGNGTG ATVAEWAILS 1020

```

```

SEQ ID NO: 222      moltype = AA length = 1016
FEATURE            Location/Qualifiers
source             1..1016
                   mol_type = protein
                   organism = Talaromyces pinophilus

```

```

SEQUENCE: 222
LPFNERVDQV LRSYSPKNLE SRSTKHGNSY QTQFSGVTWD QRNWRLQSTV LDQGHYESRG 60
SIANGYIGLN VAGAGPFPEL DTAVDGDVIN GWPLFSRRQT FAGLAGFYDL QPTTNGSNFP 120
WLSQYGDSSA ISGVPHWGGL ILDLGDGEYL DATVDNSTIS DYTTTYDYKA GLVSWDYKWT 180
PKNSKASFGI NYKIFANKLD VNQAVVQLSI TPSANGSGSV VNVIDGYSAV RTDFVSSGNE 240
SDVIYTAVKP VGVNNTAWI YAAALDGEAF DFSSAELVND KPYVHQNDSS IAQSVNVAF 300
AGTTITINKF VGAASTDAFP DPQSTAREAA MTARRRGFD LFRSHVSEWA QVMPDSDVD 360
FTLANGTLPN DTFIIESAVM AVVNPYYLLQ NTVGANALRR VNNAPVNDWS IPVGGTSDS 420
YAGQIFWAD VWMQPGLVAA FPESAKRITN YRTAKYSQAL ENAKTAYTSS QNQTSSFSSA 480
AIYSWTSRGY GNCATATPCW DYEYHLNGDI GISLVNQWV VSGDNETFKNT HPIYNSIAT 540
LYGDLKKNK SYTYLTNMTD PDEYANNVDA GGYTMTLISQ TSLNANAFRK QFGMDENTTW 600
TEMADNILLI RENDVTLEYT TMNSVAVKQ ADVILSTFPL DYTKNYTSA ALNDLDYYAL 660
KQSPDGPGMT YAFISIVAND VSPSGCSAYT YAQSYDPIY RGPFFQFSEQ LLDDYTANGG 720
THPAPFPLTG HGGANQVVLY GYLGLRLLPD DMLHIDPNLP PQIPSVKYRT FYWRGWPIQA 780
ASNYTHTTIQ RATSVAPLST ADPAYANTSI SVSVGQNTAN STTYSLPVNG SAIVVPNRQI 840
GSINTVAGNI AQCVSVLST AYQPGQYPI AVDGAASTKW QPEFAANVSS LTVDLTSSNA 900
SSVSGFYFDW AQAPPTNITV LLHNSSSAAL TSSSTHGGSS SVSLNITISN PYDASSYDAN 960
VIALSSSNTT NYTFSAPVAK PRYATLFEVQG NQALDETDK AGNGTGATVA EWAILS 1016

```

```

SEQ ID NO: 223      moltype = AA length = 1069
FEATURE            Location/Qualifiers
source             1..1069
                   mol_type = protein
                   organism = Talaromyces leycettanus

```

```

SEQUENCE: 223
TSANARINRC VKKHAGGKTP SGPSNNTYQT RPPGVTDQD NWCLSTTTLD QGHYESRGSV 60
ANGYLGINVA SVGPFFPEFD PVDGDVINGW PLFDRRMSFA TISGFWDQOP TTNGSNFPWL 120
YQYGGESVIS GVPHWGLIL DLGDNLYLDA TVDSRTISGF STTYDFKSGV LWSYQWTPA 180
GNMGSYNITY RLFAHKLYVN QAVVDMEVVS STEAKATVVN VIDGASAVRT DFVESGQDDG 240
AIYTAVRPWG IANTYAIYA NITGSDNVDM RSRALVTNKP VYVNGASSIT QAVNVHFTPG 300
KSVRITIKFV GASSDAFENP QQIAKQACST AQANGYVKS LSHVAEASV MPDSDVDFT 360
FPSNGTLPAD EYIIESQIIS VANTYYLLQN TVGKNAINAS SSTEINKDSI AVGGTSESSEY 420
AGMIFWADV WMPQGLVASH PEAAQRITNY RVAKYPOAKA NVATAYQSSK NQTNFSPDAA 480
VYSWTSARGY NCTATGPCW DEYHLNGDIG LSIINQYVAS GDTQTFKEKL FPVFDVATL 540
YSNIVQKNGS SWTLTNMTD DEYANQVDAG GYTMLPIAQT LLYANSFRQ FGLTNDTWN 600
ETADQDLVIR ENGVTLEFMT MNGSAVVKQA DVVLDYPLG YTHNYGPTDA LNDLDYYANR 660
QSPDGPAMTW AIFSVVANQI SPSGCSAYT AQYAFSPYAR APFYQLSEQL IDDASLNGGT 720
HPAYPFLTGH GGALQVNLFG YLGFRLPDN VIHIDPNLPP QIPHITYRTF YWRGWPIATA 780
STYTHTLTSL AWWVSSLDSA DPKFANASIP VHVGLSENVY VYRLPVNGTL TVPNRMVSGK 840
NTLAGNMVQC RPVQSMGDYQ PGQFPISVVD GASSTKWQPL YSANVSSVTV TLSSSAVGKS 900
VNGFYFDWAQ NPPVNAAVVF HNSSFANQNP TTFSDNPSA SGNLYSVSV LKDIQLSDPY 960
DPATTDLDVI AIPKGNNTNV TLSSPVPAAR YATLFIQGNQ ANSPA EVAAK NGTGATVAEW 1020
AILGQEVQNN GYGQIEARR LDVRGAAALS GMGSFTQRRK RKMILPRFD 1069

```

```

SEQ ID NO: 224      moltype = AA length = 1016
FEATURE            Location/Qualifiers
source             1..1016
                   mol_type = protein
                   organism = Talaromyces variabilis

```

```

SEQUENCE: 224
ALSPGARVNO VLRAHNSLPP SLTDNSTGSS YKTRFDGVTW DQRNWRLQSR VLDQGHYEAR 60
GSVANGYIGI NVASAGPFPE LDTVPVDGDI NGWPLFSRRQ TFAGLAGFYD LQPTTNSNTY 120
GWLNQYGDSE VISGIPHWAG LVLDLGNGDY LDATVDNSTI SDYTTYDYKA AGILSWDYKW 180
TPRQNSSSFR ISYQLFANKL DINQAVVKLS ITPSKSGNAS VNVIDGYSA VRTDFVKSGS 240
DGNAIYTAVS PVGVSNTVAV VYAVLDGDKA FDLSSPSRVT GKPYIHQNES SIAQAVNVEF 300
SAGKTVTIK FVGAASDAF SNPQKKAKTA ALDGKKKGFE DLLRSHVSEW AQVMPDSDVD 360
DFTLANGTLP DGSFIIQAV TAVVNPYYLL QNTVGKNALQ RVNNAPVNDW SIPVGLSSD 420
SYAGMIFWDA DVWMQPGLV APEPESAKRIT NYRAAKYQQA IANVKTAISS SQNETVFSNP 480

```

-continued

AAIFPWTSGR	YGNCTGTGPC	WDYEHNLNGD	IAISLVNQWV	VSGDTETFPQ	EHFPIYNSIA	540
TVYGDLLKKN	GSYYTLTNMT	DPDEYANNKD	AGGYTMPLIA	NLIKANEFR	RQFGMAENAT	600
WNEMAENVLI	IRENDVLTLEY	TTMNNSVAVK	QADVVLRTFP	LDYTTNYS	SSS AALNDLDYYA	660
LKQSPDGPQM	TYAIFSIIVAN	EVSPSGCSAY	TYAQYSYDPY	ARAPFFQLSE	QLIDDYTTNG	720
GTHPAYPFLT	GHGGANQVVL	YGYLGLRLVA	DEILHIDPNL	PPQIPQVITYR	TFYWRGWPIQ	780
AASNTHYTTI	HRATTVAPLD	SAEKKYAKSA	ISVQVQQQAN	STTYKLPAAG	TPLTVANRKY	840
GSTNTIKGNI	AQCQSVQSV	SYQPGQYALA	AVDGAASTKW	QPEFAANVSS	LTVSIPSGKT	900
SVSGFYFDWA	QAPPSNATV	FHNNSVSNPT	FSSFGSRKGA	RITHIDVKLS	NPYNASSNAD	960
ATVIPSNTT	NFTFSNPVPA	PRFATLFEVQ	NQGLDKVDVQ	NGNGTGATVA	EWAIL	1016

SEQ ID NO: 225 moltype = AA length = 1050
 FEATURE Location/Qualifiers
 source 1..1050
 mol_type = protein
 organism = *Aspergillus niger*

SEQUENCE: 225

LPGKNARISA	SLKRHAGRDV	PQTALNSTNV	YQTKFSGVTW	DEHLLTTT	TPDQGHYQSR	60
GSVANGYLG	IVANIGPFFE	LDEPVNGDVI	NGWPLYSRQ	SFATISGFWD	RQAHNGSNF	120
PWLSQYGGDS	VISGVPHWSG	LILDGLDDTY	LDATVDNRTI	SNFKSTYDFK	SGVLSWSTW	180
TPQGNKGSYA	ITYRLFAHKL	YVNRVAVDME	ITPLTNGNAT	VNVLDGYAA	VRTDFVASGQ	240
EEGAIFSAVR	PWGVNNVTAY	VYATLDGSDS	VDLSSRRIVT	DKPYVSTNSS	SVQAVDVMF	300
TANETVRIK	FVGGATTDF	LATQETAKAA	CLAGLADGYV	KSLQSHVGEW	ATIMHDHSD	360
RFTDPATGKL	PEDSHIVDSA	IIAVTNTYYL	LQNTAGTNAI	VAAGGIPVNV	DSCAPGGLTS	420
DSYGGQIFWD	ADLWMQPLV	ASHPESAQR	TNYRIALHYQ	AQANIEATFT	GSKNQTSSFS	480
SAAIYPWTSG	RFGNCTATGP	CWDYQYHLNG	DIGLAMINQW	VASGDTAWFK	NYLFPYDAA	540
ATLYSELVER	NGSSTLTNM	TDPDEYANSI	NAGGYTMPLI	AETLQANANKL	RKQFGLPNE	600
TWDEIAEDVL	ILRENGVTLE	YTMNGSAVV	KQADIVLNTF	PLTYESDNYT	ATNSLTDLDY	660
YANKQSADGP	AMTYAIFAIV	ASDVSPSGCS	ARTYHQYSYA	PYARGPWYQL	SEQMIDDASI	720
NGGTHPAPFP	LTGHGGANQV	ALYGYLGLRL	HPDDTIYIDP	NLPPQIPHIT	YRTFYWHGWP	780
ISAWSNYTH	TIQRDSSLAP	LASADLLFSN	VSIVKQVQGS	TASADEATII	YLPISGALT	840
PNRMIGSVNT	TPGNQVQCHP	VYSPDAYEPG	QFPISAVDGA	TSTKWQST	DLTSLVTLS	900
TTAEAGAEV	SGFYFDWSQA	PPENLTIVIF	DSPIGNPSTV	FAAAGSNSTG	YRVITSMNI	960
VQSKPYNAIS	AEELNVVSI	TANTTTITLD	APVQKARYAT	LLIAGNQANE	TAGATVAEWV	1020
ILQNSTSSS	SAQAKRKMSA	RSKATLAQLS				1050

SEQ ID NO: 226 moltype = AA length = 1056
 FEATURE Location/Qualifiers
 source 1..1056
 mol_type = protein
 organism = *Trichoderma reesei*

SEQUENCE: 226

TTLVDRVTKC	LSRHDGSDAE	SHFSKNVYKT	DFAGVTWDED	NWLLSTTQLK	QGAPEARGSV	60
ANGYLGINVA	SVGPFFVEDT	EEDGDVISGW	PLFSRRQSFA	TVAGFWDAQP	QMNGTNFPWL	120
SOYGSDDAIS	GIPHWGLVL	DLGGGTYLDA	TVSNKTISHF	RSTYDYKAGV	LSWSYKWTPK	180
GNKGSFDISY	RLFANKLHV	QAVVDMQVTA	SKNVQASIVN	VLDGFAAVRT	DFVESGEDGS	240
ATFAAVRPN	G	ANVTAYVYA	DITGSGGVNL	SSRKIVHNKP	YVHANASSIA	300
RTVRVTKFVG	AASSDAFKNP	KQVAKKAAAA	GLSNGYTKSL	KAHVEEWATV	MPRESSVDSFA	360
DPKTGKLPA	SHIVDSAI	IA VNTTYLLQN	TVGKNGIKAV	DGAPVNVDSI	SVGGLTSDSY	420
AGQIFWDADL	WMQPGLVAAH	PEAAERITNY	RLARYGQAKE	NVKTAYAGSQ	NETFFSASAA	480
VFPWTSGRYG	NCTATGPCWD	YEHYHLNGDIG	ISLVNQWVVN	GDTKDFEKNL	FPVYDVAQL	540
YGNLLRPNT	SWTLTNMTDP	DEYANHVDAG	GYTMPLIAET	LQKANSFRQ	FGIEQNKTN	600
DMASNVLVLR	ENGVTLEFTA	MNGTAVVKQA	DVIMLTYP	LS YGTNYSAQDA	LNDLDYYANK	660
QSPDGPAMTY	AFFSIVANEI	SPSGCSAYTY	AQNAFKPYVR	APFYQISEQL	IDDASVNGGT	720
HPAYPFLTH	GGAHQVVLFG	YLGRLVLPDD	VIHIEPNLFP	QIPYLRRTF	YWRGWPIAW	780
SNYTHTL	LSR AAGVAALEGA	DQRFAKPI	THAGPEQDPT	AYRLPVKGSV	VIPNKQIGSQ	840
QTYAGNLVQC	HAASSPNQYV	PGQFPAAVD	GATSTKWQPA	SADKVSSITV	SLDKEDVGS	900
VSGFHFDAQ	APPVNAVIF	HDEALADPAT	ALASAHKHNS	KYTTVTSLTN	IELSDPYVST	960
KDLNAIAIPI	GNTTNVTL	SH PVAASRYASL	LIVGNQGLDP	VDVKA	NGTG ATVAEWAIFG	1020
HGKEHSGKPS	SHSKRRLNVR	TAATLSNPRS	FMRRL			1056

SEQ ID NO: 227 moltype = AA length = 19
 FEATURE Location/Qualifiers
 source 1..19
 mol_type = protein
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 227

MLSLKTLTCT	LLTVSSVLA	19
------------	-----------	----

SEQ ID NO: 228 moltype = DNA length = 185
 FEATURE Location/Qualifiers
 source 1..185
 mol_type = genomic DNA
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 228

agtgccttta	actaagaatt	attagtcctt	tctgcctatt	ttttcatcat	agtttagaac	60
actttatatt	aacgaatagt	ttagaatct	atttaggtt	aaaaattgat	acagttttat	120
aagttacttt	ttcaagact	cgtgctgtct	attgcataat	gcactggaag	gggaaaaaaa	180
agggtg						185

-continued

SEQ ID NO: 229 moltype = AA length = 555
 FEATURE Location/Qualifiers
 source 1..555
 mol_type = protein
 organism = *Pycnoporus sanguineus* glucoamylase

SEQUENCE: 229

QSSAVDAYVA	SESPIAKQGV	LNNIGPNGSK	AHGAKAGIVV	ASPSTENPDY	LYTWTRDSSL	60
VFKLLIDQFT	SGDDTSLRGL	IDDFTSAEAI	LQQVSNPSGT	VSTGGLGEPK	FNIDETAFTG	120
AWGRPQDGP	ALRATSIIRY	ANWLLDNGNT	TYVSNLWFPV	IQLDLDFVAD	NWNQSTFDLW	180
EEVDSSSFPT	TAVQHRALRE	GATFASRIGQ	SSVVSgyTTQ	ADNLLCFLQS	YWNPSGGYVT	240
ANTGGGRSGK	DSNTVLTSIH	TFDPAAGCDA	ATFQPCSDKA	LSNLKVYVDA	FRSIYTTINNG	300
IASNAAVATG	RYPEDSYMGG	NPWYLTTSAV	AEQLYDALYV	WDQLGGLNVT	STSLAFFQQF	360
ASGLSTGTYT	ASSSTYATLT	SAIRSFADGF	LAINAKYTPA	DGGLAEQYSR	NDGTPLSAVD	420
LTWSYAAALT	AFAAREGKTY	GSWGAAGLTV	PASCSGGGGA	TVAVTFNVQA	TTVFGENIYI	480
TGSVAALQNW	SPDNALILSA	ANYPTWSITV	NLPANTVVQY	KYIRKFNQGV	TWESDPNNQI	540
TFPSGGSFTQ	NDVWR					555

SEQ ID NO: 230 moltype = AA length = 599
 FEATURE Location/Qualifiers
 source 1..599
 mol_type = protein
 organism = *Trichoderma reesei*

SEQUENCE: 230

SVDDFISTET	PIALNNLLCN	VGPDGCRAFG	TSAGAVIASP	STIDPDYYIM	WTRDSALVFK	60
NLIDRFETET	DAGLQRRIEQ	YITAQVTLQG	LSNPSGSLAD	GSGLGEPKFE	LTLPKPTGNW	120
GRPQRDGPAL	RAIALIGYSK	WLINNYYQST	VSNVIWPIVR	NDLNYVAQYW	NQTGFDLWEE	180
VNGSSFFTTA	NQHRALVEGA	TLAATLGQSG	SAYSSVAPQV	LCFLQRFWVS	SGGYVDSNIN	240
TNEGRTGKDV	NSVLTSIHTE	DPNLGCDAGT	FQPCSDKALS	NLKVVDSEFR	SIYGVNKGIP	300
AGAAVAIGRY	AEDVYNGNP	WYLATFAAAE	QLYDAIYVWK	KTGSITVTAT	SLAFFQELVP	360
GVTAGTYSSS	SSTFTNIINA	VSTYADGFLS	EAAKYVPADG	SLAEQFDRNS	GTPLSAVHLT	420
WSYASFLTAA	ARRAGIVPPS	WANSSASTIP	STCSGASVVG	SYSRPTATSF	PPSQTPKPGV	480
PSGTPYTPLP	CATPTSAVAT	FHELVSTQFG	HTVKVAGNAA	ALGNWTSAA	VALDAVNRYD	540
NHPLWIGTVN	LEAGDVVEYK	YIIVGQDGSV	TWESDPNHTY	TVPACVCTQ	VVKEDTWQS	599

SEQ ID NO: 231 moltype = AA length = 633
 FEATURE Location/Qualifiers
 source 1..633
 mol_type = protein
 organism = *Bacillus amyloliquefaciens*

SEQUENCE: 231

GPAAANAETA	NKSNKVTAAS	VKNGTILHAW	NWSPNTLTQN	MKDIRDAGYA	AIQTSPINQV	60
KEGNQGDKSM	RNWYWLQPT	SYQIGNRYLG	TEQEFKDMCA	AAEKYGVKVI	VDAVINHTTS	120
DYGAISDEIK	RIPNWTGNT	QIKNWSDRWD	VTQNSLLGLY	DWNTQNTVEQ	VYLKRFLERA	180
LNDGADGFRY	DAAKHIELPD	DGNYGSQFWP	NITNTSAEFQ	YGEILQDSAS	RDTAYANYMN	240
VTASNYGHSI	RSALKNRNLS	VSNISHYASD	VSADKLVTWV	ESHDTYANDD	EESTWMSDDD	300
IRLGWAVIGS	RSGSTPLFFS	RPEGGGNGVR	FPGKSQIGDR	GSALFKDQAI	TAVNTFHNVM	360
AGQPEELSNP	NGNNQVFMNQ	RSGKGVVLAN	AGSSSVTINT	SAKLDPGRYD	NRAGAGSFQV	420
ANGKLTGTIN	ARSAAVLYPD	DIGNAPHVFL	ENYQTGAHVS	FNDQLTVTLR	ANAKTTKAVY	480
QINNQQQTAF	KDGRRLTIGK	GDPIGTNYNI	KLGTNGEGA	ARTQYTFVK	KDPSQTNIIIG	540
YQNPDPHWGV	NAYIYKHDGG	RAIELTGSWP	GKAMTKNANG	MYTLTLPENT	DTANAKVIFN	600
NGSAQVPGQN	QPGFDYVQNG	LYNNSGLNGY	LPH			633

SEQ ID NO: 232 moltype = DNA length = 700
 FEATURE Location/Qualifiers
 source 1..700
 mol_type = genomic DNA
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 232

cacccatgaa	ccacacgggtt	agtccaaaag	gggcagttca	gattccagat	gcgggaatta	60
gcttgctgcc	accctcacct	cactaacgct	gcgggtgtcg	gatacttcat	gctatttata	120
gacgcgcgtg	tcggaatcag	cacgcgcgaag	aaccaaattg	gaaaatcgga	atgggtccag	180
aactgctttg	agtgtcggt	attggcgtct	gatttccgtt	ttgggaatcc	ttgcccgcgc	240
gcccctctca	aaactccgca	caagtcccag	aaagcgggaa	agaaataaaa	cgccaccaaa	300
aaaaaaaaaa	taaaagccaa	tcctcgaaag	gtgggtggta	ggccctggat	tatcccgtac	360
aagtatttct	caggagtaaa	aaaaccgttt	gttttggaa	ttcccatctt	gcggccacct	420
acgcgcgtat	ctttgcaaca	actatctgct	ataactcagc	aaattttgca	tattcgtgtt	480
gcagttattg	gataatggga	gtcttacttc	caacataacg	gcagaaagaa	atgtgagaaa	540
atttgcatc	ctttgctctc	gttcaagtat	ataaagtcgg	catgcttgat	aatctttctt	600
tcacatctac	attgttctaa	ttattcttat	tctcctttat	cttttcctaa	cataccaaga	660
aattaatctt	ctgtcattcg	cttaaacact	atatcaataa			700

SEQ ID NO: 233 moltype = DNA length = 240
 FEATURE Location/Qualifiers
 source 1..240
 mol_type = genomic DNA
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 233

tcagtactga	caataaaaag	attcttgttt	tcaagaactt	gtcatttgta	tagttttttt	60
atattgtagt	tggtctatct	taatcaaatg	ttagcgtgat	ttatattttt	tttcgcctcg	120

-continued

```

acatcatctg cccagatgcg aagttaagt cgcagaaagt aatatcatgc gtcaatcgta 180
tgtgaatgct ggtcgctata ctgctgtcga ttcgatacta acgccgccat ccagtgctga 240

```

```

SEQ ID NO: 234      moltype = AA  length = 18
FEATURE            Location/Qualifiers
source              1..18
                    mol_type = protein
                    organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 234
MQLLRCSIF SVIASVLA                                     18

```

```

SEQ ID NO: 235      moltype = AA  length = 274
FEATURE            Location/Qualifiers
source              1..274
                    mol_type = protein
                    organism = Thermomyces lanuginosus

```

```

SEQUENCE: 235
EVSQDLFNQF NLFAQYSAAA YCGKNNDAPA GTNITCTGNA CPEVEKADAT FLYSFEDSGV 60
GDVTFGLALD NTKNLIVLSF RGRSRIENWI ANLNFWLKKI NDICSGCRGH DGFTSSWRSV 120
ADTLRQKVED AVREHPDYRV VFTGHSLGGA LATVAGADLR GNGYDIDVFS YGAPRVGNRA 180
FAEFLTVQTG GTLYRITHTN DIVPRLPPRE FGYSHSSPEY WIKSGTLVPV TRNDIVKIEG 240
IDATGNNQP  NIPDIPAHW  YFQATDACNA GGFS                               274

```

```

SEQ ID NO: 236      moltype = AA  length = 277
FEATURE            Location/Qualifiers
source              1..277
                    mol_type = protein
                    organism = Thermomyces lanuginosus

```

```

SEQUENCE: 236
APAPVLRDVS SSSVLSELDL FAQYSAAAYC SSNIGSPGTK LTCVSGNCPR VEAADTETLI 60
EPNESSSFGD VTGYIAVDRT NSLLVLAFRG SSTVSNWEAD LDFPLTDASS LSCGCEIHSG 120
FWAAWQTVQA SITSTLESAI ASYPGYTLVF TGHSYGAALA AIAATTLRNA GYTIQLYDYG 180
QRLGNLALA  QYITAQTQGA NYRVTHTDDI VPKLPPELFG YHHFSPEYWI TSGDNVTVT 240
SDVQVVTGID STAGNDGTL  DSTSAHDWYI VYIDGCD                               277

```

```

SEQ ID NO: 237      moltype = AA  length = 594
FEATURE            Location/Qualifiers
source              1..594
                    mol_type = protein
                    organism = Penicillium emersonii

```

```

SEQUENCE: 237
SAPYDKRDLA QEIWDDIKNA VDCAGCQVVL TALKGVADLG TTALVDVLTE VCNISGKEDS 60
DVCSGLISRE GPVLDYVLQH LDIGSHTSQV ICASAFGLCQ YPEVRPYNLT FPKPKPNTTR 120
PEPSGESPIQ VVHFSATHVD LSYETGSNYN CTKPICCRPY TAEDAPGNTT TPCGPYGNTK 180
CDAPLSLEES MFAAIKALNP QPAFSIYTG  VVAHDIWLDV QNEVIEDLNA TYDRMAGLGL 240
VYAAIGNHDT APVNDLPTSN IPSEYSANWT YEALSYDFTM LTQSASAQTA ANYGSYSAIY 300
PGSYGTDLRV ISYNSIFYVY DNFWAYQDPM EFDPDGQLAW LINELQEAET AGQRVWIIAH 360
VPTGTSDFHF DYSHYFDQIV QRYEATIAAL FYGHTHIDQF QISYSNYSNR APDTATAIGY 420
IMPSLTPTSG PPTFRVYDVD PKTFVAVLDFT NYIANISDPA FQSGPSWQKY YSAKETYGSL 480
LSPPVTDPTA ELTPAFWHNV TFAFEQDNAT FQBYWARQTR GYDVSSCTGS CITQAICGLR 540
AGDAQYNCVT PTPGFNPAKR DTSNPKQALS HVEKCEGSL  LGLLRMVAD  SKSS          594

```

```

SEQ ID NO: 238      moltype = AA  length = 259
FEATURE            Location/Qualifiers
source              1..259
                    mol_type = protein
                    organism = Bacillus thuringiensis

```

```

SEQUENCE: 238
HENDGGSKIK IHRWSAEDK HKEGVNSHLW IVNRAIDIMS RNTTLVKQDR VAQLNEWRT 60
LENGIYAADY ENPYDNDSTF ASHFYDPDNG KTYIPFAKQA KETGAKYFKL AGESYKNKDM 120
KQAFYVLGLS LHYLGDVNQP MHAANFTNLS YPQGFHSKYE NFVDTIKDNV KVTGNGYWN 180
WKGTPNPDWI HGAADVAKQD YSGIVNDNTK DWFVKAASVQ EYADKWRAEV TPMTGKRLMD 240
AQRVTAGYIQ LWFDTYGDR                                     259

```

```

SEQ ID NO: 239      moltype = AA  length = 297
FEATURE            Location/Qualifiers
source              1..297
                    mol_type = protein
                    organism = Pseudomonas sp.

```

```

SEQUENCE: 239
QESPAFIDPA SWNTFPNGIA QVACHNCYEK QYANTFSSVL DSVRTLELDF WDQRDAVSGG 60
SPHHWFVRHN PGTLFQSGND NNCTGDGTGK NDLEACLNDV KNWSDKHPGH FPI TLILDKK 120
QGWSKESSGR TPKDFDELVA RVPGQKLFTP QDLATHISGS AGALQGNLKG KSWPTANDLQ 180
GKVLVLVNH  ENQKLSQYAE ARTSKAKVFI SPVTNGQNDI SGKVSMSQSS SSGYVAMNNM 240
GKGDKSWAKQ AFAYSHIGRV WGDDEVSFQA HINQKINLSA YYRFAAQSGAG GYRIRPF    297

```

```

SEQ ID NO: 240      moltype = AA  length = 625
FEATURE            Location/Qualifiers
source              1..625

```

-continued

```

mol_type = protein
organism = Kionochaeta sp.

SEQUENCE: 240
AVNPADVLSV VEKRVDPASG LEVRSIWDTI WNDIKSAADC TACEAVLTLL KGVAAFGDNF 60
FVEVLTEICD LSGAEDDDVC SGVLSLEGPI IANDIRKMSI GSKTSELFICI TFLGLCSYPA 120
VDAFTVPFPT AKSAATRPVS SGKDPIYVVH YSDIHIDPFY VAGSASNCTK PICCRDYTSA 180
SSPGNNNSPA GPYGDHNCV PISLEDSMYA AIKKLVDAF PGIRTGDIVD HAVWNTSESQ 240
NIIDMNDAYT RMKNSGMLPT IFATAGNHEA SPVNSFPPPA IGNESQWVYD TLASDWSQWI 300
GTSGASSVES IGAYSVQYGS TKLRVISLNT NMYIENFYI YEPTMEQDPA GQFAWLVSSEL 360
SAAEAAGERV WIIGHMPLGL SDAFHDPSNY FDQIVNRYEA TIAAMFFGHT HEDHFQISYS 420
DYNARTAANA RAVSYIMPSL TPTSGHPTFR VYTVDPETFG VLDATTYYAD MSQPTYQTAG 480
PAWSVYYSAK AAYGGLVDPV VAADDAEAEL TPAFWHNVTA ALAADPASFD AYYARKTRGW 540
DVAACAGACA AAEVCALRAA RAQDNQVPT PGVHFSKRAD EGTLAHRDE CGVSVARNSL 600
SSLVVQREAL EHLEGRLEK RRMVA 625

SEQ ID NO: 241      moltype = AA length = 613
FEATURE            Location/Qualifiers
source              1..613
                    mol_type = protein
                    organism = Mariannaea pinicola

SEQUENCE: 241
QEVTHDLAGI KRSLESRDWV EDLWDKFESD ATCAGCESLV LVLKGLAAIS DQAFIDVLQE 60
ICKISGAEDD DVCDSIQLE GPVIASGLRS MAIGSRTSKE FCTTFLGLCA YPAVQQWVSP 120
FSSSKSKTR PSSSGKDIK VVHYSDIHD PLYVGGNSN CTKPICRSY TKADQPGNNK 180
YPAGPNGDHN CDSPLSEKS MYNAIKEIV DAAFTIFTGD IVDHAVWNTS QSYNTEQITN 240
AYGLMSDNLG TIYGTAGNHE AHPANAFQPN SVGNVSQWVY DLLSGLWSQW ISTEAKADSE 300
KLGAYSTKYP GGNRIISLN TNMYRENRYW LYRKTIQDP SNQISWLVE LEAAETAGER 360
VYIIGHMPLG DSNFHDQSN YLDQVINRYS ATISAMFFGH THDDQFQISY SNWSNRNFSN 420
ALVTSYIGPS LPTTAGMPAF RVYDVPVTF GILDSTTYIA DMTDSAFQTT GPVWKYYSYA 480
KEVYGSLLSP AVTDSSAELT AAFWHNVTTL FEADNTAFEA FLSRKSQGWK SESCTGTCKA 540
NEICQLRAAR SENNCYTPSL GISFNKRSLN PVEERDECGI SVTRATVSAM GVRKDVLRLL 600
KKRFIEKAGE VRG 613

SEQ ID NO: 242      moltype = AA length = 259
FEATURE            Location/Qualifiers
source              1..259
                    mol_type = protein
                    organism = Fictibacillus macauensis

SEQUENCE: 242
HENEGGNKVR VIQWSAEDP HHEDTNTHLW IVRHAMEIMA NNKDVVKPGE VEQLKQWQSD 60
LEQGIYDADH ANPYDYNATF ASHFYDPDTG KSYIPLAHA KTTSVKYFKR AGEAYQKGDH 120
KQAFYNLGLA LHYIGDLNQP MHAANFTNLS YPQGFHSKYE NYVDSFKEDY AVKDGEYWH 180
WKGTPNEDWL HGTAVAAKKD YPDIVNDTTK AWFVKAASVN SYAAKWRAAV VPATGKRLTE 240
AQRILAGYMQ LWFDTYVVK 259

SEQ ID NO: 243      moltype = DNA length = 250
FEATURE            Location/Qualifiers
source              1..250
                    mol_type = genomic DNA
                    organism = Saccharomyces cerevisiae

SEQUENCE: 243
acagaagacg ggagacacta gcacacaact ttaccaggca aggtatttga cgctagcatg 60
tgtccaattc agtgtcatct atgatttttt gtagtaggat ataatatat acagcgctcc 120
aaatagtgcg gttgccccaa aaacaccacg gaacctcatc tgttctcgta ctttgttgtg 180
acaaagtagc tcactgcctt attatcacat ttctattatg caacgcttcg gaaaatacga 240
tgttgaaaaa 250

SEQ ID NO: 244      moltype = DNA length = 822
FEATURE            Location/Qualifiers
misc_feature        1..822
                    note = ARTIFICIAL SYNTHETIC DNA
source              1..822
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 244
gaggttttctc aggatctttt caatcagttc aatttggttg ctcagtactc cgcagcagct 60
tactgtggca agaataacga tgccccagcc ggaaccaaca tcacgtgtac aggtaacgct 120
tgtctgagg ttgaaaaggg agacgctacc ttctataca gtttggaaga tagtgggggt 180
ggagacgtga ctggattcct ggcaactagac aatacgaaca aattgattgt tttgtcttc 240
agaggctctc gttctatcga aaactggatt gcaaacctga acttttggtt gaagaagatt 300
aacgatatct gttctggttg tagaggacac gatggtttta catctctctg gagatccgtg 360
gccgacacct tgagacaaaa ggttgaagat gccgttcgtg aacatccaga ttacagggtc 420
gtgtttacgg gtcatagttt aggtgtgtgt ttggccacag ttgctgtgac agaccttaga 480
ggtaatgggt atcatatcga tctcttttcc tatggagcac caaggggttg taatagagct 540
ttcgtcgagt tcttgacagt ccaaaactgt gggaccttgt atagaatcac tcacaccaat 600
gatattgtgc ctgactacc cccaagagaa ttgggtatt ctcattcttc tcccaggtat 660
tggattaaga gtggaacatt agtaccggt accaggaacg acattgtcaa gattgaaggg 720
attgacgcaa ctggtggcaa caatcaaccg aatatccagc atatccagc acatcttttg 780
tactttcaag ctacagacgc ttgtaacgct ggtggattta gc 822

```


-continued

```

SEQ ID NO: 245          moltype = DNA length = 831
FEATURE                Location/Qualifiers
misc_feature           1..831
                        note = ARTIFICIAL SYNTHETIC DNA
source                 1..831
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 245
gctccagctc cagttcttag gagagatgta tcttcctctg tattgtctga gctggactta 60
ttcgctcagt actctgcgcg tgcataattgc tcttccaata tcggttctcc gggtagaag 120
cttacttgta gtgtcggtaa ttgtccccgt gttgaagctg ccgatacaga gacgttgata 180
gagtttaacg aatcatcttc ttttgggtgat gtgactggct atatcgctgt ggataggacc 240
aattcactac tgggtgtggc atttaggggc tcttccacag tgtcaaatg ggaagcagat 300
ctagattttc cattaacaga tgcctctagt ctgtgcagtg gctgtgaaat tcaactctgt 360
ttttgggctg cctggcaaac cgtacaggca tccatcacta gcacattgga gtcagctatt 420
gcttcttacc ctggttatatc ttgtgtcttc acaggtcatt catacgggtg cgctttggca 480
gcaatagcag caactacatt gagaaatgct ggttacacaa tccagttgta cgactatggc 540
cagccgaggt taggtaatct tgctctggcc cagtacatta cggcacaaac acagggtgcc 600
aactatctgt ttactcatalc cgaatgacat gtacccaaac taccocctga gttgtttggc 660
tatcaccact tctcccagc gtattggata acgtcaggcg ataagtgtac agttacaact 720
agcgatgtgc aagtagtcac tgggaattgc agtacagctg gaaatgatgg gactttgctt 780
gattctacaa gtgcccata gtgtacattg atgggtgcga t 831

SEQ ID NO: 246          moltype = DNA length = 1782
FEATURE                Location/Qualifiers
misc_feature           1..1782
                        note = ARTIFICIAL SYNTHETIC DNA
source                 1..1782
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 246
tctgctccgt atgacaaaag agacttggct caggagattt gggacgacat taagaatgcc 60
gttgaactgc ctgggtgtca agttgttttg acagctttga aaggtgttgc cgatctgggc 120
acaacagctt tagttgatgt cttaactgag gtgtgcaata tctctggtta ggaggattca 180
gatgtctgct ctggtatcat tagtagagag ggtcctgttc tagactacgt gcttcaacat 240
ctagacatcg ggtctcatalc atcacaaagt atttgtgctt cagcatttgg tttatgcaa 300
tatcccgaag ttgagacccta taacttgact tttcctaagc ccaaaccaaa cacaaccaga 360
cctgaacctt ccggtgaatc tccaattcaa gtatgacat tctccgatac gcacgttgac 420
ttaagttacg agactggttc caattacaat tgcactaagc caatttgttg tagaccttac 480
actgcccgaag atgcacctgg gaacactact actccatgtg gtccgtacgg taataccaag 540
tgtgatgccc cattatcctt ggaagagtct atgtttgctg ccatacaaac cttaaacctt 600
caaccagcat tttctatcta cacaggagat gttgtagcac atgacatttg gttagttgat 660
caaaacgagg ttatogaaga tttgaacgca acgtacgata gaatggctgg tctgggttta 720
gtctatgccg ctattggtta tcatgataca gcaccggta acgacctgcc gacgtcaaac 780
attccatccg agtatccagc aaattggact tatgaagctt tgtcttacga ctttacaatg 840
ttgactcaaa gcgccagtgc tcaaaactgt gcaaaactatg gttcctattc agccatttac 900
cctggatctt acggtacaga cctaagagtc atttcataca actctatctt ctattacgtt 960
gacaaacttt ggggttatca agatccaatg gaatttgacc cagatggcca gctagcatgg 1020
ttgataaacg agttgcaaga agctgaaact gctggccaaa gagtctggat catagcacat 1080
gttctctacg gtacttccga tcacttccat gactattctc actactttga ccaaatagtc 1140
caaaagatatg aagccacgat tgctgccttg ttctatggtc ataccacat agaccagttt 1200
cagattagtt actcaacta ttcacacaga gcatttgata ccgcaactgc catcggtcac 1260
attatgcctt ctttaacacc cactagcggc ccacctacct ttagagtta cgacgttgat 1320
ccaaagacgt tgcagtggtt aatttcacc aattacattg ccaacatttc cgatcctgca 1380
tttcaatcag gtccatcatg gcagaaatac tattccgcca aggagacctc cggctcacta 1440
ttgtctccac ctgtaaccga ccctactgct gaattgacgc ctgcattctg gcataatgtc 1500
accgtggcat ttgagcaaga caacgcaaca tttcaagagt attgggctcg tcaaacccagg 1560
ggatacagatg tctcatcatg tacaggtagt tgcataaacc aagcaaatatg cggattgcgt 1620
gctggcgacg cacaatacaa ttgtgttact cctactcccg gtttcaactt cgcaaacgct 1680
gacactagta accccaaaca agccttgta catgttgaga aatgtgaagg atccggtctt 1740
ctaggcttgt taaggcgtat ggttgacagc tctaaatctt ct 1782

SEQ ID NO: 247          moltype = DNA length = 777
FEATURE                Location/Qualifiers
misc_feature           1..777
                        note = ARTIFICIAL SYNTHETIC DNA
source                 1..777
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 247
catgagaatg atgggggttc caagattaag atcatacata gatgggtctgc cgaagataaa 60
cataaagaag ggggttaactc tcaacttatg atagtcaaca gagcaatcga cattatgtca 120
agaaatacaa ctttagtgaa caagacaga gtccgccaac taaatgaatg gagaactgag 180
ctagaaaaatg gcattttacg tgcagattat gagaacccgt attacgataa ctccacgttc 240
gcttcccatt tctatgatcc agacaatggt aagacataca tcccctttgc aaagcaggca 300
aaagaaactg gtgcacaaa cttcaagtta gctggggaa cctacaaaaa caaagatatg 360
aaacaagcct tttctatct tggctgtgtc ctgcattacc taggtgatgt caaccagcca 420
atgcacgcag ccaatttcac aaacttatct tatcctcaag gattccattc caagtatgag 480

```

-continued

```

aactttgttg atacgatcaa ggacaactac aaagtcacag atggaaacgg ttactggaat 540
tggaaagggg ccaaccggga agactggatt cactgtgctg ctgttgttgc caaacaagac 600
tattcaggaa tcgtaaatga caataccaaa gactgggttg taaagcagc agtctctcaa 660
gaatatgcag acaaatggag agccgaggtt acacccatga cgggcaagag gttaatggat 720
gcccaacgtg tcactgctgg ttacatacaa ctatgggttg atacttatgg ggataga 777

```

```

SEQ ID NO: 248      moltype = DNA length = 891
FEATURE            Location/Qualifiers
misc_feature       1..891
                   note = ARTIFICIAL SYNTHETIC DNA
source             1..891
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 248
caggaagcc cagccttcat tgatccagct agctggaaca ctccgtttaa cggtatagca 60
caagtagcat gccacaattg ttatgagaag caatacgcga acactttttc ttcagtatta 120
gactctgtta gaacatttga attggacttt tgggacccaa gggatgctgt tagtgggtgt 180
tctccgcctc actgggttgt cagacacaa cccagggactc tttttcaaa tggtaatgac 240
aacaattgca caggtgatgg tactggaaag aatgatttgg aagcctgtct gaatgatgtc 300
aagaattggg ccgataagca tccaggtcat ttcccatta cgcttattct agacaaaaag 360
caggggtgtg caaagaaag ttccaggagg actccaaaag acttcgatga acttgttgca 420
agagttttcc aaggtaaact attcaactcc caagacttag caaccacat aggatcaggc 480
gctggcgcac tacagggcaa cttgaaagga aagtcatggc ctactgcaaa tgatttaca 540
gggaaagtgt tgtagtctt gaatcatagt gaaaatcaga agttatccca gtatgctgag 600
gctagaacat ccaaggctaa gtgtttcatt tcaccagtta caaacgggca aaacgcacatt 660
tccggtaaag tgtctggtat gtcactccaa agttccggtt atgttgcaat gaacaatatg 720
ggtaagggag ataagagtgt ggcaaaagca gcttttgcac actctcacat tgggagagtc 780
tgggtgtgat ccgaagtgtc atttgcccaa cacatcaatc aaaagatcaa tcttagtgcc 840
tactacagat ttgcccgtca gtcgcaggtt ggttaccgta tacgtccatt t 891

```

```

SEQ ID NO: 249      moltype = DNA length = 1875
FEATURE            Location/Qualifiers
misc_feature       1..1875
                   note = ARTIFICIAL SYNTHETIC DNA
source             1..1875
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 249
gcagtcacac cagcagacgt tttgagtggt gtggagaaaa gactggatcc agcatccgga 60
cttgaagtac gttcccatct ggatactata tggaaatgaca tcaaatctgc tgcagactgc 120
accgcagctg aggcgcgttt gactttactt aaggggtgtg ctgctttcgg tgacaatttc 180
ttgtttgaag tatgacaga aatctgcgat ttgtcaggag ccgaagatga tgatgtttgt 240
tctgggtgtg tatcactaga aggaccaatc attgcaaacg acattaggaa aatgtcaatc 300
ggatccaaga cctccagatt gttttgtata acttttctgg gggtatgttc ttatcctgca 360
gttgatgctt ttaactgttc atttccgacc gctaagtcag ctgtaccag accggtctcc 420
tcaggaaaag atccgatcta tgcgtacac tattcagata ttcatttga tcccttctat 480
gtcgcaggga gcgcctcaaa ctgtactaag ccaatttgtt gtagagacta tacttcagct 540
tcttctcctg gcaacaacaa tagcccagca ggtccctacg gtgatcaca ttgtgacgtt 600
ccaatctcat tagaagattc catgtacgct gccatcaaaa agctagtctc agatgctgca 660
ttcgggtatc tcaactggcg tatcgtggat cagcgtgtat ggaatacttc tgaatcacag 720
aacatcatag acatgaacga tgcctatacc cgtatgaaga attctggtat gttacctacc 780
attttcgcca ctgcaggaaa tcatgaagcc tcaccagtga atagtttccc tccccctgct 840
atcggtaatg agtcacaaat ggtttatgat acactggcct cagattgggtc tcagtggatt 900
ggaaccagtg gtgccagctc tgtcgaatcc ataggtgcct attctgtgca ataccggttca 960
accaaatgtc gtgtgatttc ttgaaatacc aatatgtatt acattgagaa cttttacct 1020
tatgaaccaa ccattggaac agatccagcc ggacaattcg catgggtagt ttcagaattg 1080
agccgagctg aggcctgcag agaaaagatt tggatcattg gacatagcc gttgggctgt 1140
tccgacgctc tccacgaccc ttccaactac tttgatcaga tgcgcaatag atatgaagct 1200
accatagccg ctatgttttt cggccacaca catgaggatc acttcagat ttcttactca 1260
gactataacg ccagaactgc cgcacaacgt agggctgttt cctacattat gccttccctt 1320
actcctacca gccgacatcc gactttccgt gtctacacgg ttgatccaga aacattcgcc 1380
gtactagacg caacaacctc ctacgcagat atgagtcacac caacgtatca aactgctggg 1440
ccagcttggt ccgtctatta cagtgtctaaa gcagcatatg ggggtttggt cgatccacca 1500
gttgacgctg atgatgcagc cgcagagttg acaccagcct tttggcataa cgttactgca 1560
gcattagctg ccgatccccc ttctttcgat gcctactatg ctagaaagac cagaggttgg 1620
gatgttgctg cctgcgcagg cgcttggtgt gctgctgaag tgtgcgcctt gaggggtgcc 1680
cgtgcacaag acaattgtgt ggttccaact cctggtgtgc atttcagtaa gcgtgcagat 1740
gaaggcactt tagcccacca cagggtatga tgcggtgtat ctgtagccag gaattcactt 1800
tcctctttgg ttgtacagag ggaagcatta gaacacttag aaggcaggct ttctgagaaa 1860
agaagaatgg cagtt 1875

```

```

SEQ ID NO: 250      moltype = DNA length = 1839
FEATURE            Location/Qualifiers
misc_feature       1..1839
                   note = ARTIFICIAL SYNTHETIC DNA
source             1..1839
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 250

```

-continued

```

caggaagtaa cacatgactt ggctggcatt aagagaagtt tggaaatctag ggactgggtg 60
gaggacttat gggacaagtt cgaatcagat gcaactttgt ctggatgtga aagtttagtt 120
ttagtcttga aaggtctagc cgaatatca gaccaagcct tcatagacgt cttgcaagag 180
atatgcaaga tatctggagc cgaagatgat gatgtttgtg atgggagtat acagttagaa 240
ggcccagtta ttgcaagtgg cttgagatct atggcaattg gctctcgta tctaaagaa 300
ttctgtacaa cgttcttggg tttatgcgcc taccacgtg ttcaacaatg gagcgtgcct 360
ttttctctct ccaaaagttc taaaaccaga ccaagctctt ctggttaagga tcctatcaag 420
gtagtgcact attcagatat tcatatcgac cctctatatg ttggtggatc aaattcaaat 480
tgtactaaac caatctgctg tcttccatat actaaggcag accaacctgg caacaacaaa 540
taccagctg gcccaaacgg tgatcacaa tgtgattcac ctgtgtctct agagaagagc 600
atgtataacg ccattaagga aattgtacca gatgctgctt ttaccatctt tactggagat 660
attgtagatc acgctgtatg gaacacctca caatcctata acaccgaaca aatcacaat 720
gcatatggat tgatgtcaga caacttagga accatctatg gtacagcagg caatcacgag 780
gctcaccag ccaacgcatt tcagttggta atgtaagcca atgggtctac 840
gacttgttaa gtggactatg gttcctaatg atttctactg aagcaaaagg agacagtga 900
aagttagggt cttatgacac caataacca ggccgaaacc ttaggattat ctcatgaac 960
acaaacatgt actatagaga aaactactgg ttatacagaa agactatgat tcaggacct 1020
agcaaccaga tatcttggct agttaacgaa ttggaggcag cagaactgc aggtgaacgt 1080
gtgtacatca ttggtcatat gcccttaggc gatagcaatt cttccacga tcaatcaaac 1140
tacttggatc aagtgtacaa cagatactca gcaacaattt cagcaatgtt cttcggtcac 1200
actcacgatg atcaatttca gatttcatat tcaaatgtgt caaacggtaa cttcagtaac 1260
gccttagtca cttcctacat tggacctagt ttaactccaa ctgctgggat gccctgcttt 1320
agagtcttac acgtggatcc agtgactttt ggcatcttag attccactac ctatcgcgc 1380
gatatgaccg atagtgtttt ccaacaacca ggtccagtat ggaagaagta ttacagtgc 1440
aaagaagtgt atggcagcct actaagccca gctgttacag atagtgtgc tgaattgaca 1500
gcagccttct ggcataatgt cactacctta ttcgaagcag acaatacggc ttttgagca 1560
ttcttaagta ggaaatccag aggtctggaag tctgaatcct gtacaggtac ttgcaaggca 1620
aatgagatct gccagttgag ggctggccaga tctgaaaaca attgttacac cccttctct 1680
ggcatctctc ttacaaaagg atccttgaat ccagtggagg aaagagatga atgtgggac 1740
tccgtaacca gagcaactgt tctgtcaatg ggggtcagaa aggatgtcct taggtgtgtg 1800
aagaaaaggt tcattgaaaa ggctggggaa gtcaggggc 1839

```

```

SEQ ID NO: 251      moltype = DNA length = 777
FEATURE
misc_feature       1..777
                    note = ARTIFICIAL SYNTHETIC DNA
source             1..777
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 251
catgagaatg aagggggtta caaagttagg gtaatacagt attggtctgc cgaagatcca 60
catcatgaag acacaaacac tcacttatgg atagtccgtc atgcaatgga gattatggcc 120
aacaacaaag atgttgtgaa accaggcgaa gtcagcaaac taaaacaatg gcaatcagac 180
ctagaacaag gcattttacga tgcagatcat gccaaaccgt attacgataa cgcaacgttc 240
gcttccattt tctatgatcc agacactggt aagtcataca tccccttggc agcacacgca 300
aaaaacaacca gctcacaata cttcaaaaaga gctgggggaa cataccaaaa gggtgatcat 360
aaacaagcct ttacaatctt tggctctggc ctgcattaca taggtgatct aaaccagcca 420
atgcatgcag ccaatttcac aaacttatct tatcctcaag gattccattc caagtatgag 480
aactatgttg atagtctcaa ggaaggattat gctgtcaaa atggagaggg ttactggcat 540
tgaaaaggga ccaaccggga agactgggtg cacggtacag ctgttctgct caagaaagat 600
tatccagaca tctgtaaatga tacaacccaa gcctgggttg taaaagcagc agtctcaaac 660
tcttatgcag ctaaatggag agctgccgtt gttcccgcaa cgggcaagag gttacagaaa 720
gccaacgcta tcttgggttg ttacatgcaa ctatgggttg atacttatgt gaacaaa 777

```

```

SEQ ID NO: 252      moltype = AA length = 585
FEATURE
source             1..585
                    mol_type = protein
                    organism = Aspergillus wentii

```

```

SEQUENCE: 252
NWAESIWDVV KHAVNCAGCE TVLFALKGVA DLGEHAFQTV LTDVCDISGT EDKDVCSGLI 60
AAESPALYYN IKNLGVKSHT SKVLCAQLFG LCQPPAVRPY NLTFPPSPKPT TSRPPPSGQS 120
PIRVAHISDT HVDLSYETGS NYECSKPICC RVYTDEDAPG NTSFPCGPGY NTNCDPPLRL 180
EESMMAAIKS LNPFAFYTIG DVVAHDLWMV DKTEVLDDFN ATYSMLDQLD LVYAAGVNH 240
TTPVNLFPST QLPKDNQWA YDALTAEWKS LTNSSIQTTE YGSYSAIYEN LRIISYNSIF 300
YYQDNFYAYT DPMADPSNQ LTWLIDQLHE AESANQRVWL ISHIPTGGVD HLHDYSHYID 360
EIVQRYEATI SALFFGHTHT DLPQIAYS DY KNRAWDTASA IGYVAPSLTP TSGPPAFRIY 420
DIDPVTFAVL DTYVYIANIS NPSYQSNPKW EKYSYSAKEAY GSLLSPPVTD SSELTPSPFW 480
HNV TALMEKD DSVFQDWNSR TTRGYNVTTT TGDCAKKEIC ALRGADAYN CVTATPGFHF 540
DKREESEQKP RPECEDGLGR VLGDMIHKKD FVDLLHERTA QYQHR 585

```

```

SEQ ID NO: 253      moltype = AA length = 588
FEATURE
source             1..588
                    mol_type = protein
                    organism = Penicillium cylindrosporum

```

```

SEQUENCE: 253
TYDKRDLAQD LWNIDIKNAVD CGGCQVILTA LKGLSDLGTT VFVDVLTEVC EIGKLESDV 60
CSGTISREGP VLDYILQHL D IGSHTSQVLC ASAFGLCQYP AVRPNLFPF SPKPNTRTPA 120

```

-continued

```

PSGELPIQVV HISDTHVDLS YETGSSYNCT KPICCRPYTA ADAPGNTTTP CGPYGNTNCD 180
APIGLEESMF AAIKALQPAF SIYTGDDVAH DLWLVDQNEV LQDFNATFNR MAELGVVYAA 240
IGNHDTAPVN DLPASNIPSQ YSANWTYDAL AYDFSTLTNS ASAQTAEDYG SYSSVYRGSH 300
GTDLRVISYN SIFYVVDNFW VYQDPMLYDP DGQLAWLISE LQEAETGGQR VWLIAHVPSG 360
ISDHFHDYSH YFDQIVQRYE ATIAALFYGH THRDQFQISY SDYTNRTWNT ATAMGYITPS 420
LTPTSGPPTY RVYKVDPKTF GVLDFTNYIA NISDPAYQSG PTWQKYYSK ETYGSLLSPP 480
VTDPATAELTP AFWHNVTVLF ENDNATFQQY IARQTRGWDP SSCTGSCIDQ TICGLRAGDA 540
QYNCVTPTPG FNFARDAST GGQPEHVEK CEGSGLLALL GRMVATKS 588

```

```

SEQ ID NO: 254      moltype = AA length = 588
FEATURE            Location/Qualifiers
source              1..588
                    mol_type = protein
                    organism = Penicillium meridianum

```

```

SEQUENCE: 254
SADSWITIIW DDFKEAVDCA SCQALLGGLK YVAGFGESFM EDVLTGVCDI SGAEDSDVCS 60
GIIANEGPAV YYSLKNLELG SHTAKTFCSN LVGLCDYPAV RPYDLSFPSP KPSISRPPPS 120
GKPLKVVVHF SDTHVDLSYE PGSSFDCTKP ICCRVYSED APGNTSSPCG PWGNAKCDPP 180
HQLQQSMDMA IASLNPAFSI YTGDDVAHDV WLVNKTEVLQ DFNATYSTME STLGLVYAA 240
GNHDTAPLNL FPSTNIPSSY NPQWAYDALS TSWLTLTSDN PAISTAKEYG SYSARHKDTK 300
LRIISYNSIF YYKYNFFSYE EPMFPDPDGQ LSWLISELSA AEAASERVWL ISHIPSGNSD 360
HFRDHSYFDF QIVQRYEATI AGLFYGHHTH DEFQVSYSDY AHRSWDTATA MGYVAPSMTP 420
TSGPPSFRVY EIDPVTFGVL DFTQYIADIS DSSEPKWMPY YSAKKDYGSR LDVPVGEDME 480
LTPAFWHNVT VMEKDPPLF RDFWARTRTG FEVPGCEGDC VSKEICALRG ADAQYSCVEA 540
TPGFSFEKRG ERTVLLEKRF QPECNHAGMA PLLAKIAHRA SLAREMEA 588

```

```

SEQ ID NO: 255      moltype = AA length = 590
FEATURE            Location/Qualifiers
source              1..590
                    mol_type = protein
                    organism = Penicillium bialowiezense

```

```

SEQUENCE: 255
EVLENSIAS LWDDFKNAVD CASCQALLGG LKLVSFGFED FMIDVLTGIC DISKAEDSDV 60
CEGIKKKEGP ALHDAFQALH IGSNSTQTM ASLIGLCQYP EVRAHSLSFP SPKPNKVKPP 120
PSGKPPIKVV HFS DTHVDLY YETGSSYECs KPICCRVPED KYAPGITKTP CGPFGNPRCD 180
PPLKLQESMN AAIADINPEF SIYTGDDVAH DVWLNVQAEA LESFNSTYSQ MEKSLGMVYA 240
AIGNHDTAPL NLFPSIVPD GDNQWAYNAL AEYWLTLTSI SSVQSADEYG SYSAIHPDSN 300
LRIISYNSIF YYKYNFFMYQ EPMKDPNGQ FEWLKELQA AEDAGQRAWL ISHIPPGVAD 360
HFHDYSHYFD QIVQRYEATI AGLFYGHHTH DEFQIAYSNY KNRNHETATA MGYIAPAMTP 420
TEGPPSFRVY EIDPVTFGVL DFTQYIANIS DPTYQEKQW LPYYSAKADY GSKLSPVTD 480
PKIELTPAFW HNVTVAMENE PSIFQEFWAR RVRGYKVTDG NGDCMKTEIC ALRAADAQFN 540
CVKPKPGFNF SKRDKDTLA EQPHHCDHAG LAPLLGKIAY RARIAREMEA 590

```

```

SEQ ID NO: 256      moltype = AA length = 590
FEATURE            Location/Qualifiers
source              1..590
                    mol_type = protein
                    organism = Penicillium sclerotiorum

```

```

SEQUENCE: 256
SADSWISSIW DDFKEAVDCG SCQALLGGLK VVAAFGESFM EDVLIGVCDI SGAEDSDVCA 60
GVIANEGPAV HSYLTNLEIG SHTAKTFCAT LVGLCKYPEV RPYNLTFPSP KPPTSRRPPS 120
GKTPKVVVHF SDTHVDLSYE TGSNYDCSKP ICCRAYTEDD APGNTSSPCG PWGNTNCDPP 180
HRLQKSMNAA IADLKPAFSI YTGDDVAHDV WLVNKSEALQ DFNATYGAME DSLGRVYAA 240
GNHDAAPLNL FPSNQIPSEY NPQWAYDALA ADWMALTDIP SVETANEHGS YSAIHPDSNL 300
RIISYNSIFY YKYNFFSYTE PMEYDPNGQL EWLINELHAA ETANQVRWLI SHIPSGNSDH 360
FHDHSHYFDE IIRYQYATIA GLFFGHHTHD EFQISYSNYS NRNWDATAM GYVAPSMTP 420
SGPPSFRVYE IDPVTFGVLD FMQYIANISD PSYQKPEWV PYSAKSDYG SRLSPPVTD 480
DVELTPAFWH NVTVMEEDES SVFDEFWARR TRGFQVPACT GDCVSNEICA LRADGQYNC 540
FIEKPGFSFE KRDGTVEVS WKKRFQPECN HAGMAPLLAK IAHRASLARE 590

```

```

SEQ ID NO: 257      moltype = AA length = 592
FEATURE            Location/Qualifiers
source              1..592
                    mol_type = protein
                    organism = Rasamsonia byssochlamydoides

```

```

SEQUENCE: 257
SYNKRGLAQE IWDIVNAVD CSGCQVILTA LQGAELGTS AFVDILTVC DISGKEDSDV 60
CSGIISREGP VLDYILQNLD IGSHTSDVIC ASAFGLCQYP AVRSYNLTTP TPKPDTRPA 120
PSGESPIQVV HFS DTHVDLS YETGSSYNCT KPICCRPYTA ADAPGNTTTP CGPYGNTNCD 180
APLSLEESMF NAIKALSPQP AFSIYTGDDV AHDIWLVDQN EVIQDLNTTY DLMAELGLVY 240
AAGNHDTAP VNDLPTTNIP SEYSTNWTEY ALAYDFTMLT SSSAQTAAN YGSYSSIYKG 300
SYGTDLRIIS YNSIFYIDN FWAYQDPMPY DPGQLAWLI NELQEAETAG QRVWIIHVP 360
TGSDHFHDY SHYDQIVQR YEATIAALFY GHTHIDQFQI SYSDYSNQAW DTATAIGYIM 420
PSLTPTSGPP THFYVDVDPK TFAVLDFNTY IANISDPAYQ SGPTWQKYYS AKETYGALLS 480
PPLTDPTAEL TPAFWHNVTV AFENDNAAFQ EYWARQTRGW DVSSCTGSCI TQAICGLRAA 540
DAQYNCVTPT PGFNFARKDA SSATQAMAHV EKCEGSGLLA LLGRMVADKK SA 592

```

```

SEQ ID NO: 258      moltype = AA length = 593
FEATURE            Location/Qualifiers

```

-continued

```

source          1..593
                 mol_type = protein
                 organism = Rasamsonia eburnea

SEQUENCE: 258
TYDKRGLAQD IWNDIKNAVD CAGCQGILTA LKGLSYLGTT AFVDVLTEVC DISGVEDSDV 60
CSGIIISSEGP ALVYILKHL D IGSHTSQVIC ASVFGLCQYP AVRAYNLTFP SPKPKDKTCPE 120
PSGESPVQIV HFS DTHADLS YETGSNYNCT KPICCRSYTA EDAPGNTTTP CGPYGNPKCD 180
APMSLEESMF AAIKALSPQP AFSIYTGDDV AHDIWLVDQN EVVEDLNATY DRMAGLGLVY 240
AAIGNHDTAP VNNLPTSNI S QYSANWTYE ALEYHFSLLT KSASQAQTAEN YGSYSSVYRG 300
RYGTDLRVIS YNSIFYIAD FWAYQDPMY DPDGQLAWLI NELQEAETAG QRVWLIAHVP 360
SGTADHFHDY SHYFDQIVQ R YEATIAALFY GHTHIDQFQI SYSDYSNRAF DTATAIGYIM 420
PSMTPTSGBP TFRVYDVPK TFAVLDFNTY IANISDPAYQ SGPTWQKYYS AKEAYGSLLS 480
PPVTDATAEL TPAFWHNV T AFENDDTAFQ EYWARQTRGY AVSSCTGDCI TQAICGLRAG 540
ESQHNCVTPT PGFNFAR D V STDGQALPHI EKCEGSGLMA LLAKMVASNR QSS 593

SEQ ID NO: 259      moltype = AA length = 593
FEATURE            Location/Qualifiers
source             1..593
                   mol_type = protein
                   organism = Penicillium brefeldianum

SEQUENCE: 259
GAIENWAATI WEDFKEAVDC GSCQVLLGGL KLVADFGEGF LGDVLTVGVD ISKAEDRDVC 60
AGVVASEVPA LHYALKNMKV GSHTAKTLC S ALVGLCDFFD VRPFDLTFFS PKPATSRPPP 120
SGKPKIKVVH FSDTHVDLSY ETGSNYDCSK PICCRVYTDK DAPGTTDKPC GPWGHKPCDP 180
PHQLQESMMT AIALNPAFS IYTGDDVVAHD VWLVNKDEV L QDLNATYGAM ENHGLGVYAA 240
LGNHDAAPLN LFPSNKVPSK YNPQWAYDAL TADWMTLTGL DSVQNANKYG SYSAVHPNSK 300
LRIISYNSIF YKYNFFSYT EPMEYDPNGQ L TWIINELQA AETAGQVRWL ISHIPSGDVD 360
HFRDHSHYFD QIVQRYEAT I AGLEFGHTHT DEFQIAYS D Y NNRNWD TATA MGYVAPSMTP 420
TSGPPSFRVY DIDPETFGVM DYTQYIANIS DPSFQTKQEW VPYYTAKKDY GAKLSPPPAP 480
TGELTPAFWH NVTVAMEKDS SVFEAFWARR TRGFSIPACT GDCVKNEICA LRGADAQYSC 540
VKRTPGFSFS KRDEIESDPL LSKRFQPECN HAGMAPLLAK IAHKANVAKW NGE 593

SEQ ID NO: 260      moltype = AA length = 593
FEATURE            Location/Qualifiers
source             1..593
                   mol_type = protein
                   organism = Penicillium adametzii

SEQUENCE: 260
SVDSWITTIW DDFKEAVDCA SCQALLGGLK LVAVFGESFM EDVITGVCSI SGAEDSDVCA 60
GVIANEGPAV YHSLSNLKIG SHTAKTFCAS LVGLCDYPAV RPYSLTFPSP KPPTTRPPPS 120
GKSPIKVVHF SDTHVDLSY E TGSNYDCSKP ICCRAYTEND APGNTSSPCG PWGNSKCDPP 180
HRLQESMNEA IADLNPAFSI YTGDDVVAHD VWLVKSEALQ DFNATFSAME TLGRVYAAALG 240
NHDAAPLNLF PSNQIPSYN PQWAYDTLAS DWMGLTG I QS VETANEYGSY SAIHPNSNLR 300
IISYNSIFYY KYNFFSYTEP MEYDPNSQLQ W LINELHEAE LANQRVWLIS HIPSGDPDPHF 360
HDHSHYFDQI VQRVDTATIA LFFGHHTDQ FQISYSSYQN RTWDKATAMG YVAPSMTPS 420
GPSPFRVYEI DPTVFGVLD F TQYIANISDP TQTKPKWVPY YSAKKDYGSH LDPPVTDABA 480
ELTPAFWHNV TVSMENDNSV FQDFWARRTR GFQVPDCTGD CSSSEICALR GADAQYNCV 540
EKPFGFSFEK R DVQESDEEMW KKRFRQPECNH AGMAPLLAKI AYRASLEREQ KKG 593

SEQ ID NO: 261      moltype = AA length = 594
FEATURE            Location/Qualifiers
source             1..594
                   mol_type = protein
                   organism = Rasamsonia brevistipitata

SEQUENCE: 261
TSLTHDKRDL AQEIWN DIK AATCAGCQVI LTALKGVSDL GTSIFVDVLT EVCDISGLED 60
PDVCSGII S R EGPVLDYILQ HLDIGSHTSD VLCASAFGLC QYPAVRPYNL TFPSPKPDPT 120
RPAPSGESI QVWHISDTHV DLSYETGSSW NCTKPICCRP YTAEDAPGNT TTPCGPYGDT 180
HCDAPMSLQE SMIAAIQAL R PQPAFSIYTG DVVAHDIWL V DQNEVIEDFN ATYNRMAELG 240
LVYAAIGNHD SAPVNDLPAS NIPSQYSVNW TYEALAYDFS MLTGSASAE AENYGSYSSI 300
YKGSNGTDLR VISYNSIFYY VTNFWAFQDP MPYDPDGQLA W LINELQEA E TAGQRAWLIA 360
HVPTGTGDHF HDYSHYFDQI VQRYEATIAA LFFGHHTQDE FQIAYSNSYN QNFDTATAIG 420
YIMPSLTPTS GPPSFRVYDI DPKTFGVLD F TNYIANISDP AYQSGPTWQK YYSAKEAYGT 480
LLSPAVTDP T AELTPAFWHN LTVVFENDNA TFQEWTRKT RGHAVSNCTG SCITQSI CGM 540
RAADAQYN CV TPTPGLSFAK RDAETSTPEP HIEPCEGSG L MSLGLRMVAQ GKSS 594

SEQ ID NO: 262      moltype = AA length = 594
FEATURE            Location/Qualifiers
source             1..594
                   mol_type = protein
                   organism = Penicillium scabrosum

SEQUENCE: 262
ETLEKSISS IWEDFKNAVD CGSCQVLLGG LKFVSGFGEN FMIDVLTGLC DISKAEDSDV 60
CEGIKKKEGP ALHDAFQALK IGSHTSKTMC ANLIGLCQYP EVRPNTLSFP SPKPKDVRPT 120
TSGKPKIKVV HFS DTHVDLL YETGSNYECT KPICCRVFED KDAPGITKNP CGPFGNPKCD 180
PPQALQESMN AAI AEINPAF SIYTGDDVSH DVWLVNQEEA LESFNSTYSQ IEKSLGMVYA 240
AIGNHDTAPL NLFPSKNQPD GNNQWAYDAL AEDWLAITGI PSVQSADEYG SYSAIHPNSN 300
LRIISYNSIF YKYNFFSYT EPMEYDPNGQ FEWLKELQA AEDAGQRAWL ISHIPSGVAD 360
HFDHSQYFD QIVQRYEAT I AGMFYGH T HM DEFQIAYS D Y NNRNWD TATA MGYIAPAMTP 420

```

-continued

TEGPPSFRVY	EIDPDTFGVL	DFTQYIANIS	DPAYQKKPEW	VPYYSAKADY	GSKLSPVTD	480
PKVELSPAFAW	HNVTVSMERD	ESVFQDFWAR	RSRGYNVTAC	TGDCMKMELC	TLRAADAQYN	540
CVKPKPGPNF	SKRDGELGGL	LEQESHNSCD	HAGLATLLGK	IAHRARVARK	IEEA	594

SEQ ID NO: 263 moltype = AA length = 594
 FEATURE Location/Qualifiers
 source 1..594
 mol_type = protein
 organism = *Penicillium manginii*

SEQUENCE: 263

SADDWITTIW	DDFKNAVDCF	SCQALLGGLK	LVSGLGESFM	EDVITGVCSI	SGAEDNDVCA	60
GVIANEGPAV	YYSKLNKLG	SHTAKTFCAT	LTGLCEFPKV	RPYDISFPSP	KPSTTRPPPS	120
GEAPIKVVHF	SDTHVDLSYE	EGSNYECSKP	ICCRAYTEKD	APGNTTSPCG	PWGNSKCDPP	180
HRLQESMMSA	IADINPAFSI	YTGDDVVAHDV	WLVNKTEVLQ	DLNATYSSIE	NHLGLVYAAL	240
GNHDAAPLNL	FPSDKIPSQY	NPQWAYDALA	EDWLTLTGIP	SVQKASEYGS	YSAVHPGSKL	300
RIISVNSIFY	YKYNFFSYTE	PMEFDPNKQL	DWLIQQLQEA	EDAKQRVWLI	SHIPTGNSDH	360
FRDHSYFDQ	IIQRYDATIA	ALFFGHHTTD	EFQISYSNYK	NRNWDTATAM	GYVAPSMPT	420
SGPPSFRVYE	IDPVTFGVMD	FTQYIANITD	PSLQTEPEWK	PYYSAKADYG	SKLSPAIKDP	480
GIELTPGFWH	NVTVAMEKDA	TVFQDFWSRR	TRGFNVPGCT	GDCISNEICA	LRGADAQYSC	540
YKQAPGFSFE	KRDGSGVPYL	SEESFQQPEC	NHAGMAPLFA	KISHRAKLAR	ERGE	594

SEQ ID NO: 264 moltype = AA length = 594
 FEATURE Location/Qualifiers
 source 1..594
 mol_type = protein
 organism = *Penicillium emersonii*

SEQUENCE: 264

SAPYDKRDLA	QEIWDDIKNA	VDCAGCQVVL	TALKGVADLG	TTALVDVLTE	VCNISGKEDS	60
DVCSGIISRE	GPVLDYVLQH	LDIGSHTSQV	ICASAFGLCQ	YPEVRPYNLT	FPKPKPNTTR	120
PEPSGESPIQ	VVHFSATHVD	LSYETGSNYN	CTKPICCRPY	TAEDAPGNTT	TPCGPYGNTK	180
CDAPLSLEES	MFAAIKALNP	QPAFSIYTG	VVAHDIWLDV	QNEVIEDLNA	TYDRMAGLGL	240
VYAAIGNHDT	APVNDLPTSN	IPSEYSANWT	YEALSYDFTM	LTQSASAQTA	ANYGSYSAIY	300
PGSYGTDLRV	ISYNSIFYV	DNFWAYQDPM	EFDPDGQLAW	LINELQEAET	AGQRVWIIAH	360
VPTGTSDFHF	DYSHYFDQIV	QRYEATIAAL	FYGHTHIDQF	QISYSNYSNR	AFDTATAIGY	420
IMPSLTPTSG	PPTFRVYDVD	PKTFFAVLDT	NYIANISDPA	FQSGPSWQKY	YSAKETYGSL	480
LSPPVTDPTA	ELTPAFWHNV	TVAFEQDNAT	FQEYWARQTR	GYDVSSCTGS	CITQAICGLR	540
AGDAQYNCVT	PTPGFNFAKR	DTSNPKQALS	HVEKCEGSGL	LGLLRMVAD	SKSS	594

SEQ ID NO: 265 moltype = AA length = 595
 FEATURE Location/Qualifiers
 source 1..595
 mol_type = protein
 organism = *Rasamsonia argillacea*

SEQUENCE: 265

GVTYDKRDLA	QDIWNDDIKNA	VDCAGCQGIL	TALKGLSYLG	TTAFVDVLTE	VCDISGMEDS	60
DVCSGIISSE	GPVLDYILKQ	LDIGSHTSQV	ICASAFGLCQ	YPAVRAYNLT	FPSPKPDKTR	120
PEPSGESPMQ	IVHFSATHVD	LSYETGSNYN	CTKPICCRPY	TADDAPGNTT	TPCGPYGNTK	180
CDAPMTLEES	MFAAIKALSP	QPAFSIYTG	VVAHDIWLDV	QNEVEDLNA	TYDRMAGLGL	240
VYAAIGNHDT	APVNDLPTSN	IPSQYSANWT	YEALYHFSL	LTNSASAQTA	ENYGSYSSVY	300
PGKYGTDLRV	ISYNSIFYV	DNFWAYQDPM	LYDPDGQLAW	LINELQEAET	AGQRVWIIAH	360
VPSGTADHFF	DYSHYFDQIV	QRYETTIAAL	FYGHTHMDQF	QISYSDYSNR	AFDTATAIGY	420
IMPSMTPTSG	PPTFRVYDVD	PKTFFAVLDT	NYIANISDPA	YQSGPTWQKY	YSAKEAYGPL	480
LSPPVTDATA	ELTPAFWHNV	TVAFENDDTA	FQEYWARQTR	GYAVSNCTGN	CVTQAICGLR	540
AGESQYNCVT	PTPGFNFAKR	DVSSDGQALP	HIEKCEGSGL	LSELLAKMVAS	NGQSS	595

SEQ ID NO: 266 moltype = AA length = 595
 FEATURE Location/Qualifiers
 source 1..595
 mol_type = protein
 organism = *Penicillium parviverrucosum*

SEQUENCE: 266

HIESADNWIT	TIWDLISKQAV	DCAGCQALLG	GLKLAADLGE	TFMEDVLIGV	CNIAGVEDHD	60
VCSGIIQNEG	PAVHYSLNL	HIGSHTATTL	CASLFGLCQY	PAVRPYNLSF	PVPKPTKSRP	120
EPGQSPRI	VHFSATHVDL	SYETGSNYDC	SKPICCRPYT	EEDAPGNTST	PCGPWGNPMC	180
DPPNRLQESM	MTAIADLNPA	FSIYTGDPVA	HDIWSATKAE	ALRDFNATYG	SMEKRLGMVF	240
AALGNHDAAP	LNLFPSNKIP	SEYSPQWAYD	ALAADWLGLS	AMASVHSAIH	HGSYSVAVHSE	300
DKLRVISYNS	IFYYKDNFFM	YEEPMEDHPN	GQFAWLISEL	QSAESTSQRV	WLIAHIPSGN	360
ADHFRDFSHY	DEIVQRYDT	TIAALYFGHT	HTDTFQIAYS	NYSNRSDWTA	SAMGYVAPSM	420
TPTSGSPSFR	VVEVDPTVFG	IIDFTQYIAN	ISDPSYQINP	KWEPYYSACK	AYGSKLSPPA	480
QDPGAEMTPA	FWHNVTIAME	QDASIFQAFW	ARRTRGNKVT	SCTGNCMANE	ICALRGADAQ	540
YNCATPTVGF	RFRKRDMTSD	LSLQKEEFRP	ECNHAGMGPL	LAKIVHQAAL	ENERG	595

SEQ ID NO: 267 moltype = AA length = 595
 FEATURE Location/Qualifiers
 source 1..595
 mol_type = protein
 organism = *Penicillium flavesceus*

SEQUENCE: 267

HSSESABSWIS	DIWDHFKEAV	DCTSCQVLLG	SLKLVAAFGD	TFMVDVLTGV	CAISGAEDHD	60
-------------	------------	------------	------------	------------	------------	----

-continued

VCAGVIAGEG	PSVQYSLKNL	KIGSHTAKTF	CASLVGLCQY	PKVREYDLAF	PAPKPPNGRP	120
PPSGEPPIKV	VHFSDDTHVDL	AYEPGSNYAC	SKPICCRTRYK	ENDAPGNTSS	ACGPWGNPRC	180
DSPHRLQESM	NAAIADLNPA	FSIYTGDDVVA	HDVWLDNKFE	VLQNFNATYG	AMETTLGQVY	240
AALGNHDTAP	LNLFPSSKIP	SIYNPQWAYD	ALTENWLALT	GIPSIESADQ	YGSYSVLHPD	300
SNLRIISYNS	IFYKYNNFFA	YTEPMEYDPN	SQLKWLINEL	QAEEKASERV	WLISHIPSGN	360
SDHFDHSHY	FDQIIQRYDA	TIAALFFGHT	HLDEFQISYS	DYKSRTWDTA	TAMGYIAPSM	420
TPTSGPPSFR	VYDIDPVTFG	VLDFTQYIAN	INAPDQESLE	WVPYHSAKEA	YGSKLVSPIT	480
DPSAELSPAF	WHNVTLAMEN	DSAIFGDFWA	RRTRGYQVPS	CTGDCISGEI	CTLRGADAQY	540
NCFVQKVGPS	FEKRHDQGDS	TREERILPEC	NHAGMAPLLA	KIARLAAIAR	DMEKR	595

SEQ ID NO: 268 moltype = AA length = 596
 FEATURE Location/Qualifiers
 source 1..596
 mol_type = protein
 organism = *Penicillium hispanicum*

SEQUENCE: 268

ESAESWVASI	WDDFKVAVDC	ASCQALLGGL	KLVAEFGESF	MEDVLIGVCD	VSGVEDSDVC	60
AGVIANEGPA	VHYTLKNLEI	GSHAANTLCA	SLVGLCEYPA	VRPYNLSFSP	LKPATSRPPP	120
SGKPAIKVVH	FSDTHVDLSY	ETGSNFDCSK	PICCRVYTEE	DAPGNTSSPC	GPWGNPKCDP	180
PHRLQESMVE	AIAVLDPAFS	IYTGDDVVAHD	VWLNVKSEAL	QDFNATYGAM	ENRFGPVYAA	240
LGNHDTAPLN	LFPNKNISRE	YNPQWAYDAL	AADWAALTGI	PSVASAREYG	SYSAIHPNSN	300
LRISYNSIF	YYPNFPAYE	EPMEYDPDSQ	LAWLITELDA	AETAGQVRWL	ISHIPSGKPD	360
HFRDHSYFDP	QIVQRYDATI	AALFFGHTHK	DEFQISYSTY	TNRAWDTATA	MGYIAPSMTP	420
TSGPPSFRVY	EIDPVTFGVM	DYTQYIAKIS	DSSAQIDTTP	EWVPYYSAKA	DYGAKLAPPV	480
EGAGVELTPA	FWHNVTVAME	ADSSLFQAFW	GRRTRGYNVS	SCTGECMATE	ICALRGADAQ	540
YSCVGAKPGL	SFSKRGGKDV	DVLWQRRLLQ	ECNHAGMAPL	LGKIWTRAAAL	VWRSEA	596

SEQ ID NO: 269 moltype = AA length = 598
 FEATURE Location/Qualifiers
 source 1..598
 mol_type = protein
 organism = *Penicillium simplicissimum*

SEQUENCE: 269

GATENWISTI	WNDFKEAVDC	GSCQVLLGGL	KLVAADFGEF	MEDVLTGVCD	ISGAEDRDVC	60
AGVIASEVPA	LHYAIKNMHV	GSHTAKTLCS	ALVGLCDFPD	VRPFDLAPPS	PKPANSRPPP	120
SGKPIKVVH	FSDTHVDLSY	ETGSNYDCSK	PICCRVYTDA	DAPGTTDKPC	GPFGNTKCDP	180
PHQLQESMMT	AIAELNPAFS	IYTGDDVVAHD	VWLVTKEEVL	QDLNATYGAM	ENHLGLVYAA	240
LGNHDAAPLN	LFPNHNVPK	YNPQWAYDAL	TADWMALTGI	ENVQANNEYG	SYSAIHPNSK	300
LRISYNSIF	YKYNNFFSYT	EPMEYDPNGQ	LTLWTEELQA	AENAGQVRWL	ISHIPSGDVC	360
HFRDHSYFDP	QIIQRYEATI	AGLFFGHTHT	DEFQVSYSDY	KNRNWDTATA	MGYVAPSMTP	420
TSGSPSFRVY	DIDPVTFGVL	DFTQYIANIS	DASFQTKPTW	IPYYTAKKDY	GSKLPTIPDD	480
TAEITPAFWH	NVTVAMEKDS	AVPDEFWARR	TRGYNVPACT	GDCAKNEICA	LRGADAQYSC	540
VQRTPGFSFS	KRDGVGDGEME	EVAPLLSKRF	QPECNHAGMA	PLLAKIAHNA	NLAKMNGE	598

SEQ ID NO: 270 moltype = AA length = 599
 FEATURE Location/Qualifiers
 source 1..599
 mol_type = protein
 organism = *Penicillium vasconiae*

SEQUENCE: 270

GAVESWISTI	WNDFKEAVDC	GSCQVLLGGL	KLVAADFGEF	LEDVLTGVCD	ISGAEDRDVC	60
AGVIASEVPA	LHYAIKNMHV	GSHTAKTLCS	ALVGLCDFPD	VRPFDLAPPS	PKPAKSRPPP	120
SGKSPIKVVH	FSDTHVDLSY	ETGSNYDCSK	PICCRVYSDE	DAPGKTDKPC	GPFGNTKCDP	180
PHQLQESMMT	AIAELNPAFS	IYTGDDVVAHD	VWLNVKDEVL	QDFNSTYGAM	ENHLGLVYAA	240
LGNHDAAPLN	LFPNHNVPK	YNPQWAYDTL	TANWMTLTGI	ESVQANNEYG	SYSAIHPNSK	300
LRISYNSIF	YKYNNFFSYT	EPMEYDPNGQ	LTLWLISELQS	AENAGQVRWL	ISHIPSGNVD	360
HFRDHSYFDP	QIIQRYEATI	AGLFFGHTHT	DEFQIAYS DY	KNRNWNTATA	MGYVAPSMTP	420
TSGPPSFRVY	DIDPVTFAVI	DFTQYIANIS	DPTFQKKPDW	VPYYSAKKDY	GGLKSPRPAD	480
TAEITPAFWH	NVTVAMEKDS	SVFNEFWARR	TRGYNVPACT	GDCVKNEICA	LRGADAQYSC	540
VQRTPGFSFS	KRDDENVGDG	VGNYPILLSKR	FQPECNHAGM	APLLAKIAHK	ASLAKANGE	599

SEQ ID NO: 271 moltype = AA length = 600
 FEATURE Location/Qualifiers
 source 1..600
 mol_type = protein
 organism = *Talaromyces columbinus*

SEQUENCE: 271

YDSALVDHNL	VSDIWEIDKE	AVTCAGCQVI	LAALKGVSDL	GTTALVDTLT	GVCKLSGAAD	60
DNVCEGIIIS	EGAVLHYVLS	ELSLGSETSN	ALCASAFGLC	LYPDVRNYTL	NFPKSAKPKNI	120
TRPAPSGKPP	IQVAHFSDDH	VDLSYEVGSN	WNCTKPICCR	SFEASDAPGN	TTTPCGPFGN	180
TKCDTPLTLE	ENMLDSIKKS	DPTPAFSIYT	GDVVAHDIWL	VDKDEVLTDL	NATYSLMAEI	240
GTVYAAIGNH	DTAPLNDLPT	TQVPESYSAN	WTYQALATNF	TTLTRDSSVI	SVAKNYGSYS	300
SVFTGSYGTD	LKIIISYNSMF	YYVDNFYAFI	DPMPYDPDQ	LAWLIDELQS	AETAGQVRWL	360
IAHVPTGSSD	HFHDYSHYFD	QIVQRYDATI	AALFFGHTHT	DQFQIAYS DY	ANQNADTATA	420
IGYIMPSLTP	TSGPPAYRIY	DIDPVTFSVL	DYTVYITNIS	HPDFQKGPKW	EKYYSKADTY	480
GSLSPPVTD	PSAEMTPAFW	HNVTAVFESD	DVAFQGYWAR	QTRGYDVSDC	TDESCKNQT	540
CALRAADAQY	NCVVPISIGFN	FAKRDDTDQA	HVRAQKEKCD	DTGLVSLLGK	ILAKSKETT	600

SEQ ID NO: 272 moltype = AA length = 600

-continued

FEATURE
source Location/Qualifiers
1..600
mol_type = protein
organism = Talaromyces variabilis

SEQUENCE: 272

SSSALIDRDL	ASEIWDDIKE	AATCAGCQVI	LAALKGVSDL	GTTALIDTTL	EVCKISGAED	60
DDVCEGIIIS	EGPVLHYILS	QLSLGSETSD	ALCTTAFGLC	AYPDVRNYTL	TFPSAKPENS	120
TRPSSSGESP	IQVVFHSDTH	VDLSYETGSN	WNCTKPICCR	SFDSSDAPGN	TKTPCGPYGN	180
TKCDAPISLE	KSMVDSIKGL	SPAPAFSIYT	GDVVAHDIWL	VDEDEVLTDL	SSTYGLVQDV	240
GTVFAAIGNH	DTAPVNDLPT	TQVPSTYSAN	WTEALAGNC	TTLTGDSVM	SVAENYGSYS	300
SVFTGSHGTD	LKVISYNSIF	YYADNFYAF	DPMPYDPDQ	LAWLIDELQA	SETAGQRVWL	360
IAHVPTGSSD	HFHDYSHYLD	QIVQRYDATI	AALFFGHHT	DQFQIAYSNY	SNQNADTATA	420
IGYIAPSLTP	TSGPPAYRVY	DIDPVTFGVL	DFTVYIANIS	DPDYQNGPTW	AKYYSAKETY	480
GSLSPPPVTD	SSAELTPAFW	HNVTAVFETD	DAAFQGYWAR	QTRGYDVSNC	TDSSCKNQNI	540
CALRAADAQY	NCVVPISGFN	FAKRDETDQA	HVKAQKEKCD	DAGLVSLLGK	IISKSRDVSN	600

SEQ ID NO: 273 moltype = AA length = 600

FEATURE
source Location/Qualifiers
1..600
mol_type = protein
organism = Talaromyces rugulosus

SEQUENCE: 273

SSSALVDRDL	ASEIWDDIKE	AATCTGCQVI	LAALKGVSDL	GTTVLVDTLT	EVCKLSGAED	60
DDVCEGIIIS	EGPVLQYILS	QLSLGSETSD	ALCASFAGLC	SYPDVRSYTL	TFPSTKPENS	120
TRPSSSGQAP	IQVVFHSDTH	VDLSYETGSN	WNCTKPICCR	SFDSSDAPGN	TSTPCGPYGN	180
TKCDAPLSLE	ESMFDSIKSL	SPAPAFSIYT	GDVVAHDVWI	VDKDEVLTDL	NATYSLMAEV	240
GTVYAAIGNH	DTAPLNDLPT	SQVPSTYSAN	WTYQALATNF	TTLSGDSSIM	SVAENYGSYS	300
SVFAGSHGTD	LKVISYNSIF	YYVDNFYAF	DPMPYDPDQ	LAWLIDELQA	AESAGQRVWL	360
IAHVPTGSSD	HFHDYSHYFD	QIVQRYDATI	AALFFGHHT	DQFQIAYS DY	SNQNSDTATA	420
IGYIMPSLTP	TSGPPAYRVY	DIDPVTFGVL	DFTVYIANIS	DPDYQNGPTW	AKYYSAKETY	480
GTLSPAVTD	SSAELTPAFW	HNVTAVFETD	DTSFQGYWAR	QTRGYDVSNC	TDSSCKNQTI	540
CALRAADAQY	NCVVTPTGFN	FAKRDETDQA	HVKAQKEKCD	DTGLVSLLGK	IISSSRNVS	600

SEQ ID NO: 274 moltype = AA length = 601

FEATURE
source Location/Qualifiers
1..601
mol_type = protein
organism = Hamigera terricola

SEQUENCE: 274

QMOTSERSWG	STIWKDVEEA	VDCAGCEVIL	GALKLVADLG	KGTLETAMID	VCDLSGAEDS	60
DVCSGLITAE	IDALYYALNN	VHVGSHTSKV	LCAHLFGLCS	YPDVRSYHLI	FPSSKPATSR	120
PPSPGQKPIK	VAHISDTHVD	LSYEPGSNYE	CSKPICCRAY	TKEDSPGNT	HPCGPYGNTN	180
CDAPYRLEES	MFAAIEALDP	AFSIYTGDDV	AHDIWLVNET	EVLDLDLDTY	SLMKSGLVY	240
AAVGNHDAAP	VNLFPNSRIP	STYSPQWAYD	ALAEDWLALT	NDSSVDSARE	YGSYSVYPG	300
SNLRIISYNS	VFYKDNFWM	YEDPMEYDPN	GQLAWLINEL	QAASAGERV	WLISHIPSGN	360
SDHLYDYSHY	FDIAIVQRYEA	TIAALFFGHT	HTDLFQIAYS	DYDNRNWDTA	TAIGYIAPSM	420
TPTSGSPAPR	IYEVDPTVFG	ILDYTVYIAN	ISHPSYQTP	TWEKYSAKE	AYGSLTTPV	480
TNPSIELTPA	FWHNVTVLME	DDESVPKDFW	ARTTRGFNVS	TCVGSCTEE	ICALRSADAK	540
YNCATAPKGL	NFLKRDDVEA	QSKPKPQCE	DSGLAAVLVK	MMENTDDFAG	LLKEKALAHG	600
K						601

SEQ ID NO: 275 moltype = AA length = 601

FEATURE
source Location/Qualifiers
1..601
mol_type = protein
organism = Penicillium piscarium

SEQUENCE: 275

RTEAADSWIS	NIWNEFKEAV	DCGSCQVLLG	GLKLVADEFGE	GFLEDVLTGV	CDVSGAEDRD	60
VCAGVIANEV	PALHYALKNM	HVGSH TAKTL	CSALVGLCDF	PNVRPFDLTF	PSPKPAKSRP	120
PPSGKPKPIK	VHFS DTHVDL	SYETGSNYDC	SKPICCRVYS	DKDAPGKTDK	PCGPPGNTKC	180
DPPELRQDSM	MAAIALNPA	FSIYTGDDVA	HDVWLNVKDE	VLQDFNSTYG	AMENHLGLVY	240
AALGNHDAAP	LNLFPNSNIP	SKYNPRWAYE	ALTANWITLT	GIQSVQNANE	YGSYSAIHPN	300
SKLRIISYNS	IFYYKYNFFS	YTEPMEYDPN	GQLTWLIEEL	QAENAGQVR	WLISHIPSGN	360
VDHFRDHSY	FDQIIQRYEA	TIAGLFFGHT	HTDEFQISYS	DYKNRNWDTA	TAMGYVAPSM	420
TPTSGPPSPR	IYDIDPETFA	VMDFTQYIAN	ISEASFQTKP	NWIPYSAKK	DYGGRLTPPT	480
PNCTAELTPA	WHNVTVAMEK	DSSVFNEFWA	RRTRGYSVPA	CTGDCVKNEI	CALRGADAQY	540
SCVQRTPGFS	PSKRDDGEVE	VDLENAPLLS	KRFQPECNHA	GMAPLLAKIA	HKASI AKMNG	600
E						601

SEQ ID NO: 276 moltype = AA length = 602

FEATURE
source Location/Qualifiers
1..602
mol_type = protein
organism = Talaromyces bacillisporus

SEQUENCE: 276

GQTPTSSLVA	RDLASEIWN	IKEAATCAGC	KVILAALKGV	ADLGTTLVID	VLTEVKISG	60
EEDDDVCEGI	ISREGPVLEY	ILSQLSLSSE	TSVLCASAF	GLCSYPAVRD	YTLTFPSKPK	120
ANITRPPSPG	KSPIQVHFS	DTHVDLSYET	GSNNWNCTKPI	CCRAYEASDA	PGNTTTPCGP	180
YGNTKCDAPL	SLEQSMIDSI	KALDPAPAFS	IYTGDDVVAHD	IWIDEDEV	TDLNASYSLM	240

-continued

AETGKVFAAI	GNHDSAPVND	LPTTQVPSKY	NANWTYQALA	NNFSTLTGDS	AVLSVAEQYG	300
SYSSVFTGSY	GTDLKVSYN	SIFYYIDNFI	AFLDPMYPDP	DGQLAWLIEE	LQASETAGQR	360
VWLIHVPTS	SSDHFHDYSH	YFDQIVQRYE	ATIAALFFGH	THTDQFQISY	SNYSNQADT	420
ASAIGYIMPS	LTPTSGLPPAY	RVYDIDPVTF	GVLDFTVYIS	NISDPAFQNG	PTWSKYYSK	480
ETYGSLSP	VTDSTAELTP	AFWHNVTAVF	ETDDDAFQGY	WARQTRGYDV	SNCTDTCKNQ	540
TICGIRGADA	QYNCVVPKIG	FNFAKRDETD	QAHVKTQKEK	CDDAGLASLF	GRMVANSKNA	600
SN						602

SEQ ID NO: 277 moltype = AA length = 604
 FEATURE Location/Qualifiers
 source 1..604
 mol_type = protein
 organism = Galactomyces candidus

SEQUENCE: 277

APPTTKRSLA	SDIWDIVDA	VDCGACDTIL	EALKGLADLG	NTVFVDVLT	VCDISGAEDS	60
DVCSGTISEE	GPILRTIIG	LSVGSATSDL	FCGTLGLGCQ	NPAIASWSVP	FPKPKPNTVR	120
PPPSGQSPIS	VAHISDVHVD	LSYTTGANYD	CSKPICCRPY	TSDDAPGNTD	YPAGPYGNTN	180
CDAPLDLESS	AMAAIKKLN	AFSIFTGDVA	AHDVWLNVQA	EVELDLNTTY	NTQFTTLGTV	240
FPALGNHDA	PVNGFAPS	SSNPNIQWAY	DTNAEDWTKW	IGSTAANAEE	SFGAYSIVHG	300
NLRVISFNSI	FYYRLNFYMY	QDPLQRDPSS	QFSWLVNQLQ	AAEDAGQRAW	LISHVPSGSG	360
DYFPQYSNYF	NQIVNRYEAT	IAALFYGHTH	VDQFEISYSD	YSNQNSNTAV	AMSYITPSLT	420
PTSGSPSRFI	YSIDPVTYGV	LDYTNVIANI	SSPTYQNGPQ	WVEYYSAKAA	YGPYVSPPLT	480
SAAAEITPAF	WHNVTVAFQN	NNDLQFEYIS	RKSRGFDVSS	CTGSCQTDEI	CQLRAAESQY	540
NCVTISPGIN	FNKRQDQSNL	GAQEKHHRDGC	EGSPIRDIFA	TLMQDRRGLV	SAINEGIAKR	600
SIRA						604

SEQ ID NO: 278 moltype = AA length = 607
 FEATURE Location/Qualifiers
 source 1..607
 mol_type = protein
 organism = Penicillium megasporum

SEQUENCE: 278

NIIESWASEI	WDDIKNAVDC	AGCETIIGAL	KVVADLGKGT	LNGTLIDVCD	LSGVEDPDVC	60
TGLISSEIDA	LYYSLKMMAV	DSHTSKVLCA	GLFSLCPPFA	ARPYNLSFPT	PKPATSRPAP	120
SGQQPIKVAH	ISDTHVDLDY	EAGSNYQCSK	PICCRPYTAE	DAPGNTSHPC	GPWGNTKCDP	180
PPRLEESMVA	AVNALNPSFS	IYTGDDVVAH	IWLNVNESEV	TDLNATYSLF	QNLNSLVYAA	240
VGNHDAVAPN	LFPSNKIDSA	YNPQWYDAL	TADWLALTNG	DSSVASAKAD	GSYSAIYPGT	300
NLRIISYNSI	FYYKDNFWMY	SDPMEYDPNG	QFAWLIDELQ	AAETAGQRVW	LISHIPSGNS	360
DHLYDYSHYF	DQIVQRYEAT	IAALFFGHTH	TDLFQVAYSD	YGNRNSDTAS	AIGYVTPSMT	420
PTSGPPAFRI	YVIDPVTFGV	LDYTVYIANI	SDPAYQTGPS	WQKYSAKEV	YGSLLSPPLT	480
DPAAELTPAF	WHNVTVLMEE	DDSVFQDWWA	RTTRGFNVST	CTGSCATNEI	CALRGADAQY	540
NCVTASPGIQ	FAKRGAVDF	DPNAQPVAKP	HCDEGSGLAP	VLVKMMRNTD	DFAGLLKERA	600
ALQDSQQ						607

SEQ ID NO: 279 moltype = AA length = 610
 FEATURE Location/Qualifiers
 source 1..610
 mol_type = protein
 organism = Penicillium jensenii

SEQUENCE: 279

ETSKASTESF	ISSIWDDFKQ	AVDCGSCQAL	LGLGLVSGF	GEGFMIDVFI	GLCNLSGVED	60
PDVCRGIIIEK	EGPALHDAFQ	NLHIGSHATR	TMCASLIGLC	QYPEVRPHTL	QFPSSKPDTT	120
RPPPSGKSPI	KVVHFSATHV	DLFYETGASY	ECSKPICCRV	YEDKDAPGIT	KTPCGPFGNT	180
KCDPPHILQE	SMNAAIAKID	PDFSITYGTDV	VAHDIWLVGQ	DEALQVFNMT	YGQMEKDLGM	240
VYAAIGNHDT	APVNLFPFND	IKGKDSAQFA	YNALAEWDYA	LTGIPSVKSA	DEFGYSIAIH	300
PNSNLRIISY	NSIFYYNFNF	YMYQDPMKED	PNGQFEWLIK	ELQAAEDAGQ	RAWLISHIPS	360
GVTDHFRDYS	QYFDQIVQRY	EATIAGLFYQ	HTHMDEFQIA	YSDYNNRKWD	TATAMGYIAP	420
SMTPTSGPPS	FRVYEIDPVT	YGVLDFTQYI	ANISDPSYQT	KPEWVPYISA	KAAYGSKLSP	480
PLTDSTAELT	PAFWHNVTVT	MEKDPSIFQD	FWARRNRGWN	IAACTGDCMK	KELCTLRAAD	540
AQHNCHEPTP	GLNISINKRD	QSGGDVPLEG	EKVSQPECDH	AGMATLLGKI	AYRARLVREA	600
EERDPVRAEA						610

SEQ ID NO: 280 moltype = AA length = 610
 FEATURE Location/Qualifiers
 source 1..610
 mol_type = protein
 organism = Aspergillus stramenius

SEQUENCE: 280

ENWVNTIWE	IKETISCAGC	EGLLGTCLKV	AGLGPDLVTN	VLTDVCKLAK	VEDPDVCAGI	60
IEAEGPAAYY	VLKQLKVGSH	TSKSFCSQMV	GLCDYPEVRP	YNISFPPIKP	STHRPPPSGQ	120
PPIRVAHISD	THVDRAVETG	ANYECSKPIC	CRVYTEDDAP	GKTSFPCGPY	GHPKCDPPLR	180
LEESMMAAIA	AMDPAFSIYT	GDVVPDHDVWS	VNRTEVLHDL	NATYSLDLRL	GLVYAALGNH	240
DTAPVNLFP	ERIPVSHNPQ	WAYDALAEDW	TNLVDGPLSA	PVVHATDQFG	SYSAHPGGK	300
LRIISYNSVF	YYTYNFYAYQ	EPMEYDPNGQ	LAWLVAELQA	AETAGQRVWL	IAHIPTGAAD	360
TLRDYSHYLD	QIIQRYDATI	AALFFGHHTH	DLFQVSYANP	AHPSADSASA	VGYYITPSLTP	420
TSGPPAFRIY	DIDPVTFAVL	DYTVYTANIS	TGATPKWNKY	YSAKQTYGSL	LTPPLTDPTA	480
ELTPAFWHNV	TALMETDNTV	FQAWWARTTR	GFNVPECAQ	CARDQICSLR	AADAQYGCVR	540
GTLISITKRAG	DGDGLDLGGA	GAGGPPGRRR	HVQSARPOQE	EAGLARVLAA	VIRETDDLQ	600
LLQRAQLYI						610

-continued

SEQ ID NO: 281 moltype = AA length = 258
 FEATURE Location/Qualifiers
 source 1..258
 mol_type = protein
 organism = *Bacillus pseudomycoides*

SEQUENCE: 281
 HTNDCGNEAP ILRWSAEDRH KEGVNSHLWI VNRAIDIMSR NNTIVKPNET ALLNEWRNEL 60
 ENGIYSADYE NPYVDNSTFA SHFYDPDTQK TYIPLAKQAK ETGSKYFKLA GEAYQNKDMK 120
 QAFFYLGSL LHYLGDVNQPM HAAANFTNLSY PMGFHISKYEN FVDTIKDNKY VADGNGYWNW 180
 KGTNPPEWIH GAAAAAKQDY PGIVNDSTKS GFVKAATSQE YANKWRAEVT PATGKRLTEA 240
 QRVTAGYIHL WFDITYVNR 258

SEQ ID NO: 282 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein
 organism = *Bacillus mycoides*

SEQUENCE: 282
 QENDGGNRVN IIQYWSAEDK HTEGVNSHLW IVNRAIDIMS RNMTLVKQDQ VALLNEWRTD 60
 LENGIYSADY ENPYVDNSTF ASHFYDPDDG STYIPFAKQA KETGAKYFKL AGESYKNKDM 120
 KQAFYVLGLS LHYLGDVNQPM HAAANFTNIS YPQGFHISKYE NFVDTIKDNKY KVTGNGYWNW 180
 WKGANPEDWI HGAATAAKQD FPGIVNSNTK SWFVKAASVQ SYADKWRAEV TPMTGKRLIE 240
 AQRVTAGYIQ LWFDITYVNR 259

SEQ ID NO: 283 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein
 organism = *Bacillus thuringiensis*

SEQUENCE: 283
 HEKTEGHNVN IIQYWSAEDK HSEGVNSHLW IVNRAIDIMS RNTTRVKQDQ VLLNEWRTD 60
 VENGIYSADH ENPYVDNSTF ASHFYDPDDG STYIPFAKQA KETGAKYFKL AGESYKNKDM 120
 KQAFYVLGLS LHYLGDVNQPM HAAANFTNIS YPQGFHISKYE NFVDTIKDNKY KVTGNGYWNW 180
 WKGHPEDWI HGAATAAKQD FSGIVNRNTK SWFVQAASVQ SYADKWRAEV TPMTGKRLIE 240
 AQRVTAGYIQ LWFDITYGNR 259

SEQ ID NO: 284 moltype = AA length = 262
 FEATURE Location/Qualifiers
 source 1..262
 mol_type = protein
 organism = *Listeria innocua*

SEQUENCE: 284
 CCDEYLQAPA APHDIDSKLP HKLSWSADNP TNTDVNTHYW LFKQAEKILA KDVNHIRANL 60
 MNELKNFDKQ IAQGIYDADH KNPYYDTSTF LSHFYNPDRD NTYLPGFANA KITGAKYFNQ 120
 SVADYREGKF DTAFYKLGLA IHYYTDISQP MHANNFTAIS YPPGYHCAYE NYVDTIKHNK 180
 QATEDMVVKR FCSDDVKVWL YENAKRAKAD YPKIVNAKTK KSYLVGNSKW KKDTEVEPTGA 240
 RLRSQQTILA GFLEFWSKKT NE 262

SEQ ID NO: 285 moltype = AA length = 275
 FEATURE Location/Qualifiers
 source 1..275
 mol_type = protein
 organism = *Aspergillus egyptiacus*

SEQUENCE: 285
 APAPLTRRDV SSEVLEQLTL FAEYSAASYC PSNLDSPGTK LTCSTGNCPT VEAADTETLA 60
 EPYHADEYGD VAGYLAVDTT NQLLVVAFRG SRALDTWIAN LNFVKDSVDD LSCGCEVHGG 120
 FWQSWQVVAD SVASGVESAL QTYPDYITVF TGHSPGGAVA TLGAVELRNA GYDIELYPYG 180
 APRVGNEALA QYITDQGSNY RVTHNTNDIV RLPPMSFGFS HSSPEYWITS DDEVTPTTAD 240
 VEVIQGVGST EGNAGEFPQS TAAHSHYIID ISACE 275

SEQ ID NO: 286 moltype = AA length = 276
 FEATURE Location/Qualifiers
 source 1..276
 mol_type = protein
 organism = *Aspergillus tamarii*

SEQUENCE: 286
 TPAPLRRDVS SLLNLDLDF AQYSAAYCS ENLNTGTKL TCSVGNCPV ELASTNTLDE 60
 FDESSSYGNP AGYLAADETN KLLVLVSRGS SDLANWVANL NFGLEDASDL CSGCEVHSGF 120
 WKAWSEIADT ITSKEVESALS DHSYSLVLT GHSYGAALAA LAATALRNAG HSVQLYNYGQ 180
 PRLGNEALAT YITDQNKGAN YRVTHNTNDIV PKLPPTLLGY HHFSPEYYIS SADEATVTTA 240
 DVTEVTGIDA TGGNDGTDGT SIDAHRWYFI YISQCS 276

SEQ ID NO: 287 moltype = AA length = 277
 FEATURE Location/Qualifiers
 source 1..277
 mol_type = protein
 organism = *Aspergillus niger*

SEQUENCE: 287

-continued

APAPMQRRDI	SSTVLDNIDL	FAQYSAAAYC	SSNIESTGTT	LTCDVGNCL	VEAAGATTID	60
EFDDTSSYGD	PTGFIADVPT	NELIVLSFRG	SSDLSNWIAD	LDFGLTSVSS	ICDGCCEMHKG	120
FYEAWEVIAD	TITSKVEAAV	SSYPDYTLVF	TGHSYGAALA	AVAATVLRNA	GYTLDLNFG	180
QPRIGNLALA	DYITGQNMGS	NYRVTHTTDDI	VPKLPPELLG	YHHFSPPEYWI	TSGNDVTVT	240
SDVTEVVGVAD	STAGNDGTLL	DSTTAHRWYT	IYISECS			277

SEQ ID NO: 288 moltype = AA length = 349
 FEATURE Location/Qualifiers
 source 1..349
 mol_type = protein
 organism = Bacillus luciferensis

SEQUENCE: 288

ITSFFGNVQK	APAWSDEEDVH	NQDHSSTHFI	VNGSVKLIAD	NANPAINKPT	TLLNQFRDRW	60
EQGLYDADHI	NPFYDTGTFM	SHFYDPDTQT	NYTGASYPTA	RQSGAKYFNL	ASDYYKKGDF	120
NNAFYYLGVS	LHYFTDVTQP	LHASNISNLD	HHAPGYHSKF	ETYAESIQNE	MTVPDSGLYN	180
WIGSTDPEAW	IHQAAVQAKS	VLPQVWNTDI	INWFVQAAYS	NYYSAMWKNE	VKNPTLVQLN	240
QAERETAGFI	DMFFRVNGVE	MPVTYVYKNA	FGGVSELLGS	GNVDYDQLVK	GIGNDTISSI	300
HIAPGYQVTL	FSDANYKGAS	IVLTNDVHDL	GNFSHQVSSI	KIAKISALK		349

SEQ ID NO: 289 moltype = AA length = 258
 FEATURE Location/Qualifiers
 source 1..258
 mol_type = protein
 organism = Bacillus mycoides

SEQUENCE: 289

QTNNSENPTP	VLRWSAEDKH	NEGVSTHLWI	VNRAIDIMSR	NTAIVKPNET	ALLNEWRTDL	60
ENGIYSADYE	NPYYDNGTYA	SHFYDPDTGG	TYIPPAKQAK	ETGTYKPKLA	GEAYQNQDMK	120
QAFFYLGSL	HYLGDVNQPM	HAANFTNLSY	PMGFHISKYEN	FVDTVKDNVI	VSDSNGYWNW	180
KGTNPEDWIQ	GSAVAAKQDY	PGIVNDTTKD	WPFVKAASQ	YADKWRAEVT	PVTGKRLMEA	240
QRVTAGYIHL	WFDTYVNR					258

SEQ ID NO: 290 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein
 organism = Bacillus mycoides

SEQUENCE: 290

HENEGGNKVR	VIQYWSAEDK	HAEGVNSHLW	IVNRAIDIMS	RNTTVVKQDQ	VALLNEWRTD	60
LENGIYAADY	ENPYDNTSTF	ASHFYDPDTG	KTYIPPAKQA	KETGAKYFKL	AGEAYQKQEI	120
KQAFFYLGSL	LHYLGDVNQPM	MHAANFTNLS	YPQGFHISKYE	NFVDTIKNNY	KVADGNGYWN	180
WKGVPEDWI	HGAVAAKQD	YAGIVNGTTK	DWVFVRAVSQ	EYADKWRAEV	TLTTGKRLVE	240
AQRTVAGYIQ	LWFDTYVNR					259

SEQ ID NO: 291 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein
 organism = Bacillus sp.

SEQUENCE: 291

HENDGGHGVG	VIPRWSAEDR	HKEGVNSHLW	IVNRGIDIMS	HNTTVVKQDE	VALLNEWRTD	60
LENGIYSADY	ENPYDNTSTF	ASHFYDPDTG	TTYIPPAKQA	KETGAKYFKL	AGESYQNKDM	120
KQAFFYLGSL	LHYLGDVNQPM	MHAANFTNLS	YPQGFHISKYE	NFVDTIKNNY	KVNDGNGYWN	180
WKGVPEDWI	HASVAAKQD	FPSIVNDNTK	DWVFVKAASQ	DYANKWRAEV	TPMTGKRLME	240
AQRTVAGYIQ	LWFDTYVNR					259

SEQ ID NO: 292 moltype = AA length = 263
 FEATURE Location/Qualifiers
 source 1..263
 mol_type = protein
 organism = Bacillus drentensis

SEQUENCE: 292

ARVNHDSYD	SGIISPYWS	AEEMHTEGKN	THLWIVNRAI	DIMARDNTTV	KENEVALLNE	60
WRTDLEDGIY	TADYENPYD	NGTFASHFYD	PDTDDTYIPF	AKNAKVTVGK	YFKLAGEAYQ	120
QQAMNQAFY	LGLSLHYFGD	INQPMHASNF	TNISHPFGEH	SKYENFVDTI	KAPYSVTDGN	180
GYWNFAGETP	EEWLHTAAVA	AKQDAPGIVN	ETTISWFLQA	AFSQEYADMW	RAEVTPETGA	240
RLIEAQRAMA	GYIHLWFDY	VNL				263

SEQ ID NO: 293 moltype = AA length = 275
 FEATURE Location/Qualifiers
 source 1..275
 mol_type = protein
 organism = Aspergillus turcosus

SEQUENCE: 293

APAGLAERDV	SASVLQELSL	FAQYSAAAYC	TNNINSTGTK	LTCAGNCL	VEAANTKTLA	60
EFYDSDSFGD	TAGFLVADTT	NKLLVVSFRG	SRTLDNWIAN	LDFVLSISD	ICSGCAAHGG	120
FWKSWEVVAN	SLTTELNSAV	NTYPGYTIVF	TGHSGLAALA	TLGATTLRKA	GIPVQLYNYG	180
SPRVGNKALA	TYITAQGPNY	RVTHTNDIVP	RLPPQSFGFS	HLSPEYWITS	GDNVPVTTSD	240
ITVIQIDSN	AGNSGEDITS	IEAHNWIYGN	IDACP			275

-continued

SEQ ID NO: 294 moltype = AA length = 276
 FEATURE Location/Qualifiers
 source 1..276
 mol_type = protein
 organism = Talaromyces subinflatus

SEQUENCE: 294
 VLSPIGRRTV TTTQLDDMNL FAQYSAAAYC SANLNSTGSA LACNVGNCPL VEGADTTILY 60
 DFDESAGFGD ATGYIAVDET HKSIIILAFRG SSDLDNWIAD LDIPLVASSI CLGCEVHQGF 120
 WDTWQTVASD VTSQVEYALS AYAGYTFVVT GHSMGAALAA IAATVFRDSG YTVELYNYGQ 180
 PRIGNLILAY YITQNHGGSN YRVTHTDIV PKLPPELLGY DHFSPPEYWI SGDNVTVTDS 240
 DIDVIVGIDS ADGNDGTIDD SVEAHHWYFV YISECS 276

SEQ ID NO: 295 moltype = AA length = 277
 FEATURE Location/Qualifiers
 source 1..277
 mol_type = protein
 organism = Aspergillus tubingensis

SEQUENCE: 295
 APAPMQRRDI SSTVLNIDL FAQYSAAAYC SSNIESTGTT LTCDVGNCPL VEAAGATTID 60
 EFDDSSSYGD PTGFIADVPT NELIVLALRG SSDISNWIAD LDFGLTSVSD ICDGCEMHKG 120
 FYEAWEVIA D TITSKVEAAV SSYPDYSIVF TGHSYGAALA AIAATVLRNA GYTLDLYNFG 180
 QPRIGNLALA DYITDQNMGS NYRVTHTDI VPKLPKLLG YHHFSPPEYWI TSGNDVTVT 240
 SDVTEVVGV D STDGNDGTL DSTTAHRWYT IYISECS 277

SEQ ID NO: 296 moltype = AA length = 350
 FEATURE Location/Qualifiers
 source 1..350
 mol_type = protein
 organism = Bacillus acidicer

SEQUENCE: 296
 ITSLSFNDQK AFAWSDEVDH NQDHSSTHFI VNGSVKLIAD NTNPAINKPT TLLNQFRDRW 60
 EQGLYDADHI NPFYDTGTFM SHFYDPDTQT NYTGVSYPYA RQSGGKYFNL ASDYYKKGDF 120
 YNAFYVLGVS LHYFTDVTQP LHASNISNLD HNAPGYHSKF ENYAESI QN MAIPDSGLYN 180
 WISSTDPEAW IHQAQVQAKS VLPQVWNTDI INFFWQAAYS NYSSMWKNE VKNPTLVQLN 240
 QAERETAGFI DMFPRVNGVE MPVKVYKENA FGGASEILGL GNYDYDQFVK GIGNDTISSI 300
 HIAPGYQVTL FSDANYKGTS TVLTGDVNDL GNFNHQVSSL KIVKISAISK 350

SEQ ID NO: 297 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein
 organism = Lysinibacillus xylanilyticus

SEQUENCE: 297
 HENCYQDPPI PLKWSAESIH NEGVSSTHFW VNRAIDIMSQ NTTIVKQHET DLLNEWRTDL 60
 EBGIYSADYQ NPYDNDSTFA SHFYDPDSGK TYIPPAKQAK QTGAKYFKLA GEAYQNKDLK 120
 NAFYVLGLSL HYLGDVNQPM HAANFTNISH PFGPHSKYEN FVDTVKDNRY VTDGNGYWNW 180
 KSANPEEWHV ASAVAADKAD LLIIVNDNTES GFLKAAVSQD SADKWRAEVT PVTGKRLMEA 240
 QRITAGYIHL WFDTYVNNK 259

SEQ ID NO: 298 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein
 organism = Bacillus toyonensis

SEQUENCE: 298
 HENEDVNYNA PILRWSAEDK HKEGVNSHLW IVNRSIDMMS RNTTIVKKQV VALLNEWRTD 60
 LENGIYNADH ENPYDNDSTF ASHFYDPETG STYIPLVSTQ AKEAGSKYFK LAGESYKKN 120
 MKQAFYVLGL SLHYLGDVNQ PMHAANFTNL SYPQGFHSKY ENFVDTIKDN YKVNDGNGYW 180
 NWKGSNPGDW IHGAAVAACK DYTGIVNDTT KDWFKAAIS SEYADKWRAE VTPATGKRLM 240
 EAQRITAGYI QLWFDTYAN 259

SEQ ID NO: 299 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein
 organism = Bacillus wiedmannii

SEQUENCE: 299
 HENDGGSKIK IVHRSWAEK HKEGVNSHLW IVNRAIDIMS RNTTIVKQDR VTQLNEWRTD 60
 LENGIYAADY ENPYDNDSTF ASHFYDPDNG KTYIPLAKQA KETGAKYFKL AGESYKKN 120
 KQAFYVLGLS LHYLGDVNQ PMHAANFTNL SYPQGFHSKY ENFVDTIKDN YKVNDGNGYW 180
 WGTNPEEWI HGAAVAKQD YSGIVNDNTK DWFVKAASQ EYADKWRAE TPMTGKRLMD 240
 AQRVTAGYIQ LWFDTYGDR 259

SEQ ID NO: 300 moltype = AA length = 268
 FEATURE Location/Qualifiers
 source 1..268
 mol_type = protein
 organism = Listeria seeligeri

SEQUENCE: 300

-continued

CGDESIKQDI APHAIQNKLP SKLGWSAEHP SKDEINTHLW LFNQAEKILA KDVGTGAQLDL 60
 VRELKYNKE IAQGIFDADH KNPYYDKNTF LSHFYNPKTH KTYIPGFPNA KDTGTYFNI 120
 SVEEYQDGNF EKAFYNLGLA IHYYTDVSQP MHANNFTALS HPVGYHCAYE NYVDTPKQIF 180
 QASAESEAKW FCTDDISEWY HENAKRAQAD YPKIVNAIHK KSVIQGLSDS QKDRTWKKAV 240
 RAATGKRLRD SQETLAGFLE FWYAKTNE 268

SEQ ID NO: 301 moltype = AA length = 275
 FEATURE Location/Qualifiers
 source 1..275
 mol_type = protein
 organism = *Penicillium swiecickii*

SEQUENCE: 301
 APSRPVPRDV STSVLSQLSL FAEYSAASYC SNNINSTGNA LSCEAGNCPS VQSADTTTLW 60
 EFDRTCSYGN VAGFLAVDKT NKLLVVSFRG SRSISNWIAN INFGLTADPS LCSGCEAHSG 120
 FLESWETVAD DLTTNIKSAQ STYSGYTLVL TGHSPGGAVA ALGGTALRNG GSTLNVITYG 180
 QPRVGNLALA SYITNQGSW RVHTDDIVP KLPPSSFGFS HPSPEYWITS ENEVTVTSSD 240
 VEVIEGVGSK SGNAGTLNPD VEAHNWYLG YIDGCG 275

SEQ ID NO: 302 moltype = AA length = 276
 FEATURE Location/Qualifiers
 source 1..276
 mol_type = protein
 organism = *Talaromyces boninensis*

SEQUENCE: 302
 VPTPVTRRTV STALLDTLTL FAQYSAAYC PANFNSSSTS LACSAGNCPT VQAADTTILY 60
 SFDKSASFGD ATGYVAVDNT NQLIVIAFRG SSDLSNWIAN LDVPFTDAGN ICSGCEVHSG 120
 FYDTWQTVAS DITATVDSAL STYPGYTVVA TGHSLGGALA AIGATVLRSS GQVVQLYDYG 180
 QPRIGNLALA DFITSETAGS NYRVTHSDDI VPKLPPEFLG YAHFSPPEWI TSGDNVAVTD 240
 ADIVEVIGVD STAGNDGTFG DSINAHWYF EAISAC 276

SEQ ID NO: 303 moltype = AA length = 277
 FEATURE Location/Qualifiers
 source 1..277
 mol_type = protein
 organism = *Hamigera striata*

SEQUENCE: 303
 APAPILRRDV SASVLNLDL FAQYSAAYC SSNIGSTGNK LMCNVGNCPR VEASDVTVID 60
 EPNESASYGD VAGYIAVDNT NQLLVLSFRG SSSLSNWIAN IDVDLTDASS LCSGCEVHSG 120
 FWSAQWTQVG TITSKLESAR ASYPGYTLVF TGHSYGAALA GLAATTLRDA GWTIQLYNYG 180
 QPRLGNLALA QYITSQTQGS NYRVTHSDDI VPKLPPEFLG YDHYSPEWI TSGDNVTVTT 240
 SDVQVIEGID SVAGNDGTS DSTEAHQWYF IYISECS 277

SEQ ID NO: 304 moltype = AA length = 351
 FEATURE Location/Qualifiers
 source 1..351
 mol_type = protein
 organism = *Bacillus* sp.

SEQUENCE: 304
 ITSLEGSFQK AFAWSCDDPH NQDQSTHLFI VNNGVKLISG NADPAINKPS TLLEQFRDRW 60
 EQGLYDADHI NPFYDTSTFM SHFYDPDTQT NYAGRSYPTA RQSGAKYFNL ASDYYKNGDF 120
 YNAFYVLGVS LHYPDTATMP LHASNISNLD HQAPGYHSLK ESYSESIQDQ VTPVDSGLFN 180
 WVSSTDPELW IHQAQVQAKS VLPQVWNSDI ISWFWQAAYS NYYSMDMWKSA VKAPILNQLN 240
 QAERETAGFI DMFRLNGVE MPVTVYSETA FGGASELLGS GNYDYDQLVK GIGNDAISSI 300
 HIAPGYQVTL FADSNYSYGAS KVLTDASDL DNFNKTISSL KIEKIQPVNV H 351

SEQ ID NO: 305 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein
 organism = *Bacillus thuringiensis*

SEQUENCE: 305
 HENAGGQVRG VIPRWSAEDK HKEGVNSHLW IVNRAIDIMS RNTTIVKQDQ VALLNEWRTD 60
 LENGIYSADF ENPYDNTSTF ASHFYDPDSE KTYIPLAKQA KETGAKYFKL AGESYQNNDM 120
 KQAFPLYLGLS LHLYGDNVQP MHAANFTNLS YPQGFHSHKY NFVDTIKDNV KVTGDTGYWN 180
 WKGTNPEDWI HGAADVAKQD YSGIVNNNTK EWFVKAAVSQ EYADKWRAEV TPMTGKRLID 240
 AQRITAGYIQ LWFDTYVNR 259

SEQ ID NO: 306 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein
 organism = *Bacillus mycoides*

SEQUENCE: 306
 HENCHQDPPI ALKWSAESIH NEGVSSEHLW VNKAIDIMSQ NTTVVKQNET ALLNEWRTDL 60
 EKGISADYQ NPYDNTSTFA SHFYDPDSGK TYIPFAKQAK QTGAKYFKLA GEAYQNKDMK 120
 NAFFYLGLSL HYLGDVNQPM HAANFTNISH PFGFHSKYEN FVDTVKDNV VTDGNGYWNW 180
 QSTNPEDVWH ASASAADKAF PSIVNDNTKN WFLKAAVSQ SADKWRAEV PVTGKRLIEA 240
 QRITAGYIHL WFDTYVNNK 259

-continued

SEQ ID NO: 307 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein
 organism = Fictibacillus macauensis

SEQUENCE: 307
 HENEGGNKVR VIQWSAEDP HHEDTNTLW IVRHAMEIMA NNKDVVKPGE VEQLKQWQSD 60
 LEQGIYDADH ANPYDNDATF ASHFYDPDTG KSYIPLAAHA KTSVKYFKR AGEAYQKGDH 120
 KOAFYNLGLA LHYIGDLNQP MHAANFTNLS YPQGFHASKYE NYVDSFKEDY AVKDGEYWH 180
 WKGTNPEDWL HGTAVAAKKD YPDIVNDTTK AWFVKAASVN SYAAKWRAAV VPATGKRLTE 240
 AQRILAGYMQ LWFDTYVVK 259

SEQ ID NO: 308 moltype = AA length = 268
 FEATURE Location/Qualifiers
 source 1..268
 mol_type = protein
 organism = Listeria seeligeri

SEQUENCE: 308
 CGDESVDQI APHDIQNKLP SKLGWSAHP SKNEINTLW LFNQAEKILA KDVTGAQLDL 60
 VRELKYNKE IAQGIQDADH KNPYDKNTF LSHFYNPETH KTYIAGFPNA KDTGTYFNI 120
 SIEEYQDGNF EKAFYNLGLA IHYYTDISQP MHANNFTALS HPVGYHCAYE NYVDTFKQIF 180
 QASAESAQW FCTDDVSEWF HENAKRAQAD YPKIVNTIHK KSYIQGLSDS QKDRTWKKA 240
 RAATGKRLRD SQETLAGLLE FWYTKTNE 268

SEQ ID NO: 309 moltype = AA length = 275
 FEATURE Location/Qualifiers
 source 1..275
 mol_type = protein
 organism = Penicillium donkii

SEQUENCE: 309
 APARPVPRDI SSSLLDELTL FAEYAAASYC SNNIDSTGDA VTCSGDYCPL VQSAGAKTLY 60
 EFNDSTEWGD VAGFLAVDTT NKLIVLSFRG SRSISTWIAN LDFGLTDTSS LCDDCEAHSG 120
 FWKSWETVAD DMTAQIESAQ SSYPSTLV L TGHSPGAAVA ALGATALRNA GYTLDLYTYG 180
 QPRVGNELAL TYMTSQGSLW RVTHEDDIV KLPPMSWGFH HASPEYVWTS DSDVTVTTS 240
 VEEVVGVDST AGNAGTSGES ISAHNWFYVE IDGCD 275

SEQ ID NO: 310 moltype = AA length = 276
 FEATURE Location/Qualifiers
 source 1..276
 mol_type = protein
 organism = Hamigera paravellanea

SEQUENCE: 310
 APAAVRRDVS AGVLANDLDF AQYSAAAYCD SNLNSDGTGL TCQAGNCPLV EAADTETLDE 60
 FDMTATYGNV AGYIAVDRTN RLLVLAFRGS ASISNWIANL NLGLTDASAL CAGCRVHSGF 120
 WEAWQTAEAT MSDIISAAQ TYPGYTLVAT GHSGYGAALA IAATKFRNEG YAVELYDYGQ 180
 PRIGNLALAQ YITNQSSGGN FRVTHNDIV PKLPPDWLGY SHFGPEYWIT SGDGIPTTTA 240
 DVEVISGVDA TGGNDGAEGT SVDHRWYFV YISQCE 276

SEQ ID NO: 311 moltype = AA length = 277
 FEATURE Location/Qualifiers
 source 1..277
 mol_type = protein
 organism = Talaromyces lecycteanus

SEQUENCE: 311
 APAPVLRDV SSSVLSLDL FAQYSAAAYC SSNIGSPGK LTCVGNCPV VEAADTETLI 60
 EFNESSSGD VTGYIAVDRT NSLLVLAFRG SSTVSNWEAD LDFPLTDASS LSCGCEIHSG 120
 FWAAWQTVQA SITSTLESAL ASYPGYTLVF TGHSGYGAALA AIAATTLRNA GYTIQLYDYG 180
 QPRLGNLALA QYITAQTQGA NYRVTHDDI VPKLPPDLFG YHHSPEYWI TSGDNVTVT 240
 SDVQVVTGID STAGNDGTLT DSTSAHDWYI VYIDGCD 277

SEQ ID NO: 312 moltype = AA length = 449
 FEATURE Location/Qualifiers
 source 1..449
 mol_type = protein
 organism = Paenibacillus sp.

SEQUENCE: 312
 ITSIFGNSQD AYAWSADDPH SQDFSTHLFI VNGGVKLISG NVDSAINKSS TLLEQFRGRW 60
 EQGLYDADHL NPFYDSTTFM SHFYDPDTQT NYAGLSYPTA RQSGSKYFKV ASNYKNGDF 120
 SNAFYVLGV LHYFTDSTMP LHASDISNLD HRAPGYHAKL EEYATSIQNG INVPDSGLFN 180
 WISSTDPELW IHQAQVQAKS VMPEVFNDTI TDWFWKAAPS YYYSDMWKSA VKIPILNQLN 240
 QAERETAGYI DLFFRLNGVD MPVAVYKGT FGGALQLLGF GNYDYDQLVK GIGNDTVSSI 300
 RIAPGYQVTL PADSNSYGV S KVLTDASDL GNFNKTSSL KIEKIQPVTV YTDASFSGSS 360
 QSFSGVNHDI NEIVNRKLN TISSIRIAPG YQVTLFRDSN YSGVSTVVTG DVGSLNLD 420
 QTSSLKVEVI PTNPAPSQTK QSIFSNPLN 449

SEQ ID NO: 313 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein

-continued

```

                                organism = Bacillus toyonensis
SEQUENCE: 313
HENDGGQVRVG VIPRWSAEDK HKEGVNSHLW IVNRAMDIMS RNTTLVKQDR VALLNEW RTE    60
LENGIYAADY ENPYDYNSTF ASHFYDPDNG KTYIPYAKQA KETGAKYFKL AGESYKNKDM    120
KQAFFYLGSL LHLYGDNQNP MHAANFTNLS YLQGFHSKYE NFVDTIKDNV KVTGNGYWN    180
WKGTNPEDWI HGAAVVAKQD YAGIVNDNTK DWFVRAAVSQ EYADKWRVEV TPMTGKRLMD    240
AQRVTAGYIQ LWFDYGNR                                     259

SEQ ID NO: 314          moltype = AA length = 259
FEATURE                Location/Qualifiers
source                 1..259
                        mol_type = protein
                        organism = Bacillus thuringiensis

SEQUENCE: 314
DEHTTNKID IFQPSNEEN HKEGVNSHLW IVNGAIDIMS RNTTIVKQEN LALLQQRWTH    60
LENGLYVADY ENPYDYNSTF ASHFYDPDNG KTYIPYAKQA KETGATYFTL AGEAYQHKNI    120
QKQAFFYLGVS LHLYGDNQNP MHAANFTNLS YPFGFHSKYE HFVDTIKQNY EIMDGEGYWN    180
WKGRDPEDWI HQAAVAANQD FSDIVNSDTK NWFVKAASVQ TYADRWRAAV TPITGKRLIE    240
AQRITAGYIQ LWFDYIYHQ                                     259

SEQ ID NO: 315          moltype = AA length = 259
FEATURE                Location/Qualifiers
source                 1..259
                        mol_type = protein
                        organism = Bacillus thuringiensis

SEQUENCE: 315
HENDGGSKIK IHRWSAEDK HKEGVNSHLW IVNRAIDIMS RNTTLVKQDR VAQLNEW RTE    60
LENGIYAADY ENPYDYNSTF ASHFYDPDNG KTYIPYAKQA KETGAKYFKL AGESYKNKDM    120
KQAFFYLGSL LHLYGDNQNP MHAANFTNLS YPQGFHSKYE NFVDTIKDNV KVTGNGYWN    180
WKGTNPEDWI HGAAVVAKQD YSGIVNDNTK DWFVKAASVQ EYADKWRRAEV TPMTGKRLMD    240
AQRVTAGYIQ LWFDYIGDR                                     259

SEQ ID NO: 316          moltype = AA length = 269
FEATURE                Location/Qualifiers
source                 1..269
                        mol_type = protein
                        organism = Talaromyces rugulosus

SEQUENCE: 316
DVTAVLDDL TLFSQYSAAY YCSTNLNSTG VAVTCSVGNL PLVEAADTQI LYDFDESCKF    60
GDASGFIADV NTNLIVLSF RSHDLSNWI ANLDFFLVET ASICEGCMH GGFWEWQTV    120
AANVTEQLEA AITANPGYTL VMTGHSLGAA LAAIVATEFR NEGIDVEMYN YGQPRGLNLA    180
LAQYMTNQQT TSNYRVTHSD DIVPKLPPRV LDFDHYSPEY WITSANNVSV SDADVQVVG    240
IDSTDGNDGT ILDDIEAHRW YLGYISECS                                     269

SEQ ID NO: 317          moltype = AA length = 275
FEATURE                Location/Qualifiers
source                 1..275
                        mol_type = protein
                        organism = Penicillium sp.

SEQUENCE: 317
APARPVPRDV SSATLSELTL FAEYAAAYC SNNIDSTGDA LSCSGGYCPE VQSAGATTLY    60
EFEDSTDFGD VTGFFAVDNT NKLLVLSFRG SSSISNWIAN LDFGLTDASS LSCGCEAHSG    120
FYKSWGVDV TLTAQVASAV STYPSYTLVV TGHSLGGALA ALGGTALRNA GYTLDIYTYG    180
QPRVGNATAL DYMTNQGSGL RVTHSNDIVP KLPPASWGFT HASPEYWITS GNDVTVTSD    240
VTEVTGVGSS DGNAGSSGDS VSAHNWYIVD IDGCS                                     275

SEQ ID NO: 318          moltype = AA length = 276
FEATURE                Location/Qualifiers
source                 1..276
                        mol_type = protein
                        organism = Hamigera avellanea

SEQUENCE: 318
APAPVRRDVS AGVLANLDFL AQYSAAAYCD SNLNSDGTKL TCAAGNCPLV EAADTETLDE    60
FDMTATYGNV AGYIAVDRTN RLLVLAFRGS ASISNWIANL NLGLTDASAL CAGCEVHSGF    120
WEAWQTAET ISDIIASAAQ TYPGYTLVVT GHSYGAALAA IAATFRFRNEG YAVELYDYGQ    180
PRIGNLALAQ YITNQSGGGN FRVTHNDIVP PKLPPDWLGY SHFGPEYWIT SGDVPVTTA    240
DVEVISGVDA TGGNDGAGBT SVDHRWYFV YISQCE                                     276

SEQ ID NO: 319          moltype = AA length = 278
FEATURE                Location/Qualifiers
source                 1..278
                        mol_type = protein
                        organism = Penicillium spikei

SEQUENCE: 319
APTRLLIPRD ISSDVLAELE LFAEYSAAAY CSANIDSASA GSALTCESGN CPEVQSADTS    60
TLYEFDETD YGDVAGFFAV DKTNELLVLS FRGSRITSNW VANLDFDLTD ASSLCSDCEA    120
HSGFWKSWET VADELTTQIE SAQNSYPDYQ LVLTGHSGLA ALAALAGTAL RNAGYTLDLY    180
TFGQPRVGNL ALADYMTDQG SLWRVTHTDI IVPRVPPESE GYAHASPEYW ITSGNDVTVT    240
TSDVEEIVGV NSSAGNAGEA DLSIDAHNWI IVYIDBCE                                     278

```

-continued

SEQ ID NO: 320 moltype = AA length = 480
 FEATURE Location/Qualifiers
 source 1..480
 mol_type = protein
 organism = *Paenibacillus alginolyticus*

SEQUENCE: 320
 ITSVPGNYQN AYAWSADDVH NQDYSTHLFI VNGGVKLISG NADSAINKSS TLLEQFRDRW 60
 EQGLYDADHL NPFYDSSTFM SHFYDPDTQT NYAGLSYPTA RQSGSKYFKV ASDYYKNGDF 120
 SNAFYVLGIS LHYPDTSTMP LHASDISNLD HRAPGYHAKL EBYTTSIQNQ ITVPDSGLFN 180
 WISSTDPELW IHQAAVQAKS VMPQVFNDIS TDWFWKAAVS YYYSDMWKNT VKTQILNQLN 240
 QAERETAGYI DLFFRLNGVD MPVTVYNGTA FGGASQLLGF GNYDYNQLVK GIGNDTISSI 300
 RIAPGYQVTL PADSNYSQVS TVVTGNVYGL SNLNDQTSSL KVGVIPTNPA PSPTIQAESF 360
 SGSQGILTHS AGSGTVVUNI NSGSWLSYDN VDFGTGKTKF VASVGMDPAF AAIGKQLELR 420
 LDSPTGTLLG TFTISSSGGW DAYTTQNCYV TSASGTHKLY IITKNGSGF GNIDWFTFSS 480

SEQ ID NO: 321 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein
 organism = *Bacillus mycoides*

SEQUENCE: 321
 HENDGGSRLN IVHRWSAEDK HKEGVNSHLW IVNRAMDIMS RNTTLVKQDQ VALLNEW RTE 60
 LENGIYAADY ENPYDNTSF ASHFYDPDNG KTYIPFAKQA KETGAKYFKL AGESYKNDKM 120
 KQAFYVLGLS LHLYGDNQNP MHANFTNLS YPQGFHSKYE NFVDTIKDNV KVTGNGYWN 180
 WKGTPNPEDI HGAATAAKQD YSGIVNDNTK DWFVKAASVQ EYADKWRAEV TPMTGTRLMD 240
 AQRVTAGYIQ LWFDTYGNR 259

SEQ ID NO: 322 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein
 organism = *Bacillus bingmayongensis*

SEQUENCE: 322
 HENEGGNKIR VIQYWSAEDK HAEGVNSHLW IVNRAIDIMS RNTTIVKQDE VALLNEW RTE 60
 LENGIYAADY ENPYDNTSF ASHFYDPDSG KTYIPFAKQA KETGAKYFKL AGESYKQDEM 120
 KQAFYVLGLS LHLYGDNQNP MHANFTNLS YPQGFHSKYE NFVDTIKDNV KVIDGNGYWN 180
 WKGTPNPEDI HGAATAAKQE YAGIVNDTTK DWFVWAASVQ EYADKWRAEV TPATGKRLVE 240
 AQRVTAGYIQ LWFDTYGNR 259

SEQ ID NO: 323 moltype = AA length = 260
 FEATURE Location/Qualifiers
 source 1..260
 mol_type = protein
 organism = *Bacillus mycoides*

SEQUENCE: 323
 HENSHQNGIS FGHKWSAEAI HDEGVSTHLW IVNRSIEVMA QNKTVVQPN E ISLLNEW RAD 60
 LEKGIYSADY DNPYFDNGTF ASHFYDPDTG GTYLPALAKA KETGAKYFKL AGEAYQNNDL 120
 KNAFFYVLGLS LHLYGDNQNP MHANFTNLS LPVALHSKYE NFVDTIKDNV KVIDGNGYWN 180
 WKSVPNPEDI HASAVGAKAD FPLIVNDKTK KWFLDAAISQ DAADKWRAEV TPVTGKRLME 240
 AQRITAGYIH LWFDTYVNYK 260

SEQ ID NO: 324 moltype = AA length = 269
 FEATURE Location/Qualifiers
 source 1..269
 mol_type = protein
 organism = *Brevibacillus sp.*

SEQUENCE: 324
 KKEYVKVYHG KTITSPYKID PRWSEESPHE EGHATHLWIV NRAIDILSRT SNKDVNSKET 60
 EMLNAWRSSW EQGLYDADHT NPYYNFGTFA SHFYDPDTKS NWLDTSGTAL TEGSRYFALA 120
 GKYYQNGDKE KAFYVLGLSL HYLTDVTQPM HAANFTWLNW PTSFHKGFED YTDIIGNYA 180
 VTDGEGYWDF QDSNPEWHI QAAVDAKAEF PNIHTSDITK WFLAAVSDY YSDKWHKAVQ 240
 PTIEHRLTEA QRITAGYLHL WFKTYVDNQ 269

SEQ ID NO: 325 moltype = AA length = 275
 FEATURE Location/Qualifiers
 source 1..275
 mol_type = protein
 organism = *Penicillium vasconiae*

SEQUENCE: 325
 APARVPVPRV SSATLSELTL FSQYAAAAYC TNNVNSAGDA VSCSGGYCPE VQSAGATTLY 60
 EPDDSTDFGD VAGFFAVDAT NKLLVLSFRG SRTISNWIAN LDFGQTDASS LSCGCEAHSG 120
 FPKAWAVAD TLTAQIASAV ATYPSYTLVL TGHSFPGGAVA ALGGTALRNA GYTLDLTYTG 180
 QPRVGNATAL DYMNTQGSLS RVTHSDDIVP KLPPTSWGFT HASPEYWITS GDDVTVTTS 240
 VTEVTGVGSS GGNAGTSGDS VSAHNWYIVD IDGCQ 275

SEQ ID NO: 326 moltype = AA length = 277
 FEATURE Location/Qualifiers
 source 1..277

-continued

```

mol_type = protein
organism = Talaromyces diversus

SEQUENCE: 326
AIDPLNRRTI SESLLDELDF FAQYSAAAYC SANLDSTGSA LACDVGNCPVL VEAASTTILY 60
DFDETNDPFG ATGYIAVDTT NEYIILSFRG TDDLENWIAN LDFPLIDASD ICSGCEIHG 120
WWDSWETVAS DITAQIESAV STYPDYTLVA TGHSLGAALA AIAATVLRDL GYTVQLYNYG 180
EPRIGNLALA DYITTETMGS NYRVTHTDDI VPKLPPELLG YDHFSPEYWI TSGDDVTVL 240
TDVTEVVGVD STAGNDGTLT DSIDAHRWYF VYISECS 277

SEQ ID NO: 327      moltype = AA length = 280
FEATURE            Location/Qualifiers
source              1..280
                    mol_type = protein
                    organism = Aspergillus wentii

SEQUENCE: 327
APSPVRRDVD SSVLNNLDF AQYSAASYCL ENLSSNTKL ECSVGNCPVL EAASTVTLDE 60
FDESSSFQDV TGFIAAETN KLLVLSFRGS SDIANWIADL DFGLTDGSDL CSGCKVHSGF 120
WEAWGTVSDN ITSIIESATA KYPNYELAF TGHSYGAALAA VAAVVFNRSG YTVQLYNYGQ 180
PRIGNLALAD YITNVTDKGD NYRVTHTDDI VPKLPPELLG YHHSPEYWI TSGNNVTVT 240
ADVDVVTGVD STDGNDGTTA DSRTAHRWYF GYISECSTLY 280

SEQ ID NO: 328      moltype = AA length = 480
FEATURE            Location/Qualifiers
source              1..480
                    mol_type = protein
                    organism = Bacillus acidiceler

SEQUENCE: 328
ITSLFGNVQR AFAWSDEDVH NQDHSTHHFI VNGGVKLIAD NTDPAINKPT TLLNQFRDRW 60
EQGLYDADHI NPFYDTGTFM SHFYDPDTQT NYSGLSYPTA RQSGGKYFNL ASDYYKKGDF 120
NNAFYVLGVS LHYFTDVTQP LHASNISNLD HNAPGYHSKY ETYAESIQSQ IIVPDSGLYN 180
WTDSTDPEAW IHKAAIQAKS VPLPLVWNDTI INWFVQAAYS NYYSAMWKNE VKNPTLAQLN 240
QAERETAGFI DMFPRRLGVE MPVTVYNENA FGGASELLGS GNYDYDQLIK CIGNDTISSI 300
HIAPGYQVTL FADANYKGAS IVLTGVDVNDL GNFNHQVSSL KIEKISTNPA SSPTIQAESF 360
ISSKGLLTHN VSGSTVVGNI NSGSWIGYDN VDFGTGKTKF IARVGMDEPSY AIFDKQLELR 420
LDSPTGTIIG TFTINNTGGW DTYATQTSIL SGATGTHKLY IVSKGSGDGF GNIDWITFSP 480

SEQ ID NO: 329      moltype = AA length = 259
FEATURE            Location/Qualifiers
source              1..259
                    mol_type = protein
                    organism = Bacillus luti

SEQUENCE: 329
HEKNEGNVPI ITQYWSAEDT HSEGVNSHLW IVNRAIDIMS RNTTLVRQDE VALLNAWRD 60
LEKGIYAADY ENPYDNTSTF TSHFYDPDTG KTYVGLAKQA KETGNKYFKL AGESYKNKDM 120
KQAFFYLGSL LHYLGDVNQP MHAANFTNLS YPQGFHSKYE NFVDTIKDNV KVTGNGYWN 180
WKGMPPEEWI HGAATAKQD YSGIVNSNTK SWFVKAASVQ SYADKWRAEV TPTTGKRLME 240
AQRVTAGYIQ LWFPTYGNR 259

SEQ ID NO: 330      moltype = AA length = 259
FEATURE            Location/Qualifiers
source              1..259
                    mol_type = protein
                    organism = Bacillus pseudomycoides

SEQUENCE: 330
HTNDCGNEAP VLKWSAEDKH NEGRNSHLWI VNRAIDIMS DKTIVVKPNET ALLNEWRDD 60
ENGIYSADYE NPYFDNGTFA SHFYDPDTQK SYIPFAKHAK ETGAKYFKLA GEAYQNKDM 120
QAFFYLGSL LHYLGDVNQP MHAANFTNLS EAPIFHSKYE NFVDTIKDNV KVDGNGYWN 180
WKGTSNEDWI HGAATAKQD YSSIVNDTTT SWFLKAATSQ EYANKWRAEV TPTSGKRLIE 240
AQRVTAGYIH LWFPTYVNR 259

SEQ ID NO: 331      moltype = AA length = 260
FEATURE            Location/Qualifiers
source              1..260
                    mol_type = protein
                    organism = Bacillus mycoides

SEQUENCE: 331
HENDGQFDPPI IQRWSAESI HNEGVSLLW IVNRAIDIMS QNTTVVKQNE TALLNEWRTN 60
LEEGIYSADY KNPYDHTSTF ASHFYDPDSG KTYIPFAKQA KQTGAKYFKL AGEAYQNKDM 120
KNAFFYLGSL LHYLGDVNQP MHAANFTNIS HPFGFHSKYE NFVDTVKDNV RVTGNGYWN 180
WKSANPEEWI HASASAAKAD FPSIVNDNTK NWFLKATVSQ DSADKWRAEV TPTGKRLME 240
AQRITAGYIH LWFPTYVNNK 260

SEQ ID NO: 332      moltype = AA length = 272
FEATURE            Location/Qualifiers
source              1..272
                    mol_type = protein
                    organism = Penicillium cinnamopurpureum

SEQUENCE: 332
VPLRRDVSSD DLKQLTLFAE YASASYCTNN INSTGDALSC AEGNCPAVQS ATTKTLYEFN 60

```

-continued

DSTFPGDVAG	FLAADETNEL	LVLSPFRGSR	ISTWVANLDF	GLTDTSDLCS	GCEAHGGFWK	120
SWQTVTDDIT	SKIDAGLKSH	PGYTVVLTGH	SFGAAMATLG	GTALRNAGYK	IKLYTYGEPR	180
VGNEALAKYI	TKQGDLYRVT	HADDVVPKVP	PASFGFSHAS	PEYWITSGNN	KTVSTSDIKV	240
IQGVGSKDGN	AGTINPDIEA	HNWYIVHIDG	CQ			272

SEQ ID NO: 333 moltype = AA length = 275
 FEATURE Location/Qualifiers
 source 1..275
 mol_type = protein
 organism = Talaromyces verruculosus

SEQUENCE: 333

EPIYRRKIAT	SLLDSLDF	QYSAAAYCSA	NLDDTTGTALA	CNVGNCPAVE	AADTTILYSF	60
DSSSSFGDAT	GYIADVESHG	YIILSPRGTS	NLENFIATLD	MLIDASSIC	SGCKVHKGF	120
NTWETVASDV	TSQIKAALSA	YPDYTLVATG	HSLGAALAAI	AATVFRASGY	TVQLYNYGEP	180
RIGNLALADF	ITSETSGTNY	RVTHSNDIIP	KLPPGLLGYH	HFSPEYWITS	KDNVTVTDS	240
VVEIKGVDST	DGNDGTAGAS	IEAHTWYFVY	ISECL			275

SEQ ID NO: 334 moltype = AA length = 277
 FEATURE Location/Qualifiers
 source 1..277
 mol_type = protein
 organism = Talaromyces cellulolyticus

SEQUENCE: 334

APKPIHRRTI	PTSLLDNLDL	FAQYSAAAYC	SANLETTGTA	LACNVGNCPA	VEAADTIILY	60
SFSSSSFGD	ATGYVAVDES	NEYIILSFRG	SSNLENWIAN	LDIPLIDASS	ICSGCTVHEG	120
FWDWTETVAS	DVTSQIESAL	STYPNYTLVA	TGHSGLGALA	AIAATVFRAS	GYTVQLYNYG	180
QPRIGNLALA	DFITSETSGT	NYRVTHSDDI	VPKLPPPELLG	YHHFSPEYWI	TSNDNVTVTD	240
SDVVEIQGVD	STAGNDGTSG	DSIDAHSWYF	VSISECS			277

SEQ ID NO: 335 moltype = AA length = 280
 FEATURE Location/Qualifiers
 source 1..280
 mol_type = protein
 organism = Penicillium megasporum

SEQUENCE: 335

GPVSVLRDE	DVSASVLSEL	DFFSQYSAAA	YCSTNINSAG	TKLTCSEGIC	PLVENADTET	60
LDFDESASY	GDVAGFIAVD	RTNELLVLSF	RGSASFNSWL	ANIDLFLDDA	SSVCSGCEVH	120
SGFWDWQTV	EGQITTALGS	AMETYPGYTL	VFTGHSYGAA	LAAIAATIFR	NSGYTVELYN	180
YGQPRIGNLA	LAEYITQNK	GGNYRVTHTD	DIVPKVPPKI	TGYHHASPEY	WITSGNNVTV	240
TTSDVQLITG	VDSTSGNDGT	SDDSVAEHRW	YFVHISMCTI			280

SEQ ID NO: 336 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein
 organism = Bacillus toyonensis

SEQUENCE: 336

HEKTEGHNVN	IIQYWSAEDK	HSEGVNSHLW	IVNRAIDIMS	RNTKLVKQDQ	IILLNEWRTD	60
LENGIYSADH	ENPYDNTSFA	VSHFYDPDDG	STYIPFAKQA	KETGAKYFKL	AGESYKNKDM	120
KQAFFYLGV	LHLYGDNVQ	MHAANFTNLS	YPQGFHSGYE	NFVDTIKDN	KVTDGNGYWN	180
WKGHPEDWI	HGAAGVAKQD	FSGIVNSNTK	SWFVKAASVQ	SYADKWRAEV	TPMTGKRLIE	240
AQRVTAGYIQ	LWFDTYVNR					259

SEQ ID NO: 337 moltype = AA length = 259
 FEATURE Location/Qualifiers
 source 1..259
 mol_type = protein
 organism = Bacillus sp.

SEQUENCE: 337

HEDTHQDPPI	ALKWSAESVH	NEGVSSHLWI	VNRAIDIMSQ	NTTVVKQNET	ALLNDWRTNL	60
EEGIYSADYK	NPYYDNSTFA	SHFYDPDSEK	TYIPFAKQAK	QTGAKYFKLA	GEAYQNKDMK	120
NAFFYLGLSL	HYLGDVNQPM	HAANFTNISH	PFGPHSKYEN	FVDTVKDN	YRVTGNGYWN	180
KSANPEEWVH	ESAAAADKAD	PSIVNDNTKS	WFLKAAVSQD	SADKWRAEV	PVTGKRLIEA	240
QRITAGYIHL	WFDTYVNNK					259

SEQ ID NO: 338 moltype = AA length = 262
 FEATURE Location/Qualifiers
 source 1..262
 mol_type = protein
 organism = Bacillus manliponensis

SEQUENCE: 338

HGNHDASNDS	GISISPRWSA	EEMHAEKGNS	HLWIVNRAID	IMARDTTVVK	ENEVALLNEW	60
RTDLEDGIYT	ADYENPYDYN	STFASHFYDP	DTDDTYIPFA	KNAKVTGAKY	FKLAGEAYEQ	120
QDMQQAFFYL	GLSLHYFGDI	NQPMHASNFT	NISHPFGFHS	KYENFVDTIK	APYAVTDSKG	180
YWNFAGGTPE	EWLHTAAVAA	KKDAPGIVNE	TTKSWFLKAS	VSQYANMWR	AEVTPETGAR	240
LMEAQRAMAG	YIHLWFDTYV	NR				262

SEQ ID NO: 339 moltype = AA length = 274
 FEATURE Location/Qualifiers

-continued

```

source                1..274
                      mol_type = protein
                      organism = Penicillium simplicissimum

SEQUENCE: 339
APARSVPRDV SASVLEQFTL YAQWAAAAAYC SNNLDSTGDA ITCAGGYCPE VESSTTISLS 60
EFNDTNDFGD TAGFVAVDKT NKQIVVAFRG SKSISNWIAD LDFGLTDASN LCSGCEAHTG 120
FLEAWETVAD SITSQIGAAM KTYSGYTLVV TGHSLGGAIA AIGATVLRNA GYTLDLYTFG 180
QPRVGNLALA TFLTKQGNR MTHLNDIVPR LPPTSFGFSH SSPEYWITSA DDVTVTTSID 240
EVIEGIDSTA GNAGELIESV AAHAWYIID DGCE 274

SEQ ID NO: 340        moltype = AA length = 276
FEATURE              Location/Qualifiers
source                1..276
                      mol_type = protein
                      organism = Penicillium arenicola

SEQUENCE: 340
APFQLWSRAV TPSVLSKLDL YGQYIAAAYC DDNIASAGTE VTCSAGNCPL VQAATTNTLS 60
EFNESNEFGD VAGFPAVDTT NQALVLSFRG SHTIDNWIAN LDFGLTSVST LCSGCKAHTG 120
FWKAWNTVAS DIAAAVDAAQ DTYPSPYPIIF TGHSGAALA ALAATTMRNA GYSIELYTVG 180
QPRIGNTALA TYITNQNKGG NYRVTHNTDI VPRLVPRLLG YSHFSPEYWI TSGNNVTVTA 240
SDITLVGTID SNGGNAGELL QSVEPHYWYF VEVEDC 276

SEQ ID NO: 341        moltype = AA length = 277
FEATURE              Location/Qualifiers
source                1..277
                      mol_type = protein
                      organism = Aspergillus aculeatus

SEQUENCE: 341
APAPIERRSV STLLDQMDL FAQYSAAYC STNIDSASTA LSCSADNCPL VVAAAPTVDL 60
EFNETAEFGD TAGFVAVDST NKAIVVAFRG SSDLSNWIAN IDFGLTASS ICTGCEIHS 120
FWKAWNTVAS LHYPTDVTQP TTYSDYDVVF TGHSLGALA AIGATVLRND GYTVLDYNFG 180
QPRIGNLALA DYITDQNKGS NYRVTHNTDI VPKVPPELLG YHHFSPEYWI TSDNDVTVT 240
SDITEVTGVD STAGNDGTLT DSVSAHKFYF EYISACD 277

SEQ ID NO: 342        moltype = AA length = 349
FEATURE              Location/Qualifiers
source                1..349
                      mol_type = protein
                      organism = Bacillus acidiceler

SEQUENCE: 342
ITSFFGNQYK AFAWSEDEVDH NQDHSTHHFI VNGSVKLIAD NTNPAINKPT TLLNQFRDRW 60
EQGLYDADHI NPFYDTGTTFM SHFYDPTQT NYTGASYPTA RQSGAKYFNL ASDYYKKGDF 120
NNAFYVLGVS LHYPTDVTQP LHASNISNLD HHAPGYHSKY ETYAESIQNE MTPDPSGLYN 180
WIASTDPEAW IHQAQVQAKS VLPQVWNDTI INYFWQAAYS NYYSAMWKSE VKNPTLDQLN 240
QAERETAGFI DMFPRVNGVE MPVTYKENA FSGASELLGS GNYDYDQLVK GIGNDTISSI 300
HIAPGYQVTL FSDANYKGAS TVLTNDVHDL GNFSHQVSSI KVAKISALK 349

SEQ ID NO: 343        moltype = DNA length = 1815
FEATURE              Location/Qualifiers
source                1..1815
                      mol_type = genomic DNA
                      organism = Aspergillus wentii

SEQUENCE: 343
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagetaac 60
tgggcagaaa gtatctggga cgaatgtaaaa catgcgctga attgcgctgg ttgtgaaaca 120
gttttggttg ctctaaaaagg tgtggcagat ctgggtgagc atgcatttca aacgggtcta 180
accgatgtgt gtgacattag cgggacggaa gacaaagacg tttgttctgg ttgtatcgca 240
gctgaatcac ctgctttgtg ctacaacatc aagaatttgg gtgtcaaatc tcatacatct 300
aaagttttgt gtgcccactt gtttggttta tgccaatttc cagcagttag gccttacaac 360
ttgaccttgc cttcccctaa accaactact tctaggcctc ctccctctgg gcaatcacc 420
atcagagttg ctacacatct cgatactcac gttgatttgt catatgagac tggtagcaac 480
tatgaatgta gtaaacctat ctgttgtaga gtgtacacag acgaggatgc tccaggaaat 540
acgtcatttc cctgtgggcc atacggaaac acaactctgt acccaccttt gaggtgagg 600
gaaagtatga tggcagatct caaatcccta aatcccgctt tttcaatcta cactggggat 660
gtcgtcgctc atgacttgtg gatggctcac aaaactgaag tattggatga cttcaatgcc 720
acttattcaa tgttgatca attggatcta gtttatgctg ctgtggggaa ccatgacact 780
acctctgtta atctatttcc atctacacaa ttgccagata aggataatca atgggcata 840
gatgctttta ctgcagaatg gaaatccctt accaattctt caatccaaac tacggagtat 900
ggctcatata ggcacatata cgaaaacctg aggatcataa gctacaacag catctctac 960
tatcaagaca actctacgc atacacagac ccgatggcac atgaccttc aaatcagtta 1020
acatgggtga tagatcaact gcacgaagct gaactctgcaa atcaaagggt ttggttgata 1080
agtcacattc caactggcgg tgtggatcat ttgcacgatt acagtcatta catagatgaa 1140
atagtcacac gttacgaagc aactatctcc gcttggttct ttggtcacac acatacagac 1200
ttgtttcaaa tgccttcacg cgactacaaa aacagagctt gggacacagc ctccgccatt 1260
ggttacgtag ctcccagttt aactccaact tctggccac cagcttttag gatctacgat 1320
attgatcctg ttacctttgc cgtccttgac tatacgggtc atatcgcaa catatcaaac 1380
ccctcatacc aaagcaatcc gaaatgggag aagtactatt cagccaagga ggcctatggt 1440
agtttggtga gtcctccagt caccgattca tctctgagt tgacgccttc tttctggcat 1500
aatgtgacag cattgatgga aaaggacgac tcagttttcc aagactgggt gtcccgtag 1560

```

-continued

```

accagagggt acaatgtcac aacatgcact ggggattgcg ccaaaaagga gatatgtgct 1620
ttgagagggt cagatgtctca gtacaattgt gtgacgggcta cacctgggtt tcattttgac 1680
aaaagagagg agtccgaaca aaaaccgaga cctgaatgcg aggatggctt gggcagagtgt 1740
ttaggtgata tgatccacaa gaaagatttt gtggatttgt tgcatgaaag aaccgcacaa 1800
taccacata gataa 1815

```

```

SEQ ID NO: 344      moltype = DNA length = 1824
FEATURE
source             Location/Qualifiers
                   1..1824
                   mol_type = genomic DNA
                   organism = Penicillium cylindrosporum

```

```

SEQUENCE: 344
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctaca 60
tacgataaga gagatcttgc ccaagacttg tggaaatgaca tcaagaatgc tgttgattgt 120
ggtaggtgctc aagtgtattct tactgcattg aaagggttat cagacttggg gacaacagtt 180
ttcgtggacg ttttgacaga ggtttgtgaa attggtaagt tagaagattc agatgtttgc 240
tctgttacta tttctagaga aggtccagtt ttggattaca ttttgcaaca tctagacata 300
gggtctcata caagtcaagt attgtgtgct agtgcatttg ggtgtgtgcca atacctgtca 360
gtaagacctt acaactctacc ttccctccc ccgaaaccaa acaccactag accagctccg 420
tccggtgaac tgcgaattca agtagttcac atttcagaca ctcatgtaga tctttcatalc 480
gagactggat ccagctacaa ttgcacgaaa ccaattttgt gtctgtccta taccgtgccc 540
gatgcacctg gaaataccac aaccocatgt ggtccttacg gcaataccaa ctgtgatgct 600
ccatttggac tagaagagtc aatgtttgct gccattaagg ctctacagcc tgtttctctc 660
atctataccg gtgatgttgt cgcacatgac ttatggtagg tggatcaaaa tgaagtttta 720
caagacttta acgctacact caatcgtagt gccgaacttg gtgtcgtcta tgctgcaatt 780
ggtaatacatg acaactgtctc agtgaatgat ttacctgcat ccaacatttc ttcacagtat 840
agcgcaaaact ggacctatga tgcacttgct tatgacttct ctaccttaac aaactctgca 900
agtgtctcaga cggcagaaga ttacgggagc tacagcagtg tctatagagg aagtcacggc 960
acagacttac gtgtgatctc ctataactcc atcttttact atgtagacaa cttttgggta 1020
taccaggatc caatgtttga cgtaccagac ggtcaattag catggttgat ttctgaacta 1080
caagaggcag aaactggggg tcagcgtggt tgggtgattg ctcatgtccc aagcggcatt 1140
tccgatcact ttcatgacta ttctcattac ttgatcaaaa tagtcaaaag atacgaagcc 1200
acgattgcaa ctctgttcta tggtcacaca cacagagacc aatttcagat ttcatacagt 1260
gactatacaa atagaacctg gaatactgct acagccatgg gctacataac accatccttg 1320
acgcccacct ctggtccgcc tacatacagg gtgtacaaag tcgacccaaa gacattcggg 1380
gtgttggtatt tcaccaacta catagccaac atcagtgatc cagcctatca aagtggacca 1440
acctggcaga agtactatag tgccaaggaa acttacggat ctttactgtc cctctctgta 1500
acagatccga cggcagaatt gcacactgct ttttggcaca acgtcactgt tctatttgag 1560
aatgataacg ccacgtttca gcagtagata gccaggcaca caagagggtg ggatccttct 1620
tctgtacagc gaagtgtcat tgatcaaacg atttgtggtt tgagagctgg cgacgcacag 1680
tacaattgctg ttaccctcac tcccggtttc aatttcgcta agagagatgc ttctactggc 1740
ggtaaacagg aaacgcattg tgaagaggtg gaaggctccg gtctgttggc tttgttggga 1800
agaatggtcg caactaaatc ataa 1824

```

```

SEQ ID NO: 345      moltype = DNA length = 1824
FEATURE
source             Location/Qualifiers
                   1..1824
                   mol_type = genomic DNA
                   organism = Penicillium meridianum

```

```

SEQUENCE: 345
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagcttca 60
gccgatagtt ggattaccat catatgggat gacttcaaaag aagctgttga ttgtgcatca 120
tgtcaggcac tactgggttg cttgaagtat gttgccggtt ttggagaaaag ctttatggaa 180
gacgtgttga caggggtatg cgacatttct ggtgtgtgag attcagatgt atgttctggc 240
atcattgcaa atgaaggtcc acgtgtctac tattctttga agaacttaga actagggagt 300
cacactgcta aaaccttttg ttccaatttg gtcggattat gcgattaccc agcagtaaga 360
ccatattgat tgtctttccc tagtcccaag ccgtcaattt cccgtctctc accaagcggg 420
aagcctccac tgaaagtgtg tcacttttcc gatactcacg ttgacttacc atacgaacct 480
ggctctagtt ttgactgtac aaagccgatt tgttgttagg tttactcaga agacgatgcc 540
ccaggcaata cctcatcccc atcggtccct tggggaaaag ccaaatgcga tccacctcat 600
cagttgcaac aatctatgat ggatgcaatc gcctctctga atccagcctt ttctatctat 660
actggtgacg ttgttgtctc cgatgtttgg ttagtcaata agactgaagt attgcaagat 720
ttcaatgcaa cctattctac tatggagagc acgttaggct tgggtctacg tgctctaggt 780
aatcatgata ccgtctcatt gaacttgttt ccttcaacca acattccaag ttcatacaat 840
ccgcaatggg cctacgatgc tttatccaca tcttgggtga ctcttacatc agacaatcct 900
gccatttcca cggcaaaaaga atacgggagc tacagcgcca gacataaaga caccaagcta 960
agaatcattt cctacaattc cattttctac tacaataaca actttttcag ctatgaagaa 1020
ccgatgccat ttgaccaga cggtcaactt tcatggttga tatccagatt gtctgtgca 1080
gaagccgat ctgaacgtgt ttggttgatt tcccatattc caagtggaaa tagcgacct 1140
ttcagagacc attcacacta ctttgatcaa atagttcaaa ggtatgaagc taccattgct 1200
gggtgtttct ttgggcatac tcataccgat gagtttcagg tttcttactc agattacgcc 1260
catagatctt gggataccgc tactgctatg ggatacgtag caccatcaat gactcctact 1320
tctgttcccc catcatcaga agtatacga atcgatccag tcacctttgg ggtcctggat 1380
ttcactcaat acattgcaga catatccgac tccctcgaa caaagtggat gccctcat 1440
tccgccaaga aagattatgg cagccgtcta gatgtgcctg ttggagagga tatggagtta 1500
acacctgcct tctggcataa cgtaacctgt ggtatggaga aggacctatc tttgtttagg 1560
gacttttggg ctctagaaac gagaggtttc gaggtaccag gttgtgaagg tgatttgttt 1620
tctaagaga tatcgccctt gagaggagcc gatgcacagt attcatgtgt ggaggcaaca 1680
ccaggtttct ctttogaana gaggggtgag agaaccgttc tgttagagaa aagattccag 1740

```

-continued

```
ccagaatgta atcacgctgg tatggcacca ttgcttgcca aaatcgctca tagagcttca 1800
ttggctaggg aaatggaagc ataaa 1824
```

```
SEQ ID NO: 346      moltype = DNA length = 1830
FEATURE            Location/Qualifiers
source              1..1830
                   mol_type = genomic DNA
                   organism = Penicillium bialowiezense
```

```
SEQUENCE: 346
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctgaa 60
gttcttcttg aaaaactctat tgccttcotta tgggatgact ttaagaatgc tgtggactgt 120
gcctcttgcc aagcattgtt aggtggcetta aagttgggtg caggatttgg ggaggacttt 180
atgattgacg ttttgactgg tatttgcgac atttccaaag cagaagactc agatgtttgt 240
gaaggcatca tcaagaaaga aggtcctgct ttgcatgatg cattccaagc cttgcatatt 300
ggctcaaatt ctacacaaac catgtgtgcc tctttgattg gattatgtca atatccagaa 360
gttagggcac acagtctaag ctttccaagt cccaaaccca ataagggtgaa gcctccaccg 420
tctggtaaac ctcccactca agttgtgcac ttttcogata cgcattgtcg cttgtactat 480
gagacaggtg gcagctatga atgttccaaa ccgattttgt gcagagtgtt tgaggataag 540
tatgcaccag gcattacaaa aacaccgtgc ggtcccttcg gtaatccaag gtgtgaccct 600
ccattgaagt tacaagagag tatgaacgct gcaatcgag ataatcaatc agaattctcc 660
atctataacc gagatgttgt agctcatgac gtttggttag ttaatcaagc agaagcacta 720
gaatccttca atagcactga ctcccagatg gagaagtctt tgggaatggt ttatgcagca 780
attggtaatc acgatactgc acccttgaat ttgttctctt catccatagt gccagatggc 840
gacaatcaat gggcctacaa tgccttggtc gagtattggt tgactttgac ctctatcagc 900
tccgtacaat ctgcccagca gtacggctca tattcagcaa tcccccaga tagtaattct 960
aggatcattt cttacaactc cattttctat tacaagttca acttctatat gtaccaagaa 1020
cctatggaga aagatcccaa tgggtcaattt gagtgggtga tcaagaggtt gcaggcagct 1080
gaagacgctg ggcagagagc ctggttgatt tcccacattc cgcctggtgt cgcagaccat 1140
tttcatgact atttctacta ctttgatcaa attgttcaaa ggtagagggc cacaatagct 1200
gggttggttc atggccatca gcataatggat gaattccaaa ttgcctactc caactacaag 1260
aataggaatc atgaaactgc aaccgcaatg ggctacatag ctctgctgat gacccccact 1320
gaagggcctc ctagttttccg tgtatatgaa atcgatccgg aaacggtttg ggttttagac 1380
tatacacagt atatcgcaaa catctcagat cccacttacc agggaaaaac ccagtgggtta 1440
ccctattaca tgcgccaaggc agactacggt tccaagttat cacctcctgt aacagaccgc 1500
aagattgagt taaccccagc tttttggcac aacgttactg tcgcaatgga aaatgagccg 1560
agcatttttc aagaattctg ggcaagaaga gttagaggct ataaggttac agattgtaat 1620
ggcgactgta tgaaaacgga gatattgtct ttaagagccg cagatgcaca gtttaactgt 1680
ctcaagccca agccaggctt taactttagc aaaagagatg gtaaggcac tttggctgaa 1740
cagccacatc attgtgatca tgcgggatta gctcccttat tgggcaagat tgcttacagg 1800
gctagaatag caagagagat ggaagcttaa 1830
```

```
SEQ ID NO: 347      moltype = DNA length = 1830
FEATURE            Location/Qualifiers
source              1..1830
                   mol_type = genomic DNA
                   organism = Penicillium sclerotiorum
```

```
SEQUENCE: 347
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagcttct 60
gcagactcct ggatctcttc tatttgggat gacttcaaa aagcagtaga ttgcggcag 120
tgccaagcct tggtaggttg tctgaaagta gtcgctgcct ttggagaaag cttcatggaa 180
gacgtattga ttggtgtttg tgatatctca ggtgcagaag attcagacgt ttgtgctggt 240
gttattgcaa atgaaggctc agctgtccac tattcattga caaacctaga gattgggtta 300
cacacagcaa aaacattttg cgcacacactg gttggcttgt gcaagtatcc agaagttaga 360
ccatacaatt tgacctctcc gtacacaaaa ccccactt ctctgcccgc tccatctggg 420
aaaaactcca tcaaaagtct tcaattctca gatacacatg tggatctatc ctatgaaaca 480
ggctccaatt acgactgttc taaacctatt tgcgttagag cctatacgga agacgatgca 540
cctggcaata cctcttcacc atgtgtgccc tggggcaaca caaattgtga cccacctcac 600
agacttcaga aatcaatgaa cgctgccatt gcagacttga aaccggcttt ttccatctat 660
actggtgacg tagttgtctc tgatgtatgg ttggtcaaca aatctgaagc attgcaagac 720
ttcaatgcta cttatgtgtc catggaggat tccctaggta gactctatgc tgctttagg 780
aatcatgatg ccgctccctt gaacttggtt ccatcaaatc aaatacctag cgaatacaac 840
ccccaatggg catacgatgc tcttgccgca gactggatgg ccttgacaga cattcccagc 900
gtagaaacag ccaacgagca cgggtcctat tcagctatac acccagacag taacttacgt 960
atcatatcct ataatccat cttttactac aaatacaact tcttttcata cacggaacca 1020
atggaatacg atccaaacgg tcaattagaa tgggtgatca atgaactaca tgcgcagaa 1080
actgccaatc agcgtgtttg gttgatttct catataccaa gcggcaatag tgatcacttt 1140
catgatcatt ctctactact cgatgagatc attcaaagat acgaagccac gattgcagg 1200
ctgttctttg gtcacacaca cacagacgaa tttcagattt catactccaa ctattcaaat 1260
agaaactggg atactgtctc agccatggga tatgtagcac caagtatgac gccacctct 1320
ggtccgctct cattcagggt gtacgaaata gaccagtgta cattcggtgt gttggatttc 1380
atgcagtaca tagccaacat ctccgatcca agaatccaa agaagccaga gtgggttccc 1440
tactattccg ccaagagtga ttacggatct aggtctgagc ctctgtaac agatgccgac 1500
gtagaattga cacctgtttt ttggcacaaac gtcactgttt tgatggagga agattccagc 1560
gtgtttgacg agttttgggc caggagaaca agaggttttc aagtaccagc ttgtcacgga 1620
gactgcgttt caaatgagat ttgtgctttg agaggtgccc acggacagta caattgcttc 1680
atagaaaaac ccggtttctc ttccgaaaag agagatggtta ctgtcgacga agtttcttgg 1740
aagaaaagat ttcaaccaga atgcaacctat gccggtatgt ctcccttatt ggctaagata 1800
gctcatagag catcactggc aagagaataa 1830
```

-continued

```

SEQ ID NO: 348      moltype = DNA length = 1836
FEATURE            Location/Qualifiers
source              1..1836
                    mol_type = genomic DNA
                    organism = Rasamsonia byssochlamydoidea

SEQUENCE: 348
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagcttcc 60
tacaacaaaa gaggattagc tcaagagatt tgggatgata tcgtcaacgc cgttgactgt 120
tcagggtgtc aggcattttt gactgcctta caagggtgctg ccgagttagg gacaacgcgt 180
tttgttgata ttcttacaga agtatgtgac attagtggta aagaagattc agatgtttgt 240
tcaggcatca ttccagaga aggtccagtc ctggattaca ttttgcaaaa cttagatatt 300
ggatctcaca catccgagct catatgtgca tccgcatttg gtttatgtca ataccctgca 360
gtcaggagct acaatttgac ctttccaact ccgaagccag atactacgag accagctcct 420
cttgccgaat ctccgattca agtagtacac ttttccgaca ctcatgtaga cctatcatac 480
gagactggtt catcctataa ctgtacaaaag ccaatctgtt gcagacctta taccgctgcc 540
gatgcaccag gaaacaccac taccccttgt ggaccatacg gtaaacccaa ttgcgatgct 600
cctctgagtt tggaggaaag tatgttcaat gccatcaaaag ctctatctcc acaacctgca 660
ttttccatct atacaggcga tgtttagct catgacatat ggtagtcga tcaaaatgaa 720
gtgatccagg atttgaatac tacctatgat ttgatggctg aattaggatt ggtttatgcc 780
gctatttgaa atcatgatac tgcctctgtt aatgatctgc caacaaccaa cattccatca 840
gagatttcta cgaactggac ctatgaagcc ttggcatacg atttccagat gttacaacagc 900
agcagctctg cccagacagc tgcaaacctac ggatcctaca gttcaatcta caaaggttct 960
tacggcacag atcttaggat cataagctac aattcaatct tctactacat agataacttt 1020
tgggcctatc aagatcctat gccctatgat ccagacggtc agttagcttg gttgattaac 1080
gagttgcagg aagctgaaac agcaggtcag agggatgga ttatgcaca tgcctcact 1140
ggatcctccg atcatttcca tgactatagt cattacttcg atcaaatgtg gcaagatat 1200
gaggtacaga tagcgcgttt gttctatggt cacaccata ttgatcaatt tcagatttct 1260
tattcagatt actccaatca accatgggat accgctacgg ccacgggcta tatcatgct 1320
agtttgacgc ctacttcttg ccacccgaca ttcagagtg atgatgtgga tccaaaaaca 1380
ttcgcagtgct ttgactttac caactatata gccaatatca gtgatcctgc ttatcagct 1440
gggcctactt ggcagaaata ctactccgca aaggaaacat atgggtgcctt attgagcct 1500
cctttgacag atcccaactgc cgaattgaca cctgcctttt ggcaaacagt gactgtggct 1560
tttgaacacg ataattgtgc ttttcaagaa tactgggcta gacaacacg tgggtgggat 1620
gtaagctcat gtactgtgtc atgtatcacc caagcaattt gtggttgag ggcagccgac 1680
gctcagtata actgtgtgac accaactcca ggcttcaact ttgctaaccg tgatgcttct 1740
tccgctactc aggcaatggc ccatgttgag aaatgtgaag gttccggatt gttggccttg 1800
ttaggaggga tgggtgcaga caagaaatcc gcttaa 1836

```

```

SEQ ID NO: 349      moltype = DNA length = 1839
FEATURE            Location/Qualifiers
source              1..1839
                    mol_type = genomic DNA
                    organism = Rasamsonia eburnea

SEQUENCE: 349
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctact 60
tatgacaaaa gagggttggc tcaagacatt tggaaacgaca taaagaatgc cgttgattgt 120
gctgggtgtc aaggcatttt gacagcattg aaggggctgt catacttagg gactactgct 180
tttgttgacg tcttaacaga agtatgtgac atttcagggt tagaagattc agatgtttgt 240
ctcggcatca ttctttcaga aggacgggca ctggtctaca ttcttaaaaa ttggatatt 300
ggcagtcata cgtctcaggt catatgtgca tcagtttttg gtttgtgcca atatcccgca 360
gtgagagctt acaacttgac tttccgagc cctaaaccag acaaaacttg tctgaacct 420
ctcgttgagt ctcagatata aatcgtacac ttctcagata cacacgcaga tcttagctat 480
gaaacaggat caaactacaa ttgcacaaaag ccaatatgtt gcagatccta taccgctgaa 540
gatgcaccag gcaataccac aactccttgt ggccatacag gaaatcccaa atgtgatgct 600
cccatgagct tagaggagtc aatgttcgct gccattaagg ctttgtctcc tcaaccagct 660
ttttccatat acactgggtg tgttgttct catgacatat ggtagttga tcaaaacgag 720
gtggttagagg atcttaacgc tacttacgat agaatggcag gattgggttt ggtgatgct 780
gcaataggta atcatgatac agctcctgta aacaatctac ccactccaa cataccctca 840
caatacagcg caaactggac ttacgaagca cttgaatacc acttctcctt attgacgaag 900
agcgcctccg cacaacacag agagaactac ggatcctatt ccagcgtcta tagagggaag 960
tatggtaccg acttaagggt gatatcctat aacagcatct tttactatat cgcagatttc 1020
tgggcttata aagaccccat gttgtatgac ccagacggtc aattggcctg gttgatcaat 1080
gagttacaag aagcagaaac agctggccaa aggggtttgtg tgattgcaca tgttccaagt 1140
ggaaactcag accattttca tgactattcc cactactttg atcagatagt tcaaaaggat 1200
gaggcaacca ttgctgcctt gttttacggc catactcata ttgatcagtt tcagatttca 1260
tactccgact attcaaatag ggctttcgat acggctacgg ctattgggta cataatgcct 1320
agcatgactc ctacctcagg gccctccgact ttcagagtat acgatgtaga tccaaagacc 1380
ttcgcggttt tagactttac aaactacatt gccaacattt cagatccagc ctaccagagt 1440
ggtcgcagat ggcaaaagta ctatagtgca aaagaagcct acggttctct gttaaagtcca 1500
ccagtaacgg acgccaactgc tgagttaacc ccagcattct ggcaacaatg gactgttgca 1560
ttcgaaaatg atgatactgc ttttcaagag tattgggcta gacagacgag aggttatgcc 1620
gtttcatctt gtacagggtg ctgcataaca caagctatct gtggactaag agccggtgag 1680
ctcagcataa actgtgttac tccaactcca ggcttcaact ttgcaaaaag agatgtctct 1740
acagacggac aagccttgcc acatattgag aaatgtgaag gatctggctt aatggcactg 1800
ctagccaaaa tggctgcctc aaacagacaa agttcataa 1839

```

```

SEQ ID NO: 350      moltype = DNA length = 1839
FEATURE            Location/Qualifiers
source              1..1839

```

-continued

```

mol_type = genomic DNA
organism = Penicillium brefeldianum

SEQUENCE: 350
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctggt 60
gctatttgaga attggggtgc tactatatgg gaagacttca aggaggctgt ggattgtggt 120
tcttgccaag tactattagg tggcttgaaa ctggtggccg atttcggtga gggctttttg 180
ggatgatgtc taacaggagt atgtgatatc tctaaagctg aagatagaga tgtttgtgct 240
gggtgtgtcg cttctgaagt tectgacctc cactatgcct taaagaatat gaaagtaggt 300
agtcatacag ccaaaacctt gtgctccgca ttagttggtt tgtgtgactt tccagacgtc 360
cgtccatttg accttacttt tccaagcccg aagccagcta cgtctcgtcc acctccaagt 420
ggtaaacctc ccatcaaaagt agtgcatctt agtgatacac atgtggactt aagctacgaa 480
actggttcca attacgattg tagtaaacct atttgctgta ggggtatacac agacaaagat 540
gctcctggga ccacagacaa gccatgtggt ccttggggac acccaaagtg tgatcctccg 600
caccactgc aagagagtat gatgacggct attgcaaac tgaatccagc attttccatc 660
tatactggag acgttggtgc acatgatgtt tgggtggtaa acaaatgata agttctgcag 720
gacttaaacg ctacctacgg tgctatggaa aatcatttag gattggttta tgccgctttg 780
ggaaatcatg atgctgctcc cttgaactta ttcccatcca acaaggtacc atcaaatgac 840
aatccccagt gggcctatga tgccttgaca gccgattgga tgacattaac agggctggac 900
agcgtgcaga atgccataaa gtacggatcc tacagtgtcg tgaccccaaa ttctaagcta 960
aggattatct catacaacag catcttctat tacaagtaca atttctttag ttatactgag 1020
ccaatggaat acgaccccaa tggtaaatgt acttggtaca tcaatgattt gcaagccgct 1080
gagacagctg gtcacaagggt ttggttgatt tcccatatcc cttcaggaga tgttgatcac 1140
ttcagagatc actcacacta ttctgacag atcgtacaaa gatacgaagc tactatagct 1200
gggttattct ttggttcacac acataccgac gaatttcaaa ttgcttactc agactacaat 1260
aacagggaatt gggatacagc aactgcaatg gggtagctcg ctccaagcat gacacccaca 1320
agcgtccac ctagtcttctg tgtctacgat atcgatcctg agacttttgg tgtaattggac 1380
tacacccagt atatcgccaa tatctccgat cccagttttc aaactaaaca ggagtgggtc 1440
ccatactaca ctgcaaaagaa ggattacgga gctaagctga gccctccacc ggcacctact 1500
ggtgaaacta ctcccgcttt ttggcataac gttacagtgg ccatggaaaa ggattcttcc 1560
gtgttcgaag ctttttgggc tagaaggact agaggttttt ccattccggc ttgtacaggg 1620
gattgtgtca agaataaagt tctgcgacta agaggtgccc atgcacaata cagttgtgtc 1680
aaaaggactc cggggttttc ttttagtaag agagatgaaa ttgagtcaga tccactattg 1740
tctaagaggt ttcagccaga atgtaatcat gctggtatgg caccattgtt agccaaaatc 1800
gcacataaag caaacgtcgc taagtggaaat ggagaataa 1839

```

```

SEQ ID NO: 351      moltype = DNA length = 1839
FEATURE            Location/Qualifiers
source              1..1839
                    mol_type = genomic DNA
                    organism = Penicillium adametzii

```

```

SEQUENCE: 351
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagcttct 60
gttgattctc ggatttactac aatttgggat gatttcaagg aagccgttga ctgtgctact 120
tgtagcaggt tactaggttg gttaaagttg gtggccgtgt ttggggaaag ctttatggaa 180
gatgttatca caggtgtatg ttccattagt ggtgctgaag attcagatgt ttgtgctggt 240
gttattgcaa atgaagttcc agcagtgtag cactctttgt caaaccttga gattggatct 300
cacacagcaa agactttctg tgcactctta gtaggtttat gtgattaccc tgcagtcagg 360
ccgtattcat tgacctttcc atctccgaag cctccaacta cgcgtctctc acctctggtc 420
aaatctccga tcaaatgtag aacttttcc gacactcatg tagacctatc atacgagact 480
ggttccaact atgactgttc aaagccaatc tgttgtagag catataccga aaacgatgca 540
caggaaacaa cctctagctc ttgtggacca tggggttaact ccaaatgcga tccacctcac 600
agggtgcagg aaagtattga tgaagccatt gcagatctaa atccagcttt ctccatctat 660
acaggtgatg ttggttgctc tgatgtttgg ctagtgtata agtcagaagc ttgtcaggac 720
ttcaatgcca ctttcagtgc tatggaaaca ttgggttagag tctatgctgc tctgggtaat 780
catgatgcag ctccattgaa cttgtttcct agcaatcaaa tccctccaca gtataacca 840
caatgggctc atgacacatt agcctcagat tggatggggt tgacaggaat acagagcgtg 900
gagacggcca atgaatatgg aagttactcc gcaatccatc ccaactccaa tttaggagtc 960
atctcttaca attccatatt ctactacaaa tacaacttct ttctctatc cgaaccaatg 1020
gaatatgac caaacagtca attgcaatgg ttgatcaatg aattgcacga agccgagttg 1080
gccaatcaaa gagtctgggt aatcagtcac ataccaagcg gagatccaga tcattttcac 1140
gatcattccc actactttga tcagatttgt cagagatatg atgccacaat tgctgacctt 1200
ttctttggcc atactcacac agatcagttc cagatttctt attctagtta tcagaataga 1260
acatgggata aagctacagc aatgggatat gttgctccct ctatgacgcc tacaagtggg 1320
ccgctctctt tcaggggttta cgaatccgat ccgggttacat tgggtgtatt ggatttcacc 1380
cagtagactg caaacatttc agaccctact cagaccaaac ctaaatgggt gccgtactac 1440
tctgccaaga aagattacgg atcacactta gatcctccgg tgacagatgc ccatgcagag 1500
ttaacaccag cctctctggca taacgttaacc gtttccatgg agaacgataa ctctgttttc 1560
caggactttt gggctagaag aacaaggggt tttaagttc cagactgcac tgggtgattg 1620
tcatcatctg agatttgtgc cctaagagga gcagatgccc aatacaactg cttcgttgag 1680
aaactcgtgt tctcttttga gaaaagggac gttcaagaat cagacgaaga aatgtggaag 1740
aaacgtttcc aaccgggaat taatcatgca ggtatggcac ctttgttggc caagatagcc 1800
tacagagctt ctctagagag agaacaaaaa ggaaaaataa 1839

```

```

SEQ ID NO: 352      moltype = DNA length = 1842
FEATURE            Location/Qualifiers
source              1..1842
                    mol_type = genomic DNA
                    organism = Rasamsonia brevistipitata

```

SEQUENCE: 352

-continued

```

atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctact 60
agtttaaacac atgacaaaag ggatttggcc caggaaatct ggaacgacat caagaatgcc 120
gctacatgcg ctgggtgtca agttatcttg acggctctga aagggttttc agatttgggt 180
acctccattt tcgtagatgt tcttactgaa gtttgtgaca tttctggttt agaagatcca 240
gatgtttgtt caggcataat cagtcgtgaa ggtcctgtct tagactacat attgcagcat 300
ttagacattg gctctcatic atcagatgtc ttatgtgcta gtgcattcgg cctatgocaa 360
tatccagccg ttagaccata caacttaact ttccgtctc caaaaccaga taccactaga 420
ccggctccaa gtgggtgagtc acctatccaa gtagtacaca tatcagacac gcatgttgac 480
ttatcttacg aaacaggctc aagctggaac tgtacaaaac ctatttgcgt tagaccatac 540
actgcccgaag acgcaccagg aaataccact accccttgtg gcccatatgg tgaccoccat 600
tgcatatgcac ccatgtctct gcaagaatct atgattgctg ctatacaagc tttgagacca 660
caaccgcgtt tttccatcta tactggagat gtagtcgctc atgacatttg gctgggtggat 720
cagaatgagg tcattgaaga ctccaatgct acatacaata gaatggccga attgggatta 780
gtatatgctg ctataggtaa tcatgactct gcaccgctca atgatttacc cgttccaaac 840
atcccatccc agtatagcgt taactggact tacgaggcat tagcatacga cttcagcatg 900
cttacaggtt cagcatccgc acaggaagcc gaaaactacg gatcatatag ttccatctat 960
aagggtagta acgggacaga ttacgtgtg atttcttaca attccatctt ttactatgtt 1020
accaatttct ggggttttca agaccaatg ccttatgctc cggatgggca gttggcctgg 1080
ttgattaacg aattacagga ggcagaaact gccggacaaa gggcatgggt gatagcacat 1140
gtccctacag gtacagcgca tcacttccat gactatagcc actactttga ccaaatcgta 1200
cagagatatt aggcaacaat tgccgcattt ttctttgttc aactcatca agatgaattc 1260
caaatagctt attccaatta ctcaaacagg aactttgata cggcaacagc catagggtac 1320
attatgccct cattgacacc gactagcggg cctccctctt ttagagtata cgacatcgac 1380
ccaaaaacct ttggttgtct agatttcacc aactacattg caaacatcag cgatccagcc 1440
taccaaagtg ggcctacatg gcagaaatac tacagcgcaa aagaggcata cggtacttta 1500
ctttctcccg ctgttacaga tcccacagca gagttgacgc cagccttttg gcacaactta 1560
actgtttgtt ttgaaaatga taatgtctaca ttccaagagt actggacaag aaagactaga 1620
ggtcatgcag tttccaactg tacaggttct tgcattacc aatccatctg tggcatgagg 1680
gctgccgatg ctcagatataa ctgcgttact ccaactccgg gtttgtcttt cgctaaccgt 1740
gatgcagaga cttcaactcc agaaccacat atagaacctat gtgaaggatc agggttgatg 1800
tctttgctag gcagaatggg tgctcaaggt aaatcttcat aa 1842

```

```

SEQ ID NO: 353      moltype = DNA length = 1842
FEATURE            Location/Qualifiers
source              1..1842
                    mol_type = genomic DNA
                    organism = Penicillium scabrosum

```

```

SEQUENCE: 353
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctgaa 60
acaacatttag agaaatctat atcatctatc tgggaagact ttaagaatgc agttgattgc 120
ggttcatgcc aagtggtgtt aggtgggctg aagtttgtgt cagggttttg tgagaacttt 180
atgattgatg tcttaacagg tctttgtgac atttcaaaag ctgaagattc agatgtttgt 240
gaaggcatca tcaaaaagga gggaccggct ttgcatgacg catttcaagc attgaagata 300
ggaagccatt ccactaagac aatgtgtgcc aatttgattg gcttatgcca gtaccocgga 360
gttagggcca ataacttgag ctcccttcca ccaaagccaa aagatgtaag acctactact 420
agcggtaaac ctctatcaa agtcgtgcat ttcatgtgaca cacatgtgga cttgttgat 480
gaaactgggt ccaattacga atgcactaag cctatatgtt gcagagtgtt tgaagataag 540
gatgccccag gcataaccac gaatccctgt ggaccttttg gcaatccaaa atgcgaccca 600
ccgcaagcac ttcaagaaag catgaatgct gcaattgctg agatcaatcc agctttttcc 660
atatacactg gtgatgttgt ttccatgatg gtatggtag ttaatacaga ggaggcctta 720
gagttcttca actctactta ctcaaaaatt gaaaagtctt taggtatggt ttatgcagca 780
attggtaatc atgatactgc acctctgaat ctattccctc ccaagaatca gccagatggc 840
aacaatcagt gggcttacga tgcacttgct gaggactggg tagcaatcac ggggatcccg 900
tcogtacaat cagcagacga atacggatcc tattccgcca tacatccaaa ctccaatttg 960
aggatcatat cctacaattc cattttctac tacaaaattca atttctatat gtaccaagaa 1020
ccaatagaga aggatccaaa tggccaattc gaatggttga tcaaagagct tcaagccgca 1080
gaagatgcag gacaacgtgc atgggtgatt agccatattc ctagtgtgtg tgcagatcat 1140
ttccacgata attcacagta ttccgaccaa atagtccaga gatatgaagc aacaatcgca 1200
ggcatgtttt acggtcatic ccacatggat gaatttcaaa tagcttattc agattacaag 1260
aatagaaatt gggaaactgc aactgccatg ggctacattg cccctgcaat gacaccgaca 1320
gaaggacccc caagcttccg tgtttacgaa atagaccag atacttttgg cgtccttgac 1380
tttacacagt acattgocaa catttccgac ccagcatatc aaaagaagcc ggaatgggtg 1440
ccctattaca gcgcaaaagc agattatgga agcaagttat ctccgctgt cagagatcct 1500
aagggtgagt tgtctctctg atctggcat aacgtgacgg ttagcatgga aagggacgaa 1560
tctgtttttc aagacttttg ggctagaaga tcaagaggct acaatgtaac agcctgtaca 1620
ggtgattgta tgaagatgga actatgtact cttagagcag cagacgcccc atacaattgc 1680
gttaagccta accagggttt caacttttct aaaagagacg gtgaattggg aggtttgctt 1740
gaacaagagt cctattcaaa ttgtgatcat gcaggattgg ctacctgtt agggaagatt 1800
gctcataggg ccagatgcgc taggaagatt gaagaagcat aa 1842

```

```

SEQ ID NO: 354      moltype = DNA length = 1842
FEATURE            Location/Qualifiers
source              1..1842
                    mol_type = genomic DNA
                    organism = Penicillium manginii

```

```

SEQUENCE: 354
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagcttct 60
gcagatgatt ggtactactac aatttgggac gacttcaaga atgcggttga ttgttttagc 120
tgtcaagcat tgctgggagg attgaagctg gtgtcagggt taggggaatc ctttatggaa 180

```


-continued

```

gacgtcatta cagggtgatg ttccatttca ggtgctgaag ataattgatgt ttgtgctggt 240
gtcatttgcaa atgaaggacc ggacgtgtac tattcattga agaatttgaa attgggcagt 300
catacggcta agactttctg tgcacatttg actggtttgt gcgaattccc caagggtgaga 360
ccatacgaca tttcttttcc gagccctaaa ccaagcacta ctgacctccc acctctggt 420
gaggctccca tcaaagtggt acacttctca gatacacacg tcgatcttag ctatgaagaa 480
ggatcaaact atgaatgctc aaagccaata tgttgccagag catataccga aaaagatgca 540
ccaggcaata ccacatctcc ttgtggccca tggggaaatt ccaaattgta tccaccccat 600
cgtttacagg agtcaatgat gtctgtctatt gcagatatca atcctgcatt ttccatctat 660
acaggcgatg ttgtagctca tgatgtctgg tttagcaaca aaactgaagt gctgcaggat 720
cttaattgcta cctattctag tattgaaaa catctaggtt tagtttacgc tgcttttagga 780
aatcatgatg ctgcaccctt gaatctggtt ccttcgcaga agatttcctt ccagtataac 840
ccgcaatggg cttatgacgc tctagccgaa gactgggtga cattaacggg gatcccagac 900
gtgcaaaaag ctagtgagta cggcagttac tcagccgtgc atcctggtag taagttaaga 960
atcattagtt acaactccat cttttactac aagtataact ttttctctta taccgaacca 1020
atggaattcg accccaacaa acagttggac tgggtgattg cccaactgca ggaagccgaa 1080
gatgtaagc aaagagtgtg gttgatctct cacataccta caggaaattc agaccacttt 1140
agggaccatt cacattactt tgaccagatc atacaacgtt atgatgctac cattgcccga 1200
ttgttctttg ggcatacaca caccgatgaa ttccagattt cctattccaa ctataagaac 1260
agaaattggg atacagctac cgctatgggc tacgttgctc cgagtatgac gcctacgagt 1320
ggctctccct catcaccggg gtatgagatt gatcccgta catttgaggat gatggacttc 1380
actcaatata ttgcaaacat aacagacccc tctttacaaa ctgagccgga gtggaaccg 1440
tactatagtg ccaaggcaga ctatggtagt aagctgtcac cagccattaa ggatccagga 1500
atcgagctta ctcaggattt ttggcataat gtgacgggtg cattggaaaa agatgccaca 1560
gtttttcaag acttttggag tagaagaaca agaggattca acgttctctg atgcaactgga 1620
gattgtattt caaacgagat atgtgctttg aggggtgacg acgcccagta ttctgtttac 1680
aaacaggcac caggtttttc cttcgagaaa agagatgggt ctggtgttcc ctatttgtcc 1740
gaagaaagct ttcaacaacc agaattgaat catgcaggtg tggcaccctt attcgccaag 1800
atttctcaca gggctaagtt ggctagagaa agaggtgaat aa 1842

```

```

SEQ ID NO: 355      moltype = DNA length = 1842
FEATURE             Location/Qualifiers
source              1..1842
                    mol_type = genomic DNA
                    organism = Penicillium emersonii

```

```

SEQUENCE: 355
atgagatttc cttcaatttt tactacagtt ttattcgacg catcctccgc attagcttct 60
gtcccgtagt acaaaagaga cttggctcag gagatttggg acgacattaa gaatgcggtt 120
gactgcgctg gctgtcaagt tgttttgaca gctttgaaag gtgttgccga tctgggcaca 180
acagctttag ttgatgtctt aactgaggtg tgcataatct ctggttaagg ggattcagat 240
gtctgctctg gtatcattca tagagagggg cctgttctag actacgtgct tcaacatcta 300
gacatcgggt ctcatacatc acaagtcatt tgtgcttcag catttggttt atgccaatat 360
ccggaagtga gacctataa ctgacttttt cctaagccca aacaaaacac aaccagacct 420
gaaccttccg gtgaatctcc aattcaagta gtgcacttct ccgatacgca cgttgactta 480
agttacgaga cttggttccaa ttacaattgc actaagccaa tttgtgttag accttacct 540
gccgaagatg cacctgggaa cactactact ccatgtgtgc cgtacggtaa taccaggtgt 600
gatgccccat tatccttgga agagtctatg ttgtctgcca tcaagccctt aaacctcaa 660
ccagcatttt ctatctacac aggagatgtt gtagcacatg acatttggtt agttgatcaa 720
aacgaggtta tcgaagattt gaacgcaacg tacgatagaa tggctgtgct gggtttagtc 780
tatgcccgtc tatgctaata tgatacagca ccggttaacg acctgcccgc gtcaaacatt 840
ccatccgagt attcagcaaa ttggacttat gaagctttgt cttacgactt tacaatgttg 900
actcaaacgc ccagtgtctc aactgtgca aactatggtt cctattcagc catttaccct 960
ggatcttacg gtacagacct aagagtcttt tcatacaact ctatcttcta ttacgttgac 1020
aacttttggg cttatcaaga tccaatggaa tttgaccagc atggccagct agcatgggtg 1080
ataaacgagt tgcaagaagc tgaactgtct ggccaaagag tctggatcat agcacatgtt 1140
cctacgggta cttccgatca cttccatgac tattctcact accttgacca aatagtccaa 1200
agatatgaag ccacgattgc tgccttgctc tatggtcata cccacataga ccagtttcag 1260
attagttact caaactattc aaacagagca tttgataccg caactgccat cggctacatt 1320
atgccttctt taacacccac tagcgggtcca cctaccttta gattttacga cgttgatcca 1380
aagacgttgc cagtgttaga ttccaccaat tacattgcca acatttccga tctgtcattt 1440
caatcaggtc catcatggca gaaatactat tccgccaaag agacctacgg ctcaactattg 1500
tctccacctg taaccgacct tactgtgaa ttgacgcctg cattctggga taatgtcacc 1560
gtggcatttg agcaagacaa cgcaacattt caagagtatt gggctcgtca aaccagggga 1620
tacgatgtct catcatgtac aggtagtgtc ataaccacag caatatgcgg attgctgtct 1680
ggcgacgcac aatacaattg tgttactcct actcccggtt tcaacttcgc aaagcgtgac 1740
actagtaacc ccaacaagc cttgtcacat gttgagaaat gtgaaggatc cggctcttcta 1800
ggcttggtta ggcgtatggt tgcagactct aaatcttctt aa 1842

```

```

SEQ ID NO: 356      moltype = DNA length = 1845
FEATURE             Location/Qualifiers
source              1..1845
                    mol_type = genomic DNA
                    organism = Rasamsonia argillacea

```

```

SEQUENCE: 356
atgagatttc cttcaatttt tactacagtt ttattcgacg catcctccgc attagctggt 60
gtcacctacg ataagcgtga tttagcacia gacatatgga acgacataaa gaatgctgta 120
gattgtgctg gttgtcaagg tatcttgacc gctctaaaag gtctttcata tcttgccact 180
acggcctttg ttgatgtttt gaccgaagtc tgtgacatct ccggaatgga ggactcagac 240
gtttgttccg gattcctaag cagcgaaggc ccagtgtgtg actatacct gaagcaatta 300
gatatcgggt ctcatacttc tcaagtgtac tgcgcctcag cttttgtgtc gtgccagtat 360

```

-continued

```

cctgctgttc gtgcttacaa ccttacattt ccttcaccaa agccagataa gactcgtcct 420
gaaccgagtg gagaatcacc aatgcaaat gtctacttta gcgatacaca tgtggactta 480
tcataatgaa caggtagcaa ctacaattgc actaagccta tttgtcgcag accgtatacg 540
gcagacgatg cacttggtta caccactact ccatgtggac cctacggaaa tacaaaaatgt 600
gaacgaccta tgacottaga ggaatcaatg ttgcgagcca tcaaaagcact ttcacctcaa 660
cccgctttca gcactatata cggtagtgta gtagccacag acatatggtt agtagaccaa 720
aacgaagtgc ttgaagactt gaatgaaca tacgatagaa tggcagggtt gggcttggtt 780
tacgctgcca ttggcaacca cgatactgct cccgttaagt atcttcttac ttccaacatt 840
ccctctcagt attcagcaaa ttggacgtat gaagccttag aataccattt ctctttgtta 900
acaaactcag cctctgcccc aactgcccag aactatggtt ctactcacc cgtctatcct 960
ggcaagtatg gcacggattt gagagtgatt tcatacaatt ccataattcta ctatgtcgac 1020
aacttctggg catatcaaga tccgatgttg tacgatccag acggccaact ggcttggttg 1080
atcaatgaat tacaagaggg cgaactgctt ggtcaaagag tctggttgat agtcatgtt 1140
cggagtggga ctgcccagca ttcccacgat tactccactt actttgacca aattgttcag 1200
aggtagaaaa ccacgattgc tgccttggtt tacggtcata cccatatgga tcaatttcag 1260
atttcttatt cagactattc caatagggct ttgatacag caaccgctat aggttacatt 1320
atgctttcca tagcgctcac atctggtccg cctacattca gagtttatga tgttgaccca 1380
aaaacattcg ctgtcttgga cttcaccaac tacatagcca acatttagta tcccgcatac 1440
cagagtggtc caacctggca aaagtactat agtgcaaaag aagcttatgg tccacttctg 1500
tctctccag tactgcagag ttaactccag cattttggca taatgttacc 1560
gtggctttcg aaaatgacga tacggctttc caagagtact gggctagaca gacgaggggt 1620
tacgcagttt caaattgcac cggtaattgc gttaccagg ctatctgtgg cctgagggca 1680
ggtagaaagt aatacaattg tgtcactcca acaccaggct tcaacttgc aaaaagagat 1740
gtctcttcag atggtcaagc acttccccac atagagaagt gtgaagggtt tggcctacta 1800
tccttattgg ccaaaatggt agcctcaaac ggtcagtcac gtttaa 1845

```

```

SEQ ID NO: 357      moltype = DNA length = 1845
FEATURE            Location/Qualifiers
source              1..1845
                    mol_type = genomic DNA
                    organism = Penicillium parviterrucosum

```

```

SEQUENCE: 357
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcat 60
attgaatcag cagacaattg gatcaccaca atctgggact tgtcaaaaca agccgtagat 120
tgcgctgggt tccaagctct gttagggggt ctgaaactgg ctgcagattt gggtagagac 180
tttatggaag atgttcttat tgggtgttgc aacattgctg gtgttgaaag tcatgatgtt 240
tggtcaggca taatccaaaa cgaaggctct gccgtacact attcattgct taacttacac 300
attggctctc atacagcaac aactttatgt gcttcttat tgggcctatg ccaatatcca 360
gccgttagac catacaactt atcatttccg gttccaaaaa ctactaagtc tagaccggaa 420
ccatccggtc agtcacctat cagaatagta cacttctcag acacgcatgt tgacttatct 480
tacgaaacag gctcaaaacta tgactgttcc aaacctattt gctgtagacc atacacggag 540
gaagacgcac caggaaatata ctctacccct tgtggcccat ggggtaaccc catgtgtgat 600
ccaccaataa gactgcaaga atctatgatg actgctatag cagatttgaa tccagcattc 660
tccatctata ctggtgatgt cccagcacac gacatttggg ctgccacgaa ggcagaggct 720
cttaggggact ttaacgctac ttatggatcc atggaaaaac gtttgggtat ggtatttgc 780
gctttgggaa atcatgatgc cgctcctctt aacttgttcc cttccaaaaa aatcccgctc 840
gagtattcac cgcaatgggc ctatgacgcc ctagecagcag actggctggg gctgagtga 900
atggcttccg tgcactccgc aatccatcac gggctcctatt ctgcagtcac ttctgaggat 960
aaactacgtg tcattttcata caactccatt ttctattaca aagataaact ctttatgtac 1020
gaggagccaa tggaaacacga tccgaatggt caattcgctt ggttgattag tgaattgcaa 1080
tccgcagaaa gcacctcaca aagagtgtgg ttgatagcac atattccttc cggtaaacga 1140
gatcacttta gggattttcag ccactacttt gacgaaattg ttcagagata cgataccact 1200
attgctgcct tatacttcgg acatactcat acagacactt ttcaaatgac atactccaac 1260
tattcaataa gatcctggga caccgcctct gctatgggat atgtggcacc gtcctagaca 1320
cccactagcg gaagcccaag ctttaggggt tatgaagttg atcccgctac cttggcatt 1380
atcgacttta cccagtatat cgccaacatt tcagatccca gctaccagat taaccctaaa 1440
tgggagccct attactcagc taagaaggcc tatggtagca aactatctcc acctgcccag 1500
gaccaggtg cagaaatgac acctgccttc tggcataacg ttactatagc catggaacaa 1560
gatgcttcca tattccaaagc tttttgggca cgtcgtacta gaggaaacaa agtaacttct 1620
tgtactggca actgtatggc aaatgagatt tgcgcactta ggggtgcga tgcacaatac 1680
aattgtgcta ctcccaccgt tggtttttaga ttcagaaaaa gagatagac tagtgactta 1740
agcttgcaaa aagaagagtt tagacctgaa tgtaatcatg caggaatggg tccattgttg 1800
gcaaagattg tccatcaagc agcttgggaa aatgaacgtg ggtaa 1845

```

```

SEQ ID NO: 358      moltype = DNA length = 1845
FEATURE            Location/Qualifiers
source              1..1845
                    mol_type = genomic DNA
                    organism = Penicillium flavesens

```

```

SEQUENCE: 358
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcat 60
agtgaatccg ctgagagctg gattagtgc atattgggacc acttcaaaaga agctgttgat 120
tgtacttctt gtcaggtttt acttggttcc ttaaagttag tagccgcat tggcgacact 180
ttcatggtag atgtattgac tggcgatgt gctatcagtg gcgcagaaaga ccacgatgtg 240
tgtgctggcg ttattgcagg ggagggtcct tctgtgcagt attcactgaa gaacttgaag 300
attggcagcc atactgctaa aactttctgc gccagtttag ttggtttgtg ccagtaccca 360
aaggttagag agtatgacct ggccttccct gcacccaaaac cgccaaatgg aagaccgcct 420
cctagcggag aacctccact caaagtattt catttctcag acacacatgt agacttggca 480
tatgaaccag gttcaacta tgcattgtagc aaacccatat gttgcaggac ttacaaagag 540

```

-continued

```

aatgacgctc ctggcaatc cagttctgct tgtggctcct ggggcaatcc tagatgtgat 600
agccctcata gactacaaga gtctatgaat gccgcaatcg cagacttaaa cccagcattt 660
tccattttaca ctggcgatgt ggttgctcat gatgtgtggc tagacaacaa gtttgagtt 720
ctacaaaact tcaatgaac atattggagct atggaaacta ccctagggtca agtctatgct 780
gccttaggca atcacgatac ggcacctctt aacttggttc caagttccaa gattccatca 840
atctataacc cacaatgggc ttatgacgct ttgaccgaga attgggttagc attgaccgga 900
ataccttcaa tcgaaagcgc agaccagtac ggatcttatt ctgtcttaca cccagattcc 960
aatttgagaa tcatctccta caattccatc ttttactaca aatacaactt ctttgcttac 1020
accgaaccta tggaaataga tccgaattct cagttgaaat ggttgataaa cgagttgcaa 1080
gctgctgaaa aagctagtga gagagtttgg ttgatttcac acataccttc cgtaactca 1140
gaaccattcc acgaacctc tcactacttc gaccagatca tccaaaggta tgatgctacc 1200
atagctgctc tattctttgg tcatactcat ctggatgaat tccaaatctc ctattcagac 1260
tacaaatcta gaacttggga cactgccact gctatgggtt acatcgctcc ttcaatgaca 1320
cctaccagtg gtccgccttc atttagagtg tacgatatag atccagttac ttttggcgtt 1380
ttagatttca ctacgtacat agccaacatc aatgcaccag atcaagaaag tcttgagtgg 1440
gtgccttata atagtgttaa agaagcttat ggcagtaaac tagttctctc aattaccgac 1500
ccatcagctg aattgagccc agctttctgg cataaatgta cattagctat ggaaatgac 1560
agcgctatct tcggagactt ttgggctagg agaaccaggg gttatcaagt gccgagttgt 1620
accggagatt gtattagtgg cgagatttgc acacttaggg gtgcccagcg ccagtacaac 1680
tgtttcgctc aaaggttgg ttctctttt gaaaagagag atcatcaagg agattctact 1740
agggaaagaa ggatcttacc tgagtgcac catgctggca tggctccatt gcttgcaaa 1800
attgcaagat tagctgctat agccagagac atggagaaaa ggtaa 1845

```

```

SEQ ID NO: 359      moltype = DNA length = 1848
FEATURE            Location/Qualifiers
source              1..1848
                    mol_type = genomic DNA
                    organism = Penicillium hispanicum

```

```

SEQUENCE: 359
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagctgaa 60
tcagctgaat catgggtggc atccatctgg gacgacttca aagttgccgt agattgcgct 120
agctgtcaag ctctgttagg ggggtctgaa ctgggtgctg aatttggtga gtcctttatg 180
gaagatgttc ttattgtgtg ttgtgatgta tctggtgttg aagattcaga tgtttgtgct 240
ggtgtcatcg caaacgaagg tctgcccgtc cactatacat tgaagaactt agagattggc 300
tctcatgcag caaatacttt atgtgcttcc ttagtggccc tatgcgaata tccagccggt 360
agaccataca acttatcatt tccgtctttg aaacctgcta cctctagacc tccaccatcc 420
ggtaagcctg caatcaaaat agtacacttc tcagacacgc atgttgactt atcttaacga 480
acaggctcaa actttgactg ttccaaacct atttgctgta gagtttacac ggagggaagac 540
gcaccaggaa atacctcttc ccttgtgtgg ccatggggta accccaaatg cgatccaccc 600
catagactgc aagaatctat ggttgaagct atagcagttt tggatccagc attctccatc 660
tatactggtg atgtogttgc acacgatgtt tgggtgggta acaagtcoga ggctcttcag 720
gactttaacg ctacttatgg agcaatggaa aatcgttttg gtccgtgata tgctgctttg 780
ggaaatcatg atacgggtcc tcttaacttg ttctcttcca acaaaatcag cagagagtac 840
aatccgcaat gggcctatga cgccttagca gcagactggg ctgccttgac cggaaattcca 900
tccgtggcat ccgcaaggga atacgggtcc tattctgcaa tacatccaaa ctcaaatcta 960
aggatcattt catcaaacct cattttctat tacagggtta acttctttgc ttacgaggag 1020
ccaatggaat acgatccgga ttctcaactg gcttgggtga ttaccgaatt ggatgcagca 1080
gaaacggcag gtcaaaagag gtgggtgata tcccatattc ctccggtaa gccagatcac 1140
tttagggatc acagccacta ctttgaccaa attgttcaga gatacagatg cactattgct 1200
gccttattct tcggacatac tcataaagac gaatttcaga tttctactc cacatataca 1260
aatagagcat gggacaccgc cactgctatg ggctatatcg caccgtccat gacacccact 1320
agcggaccgc aagctgttag ggtgtatgaa attgatcccg tcacctcgg tgtaatggac 1380
tataccagat atatcgccaa gatctcagat agttctgccc aaattgacac aactcctgag 1440
tgggtaccct actattccgc caaggcagat tacgggtgca agttagctcc accggtcgaa 1500
ggtgcaggag tccaggttaac tccagccttt tggcataatg tcacggtagc tatggaagca 1560
gattcatcct tgtttcaagc tttttggggg agaagaacac gtggctacaa tgtttcatct 1620
tgcaccggag aatgtatggc cactgagata tgcgccttga ggggtgcaga tgcacagat 1680
tcttgtgttg gcgctaaacc aggtctgtct ttttctaaga gaggtggcaa ggatgtggat 1740
gttttgtggc aaagaagact acaaccagaa tgtaatcatg ctggcatggc tccactactt 1800
ggtaagattt ggacaagagc tgctttagtt tggcggtcag aagcttaa 1845

```

```

SEQ ID NO: 360      moltype = DNA length = 1854
FEATURE            Location/Qualifiers
source              1..1854
                    mol_type = genomic DNA
                    organism = Penicillium simplicissimum

```

```

SEQUENCE: 360
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagctggt 60
gtacccgaga actggattag tactatatgg aatgacttca aagaagctgt ggattgtggt 120
agctgccagg ttctgttagg ggggttgaaa ttgggtgccg attttgagga aggcctttatg 180
gaagatgtct taactgtgtg ttgtgacatt tctggtgcag aagatagaga tgtctgcgca 240
ggggttattg caagtgaagt accagcattg cactacgcaa tcaaaaacat gcatgtagg 300
tcacatactg ctaaaacctt atcagtgca ctgggtgtgc tatgcgattt tccagacgtg 360
aggccatttg acttggtttt tcttctctct aaaccagcca attctagacc tcccccttcc 420
ggcaaacctc ctatcaaggt tgtgcacttt tcagatacac atgtggactt aagttatgag 480
actggctcaa actatgactg tagtaacct atttgttgca gggtttacac agatgcagat 540
gccccaggaa ccacagacaa accatgtggg ccttttggga ataccaagtg tgatccgcca 600
catcaactac aagaatctat gatgactgct attgtgaac ttaatccagc attttccatt 660
tacactggag acgttgttgc tcatgacgtg tggttagtga ctaaggagga agttctacaa 720

```

-continued

```

gatttgaatg caacttatgg tgccatggaa aatcatttag gtttggttta tgctgcatta 780
gggaatcacg atgocgcacc cttaaacttg tttccacgcc ataacgttcc aagcaaatac 840
aaccacacagt gggcatatga tgcccttgact gccgactgga tggccttaac cggcatagaa 900
aatgtgcaaa acgccaatga atattggatcc tattctgcca ttcacccaaa ctccaagtta 960
cgtatcattt cttacaattc catcttttac tacaagtaca atttcttctc ttatactgag 1020
ccaatggaat atgatccgaa tgggcagttg acctgggtga ctgaagaatt acaggcagca 1080
gaaaatgccg gacaaagggt atgggtgata tcccatatac cttcaggtga tgtcgatcac 1140
ttcagagatc atagocacta ctttgaccaa atcatacaga gatatgaggg aacaattgcc 1200
ggattgttct ttggtcacac tcatacagat gaattccaag tatcttattc cgattacaaa 1260
aacaggaact gggatacggc aacagccatg gggtagcttg caccctcaat gacaccgact 1320
agcggttccc cctcttttag agtatacga atcgaccag taacctttgg tgtgctagat 1380
ttcaccacgt acattgcaaa catcagcgat gctagcttcc aaacaaagcc tacatggata 1440
ccatactaca cggccaaaaa ggattacggg tcaaagcttc cgaccattcc agatgatacc 1500
gctgaactta ctccggcttt ctggcataac gtgacagttg ctatggaaaa agattccgct 1560
gtttttgatg aattctgggc tagaagaacg agaggctaca atgtaccagc ctgcacagga 1620
gattgtgcca aaaacgaaat ctgcgctctt agaggcgagc acgcacagta ttcttgctgt 1680
caaaagaactc cagggttttc tttttccaaa agagacggag tagacgttga aatggaagaa 1740
gttgctcctc tatgtctaa aagatttcag cctgaatgta atcatgccg tatggctccc 1800
ttgttagcca agatagctca caacgccaat ttggctaaga tgaatggcga ataa 1854

```

```

SEQ ID NO: 361      moltype = DNA length = 1857
FEATURE
source             Location/Qualifiers
                   1..1857
                   mol_type = genomic DNA
                   organism = Penicillium vasconiae

```

```

SEQUENCE: 361
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagctggc 60
gcagtcgagt cttggatttc aactatttgg aatgatttca aagaggcagt agattgcggg 120
agctgtcagg tactgttggg tggtttgaaa cttgtagcag attttggaga aggccttttg 180
gaagatgtct taactgtgtg ttgtgacatt agtgggtgctg aagataggga cgtctgcgct 240
ggtgtaatcg caagtgaagt accagcccta cattacgctt tgaaaaacat gcatgtagg 300
tcacatactg caaaaaacct ttgtagcgct ttggtcggtt tgtgtgattt cccggatgtt 360
agaccgtttg gtttgacatt tccttctcca aagccagcta aatcacgtcc tcctccctct 420
ggaaaaagcc cattaagggt ggttcacttt agtgatacgc atgtagactt aagctatgag 480
acaggatcaa actatgactg ttcaaaacct atctgctgtc gtgtctatag tgatgaggat 540
gcacctggta agacagataa accgtgcggt ccgtttggca acacaaaatg cgtaccacca 600
caccgaattg aagaaagtat gtagcacgct atagccgatt tgaatccagc tttttccatc 660
tataccggtg atgttgttgc ccacgacgta tggttgggtc acaaagatga agtattgcaa 720
gatttcaatt caacttacgg tgctatggaa aatcatttgg gtttagttta cgcagccttg 780
ggaaaccacg atgctgcaac ctgaaatctt ttcccatcca agaattgtgc aagtaagtat 840
aaaccgcaat gggcatacga cacgttgacg gccaatggga tgacccttac cggcatagag 900
tcagtgcaga atgcaaatga atacggttct tactcagcaa tacacccaaa ttcaaaagct 960
aggatcatac cttacaattc tatcttctac tacaataaca acttcttttc atacactgag 1020
cctatggaat acgatccgaa cggtcagtta acttgggtga tatcagagct acaaagtgtc 1080
gagaatgctg gacaaagagt ttgggtgatt agtcacattc catcaggcaa tgtggaccac 1140
tttagagacc atttctcata ctttgaccaa atcattcaac gttatgaagc taccatagct 1200
ggtttgtttt tcggtcatac tcatacagac gaatttcaaa tagcatatag tgattacaag 1260
aacagaaatt ggaatacagc cacagccatg ggttacgtcg ctccgagtat gactccgact 1320
agtggccca cactcttttc tgctacgac attgatcccg ttactttcgc agttatcgac 1380
tttacacagt atatcgccaa catatcagac cctacttttc agaagaaacc agattgggtc 1440
ccgtactatt ctgccccaaa ggattatgga ggcaagcttt ctccaaggcc tgcgataca 1500
gccaggttga ctccggcttt ttggcacacac gttacagttg ctatggaaaa ggattcttca 1560
gtctttaacg agttttgggc acgtagaaca agaggctaca atgtgcctgc atgtactggt 1620
gactgtgtca agaacgagat ttgcgctcta agaggggacg atgtcctaata ctctgtgtt 1680
caaaagaacgc caggttttag tttttctaag agggatgacg aaaatgttga tggagatgtt 1740
ggaaactatc ccttgttatc aaagagggtt caaccagaat gcaatcatgc tggatatggc 1800
ccattgttag ctaagatcgc tcataaagct tccttggcta aggcaaatgg tgaataa 1854

```

```

SEQ ID NO: 362      moltype = DNA length = 1860
FEATURE
source             Location/Qualifiers
                   1..1860
                   mol_type = genomic DNA
                   organism = Talaromyces columbinus

```

```

SEQUENCE: 362
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagcttac 60
gactctgcct tggtagatca taacttagta tccgatatct gggaagatat caaggaggct 120
gtgacatgcg tgggttgta agttatctta gctgccttga aaggtgtctc agatttgggt 180
actacagctt ttgttgatac ttgacaggt gtttgaatac taagcggagc cgcagatgat 240
aacgtatgtg aaggcattat ctccaggggag ggtgcagttc tacattatgt attgtcagaa 300
ttgtccttag gaagtgaac gtcaaaatga ctttgtgctt ctgccttcgg attatgtctg 360
tatccagatg taagaaacta tactctgaac ttctctcag ccaaacctaa gaacataaca 420
cgtcctgtct cgagcggtaa acctctctatt caggtagctc actttagcga taccatgtc 480
gatctatcat acgaagtagg ttccaactgg aattgcacaa aacctatttg ctgtagggtc 540
tttaggctct cagatgctcc cggtaacaca actaccctct gtggggcatt tggtaaacat 600
aaatgtgaca caccattgac ttgggaagag aatatgttag attccattaa gaaatcagac 660
ccaactccag ctttcagcat ctatacgggt gacgtcgttg ctacgacat ttggttagtc 720
gataaagatg aggttttgac agacttgaat gctacttatt ccttaatggc cgaatatagg 780
accgtatatg cagccattgg taaccacgat accgctccac tgaatgatct tccaaccacg 840
caagtacctg agagctacag cgccaactgg acatatcaag ccctagcaac aaactttacc 900

```

-continued

```

acgttaacaa gagattcttc agttatctca gtagcaaaaga actatggtag ctactcttct 960
gtctttaccg gttcttaacg tacagactta aagatcattt cctataactc tatgtttctac 1020
tatgtggata acttctatgc ctttttggat cctatgccat acgaccocaga tggtcagtta 1080
gcttggttga ttgatgaact acaatccgca gaaacagctg gccaacgtgt gtggttgatt 1140
gcccattgtc caacoggtag ctccgacctt ttcccgact atagtcacta ctttgatcaa 1200
attgtccaaa gatatgatgc aacctagca gccctgttct ttggacatac gcacaccgat 1260
caattccaaa tagcctacag cgattatgca aacaaaaacg cagacactgc cactgcaatc 1320
ggctacatta tgcccagttc taccccaaca agtgggtccac ccgcatatag gatctatgat 1380
attgatcccg tgacgttctc tgtgctagat tacacgggtc acataaccaa catatctcac 1440
ccagatttcc aaaaggggtc aaagtgggag aagtactatt ccgcaaagga tacatacggg 1500
agcttgctgt ctccacgtgt gacagatcca tctgctgaaa tgacaccagc tttttggcat 1560
aacgtgactg cagtattcga atcagatgat gtacgtttcc agggatactg ggctaggcaa 1620
accaggggct acgatgtctc agactgtacg gatgaatctt gtaagaacca aactatttgt 1680
gccttgagag ctgcagatgc tcagtacaat tgcgttgctc catctatagg ctttaacttc 1740
gctaaaagag atgatacgga ccaagctcat gtcaagggtc aaaaagagaa atgtgatgat 1800
acaggtcttg tgtccttatt gggttaagatt ttggcaaaat caaaagaaac tacaactaa 1860

```

```

SEQ ID NO: 363      moltype = DNA length = 1860
FEATURE            Location/Qualifiers
source              1..1860
                    mol_type = genomic DNA
                    organism = Talaromyces variabilis

```

```

SEQUENCE: 363
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagcttcc 60
agctctgcct tgatagatag agacttagca tccgaaatct gggatgatat caaggaggct 120
gccacatgag ctggttggtc agttatctta gctgccttga aaggtgtctc agatttgggt 180
actacagctt tgattgatca ttggacagaa gtttgcaaga taagcggagc cgaagatgat 240
gacgtatgtg aaggcattat ctccagggag ggtcctgttc tacactacat ttgtcaca 300
ttgtccttag gaagtgaaac gtcagatgca ctttgtacta ctgccttcgg attatgtgcc 360
tatccagatg taagaaacta tcttttgact ttcccttcag ccaaactcga gaattccaca 420
cgtccttcaa gcacgggtga atcacctatt caggtagtac actttagcga taccatgtc 480
gatctatcat acgaaacagg ttccaactgg aattgcacaa aacctatttg ctgtagggtc 540
tttgactcat cagatgtctc cggtaacaca aagacccctt gtgggccata tggtaaacct 600
aaatgtgacg caccatttcc attggaanaa tctatggtag attccattaa gggtttaagt 660
ccagctccag ctttcagcat ctatacgggt gacgtcgttg ctacagcat ttggtagtc 720
gatgaagatg aggttttagc agacttatct tctacttatg gttagtaga ggatgtcggc 780
accgtatttg caccatttgc taaccacgat accgctccag tgaatgatct tccaaccacg 840
caagtaccta gcacgtacag cgccaactgg acatatgaag ccctagcagg aaactgtacc 900
acgttaacag gtgattcctc agttatgtca gtagcagaga actatggtag ctactcttct 960
gtctttaccg gttcttaacg tacagactta aaggtgattt cctataactc tatcttctac 1020
tatgcccaga actctctatgc ctttttggat cctatgccat acgaccocaga tggtcagtta 1080
gcttggttga ttgatgaact acaagcatcc gaaacagctg gccaacgtgt gtggttgatt 1140
gcccattgtc caacoggtag ctccgacctt ttcccgact atagtcacta ttggatcaa 1200
attgtccaaa gatatgatgc aacctagca gccctgttct ttggacatac gcacaccgat 1260
caattccaaa tagcctacag caactattcc aacaaaaacg cagacactgc cactgcaatc 1320
ggctacatta tgcccagttc taccccaaca agtgggtccac ccgcatatag ggtgatgat 1380
attgatcccg tgacgttcgg tgtgctagac ttacgggtc acatagccaa cataagcgac 1440
ccagattacc aaaacgggtc aacatggggt aagtactatt ccgcaaagga aacatacggg 1500
agcttgctgt ctccacgtgt gacagattca tctgctgagt taacaccagc tttttggcat 1560
aacgtgactg cagtattcga aacagatgat gcagctttcc agggatactg ggctaggcaa 1620
accaggggct acgatgtctc aaactgtacg gattcatctt gtaagaacca gaactattgt 1680
gccttgagag ctgcagatgc tcagtacaat tgcgttgctc catctatagg ctttaacttc 1740
gctaaaagag atgaaacgga ccaagctcat gtcaagggtc aaaaagagaa atgtgatgat 1800
gcaggtcttg tgtccttatt gggttaagatt attagtaaat caagagacgt tcaactaa 1860

```

```

SEQ ID NO: 364      moltype = DNA length = 1860
FEATURE            Location/Qualifiers
source              1..1860
                    mol_type = genomic DNA
                    organism = Talaromyces rugulosus

```

```

SEQUENCE: 364
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagcttct 60
tccagtgcct tggtcgacag agatctagca agtgagatat gggatgacat caaagaagct 120
gctacttgta ctggttggtc agtgattctt gctgccttga aaggcgtatc agaccttgg 180
acgaccgtat tggtagatag tcttacagaa gtttgcaagc ttagtggagc cgaggatgac 240
gatgtatgag aaggcataat cagcagagag ggtccgggtc tgcaatacat cctttcacag 300
ttatccctgg gttctgaaac ttacagcgcc ctgtgcgcac cagcttttgg gctttgctca 360
taccagatg tgagaagcta cacttgaca ttccatcta cgaacactga gaatagcaca 420
aggccgagta gttcaggaca agctcctatt caagtcttc acttttcaga taccatgta 480
gatttgctct atgaaacggg tagtaattgg aattgcacaa aacgatctg ttgcagatcc 540
tttgactcta gtgacgtccc agccaatact tcaactcctt gtggccctta tggaaataca 600
aagtgtgatg cacttttaag cttggaagag tccatgttcg attccataaa gtcattatct 660
cccgtccag cctctcaat ctatactgga gatgtgggtc cacatgatgt ttggatagtt 720
gacaaagatg aagctttaac agacttgaat gccacttatt cttgatggc tgagggttgg 780
acggatagc gtcccatagg gaatacagac accgctccct taaacgattt gccaaactca 840
caggttcctt ctacatatcc agcaaacctg acttaccagg ctttagctac caatttcaca 900
acactatcag gcgatagctc tatcatgtcc gtagctgaga attacgggtc ctactcatcc 960
gtttttgctg gctcccatgg tacagacttg aaagtgatta gttacaactc cattttctac 1020
tacgtcgaca actctatgc attcttagat cctatgccat acgaccocaga cggccaatta 1080

```

-continued

```

gcttggttga tagatgagct gcaagctgct gaattctgctg gccagagagt ctggttgatt 1140
gcccacgtgc cgacogggtc tagcgacctat ttccacgact actctcacta ctttgatcag 1200
atagtccaac ggtacgatgc tactatttga gccttggttt tcggctcacac acatacagat 1260
caatttcaaa tagcttattc cgattactca aatcagaata gtgacactgc aacagcaatt 1320
ggatacatca tgcggtcatt aactccaacg agtggacctc cagcttatag agtttaogat 1380
attgatccgg ttacctttgg cgtggttagac ttacagctct acattgcaaa catctccgac 1440
ccggattatc aaaaacgggtc aacctgggct aagtactata gtgctaaaga aacttaacggg 1500
accttattga gtccagcgct tacagactct agtgctgagt taacaccagc tttttggcac 1560
aacgttaccg cagtattcga aacggatgat acttccttcc aagggtattg gccacggcaa 1620
accagggggt atgatgtgtc caattgtacc gattcctcct gtaagaacca gacgatctgc 1680
gcactaagggt ctgocgatgc tcagtacaat tgtgtcgtac ctactattgg cttcaacttt 1740
gccccaaagag atgaacacaga tcaggcacac gtaaaaggccc aaaaagagaa atgcgacgat 1800
actggtttgg tatccttgtt agggaaaatc atttcctcaa gtcgtaattg ttctctttaa 1860

```

```

SEQ ID NO: 365      moltype = DNA length = 1860
FEATURE            Location/Qualifiers
source              1..1860
                    mol_type = genomic DNA
                    organism = Hamigera terricola

```

```

SEQUENCE: 365
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcaa 60
atgcaaacat ctgaacgttc atggggatct acgatatgga aagacgtgga ggaagccgta 120
gactgtgctg gttgtgaagt catcttagga gccttgaaac ttgttgacga tttaggtaaa 180
ggtacttttag aaactgctat gattgatgtt tgtgatctgt ctggggctga agatagtgat 240
gtatgttcag ttatgtatac agccgaaata gatgccttgc actatgcctt gaacaatgtc 300
catgtcggct cccacacatc aaagggtgtg tgtgtcacc tgtttggttt gtgctcttat 360
ccagatgtta ggtcttatca cttgatcttt cctagtcca aaccogctac atcccgctct 420
tcaccgagcg gtcaaaaacc catlaaggta gctcacatta gcgataccca tgcgatcta 480
tcatacgaac ctgggttccaa ctacgaatgc tccaaaccta tttgctgtag ggcataatac 540
aaagaagatt ctcccggtta cacaccccac cctgtgtggc catatggtaa cacaaattgt 600
gacgcacatc atagattgga agagtctatg tttgctgcta tgaaagcttt agaccagct 660
ttttctatct acacgggaga cgttgtcgca catgacatat gggtgggtta tgaaactgaa 720
gttctggatg atttagatgc aacttattcc ttaatgaaat ccctgggtct agtctacgca 780
gctgtaggca atcatgatgc cgtcccggtt aatctgttcc ctagttaatc tatcccttca 840
acatactctc cgcagtggtc ctatgatgct ctagccgaag actggcttgc tcttacgaat 900
gattcttccg tcgattctgc cagagaatac gggctcatatt cagccgtgta tccaggatca 960
aaacttgagaa tcatttcccta caatagtgtg ttctactata aggataactt ttggatgtac 1020
gaagacccca tggagatgta tccaaacgga caactggcat gggtgattaa cgaattacaa 1080
gcagctgaat ccgaggtgga aagggatagg ttaactctcc acatcccttc aggttaactca 1140
gatcatctat atgactatag tcactatttc gatgctatcg tgcaaaagata tgaagctacc 1200
attgtgcctt tattcttcgg acatactcat acagatctat ttccagatgc ctattcagac 1260
tatgataaca ggaactggga tacagccact gccatagggg atatcgctcc tagtatgaca 1320
ccaaacaagt gcagtccgcg ttttaggata tatgaggtcg atcccgtagc ttttggattt 1380
cttgactaca cagtttatcc agccaatata tcccactcct catatcaaac gcagcctact 1440
tgggaaaagt actattccgc caaggaggct tatggatcct tgtaactccc accggtgaca 1500
aaaccatcta tagagttaac tcacgttttt tggcataacg ttactgtcct aatggaggac 1560
gatgaatcag tgttcaaaaga cttttgggct cgtaccacaa ggggctttta cgtttccacc 1620
tgctgcggtt gttgtatgac tgaggagatt tgtgtctcta ggtctgcaga tgctaagtac 1680
aattgtgcaa gttctgaagc ttgtctgaac tttttgaaaa gagatgatgt cgaagctcaa 1740
tcaaaacctg tgaagccaca atgtgaagac tctggattgg ctgcagtttt ggtaaaaatg 1800
atggagaata cagatgattt cgtctgtttg ttgaaagaaa aggcctttagc acatggctaa 1860

```

```

SEQ ID NO: 366      moltype = DNA length = 1860
FEATURE            Location/Qualifiers
source              1..1860
                    mol_type = genomic DNA
                    organism = Penicillium piscarium

```

```

SEQUENCE: 366
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctaga 60
accgaagcag cagacagctg gatttccaac atatggaacg agttcaaaaga agctgttgat 120
tgttggttct gtcaagtttt acttggtggc ttaaaagttag tagccgattt tggcaggggt 180
ttccttgaag atgtattgac tggcgtatgt gatgtgtctg gcgcagaaga cagggatgtg 240
tgtgtcggcg ttattgcaaa cgaggtaact gctctgcact atgcactgaa gaacatgcat 300
gtaggcagcc atactgctaa aactctgtgc agcgccttag ttggtttgtg cgacttccca 360
aacgttagac cgtttgactt gaccttccct tcacccaaac cggctaaaag tagaccgcct 420
cctagcggaa aacctcccat caaagtagtt cattctcag acacacatgt agacttgtca 480
tatgaaactg gttcaaaacta tgattgtagc aaaccatata gttgcagggt ttactcagac 540
aaagacgctc ctggcaaaac cgacaaacca tgtgtcctt tcggcaatac aaaaatgtat 600
ccgcttgaaa gactacaaga ctctatgatg gctgcaatcg cagacttaaa cccagcattt 660
tcacattaca ctggcgatgt ggttgcctcat gatgtgtggc tagtcaacaa agatgaagtt 720
tcacaagact tcaatagtac atatggagct atggaaaatc acctaggttt ggtctatgct 780
gccttaggca atcacgatgc cgcacctctt aacttgtttc catccaacaa cattccatca 840
aagtataacc caagatgggc ttatgaggct ttgaccgcaa attggattac cttgaccgga 900
atacaatcag tccaaaacgc aaacagtagc ggatcttatt ctgcaattca cccaaattcc 960
aaattgagaa tcactctcta caattccatc ttttactaca aatacaactt cttttccctac 1020
accgaacctc tggaaatgta tccgaatggt cagttaactt ggcttataga ggagttgcaa 1080
gctgtgaaac atgctggcca gagagtttgg ttgatttccac acatacctag tggttaacgtt 1140
gaccacttcc gtgaccactc tcactacttc gaccagatca tccaaaggta tgaagctacc 1200
atagctggtc tattcttttg tcatactcat acggatgaat tccaaatcag ttattcagac 1260

```

-continued

tacaagaata	gaaattggga	cactgccact	gctatgggtt	acgtgggtcc	ttcaatgaca	1320
cctacotccg	gtccgccttc	atttagaatc	tacgatatag	atccagaaac	ttttgcgtt	1380
atggatttca	ctcagtagat	agccaacatt	tctgaagcct	cttttcaaac	caagccaaac	1440
tggtacacct	attactctgc	taagaaagat	tatggcggtg	gggtgactcc	accaaccccg	1500
aatacagctg	aattgacgcc	agctttctgg	cataatgtta	cagtagctat	ggaaaaagac	1560
agctctgtct	tcaatgagtt	ttgggctagg	agaaccaggg	gttattctgt	tccagcctgt	1620
accggagatt	gtgtaaagaa	cgagatttgc	gctcttaggg	gtgccgacgc	ccagtacagt	1680
tgtgtccaaa	gaacgccagg	tttctctttt	tcaaagagag	atgatgggtg	agttgaagtt	1740
gacttggaag	atgctccttt	actttccaag	agattccagc	cagaatgtaa	tcacgcaggg	1800
atggcacctt	tgtagccaa	aatcgctcac	aaagcctcaa	tagccaaaat	gaacgggtta	1860

SEQ ID NO: 367 moltype = DNA length = 1866
 FEATURE Location/Qualifiers
 source 1..1866
 mol_type = genomic DNA
 organism = Talaromyces bacillisporus

SEQUENCE: 367

atgagatttc	cttcaatttt	tactacagtt	ttattcgcag	catcctccgc	attagctggt	60
aaactccaa	cgatcatcact	agttgccaga	gatttggcca	gcgagatttg	gaatgacatc	120
aaagaagctg	ctacotgtgc	tggtatgtaa	gttatcttgg	ctgccttgaa	aggtgttgca	180
gatttgggta	ctacagtttt	gattgatgtc	ttgacagagg	tttgcaaaat	ctcaggagaa	240
gaagatgacg	atgtatgcga	agggatcata	agttagaag	gtccagtttt	ggagtacata	300
cttagtcagt	tatactatc	ctctgaaact	agcgacgtct	tatgtgcctc	tgcttttga	360
ttgtgctctt	atccggctgt	tagagattac	acattgacct	ttcctagtcc	caagccagcc	420
aacattacta	gacctttctc	gtcaggtaag	tcaccgattc	aggtagtcca	tttctcagac	480
actcatgttg	atctgtccta	tgagacaggt	agtaattgga	attgcaccaa	acccatttgt	540
tgctgtgcct	atgaggcttc	cgacgctcca	gggaatacta	ctacccccatg	tggtccttac	600
ggtaatacaa	aattgtgacgc	tccattatct	ttggaacaat	ctatgatcga	ttccatcaaa	660
gccctggacc	ctgcccagc	cttctccatt	tacacagggtg	atgtagtgcg	tcagtatgc	720
tggtattgtg	atgaagatga	agctttaaca	gacttaaacg	ctagttaactc	cttaattggca	780
gagacaggta	aggtgtttgc	cgctattggg	aatcatgact	ctgctccggt	aaatgatctg	840
ccgacgacgc	aggtgccttc	caaatataca	gcaaacctgga	cctatcaggg	cttagccaat	900
aaactctcaa	ctcttacagg	agacagcgca	gtcttaagcg	tggtctaaca	atatggctct	960
tattcctcag	tcctttacagg	ctcctatggg	acagacttga	aggtcatttc	ttacaatagc	1020
atcttttact	acattgataa	cttctatgcc	ttcttagatc	caatgcctta	tgatccagat	1080
ggtaactagt	cttgggtgat	tgaagaactt	caagccagcg	aaacagctgg	gcaaaagatt	1140
tggttaactg	cacagcttcc	aacgtcatca	tccgaccatt	ttcatgacta	ttccactac	1200
tttgatcaaa	tcgtacaaag	atatgaagct	accattgctg	ccttggtctt	tggtgcatac	1260
cacacagacc	aatttcaaat	ctcatacagt	aactatagca	atcaaaacgc	agatactgca	1320
tcgcaactcg	gctacattat	gccatctctt	accccaacct	ccggtcctcc	agcttaccgt	1380
gtgtacgata	tcgatccagt	aacgtttgga	gtcttagatt	tcacgggtga	cattagcaac	1440
atatcagatc	ctgccttcca	gaatggctct	acatgggtcaa	agtactacag	cgctaaggaa	1500
acttaaggga	gtttgttatc	tccaccagta	acagattcta	cagctgaact	tacgcccagca	1560
ttttggcata	atgttactgc	agttttcgag	acagatgatg	atgcctttca	aggctactgg	1620
gctaggcaaa	ccagaggggtg	tgtatgttca	aactgcacag	atacttgtaa	gaatcaaaact	1680
atatgtggta	tcaggggtgc	gaatgctcag	tacaattgtg	ttgtcccaaa	gattggcttc	1740
aatctcgcaa	aaagagatga	aacccgatcaa	gctcatgtca	aaactcaaaa	agaaaagtgc	1800
gatgatgctg	gtctagcttc	tctatttggg	agaatgggtg	caaactcaaa	gaatgcctcc	1860
aattaa						1866

SEQ ID NO: 368 moltype = DNA length = 1872
 FEATURE Location/Qualifiers
 source 1..1872
 mol_type = genomic DNA
 organism = Galactomyces candidus

SEQUENCE: 368

atgagatttc	cttcaatttt	tactacagtt	ttattcgcag	catcctccgc	attagctgcc	60
ccactacca	ctaagagatc	cttggccagc	gacatttggg	atgacattgt	tgatgctggt	120
gactgtgggtg	catgtgatac	tatcttggaa	gccttgaaag	gtttggcaga	tttgggtaat	180
acagtttttg	ttgatgtctt	gacagacgtt	tgtgatctct	caggagctga	agatagtgtat	240
gtatgctctg	gcactatctc	cgaagaaggt	cccattttga	ggaccataat	caagggggtta	300
tcagtcggct	ctgctactag	cgacttggtt	tgtgggactt	tggtgggatt	gtgccaaaat	360
ccggctattg	catcatgggt	agtcctcttt	cctaaaccca	agccaaacac	ggttcgtcca	420
cctccaagcg	gtcacaagtcc	aactctctgc	gcacacataa	gtgatgtgca	tgtagatcta	480
agctacacta	cgaggagccaa	ctatgactgt	tccaagccaa	tctgtttag	accgtacact	540
agcgatgatg	cccttggtta	cacagattat	cctgtctgtc	cataccgaaa	tacaaattgt	600
gatgccccat	tggtatttga	atcatccgct	atggcagcca	tcaagaaact	gaatccggca	660
ttctctatct	tcactgggtg	cgtagctgct	catgacgttt	ggttgggtga	tcaagcagaa	720
gtagaatttg	acttgaatc	aacatacaat	acacagtcca	caactttagg	cacagttttt	780
ccggctcttg	gtaatcacga	tgttgcacca	gtgaacggat	ttgcacgcgc	tggtgggtgagc	840
agtaatccca	acatacagtg	ggcatacgc	acaaacgcgc	aagattggac	aaagtggatt	900
ggctcaacag	cagccaatgc	tgaagagagc	tttgggtgcat	atagtattgt	tcacggcaac	960
ttaagagtc	tctcctcaa	ttccattttc	tactatagac	tgaatttcta	catgtaccaa	1020
gatccactac	aaagagaccc	ttcatcccaa	ttctcctggc	tagtaaatca	actacaagct	1080
gcagaagatg	ctggctcagag	agcctgggtg	atatcacacg	taccatctgg	ctcaggggat	1140
tactttccac	agtattcaaa	ctacttcaat	caaattgtta	acagatatga	agctacaatc	1200
gcagcattgt	tttacgggtc	caactcagtc	gaccaatttg	agatatacata	ctccgactac	1260
tcaaacacga	atagcaaac	ggctgtggct	atgtcctaca	ttactcctag	tttgaccccc	1320
acttcaggct	ctccaagttt	cagaatctac	tctatttgatc	cagtgcagta	cggagtctctg	1380

-continued

```

gactatacaa actacatagc caacatttcc agcccgacat atcagaatgg accacaatgg 1440
gtggagtact attctgcaaa agccgcttat ggtccgtacg ttagccacc attgacgtcc 1500
gctgctgctg agttaacacc tgcttttttg cataatgtga cggttgcatt tcagaacaac 1560
aatgacttgt ttcaggagta catatcaaga aagtctaggg gcttcgacgt tagttcttgc 1620
actgggtcat gtcaaacaga cgagatttgt caattgcgtg ctgctgaatc ccagtataac 1680
tgtgtaacga tttcaccagg catcaacttt aacaagagag atcaacaatc aaacctaggt 1740
gctcaagaaa agcataggga tggttgtgaa ggttctccta ttcgtgacat ctttgcact 1800
ttgatgcaag acagaagagg tctagtttca gccattaacg aagggatcgc taagagatct 1860
attagagcct aa 1872

```

```

SEQ ID NO: 369      moltype = DNA length = 1881
FEATURE            Location/Qualifiers
source              1..1881
                    mol_type = genomic DNA
                    organism = Penicillium megasporum

```

```

SEQUENCE: 369
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctaatt 60
atcatagaat cttgggcttc agagatatgg gatgacataa agaatgcagt cgattgtgct 120
ggttgtgaaa ccatcattgg tgccctgaag gtatgcgcag atcttggtaa aggcaccttg 180
aacggcacat tgatagatgt ttgcgatcta tcagggggtg aggatccaga cgtctgtacc 240
ggcttaactc cttcagaaat cgacgcattg tattactcct tgaaaaacat ggcagtagat 300
tcacacacgt ctaaagtttt gtgtgctggc ctgttctcat tatgtccatt tccagccgct 360
aggccctata acttgagett cccacacact aaaccagcta cgtcaagacc tgcctcgagt 420
ggacagcaac ccatcaaatg agctcacatt tcagacactc acgttgattt agactatgaa 480
gctggttcca actatcagtg tagtaaacgg atttgctgta gaccgtacac tgccgaagac 540
gctcccggtg atacctctca ccttctgtgt ccatggggaa acaccaaatg tgatccacct 600
ttcagattag aagaatcaat ggttgcagcc gtgaatgcct tgaatccttc cttttccatt 660
taacctgggg acgtcgttgc ccatgacata tggctgggtc acgaatctga agtattgaca 720
gacttgaacg ctacttactc cttgtttcaa aacttgaatt ccctagttta tgccgctgtt 780
gggaatcacg atgtgcacc ggttaaacctt ttcccattca ataagataga ttctgcctat 840
aaaccacaat gggcttatga tgccttgaca gcagactggt tggcattaac caatggcgat 900
agtagtgtag cctctgcaaa agccgacggc tcataattctg ccactctatc tggaacaaac 960
ttgaggatta tctctacaaa tagtatcttc tattacaaag ataactctg gatgtacagc 1020
gacctatggt aatattgacc aaacggtcaa ttcgcttggt tgatcgatga gcttcaagca 1080
gctgaaactg caggctcagc gtgttgggtg atttctcata tcccagtggt aaatagtgtat 1140
cacctatacg actacagcca ttacttcgat cagatcgtgc agagatacga agctactatc 1200
gcagctttgt tcttcgttca tactatatac gacttggttc aggtagccta ttcagattat 1260
ggtaatatga attcagatag cgcttccgct attggatatg tcaccccaag tatgacctct 1320
acctctggcc caccgcatc taggattttac gaaatagacc cagttacttt tgggtgtctg 1380
gactatacgg tttacattgc aaacatttca gatcctgctt atcaaacagg tccctcctgg 1440
cagaagtact attcgcgtaa agaagtttat ggctccttgc tatctccacc attgaccgac 1500
ccagcagctg aattgacgcc agctttctgg cataatgtta cagtattgat ggaagaagac 1560
gactctgtct tccaagatgc gtgggctagg actaccaggg gcttcaatgt gagcacctgt 1620
accggatctt gtgctacgaa cgagatttgc gctcttaggg gtgcccagcg ccagtacaac 1680
tgtgtcaccc ctagcccagg tatccaattt gctaagagag gaggtgctgt agattttgat 1740
cccaatgcac aaccagtac aaagcccac tgtgacgagg gttctggttt agcccctgtc 1800
ttgggtcaaaa tgatgagaaa tacggatgac tttgcccgtc tactgaaaga acgtgctgca 1860
ttgcaagatt ctcaacagta a 1881

```

```

SEQ ID NO: 370      moltype = DNA length = 1890
FEATURE            Location/Qualifiers
source              1..1890
                    mol_type = genomic DNA
                    organism = Penicillium jensenii

```

```

SEQUENCE: 370
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctgaa 60
acctccaagg ctagtacgga atcattcata tctcccatat gggatgattt caaacaaagt 120
gttgattgtg gttctctgca agccttgctt ggtggcttga aacttgatc aggtctcggt 180
gagggtctta tgatagatgt tttcattgga ttatgcaacc tttccggagt cgaggatccc 240
gatgtatgca gaggcataat cgagaaagag ggtccagcct tgcattgacgc cttccagaac 300
ttacacatcg gttctcatgc tactcgtacc atgtgcgcac cattgattgg gctttgcaa 360
taccacagaag tgagaccgca caccctacaa tttccatcta gcaaacacga tactacgagg 420
cctccgcctt ccggtaaatc acccatcaaa gttgtccatt tctcagatac acacgttgac 480
ttgttctatg aaactggggc ttctatgag tgtagtaagc caatctgctg tctgttttac 540
gaagacaaag atgcaccagg cataactaaa acaccatgtg gcccttttgg taatactaaa 600
tgcatcctc ctcacattct gcaagaaagt atgaatgccg ctattgcaaa gattgatcca 660
gactttagca tctacactgg cgatgttcta gcacacgaca tatggttggt tggacaagat 720
gaggctctac aagtcttcaa tgatacatat ggccaaatgg agaaggattt gggcatgggt 780
tacgctgcca ttggcaacca cgatactgct cccgtcaact tattccctcc aaatgacatt 840
aagggttaag attcagctca atttgcttac aatgccttag ctgaagattg gtatgcctta 900
acaggaatac cctctgtgaa atctgcccag gaatttggtt ctactcagc aatacatcca 960
aattcaaatc ttaggatcat ttctacaaat tcaatcttct attacaactt caatttctac 1020
atgtaccaag atcctatgga aaaggatcca aacggtcaat tccagtggtt gatcaagaa 1080
ttgcaagctg cagaagatgc cgttgggtta cagtcataat accatccggt 1140
gtgaccgacc acttccgtga ttacagtcac tacttcgacc agattgtcca aagatatgag 1200
gcaactattg caggtctgtt ctatggctcat acgcatatgg acgaattcca aattgcttat 1260
tcagactaca acaatagaaa gtgggacact gctacagcta tgggctacat agctccctca 1320
atgaccccca caagcggtcc tccatctttc agagtctacg aaattgatcc agttacocat 1380
ggtgtactgg atttactca gtacattgcc aacatcatcag atccatccta ccagaccaag 1440

```


-continued

```

ccggaatggg  ttccctacta  ttccgctaaa  gcagcatatg  gatccaaatt  gtctccaccg  1500
ttgacagatt  ccactgcoga  attgacccca  gcatttttgg  ataatgttac  tgttactatg  1560
gagaagacc  cttccatctt  tcaggatttc  tgggcaagga  gaaatagagg  gtggaacatt  1620
gcagcatgta  ctggtgactg  catgaaaaag  gaactttgca  ctcttcgtgc  tgccgacgca  1680
caacacaact  gtcacgaacc  aaccccaggc  ttaaacatat  ccatcaataa  gagagatcaa  1740
ggctctggtg  atgttccttt  agaggcgag  aaagtgagtg  gtccagaatg  tgaccacgca  1800
ggtatggcta  ccttgtagg  aaagattgcc  tatagggcaa  ggctggttag  agaggctgaa  1860
gaaagagatc  cagtcaaggc  tgaaacataa

```

```

SEQ ID NO: 371      moltype = DNA length = 1890
FEATURE            Location/Qualifiers
source              1..1890
                    mol_type = genomic DNA
                    organism = Aspergillus stramenius

```

```

SEQUENCE: 371
atgagatttc  cttcaatttt  tactacagtt  ttattcgcag  catcctccgc  attagctgaa  60
aactgggtca  ataccatctg  ggatgagata  aaggaaacca  tctcttgtgc  aggttgtgaa  120
ggtttggttg  gtacctgtaa  acttgttgcc  ggtctaggac  cagacgtctt  aaccaacgtt  180
ttgaccgatg  tttgcaaat  ggccaaagtc  gaagaccgcg  acgtctgtgc  tggcattatc  240
gaagctgagg  gtccctgcagc  ctactatgtc  ttgaagcagt  tgaaagtggg  gtcccataca  300
tcaaagagct  tttgttctca  aatggtgggc  ctgtgcgatt  atccagaagt  gcgtccctat  360
aacatttcat  tcccaatccc  gaagccttca  acacatagac  ctccacctag  tggtcagcct  420
cctatcaggg  tagctcacat  ttcagatact  catgtcgatc  gtgcttatga  aaccggtgcc  480
aactatgaat  gttctaagcc  aatattgtgc  agagtctaca  cagaggacga  tgcccaggcg  540
aaaaacctct  ttcttctgtg  cccttatggg  catccaaat  gcgaccctcc  tttaaagatta  600
gaggaaagta  tgatggctgc  cattgcagct  atggatccgg  cttttagtat  ctatacaggt  660
gacgttgtgc  cgcacgatgt  gtggtctgta  aatcgtaacc  aggtcttgca  tgatctaaat  720
gcaacttact  ctttgttaga  tagattaggt  cttgtttatg  cgcacttggg  taaccatgat  780
acggctcccg  ttaacctatt  cctagttag  aggattcctg  tttcccataa  tccacaatgg  840
gcctatgatg  ctttagctga  agattggaca  aatcttgtag  atggaccact  atctgcccct  900
gtagtccatg  ccacagacca  attcgctcc  tattctgctt  tgcaccctgg  cgaaaaacta  960
aggatcattt  actacaactc  tgtattctac  tatacgtaca  atttctacgc  ctatcaggag  1020
cctatggaat  atgatccaaa  cgggcaattg  gcatggttgg  tggccgaact  gcaggctgca  1080
gaaacagctg  gacaaaggg  gtggttgatt  gctcatattc  ccaccggagc  agcagatact  1140
ctgagggact  actcccacta  tctggacca  atcatccaga  ggatgacgc  tactattgcc  1200
gcattgtttt  tcggccatc  tcatacagat  ctgtttcaag  tctcatacgc  aaatccagca  1260
catccatctg  cagatttcag  ctctgcccgt  ggctacatta  cccctctttt  aacacccaca  1320
agtgttcccc  cagcctttag  gatctatgac  atagaccctg  ttacatttgc  tgttcttgac  1380
tatacagtg  ataccgcaaa  catctcaaca  ggtgcaactc  caaaatggaa  caagtactac  1440
agcgcaaaa  aaacctatgg  tagttgttg  accctccat  tgacagatcc  aactgcgcag  1500
ttgacaccag  cattttggca  taatgtcact  gcattgatgg  aaacagacaa  tactgttttc  1560
caggcttggt  gggcaaggac  tcagagaggc  ttaacgtcc  cagaatgtaa  cgcccagtg  1620
gcaagggatc  agatttgctc  cttagagcc  gcagacgccc  aatcgggtg  cgtgaggggc  1680
acactaagta  ttacgaaaag  agccggtgat  ggtgacggtt  tagatttggg  aggtgctggt  1740
gccggtggtc  ctccaggaag  acgtaggcac  gtacagtcgc  ctagaccaca  atgcgaagaa  1800
gctggtttgg  caagagtttt  agccgtgtg  attagggaaa  cagacgattt  gcagggtcta  1860
ttgttgcaaa  gagcacagtt  gtacatataa

```

```

SEQ ID NO: 372      moltype = DNA length = 834
FEATURE            Location/Qualifiers
source              1..834
                    mol_type = genomic DNA
                    organism = Bacillus pseudomycoides

```

```

SEQUENCE: 372
atgagatttc  cttcaatttt  tactacagtt  ttattcgcag  catcctccgc  attagctcat  60
accaacgact  gtggcaacga  agctcctata  ctaagggtga  gtgctgaaga  cagacataaa  120
gaaggtgtca  attcccactt  atggattgtc  aatagggcaa  ttgacattat  gagttaggaat  180
aacaccattg  ttaagccaaa  tgagacagca  ttactgaacg  agtggagaaa  cgagttagag  240
aatggtatct  attcagcaga  ctacgaaaac  ccatactacg  acaatagcac  gttcgcatac  300
catttctacg  atccagatc  tcaaaagacc  tacatccctt  tagctaaaca  ggccaaggaa  360
acaggttcaa  agtatttcaa  gcttgaggt  gaagcttatc  agaacaaaga  tatgaagcaa  420
gcattcttct  acttaggatt  atctttacat  tatcttggtg  acgttaatca  acccatgcat  480
gctgccaaat  ttacaaactt  gtccatccg  atgggctttc  atagcaagta  tgagaatttc  540
gttgacacca  taaaggataa  ctacaaagtt  gcagatggaa  acgctattg  gaattggaaa  600
gggacaaatc  ctgaagaatg  gattcatggt  gccgcagccg  ctgctaaaca  ggattaccca  660
ggcatttgta  acgactcaac  caaatctgga  tttgtgaagg  cagcaacttc  acaagaatat  720
gccaataagt  ggagagcaga  agttacacca  gccactggta  agagggtgac  tgagggtcaa  780
cgtgtaacgt  ctggtacat  ccatctatg  ttcgatacat  atgtcaacag  ataa  834

```

```

SEQ ID NO: 373      moltype = DNA length = 837
FEATURE            Location/Qualifiers
source              1..837
                    mol_type = genomic DNA
                    organism = Bacillus mycoides

```

```

SEQUENCE: 373
atgagatttc  cttcaatttt  tactacagtt  ttattcgcag  catcctccgc  attagctcag  60
gagaatgacg  gaggtaatat  agtgaacatc  attcagtatt  ggagtgtgta  ggacaagcac  120
accgaaggtg  tcaactctca  cttatggatc  gtcaatagag  caatagacat  tatgtccagg  180
aatatgacct  tggtaaagca  agatcaagtg  gcactgttaa  acgagtggag  aacagatctt  240

```

-continued

```

gaaaacggca tctattcagc agattatgag aatccgtatt acgataacag cacttttgcc 300
tctcacttct atgatccaga cgaacggttct acctatatcc catttgccaa acaagcaaaag 360
gaaactgggg ctaagtactt taagcttgct ggagaatctt acaagaataa ggataggaag 420
caagccttct tctatcttgg cctatcttta cattatctgg gtgatgtaaa ccaacctatg 480
catgcagcca accttaccac catatcttat cccaaggct tccactccaa atacgagaac 540
ttgttgata ccatcaagga caattacaaa gttacagatg gtaatggcta ttggaactgg 600
aaaggcgcaa atcctgaaga ttggatacat ggcgctgctg tcgcagccaa gcaagatttt 660
cctggatttg tcaatagcaa cacaaagtct tggttcgta aggctgctgt ttccaatca 720
tatgcagata agtggagagc agaagtaacc ccaatgaccg gaaaaagggt gattgaagca 780
caaagagtta cagcaggata catacagtta tggttcgata cctatgtgaa cagataa 837

```

```

SEQ ID NO: 374      moltype = DNA length = 837
FEATURE
source             Location/Qualifiers
                   1..837
                   mol_type = genomic DNA
                   organism = Bacillus thuringiensis

```

```

SEQUENCE: 374
atgagatttc cttcaatttt tactacagtt ttattcgcat catcctccgc attagctcac 60
gagaaaaccg aaggtcatab tctgaacatc attcagattt gggtccgctga ggacaagcac 120
tccgaaggtg tcaactctca cttatggatc gtcaatagag caatagacat tatgagtaga 180
aatactacca gagtaaagca agatcaagtg gtccgtgtaa acgagtggag aacagatgtg 240
gaaaacggca tctatcttgg cgtctcttta cattatctgg gtgatgtaaa ccaacctatg 480
tctcacttct atgatccaga cgaacggttct acctatatcc catttgccaa acaagcaaaag 360
gaaactgggg ctaagtactt taagcttgct ggagaatctt acaagaataa ggataggaag 420
caagccttct tctatcttgg cgtctcttta cattatctgg gtgatgtaaa ccaacctatg 480
catgcagcca accttaccac tctatcttat cccaaggct tccactccaa atacgagaac 540
ttgttgata ccatcaagga caattacaaa gttatggatg gtaatggcta ttggaactgg 600
aaaggcattc atcctgaaga ttggatacat ggcgctgctg tcgcagccaa gcaagatttt 660
tcaggtattg ttaatcgtaa cacaaagtct tggttcgta aggctgctgt tagtcaatca 720
tatgcagata agtggagagc agaagtaacc ccaatgaccg gaaaaagggt gattgaagca 780
caaagagtta cagcaggata catacagtta tggttcgata cctatgggaa cagataa 837

```

```

SEQ ID NO: 375      moltype = DNA length = 846
FEATURE
source             Location/Qualifiers
                   1..846
                   mol_type = genomic DNA
                   organism = Listeria innocua

```

```

SEQUENCE: 375
atgagatttc cttcaatttt tactacagtt ttattcgcat catcctccgc attagcttgt 60
tgtgacgagt atttgcaggc tccagccgct cccacgcaca tcgacagtaa actacatcat 120
aagttatcat ggtctgcaga taacctaca aatacagatg tcaatactca ctattgggtg 180
tcaaaacaag ccgagaagat acttgccaaa gatgttaatc acataagggc caatttgatg 240
aatgagttaa agaatttctg taagcaataa gcccaaggta tctatgatgc agatcacaaa 300
aaccctactc atgatacttc cacatttctg tctcactttt acaaccacga cagagacaat 360
acataccttc caggatttgc aaacgccaag ataactggtg ccaataactt caatcaatcc 420
gtcgagactc atagagaagg gaagtttgat acagcattct acaaaactagg tttagctatt 480
cattactata cagacatttc ccagcctatg catgccaaac actttactgc tatatcctac 540
cctccagggg atcattgtgc atatgaaaac tatgttgaca ccattaaagca caactatcaa 600
gtcacggaag acatggtttg taagagggtc tgctcagatg atgtaaaagt ttggtgtgac 660
gagaatgcaa aaagagcaaa ggcgatttat ccaaagattg ttaatgctaa gaccaagaaa 720
tctacttagc tcggtaacag taaatggaag aaagatacag tagaaccaac tggagcaagg 780
ctgagagatt cacagcagac acttgcaggg ttcttggaat tctggagcaa gaaaaccaat 840
gaataa 846

```

```

SEQ ID NO: 376      moltype = DNA length = 885
FEATURE
source             Location/Qualifiers
                   1..885
                   mol_type = genomic DNA
                   organism = Aspergillus egyptiacus

```

```

SEQUENCE: 376
atgagatttc cttcaatttt tactacagtt ttattcgcat catcctccgc attagcttgt 60
cctgcacctt taactagaag ggatgtttct agtgaagtat tagaacagtt aacgctattc 120
gcagagtact ctgctgcttc ttactgccct tccaatttgg attcaccagg aacgaaattg 180
acttgcctca cagggaattg tccactgta gaagctgcag ataccgagac tttggcagaa 240
ttctatcatg ccgatgaata tggagacggt gccggttact tggcagtcga cactacaaat 300
caactattag tagtcgcttt tcttggaagc cgtgctttgg acacctggat tgccaatttg 360
aactttggta agattccgct cgatgatcta tgttcagggt gtgaagtcca tgggtggctt 420
tggcagtcct ggcaggtcgt agcagattct gttgcaagtg gactcgaaag cgctttgcaa 480
acataccagc actatactat agtctttacc ggtcattcat ttggtggggc agtggccact 540
ttaggcgcag tcgagttaag aaatgctggt tatgatattg agctatatcc ttatggagca 600
cctagggttg gtaatgaagc attagctcaa tacataacag atcaaggatc aaactatcgt 660
gtcacacata caaatgatat tgtacctaga ttacctccta tgagctttgg atttagtcac 720
tccagtctcg agtactggat taacctcgac gacgaagtca caccaaccac agcagatgtt 780
gaggtaatcg aagggtgttg ttctacagag ggtaacgctg gtgaatttcc tcagagtaca 840
gcagccattc ctcactacat cattgacatt tcagcctgtg aataa 885

```

```

SEQ ID NO: 377      moltype = DNA length = 888
FEATURE
source             Location/Qualifiers
                   1..888

```

-continued

```

mol_type = genomic DNA
organism = Aspergillus tamarii

SEQUENCE: 377
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctact 60
cccgacacct tgaggcgtga tgtatccagt agtctactaa acaacttaga cttgtttgct 120
cagtatctcg ctgcagctta ttgctctgag aatttgaatt caaccggtac aaagcttact 180
tgctccgtag gtaattgtcc acttgtagaa ttagcctcca ccaacacact tgacgaattc 240
gatgaatcct cttcatacgg gaacctgca ggttaccttg ctgcagacga gaccaacaag 300
ttacttgtgt tgtcttttag aggtagctcc gaccttgcaa attgggttgc caacttgaat 360
ttcggtttag aggatgccag gcacctatgt tcaggttgtg aggttcatag tggtttctgg 420
aaggcctgga gtgaaatagc agatacaatt acatccaaag tggaatcagc actttcagat 480
cattcagact attctctggt ctttaactggg cattcctacg gtgctgcctt agcgcctctt 540
gctgccactg cactaagaaa ggcagggtcat tcagttcaac tttacaacta tggacaacct 600
cgtttaggaa atgaagctct ggctacatgc atccacagacc agaacaagg tgccaacctat 660
cgtgttacgc acaccaatga cattgttccc aagttacccc ctacattggt aggataccat 720
cactttagtc gacaaatacta catttcacga gccgatgaag ccacagttac taccgcagat 780
gtgacagcag ttacgcgaat cgaatgcaat ggcggtaatg atggcacaga tggtaactta 840
atcgatgcac atagatggta ctttatctac atttcccaat gtagttaa 888

SEQ ID NO: 378      moltype = DNA length = 891
FEATURE            Location/Qualifiers
source              1..891
                    mol_type = genomic DNA
                    organism = Aspergillus niger

SEQUENCE: 378
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctgca 60
ccagctccta tgcaaaagaag ggacataaagt tctacagtac ttgacaacat agaccttttt 120
gctcagattt tggcagcagc tttattgttca tccaacatag agtctacggg taccacattg 180
acgtgtgatg tgggtaattg tccattgggt gaagccgctg gtgcaactac cattgatgag 240
tttgatgata ccagtttctta cgttgaccca acaggattca ttgcagttga tccatacaat 300
gagttgatgg tgttaagctt cagaggttcc tccgacttgt caaattggat cgcagattta 360
gatttcggct tgtaactagt ttcatacaatt tgtgacgggt gtgagatgca caagggcttc 420
tatgaggctt ggggaagtta tgcagacaca ataactagca aagttgaagc agctgtttcc 480
tcataccagc actatacctt agtctttact ggtcacagtt atggcgctgc tcttgacgcc 540
gtggcagcta cagtattgag aaatgctggg tacacactag acttatacaa cttcgggtcaa 600
ccaaggattg gcaacttagc actagcagac tacataacag gccaaaacat gggatccaac 660
tatagagtta cacatacaga tgatatagtc ccgaagcttc caccgaggtt acttggttat 720
caccactttt cccctgagta ctggatcaca tcaggcaatg atgttactgt gacaaacctca 780
gatgttactg aagttgtggg tgtggattca actgctggca acgacggtac cctgttagat 840
tcaacaactg ctcatagatg gtacaccatt tacatttccg agtgtagtta a 891

SEQ ID NO: 379      moltype = DNA length = 1107
FEATURE            Location/Qualifiers
source              1..1107
                    mol_type = genomic DNA
                    organism = Bacillus luciferensis

SEQUENCE: 379
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctatt 60
acctccttct ttggcaacta tcaaaaggca ttgcgatgga gtgatgaaga cgttcataat 120
caagatcatt ctacacatca tttcattgtc aatggctcag taaagttgat agccgacaat 180
gccaacccag ccatacaaca accccaaca ttaactgaacc agttcagaga caggtgggag 240
caaggtcttt atgatgcaga ccacattaac ccattctacg atacagggac gttcatgtca 300
catttctacg atccagatca tcaaacgaac tacaccggtg cttcttatcc gaccgccaga 360
caatctggtg ccaataactt caacctagca tcagactatt acaagaaagg agatttcaac 420
aacgccttct actacttagg tgtttcattg cattacttta cggatgtaac acagccatta 480
catgcttcca acattagcaa cttagatcac catgacacct gttaccactc taagtttgag 540
acttacgctg aaagtattca gaacgaaatg acagttccag acagtggctt atacaattgg 600
atcggatcca cagaccaga ggcttggatc caccaagccg ctgttcaagc caaatccgtt 660
cttcacacag tgtggaatga caccatcac aattgggtct ggcaagctgc ttattccaat 720
tactattccg ctatgtggaa gaatgaagtt aagaatccaa ccctagtcca acttaataa 780
gccgaacgtg aaacgcgagg gttcatagac atgttcttta gagtcaacgg tgttgaaatg 840
cccgttacag tctacaaaga aaatgccttt ggtggcgtat ctgaattact aggtagtggc 900
aattacgatt acgatcagtt ggtgaagggc ataggtaatg atacaattag ttccattcac 960
attgctccag gctaccgaat tacattgttt tcagacgcca attacaaagg cgctagtatc 1020
gtcttgacaa atgatgttca tgacttaggt aacttttctc accaggttaag ctccatcaa 1080
attgccaaga tatctgcact gaaataa 1107

SEQ ID NO: 380      moltype = DNA length = 834
FEATURE            Location/Qualifiers
source              1..834
                    mol_type = genomic DNA
                    organism = Bacillus mycoides

SEQUENCE: 380
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcaa 60
acgaacaact ctgaaaaccc aactcctgtg ctaagatggg ccgctgagga caaacacaac 120
gaggggggtc gtacacacct atggatcgct aatagggtca ttgacataat gtcaagaat 180
acagctattg ttaagccaaa tgaactgca ttacttaacy agtggaggac cgatttggaa 240
aatggcatat actctgcaga ttatgaaat ccttactatg acaatggcac atacgcctc 300
catttctatg atccagatca tgggtgaacc tacattcctt ttgccaacaa agccaagaa 360

```

-continued

```

acaggaacca aatactttta gttggcagga gagggcttacc aaaatcaaga tatgaagcaa 420
gctttctttt acctgggttt gtccctacac tacttaggtg atgtgaatca acctatgcat 480
gcagccaatt tcaccaactt atcataccct atgggttttcc attccaagta cgagaatttc 540
gtagatacag ttaaagacaa ctacatagtt agcgattcaa atggatattg gaactggaaa 600
ggcacaatat ccgaagattg gattcaaggg agtgccgtcg ctgctaagca agactaccct 660
ggtatagtaa atgatactac taaggactgg ttcggttaaag cagcagtcct tcaagaatac 720
gcagacaaat ggagggcaga agttacacct gtaaccggta agaggttaat ggaagcacia 780
agagtaacag ctggttacat acacctgtgg ttgataacct acgtaaacag ataa 834

```

```

SEQ ID NO: 381      moltype = DNA length = 837
FEATURE            Location/Qualifiers
source             1..837
                  mol_type = genomic DNA
                  organism = Bacillus mycoides

```

```

SEQUENCE: 381
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcac 60
gagaatgaag gaggcaacaa agtgagggtta attcagattt ggagtgcctga ggacaagcac 120
gcagaagggtg tcaactctca cttatggatc gtcaatagag caatagacat tatgtccaga 180
aatactaccg ttgtaaagca agatcaagtg gcactgttaa acgagtggag aactgaaatt 240
gaaaacggca tctatgcagc agattatgag aatccgtatt acgataacag cacttttgcc 300
tctcactttc atgatccaga caccggtaaa acctatatcc catttgccaa acaagcaaaag 360
gaaactgggg ctaagtactt taagcttgct ggagaagcct accaaaaaca ggagattaag 420
caagccttct tctatcttgg cctatcttta cattatctgg gtgatgtaaa ccaacctatg 480
catgcagcca accttacaaa tctatcttat ccccaaggct tccactccaa atacgagaac 540
ttgttgata ccatcaagaa caattacaaa gttgcagatg gtaatggcta ttggaactgg 600
aaaggcgtaa atcctgaaga ttggatcacat ggcgtctgtg tcgcagccaa gcaagattat 660
gctggatttg ttaatgggac cacaaaggat tgggtcgtca gggctgtgtt ttcccaagaa 720
tatgcagata agtggagagc agaagtaacc ttgacgaccg gaaaaagatt agttgaagca 780
caaagagtta cagcaggata catcacagtta tgggtcgtga cctatgtgaa cagataaa 837

```

```

SEQ ID NO: 382      moltype = DNA length = 837
FEATURE            Location/Qualifiers
source             1..837
                  mol_type = genomic DNA
                  organism = Bacillus sp.

```

```

SEQUENCE: 382
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcac 60
gagaatgacg gaggtcatgg tgtgggcgta attcccagat ggtccgctga ggacaggcac 120
aaggaagggtg tcaactctca cttatggatc gtcaatagag gcataagacat tatgagtcac 180
aatactaccg ttgtaaagca agatgaagtg gcactgttaa acgagtggag aacagatcct 240
gaaaacggca tctattcagc agattatgag aatccgtatt acgataacag cacttttgcc 300
tctcactttc atgatccaga caacggtact acctatatcc catttgccaa acaagcaaaag 360
gaaactgggg ctaagtactt taagcttgct ggagaatcct accagaataa ggatataag 420
caagccttct tctatcttgg cctatcttta cattatctgg gtgatgtaaa ccaacctatg 480
catgcagcca accttacaaa tctatcttat ccccaaggct tccactccaa atacgagaac 540
ttgttgata ccatcaagaa caattacaaa gttaatgatg gtaatggcta ttggaactgg 600
aaaggcacia atcctgaaga ttggatcacac gccctctgtg tcgcagccaa gcaagatttt 660
cttctatttg ttaatgacaa cacaaaggat tgggtcgtca aggtctgtgt tagtcaagat 720
tatgcaacaa agtggagagc agaagtaacc ccaatgaccg gaaaaaggtt aatggaagca 780
caaagagtta cagcaggata catcacagtta tgggtcgtga cctatgtgaa cagataaa 837

```

```

SEQ ID NO: 383      moltype = DNA length = 849
FEATURE            Location/Qualifiers
source             1..849
                  mol_type = genomic DNA
                  organism = Bacillus drentensis

```

```

SEQUENCE: 383
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctgcc 60
agggtaaatc acgatatagtag ttatgacagc ggcatacatca tatcacctta ctggagcgct 120
gaagagatgc atacagaggg caagaatact catttgtgga ttgtcaatag agcaattgac 180
attatggcac gtgataacac cgttgtttaa gaaaacgaag tcgccttatt gaatgaatgg 240
agaacgggatc tgggaagacgg catatacact gcagactatg aaaatcccta ctatgacaac 300
ggtacttttg ctagtcaactt ttacgatcca gataccgacg atacgtatat cccatttgca 360
aagaatgcac aagttacagg tttaaagtac tttaagttag ctggggaagc ttaccagcag 420
caagcaatga atcaggcctt cttttaccta ggcttgcctc tacactactt tggtagacac 480
aatcaaccca tgcacgcac ccaactttacc aacatttcac atccattcgg ctccactct 540
aagtatgaga actttgttga taccatcaaa gcaccatata gtgtaacaga tggtaatggg 600
tattggaaat ttgctggcga aactccggaa gaatgggtac atactgctgc tgtggcagca 660
aagcaagatg cacctggcat tgttaatgaa acaacaattt catggttctt acaagcagca 720
tttagtcaag agtacgcaga tatgtggaga gctgaagtaa ctccagaaac aggagccagg 780
ttgattgagg cacaaagggc aatggctggc tacatacatc tgtgggttga cacatatgtc 840
aacttataa 849

```

```

SEQ ID NO: 384      moltype = DNA length = 885
FEATURE            Location/Qualifiers
source             1..885
                  mol_type = genomic DNA
                  organism = Aspergillus turcosus

```

```

SEQUENCE: 384

```

-continued

```

atgagatttc cttcaat tttt tactacagtt ttattcgcag catcctccgc attagetgct 60
ccgcagagat tggcagagag agatgtcagt gcaagtgtcc ttcaagaact atctttgttt 120
gctcagttatt ctgcagctgc ttactgtacc aataacatca attccacagg aaccaagttg 180
acgtgctcag ctggaaattg tcccttagtc gaagctgccac acacaaaaac ccttgctgag 240
ttctatgact cagatagttt cggggatacc gctgggttct tggtagcga cactaccaac 300
aaacttcttg ttgtttcttt tagagggtcc cgtaccttag acaattggat cgccaactta 360
gattttgttc tggatagcat ctccgacatt tgttcagggt gcgctgcaca cggtaggttc 420
tgggaagagt ggggaagtag agcaaatctt ttaaccactg agttgaattc agccgtcaat 480
acttatccag gttatagcat tgtctttacg gggtcacagt tagtgccgcg tctgggtacg 540
ttaggcgcta caaccttgag gaaggtcgtt attccagtag agttgtacaa ttacggtagt 600
ccgagagtag gtaacaaagc cttggcaacc tacataacgg ctcaaggacc caattacaga 660
gtgactcaca cgaacgatat cgttccaaga cttccacccc agagtttttg ttttagtcat 720
ctatctcctg agtattggat tacatcaggt gacaatgttc ccgtcactac ttccgacatt 780
acggtaatcc aaggcataga ctcaaatgct ggcaatagtg gtgaggacat tacttctatt 840
gaggctcaca attggtacat tggtaacatt gatgcctgtc cataa 885

```

```

SEQ ID NO: 385      moltype = DNA length = 888
FEATURE            Location/Qualifiers
source              1..888
                    mol_type = genomic DNA
                    organism = Talaromyces subinflatus

```

```

SEQUENCE: 385
atgagatttc cttcaat tttt tactacagtt ttattcgcag catcctccgc attagetgct 60
ttatcaccaa ttggaagaag aactgtcact acaacacaa tagatgacat gaacttattc 120
gcccagttatt ctgctgccgc atactgttca gccaaactga attcaacggg tagcgctctt 180
gcatgtaacg ttggtaaactg tcccttggtt gaaggtgcag acactaccat cttatagcat 240
tttgacgaat ctgcagagatt cggtagcagct accggtcaca tagccgttga tgaacctcac 300
aaatccatca ttttggcctt tcgtgggtct tcagacctag acaattggat tgcagacttg 360
gatatacctc tagtcgcttc tagtatttgt tttagttgtg aagttcatca gggtttcttg 420
gacacctggc agacagttgc atcagatgtc accagtcagg ttgagtagtc ttttaagtgt 480
tatgcaggtt atacctttgt tgtcacaggt cattctatgg gtgcagccct tgctgctatc 540
gcagccacgg tctttagaga tcttggttat actgtggagt tatacaacta tggcccaacc 600
agaatttgta acttgatttt ggcttactac ataacaatc aaaatcacgg tagcaactat 660
agagttacac atacagatga tattgttctt aaactaccgc ctgaattact tggctacgac 720
catttctccc cagaatactg gatcacgtca ggcgataatg tcacagtaac agattccgat 780
atcgatgtta ttgttggtat tgactctgcc gatggtaatg atggcaccat cgatgacagc 840
gtagaagctc atcattggta ctttgtttac atttctgaat gttcctaa 888

```

```

SEQ ID NO: 386      moltype = DNA length = 891
FEATURE            Location/Qualifiers
source              1..891
                    mol_type = genomic DNA
                    organism = Aspergillus tubingensis

```

```

SEQUENCE: 386
atgagatttc cttcaat tttt tactacagtt ttattcgcag catcctccgc attagetgca 60
ccagctcccta tgcaagaag ggcataaagt tctacagtag ttgacaacat agaccttttt 120
gctcagttatt ccgcagcagc ttattgttca tccaacatag agtctacggg taccacattg 180
acgtgtgatg tgggtaattg tccattggtt gaagccgctg gtgcaactac catttgatgag 240
tttgatgata tgagttctta cggtagacca acaggattca ttgcagttga tctacaaaat 300
gagttgattg tttagccctt gagaggttcc tccgacattt caaattggat cgcagattta 360
gatttcggct tgactagtgt ttcagacatt tgtgacggtt gtgagatgca caagggcttc 420
tatgaggctt ggggaagttat tgcagacaca ataactagca aagttgaagc agctgtttcc 480
tcataccagc actattccat tgtctttact gggtcacagt attggcgctgc tcttgacgac 540
atcgagctga cagttatgag aaatgctggg tacacactag acttatacaa cttcggtcaa 600
ccaaggattg gcaacttagc actagcagac tacataacag accaaaacat gggatccaac 660
tatagagtta cacatacaga tgatatagtc ccgaagcttc caccacaggt acttggttat 720
caccactttt cccctgagta ctggatcaca tcaggcaatg atgttactgt gacaacctca 780
gatgttactg aagttgtggg ttgtgattca acagatggca acgacggtac cctgttagat 840
tcaacaactg ctcatagatg gtacaccatt tacatttccg agttagtagta a 891

```

```

SEQ ID NO: 387      moltype = DNA length = 1110
FEATURE            Location/Qualifiers
source              1..1110
                    mol_type = genomic DNA
                    organism = Bacillus acidiceles

```

```

SEQUENCE: 387
atgagatttc cttcaat tttt tactacagtt ttattcgcag catcctccgc attagetatc 60
acatccttgt tttcaaacga tcagaaggct tttgcctggt ccgacgaaga tgtacataat 120
caagatcata gcaacccatca cttcatagtc aacggctctg tcaaatgatg tcgagacaat 180
acaaaccctg ccatcaacaa acctactacg ttgttgaatc agtttagaga tagatgggaa 240
caaggcttat acgacgcaga tcatatcaac cctttctatg acacgggtac tttcatgagc 300
cactttttac atccagacac acaaaccaat tacacaggcg tttcatatcc aactgctcgt 360
caaagcggtg gcaagtattt caaccttgca tccgactact acaagaaagg tgacttctac 420
aatgccttct attacttggg cgtgtcacta cactacttta cagatgtgac ccaaccttta 480
catgcctcca acattttcaa cctggaccat aacgctcccg gataccactc taagtttgag 540
aactatgcag agtctattca gaatcagatg gccattccag attccggttt gtacaactgg 600
attagtagca ccgatcctga ggcttgata caccaagccg ctgtacaagc taaatctgtt 660
ctgccacagg tttggaatga caccatcatc aactttttct ggacggctgc atattccaac 720
tattacagtt ccatgtggaa gaatgaagta aagaatccaa ctttagttca attgaatcaa 780

```

-continued

```

gccgagaggg agactgctgg ttctattgat atgttctttc gtgtcaatgg cgtagagatg 840
cccgtagaag tgtacaaaga gaatgctttt ggtggtgcct cagagatttt gggcttaggc 900
aactatgatt atgaccaatt cgtaaaagga ataggaaatg atactatatt ttcaattcat 960
attgccccag gctaccaagt caccttattc tccgacgcca actacaaagg cactagcacc 1020
gttttgactg gggatgtaaa cgacttgggc aactttaacc atcaagttag tagtcttaag 1080
attgtcaaga tatcgcaat atcaaaaata 1110

```

```

SEQ ID NO: 388      moltype = DNA length = 837
FEATURE            Location/Qualifiers
source              1..837
                    mol_type = genomic DNA
                    organism = Lysinibacillus xylanilyticus

```

```

SEQUENCE: 388
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagctcac 60
gagaattgct accaagatcc tcccatacct ttgaagtggg gtgcccgaag tatccacaac 120
gagggcgctat cctcccatct gtggattgtc aatagagcta tagacataat gtctcagaat 180
actactatcg ttaacaacaa tgaacacagc ctactgaatg agtggagaac agatttggag 240
gaaggcatct attctgcaga ttatcaaaac ccatactatg acaattccac gtttgctagt 300
cacttctatg atccagattc cggcaaaacc tacattccct ttgctaagca agctaataaa 360
accggtgcca aataactcaa actagccggt gaagcttacc agaacaaga cttgaaaaac 420
gcattcttct atttgggcct aagtttgcac tatttggggg atgttaatca acctatgcat 480
gtgccaact ttaccaact atcccatcca ttcggtttcc actccaagta tgagaacttt 540
gtagatactg tcaaggataa ctactgtgt acagatggaa atggttactg gaattggaaa 600
tctgccaact cagaagaatg ggtgcatgcc agtgcgtgtg ccgcaaaagg cgacttcttg 660
ttgattgtta atgataacac cgaagtgggt ttctaaaag ccgctgtttc tcaagattca 720
gcagataagt ggagagctga agtaaacact gttacgggaa aaagattgat ggaagctcaa 780
agaattactg caggatacat ccacttatgg ttcgacacat acgtaaaaaa caataaa 837

```

```

SEQ ID NO: 389      moltype = DNA length = 837
FEATURE            Location/Qualifiers
source              1..837
                    mol_type = genomic DNA
                    organism = Bacillus toyonensis

```

```

SEQUENCE: 389
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagctcac 60
gagaatgaag acgtcaacta taacgcacct attcttagat ggagtgtgta ggacaagcac 120
aagggaagggt tcaactctca cttatggatc gtcaatagaa gtatagatat gatgtccaga 180
aatactacca ttgtaaaaga gaatacagtg gcaactgtta acgagtggag aactgaactt 240
gaaaacggca tctacaatgc agatcatgag aatccgtatt tcgataaact cacttttgcc 300
tctcaacttc atgatccaga aaccggttct acctatatcc cattagtgtc tactcaagca 360
aagggaagcgg gttccaagta tttcaagttg gcaggtgaat cctacaaaaa gaatgatatg 420
aaacaggcct ccttttacct aggtctgtca ttgcattact tgggtgatgt caaccagcca 480
atgcatgctg ccaactttac aaacttgtct taccacaggg gcttccactc taagtatgag 540
aactttgttg acaccataaa ggataactac aaagtaaatg atggtaacgg ttattggaat 600
tggaagggtt caaatccagg tgactggatc cagcgtgtcg ccgttgtgtc caagaagac 660
tatactggta ttgttaacga cacaacaaa gattggttcg tcaagctgtc tatatcatca 720
gaatatgcag acaaatggag agcagaagtc actccggcaa caggtaaaag attgatggaa 780
gcacaaagaa taacagcagg gtacatacaa ctttgggttg acacatacgc caattaa 837

```

```

SEQ ID NO: 390      moltype = DNA length = 837
FEATURE            Location/Qualifiers
source              1..837
                    mol_type = genomic DNA
                    organism = Bacillus wiedmannii

```

```

SEQUENCE: 390
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagctcac 60
gagaatgacg gaggttccaa aatcaagatt gttcacagat ggagtgtgta ggacaagcac 120
aagggaagggt tcaactctca cttatggatc gtcaatagaa caatagacat tatgtccaga 180
aatactacct tggtaaaaga agatagagtg acccagttaa acgagtggag aactgaactt 240
gaaaacggca tctatgcagc agattatgag aatccgtatt acgataacag cacttttgcc 300
tctcaacttc atgatccaga caacggtaaa acctatatcc cattagccaa acaagcaaa 360
gaaactgggg ctaagtactt taagcttgcg ggagaatctt acaagaataa ggatatgaag 420
caagccttct tctatcttgg cctatcttta cattatctgt gtgatgtaaa ccaacctatg 480
catgcagcca actttacaaa tctatcttat ccccaaggct tccactccaa atacgagaac 540
tttgttgata ccatcaagga caattacaaa gttacagatg gtaatggcta ttggaactgg 600
aaaggcacia atcctgaaga atggatcatc ggcgctgtcg tctgcgcaa gcaagactat 660
tcaggtattg ttaatgacaa acaaaaggat tgggttcgta aggtctgtgt ttcccaagaa 720
tatgcagata atggagagc agaagtaacc ccaatgacgg gaaaaagggt aatggatgca 780
caaagagtta cagcaggata catacagtta tgggtcgata cctatgggga cagataa 837

```

```

SEQ ID NO: 391      moltype = DNA length = 864
FEATURE            Location/Qualifiers
source              1..864
                    mol_type = genomic DNA
                    organism = Listeria seeligeri

```

```

SEQUENCE: 391
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagcttgt 60
ggcgacaggt ccattaaagga ccaaatcgct ccacacgcta ttcaaaacaa gttaccatct 120
aagttggggt ggagtgtgta acatccatcc aaagacgaga tcaataccca cttgtgggtg 180

```

-continued

```

ttaaaccaag cagagaagat cttagctaaa gatgtgactg gggctcagtt agatctgggt 240
agagagttga agaactacaa caaagaaatc gctcaaggca ttttcgatgc agaccacaag 300
aatccttact atgacaaaaa cactttttcta agtcattttc acaatcctaa aacacacaaa 360
acttacatac cagggtttcc aaacgcaaa gatacaggta ctaagtactt caacatatcc 420
gtcgaagaat accaagatgg caatttcgaa aaagctttct ataacctagg cctagctatc 480
cactactata cagatgtcag tcaacctatg cagcccaaca actttacagc tctatcacat 540
cctgttggtt accattgtgc atatgaaaac tatgtggaca cctttaagca aatcttccaa 600
gcttcagctg aatctgaagc taagtgggtt tgcacagatg acataagtga atgggtatcac 660
gaaaatgcta aaagggcaca agcagattat cctaagattg tgaacgccat catcaagaaa 720
tcctacatac aaggtctatc agacagccaa aaagatagga cctggaagaa agccgtcaga 780
gctgcaacag ggaagagact tagagatagt caagaaactt tagccggatt cttagagttt 840
tggtagtcta agacaaaaga ataa 864

```

```

SEQ ID NO: 392      moltype = DNA length = 885
FEATURE            Location/Qualifiers
source              1..885
                    mol_type = genomic DNA
                    organism = Penicillium swiecickii

```

```

SEQUENCE: 392
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctgct 60
cctagtagac ctgttccaag ggaatgtttc acctctgtct tatctcagtt aagtctattc 120
gcagagtact ctgctgcttc ttactgtctc aacaacatca attcaactgg aaacgctctg 180
tcttgccaag cagggaattg tccctctgta caatctgcag atactaccac ttgtgggaa 240
tcgatagaa cgtgttctta tggaaacgtt gccggtttct tggcagtcga caaaaccaac 300
aaactattag tagtctcttt tctggaagc cgtttctatta gtaactggat tgccaacatt 360
aaccttgggt taacagatgc accatctcta tgttcagggt gtgaagcaca ttctggcttt 420
ctagagtcct gggaaaccgt agcagatgac ttaactacca acatcaaaag cgctcaatca 480
acataattcg gctatactct agttttgacc ggtcattcat ttggtggggc agttgcccgt 540
ttaggcggaa ccgccttaag aaatgggtgt tctactttga acgtctatca ctatggacag 600
cctaggggtg gtaatggagc attagcttca tacataacaa atcaaggatc actatggcgt 660
gtcacacata cagatgatat tgtacccaag ttacctcctt ccagctttgg atttagtcac 720
cctagtcccg agtactggat tacctccgag aacgaagtca cagttacctc atccgatgtt 780
gaggtaatcg aaggtgttgg ttctaaaagc ggtaacgctg gtaccttaaa ccccgacgta 840
gaggccacac attggtattt gggctacatt gatgggtgtc aataa 885

```

```

SEQ ID NO: 393      moltype = DNA length = 888
FEATURE            Location/Qualifiers
source              1..888
                    mol_type = genomic DNA
                    organism = Talaromyces boninensis

```

```

SEQUENCE: 393
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctgct 60
cctacaccag ttacaagaag aactgtctct acagcattgt tagatacgtc agacttattc 120
gccagattat ctgctgcgcg atactgtcct gccaaactca attcaagctc tacgtctctt 180
cagtgtagcg ctggttaactg tctactgtgt caagctgcag acactaccat cttatactca 240
tttgacaaat ctgcaagttt cggtagacgt accggttatg tgcgcgttga caatacaaac 300
cagttgatag ttattgcctt tctggtgtct tcagacctat ccaattggat tgcaaaactg 360
gatgtacctt tcaccgatgc tggcaacatt tgttcaggat gcgaagtcca ctctggcttc 420
tacgacactt ggcacactgt agcttcagac atcacagcaa ctgtggattc tgcattatca 480
acttaccctg gttacactgt tgtcgtactt ggacattcac tgggaggtgc cttggctgcc 540
atagtgccca cagtcttaag atcttctgtt caagtggtcc aactatatga ttacggacag 600
ccaagaattg gcaacttagc tttagcagac ttcatataca gtgaaccgcg tgggtccaac 660
tatagagtaa ctcatagtga tgatattgtc cctaagctgc ctctgaattt cctaggatac 720
gcacacttct ctccagagta ttggatcaca tccggtgata acgtagcagt tacagacgca 780
gacattgttg aagttatttg tttggatagc actgcaggta atgacggcac gtttggcgac 840
tccatcaatg cacattttgt gttcttttag gctattatgt cttgctaa 888

```

```

SEQ ID NO: 394      moltype = DNA length = 891
FEATURE            Location/Qualifiers
source              1..891
                    mol_type = genomic DNA
                    organism = Hamigera striata

```

```

SEQUENCE: 394
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctgct 60
cccgcacctt ttctacgtag agatgtcagt gcaagtgtcc ttaatgaact agatttgttt 120
gtcagattat ctgcagctgc ttactgttcc tcaaacatag gttccacagg aaacaagttg 180
atgtgcaatg ttggaaattg tcccagagtc gaagcctccg acacagtaac catcgatgag 240
ttcaatgagt ctgcaagtta cggggatgtc gctgggtaca ttgcagtgga caacacaaat 300
caacttctgg ttttgtcttt tagaggggtc agctccttat caaattggat cgccaacatt 360
gatgttgatt tgacagacgc ctccagttta tgttcagggt gcgaagtaca ctctgggttc 420
tggagcgcac ggcaaacagt acaaggaaat attacctcta agttagaatc agccagagca 480
tcataccagc gttatacgtt ggtctttacg ggtcacagtt atggtgccgc tctggctggc 540
tagccgctca caaccttgag ggacgctggt tggactatac agttgtacaa ttacggctca 600
ccgagattag gtaacttagc cttggcacag tacataacgt ctcaaacaca gggttccaat 660
tacagagtga gtcacacaga cgatattgta cctaagcttc ctctgagttt cttaggatat 720
gaccattact ctccagagta ttggataaca tccggtgata acgtcactgt taccacttca 780
gacgtacagg ttatcgaagg gattgactcc gtcgctggaa atgacggaac ttcagatgat 840
tccactgaag ctcatcaatg gtatttcatc tatatcagtg aatgttcata a 891

```

-continued

```

SEQ ID NO: 395      moltype = DNA length = 1113
FEATURE            Location/Qualifiers
source             1..1113
                  mol_type = genomic DNA
                  organism = Bacillus sp.

SEQUENCE: 395
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctatt 60
acatccttat tcggaagcgt ccaaaaagcc ttgcctgggt catgtgatga tccacataat 120
caagaccagt caaccacttt attcatagtc aacaatggtg tcaagttgat ttcaggaaac 180
gcagatcctg ccataaacia accgtccaca ttgttggaac agtttagaga tagatgggaa 240
caaggactgt atgatgcaga ccacataaac cctttctacg atactagtac attcatgtca 300
cacttctatg acccagatag tcaaaaccaat tacgctggtg gatcttatcc aacggctcgt 360
caatcagggtg ctaagtactt caatctagcc tccgactact acaagaatgg agacttttac 420
aatgcttttc attaccttgg tgtctcatta cactacttta ccgacgcaac aatgccactt 480
catgcatcca atatcagcaa cttagaccat caggcaccag ggtatcattc taagttggaa 540
tcatactctg aaagcatttc ggaccaggtt acagtccccc attctggact attcaattgg 600
gtgagtttct cagacccgga actatggatt caccaagcag ctgttcaagc taaaagcgtt 660
cttcacacag tttggaatga ctccatcatt tctctggttc ggcaagctgc atactcaaac 720
tattactcag atatgtggaa atccgcagtt aaagctccca tttgaaatca gttaaaccag 780
gcagagagag aaactgcagg ctccatagac atgtttctca gactaaatgg ggtcgagatg 840
ccagtgcacg tttactcaga aactgctttt gggggagcgt ctgaattgct aggcctcaggc 900
aaactagact atgaccaatt ggtaaaagga ataggaaatg atgctatttc atctattcac 960
attgctcccc gttaccaggt aacgcttttc gccgattcca attactcagg tgctctctaa 1020
gttttgacag cagatgcctc agatttggat aacttcaaca aaaccataag ttccttaaa 1080
attgaaaaga tacaacctgt aaacgttcac taa 1113

SEQ ID NO: 396      moltype = DNA length = 837
FEATURE            Location/Qualifiers
source             1..837
                  mol_type = genomic DNA
                  organism = Bacillus thuringiensis

SEQUENCE: 396
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcac 60
gagaatgcag gaggtcaaaag agtgggcgta attccagat ggagtgcgtg ggacaagcac 120
aaggaaggtg tcaactctca cttatggatc gtcaatagag caatagacat tatgtccaga 180
aatactacca ttgtaaagca agatcaagtg gcaactgtta acgagtgagg aacagatctt 240
gaaaacggga tctattcagc agattttgag aatccgtatt acgataacag cacttttgcc 300
tctcacttct atgatccaga cagtgagaaa accatatccc cattagccaa acaagcaaa 360
gaaactgggg ctaagtactt taagcttgct ggagaaatct accagaataa cgatattgaag 420
caagccttct tctatcttgg cctatcttta cattatctgg gtgatgtaaa ccaacctatg 480
catgcagcca accttaacaa tctatcttat ccccaaggct tccactccaa atacgagaac 540
ttgttgata ccatcaagga caattacaaa gttacagatg gtactggcta ttggaactgg 600
aaaggcacia atcctgaaga ttggatacat ggcgctgctg tctgcgccaa gcaagacat 660
tcagggtatt tcaataacaa cacaaaaggaa tgggtcgtca aggctgctgt ttcccaagaa 720
tatgcagata agtggagagc agaagtaacc ccaatgaccg gaaaaagggt gattgatgca 780
caaagaatta cagcaggata catacagtta tggttcgata cctatgtgaa cagataa 837

SEQ ID NO: 397      moltype = DNA length = 837
FEATURE            Location/Qualifiers
source             1..837
                  mol_type = genomic DNA
                  organism = Bacillus mycoides

SEQUENCE: 397
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcac 60
gagaattgtc accaagatcc tcccatagca ttgaaagtgt ctgcggaatc catccacaac 120
gagggcgtaa gtagtcatct gtggattgtc aacaaaagcta tagacataat gtctcagaat 180
actactgtcg ttaaacaaaa tgaacagacc ctactgaatg agtggagaac agatttgagg 240
aagggcatct attctgcaga ttatcaaaac ccatactatg acaatagtac gtttgcttcc 300
cacttctatg atccagatag tggcaaaacc tacattccct ttgctaagca agctaataca 360
accgtgcca aatacttcaa actagccggt gaagcttacc agaacaaga catgaaaaac 420
gcattcttct atttggccct atccttgcac tatttggggg atgttaatca acctatgcat 480
gtgccaact ttaccaacat aagtcacca ttcggtttcc acagtaagta tgagaacttt 540
gtagatactg tcaaggataa ctatcgtgtt acagatggaa atggttactg gaattggcaa 600
tctaccaatc cagaagattg ggtgcatgcc tccgcttctg ccgcaaaagg cgattttcct 660
tcaattgtta atgataacac caaaaactgg tttctaaaag ccgctgtttc tcaagattca 720
gcagataagt ggagagctga agtaaacact gttacgggaa aaagattgat agaagctcaa 780
agaattactg caggatacat ccacttatgg ttcgacacat acgtaaacaa caaataa 837

SEQ ID NO: 398      moltype = DNA length = 837
FEATURE            Location/Qualifiers
source             1..837
                  mol_type = genomic DNA
                  organism = Fictibacillus macauensis

SEQUENCE: 398
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcat 60
gagaatgaag ggggtaacaa agtttaggta atacagtatt ggtctgcga agatccacat 120
catgaagaca caaacactca cttatggata gtccgtcatg caatggagat tatggccaac 180
aacaagatg ttgtgaaacc agcggaagtc gagcaactaa acaaatggca atcagacct 240
gaacaaggca tttacgatgc agatcatgcc aaccgtatt acgataacgc aacgttcgt 300

```


-continued

```

tcccatttct atgatccaga cactggtaag tcatacatcc ccttggcagc acacgcaaaa 360
acaaccagcg tcaaatactt caaaagagct ggggaagcat accaaaaggg tgatcataaa 420
caagcctttt acaatcttgg tctggctctg cattacatag gtgatctaaa ccagccaatg 480
catgcagcca atttcacaaa cttatcttat cctcaaggat tccattccaa gtatgagaac 540
tatgttgata gcttcaagga ggattatgct gtcaaagatg gagagggtta ctggcattgg 600
aaagggaaca acccggaaga ctggttgcac ggtacagctg ttgctgccaa gaaagattat 660
ccagacatcg taaatgatac aaccaagcc tggttgttaa agcagcagc ctcaaactct 720
tatgcagcta aatggagagc tgcggttgt cccgcaacgg gcaagaggtt aacagaagcc 780
caacgtatct tggctgtgta catgcaacta tggtttgata cttatgtgaa caataaa 837

```

```

SEQ ID NO: 399      moltype = DNA length = 864
FEATURE            Location/Qualifiers
source              1..864
                    mol_type = genomic DNA
                    organism = Listeria seeligeri

```

```

SEQUENCE: 399
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagcttgt 60
ggcgacgaga gtgtaaaagga ccaaatcgct ccacacgata ttcaaaacaa gttaccatct 120
aagttggggtt ggtccgctga acatccatcc aaaaacgaga tcaataccca cttgtgggtg 180
tttaaccaag cagagaagat cttagctaaa gatgtgactg gggctcagtt agatctgggt 240
agagagttga agaactacaa caaagaaatc gctcaaggca ttttcgatgc agaccacaag 300
aatccttact atgacaaaaa cacttttcta tcccatttct acaatcctaa aacacacaaa 360
acttacatag ctgggttttc aacgcaaaag gatacaggtta ctaagtactt caacataagt 420
atagaagaat accaagatgg caatttcgaa aaagctttct ataacctagg cctagctatc 480
cactactata cagacataac ccaacctatg cagcccaaca actttacagc tctatcacat 540
cctgttggtt accattgtgc atatgaaaac tatgtggaca cctttaagca aatcttccaa 600
gcttcagctg aatctgaagc taagtgggtt tgcacagatg acgtctccga atgggtttcac 660
gaaaatgcta aaagggcaca agcagattat cctaagattg tgaacacccat catcaagaaa 720
tctacatcac aaggtctatc agacagccaa aaagatagga cctggaagaa agccgtcaga 780
gctgcaacag ggaagagact tagagattcc caagaaactt tagccggact tttagagttt 840
tggtatacaa agacaaacga ataa 864

```

```

SEQ ID NO: 400      moltype = DNA length = 885
FEATURE            Location/Qualifiers
source              1..885
                    mol_type = genomic DNA
                    organism = Penicillium donkii

```

```

SEQUENCE: 400
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctgct 60
cctgcaagac ctgttccaag ggacatttct tctccttgt tagatgagtt aacgctattc 120
gcagagtacg ctgctgcttc ttactgctcc aacaacattg attcaactgg agacgctgtg 180
acttgctcag gagactattg tcccttggtta caatctgctg gtgcaaaagc tttgtatgaa 240
tccaatgata gcactgaaatg gggagacggt gccggtttct tggcagtcga cactaccaac 300
aaattgattg tactatcttt tcgtggaagc cgttctattt ccacctggat tgccaatttg 360
gactttggtt taacagatac ctcttctcta tgtgatgatt gtgaagcaca ttctggcttt 420
tggaagagtt gggaaaccgt agcagatgat atgactgccc aaatagaaaag cgctcaatca 480
agttatccat cctatactct agttttgacc ggtcattcat ttggtgccgc agttgccgct 540
taggcgcaaa ccgccttaag aaatgctggt tatactttgg acttgatata ctagggacag 600
cctaggggtg gtaatgaagc attagctact tacatgacat ctcaaggatc actatggcgt 660
gtcacacatg aagatgatag tgtacccaag ttacctccta tgagctgggg attttccac 720
gcatccccct agtactgggt taccaagtgc agcgatgtca cagtaccac aagtgtatgt 780
gaggaagtggt ttggtgttga ttctacagcc ggtaacgctg gtacttcagg cgagtccatt 840
tctgcccaca attggtactt tgttgaattt gatgggtgtg attaa 885

```

```

SEQ ID NO: 401      moltype = DNA length = 888
FEATURE            Location/Qualifiers
source              1..888
                    mol_type = genomic DNA
                    organism = Hamigera paravellanea

```

```

SEQUENCE: 401
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctgct 60
cccgacgagc ttaggcgtga tgaagtgcga ggagtcctag ccaacttaga cttgtttgct 120
cagtattctg ctgcagctta ttgcgattcc aatttgaatt cagacggtac aaagcttact 180
tgccaggtcg gtaattgttc acttgtagaa gctgcccaga ccgaaacact tgacgaattc 240
gatatgaccg ctacatacgg gaacgtagca ggttacatcg ctgtagaccg taccaataga 300
ttacttgtgt tggcttttag aggtagcgcc agcatctcaa attggattgc caacttgaac 360
ttagggttaa cggatgcctc tgccctatgt gcaggttgct gtgttcattc cggttttctg 420
gaagcctggc aaacagcaga agctacaatg tcagacatta tcgcttcagc agcccaaaaca 480
tatccaggtt atactctggt tgcaactggg catagttagc gtgctgcctt agccgctatc 540
gctgtacca aattcagaaa cgagggttat gctgttgaa tttacgatta tggacaacct 600
cgatatagaa atctagctct ggctcaatac atcacaaacc aatcttctgg tggcaacttt 660
cgtgttacgc acaccaatga cattgttccc aagttacccc cagattgggtt aggatactct 720
cactttggac cagaatactg gattacatca ggcgatgtta tccctgttac taccgcagat 780
gtggaagtga ttagtggagt ggatgcaact ggcggtaatg atggcgacga aggtacttca 840
gtggatgcac atagatggta ctttgtgtac attagtcaat gtgaataa 888

```

```

SEQ ID NO: 402      moltype = DNA length = 891
FEATURE            Location/Qualifiers
source              1..891

```

-continued

```

mol_type = genomic DNA
organism = Talaromyces leycettanus

SEQUENCE: 402
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagetgct 60
ccagctccag ttcttaggag agatgtatct tcctctgtat tgtctgagct ggacttattc 120
gctcagttact ctgcgcgtgc atattgctct tccaatatcg gttctccggg tacgaagctt 180
acttgtagtg tcggtaattg tccccgtgtt gaagctgccg atacagagac gttgatagag 240
tttaacgaat catcttcttt tggtgatgtg actggctata tcgctgtgga taggaccaat 300
tcactactgg tgttggcatt taggggctct tccacagtgt caaattggga agcagatcta 360
gattttccat taacagatgc ttctagtctg tgcagtggct gtgaaattca ctctggtttt 420
tgggctgcct ggcaaacctg acaggcatcc atcactagca cattggagtc agctatttgc 480
tcttaccctg gttatacttt ggtcttcaca ggtcattcat acggtgccgc tttggcagca 540
atagcagcaa ctacattgag aaatgctggg tacacaatcc agttgtacga ctatggccag 600
ccgagggttag gtaactcttg tctggcccag tacattacgg cacaacacac gggtgccaac 660
tatcgtgtta ctcataccga tgacatcgta cccaaactac cccctgagtt gtttggtgat 720
caccacttct ccccgagta ttggataacg tcaggcgata atgttacagt tacaactagc 780
gatgtgcaag tagtcaactg aattgacagt acagctggaa atgatgggac tttgcttgat 840
tctacaagtg cccatgattg gtacattgtc tacattgatg ggtgcgatta a 891

SEQ ID NO: 403      moltype = DNA length = 1407
FEATURE             Location/Qualifiers
source              1..1407
                    mol_type = genomic DNA
                    organism = Paenibacillus sp.

SEQUENCE: 403
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagetatt 60
acttctatct ttggcaatag ccaagacgct tacgcatggt cagcagacga cccacattct 120
caggactttt ccacacactt gttcattgta aatggaggcg tcaaatgatg ttctggtaat 180
gttgatagcg ccataaaacaa atcttctacc ttggtggaac agtttctgtg aagatgggaa 240
cagggccttt acgatgccga tcatcttaat ccttctcatg atagctcaac ctttatgtca 300
cacttctatg atccagatac caaacgaac tatgcaggat tatcataccc aacagctaga 360
caatctgggt ccaaatactt caaagtcgcc tcaaaactact acaagaatgg tgatttctca 420
aacgccttct actatctggg agtaagtctt cactacttta cagactcaac tatgcctcta 480
cacgcatcag acattagtta cctagatcat agagccccag gctaccacgc caaattggaa 540
gaatatgcca catccatata aaatcagatc aatgttccgg atagcggctt attcaattgg 600
atttccagta cagatcctga actatggatt catcaggctg ctgtacaagc caaatctgtg 660
atgccagaag tcttcaatga cacgattacc gattggtttt ggaagccgcg ttttctcat 720
tactattccg atatgtggaa atccgctgtc aaaatcccta tactaaatca gcttaataca 780
cgagaagag aaacagccgg ttacattgac ttgtcttcta gggttaaacg tgtggatatg 840
ccggttgccg tgtacaaagg gactgcattt ggtggggcac tacaacttct gggtttcgce 900
aactatgatt tgcaccaact gttaaagggt ataggtaatg atacagttag ttcaattagg 960
atagcaccgg gttaccaagt aactttgttc gcagattcca attactcagg agttagcaag 1020
gtgttaacgg cagacgtctc agatttgggt aacttcaata agacaacatc ttcttgaag 1080
attgagaaga tccaaccagt tactgtctat acagatgctt catttagtgg gtagtagcaa 1140
tattttccg taggaatca tgattacaac gaaatcgtaa acagaaagt gaatgatact 1200
atttcttcta taagaatagc cctatgggtat caagtcacct tgtttagaga cagcaactat 1260
tctgggggtt ctactgtagt taccggtgat gtgtatgggt tatcaaaact aaacgaccaa 1320
acaagtagcc taaagtgga ggttattcca acaaatccgg caccatccca aacgaacag 1380
tccatctttt ccaatccctt gaattaa 1407

SEQ ID NO: 404      moltype = DNA length = 837
FEATURE             Location/Qualifiers
source              1..837
                    mol_type = genomic DNA
                    organism = Bacillus toyonensis

SEQUENCE: 404
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagetcac 60
gagaatgacg gaggtcaaaag agtgggcgta attcccagat ggtccgctga ggacaagcac 120
aagggaagtg tcaactctca cttatggatc gtcaatagag caatggacat tatgagtaga 180
aatactacct tggtaaagca agatagagtg gcactgttaa acgagtgagg aactgaactt 240
gaaaacggca tctatgcagc agattatgag aatctgtatt acgataacag cacttttgcc 300
tctcacttct atgatccaga caacggtaaa acctatatcc catatgccaa acaagcaaa 360
gaaactgggg ctaagtactt taagcttgct ggagaactct acaagaataa ggatagaa 420
caagccttct tctatcttgg cctatcttta cattatctgg gtgatgtaa ccaacctatg 480
catgcagcca actttacaaa tctatcttat cttcaaggct tccactccaa atacgagaac 540
tttgttgata ccatcaagga caattacaaa gttacagatg gtaatggcta ttggaactgg 600
aaaggcacia atcctgaaga ttggatacat ggcgctgctg tcgtcgccaa gcaagattat 660
gctgtgattg ttaatgacaa cacaaaggat tggttcgta gggctgctgt tagtcaagaa 720
tatgcagata agtggagagt agaagtaacc ccaatgaccg gaaaagggtt aatggatgca 780
caaagagtta cagcaggata catacagtta tggttcgata cctatgggaa cagataa 837

SEQ ID NO: 405      moltype = DNA length = 837
FEATURE             Location/Qualifiers
source              1..837
                    mol_type = genomic DNA
                    organism = Bacillus thuringiensis

SEQUENCE: 405
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagetgat 60
gagcacacca caaacaacaa gattgacata ttccaacctt ggtcaaatga agaaaatcat 120

```

-continued

```

aaagaaggca agaattcaca tctttggatc gtaaacggag caatcgacat aatgagtaga 180
aacaccacaa ttgtaaaaca agagaatcta gctctgcttc agcaatggag gacctatctt 240
gaaaatggct tgtatgtcgc cgactatgag aatccgtact atgatatcgg gacctttgca 300
tcacacttct acaatccaga tacagacagt acgtacttac catttgctaa gcacgcaaaa 360
gaaactgggt ccacttactt caccctagca ggtgaagctt accaacataa gaatatocaa 420
caagccttct tctacttagg tgtttcattg cattacttgg gggacatcaa tcagccaatg 480
catgctgcca accttacgaa cttatcttac ccattcggat ttcactccaa atacgaacac 540
tttgtcgata ccatacaaca gaactatgag attatggatg gcgagggata ttggaattgg 600
aaggggaagg atcctgaaga ctggatacac caggcagccg ttgctgcaaa ccaagacttt 660
agtgatatcg tgaactcaga cccaagaat tgggtcgtca aggcagctgt ttcacaaact 720
tacgcagata gatggagagc tcgtgtaaca cccattactg gcaagagatt gatcgaagct 780
cagagaatca ctgcgggata cattcaactt tggtttgata catcacata tcaataa 837

```

```

SEQ ID NO: 406      moltype = DNA length = 837
FEATURE            Location/Qualifiers
source              1..837
                    mol_type = genomic DNA
                    organism = Bacillus thuringiensis

```

```

SEQUENCE: 406
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcat 60
gagaatgatg ggggttccaa gattaagatc atacatagat ggtctgccga agataaacat 120
aaagaagggg ttaactctca cttatggata gtcaacagag caatcgacat tatgtcaaga 180
aatacaactt tagtgaacaa agacagagtc gcccaactaa atgaatggag aactgagcta 240
gaaaatggca tttacgctgc agattatgag aacccgattt acgataaact cactgtcgcg 300
tcccatttct atgatccaga caatggtaag acatacatcc cctttgcaaa gcaggcaaaa 360
gaaactgggt ccaaatactt caagttagct ggggaatcct acaaaaaaaa agatatgaaa 420
caagcctttt tctatcttgg tctgtctctg cattacctag gtgatgtcaa ccagccaatg 480
catgcagcca atttcacaaa cttatcttat cctcaaggat tccattccaa gtatgagaac 540
tttgttgata cgatcaagga caactacaaa gtcacagatg gaaacggtta ctggaattgg 600
aaagggacca acccggaaga ctggattcac ggtgctgctg ttgttgccaa acaagactat 660
tcaggaatcg taaatgacaa taccaaagac tggtttgtaa aagcagcagt ctctcaagaa 720
tatgcagaca taatggagagc cgagggttaca cccatgacgg gcaagagggt aatggatgcc 780
caacgtgtca ctgctggtta catacaacta tggtttgata cttatgggga tagataa 837

```

```

SEQ ID NO: 407      moltype = DNA length = 867
FEATURE            Location/Qualifiers
source              1..867
                    mol_type = genomic DNA
                    organism = Talaromyces rugulosus

```

```

SEQUENCE: 407
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctgac 60
gtcaccacag cagtcttggc cgacttaacc ttgttttcac agtattcagc tgctgcatat 120
tgttctacca acttgaattc tacagggtgt gcagtgacat gcagtggttg taattgtccc 180
ctagtcgaag cagcagatag acaataactg tatgactttg acgaatcctg caagtttggc 240
gatgcatcgt gcttcattgc agtggaacat accaataact tgatttgtgt gtcttttagg 300
ggatctcacg atctttcaaa ttggatcgca aacttagatt tcttcttagt cgaactgcc 360
tcaatctcgc aaggttgcta catgcatggt ggtttttggg aaacttggca aacagtcgct 420
gcaaacgtta cagaacaatt agaagctgct atcactgcaa accctggata tacttagtt 480
atgactggtc acagtctagg agctgccttg gctgcaatag ttgctaccga gtttaggaat 540
gaaggcattg atgtagaat gtacaattac ggccaaccaa ggctaggcaa cttagcttta 600
gctcagtaga tgacaaaacc gacacagact tcaactaca gggttactca ttccgatgat 660
attgttccta agcttccacc tagagtacta gatttcgacc actattctcc tgagtattgg 720
ataacttcag ccaataacgt gtcagtatca gacgcagatg tcgtacaggt cgttggtata 780
gatagcacgg atggcaatga tggcaccatc ttagatgaca tagaggctca tagatggtag 840
ttaggtgata tctctgaatg ttcataa 867

```

```

SEQ ID NO: 408      moltype = DNA length = 885
FEATURE            Location/Qualifiers
source              1..885
                    mol_type = genomic DNA
                    organism = Penicillium sp.

```

```

SEQUENCE: 408
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctgct 60
cctgcaagac ctgttccaag gtagtgttct agtgctacat tatctgagtt aacgctattc 120
gcagagtacg ctgctgctgc ttactgctcc aacaacattg attcaactgg agacgctctg 180
tcttgctcag gagggatttg tcccgaagta caatctgctg gtgcaaccac ttgtatgaa 240
ttcgaagata gcacagattt tggagacggt acgggtttct ttgcagtcga caataccaac 300
aaactattag tactatcttt tcgtggaagc agctctatta gtaactggat tgccaatttg 360
gactttggtt taacagatgc atcttctcta tgttcagggt gtgaagcaca ttctggttt 420
tacaagtctt ggggtgtcgt agcagatact ttaactgccc aagtcgctag cgctgtttca 480
acatatccaa gttatactct agttgttacc ggtcattcat tgggtggggc attagccgct 540
ttaggcggaa ccgccttaag aaatgctggt tatactttgg acatctatac ctatggacag 600
cctaggggtg gtaataccgc attagcagat tacatgacaa atcaaggatc actatggcgt 660
gtcacacatt caaatgatat ttaccacca ttacctcctg caagctgggg atttaccac 720
gcaagtctcg agtactggat tactctcggg aacgatgtca cagttaccac atccgatgtt 780
accgaagtga ctggtgttgg ttcttcagac ggtaacgctg gttcatcagg cgacagtgt 840
agtgccacaa attggtacat tgttgatatt gatgggtgtt cttaa 885

```

```

SEQ ID NO: 409      moltype = DNA length = 888

```

-continued

FEATURE
source

Location/Qualifiers
1..888
mol_type = genomic DNA
organism = *Hamigera avellanea*

SEQUENCE: 409

atgagatttc	cttcaatttt	tactacagtt	ttattcgcag	catcctccgc	attagctgct	60
cccgacctg	ttaggcgtga	tgtaagtgc	ggagtcctag	ccaacttaga	cttgtttgct	120
cagtatctg	ctgcagctta	ttgcgattcc	aatttgaatt	cagacggtag	aaagcttact	180
tggtccgctg	gtaattgtcc	actttagtaa	gctgccgaca	ccgaacact	tgacgaattc	240
gatatgaccg	ctacatacgg	gaacgtagca	ggttacatcg	ctgtagaccg	taccaataga	300
ttactttgtg	tggctttttag	aggtagcgcc	agcatctcaa	attggattgc	caacttgaac	360
ttaggtttaa	cggatgcctc	tgccctatgt	gcaggttggt	aggttcattc	cggttttctg	420
gaagcctggc	aaacagcaga	agctacaatt	tcagacatta	tcgcttcagc	agcccaaa	480
tatccaggtt	atactctggt	tgtaactggg	catagttacg	gtgctgcctt	agccgctatc	540
gctgccacta	gattcagaaa	cgaagggttat	gctgttgaa	tttacgatta	tggaacacct	600
cgtataggaa	atctagctct	ggctcaatac	atcacaaacc	aatctggagg	tggcaacttt	660
cgtgttacgc	acaccaatga	cattgttccc	aagttacccc	cagattgggt	aggataactc	720
cactttggac	cagaatactg	gattacatca	ggcgatgggt	tgctctgtac	taccgcagat	780
gtggaagtga	ttagtggagt	ggatgcaact	ggcggtaatg	atggcgacga	aggtaactta	840
gtggatgcac	atagatggta	ctttgtgtac	attagtcaat	gtgaataa		888

SEQ ID NO: 410

FEATURE
source

moltype = DNA length = 894
Location/Qualifiers
1..894
mol_type = genomic DNA
organism = *Penicillium spikei*

SEQUENCE: 410

atgagatttc	cttcaatttt	tactacagtt	ttattcgcag	catcctccgc	attagctgct	60
cctactagat	tgttatttcc	tagggacata	tcaagtgcag	tggttagcga	gttgactttg	120
tttgccgaat	actctgcagc	tgcatattgt	tcagccaata	tcgatagtgc	atctgcgggt	180
ttgccttga	catgtgaatc	tggttaactgc	ccagaagtac	aaagtgcaga	tacgtctacc	240
ttatccgaat	ttgacgaaac	cacagattac	ggcgatgtag	ctggtttctt	tgctgtcgac	300
aaaacgaatg	agctgttggt	tcttagtttc	agaggtttcta	gaacgatttc	aaattgggtc	360
gccaatttgg	attttgactt	aacagatgca	tcttctcttt	gtagtgaact	cgaagctcac	420
tcagggtttt	ggaaatcctg	ggaaacagtt	gccgatgagt	taacaacaca	aatagaatct	480
gcacagaata	gttatccaga	ttatcaactg	gtattgaccg	gtcatagtct	agggtccgca	540
ttggcagctt	tggtctgtac	tgccctaaga	aacgctggat	acacccttga	ttgtatatac	600
tttgggtcaac	caaggggtcgg	caacttagca	ttagcagatt	acatgacaga	tcaaggatca	660
ttatggagag	ttactcatat	agacgacata	gtaccaaggg	tcctcccgga	aagcttcggg	720
tatgcacatg	ccagccctga	gtattggatc	acatccggta	atgacgtaac	cgttactaca	780
tcagatgttg	aggaaatcgt	tggtgtcaat	tccagtgcgg	gtaatgcggg	agaagcagat	840
ttgtcaatag	atgctcacia	ttggtacatt	gtctacattg	atgaatgtga	ataa	894

SEQ ID NO: 411

FEATURE
source

moltype = DNA length = 1500
Location/Qualifiers
1..1500
mol_type = genomic DNA
organism = *Paenibacillus alginolyticus*

SEQUENCE: 411

atgagatttc	cttcaatttt	tactacagtt	ttattcgcag	catcctccgc	attagctatt	60
acctccgtct	ttggcaacta	tcaaaacgca	tacgcagatg	gtgcagatga	cgttcataat	120
caagactatt	ctacacactt	attcattgtc	aatggcggag	taaagttagt	aagtggcaat	180
gccgactcag	ccatcaacaa	aagttcaaca	ttactggagc	agttcagaga	cagggtggag	240
caaggtcttt	atgatgcaga	ccacttaaac	ccattctacg	attcaagcac	gttcattgtc	300
catttctacg	atccagatag	tcaaacgaac	tatgcccgtt	tatcttatcc	gaccgcca	360
caatctgtga	gcaaatactt	caaggtcgca	tcagactatt	acaagaatgg	agattttctc	420
aacgccttct	actacttagg	tatttcattg	cattacttta	cggattcaac	aatgccatta	480
catgcttcag	acattagcaa	cttagatcac	agagcacctg	gttaccacgc	taagttggag	540
gaatacacta	caagtattca	gaaccagatt	acagttccag	acagtggctt	gttcaattgg	600
atctcctcca	cagaccgaga	gttggtggat	caccaagccg	ctgttcaagc	caaatccgtt	660
atgccacagg	tggtccaatga	ctctattaca	gattgggtct	ggaaggctgc	tgtatcctat	720
tactattccg	atatgtggaa	gaatacagtt	aagactcaaa	tcctaaatca	acttaataca	780
gccgaacgtg	aaaccgcagg	gtacatagac	ttgttcttta	gacttaacgg	tggtgatatg	840
ccggttacag	tctacaatgg	tactgccttt	ggtggcgcat	ctcaattact	aggtttccgc	900
aattacgatt	acaatcagtt	gggtgaaggc	ataggtaatg	atacaattag	ttccattagg	960
attgctccag	gctaccaagt	tacacttttt	gcagactcca	attactctgg	cgttagtagc	1020
gtcgtcacag	gaaatgttta	cgggttatca	aacttgaatg	accagacaag	cagtttgaaa	1080
gttggtgtta	taccaaccaa	ccagctcca	tctccaacaa	tccaggcaga	atctttctct	1140
gggtcccagg	ggatcttgac	tcacagtgca	gggagtgagg	ccgttgtcgg	caacattaac	1200
agtggttctt	ggctttctta	tgataatgta	gacttttgta	ccggttaagc	caagtttgta	1260
gcttctgttg	gaatggatcc	cgttttcgct	gctataggta	agcagttgga	gttaagggtta	1320
gacagtccca	cagggacctt	gatttggtac	tttacaatat	cttcactctg	tggttgggat	1380
gcttatacga	ctcagaattg	ctacgtaaca	tctgcatctg	gtactcataa	gctttacatc	1440
atcacaaaag	gtaacggctt	tggatttggg	aacatagatt	ggtttacttt	ttctagttaa	1500

SEQ ID NO: 412

FEATURE
source

moltype = DNA length = 837
Location/Qualifiers
1..837
mol_type = genomic DNA

-continued

```

organism = Bacillus mycoides

SEQUENCE: 412
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcat 60
gagaacgacg gtggcagcag gattaacata gtccataggt ggtctgcaga agataaacat 120
aaagaagggtg tcaattcaca tctttggatc gtaaacacag caatggacat aatgagtaga 180
aacaccacat tggtaaaaca agaccaagtc gctctgccta acgaatggag gaccgaactt 240
gaaaatggca tctatgcagc cgactatgag aatccgtact atgataacag cacctttgca 300
tcacacttct atgatccaga taatgggaag acgtacattc catttgctaa gcaggccaaa 360
gaaactgggtg ccaataactt caagctagca ggtgaatctt acaagaataa ggataggaag 420
caagccttct tctacttagg ttgtcattg cattacttgg gggatgtaaa tcagccaatg 480
catgctgcca actttacgaa cttatcttac ccacagggat ttcactccaa atacgaaaac 540
tttgcgata ccatcaaga caactacaaa gttacagatg gcaacggata ttggaattgg 600
aagggacaaa atcctgaaga ctggatacat ggcgagccg ttgctgcaaa gcaagactat 660
agtggtatcg tgaacgataa cactaaagat tggctcgta aggcagctgt ttcacaagaa 720
tacgcagaca aatggagagc tgaagtaaca cccatgactg gcacaagatt gatggatgct 780
cagagagtga ctgccggata cattcaactt tggtttgata catacggaaa tagataa 837

SEQ ID NO: 413      moltype = DNA length = 837
FEATURE            Location/Qualifiers
source              1..837
                    mol_type = genomic DNA
                    organism = Bacillus bingmayongensis

SEQUENCE: 413
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcac 60
gagaatgaag gaggcaacaa aatcagggtg attcagtagt ggtccgctga ggacaagcac 120
gcagaagggtg tcaactctca cttatggatc gtcaatagag caatagacat tatgagtaga 180
aatactacca ttgtaaagca agatgaagtg gcaactgtta acgagtgagg aactgaactt 240
gaaaacggca tctatgcagc agattatgag aatccgtatt acgataacag cacttttgcc 300
tctcacttct atgatccaga ctccggtaaa accatatccc catttgccaa acaagcaaaag 360
gaaactgggg ctaagtactt taagcttgct ggagaatctt accaaaaaca ggaatggaag 420
caagccttct tctatcttgg cctatcttta cattatctgg gtgatgtaaa ccaacctatg 480
catgtagcca actttacaaa tctatcttat ccccaaggct tccactccaa atacgagaac 540
tttgttgata ccatcaagga caattacaaa gctattgatg gtaatggcta ttggaactgg 600
aaaggcacia atcctgaaga ttggatacat ggcgctgctg tcgcagccaa gcaagaatat 660
gctggtattg ttaatgacac ccaaaaggat tggctcgctt gggctgctgt tagtcaagaa 720
tatgcagata agtggagagc agaagtaacc ccagctaccg gaaaaagatt agttgaagca 780
caaagagtta cagcaggata catacagtta tggttcgata cctatgggaa cagataa 837

SEQ ID NO: 414      moltype = DNA length = 840
FEATURE            Location/Qualifiers
source              1..840
                    mol_type = genomic DNA
                    organism = Bacillus mycoides

SEQUENCE: 414
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcat 60
gagaacagtc atcagaacgg aatttccttt ggacataagt ggtctgcaga agctattcat 120
gatgaagggtg ttagtactca tctttggatc gtaaacagat caatcgaggt aatggcccaa 180
aacaagacag ttgtacaacc aaacgagata tcaactgcta acgaatggag agccgatctt 240
gaaaagggca tctatagtgc cgactatgac aatccgtact ttgataacgg gacctttgca 300
tcacacttct atgatccaga tactgggtgc acgtacttac cattggctaa gcacgccaaa 360
gaaactgggtg ccaataactt caagctagca ggtgaagctt accagaataa cgatctgaag 420
aacgccttct tctacttagg ttgtcattg cattacttgg gggatgtaaa tcagccaatg 480
catgctgcca actttacgaa tgtatctctg ccagtcgcat tacactccaa atacgaaaac 540
tttgcgata cagtcaaaaga caactacaaa gttaaagatg gcaacggata ttggaattgg 600
aagagtgtaa atcctgaaga ctgggtacac gcctcagccg ttggtgcaaa ggcagatttt 660
cccttgatcg tgaacgataa gaccaagaaa tgggtccttg acgcagctat ttcacaagat 720
gcagcagaca aatggagagc tgaagtaaca cccgttactg gcaagagatt gatggaagct 780
cagagaatca ctgccggata cattcacctt tggtttgata catacgtcaa ctacaaataa 840

SEQ ID NO: 415      moltype = DNA length = 867
FEATURE            Location/Qualifiers
source              1..867
                    mol_type = genomic DNA
                    organism = Brevibacillus sp.

SEQUENCE: 415
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctaa 60
aaggagtata aggtcaaaat ccatggtaag actattacaa gtccctacaa gattgatcca 120
agatggagtg aagaatcccc acatgaagag ggccacgcaa cccacttggt gattgttaac 180
agggcaattg acatcttatc aagaacaagc aataaggatg tgaattccaa ggaaactgag 240
atgttgaatg cttggagatc cagttgggaa caaggtcttt atgatgcga tcatacaaac 300
ccctattaca attcoggtac attcgcctct cacttctatg atccagatc caaatcaaac 360
tggctggaca cttctggcac cgcattgaca gaaggtagca ggtactttgc actagcaggc 420
aaatactatc aaaatggtga taagggaaaa gccttttact acctaggatt gagtttacac 480
tatttgacag acgtcacaca accaatgcat gctgccaaact ttacttgggt gaattggcca 540
actagtttct atggcaagtt tgaggattac accgatgata tccaaggcaa ctatgctgta 600
acagatgggt aaggttactg gattttccaa gatagtaacc ccgaacattg gatacatcaa 660
gctgctgtcg acgcacaaagc agaatttccc aatatccaca catcagacat aactaaatgg 720
ttttggctcg cagcagtcag cgactactat agtgacaaat ggcatagggc cgttcaacct 780
acaatcgagc ataggttaac tgaggcacia agaattaccg caggctactt acatttgggt 840

```

-continued

ttaaagactt acgttgataa tcaataa 867

SEQ ID NO: 416 moltype = DNA length = 885
 FEATURE Location/Qualifiers
 source 1..885
 mol_type = genomic DNA
 organism = *Penicillium vasconiae*

SEQUENCE: 416

atgagatttc	cttcaatttt	tactacagtt	ttattcgcag	catcctccgc	attagctgct	60
cccgcaaggc	cagtcgccag	agatgtcagt	agtgaacccc	ttctgaatt	gactttgttt	120
tcacaatatg	ctgcagctgc	ttactgtacc	aataacgtca	attccgcagg	agacgcggtt	180
agctgctcag	gtggatattg	tcccgaagtc	caatctgctg	gcgcaacaac	cctttatgag	240
tttgatgact	ctacagattt	cggggatgtc	gctgggttct	ttgcagtgga	cgcaaccaac	300
aaacttctgg	ttttgtcttt	tagaggggtc	cgtaccattt	caaattggat	cgccaaacta	360
gatttttggtc	agacagacgc	ctccagttta	tgttcaggtt	gcgaagcaca	ctctgggttc	420
ttcaagggcat	gggaagcagt	agcagatact	ttaaccgctc	agattgcac	agccgttgca	480
acttatccat	catatacgtt	ggtattgacg	ggtcacagtt	ttggtggcgc	tgtggctgcc	540
ttaggcggtta	cagcattgag	gaacgctggt	tatactttag	acttgatatac	atacgggtaa	600
ccgagagtag	gtaacacagc	cttggcagac	tacatgacga	atcaaggatc	cttatggaga	660
gtgactcaca	gcgcagatat	cgttcccaag	cttccaccca	ccagttgggg	ttttacccat	720
gcatctcctg	agtattggat	tacatcaggt	gacgatgtta	ccgtcaactac	ttccgatgtt	780
acggaagtga	ctggcgctcg	gtcaagtggg	ggcaatgcag	gtaccagtgg	tgattctggt	840
tcgcctcaca	attgggtacat	gtttgacatt	gatggctgtc	aataa		885

SEQ ID NO: 417 moltype = DNA length = 891
 FEATURE Location/Qualifiers
 source 1..891
 mol_type = genomic DNA
 organism = *Talaromyces diversus*

SEQUENCE: 417

atgagatttc	cttcaatttt	tactacagtt	ttattcgcag	catcctccgc	attagctgct	60
atcgaccctt	tgaaccgtag	aacaataagt	gaaagtctac	ttgatgaact	agatttggtt	120
gctcagattt	ctgcagctgc	ttactgttcc	gcaaaccttag	attccacagg	atccgccttg	180
gctcgcatg	tggaatttgc	tcccttagtc	gaagctgcct	ccacaacaat	cctttatgac	240
tttgatgaga	caaatgattt	cggggatgca	actgggtgac	ttgcagtgga	caccacaaat	300
gaatacatca	ttttgtcttt	tagagggacc	gacgacttag	agaattggat	cgccaaacta	360
gattttccat	tgattgacgc	ctccgacatt	tgttcaggtt	gcgaataaca	cgaagggtgg	420
tgggacagtt	gggaaacagt	agcttcagac	attaccgctc	agattgaatc	agccgtttca	480
acttatccag	actatacgtt	ggtagctacg	ggtcacagtt	taggtgcccgc	tctggctgcc	540
attgcccgtca	cagctcttag	gctagatggt	tatactgtac	agttgtacaa	ttacgggtgaa	600
ccgagaatag	gtaacttagc	cttggcagac	tacataacga	ctgaaacaat	gggttccaat	660
tacagagtag	gcgcacacaga	cgatattgta	cctaagcttc	ctcctgagtt	gttaggatat	720
gaccatttct	ctccagagta	ttggataaca	tccggtgatg	acgtcactgt	tcttgataca	780
gacgtaaccg	aagtcgctcg	ggtgactacc	accgctggaa	atgacggaac	tttgttggat	840
tccattgatg	ctcatagatg	gtatttcgtc	tatatcagtg	aatgttcata	a	891

SEQ ID NO: 418 moltype = DNA length = 900
 FEATURE Location/Qualifiers
 source 1..900
 mol_type = genomic DNA
 organism = *Aspergillus wentii*

SEQUENCE: 418

atgagatttc	cttcaatttt	tactacagtt	ttattcgcag	catcctccgc	attagctgct	60
cccagtcctg	ttaggcggtga	tgtagacagt	agtgtcctaa	acaacttaga	cttggttgct	120
cagtattctg	ctgcatacata	ttgcttggag	aatttgaatt	catccaatac	aaagcttgaa	180
tgctccgtag	gtaattgtcc	acttgtagaa	gctgctccca	ccgtcacact	tgacgaattc	240
gatgaatcct	cttcattcgg	ggacgtaacc	ggtttcatcg	ctgcagacga	gaccaacaag	300
ttacttgtgt	tgtcttttag	aggtagctcc	gacatcgcaa	attggattgc	cgaccttgat	360
ttcggtttaa	cggatgggag	cgacctatgt	tcagggttga	aggttcatag	tggtttctgg	420
gaagcctggg	gaacagtatc	agacaacatt	acatccatta	tcgaatcagc	aaccgccaag	480
tatcccaact	atgaactggc	ttttactggg	cattcctacg	gtgctgcctt	agccgctgtg	540
gctgccgttg	tattcagaaa	ctccggttat	acagttcaac	tttacaacta	tggacaacct	600
cgtataggaa	atctagctct	ggcagattac	atcacaaaac	ttacagataa	gggtgacaat	660
tacagagtag	gcgcacacaga	cgatattgta	cctaagcttc	ctcccaagtt	gttaggatat	720
caccatgcat	ctccagagta	ttggataaca	tccggcaata	acgtcactgt	taccactgca	780
gacgtagacg	ttgtcacagg	ggtagactcc	accgatggaa	atgacggaac	tactgcagat	840
tccagaacag	ctcatagatg	gtatttcggc	tatatcagtg	aatgttcaac	cttgactata	900

SEQ ID NO: 419 moltype = DNA length = 1500
 FEATURE Location/Qualifiers
 source 1..1500
 mol_type = genomic DNA
 organism = *Bacillus acidiceles*

SEQUENCE: 419

atgagatttc	cttcaatttt	tactacagtt	ttattcgcag	catcctccgc	attagctata	60
accagcttgt	tggcaatta	ccaaagagct	tttgcttggt	cagatgaaga	tgttcacaac	120
caagatcatt	caacccatca	cttcattgta	aatggaggag	tcaaattgat	tgacagacaat	180
acagaccag	ccatcaataa	gccacaacg	cttctgaacc	agtttagaga	cagatgggag	240
caaggtttgt	acgatgcaga	tcacatcaat	cccttttacg	atactggtac	cttcagtgtc	300

-continued

```

catttctatg atccccgatac acaaacaaac tattccggac tatcctatcc aacagctaga 360
cagtcagggtg gcaagtaactt caacttagct agcgattact acaaaaaggg tgactttaac 420
aacgcctttt actatctagg ggtttctctt cactacttta cagacgttac tcagccgttg 480
catgcttcca atatcagcaa tctagatcat aatgctccag gctatcattc caaatacgaa 540
acctacgctg aatctataca atcacaaatc attgtoccag attccggctt gtacaactgg 600
acagatagta cggatcccgga agcctggata cataaagcag ccattcagggc aaaaagcgtt 660
ctgcccttgg tttggaatga taccatcatc aactggtttt ggcaggctgc ctattccaac 720
tattactccg caatgtggaa gaatgaagtc aagaatccaa ctctagctca acttaacca 780
gccgaacgtg aaacagctgg ttccattgat atgtttttca ggttgaatgg ggtggagatg 840
ccggtaacgg tctacaatga aaatgccttt ggtggagcaa gtgaactttt gggttctggc 900
aactatgatt atgatcaatt gattaaagtg attggtaatg atacgataag ctctattcat 960
atcgacactg gttatcaagt cactctattc gcagacgcaa actacaaagg tgcttcaatt 1020
gttttgacag gagacgtcaa cgatttgggc aacttcaatc atcagggtgag ttcattgaag 1080
attgagaaga tctctaccaa tccagcctct tctctacaa tccaagcaga gtctttcata 1140
agcagtaaa gcatcttgac tcataatgtt gggtcaggtc ccgttgggg taacataaac 1200
tcaggctcat ggattggata tgataacgtc gatttcggaa ctggaaaaac caaattcata 1260
gctagggttg gtatggacc cttttacgcc atattcgata aacagttgga attgaggttg 1320
gattctccta caggtaccat catttgaacc tttaccatca acaatactgg aggttggggc 1380
acatatgcta ctcagacgag cactatttct ggcgcaactg gtaccacaa attgtacata 1440
gtttcaaaag cgtctggtga tggatttggc aacattgatt ggataacatt ctctccttaa 1500

```

```

SEQ ID NO: 420      moltype = DNA length = 837
FEATURE             Location/Qualifiers
source              1..837
                    mol_type = genomic DNA
                    organism = Bacillus luti

```

```

SEQUENCE: 420
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagctcac 60
gagaaaaacg aaggtaatgt acccatcatt actcagatt ggagtgtga ggacacccac 120
agtgaaggtg tcaactctca cttatggatc gtcaatagaa caatagacat tatgtccaga 180
aatactacct tggtaaggca agatgaagtg gcactgttaa acgcctggag aacagatctt 240
gaaaagggtc tctatgcagc agattatgag aatccgtatt acgataacag cacttttacc 300
tctcacttct atgatccaga caccgttaaa acatatgtcg gtttagccaa acaagcaaa 360
gaaactggga ataagtaact taagcttgcg ggagaatctt acaagaataa ggatataag 420
caagccttct tctatcttgg cctatcttta cattatctgg gtgatgtaaa ccaacctatg 480
catgcagcca actttacaaa tctatcttat ccccaaggct tccactccaa atacgagaac 540
tttgttgata ccatcaagga caattacaaa gttacagatg gtaatggcta ttggaactgg 600
aaaggcctga atcctgaaga atggatatac ggcgctgctg tcgcagccaa gcaagactat 660
tcagggtatt tcaatagcaa cacaaagtct tgggtcgtca aggtcgtgtg ttcccaatca 720
tatgcagata agtggagagc agaagtaacc ccaacgaccg gaaaaaggtt aatggaagca 780
caaagagtta cagcaggata catcacagta tgggtcgata cctatgggaa cagataa 837

```

```

SEQ ID NO: 421      moltype = DNA length = 837
FEATURE             Location/Qualifiers
source              1..837
                    mol_type = genomic DNA
                    organism = Bacillus pseudomycoides

```

```

SEQUENCE: 421
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagctcat 60
accaacgact gtggcaacga agctcctgta ctaaaaggtt ccgctgaaga caaacataat 120
gaaggtagaa atagtcactt atggattgtc aatagggcaa ttgacattat gtccagggat 180
aagaccgtag ttaagccaaa tgagacagca ttactgaacg agtggagaga cgacttagag 240
aatggtatct attcagcaga ctacgaaaaa ccatacttcg ataattgggac gttcgcatac 300
catttctacg atccagatac tcaaaaagtc tacatccctt ttgctaaaca cgccaaggaa 360
acaggtgcta agtatttcaa gcttgcaggt gaagcttata agaacaagaa tatgaagcaa 420
gcattcttct acttaggatt atctttacat tatcttgggt acgttaatca acccatgcat 480
gctgccaaat ttacaaaact tagtactgag gctcccatct ttcactccaa atacgaaaac 540
tttgtcgata ccatcaagga caactacaaa gttgcagatg gcaacggata ttggaattgg 600
aaggaacaaa atagtgaaga ctggatatac ggcgcagccg ttgctagtaa gcaagactat 660
tcctctatcg tgaacgatac cactactagt tggttcctta aggcagctac atcacaagaa 720
tacgcaaaaa aatggagagc tgaagtaaca ccacttctg gcaagagatt gatcgaagct 780
cagagagtga ctgccggata cattcacctt tggtttgata catacgtcaa tagataa 837

```

```

SEQ ID NO: 422      moltype = DNA length = 840
FEATURE             Location/Qualifiers
source              1..840
                    mol_type = genomic DNA
                    organism = Bacillus mycoides

```

```

SEQUENCE: 422
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagctcat 60
gaaaacgagt gtcgaattga cctccaatc gctcagaggt ggagtgcga gagcatccac 120
aatgaaggcg ttagtccaca tctttggatt gtaaacagag caattgacat aatgtctcaa 180
aacactacag tagttaaaca aaacgagacg gcaacttcta atgaatggag aaccaacct 240
gaagaaggca tctattcagc agactataag aatccctatt acgaccattc tacctttgca 300
tctcacttct atgacccgga ttctggaaaa acctacattc cgtttgctaa acaagcaaaa 360
cagactgtgtg ccaataactt caaactagct ggtgaagctt atcagaacaa agacatgaag 420
aacgcctttt tctacctagg tttatctttg cactatttgg gagatgtcaa ccaaccaatg 480
catgctgcca atttcaacca cattagtcat cctttcggct tccattccaa gtatgagaac 540
tttgtcgaca cagtcaaaaga caactataga gtaacagatg gcgatggtta ctggaattgg 600

```

-continued

```

aaatctgcaa atccagagga atgggtccat gctagtgcaa gtgcccgaag agcagatttt 660
ccatctattg tgaacgataa caccacaaac tgggtcttga aagctacggt atcacaagat 720
tccgcagaca aatggagagc agagggttac cctgtaactg gtaaaagggtt aatggaagct 780
caaaggataa ctgctggcta cattcaccta tgggtcgata cgtacgtcaa taacaaataa 840

```

```

SEQ ID NO: 423      moltype = DNA length = 876
FEATURE            Location/Qualifiers
source              1..876
                    mol_type = genomic DNA
                    organism = Penicillium cinnamopurpureum

```

```

SEQUENCE: 423
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagctggt 60
ccgttaagaa gggatgtttc atccgatgat ttgaaacaat tgactttgtt cgctgaatac 120
gctagtgcac cctactgcac aaacaacatc aattctactg gcgacgcatt gtcagtgtga 180
gaaggtaatt gccagccggt tcagtcagca acgactaaaa ccttgatga gttcaatgat 240
agtaccgaat ttggagatgt cgctgggttc ttggctgcag acgaaacaaa cgaacttttg 300
gtactatcct ttaggggttc taggacaatt tccacttggg tggcaaacct ggattttggc 360
ttgaccgata catcagatct gtgttctgga tgcgaagcac atggtggatt ttggaatca 420
tggcagactg ttaccgatga cataacctcc aaaatcgatg ctggtttgaa aagtcaccct 480
ggatatacgg ttgtottaac aggtcatttc ttgtgtgcag ctatggccac attgggaggt 540
acagccttac gtaacgctgg ttacaagatc aagttgtata catatggtga accaagagtg 600
ggcaatgagg ctttagcaaa gttacattact aaacaagggt atttgtatag ggtcacacat 660
gcagatgatg tcgttccgaa ggttccacca gcatcatttg gattttctca tgcacacct 720
gagatttgga ttacctccgg aaacaataag accgtttcta ccagtgcatt caaagtcatt 780
caaggcgtag gatcaaaaga cggaacgctt ggtaccatca atccagacat tgaggctcac 840
aattggtaca tcgtgcatat agcggatgtt caataa 876

```

```

SEQ ID NO: 424      moltype = DNA length = 885
FEATURE            Location/Qualifiers
source              1..885
                    mol_type = genomic DNA
                    organism = Talaromyces verruculosus

```

```

SEQUENCE: 424
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagctgaa 60
cccatctata gaaggaagat agctacttct cttttggata gtttggactt attcgacag 120
tattctgcgc ctgcttattg ttccgcaaac ttggatacta caggtagctgc actggcttgc 180
aatgtgggta actgccagc tgtggaaagca gcagatacta ccatattgta ttcttttgat 240
tcccatcaaa gctttggtga tgcacccggt tacattgccc ttgatgaatc ccatggctac 300
atcatcttat catcagaggt tacgtcaaac ctggagaatt tcatcgcaac tttagatag 360
cagttgattg atgcaagttc catttgtagt ggttgtaaaag ttcataaggg tttttggaat 420
acctgggaaa accgttgctc agatgtaact tcacaaatca aggcagctct gctctgctat 480
ccagactata ccttagttgc aactggtcac tcatggggag ctgctctggc agccatcgct 540
gctaccgtct tccgtgcatc tggatatact gttcaattgt acaactatgg cgaacctagg 600
ataggcaatt tggcttagc agacttcatt acaagtgaat cttcaggaaac aaactatcgt 660
gtcacacatt caaatgacat cataccacaa ttacctctcg gactgttagg atatcaccac 720
ttttcccttg agtactggat taccagtaag gacaatgtca cagttaccga tagtgatgtt 780
gtggaaatca aaggtgttga ttctacagac ggtaacgatg gtactgctgg tgctccatt 840
gaggcccata cttggtactt tgtctacatt tcagagtgtt tgtaa 885

```

```

SEQ ID NO: 425      moltype = DNA length = 891
FEATURE            Location/Qualifiers
source              1..891
                    mol_type = genomic DNA
                    organism = Talaromyces cellulolyticus

```

```

SEQUENCE: 425
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagctgca 60
ccaaaacctt ttcatagaag gactatacct acttccttac ttgacaactt agaccttttt 120
gctcagatata ttgcagcagc ttattgttca gccaaacttag agactacggg taccgcacta 180
gcctgtaatg tgggtaattg tccagctggt gaagccgcag ataccattat cttgtatagc 240
tttgattcat cctcctcttt cggtagcgtc acaggatacg ttgcagttga tgaatcaaat 300
gagtagacatta tcttaagctt cagaggtagt agtaacttgg agaattggat cgccaaacta 360
gatatacctt tgattgatgc ttcatcaatt tgttccggtt gtaccgttca cgagggcttc 420
tgggacacat gggaaactgt tgcaagtgat gtcactagcc aaattgaatc cgctttgagt 480
acatatccca actatacctt agttgcaact ggtcactcct taggcggtgc tcttgagcc 540
atcgagctca cagttcttag agcttctggg tacacagtc agttatacaa ctacggtcaa 600
ccaaggattg gcaacttagc actagcagac ttcataacaa gtgaaacgct cggaaccaac 660
tatagagtta cacattcaga tgatatagtc ccgaagcttc caccagagtt acttggttat 720
caccacttta gtcctgagta ctggatcaca tcaaacgata atgttactgt gacagactca 780
gatgttggtg aaattcaggg tgtggattca actgctggca acgacggtac cagcggagat 840
tcaattgatg ctcattcttg gtactttgtt tccattagtg agtgttccta a 891

```

```

SEQ ID NO: 426      moltype = DNA length = 900
FEATURE            Location/Qualifiers
source              1..900
                    mol_type = genomic DNA
                    organism = Penicillium megasporum

```

```

SEQUENCE: 426
atgagatttc cttcaatttt tactacagtt ttattcgag catcctccgc attagctggt 60
cccgtttcag tattgagaag agacgaggac gtatccgcct ctgttttgct tgagctagac 120

```


-continued

```

ttctttttctc agtattcagc agccgcttac tgttccacaa acatcaattc tgcggtacc 180
aaattgactt gtagtgaggg tatttgtcca cttgttgaaa acgcagatac agaaacottg 240
gacgaattcg atgaaagtgc ctcttacggg gatgtcgctg gtttcattgc tgtggatcgt 300
acgaatgaac ttctagtctt gtcatttaga ggggtctgcta gtttcagtaa ctgggtggcc 360
aacattgact tattoctaga tgatgcctct tccgtctgca gtggttgoga agttcattct 420
ggtttctggg atgcatggca gactgttgaa ggacagatta ccacagctct aggttctgcc 480
atggaacat atccagggtta tacgcttgte tttaccggac acagctatgg tgcagctttg 540
gctgctattg ccgcaacccat ctttagaaac agtggctata ctgttgagtt gtacaactat 600
ggacaacctt gaattggaaa tctagctctg gcagagtaca ttacaaatca gaacaagggt 660
ggtaactata ggggtgacaca tacagacgac atagtcctta aggtccacc aaagataacg 720
gggtatcacc atgcttcccc tgagtattgg atcacatccg gcaacaatgt tactgttacg 780
actagcgatg tacagttgat aactggcgtg gacagcacat ctggtaatga tggtagatca 840
gatgactctg ttgaagctca caggtgggtac tttgtacaca tttctatgtg cacaataata 900

```

```

SEQ ID NO: 427      moltype = DNA length = 837
FEATURE
source             Location/Qualifiers
                   1..837
                   mol_type = genomic DNA
                   organism = Bacillus toyonensis

```

```

SEQUENCE: 427
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcac 60
gagaaaaacc aaggtcataba tgtgaacatc attcagtagt ggagtgtgga ggacaagcac 120
agtgaagggt tcaactctca cttatggatc gtcaatagag caatagacat tatgtccaga 180
aatactaagt tggtaaaagc agatcaaatc atactgttaa acgagtggag aacagatctt 240
gaaaacggga cttattcagc agatcatgag aatccgtatt acgataacag cactttggtg 300
ttctacttct atgatccaga cgacggttct acctatatcc catttgccaa acaagcaaaag 360
gaaactgggg ctaagtactt taagcttgct ggagaatctt acaagaataa ggatattgaag 420
caagccttct tctatcttgg cgtctcttta cattatctgg gtgatgtaaa ccaacctatg 480
catgcagcca accttacaac tctatcttat ccccaaggct tccactccaa atacgagaac 540
tttgttgata ccatcaagga caattacaaa gttacagatg gtaatggcta ttggaactgg 600
aaaggcattc atcctgaaga ttggatacat ggcgctgctg tcggagccaa gcaagacttt 660
tcaggtattg tcaatagcaa cacaaagtct tgggtcgtca aggtctgctg tccccaatca 720
tatgcagata agtggagagc agaagtaacc ccaatgaccg gaaaaagggt gattgaagca 780
caaagagtta cagcaggata catacagtta tgggtcgata cctatgtgaa cagataa 837

```

```

SEQ ID NO: 428      moltype = DNA length = 837
FEATURE
source             Location/Qualifiers
                   1..837
                   mol_type = genomic DNA
                   organism = Bacillus sp.

```

```

SEQUENCE: 428
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcac 60
gaggataacc accaagatcc tcccatagca ttgaagtgga gtgcggaagg tgtgcacaac 120
gagggcggtat cctcccatct gtggattgtc aatagagcta tagacataat gtctcagaat 180
actactgtcg ttaacaaaaa tgaacacagc ctactgaatg actggagaac caatttggag 240
gaaggcatct attctgcaga ctacaaaaac ccatactatg acaattccac gtttgcagat 300
cacttctatg atccagatcc cgagaaaaac tacattccct ttgctaagca agctaataca 360
accggtgcca aatacttcaa actagccggt gaagcttacc agaacaaga catgaaaaac 420
gcattcttct atttgggccc aagtttgccac tatttggggg atgttaatca acctatgcat 480
gtgccaact ttaccaacat atcccatcca ttcggtttcc actccaagta tgagaacttt 540
tagatactg tcaaggataa ctatcgtgtt acagatggaa atggttactg gaattggaaa 600
tctgccaatc cagaagaatg ggtgcatgag agtgctgctg ccgcaaggcc cgattttcct 660
tcaattgtta atgataacac caaaaagtgg tttctaaaag ccgctgtttc tcaagattca 720
gcagataagt ggagagctga agtaaacact gttacgggaa aaagattgat agaagctcaa 780
agaattactg caggatacat ccacttatgg ttcgacacat acgtaaacaa caataa 837

```

```

SEQ ID NO: 429      moltype = DNA length = 846
FEATURE
source             Location/Qualifiers
                   1..846
                   mol_type = genomic DNA
                   organism = Bacillus manliponensis

```

```

SEQUENCE: 429
atgagatttc cttcaatttt tactacagtt ttattcgcag catcctccgc attagctcat 60
ggcaaccacg atgcaagcaa tgattccgga ataagtatat cccaagatg gtctgctgaa 120
gaaatgcatg ctgaaggcaa aaactctcac ttatggatag taaacagagc aatcgacatt 180
atggccagag atacaactgt tgtgaaagaa aatgaggtgg cttgtgtaaa cgagtggagg 240
acagatcttg aagatggaat ctacacagca gattacgaga atccttacta cgacaattca 300
acattcgctc ctactctcta tgatccggac acggacgata cctacattcc gttcgccaaa 360
aacgctaaag tgactgtgtc caagtacttc aagttagctg gggaagcata cgaacagcaa 420
gatatgcaac aagcctctct ctacttaggt ttatccttgc actactttgg agacatcaat 480
caaccaatgc atgctagcaa ctttacaac atttcccatc cattcggtct tcactctaag 540
tatgagaact ttgtcgatgc catcaaagct ccatatgcag tcaccgattc aaaagggttac 600
tggaaacttt caggaggtac gccagaagag tggttacaca ctgctgcagt ggctgccaaa 660
aaggatgctc caggcattgt gaatgaaaca acaaagtcct ggttcttgaa agcaagtgtt 720
tcccaagaat atgccaatat gtggagagct gaggttactc ccgaaaactgg tgccagattg 780
atggaagccc aaagggcaat ggcaggatgc atccacttat ggttcgatac ttatgttaat 840
cgtaa 846

```

```

SEQ ID NO: 430      moltype = DNA length = 882

```

-continued

FEATURE
source

Location/Qualifiers
1..882
mol_type = genomic DNA
organism = *Penicillium simplicissimum*

SEQUENCE: 430

atgagatttc	cttcaatttt	tactacagtt	ttattcgcag	catcctccgc	attagctgcc	60
ccagctagat	cgtacctag	agatgtatct	gcttctgtct	tggaacagtt	taccttatac	120
gcacagtggg	ctgcogctgc	ttattgttcc	aacaacttgg	attctacagg	tgatgcaatc	180
acttgcgctg	gcggttactg	cccagaagtg	gaatcatcta	ctactattag	tctttctgag	240
ttcaatgaca	caaatgactt	tggtgatacc	gcaggtttcg	ttgcogttga	taagaccaat	300
aagcaaatgg	tcgttgcttt	cagaggtagc	aaaagcatct	caaattggat	cgcagattta	360
gattttgggt	tgacagatgc	aagtaacttg	tgtagtggtt	gtgaagctca	tacoggtttt	420
cttgaagcat	gggaaccgtg	tgccgattct	attacttcac	aaatcggagc	agctatgaaa	480
acttatagtg	gttatacctt	agttgtcact	ggtcactcat	tgaggaggtg	tatcgcagcc	540
atcggtgcta	cogtccctcg	taatgctgga	tatactttgg	attgttacac	ctttggccaa	600
cctagggctg	gcaatttggc	cttagcaacc	ttcttaacaa	agcaaggcaa	caatagaaatg	660
actcacctaa	atgatatttg	accaagatta	cctccaacta	gttttgattg	ctcacattca	720
tctcccgaat	actggatcac	ctctgcagac	gatgtgacgg	taaccacatc	agacattgaa	780
gttattgagg	gaatcgattc	aacagctggg	aatgcccggg	agttgattga	atcagttgct	840
gcccattgat	ggtacatcat	tgatattgat	ggatgtgaat	aa		882

SEQ ID NO: 431

FEATURE
source

moltype = DNA length = 888
Location/Qualifiers
1..888
mol_type = genomic DNA
organism = *Penicillium arenicola*

SEQUENCE: 431

atgagatttc	cttcaatttt	tactacagtt	ttattcgcag	catcctccgc	attagctgca	60
ccttttcaat	tgtggcttag	agctgtcact	ccttcogtct	tatcaaagct	agacttatac	120
ggccaatatg	ctattgcgcg	atactgtgat	gacaacattg	cttcagccgg	tacgggaagtg	180
acatgtagcg	ctggtaactg	tcctttgggt	caagctgcaa	ccacaaatac	attaagttag	240
tttaacgaat	caaatgaatt	cggtagcgtt	gctggctttt	tcgcccgtga	tacaacaaac	300
caagctctag	ttttgagctt	tcgtgggtct	catacagctg	acaattggat	tgcaaaacttg	360
gattttggat	tgacctctgt	ttccacttta	tgttcaggat	gcaaagctca	cactggcttc	420
tggaaggctt	ggaatactgt	agcttcagac	atcgcagcag	ctgtggatgc	tgcaacaagt	480
acttaccctt	cttaccctat	tatctttact	ggacattcat	acggagctgc	cttggctgcc	540
ctagctgcca	caactatgag	aaatgctggg	tatagcatag	aattgtatac	atacggacag	600
ccaagaattg	gaaatacagc	tttagcaacc	tacattacaa	atcaaaacaa	aggtggcaac	660
tatagagtaa	ctcatacaaa	tgatattgtc	cctaggctgg	tacctagact	tctaggatac	720
tcacacttct	ctccagagta	ttggatcaca	agtggcaata	acgtaacagt	tactgcctcc	780
gacattactt	tagttactgg	tctcgatagc	aatgggtgga	atgcaggcga	gttactacag	840
tcogttgaac	ctcactattg	gtactttgtg	gaagttgaag	attgctaa		888

SEQ ID NO: 432

FEATURE
source

moltype = DNA length = 891
Location/Qualifiers
1..891
mol_type = genomic DNA
organism = *Aspergillus aculeatus*

SEQUENCE: 432

atgagatttc	cttcaatttt	tactacagtt	ttattcgcag	catcctccgc	attagctgct	60
cccgacacct	ttgagcgtag	atcagctctc	acaacactac	ttgatcaaat	ggatttggtt	120
gctcagtagt	ctgcagctgc	ttactgtagt	acaaacatag	atagtgcata	aaccgccttg	180
agctgctcag	cagacaattg	tccttagtgc	gttgctgctg	cccctacagt	ccttgatgag	240
ttcaatgaga	ctgcagaatt	cggggatacc	gctgggttcg	ttgcagtgga	cagtaccaac	300
aaagccatcg	ttgttgcttt	tagagggagt	agcgacttat	caaattggat	cgcacaacatt	360
gattttgggt	tgacagacgc	cagttccatt	tgtacaggtt	gcgaaataca	ctctgggttc	420
tggaaggcat	gggaacacgt	agcttcaact	attgcctcta	aggttgaagc	agccgttaca	480
acttattcag	attatgacgt	tgtctttacg	ggctactcct	taggtgccgc	tctggctgcc	540
attggcgcta	cagtcttgag	gaacgatggg	tatactgtag	acttgtacaa	ctttgggtcaa	600
ccgagaatag	gtaacttagc	cttggcagac	tacataacgg	atcagaataa	gggttccaat	660
tacagagtga	cgcacacaga	cgatattgta	cctaaggctc	ctcctgagtt	gttaggatat	720
caccatttct	ctccagagta	ttggataaca	agtataatg	acgtcactgt	taccacttca	780
gacattaccg	aagtcacagg	ggtagacagt	accgctggaa	atgacggaac	tttgttggat	840
agtgtttccg	ctcacaagtt	ctatttcgag	tatatctccg	cttgtgatta	a	891

SEQ ID NO: 433

FEATURE
source

moltype = DNA length = 1107
Location/Qualifiers
1..1107
mol_type = genomic DNA
organism = *Bacillus acidiceles*

SEQUENCE: 433

atgagatttc	cttcaatttt	tactacagtt	ttattcgcag	catcctccgc	attagctatt	60
acttccttct	ttggcaacta	tcagaaagcc	ttcgcttgga	gtgatgagga	cgtacataat	120
caagatcatt	ctactcacca	cttcataggt	aatggctccg	tcaaatgat	tgacagacaat	180
acaaatcctg	ccatcaacaa	acctacaacc	ttgctgaacc	agtttagaga	tagatgggaa	240
caaggcttat	acgatgcaga	ccacatcaat	ccctctctac	acacagggac	ctttatgagt	300
catattctacg	atccccgatac	tcagaccaac	tatactggcg	caagttatcc	aacagctaga	360
caaagcgggtg	ccaagtactt	caacttagcc	agtgattact	acaagaaagg	tgacttcaac	420
aatgcctttt	actatttggg	cgttagtctt	cattacttta	cagatgttac	tcagccttta	480

-continued

```

catgcatcaa acatttccaa ttggatcac cagctcccg gttaccacag caagtatgaa 540
acttacgctg aatcaatcca aaatgagatg acgatgcccg attccggatt gtacaattgg 600
atagcctcaa cagaccccg agcatggatc catcaagccg ctgtgcaagc aaaatctgtt 660
ttgccacagg tttggaatga cactatcatt aactactttt ggcaagctgc ctattccaat 720
tactactctg ccattgtggaa atctgaagtc aagaatccta ctttggatca attgaatcaa 780
gctgaacgtg agaccgcagg cttcattgac atgttctttc gtgttaacgg agtagaaatg 840
ccggtcaccg tgtataagga aaatgccttt tcaggtgctt ctgagttatt gggttctggt 900
aactacgatt atgatcaact tgtaagggt attggaaatg atacaatatc ttccattcat 960
attgctccag gctatcaggt gactctattc tcagacgcca actacaaagg tgctagtact 1020
gtgcttacia atgatgtgca cgatctaggc aacttttctc atcaggtttc ttccataaag 1080
gtcgccaaaa tctcagcact taagtaa 1107

```

```

SEQ ID NO: 434      moltype = DNA length = 21
FEATURE            Location/Qualifiers
misc_feature        1..21
                    note = Artificial DNA Primer
source              1..21
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 434
tcttttcgcg ccctggaaag g 21

```

```

SEQ ID NO: 435      moltype = DNA length = 49
FEATURE            Location/Qualifiers
misc_feature        1..49
                    note = Artificial DNA Primer
source              1..49
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 435
tgttttatat ttgttgtaaa aagtagataa ttacttcctt gatgatctg 49

```

```

SEQ ID NO: 436      moltype = DNA length = 22
FEATURE            Location/Qualifiers
misc_feature        1..22
                    note = Artificial DNA Primer
source              1..22
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 436
gtgacaacaa cagcctgttc tc 22

```

```

SEQ ID NO: 437      moltype = DNA length = 21
FEATURE            Location/Qualifiers
misc_feature        1..21
                    note = Artificial DNA Primer
source              1..21
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 437
agctaattgag gaggatgctg c 21

```

```

SEQ ID NO: 438      moltype = DNA length = 22
FEATURE            Location/Qualifiers
misc_feature        1..22
                    note = Artificial DNA Primer
source              1..22
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 438
acagaagacg ggagacacta gc 22

```

```

SEQ ID NO: 439      moltype = DNA length = 18
FEATURE            Location/Qualifiers
misc_feature        1..18
                    note = Artificial DNA Primer
source              1..18
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 439
ggggtcgcaa cttttccc 18

```

309

The invention claimed is:

1. A recombinant *Saccharomyces* yeast cell comprising a heterologous polynucleotide encoding a phospholipase and a heterologous polynucleotide encoding a glucoamylase,

wherein the phospholipase has a mature polypeptide sequence at least 90% sequence identity to the amino acid sequence of SEQ ID NO: 236, and

wherein the cell is capable of higher ethanol yield and/or reduced foam accumulation when compared to fermentation using the same process and an identical cell without the heterologous polynucleotide encoding the phospholipase under the same conditions.

2. The recombinant yeast cell of claim 1, wherein the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase.

3. The recombinant yeast cell of claim 1, wherein the fermenting organism comprises a heterologous polynucleotide encoding a protease.

4. The recombinant yeast of claim 1, wherein the cell is a *Saccharomyces cerevisiae* cell.

5. The recombinant yeast of claim 1, wherein the phospholipase has a mature polypeptide sequence having at least 95% sequence identity to the amino acid sequence of SEQ ID NO: 236.

6. The recombinant yeast of claim 1, wherein the phospholipase has a mature polypeptide sequence having at least 97% sequence identity to the amino acid sequence of SEQ ID NO: 236.

7. The recombinant yeast of claim 1, wherein the phospholipase has a mature polypeptide sequence having at least 98% sequence identity to the amino acid sequence of SEQ ID NO: 236.

8. The recombinant yeast of claim 1, wherein the phospholipase has a mature polypeptide sequence having at least 99% sequence identity to the amino acid sequence of SEQ ID NO: 236.

9. The recombinant yeast of claim 1, wherein the phospholipase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of SEQ ID NO: 236.

10. A method of producing a fermentation product from a starch-containing or cellulosic-containing material comprising:

(a) saccharifying the starch-containing or cellulosic-containing material; and

(b) fermenting the saccharified material of step (a) with the recombinant yeast of claim 1.

11. The method of claim 10, wherein saccharification of step (a) occurs on a starch-containing material, and wherein the starch-containing material is gelatinized.

12. The method of claim 10, wherein saccharification of step (a) occurs on a starch-containing material, and wherein the starch-containing material is ungelatinized.

13. The method of claim 10, wherein saccharification of step (a) occurs on a starch-containing material, and wherein the method comprises liquefying the starch-containing material by contacting the material with an alpha-amylase prior to saccharification.

14. The method of claim 10, wherein liquefying the starch-containing material and/or saccharifying the starch-containing material is conducted in presence of exogenously added protease.

15. The method of claim 10, wherein the recombinant yeast comprises a heterologous polynucleotide encoding an alpha-amylase.

16. The method of claim 10, wherein the recombinant yeast is a *Saccharomyces cerevisiae* cell.

17. The method of claim 10, wherein the method results in higher yield of fermentation product and/or reduced foam accumulation when compared to the same process using an identical cell without the heterologous polynucleotide encoding the phospholipase under the same conditions.

18. The method of claim 10, wherein the phospholipase of the recombinant yeast has a mature polypeptide sequence having at least 95% sequence identity to the amino acid sequence of SEQ ID NO: 236.

19. The method of claim 10, wherein the phospholipase of the recombinant yeast has a mature polypeptide sequence having at least 97% sequence identity to the amino acid sequence of SEQ ID NO: 236.

20. The method of claim 10, wherein the phospholipase of the recombinant yeast has a mature polypeptide sequence that comprises or consists of the amino acid sequence of SEQ ID NO: 236.

* * * * *